Gauge-Covariant Variational Free Energy and Renormalization

Covariant Information Geometry, Timeless Inference Histories, Local–Collective Bounds, and Exact Effective Scale Theory

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General Gauge-Variational Foundations

Chapter 1

Scope, Claims, and Reading Map

This document develops a gauge-covariant variational theory from typed probability objects and then asks which parts survive coarse description. Its ambient geometry consists of one principal gauge bundle $P \to \mathcal{C}$ with structure group G, and two associated law bundles $\mathcal{E}_b = P \times_{\widehat{\rho}_b} \mathcal{B}_b$ and $\mathcal{E}_m = P \times_{\widehat{\rho}_m} \mathcal{B}_m$. The representations ρ_b and ρ_m may be inequivalent and may act on fibers of different dimensions; the hats denote their induced actions on probability laws. Belief and model coordinates may use different local sections of P. Their unique relative principal-frame field is a G-valued comparison, not a map between the two associated fibers. The declared cross-channel morphisms

$$\Phi : \mathcal{E}_b \longrightarrow \mathcal{E}_m, \qquad \widetilde{\Phi} : \mathcal{E}_m \longrightarrow \mathcal{E}_b$$

cover $\mathrm{id}_{\mathcal{C}}$; they are neither principal-bundle morphisms nor assumed inverses. Separate connections induce same-channel parallel transports Ω_γ on $\mathcal{E}_b$ and $\widetilde{\Omega}_\gamma$ on $\mathcal{E}_m$. Parallelness of either cross map is an additional hypothesis, measured otherwise by its covariant defect. The two channels may use different connections on the same P; neither a common local frame nor an equal connection is assumed. Independently declared graph links are denoted Θ_e^b, Θ_e^m on oriented edge copies (abbreviated by endpoints when the copy is unique), and are identified with base parallel transport only after a curve assignment and an equality hypothesis. These distinctions are the type contract of Chapter 2. [DEFINITION]

On a finite design the theory fixes one normalized generative joint $P_\theta(do, dY \mid X)$ and one correlated recognition law $Q_X(dY \mid o)$. Under the stated absolute-continuity and integrability hypotheses, there is one exact identity,

$$\log p_\theta(o \mid X) = \mathcal{L}(Q_X; X) + D_{\mathrm{KL}}\left(Q_X \| P_\theta(\cdot \mid o, X)\right). \tag{1.1}$$

It is invariant under independent belief- and model-coordinate pushforwards. These passive re-expressions form a product of frame-coordinate changes; they must not be confused with replacing the ambient diagonal G-gauge symmetry by a principal $G \times G$-bundle. Equality holds exactly when the recognition law is the posterior as a measure; matching coordinate marginals is insufficient. Excluding the posterior makes every pointwise gap positive, but does not by itself make the optimized infimum positive or attained. [ESTABLISHED]

The logical order is therefore

$$\begin{aligned}\text{typed bundles and kernels} &\longrightarrow \text{normalized joint}\\ &\longrightarrow \text{local and collective ELBOs}\\ &\longrightarrow \text{covariant informational pullbacks}\\ &\longrightarrow \text{relational inference histories}\\ &\longrightarrow \text{general coarse maps} \longrightarrow \text{exact effective RG}\\ &\longrightarrow \text{Gaussian realization.}\end{aligned}$$

The general coarse layer distinguishes three operations that are often conflated. A normalized Markov kernel pushes laws forward, preserves the observation evidence when it acts only on latent variables, and contracts relative entropy. Energy precomposition restricts an unnormalized energy along a sample-space map and requires a separately declared coarse reference measure and finite normalizer. Parameter aggregation is a conditional closure statement inside a selected graph-exponential family. A renormalization further requires comparison or rescaling maps that type different scales; without them there is only a changing-space coarse diagram, not a fixed point problem. [ESTABLISHED]

For an arbitrary finite agent hypergraph, normalized interaction-record kernels give one collective VFE and, by disintegration, a conditional VFE for every agent or block. A unilateral local change equals the corresponding change of the collective VFE, so local evolution is a blockwise collective descent. The local objectives do not sum symmetrically because shared factors occur in every incident coordinate objective. A categorical source label places attention inside the fixed joint, and a standard-Borel observation kernel is equivalent to a message from an operational environment node. These results are proved in Chapter 7. [ESTABLISHED]

A chosen connection turns the derivative of a smooth statistical section into a vertical first jet. Pulling the vertical Fisher and Amari tensors back by that jet gives gauge-invariant, connection-relative tensors on the contextual base; the Fisher tensor may be degenerate. This tensor pullback is distinct both from restricting a parameter manifold to a submodel and from a dynamical-systems pullback attractor. The subsequent relational construction quotients regular natural-gradient solution curves by orientation-preserving reparameterization and measures them by Fisher length. The resulting histories and durations are information-geometric objects on a declared configuration space. These distinctions are developed in Chapters 8 and 9. [ESTABLISHED]

The construction does not identify those histories or durations with a physical clock. [NOT-CLAIMED]

The exact law-level RG of Chapter 12 pushes the baseline, evidence submeasure, recognition law, posterior, and joint attention-event law through the same structural channel. Exact closure retains induced hyperedges, simultaneous root-framed holonomy and boundary data, and path memory. Cross-scale agent–meta kernels follow from posterior bridges, while differential beta functions and fixed points are typed only after an explicit rescaling returns the changing scale spaces to one common space. [ESTABLISHED]

The ordered path coordinate is neither a relational inference history nor a physical-time coordinate, and RG depth is a scale index. [NOT-CLAIMED]

The multivariate Gaussian enters only after that general structure. It realizes the smooth statistical tier in information coordinates and supplies explicit determinant gaps, Fisher charts, matrix-weighted interaction operators, aggregation formulas, Kron counterexamples, and generalized spectra. The Laplacian-plus-self-term interaction cone is a declared subfamily, not a consequence of every linear-Gaussian directed model. [HYPOTHESIS]

Within its exact domain, aggregation is closed, but closure alone does not select that cone or a nondegenerate partition. [ESTABLISHED]

Unrestricted matrix-weighted Kron closure fails, whereas the later Gaussian coarse chapter records the exact closed congruence-diagonal domain and its qualified maximality boundary. [ESTABLISHED]

The RG conclusions are deliberately narrower. Rescaling is declared rather than derived. [DEFINITION]

Bi-additive Gaussian parameters are closed under additive pushforward, and an identified homogeneous hierarchy gives an operator fixed ray, but the ray lies on the boundary of the full-space Gaussian probability domain. [ESTABLISHED]

Infinite-hierarchy attraction, uniqueness within a basin, and blocking-scheme independence remain open. [OPEN]

Finite numerical gates test specified diagnostics and controls; they do not prove those asymptotic claims. [NUMERICAL]

Identifying any repaired limiting object with a physical law remains the semantic claim B.1; it requires a complete probability repair, a target system, a validation criterion, and discriminating evidence. [OPEN]

Epistemic contract

Every nontrivial statement carries one of the following tags. The central open-obligation appendix collects unresolved conjectures and problems once; established claims remain with their hypotheses or proofs, and numerical claims remain bound to the numerical-provenance appendix.

A gesture is either discharged by proof, calculation, or a checked source, or converted into an open obligation with exact hypotheses. Likewise, `NOT-CLAIMED` is never used for a refuted statement. The obstruction chapter contains exact negative results; the interpretation chapter marks readings and empirical commitments without promoting them to theorems.

A physicist's fast path

A first reading may take the object and frame-covariance contract from Chapters 1 and 2, then read only the normalized-joint and ELBO identities in Chapters 3 to 5. The next pass should

Table 1.1: Status taxonomy: tags used throughout.

Tag	What it promises the reader
[ESTABLISHED]	Proved here, or a standard result cited to a source that has been checked.
[DEFINITION]	A declared type, construction, or convention. Nothing is being proved and the text says so.
[HYPOTHESIS]	A restriction the development adopts by choice. What it excludes, and where it is used, are stated.
[CONJECTURE]	Believed and precisely stated, but not proved. Stated sharply enough to be attacked.
[NUMERICAL]	Supported by computation only, with its measurement, seed, and control reported. Computation is not proof.
[OPEN]	Unsettled. What would settle it, and what obstructs it, are named.
[NOT-CLAIMED]	A statement the development deliberately declines to make. Declining is not refuting, and the text says which it is doing.

read the local–collective potential theorem and interaction-label attention in Chapter 7, then the covariant pullback and relational-history constructions in Chapters 8 and 9. The next pass should take the three-operation distinction, Markov coarse theorem, exact effective hypergraph/path-space closure, and typed scale diagram in Chapters 10 to 12. The Gaussian spine then consists of the interaction cone, recognition costs and Fisher geometry in Chapters 14 to 16, followed by Gaussian aggregation, congruence-diagonal Kron closure, holonomy, fixed-ray qualifications, generalized spectra, and numerical gates in Chapters 17 and 18. Finally, Chapters 19 and 20 separate exact no-go results and repairs from empirical and interpretive commitments. Disintegration proofs, convex-duality details, long rational witnesses, ledger rows, and protocol metadata can be recovered through their semantic cross-references on a second reading.

Chapter 2

The Geometric Setting

The ambient geometry has one principal gauge bundle over the contextual base and two associated statistical bundles, one for belief laws and one for model laws. The two associated actions of the same group may be inequivalent and may act on fibers of different dimensions. Belief and model frames may nevertheless be chosen separately as local sections of the common principal bundle, and the two channels may carry different connections. Cross-channel maps are additional associated-bundle morphisms; neither a relative frame nor a shared structure group supplies such a map by itself. A product-gauge theory with two independent principal bundles is recorded only as the optional extension in Section 2.12.

2.1 The contextual base

Definition 2.1 (Contextual base). Let $\mathcal{C}$ be a finite-dimensional smooth, second countable, Hausdorff manifold whose points c are called *contexts*. [DEFINITION]

This declaration fixes a type and proves nothing. The base is neither the population index set $V = \{1, \dots, N\}$ nor its interaction graph, neither latent sample space, and not physical spacetime. Its smooth structure supports curves and differential forms. A connection on a bundle over $\mathcal{C}$ defines transport in that bundle along curves in $\mathcal{C}$; it does not, unless separately specified as a connection on $T\mathcal{C}$, define parallel transport of tangent vectors of the base. The finite design $D = \{c_a\}_{a=1}^{M}$ used later is a finite subset of $\mathcal{C}$, not a continuum limit or a discretization theorem.

Throughout this development, $\mathcal{C}$ is fixed and timeless. A curve in $\mathcal{C}$ orders contexts and does not by itself represent an agent's evolution. Belief or model change at a fixed context is a curve inside one associated-bundle fiber, while simultaneous change over all contexts is a curve in a declared space of sections. A trajectory whose projection moves in $\mathcal{C}$ is a different construction and requires the horizontal–vertical distinction made by a chosen connection. These are typing declarations rather than physical conclusions. [DEFINITION]

2.2 One principal bundle and two associated statistical bundles

Definition 2.2 (Common principal gauge bundle). Let

$$\pi : P \longrightarrow \mathcal{C} \tag{2.1}$$

be a smooth principal right G-bundle for a Lie group G. For each agent i, choose an open support $\mathcal{C}_i \subseteq \mathcal{C}$ and two local principal-frame sections

$$u_i^b : \mathcal{C}_i \longrightarrow P, \qquad u_i^m : \mathcal{C}_i \longrightarrow P. \tag{2.2}$$

[DEFINITION]

The superscripts distinguish two choices of local section, not two principal bundles. They may be unequal. Because each principal fiber is a right G-torsor, there is a unique smooth map

$$h_i : \mathcal{C}_i \longrightarrow G, \qquad u_i^m = u_i^b \cdot h_i, \tag{2.3}$$

called the *relative principal-frame field*. Its existence uses only the common principal bundle and is independent of all associated representations. [ESTABLISHED]

Let $(\mathsf{K}, \mathscr{K})$ and $(\mathsf{M}, \mathscr{M})$ be the belief and model sample fibers in local coordinates, and declare normalized-law fibers

$$\mathcal{B}_b \subseteq \mathcal{P}(\mathsf{K}), \qquad \mathcal{B}_m \subseteq \mathcal{P}(\mathsf{M}). \tag{2.4}$$

A law fiber is not a kernel fiber. A generative transition belongs to a separately declared collection of normalized Markov kernels between specified measurable spaces. No density chart, common dominating measure, affine structure, or Fisher metric follows from (2.4).

Suppose the same group acts measurably on the two sample fibers through representations

$$\rho_b : G \longrightarrow \mathrm{Aut}(\mathsf{K}), \qquad \rho_m : G \longrightarrow \mathrm{Aut}(\mathsf{M}). \tag{2.5}$$

The representations need not be equivalent, faithful, or of equal dimension. Denote their induced actions on laws by

$$\widehat{\rho}_b(g)q = (\rho_b(g))_{\#}q, \qquad \widehat{\rho}_m(g)s = (\rho_m(g))_{\#}s. \tag{2.6}$$

[DEFINITION]

Thus $(\widehat{\rho}_b(g)q)(A) = q(\rho_b(g)^{-1}A)$, and similarly in the model channel. The hats distinguish actions on laws from actions on samples. Multiplying a matrix into a density is not the pushforward in (2.6). [ESTABLISHED]

Hypothesis 2.3 (Selected smooth tier). When differential statements are made, $\mathcal{B}_b$ and $\mathcal{B}_m$ are finite-dimensional smooth parametrized-measure models, each is closed under its G-action, and

$$(g, q) \longmapsto (\rho_b(g))_{\#}q, \qquad (g, s) \longmapsto (\rho_m(g))_{\#}s \tag{2.7}$$

are smooth. [HYPOTHESIS]

A finite-dimensional statistical interpretation additionally requires differentiability in quadratic mean, square-integrable scores, and the integrability required by each tensor. The Fisher tensor is a

Riemannian metric only when it is nondegenerate. Stratified and infinite-dimensional fibers require their own categories and are not included silently.

Definition 2.4 (Belief and model associated bundles). On invariant law fibers define

$$\mathcal{E}_b = P \times_{\widehat{\rho}_b} \mathcal{B}_b, \qquad \mathcal{E}_m = P \times_{\widehat{\rho}_m} \mathcal{B}_m, \tag{2.8}$$

using the quotient convention

$$(u \cdot g, \beta) \sim (u, g \cdot \beta). \tag{2.9}$$

[DEFINITION]

At the general tier these are associated measurable bundles. Under Hypothesis 2.3 they are smooth associated bundles. Smoothness of P alone does not upgrade an arbitrary law fiber.

The field h_i compares the two principal sections, even when ρ_b and ρ_m are inequivalent. It then produces two different represented operators, $\widehat{\rho}_b(h_i)$ on $\mathcal{B}_b$ and $\widehat{\rho}_m(h_i)$ on $\mathcal{B}_m$. It does not canonically produce an operator from $\mathcal{B}_b$ to $\mathcal{B}_m$. If "frame" instead means an arbitrary basis of an associated vector fiber that is not induced from a section of this same P, the existence of a G-valued relative field is not guaranteed. Moreover, when a representation is nonfaithful, represented frame data alone determine at most a coset modulo its kernel; the principal sections, rather than their represented images, are what make h_i unique.

2.3 Frame choices, gauge symmetry, and relative frames

Independent changes of the two local trivializations may be written

$$u_i^{b\prime} = u_i^b \cdot a_i, \qquad u_i^{m\prime} = u_i^m \cdot b_i, \qquad a_i, b_i : \mathcal{C}_i \to G. \tag{2.10}$$

They change associated coordinates by $\beta_b' = \widehat{\rho}_b(a_i)^{-1}\beta_b$ and $\beta_m' = \widehat{\rho}_m(b_i)^{-1}\beta_m$. The unique relative field becomes

$$h_i' = a_i^{-1} h_i b_i. \tag{2.11}$$

[ESTABLISHED]

These two passive coordinate choices must not be confused with a product gauge symmetry. The principal gauge group of the ambient theory is the single group of G-equivariant automorphisms of P over $\mathrm{id}_{\mathcal{C}}$, and each such automorphism acts simultaneously on both associated bundles. For such an automorphism F, define its local functions by $F(u_i^b) = u_i^b \cdot k_i^b$ and $F(u_i^m) = u_i^m \cdot k_i^m$. Equivariance and (2.3) imply

$$k_i^m = h_i^{-1} k_i^b h_i. \tag{2.12}$$

Thus the two local descriptions of one gauge transformation are related, not independent. By contrast, (2.10) simply permits two atlases to be replaced independently. Applying one principal gauge automorphism to both sections leaves their relative principal element unchanged, since equivariance gives $F(u_i^m) = F(u_i^b \cdot h_i) = F(u_i^b) \cdot h_i$.

2.4 Cross-associated-bundle morphisms

Definition 2.5 (Cross morphisms). At the general measurable tier, declare fiber-preserving measurable maps

$$\Phi : \mathcal{E}_b \longrightarrow \mathcal{E}_m, \qquad \widetilde{\Phi} : \mathcal{E}_m \longrightarrow \mathcal{E}_b, \qquad \varpi_m \circ \Phi = \varpi_b, \quad \varpi_b \circ \widetilde{\Phi} = \varpi_m, \tag{2.13}$$

where ϖ_b and ϖ_m are the associated-bundle projections. Both maps cover $\mathrm{id}_{\mathcal{C}}$. [DEFINITION]

These maps are extra structure. They are not principal-bundle morphisms, gauge automorphisms, same-channel transports, or assumed inverses. Under Hypothesis 2.3, differential statements additionally assume that they are smooth. In the frames u_i^b, u_i^m, let ϕ_i and $\widetilde{\phi}_i$ denote their local fiber representatives. The rechoices in (2.10) give

$$\phi_i' = \widehat{\rho}_m(b_i)^{-1} \circ \phi_i \circ \widehat{\rho}_b(a_i), \tag{2.14}$$
$$\widetilde{\phi}_i' = \widehat{\rho}_b(a_i)^{-1} \circ \widetilde{\phi}_i \circ \widehat{\rho}_m(b_i). \tag{2.15}$$

[ESTABLISHED]

Proof. Write $\beta_m = \phi_i(\beta_b)$ and substitute $\beta_b = \widehat{\rho}_b(a_i)\beta_b'$ and $\beta_m' = \widehat{\rho}_m(b_i)^{-1}\beta_m$. The second law follows by interchanging the channels. □

Proposition 2.6 (Equivariant fiber maps induce cross morphisms). Suppose maps $\phi : \mathcal{B}_b \to \mathcal{B}_m$ and $\widetilde{\phi} : \mathcal{B}_m \to \mathcal{B}_b$ satisfy, for every $g \in G$,

$$\phi \circ \widehat{\rho}_b(g) = \widehat{\rho}_m(g) \circ \phi, \qquad \widetilde{\phi} \circ \widehat{\rho}_m(g) = \widehat{\rho}_b(g) \circ \widetilde{\phi}. \tag{2.16}$$

Then

$$[p, \beta_b] \longmapsto [p, \phi(\beta_b)], \qquad [p, \beta_m] \longmapsto [p, \widetilde{\phi}(\beta_m)] \tag{2.17}$$

are well-defined cross morphisms. [ESTABLISHED]

Proof. The quotient representatives $(p \cdot g, \beta)$ and $(p, g \cdot \beta)$ have the same image under (2.17) exactly when the corresponding equality in (2.16) holds. □

Equivariance is sufficient for the representative-independent construction (2.17). A general bundle morphism is instead characterized by local maps satisfying the overlap laws

$$\begin{aligned} \phi_i \circ \widehat{\rho}_b(T_{ij}^b) &= \widehat{\rho}_m(T_{ij}^m) \circ \phi_j, \\ \widetilde{\phi}_i \circ \widehat{\rho}_m(T_{ij}^m) &= \widehat{\rho}_b(T_{ij}^b) \circ \widetilde{\phi}_j. \end{aligned} \tag{2.18}$$

In any selected regularity category admitting the two mapping objects and the composition actions displayed below, their local representatives first require conversion to one principal-frame atlas. In the belief atlas define

$$f_i := \widehat{\rho}_m(h_i) \circ \phi_i, \qquad \widetilde{f}_i := \widetilde{\phi}_i \circ \widehat{\rho}_m(h_i^{-1}). \tag{2.19}$$

Then f_i is the local representative of a section of the associated map bundle with action $f \mapsto \widehat{\rho}_m(g) \circ f \circ \widehat{\rho}_b(g)^{-1}$, and $\widetilde{f}_i$ obeys the analogous law with the two channels interchanged. Indeed, (2.18) and $T_{ij}^m = h_i^{-1} T_{ij}^b h_j$ give $f_i \widehat{\rho}_b(T_{ij}^b) = \widehat{\rho}_m(T_{ij}^b) f_j$ and $\widetilde{f}_i \widehat{\rho}_m(T_{ij}^b) = \widehat{\rho}_b(T_{ij}^b) \widetilde{f}_j$. Thus the compatible family is a map-bundle section only after the displayed conversion, and h_i does not replace the cross-fiber map. For example, if Φ is induced by the first map in (2.17), then $\phi_i = \widehat{\rho}_m(h_i^{-1}) \circ \phi$ and hence $f_i = \phi$. [ESTABLISHED]

2.5 Chosen connections, parallel transport, and covariant defects

Definition 2.7 (Two connections on the common principal bundle). Under Hypothesis 2.3, choose

$$\omega_b, \omega_m \in \Omega^1(P, \mathfrak{g}) \tag{2.20}$$

as principal connections on P. They induce horizontal distributions H^b, H^m on the two associated bundles and, for every piecewise-smooth $\gamma : [0,1] \to \mathcal{C}$, parallel transports

$$\Omega_\gamma : (\mathcal{E}_b)_{\gamma(0)} \longrightarrow (\mathcal{E}_b)_{\gamma(1)}, \qquad \widetilde{\Omega}_\gamma : (\mathcal{E}_m)_{\gamma(0)} \longrightarrow (\mathcal{E}_m)_{\gamma(1)}. \tag{2.21}$$

[DEFINITION]

Explicitly, ω_x selects the unique horizontal lift of γ through a chosen point of $P_{\gamma(0)}$; carrying the associated-law coordinate along that lift defines the corresponding map in (2.21). This is transport *over* the base curve, not transport of vectors in $T\mathcal{C}$. A tangent-vector transport would require a separately chosen affine connection on $T\mathcal{C}$, unless P and its representation have explicitly been identified with the frame bundle of $\mathcal{C}$.

The two connections need not agree. Their difference $\omega_m - \omega_b$ is horizontal and Ad-equivariant, hence descends to an $\mathrm{Ad}(P)$-valued one-form on $\mathcal{C}$. In an associated vector-bundle realization, the induced horizontal distributions also define covariant derivatives ∇^b, ∇^m. No linear covariant derivative is assumed on a merely measurable law fiber.

Let $A_i^b = (u_i^b)^* \omega_b$ and $A_i^m = (u_i^m)^* \omega_m$. Under (2.10),

$$A_i^{b\prime} = \mathrm{Ad}_{a_i^{-1}} A_i^b + a_i^{-1} da_i, \tag{2.22}$$

$$A_i^{m\prime} = \mathrm{Ad}_{b_i^{-1}} A_i^m + b_i^{-1} db_i. \tag{2.23}$$

If endpoint coordinates are changed by a_0, a_1 in the belief frame and b_0, b_1 in the model frame, the represented transports obey

$$\Omega'_\gamma = \widehat{\rho}_b(a_1)^{-1} \circ \Omega_\gamma \circ \widehat{\rho}_b(a_0), \tag{2.24}$$

$$\widetilde{\Omega}'_\gamma = \widehat{\rho}_m(b_1)^{-1} \circ \widetilde{\Omega}_\gamma \circ \widehat{\rho}_m(b_0). \tag{2.25}$$

[ESTABLISHED]

These are the standard connection and endpoint laws for the quotient convention (2.9) [43, 54]. They also show why a same-point frame comparison and transport along a nonconstant curve are different objects.

Definition 2.8 (Covariant defects). Let $e \in \mathcal{E}_b$ and $f \in \mathcal{E}_m$ lie over $c \in \mathcal{C}$, and let $X \in T_c\mathcal{C}$. For smooth associated fibers, define the vertical-valued tensors

$$(\mathcal{D}\Phi)_e(X) = T_e\Phi\big(H^b_e X\big) - H^m_{\Phi(e)}X, \tag{2.26}$$

$$(\mathcal{D}\widetilde{\Phi})_f(X) = T_f\widetilde{\Phi}\big(H^m_f X\big) - H^b_{\widetilde{\Phi}(f)}X. \tag{2.27}$$

[DEFINITION]

Each difference is vertical because its two terms project to the same X. The maps are parallel exactly when these defects vanish, equivalently when every base curve satisfies

$$\widetilde{\Omega}_\gamma \circ \Phi_{\gamma(0)} = \Phi_{\gamma(1)} \circ \Omega_\gamma, \tag{2.28}$$

$$\Omega_\gamma \circ \widetilde{\Phi}_{\gamma(0)} = \widetilde{\Phi}_{\gamma(1)} \circ \widetilde{\Omega}_\gamma. \tag{2.29}$$

[ESTABLISHED]

Proof. Vanishing defect says that the image of every horizontal lift is horizontal. Integration along γ gives the two commutative squares, and differentiation at the initial point gives the converse. □

Only in a fiberwise-linear associated-vector-bundle specialization do the defects reduce to

$$D\Phi = \nabla^m \circ \Phi - \Phi \circ \nabla^b, \qquad D\widetilde{\Phi} = \nabla^b \circ \widetilde{\Phi} - \widetilde{\Phi} \circ \nabla^m. \tag{2.30}$$

Write V_b, V_m for the two representation spaces in this linear specialization. The old-to-new coordinate maps induced by (2.10) are, by definition,

$$R^b_i := \rho_b(a_i)^{-1} : V_b \longrightarrow V_b, \qquad R^m_i := \rho_m(b_i)^{-1} : V_m \longrightarrow V_m, \tag{2.31}$$

so that $v'_b = R^b_i v_b$ and $v'_m = R^m_i v_m$. These are invertible coordinate maps, not cross-channel morphisms or coarse restriction operators. In these coordinates the defect matrices obey

$$(D\Phi)'_i = R^m_i (D\Phi)_i (R^b_i)^{-1}, \qquad (D\widetilde{\Phi})'_i = R^b_i (D\widetilde{\Phi})_i (R^m_i)^{-1}. \tag{2.32}$$

[ESTABLISHED]

The matrix laws are not asserted for nonlinear law manifolds. The connections in (2.20) are chosen data; no curvature or transport is inferred from the agent frames. [NOT-CLAIMED]

2.6 Agents and finite-moment pushforward

Definition 2.9 (Agent). Agent i is the section-bearing object

$$\mathcal{A}^i = \big(\mathcal{C}_i; q_i, s_i, u_i^b, u_i^m\big), \tag{2.33}$$

where

$$q_i \in \Gamma(\mathcal{C}_i, \mathcal{E}_b|_{\mathcal{C}_i}), \qquad s_i \in \Gamma(\mathcal{C}_i, \mathcal{E}_m|_{\mathcal{C}_i}). \tag{2.34}$$

[DEFINITION]

The sections are smooth only in the selected smooth tier. A sample

$$Y(c) = \big((k_1(c), m_1(c)), \ldots, (k_N(c), m_N(c))\big) \tag{2.35}$$

is a random element of a product of sample fibers, not a tuple of agents. The continuum sections above and the finite recognition law introduced later are separately declared objects; their compatibility is a hypothesis in the probability chapter.

Proposition 2.10 (Finite-second-moment congruence). Let a belief law q on $\mathbb{R}^K$ have finite mean μ and covariance Σ, and let $R = \rho_b(g)$ be invertible and linear for $g \in G$. Then

$$\mu' = R\mu, \qquad \Sigma' = R\Sigma R^\top. \tag{2.36}$$

If $\Sigma \succ 0$ and $J = \Sigma^{-1}$, the pushed law has precision

$$J' = (\Sigma')^{-1} = R^{-\top} J R^{-1}. \tag{2.37}$$

The model channel has the same statement in its own dimension. [ESTABLISHED]

Proof. For $X \sim q$, linearity gives $\mathbb{E}[RX] = R\mu$ and $\mathbb{E}[(RX - R\mu)(RX - R\mu)^\top] = R\Sigma R^\top$. Inverting the latter identity gives (2.37). $\square$

2.7 Pointwise frame comparisons

Fix $c \in \mathcal{C}_i \cap \mathcal{C}_j$. Freeness and transitivity give unique elements of the same group G such that

$$u_j^b(c) = u_i^b(c) \cdot T_{ij}^b(c), \qquad u_j^m(c) = u_i^m(c) \cdot T_{ij}^m(c). \tag{2.38}$$

They act through $\rho_b(T_{ij}^b)$ and $\rho_m(T_{ij}^m)$ in the two associated fibers. Under (2.10),

$$T_{ij}^{b\prime} = a_i^{-1} T_{ij}^b a_j, \qquad T_{ij}^{m\prime} = b_i^{-1} T_{ij}^m b_j. \tag{2.39}$$

Moreover, the relative fields force

$$T_{ij}^m = h_i^{-1} T_{ij}^b h_j. \tag{2.40}$$

[ESTABLISHED]

Proof. The transformation law follows by substitution and uniqueness in the principal fiber. The two expressions $u_j^m = u_j^b h_j = u_i^b T_{ij}^b h_j$ and $u_j^m = u_i^m T_{ij}^m = u_i^b h_i T_{ij}^m$ give (2.40). □

The comparisons are same-point changes of coordinates. They involve no curve and no connection. The two group elements are related by (2.40), while their represented actions can be inequivalent.

2.8 The Čech picture and bundle topology

Let $\mathcal{C}_0 = \bigcup_i \mathcal{C}_i$, and let $\mathcal{N}$ be the nerve of this open cover:

$$\{i_0, \ldots, i_r\} \in \mathcal{N} \quad \Longleftrightarrow \quad \mathcal{C}_{i_0 \cdots i_r} = \bigcap_{a=0}^{r} \mathcal{C}_{i_a} \neq \varnothing. \tag{2.41}$$

For either frame atlas $x \in \{b, m\}$, the maps $T_{ij}^x : \mathcal{C}_{ij} \to G$ obey

$$T_{ii}^x = e_G, \qquad T_{ji}^x = (T_{ij}^x)^{-1}, \qquad T_{ij}^x T_{jl}^x = T_{il}^x. \tag{2.42}$$

[ESTABLISHED]

Proof. On a triple overlap, composing the unique change from l to j with the unique change from j to i gives the unique change from l to i. □

Let $\mathcal{G}$ be the sheaf of smooth G-valued maps. There is one principal bundle and hence one nonabelian Čech class:

$$[T^b] = [T^m] = [P|_{\mathcal{C}_0}] \in \check{H}^1(\{\mathcal{C}_i\}; \mathcal{G}). \tag{2.43}$$

Indeed, (2.40) is exactly the coboundary equivalence between the two cocycles. The class is distinguished exactly when $P|_{\mathcal{C}_0}$ is trivial, equivalently when, for either atlas, there are smooth $U_i^x : \mathcal{C}_i \to G$ such that

$$T_{ij}^x = U_i^x (U_j^x)^{-1}. \tag{2.44}$$

[ESTABLISHED]

Proof. If (2.44) holds, the rechoice $u_i^{x\prime} = u_i^x U_i^x$ makes all transitions identity, so the sections glue to a global section. A global section trivializes a principal bundle. Conversely, a global trivialization supplies the U_i^x. This is the standard global-section characterization of a trivial principal bundle [36, 43, 68]. □

The single principal class does not make the two associated bundles isomorphic. Inequivalent representations can send the same principal cocycle to nonisomorphic associated bundles and to different induced characteristic data.

Hypothesis 2.11 (Common principal trivialization). Whenever global principal-frame coordinates are used, declare $P|_{\mathcal{C}_0}$ trivial and choose one reference section σ_0 such that

$$u_i^b = \sigma_0 \cdot (U_i^b)^{-1}, \qquad u_i^m = \sigma_0 \cdot (U_i^m)^{-1}, \qquad h_i = U_i^b (U_i^m)^{-1}. \tag{2.45}$$

[HYPOTHESIS]

This is one topological hypothesis, not two channelwise hypotheses. The general theory retains local frames and their common Čech class. A global section of an associated bundle does not in general trivialize P: if an associated fiber contains a G-fixed point β_0, then $c \mapsto [p, \beta_0]$ is well defined for any P. [ESTABLISHED]

2.9 Channel-specific graph links and trivializing subfamilies

Definition 2.12 (Typed graph links). Let $\mathfrak{I} = (V, E, ^{-})$ be a finite interaction multigraph declared independently of $\mathcal{C}$. Its oriented edge copies have a fixed-point-free reversal involution, and parallel copies are allowed. For $e : j \to i$, declare

$$\Theta_e^b, \Theta_e^m \in G, \qquad \Theta_{\bar{e}}^x = (\Theta_e^x)^{-1}. \tag{2.46}$$

Under independently changed channel coordinates at the vertices,

$$\Theta_e^{b\prime} = (a_i^b)^{-1} \Theta_e^b a_j^b, \qquad \Theta_e^{m\prime} = (a_i^m)^{-1} \Theta_e^m a_j^m. \tag{2.47}$$

Here the $a_i^x \in G$ are passive vertex-coordinate choices. They are induced by smooth frame rechoices only after vertices are anchored at contexts $c_i \in \mathcal{C}_i$, in which case $a_i^b = a_i(c_i)$ and $a_i^m = b_i(c_i)$. Allowing two coordinate descriptions of two graph-link data sets does not enlarge the principal gauge group to $G \times G$. In the old-to-new sample-coordinate convention of Section 4.6, the contextual elements denoted there by h_i^x satisfy $h_i^x = (a_i^x)^{-1}$. The superscripted h_i^x is a discrete contextual coordinate re-expression and is unrelated to the relative principal-frame field h_i of (2.3). [DEFINITION]

The links are channel-specific data. Neither is determined by the other, by h_i, by Φ or $\widetilde{\Phi}$, by pointwise frame comparison, or by a principal connection. For a closed edge-labeled walk $\gamma = (e_0, \ldots, e_{r-1})$, where $e_a : i_{a+1} \to i_a$ and $i_r = i_0$, define

$$H^x(\gamma) = \prod_{a=0}^{r-1} \Theta_{e_a}^x. \tag{2.48}$$

Its conjugacy class and the condition $H^x(\gamma) = e_G$ are invariant under the corresponding coordinate changes. [ESTABLISHED]

Proposition 2.13 (Trivializing criterion). Fix a channel x, a subfamily $F \subseteq V$, and a G-valued comparison assignment Ξ_e^x on its internal oriented edge copies. The following are equivalent:

$$\begin{gathered}\exists V_i^x \in G \ (i \in F) \quad \Xi_e^x = V_i^x (V_j^x)^{-1} \quad \text{for every internal } e : j \to i, \\ \prod_{a=0}^{r-1} \Xi_{e_a}^x = e_G \quad \text{for every closed internal walk.}\end{gathered} \tag{2.49}$$

[ESTABLISHED]

Proof. The first condition telescopes around every closed walk. Conversely, choose a base vertex in each connected component and define V_i^x by an ordered path product. Two choices differ by a closed walk and hence agree. Appending $e : j \to i$ gives the required edge factorization. ☐

A subfamily is *jointly trivializing* when the criterion holds separately for both link data sets. For T^x evaluated at one common context the criterion is automatic by the Čech cocycle law. Its nontrivial graph content concerns the independently declared Θ^x. Pointwise graph triviality is weaker than triviality of P, which requires smooth overlap data on the entire cover.

Hypothesis 2.14 (Graph links realized by base transport). Choose contexts $c_i \in \mathcal{C}_i$ and, for every $e : j \to i$, a piecewise-smooth curve γ_e from c_j to c_i. In the chosen endpoint frames impose

$$\widehat{\rho}_b(\Theta_e^b) = \Omega_{\gamma_e}, \qquad \widehat{\rho}_m(\Theta_e^m) = \widetilde{\Omega}_{\gamma_e}. \tag{2.50}$$

[HYPOTHESIS]

Without the curve assignment and these equalities, graph holonomy and base-connection holonomy are unrelated. Under the hypothesis, represented graph-loop products equal the corresponding parallel transports around concatenated base loops. This is a connection-holonomy statement, not a claim of nontrivial principal-bundle topology.

2.10 Flat graph links and residuals

Hypothesis 2.15 (Pointwise flat-link specialization). At one common context, declare separately for the two link data sets that

$$\Theta_e^b = T_{ij}^b = U_i^b (U_j^b)^{-1}, \qquad \Theta_e^m = T_{ij}^m = U_i^m (U_j^m)^{-1}, \qquad e : j \to i. \tag{2.51}$$

[HYPOTHESIS]

This excludes represented graph holonomy in either channel. It does not assert that either principal connection is flat or that the two channel frames coincide. Let μ_i be a belief-channel vector coordinate, set $z_i = \rho_b(U_i^b)^{-1}\mu_i$, and suppress ρ_b only in the defining representation. Then

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(U_i^b)(z_i - z_j). \tag{2.52}$$

For an independent graph link,

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(U_i^b)(z_i - H_e^b z_j), \qquad H_e^b = \rho_b(U_i^b)^{-1}\rho_b(\Theta_e^b)\rho_b(U_j^b). \tag{2.53}$$

More generally, if $\Theta_e^b = V_i^b(V_j^b)^{-1}$ on a belief-trivializing subfamily and $z_i = \rho_b(V_i^b)^{-1}\mu_i$, then

$$\mu_i - \rho_b(\Theta_e^b)\mu_j = \rho_b(V_i^b)(z_i - z_j). \tag{2.54}$$

[ESTABLISHED]

These identities follow by substitution. Belief-link trivialization says nothing about model links, either cross morphism, or equality of the belief and model frames.

2.11 Common-frame and common-connection specializations

The one-principal-bundle construction is already the ambient theory. A *common-frame* specialization makes the stronger declaration

$$u_i^b = u_i^m = u_i, \qquad h_i = e_G, \qquad T_{ij}^b = T_{ij}^m = T_{ij}. \tag{2.55}$$

Even then the represented frame changes generally differ: choose a common old-to-new sample-coordinate re-expression $g_i : \mathcal{C}_i \to G$, so that the corresponding principal-section rechoice is $u_i' = u_i \cdot g_i^{-1}$. Then

$$R_i^b = \rho_b(g_i), \qquad R_i^m = \rho_m(g_i). \tag{2.56}$$

Thus R_i^b and R_i^m have the same principal source g_i but act within different representation spaces; neither is a map between the belief and model fibers. An independent *equal-connection* specialization declares $\omega_b = \omega_m = \omega$. The underlying principal holonomy is then common, while its actions on belief and model laws remain $\widehat{\rho}_b$ and $\widehat{\rho}_m$. A further common-link specialization declares $\Theta_e^b = \Theta_e^m = \Theta_e \in G$. None of these three stronger hypotheses follows from the others. [HYPOTHESIS]

The cross maps remain extra equivariant structure governed by (2.16); they are not forced to be inverses or parallel. Thus inequivalent representations do not obstruct a common principal frame. In a linear associated-vector-bundle specialization they may obstruct a nonzero sample-fiber intertwiner. Law-level equivariance is weaker: an equivariant Markov kernel may induce an equivariant map on laws without supplying a linear map between the sample fibers. [ESTABLISHED]

2.12 Optional product-gauge extension

If the application genuinely requires independently nonisomorphic principal bundles or two independent gauge symmetries, replace the ambient P by

$$\pi_b : P_b \to \mathcal{C}, \qquad \pi_m : P_m \to \mathcal{C}, \qquad Q = P_b \times_{\mathcal{C}} P_m, \tag{2.57}$$

where Q is a principal $G_b \times G_m$-bundle. The belief and model associated bundles are then formed using the two projections. [DEFINITION]

There is no canonical relative field valued in one common group between frames in P_b and P_m without additional comparison data. When $G_b = G_m = G$, sufficient principal-level data are a chosen principal G-bundle isomorphism $B : P_b \to P_m$ over $\mathrm{id}_{\mathcal{C}}$ satisfying $B(p \cdot g) = B(p) \cdot g$. For possibly different groups, the corresponding declared data are a homomorphism $\kappa : G_b \to G_m$ and a bundle map $B : P_b \to P_m$ over $\mathrm{id}_{\mathcal{C}}$ satisfying $B(p \cdot g_b) = B(p) \cdot \kappa(g_b)$. The latter data alone do not create a canonical relative field valued in one common group: no such common group or inverse identification has been declared. [ESTABLISHED]

In a category with the required mapping object, a belief-to-model morphism is a section of

$$Q \times_{G_b \times G_m} \mathrm{Map}(\mathcal{B}_b, \mathcal{B}_m), \qquad (g_b, g_m) \cdot f = \widehat{\rho}_m(g_m) \circ f \circ \widehat{\rho}_b(g_b)^{-1}. \tag{2.58}$$

[DEFINITION]

A constant fiber map descends only if it is fixed by every independent pair. Given the κ-equivariant map B above, a fiber map $\phi : \mathcal{B}_b \to \mathcal{B}_m$ satisfying $\phi \circ \widehat{\rho}_b(g_b) = \widehat{\rho}_m(\kappa(g_b)) \circ \phi$ induces a compatible cross morphism. These principal and fiber data are sufficient, but a general mapping-bundle section need not arise that way. Thus the product bundle supplies neither a relative field nor either cross morphism automatically. [ESTABLISHED]

This extension carries additional topology and an independent product gauge freedom. It should therefore be invoked only when those degrees of freedom are intended, not merely because belief and model use different associated representations of one group.

Chapter 3

Probability Semantics

Chapter 2 fixed a geometry. This chapter fixes the measurable typing on which every probabilistic statement rests, and it fixes it before any density is written. The order is a substantive commitment: probability measures and normalized kernels are declared first, densities appear afterward as Radon-Nikodym derivatives against explicitly declared reference measures, and logarithms of density ratios appear only under an absolute-continuity convention stated in advance. That convention looks like bookkeeping and is not. It is what makes the exact evidence bound of Chapter 5 an identity rather than an inequality with hidden exceptions, and it is what later forbids degenerate recognition laws whose support is a null set of the posterior. A development that introduces densities first and normalizes later can write those degenerate objects down without noticing, and the resulting statements are false rather than merely imprecise.

Throughout, $\mathcal{P}(\mathsf{S}, \mathscr{S})$ denotes the set of probability measures on a measurable space $(\mathsf{S}, \mathscr{S})$; the calligraphic symbol with a measurable-space argument is unrelated to the total space P of the principal bundle in Chapter 2.

3.1 The finite design and its evaluation

Definition 3.1 (Finite design and incidence). A *design* is a finite set of distinct contexts

$$D = \{c_a\}_{a=1}^{M} \subseteq \mathcal{C}, \tag{3.1}$$

indexed by $a \in \{1, \ldots, M\}$. The *incidence set* is $\mathfrak{A} = \{(i, a) : c_a \in \mathcal{C}_i\}$, where $\mathcal{C}_i$ is the support of agent i from Definition 2.2. For the reference presentation $D \subseteq \bigcap_{i \in V} \mathcal{C}_i$, so $\mathfrak{A} = V \times \{1, \ldots, M\}$ is rectangular and every listed agent is defined at every design point. [DEFINITION]

Nothing is proved by this. The sparse-support case replaces the rectangular index set by $\mathfrak{A}$ throughout, and it is stated this way to make one point explicit: a sparse design is handled by indexing over declared incidences, never by writing a product over all (i, a) and then deleting the coordinates that happen to be undefined. Deleting coordinates from a product silently changes the sample space, and a bound computed on one sample space is not a bound on another.

Definition 3.2 (Sample spaces and product sigma-algebras). At each admissible (i, a) declare measurable state, model, and observation fibers $(\mathsf{K}_{i,a}, \mathscr{K}_{i,a})$, $(\mathsf{M}_{i,a}, \mathscr{M}_{i,a})$, and $(\mathsf{O}_{i,a}, \mathscr{O}_{i,a})$. The finite latent and observation spaces are

$$\mathsf{Y}_D = \prod_{a=1}^{M} \prod_{i \in V} \big(\mathsf{K}_{i,a} \times \mathsf{M}_{i,a}\big), \qquad \mathscr{Y}_D = \bigotimes_{a=1}^{M} \bigotimes_{i \in V} \big(\mathscr{K}_{i,a} \otimes \mathscr{M}_{i,a}\big), \tag{3.2}$$

$$\mathsf{O}_D = \prod_{a=1}^{M} \prod_{i \in V} \mathsf{O}_{i,a}, \qquad \mathscr{O}_D = \bigotimes_{a=1}^{M} \bigotimes_{i \in V} \mathscr{O}_{i,a}. \tag{3.3}$$

The stacked latent is written $Y \in \mathsf{Y}_D$ and the realized observation $o \in \mathsf{O}_D$. In the Gaussian realization $\mathsf{K}_{i,a} = \mathbb{R}^K$ and $\mathsf{M}_{i,a} = \mathbb{R}^{d_m}$ with their Borel sigma-algebras. [DEFINITION]

Evaluation and section evaluation have different codomains, and the difference is the one Chapter 2 insisted on. For a sample-field realization $y = ((k_i, m_i))_{i \in V}$ defined on D, the evaluation map

$$\mathrm{Ev}_D(y) = \big((k_i(c_a), m_i(c_a))_{i \in V}\big)_{a=1}^{M} \in \mathsf{Y}_D \tag{3.4}$$

returns a point of the latent space. Evaluating a distribution-valued section returns something else entirely: $q_i^{o,X}(c_a)$ is a probability measure on $\mathsf{K}_{i,a}$ and $s_i^{o,X}(c_a)$ is a probability measure on $\mathsf{M}_{i,a}$, and neither is an element of Y_D. The finite random element Y takes values in Y_D; the section values are laws. Section 3.7 is where the two are related, and it relates them by hypothesis.

Missing or aggregated observations are not represented by removing latent coordinates from (3.2). They require a separately declared observation map or kernel into a coarser observation space, and the declaration changes O_D rather than Y_D.

3.2 Structural data and measurable typing

Definition 3.3 (Structural configuration and kernel signatures). Fix a measurable configuration space $(\mathsf{X}, \mathscr{X})$. Every $X \in \mathsf{X}$ uses the same design D, the same agent index set V, the same incidence set $\mathfrak{A}$, and the same measurable identifications of the fibers in (3.2) and (3.3); therefore $(\mathsf{Y}_D, \mathscr{Y}_D)$ and $(\mathsf{O}_D, \mathscr{O}_D)$ do not vary with X. A configuration may carry the finite interaction complex $\mathfrak{I}$ of Definition 2.12 with its edge set, the frames U_i or the links L_{ij}, and the parameters of the observation and transition mechanisms. The generative and recognition objects have the signatures

$$P_\theta : (\mathsf{X}, \mathscr{X}) \longrightarrow \mathcal{P}\big(\mathsf{O}_D \times \mathsf{Y}_D, \mathscr{O}_D \otimes \mathscr{Y}_D\big), \qquad X \longmapsto P_\theta(\cdot \mid X), \tag{3.5}$$

$$Q : (\mathsf{O}_D \times \mathsf{X}, \mathscr{O}_D \otimes \mathscr{X}) \longrightarrow \mathcal{P}\big(\mathsf{Y}_D, \mathscr{Y}_D\big), \qquad (o, X) \longmapsto Q_X(\cdot \mid o), \tag{3.6}$$

each measurable in its argument: for every $A \in \mathscr{O}_D \otimes \mathscr{Y}_D$ the map $X \mapsto P_\theta(A \mid X)$ is $\mathscr{X}$-measurable, and for every $B \in \mathscr{Y}_D$ the map $(o, X) \mapsto Q_X(B \mid o)$ is $\mathscr{O}_D \otimes \mathscr{X}$-measurable. [DEFINITION]

Signature (3.5) is the typing on which several later results depend, and it is worth reading for what it omits. The arguments of the generative kernel are θ and X and nothing else. A recognition law, a recognition parameter, and a posterior are not among them, so no factor of the generative model may read one. This is a definitional requirement of the fixed-joint construction rather than a modeling preference: the joint must be a single measure fixed before an observation is seen, and a factor that reads a posterior makes the joint a function of the object it is supposed to be conditioned into. Nothing is proved here beyond the type; the consequences are drawn in Chapter 4, where the joint is built, and again in Chapter 19, where a construction that violates the typing is examined and found inadmissible.

Later chapters may make X random, but only within this fixed measurable space, and at a different probability level. Doing so introduces a configuration prior in $\mathcal{P}(\mathsf{X}, \mathscr{X})$ and a configuration recognition kernel from $(\mathsf{O}_D, \mathscr{O}_D)$ into $\mathcal{P}(\mathsf{X}, \mathscr{X})$; neither changes D, $\mathfrak{A}$, or the identified fibers, and neither is denoted by any symbol reserved above for a state-level object.

3.3 Reference measures

Definition 3.4 (Declared reference measures). Each continuous Euclidean coordinate carries its Borel sigma-algebra and Lebesgue base measure. Each discrete coordinate carries its power-set sigma-algebra and counting base measure. Each mixed real coordinate carries the measure declared in Corollary 3.6. Any other fiber carries an explicitly declared sigma-finite base measure. Writing $\nu^k_{i,a}$, $\nu^m_{i,a}$, and $\nu^o_{i,a}$ for the component measures, the finite product reference measures are

$$\nu^Y_D = \bigotimes_{a=1}^{M} \bigotimes_{i\in V} (\nu^k_{i,a} \otimes \nu^m_{i,a}), \qquad \nu^O_D = \bigotimes_{a=1}^{M} \bigotimes_{i\in V} \nu^o_{i,a}. \tag{3.7}$$

[DEFINITION]

Sigma-finiteness of every component is required and is not decoration. It is the hypothesis of the Radon-Nikodym theorem, without which the densities of Section 3.4 need not exist, and it is the hypothesis of the Fubini-Tonelli theorem, without which the iterated integrals used to marginalize Y out of the joint need not agree with the product integral. A finite product of sigma-finite measures is sigma-finite, so (3.7) inherits the property from its factors; this is the reason the construction is stated on a finite design and is one of the reasons the continuum question of Section 3.8 stays open.

Mixed coordinates are the case where a declaration is genuinely needed rather than conventional. A reference measure used by a parameterized family must be fixed for the whole family; choosing a new atomic reference after seeing the parameter would not define one density model.

Theorem 3.5 (Common domination for a mixed family). Let $(\mu_t)_{t\in T}$ be any family of Borel probability measures on $\mathbb{R}$ with no singular-continuous part, written

$$\mu_t = f_t\lambda + \sum_{a\in A_t} p_t(a)\delta_a, \qquad A_t := \{a \in \mathbb{R} : \mu_t(\{a\}) > 0\}.$$

There exists a single sigma-finite Borel measure dominating every μ_t if and only if $A_* := \bigcup_{t\in T} A_t$ is countable. When $A_* = \{a_1, a_2, \dots\}$, one such parameter-independent measure is

$$\nu_{\mathrm{mix}} := \lambda + \sum_{n\geq 1} 2^{-n}\delta_{a_n}. \tag{3.8}$$

In particular, the moving-atom mixed family $\mu_t = \frac{1}{2}\mathcal{N}(0,1) + \frac{1}{2}\delta_t$, $t \in \mathbb{R}$, has no common sigma-finite dominating measure. [ESTABLISHED]

Proof. If A_* is countable, (3.8) is sigma-finite and a ν_{mix}-null set is both Lebesgue null and contains no point of A_*. Hence every continuous and atomic term of every μ_t vanishes on it, so $\mu_t \ll \nu_{\mathrm{mix}}$. Conversely, suppose a sigma-finite ξ dominates every μ_t. For each $a \in A_*$ some t has $\mu_t(\{a\}) > 0$, whence $\xi(\{a\}) > 0$. A sigma-finite measure has at most countably many positive-mass point atoms: on each finite-measure member of a countable cover, the points of mass at least $1/n$ are finite, and the countable union over the cover and n exhausts them. Thus A_* is countable. The displayed moving-atom family has $A_* = \mathbb{R}$, proving the last assertion. □

Corollary 3.6 (Dominating measure for one mixed coordinate). Let μ be a Borel probability measure on $\mathbb{R}$ whose Lebesgue decomposition has no singular continuous part, so that $\mu = \mu_{\mathrm{ac}} + \mu_{\mathrm{d}}$ with $\mu_{\mathrm{ac}} \ll \lambda$ and μ_{d} carried by a countable set A. Put

$$\nu = \lambda + \sum_{x\in A} \delta_x. \tag{3.9}$$

Then ν is sigma-finite and $\mu \ll \nu$. [ESTABLISHED]

Proof. Apply Theorem 3.5 to the singleton family $\{\mu\}$. The measure displayed here and the weighted atomic reference in (3.8) have the same null sets, so either dominates μ. □

The hypothesis excluding a singular-continuous part is a restriction on this particular Lebesgue-plus-atoms formula, not an intrinsic nondominability claim. A family carrying Cantor-type or other singular-continuous components can be dominated when a common sigma-finite singular reference is available, but that reference must be declared as part of the family before any density is written. Equation (3.8) supplies no such component.

3.4 Kernels first, densities second

Definition 3.7 (Normalized kernels; densities as derivatives). The objects (3.5) and (3.6) are probability kernels, hence normalized by type. When $P_\theta(\cdot \mid X) \ll \nu_D^O \otimes \nu_D^Y$, its density is the Radon-Nikodym derivative $p_\theta(o, y \mid X)$ determined by

$$P_\theta(do, dY \mid X) = p_\theta(o, y \mid X)\nu_D^O(do)\nu_D^Y(dy), \tag{3.10}$$

and the evidence is the normalized marginal

$$p_\theta(o \mid X) = \int_{\mathsf{Y}_D} p_\theta(o, y \mid X)\nu_D^Y(dy). \tag{3.11}$$

When $Q_X(\cdot \mid o) \ll \nu_D^Y$, its density is written $q_X(y \mid o)$. Densities are determined only up to a null set of the declared reference measure, and any two versions agreeing outside such a set denote the same density. [DEFINITION]

The order in Definition 3.7 is the content. A normalized kernel is declared, and only then is a density extracted; the reverse order, in which a nonnegative function is written down and a normalizer is invoked afterward, is a different construction whose normalizer must be shown finite before anything it appears in denotes a probability. This document uses the first order everywhere, and where a later chapter writes an unnormalized energy it says so and exhibits the normalizer.

Regular-conditional convention. Whenever a later result conditions on a realized observation, assume in addition that $(\mathsf{O}_D, \mathscr{O}_D)$ and $(\mathsf{Y}_D, \mathscr{Y}_D)$ are standard Borel spaces. This holds for the finite products of Euclidean and countable discrete fibers used in the worked construction; it is an additional hypothesis for a general measurable fiber. For fixed (θ, X), write

$$P_{\theta,X}^O(A) := P_\theta(A \times \mathsf{Y}_D \mid X), \qquad A \in \mathscr{O}_D, \tag{3.12}$$

for the observation marginal. The standard-Borel hypothesis supplies a regular conditional probability

$$\Pi_{\theta,X}(o, B) = P_\theta(B \mid o, X), \qquad B \in \mathscr{Y}_D, \tag{3.13}$$

unique only for $P_{\theta,X}^O$-almost every o.

When the domination in (3.10) holds, Tonelli's theorem makes the function in (3.11) a density of $P_{\theta,X}^O$ relative to ν_D^O. After altering it on a ν_D^O-null set if necessary, there is a measurable set $\mathsf{O}_{\theta,X}^{\mathrm{reg}}$ with

$$P_{\theta,X}^O\left(\mathsf{O}_{\theta,X}^{\mathrm{reg}}\right) = 1, \qquad 0 < p_\theta(o \mid X) < \infty, \tag{3.14}$$

such that, for every $o \in \mathsf{O}_{\theta,X}^{\mathrm{reg}}$ and $B \in \mathscr{Y}_D$,

$$\Pi_{\theta,X}(o, B) = \frac{\int_B p_\theta(o, y \mid X)\nu_D^Y(dy)}{p_\theta(o \mid X)}. \tag{3.15}$$

The displayed version is obtained directly from the density on the positive-finite evidence set and completed arbitrarily off that set. Fubini's theorem shows why the qualification is unavoidable: changing a joint density on a $(\nu_D^O \otimes \nu_D^Y)$-null set can change an entire slice at an exceptional observation, while it changes (3.15) only on an observation-marginal-null set. An exact witness is available on $\mathbb{R}^2$. Let ϕ be the standard normal density and let ψ be any different probability density.

The functions

$$p(o,y)=\phi(o)\phi(y), \qquad \widetilde{p}(o,y)=\begin{cases}\phi(o)\phi(y), & o\neq 0,\\ \phi(0)\psi(y), & o=0\end{cases}$$

are versions of the same joint density because they differ only on the Lebesgue-null slice $\{0\}\times\mathbb{R}$. Both displayed evidence representatives equal $\phi(0)$ at $o=0$, but their density-ratio conditionals there are $\phi(y)dy$ and $\psi(y)dy$. The joint measure therefore determines neither pointwise posterior at that exceptional observation. Accordingly, every later fixed-observation posterior, ELBO, or free-energy statement is asserted for $P^{O}_{\theta,X}$-almost every o, represented by $o\in \mathsf{O}^{\mathrm{reg}}_{\theta,X}$. An everywhere pointwise statement instead requires a particular jointly measurable density, evidence representative, and regular-conditional version to be declared as part of the model data; its exceptional-point values then belong to that declaration and are not determined by the joint measure alone. [HYPOTHESIS]

The same order is what makes the group action of Chapter 2 well typed. The action (2.6) is defined on measures. A frame change that does not preserve the reference measure changes the density and the base-measure Jacobian together, and only their combination is invariant, so a rule that multiplies a matrix into a density and stops is not the represented action unless the Jacobian is identically one. [ESTABLISHED]

Theorem 3.8 (Jointly measurable Radon–Nikodym version for a dominated kernel). Let $(\mathsf{E},\mathscr{E})$ be a measurable space, let $(\mathsf{F},\mathscr{F})$ be standard Borel, and let K and L be probability kernels from E to F satisfying $K(x,\cdot)\ll L(x,\cdot)$ for every x. Then there is a finite-valued $\mathscr{E}\otimes\mathscr{F}$-measurable $r:\mathsf{E}\times\mathsf{F}\to[0,\infty)$ such that

$$K(x,B)=\int_B r(x,y)L(x,dy) \quad \text{for every } x\in\mathsf{E}, B\in\mathscr{F}.$$

The theorem supplies one simultaneous version; independently selecting an RN derivative for each x does not establish joint measurability. [ESTABLISHED]

Proof. Standard Borelness gives finite refining measurable partitions $(\mathcal{P}_n)_{n\geq 1}$ whose generated sigma-algebras increase to $\mathscr{F}$. With $0/0:=0$, set

$$r_n(x,y)=\sum_{A\in\mathcal{P}_n}\frac{K(x,A)}{L(x,A)}\mathbf{1}_A(y), \qquad r(x,y)=\begin{cases}\limsup_{n\to\infty} r_n(x,y), & \limsup_n r_n(x,y)<\infty,\\ 0, & \limsup_n r_n(x,y)=+\infty.\end{cases} \tag{3.16}$$

Every r_n and therefore the finite-valued r is jointly measurable. For each fixed x, if $h_x = dK(x,\cdot)/dL(x,\cdot)$, then $r_n(x,\cdot)$ is the conditional expectation of h_x on the finite sigma-algebra generated by $\mathcal{P}_n$. Martingale convergence gives $r_n(x,\cdot)\to h_x$ at $L(x,\cdot)$-almost every point. Because h_x is finite $L(x,\cdot)$-almost everywhere, the second branch of (3.16) changes only an $L(x,\cdot)$-null set, and $r(x,\cdot)$ is the required derivative for every x. $\square$

Corollary 3.9 (Joint densities against one reference). Let ν be a fixed sigma-finite measure on a standard Borel $(\mathsf{F}, \mathscr{F})$, and let K be a probability kernel from $(\mathsf{E}, \mathscr{E})$ to F with $K(x, \cdot) \ll \nu$ for every x. Then K has a finite jointly measurable density $k(x, y)$ against ν. Consequently every finite or countable declared collection of kernels dominated by the same fixed ν may be represented by jointly measurable densities against that common reference. In particular this applies to the generative and recognition kernels in (3.5)–(3.6) when they are dominated by the declared product references. [ESTABLISHED]

Proof. Sigma-finiteness supplies measurable w with $0 < w < \infty$ ν-almost everywhere and $\int w d\nu = 1$. The constant probability kernel $L(x, dy) = w(y)\nu(dy)$ is equivalent to ν. Apply Theorem 3.8 to (K, L) to obtain finite jointly measurable $r = dK/dL$ and set $k(x, y) = r(x, y)w(y)$, completing w arbitrarily on its ν-null exceptional set. Applying this construction to each member proves the collection statement. □

Proposition 3.10 (Integration against a fixed sigma-finite measure). If $f : \mathsf{E} \times \mathsf{F} \to [0, \infty]$ is $\mathscr{E} \otimes \mathscr{F}$-measurable and ν is a fixed sigma-finite measure on $(\mathsf{F}, \mathscr{F})$, then

$$x \longmapsto \int_{\mathsf{F}} f(x, y)\nu(dy)$$

is $\mathscr{E}$-measurable. For an extended signed integrand the same conclusion holds whenever at least one of the positive- and negative-part integrals is finite; no expression of the form $+\infty - \infty$ is defined. [ESTABLISHED]

Proof. Choose a measurable disjoint partition $\mathsf{F} = \bigsqcup_{m \geq 1} F_m$ with $\nu(F_m) < \infty$. For fixed m, let

$$\mathscr{D}_m = \{C \in \mathscr{E} \otimes \mathscr{F} : x \longmapsto \nu(C_x \cap F_m) \text{ is } \mathscr{E}\text{-measurable}\}.$$

This class contains every measurable rectangle. It is a Dynkin system: $\nu((C^c)_x \cap F_m) = \nu(F_m) - \nu(C_x \cap F_m)$, where finiteness of $\nu(F_m)$ makes the complement step legitimate, and section measures of a countable disjoint union are the corresponding pointwise sums. The π–λ theorem therefore gives $\mathscr{D}_m = \mathscr{E} \otimes \mathscr{F}$. Hence, for every product-measurable C,

$$x \longmapsto \nu(C_x) = \sum_{m \geq 1} \nu(C_x \cap F_m)$$

is measurable. This proves the assertion for indicators; linearity proves it for nonnegative simple functions, and monotone convergence proves it for every nonnegative measurable f. Apply that result to f^+ and f^- for the signed statement. On the measurable domain where they are not both infinite, their extended-real difference is measurable and no infinity cancellation is used. □

Corollary 3.11 (Measurability of kernel relative entropy). Let κ and η be probability kernels from $(\mathsf{E}, \mathscr{E})$ to a standard Borel space $(\mathsf{F}, \mathscr{F})$. Then $x \mapsto D_{\mathrm{KL}}(\kappa(x, \cdot) \| \eta(x, \cdot))$ is $\mathscr{E}$-measurable as an $[0, \infty]$-valued map. More explicitly, for finite refining partitions $(\mathcal{P}_n)$ generating $\mathscr{F}$,

$$D_{\mathrm{KL}}(\kappa(x,\cdot)\|\eta(x,\cdot)) = \sup_{n\geq 1} \sum_{A\in\mathcal{P}_n} \kappa(x,A)\log\frac{\kappa(x,A)}{\eta(x,A)}, \tag{3.17}$$

with $0\log(0/b) = 0$ and $a\log(a/0) = +\infty$ for $a > 0$. [ESTABLISHED]

Proof. Put $L = (\kappa+\eta)/2$. Applying Theorem 3.8 to (κ, L) and (η, L) gives finite jointly measurable densities $a = d\kappa/dL$ and $b = d\eta/dL$. Fix x and abbreviate $L_x = L(x,\cdot)$. If $\mathscr{G}_n = \sigma(\mathcal{P}_n)$, then the cellwise ratios in (3.17) are versions of

$$a_n = \mathbb{E}_{L_x}[a(x,\cdot) \mid \mathscr{G}_n], \qquad b_n = \mathbb{E}_{L_x}[b(x,\cdot) \mid \mathscr{G}_n].$$

Define the nonnegative lower-semicontinuous convex perspective

$$\Phi(u,v) = u\log(u/v) - u + v,$$

with $\Phi(0,v) = v$, $\Phi(u,0) = +\infty$ for $u > 0$, and $\Phi(0,0) = 0$. Because both kernels have unit mass, the nth partition expression $D_n(x)$ equals $\int \Phi(a_n, b_n)dL_x$. Conditional Jensen under refinement makes $D_n(x)$ nondecreasing, while the log-sum inequality gives

$$D_n(x) \leq D_{\mathrm{KL}}(\kappa(x,\cdot)\|\eta(x,\cdot)).$$

Martingale convergence gives $(a_n, b_n) \to (a, b)$ L_x-almost everywhere. Fatou's lemma and the preceding upper bound yield

$$D_{\mathrm{KL}}(\kappa(x,\cdot)\|\eta(x,\cdot)) = \int \Phi(a,b)dL_x \leq \liminf_n D_n(x) \leq \limsup_n D_n(x) \leq D_{\mathrm{KL}}(\kappa(x,\cdot)\|\eta(x,\cdot)).$$

Thus $D_n(x) \uparrow D_{\mathrm{KL}}(\kappa(x,\cdot)\|\eta(x,\cdot))$, proving (3.17) without a cofinality assertion. Each D_n is measurable in x, so its countable supremum is measurable. □

Proposition 3.12 (Kernel integration preserves measurability). Let κ be a probability kernel from $(\mathsf{E}, \mathscr{E})$ to $(\mathsf{F}, \mathscr{F})$ and let $f : \mathsf{E}\times\mathsf{F} \to [0,\infty]$ be $\mathscr{E}\otimes\mathscr{F}$-measurable. Then

$$x \longmapsto \int_{\mathsf{F}} f(x,y)\kappa(x,dy) \tag{3.18}$$

is $\mathscr{E}$-measurable. Consequently $X \mapsto p_\theta(o \mid X)$ and the Q_X-expectations appearing in the bound of Chapter 5 are measurable in (o, X) whenever their integrands are jointly measurable. [ESTABLISHED]

Proof. Let $\mathcal{D}$ be the collection of $C \in \mathscr{E}\otimes\mathscr{F}$ for which $x \mapsto \kappa(x, C_x)$ is $\mathscr{E}$-measurable, where $C_x = \{y : (x,y) \in C\}$. Measurable rectangles $A \times B$ lie in $\mathcal{D}$, since $\kappa(x, (A\times B)_x) = \mathbf{1}_A(x)\kappa(x,B)$ and $x \mapsto \kappa(x,B)$ is measurable by the definition of a kernel; the rectangles form a π-system generating $\mathscr{E}\otimes\mathscr{F}$. The collection $\mathcal{D}$ contains $\mathsf{E}\times\mathsf{F}$, is closed under proper differences because

$\kappa(x,\cdot)$ is a finite measure, and is closed under increasing limits by continuity from below, hence is a λ-system; Dynkin's lemma gives $\mathcal{D} = \mathscr{E} \otimes \mathscr{F}$. The claim therefore holds for indicators, hence for nonnegative simple functions by linearity, hence for all nonnegative measurable f by monotone approximation and monotone convergence. The full development of kernels and their integrals is standard [39]. □

This proposition is stated because the measurability of the evidence and of the bound in their arguments is used implicitly whenever one writes an expectation over configurations, and an implicit use of an unproved measurability claim is a gap. It is discharged here rather than assumed.

3.5 The absolute-continuity convention

Definition 3.13 (Standing convention on logarithmic density ratios). Wherever this document writes a logarithmic density ratio, or an expectation of one, the measure appearing in the numerator is absolutely continuous with respect to the measure appearing in the denominator. Relative entropy is understood in the measure-theoretic sense,

$$D_{\mathrm{KL}}(Q\|P) = \begin{cases} \displaystyle\int \log\Big(\frac{dQ}{dP}(y)\Big) Q(dy), & Q \ll P, \\ +\infty, & \text{otherwise}, \end{cases} \tag{3.19}$$

following the measure-theoretic formulations of Kullback and Leibler [44] and Csiszár [22]. [DEFINITION]

The convention is not a restriction on what may be written; the second branch of (3.19) assigns a value in every case. It is a statement about which of those values is finite, and it is load-bearing precisely because the infinite branch is not an edge case to be ignored but the mechanism by which whole classes of candidate recognition laws are excluded. The next statement makes the criterion checkable in terms of densities, which is the form in which it is applied later.

Proposition 3.14 (Density criterion for absolute continuity). Let Q and P be measures on a common measurable space with $Q \ll \nu$ and $P \ll \nu$ for a sigma-finite ν, with densities q and p. Write $Z = \{p = 0\}$. Then

$$Q \ll P \iff Q(Z) = 0 \iff q = 0 \quad \nu\text{-almost everywhere on } Z. \tag{3.20}$$

[ESTABLISHED]

Proof. First, $P(Z) = \int_Z p(x)\nu(dx) = 0$. If $Q \ll P$ then $Q(Z) = 0$ follows at once. Conversely suppose $Q(Z) = 0$ and let A satisfy $P(A) = 0$, that is $\int_A p(x)\nu(dx) = 0$ with $p \geq 0$; then $\nu(A \cap \{p > 0\}) = 0$, so $Q(A \cap \{p > 0\}) = \int_{A\cap\{p>0\}} q(x)\nu(dx) = 0$, while $Q(A \cap Z) \leq Q(Z) = 0$. Adding gives $Q(A) = 0$. The second equivalence is immediate from $Q(Z) = \int_Z q(x)\nu(dx)$ and $q \geq 0$. □

The consequence used later has two cases that must not be conflated. Proposition 3.14 applies when both measures are dominated by the same ν. If a candidate Q is singular relative to that reference measure, its failure of absolute continuity must instead be checked directly from sets. Namely, if there is a measurable S with

$$Q(S) = 1, \qquad P(S) = 0, \tag{3.21}$$

then $Q \not\ll P$ by the definition of absolute continuity, and (3.19) gives $D_{\mathrm{KL}}(Q\|P) = +\infty$. For example, let P be a nondegenerate Lebesgue-dominated law on $\mathbb{R}^d$ and let Q be supported on a proper affine subspace S. The subspace is Lebesgue null, hence $P(S) = 0$, whereas $Q(S) = 1$; the direct argument (3.21), not Proposition 3.14, proves the claim. This distinction matters because a subspace-supported Q has no Lebesgue density to which the density criterion could be applied. The specific instance, in which an identification of latent coordinates across a cluster is shown to be a generative construction rather than a restriction on the recognition law, is developed in Part II.

Hypothesis 3.15 (Regular positive-finite observation). For fixed (θ, X), the selected observation satisfies $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,X}$ under the regular-conditional convention above, and

$$0 < p_\theta(o \mid X) < \infty. \tag{3.22}$$

[HYPOTHESIS]

This hypothesis selects a pointwise posterior version and a finite real log evidence. It imposes no support or entropy condition on a candidate recognition law; those conditions are needed only for the classical split representation below.

Hypothesis 3.16 (Classical split-ELBO domain). In addition to Hypothesis 3.15, the recognition law satisfies

$$Q_X(\cdot \mid o) \ll \Pi_{\theta,X}(o, \cdot)$$

and

$$\mathbb{E}_{Q_X}\Big[\big|\log p_\theta(o, Y \mid X)\big| + \big|\log q_X(Y \mid o)\big|\Big] < \infty. \tag{3.23}$$

[HYPOTHESIS]

These are choices, and each excludes something identifiable. Membership in $\mathsf{O}^{\mathrm{reg}}_{\theta,X}$ makes the posterior the selected regular-conditional version and makes the statement invariant under changes of density version outside a marginal-full set. The lower bound in (3.22) excludes observations the selected representative deems impossible, for which the log evidence is $-\infty$. The separate absolute-continuity requirement in Hypothesis 3.16 excludes exactly the degenerate laws just described. Condition (3.23) excludes recognition laws whose tails make the two classical expectations individually infinite even when their formal difference suggests cancellation. Chapter 5 first defines an extended functional for every probability law and invokes Hypothesis 3.16 only when splitting it into separate expected-log-joint and entropy terms.

3.6 Recognition marginals and what they fail to determine

Definition 3.17 (Coordinate marginals of the recognition law). With $\mathrm{pr}^k_{i,a}$ and $\mathrm{pr}^m_{i,a}$ the coordinate projections out of Y_D, define the marginalization maps $\mathrm{Marg}^k_{i,a}(R) = (\mathrm{pr}^k_{i,a})_\# R$ and $\mathrm{Marg}^m_{i,a}(R) = (\mathrm{pr}^m_{i,a})_\# R$ on $\mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D)$, and set

$$q^{o,X}_{i,a} = \mathrm{Marg}^k_{i,a}\big(Q_X(\cdot \mid o)\big), \qquad s^{o,X}_{i,a} = \mathrm{Marg}^m_{i,a}\big(Q_X(\cdot \mid o)\big). \tag{3.24}$$

[DEFINITION]

The recognition law is not required to factor across agents, design points, or channels. In the Gaussian realization with at least two nondegenerate real coordinates, the marginals (3.24) are therefore a lossy summary, and the next statement exhibits the loss. No such nonidentifiability is asserted for an arbitrary collection of fibers: if the finite latent space is a singleton, its unique law is determined by every marginal.

Proposition 3.18 (Coordinate marginals do not determine a sufficiently rich joint). Suppose Y_D contains at least two nondegenerate real coordinates, as it does in the nondegenerate Gaussian realization. Then there exist distinct probability measures on Y_D with identical coordinate marginals. Consequently the family $\{q^{o,X}_{i,a}, s^{o,X}_{i,a}\}$ does not determine $Q_X(\cdot \mid o)$ under this richness hypothesis. Without it the conclusion can fail: for one agent and one design point with $\mathsf{K}_{i,a} = \mathsf{M}_{i,a} = \{0\}$, the space Y_D is a singleton and its unique law is determined by its marginals. [ESTABLISHED]

Proof. Take $Q^{(0)} = \mathcal{N}(0, I_2)$ and $Q^{(r)} = \mathcal{N}(0, \Sigma_r)$ with $\Sigma_r = \left(\begin{smallmatrix} 1 & r \\ r & 1 \end{smallmatrix}\right)$ for any $r \in (-1, 1)$ with $r \neq 0$; both are probability measures, since Σ_r has eigenvalues $1 \pm r > 0$. The coordinate marginals of a centered Gaussian $\mathcal{N}(0, \Sigma)$ on $\mathbb{R}^2$ are $\mathcal{N}(0, \Sigma_{11})$ and $\mathcal{N}(0, \Sigma_{22})$, so both measures have standard normal marginals in both coordinates. They are distinct because a Gaussian law is determined by its mean and covariance and the covariances differ. Embedding this pair in any two coordinates of Y_D and keeping the remaining coordinates fixed produces two distinct elements of $\mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D)$ with the same family (3.24). □

3.7 Continuum sections versus finite evaluation

Chapter 2 gave each agent continuum sections $q^{o,X}_i$ and $s^{o,X}_i$ over its support. This chapter has given the finite recognition law Q_X and its marginals. There are now two ways to obtain a probability measure on $\mathsf{K}_{i,a}$: sample the continuum section at c_a, or marginalize the finite law at (i, a). Write the first as

$$\mathrm{Samp}_{D,i}\big(q^{o,X}_i\big) = \big(q^{o,X}_i(c_a)\big)_{a=1}^M, \qquad \mathrm{Samp}_{D,i}\big(s^{o,X}_i\big) = \big(s^{o,X}_i(c_a)\big)_{a=1}^M. \tag{3.25}$$

The two constructions have different domains. The map (3.25) acts on sections in $\Gamma(\mathcal{C}_i, \mathcal{E}|_{\mathcal{C}_i})$; the maps of Definition 3.17 act on $\mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D)$. Neither is defined in terms of the other, so their outputs coincide only if that is declared. The following statement shows that the declaration is substantive in one direction and empty in the other, so that calling it a consequence would be wrong in both.

Proposition 3.19 (Compatibility determines neither side). Suppose the compatibility relation

$$q_i^{o,X}(c_a) = q_{i,a}^{o,X}, \qquad s_i^{o,X}(c_a) = s_{i,a}^{o,X} \qquad \text{for every } (i,a) \in \mathfrak{A} \tag{3.26}$$

holds. Then (i), under the two-nondegenerate-coordinate richness hypothesis of Proposition 3.18, (3.26) does not determine $Q_X(\cdot \mid o)$; and (ii), if some support $\mathcal{C}_i$ contains a point outside the finite design D and the selected section class admits a nonzero smooth deformation supported away from D, the relation does not determine the sections $q_i^{o,X}$ and $s_i^{o,X}$. The positive-dimensional Gaussian tier supplies such deformations. If every support is zero-dimensional and exhausted by its design points, the finite values can instead determine a section pointwise. [ESTABLISHED]

Proof. (i) By Proposition 3.18 there are distinct laws in $\mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D)$ with the same coordinate marginals, and (3.26) constrains Q_X only through those marginals; both such laws satisfy it against the same sections. (ii) Fix $c^\star \in \mathcal{C}_i \setminus D$. In the positive-dimensional Gaussian realization, choose a neighborhood of $c^\star$ disjoint from the finite set D and a smooth bump function $\chi : \mathcal{C}_i \to [0,1]$ supported there with $\chi(c^\star) = 1$. The section obtained from $q_i^{o,X}$ by replacing $\mu_i(c)$ with $\mu_i(c) + \chi(c)v$ for any fixed $v \neq 0$ is a smooth section of the same bundle, differs from $q_i^{o,X}$ at $c^\star$, and agrees with it on all of D; it therefore satisfies (3.26) against the same marginals. By contrast, if $\mathcal{C}_i = D \cap \mathcal{C}_i$ as a zero-dimensional support, equality at those points is pointwise equality of the section, so the nonuniqueness conclusion has no such witness and can fail. $\square$

Hypothesis 3.20 (Sampling and marginalization are compatible). Where a result requires the continuum sections and the finite recognition law to describe the same coordinate laws, relation (3.26) is adopted. [HYPOTHESIS]

This is a choice and is named as one. It excludes configurations in which an agent's continuum section and the finite law disagree at a design point, which is a genuine possibility because the two objects are declared independently. It is invoked only where a statement about the finite law is transferred to the geometric layer, or the reverse; a result stated entirely about the finite construction does not need it. Proposition 3.19 records that, under its richness hypothesis, adopting compatibility does not reconstruct the joint from its marginals and, when the support extends beyond the design and the selected fiber is locally deformable, does not reconstruct a continuum section from finite samples. Singleton finite fibers and zero-dimensional supports exhausted by the design are the explicit exceptions.

3.8 What the finite construction does not supply

Open Problem 3.21 (Continuum probabilistic theory). The finite construction supplies probability spaces and kernels associated with the declared design D. It supplies no probability measure on a space of sections over $\mathcal{C}$, no unique extension of the finite marginals to such a space, and no continuum limit. Whether one exists for this construction is not settled here. [OPEN]

What would close it can be listed exactly, and listing it is the point of declaring the gap rather than gesturing at it. One would need a refining family of designs; a compatible, that is projective, system of finite laws indexed by those designs, so that a Kolmogorov-type extension theorem applies [39]; a declared topology and sigma-algebra on the candidate space of sections, since the extension theorem produces a measure on a product space and further work is needed to place it on a space of smooth sections; tightness or compactness sufficient to control the limit; convergence of the functionals and observables that the finite theory computes; control of the gauge choices of Chapter 2 and of the reference measures of Section 3.3 under refinement, since neither is canonical; and a proof that any proposed continuum interaction energy is normalizable.

The last item is where the obstruction is concrete rather than merely laborious. The reference measures (3.7) are finite products, and sigma-finiteness was inherited from finitely many sigma-finite factors; the same construction does not extend to infinitely many non-probability factors, so a continuum theory cannot obtain its reference measure by the route used here and would have to declare a measure on the section space directly. Until that is done, a continuum energy has no density to be the exponent of, and the objects that the finite theory calls densities have no continuum counterparts. Every evidence-bound claim in this document is accordingly a finite-design claim, and none of them is asserted in a continuum limit.

Chapter 4

The Fixed Normalized Generative Law

This chapter fixes one normalized joint before inference begins. The latent channels inherit the two representations of the common principal gauge group from Chapter 2. Their local coordinates may be re-expressed independently, so the covariance formulas carry a passive product of frame changes without introducing a second principal bundle. The linear-Gaussian realization is deferred to Chapter 13.

4.1 Parameters, structure, and kernel types

Fix the design $D = \{c_a\}_{a=1}^{M}$, the agent set $V = \{1, \dots, N\}$, and the measurable spaces and sigma-finite reference measures of Chapter 3. Structural data $X \in \mathsf{X}$ contain a finite directed acyclic graph $\Gamma = (V, E)$, design incidence, and the geometric data. Numerical parameters that a learning coordinate may move are denoted θ. This split is declared rather than derived; all typing requirements below apply to both symbols. [DEFINITION]

Write $\mathrm{pa}(i)$ for the parents of i and $\mathcal{R} = \{r : \mathrm{pa}(r) = \varnothing\}$. At one design point, declare normalized Markov kernels

$$P^{m}_{\theta,i}(dm_i \mid m_{\mathrm{pa}(i)}, X), \qquad P^{k}_{\theta,i}(dk_i \mid k_{\mathrm{pa}(i)}, m_i, X), \qquad P^{o}_{\theta,i}(do_i \mid k_i, m_i, X). \tag{4.1}$$

At a root, the first factor is a proper model law and the second is a normalized bridge from the same-agent model coordinate to the belief coordinate. [DEFINITION]

The channels are asymmetric by declaration. At a root, write $K_i(m, A) = P^{k}_{\theta,i}(A \mid m, X)$. This kernel induces a map on laws,

$$\widetilde{\phi}_i(s)(A) = \int_{\mathsf{M}} K_i(m, A) s(dm). \tag{4.2}$$

Only this pushforward-on-laws map can represent the local associated-bundle morphism $\widetilde{\Phi} : (\mathcal{E}_m)_c \to (\mathcal{E}_b)_c$, and only if its contextual and gauge compatibility is declared. At a nonroot, the kernel is indexed by the parent belief coordinate; it does not define a map $\mathcal{E}_m \to \mathcal{E}_b$ unless that parent is fixed or the domain is enlarged. Otherwise it is a separately declared kernel-valued bridge. In every case the generative dependence runs from model samples to belief samples. The geometric existence of $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ neither inserts a reverse factor into (4.1) nor forces cyclic generative dependence. A model using both directions would need a separately normalized joint and cannot infer its factorization from the two associated-bundle morphisms.

4.2 Iterated composition and densities

Choose a topological ordering $v_1, \ldots, v_N$ of Γ. The one-design-point law is the successive kernel composition

$$P_{\theta,a}(do_a, dY_a \mid X) = \prod_{\ell=1}^{N} P^m_{\theta,v_\ell}(dm_{v_\ell} \mid m_{\mathrm{pa}(v_\ell)}, X) P^k_{\theta,v_\ell}(dk_{v_\ell} \mid k_{\mathrm{pa}(v_\ell)}, m_{v_\ell}, X) P^o_{\theta,v_\ell}(do_{v_\ell} \mid k_{v_\ell}, m_{v_\ell}, X). \tag{4.3}$$

The product denotes ordered application of kernels, not multiplication of measures. For bounded measurable f, its meaning is

$$\int f dP_{\theta,a} = \int P^m_{\theta,v_1}(dm_{v_1} \mid X) \int P^k_{\theta,v_1}(dk_{v_1} \mid m_{v_1}, X) \int P^o_{\theta,v_1}(do_{v_1} \mid k_{v_1}, m_{v_1}, X) \cdots f(o_a, y_a). \tag{4.4}$$

Every conditioning coordinate has been introduced before its kernel is applied.

Construction 4.1 (Finite directed law). Equation (4.4) defines a probability kernel $X \mapsto P_{\theta,a}(\cdot \mid X)$. [ESTABLISHED]

Proof. Integrating a bounded measurable function against a probability kernel preserves boundedness and measurability in the remaining variables. Finite iteration therefore defines a measure, and $f \equiv 1$ gives total mass one. This is the finite kernel construction underlying directed graphical models [42, 47, 75]. □

Distinct design points are conditionally independent given X:

$$P_\theta(do, dY \mid X) = \bigotimes_{a=1}^{M} P_{\theta,a}(do_a, dY_a \mid X). \tag{4.5}$$

This modeling hypothesis excludes residual cross-design dependence and is used in the finite-design normalization and factor count. [HYPOTHESIS]

When component kernels have densities relative to the declared component reference measures,

$$p_{\theta,a}(o_a, y_a \mid X) = \prod_{i \in V} p^m_{\theta,i}(m_i \mid m_{\mathrm{pa}(i)}, X) p^k_{\theta,i}(k_i \mid k_{\mathrm{pa}(i)}, m_i, X) p^o_{\theta,i}(o_i \mid k_i, m_i, X), \tag{4.6}$$

and

$$p_\theta(o, y \mid X) = \prod_{a=1}^{M} p_{\theta,a}(o_a, y_a \mid X), \qquad P_\theta(do, dY \mid X) = p_\theta(o, y \mid X)\nu_D^O(do)\nu_D^Y(dy). \tag{4.7}$$

The measure P_θ is authoritative; p_θ denotes only its Radon–Nikodym density.

4.3 The typing prohibition

Requirement 4.2 (Typing prohibition). *Each factor in (4.1) is a function only of θ, X, and its displayed parent and same-agent latent coordinates. No generative kernel may take as an argument the recognition kernel Q_X, one of its marginals or parameters, or a posterior of the model.* [DEFINITION]

This requirement defines the fixed-joint construction and proves nothing. A transition built from a live peer recognition marginal is a map from recognition laws to kernels, not a factor of a joint fixed before inference.

Proposition 4.3 (No distinguished target). Let $\mathcal{Q}$ be a nonempty family of recognition laws. If a factor in (4.3) is replaced by a nonconstant Q-indexed family, producing a normalized joint $P_{\theta,Q}$ for each $Q \in \mathcal{Q}$, assume at the displayed observation that $0 < p_{\theta,Q}(o \mid X) < \infty$, that $Q \ll P_{\theta,Q}(\cdot \mid o, X)$, and that the log-density terms defining the ELBO and KL divergence are integrable. Then for each fixed Q

$$\log p_{\theta,Q}(o \mid X) = \mathcal{L}(Q; X) + D_{\mathrm{KL}}\left(Q \| P_{\theta,Q}(\cdot \mid o, X)\right). \tag{4.8}$$

The family itself selects no joint as "the model." Writing $e(Q) = \log p_{\theta,Q}(o \mid X)$, two members satisfy

$$\begin{aligned} \mathcal{L}(Q'; X) - \mathcal{L}(Q; X) &= e(Q') - e(Q) \\ &\quad - \left[D_{\mathrm{KL}}\big(Q' \| P_{\theta,Q'}(\cdot \mid o, X)\big) - D_{\mathrm{KL}}\big(Q \| P_{\theta,Q}(\cdot \mid o, X)\big)\right]. \end{aligned} \tag{4.9}$$

Thus a divergence decrease certifies improvement relative to one fixed reference value exactly when the two members lie on one level set of e. [ESTABLISHED]

Proof. Apply the exact ELBO identity to each fixed pair $(Q, P_{\theta,Q})$ and subtract the two instances. No member is distinguished because no selection rule was declared. □

Proposition 4.4 (The stronger consequences fail). Nonconstant dependence of $P_{\theta,Q}$ on Q does not imply that e varies, that no ascent can improve a fixed quantity, or that no single member's evidence bounds the family of bounds. [ESTABLISHED]

Proof. Take one binary latent y, one binary observation o, and $Q_\beta = \mathrm{Ber}(\beta)$ for $\beta \in (0, 1)$. Let $g(\beta) = \beta/2 + 1/4$ and define

$$P_{\theta,Q_\beta}(o, y \mid X) = \tfrac{1}{2}\,\mathrm{Ber}\big(g(\beta)\big)(y). \tag{4.10}$$

The joint is normalized and varies injectively with β, but $e(Q_\beta) = \log(1/2)$ for every β. The posterior is $\mathrm{Ber}(g(\beta))$, so the gap at $\beta = 9/10, 7/10, 1/2$ is respectively

$$\tfrac{9}{10}\log\tfrac{9}{7} + \tfrac{1}{10}\log\tfrac{1}{3} = 0.1163218\ldots, \qquad \tfrac{7}{10}\log\tfrac{7}{6} + \tfrac{3}{10}\log\tfrac{3}{4} = 0.0216009\ldots, \qquad 0. \tag{4.11}$$

The bound therefore increases to the fixed evidence, and the member at $\beta = 1/2$ has evidence equal to the supremum of all these bounds. □

These are closed-form relative entropies, not numerical experiments. The development declines the three stronger statements as false in this witness, while retaining the absence of a distinguished member. [NOT-CLAIMED]

The typing prohibition does not ban population coupling through latents inside a fixed joint; it bans a generative factor from reading the posterior that the joint is supposed to determine. [DEFINITION]

4.4 Exact normalization

Proposition 4.5 (Normalization without a partition function). Under the domination assumptions of Chapter 3 and Construction 4.1,

$$P_\theta(\mathsf{O}_D \times \mathsf{Y}_D \mid X) = 1, \qquad \int p_\theta(o, y \mid X)\nu_D^O(do)\nu_D^Y(dy) = 1 \tag{4.12}$$

for every admissible (θ, X), without a global normalizing constant. [ESTABLISHED]

Proof. At one design point, Tonelli's theorem permits reverse-topological integration. Integrate each observation factor first. Then remove each nonroot in reverse topological order by integrating its normalized belief kernel and then its normalized model kernel; all descendant factors containing those coordinates have already disappeared. Finally integrate each root bridge and proper root law. Every integral contributes one. The same induction proves the measure statement without domination, and (4.5) multiplies M unit masses. □

4.5 The normalized undirected alternative

An undirected cyclic model is valid when its global normalizer exists. For nonnegative node and edge potentials, define

$$\Psi_X(o, y) = \prod_{i \in V} \psi_i(o_i, k_i, m_i; X) \prod_{\{i,j\} \in E} \psi_{ij}(k_i, m_i, k_j, m_j; X), \tag{4.13}$$

$$Z_X = \int \Psi_X(o, y)\nu_D^O(do)\nu_D^Y(dy), \qquad 0 < Z_X < \infty, \tag{4.14}$$

$$p_\theta^{\mathrm{Gibbs}}(o, y \mid X) = Z_X^{-1}\Psi_X(o, y). \tag{4.15}$$

Finiteness and positivity of Z_X are model-existence conditions. [DEFINITION]

Proposition 4.6 (Local normalization is insufficient). Normalized node potentials and an everywhere positive, finite, continuous edge potential need not give finite Z_X. [ESTABLISHED]

Proof. On $\mathbb{R}^2$ take

$$\psi_i(y_i) = (2\pi)^{-1/2} e^{-y_i^2/2}, \qquad \psi_{12}(y_1, y_2) = e^{c y_1 y_2}, \qquad c \geq 1. \tag{4.16}$$

The quadratic precision is

$$\Lambda_{\text{ex}} = \begin{pmatrix} 1 & -c \\ -c & 1 \end{pmatrix}, \qquad \operatorname{spec} \Lambda_{\text{ex}} = \{1 - c, 1 + c\}. \tag{4.17}$$

In orthonormal sum-and-difference coordinates the integral factorizes, and the factor in the $1 - c$ direction diverges. □

The document chooses the directed construction because its normalization does not depend on the parameters, not because normalized cyclic Gaussian Markov random fields are invalid.

4.6 Covariance under independent channel-frame re-expression

At each design point c_a and vertex, independently re-express the belief and model frame coordinates by $g^b_{a,i} = g^b_i(c_a) \in G$ and $g^m_{a,i} = g^m_i(c_a) \in G$. Define the old-to-new represented sample-coordinate maps

$$R^b_{a,i} := \rho_b(g^b_{a,i}) : \mathsf{K} \longrightarrow \mathsf{K}, \qquad R^m_{a,i} := \rho_m(g^m_{a,i}) : \mathsf{M} \longrightarrow \mathsf{M}. \tag{4.18}$$

With the passive-section convention of (2.10), this sample-coordinate transformation corresponds to the section rechoice $u^{x\prime}_i = u^x_i \cdot (g^x_i)^{-1}$, whose value at c_a uses $(g^x_{a,i})^{-1}$. Thus $R^x_{a,i}$ acts within the channel's sample fiber, while the group element $(g^x_i)^{-1} \in G$, not the represented operator $(R^x_{a,i})^{-1}$, multiplies the principal frame on the right. This distinction remains necessary when the representation is not faithful. Within a formula for one design point, suppress the index a and write g^x_i, R^x_i. They act on latent coordinates as $k'_i = R^b_i k_i$ and $m'_i = R^m_i m_i$, while observations are gauge inert. [DEFINITION]

To permit all such independent coordinate changes, the finite-design structural data carry graph-link copies $\Theta^x_{a,ij}$, one for each design incidence. At fixed a write Θ^x_{ij}; their discrete vertex frame changes are

$$h^x_{a,i} := g^x_{a,i} = g^x_i(c_a), \qquad a^x_{a,i} := (h^x_{a,i})^{-1}, \tag{4.19}$$

in the notation of (2.47). The contextual coordinate element $h^x_{a,i}$, abbreviated h^x_i at one design point, is unrelated to the relative principal-frame field h_i defined by (2.3). The suppressed one-design-point law is

$$\Theta^{b\prime}_{ij} = h^b_i \Theta^b_{ij} (h^b_j)^{-1}, \qquad \Theta^{m\prime}_{ij} = h^m_i \Theta^m_{ij} (h^m_j)^{-1}. \tag{4.20}$$

Consequently the represented links obey

$$\rho_x(\Theta^{x\prime}_{ij}) = R^x_i \rho_x(\Theta^x_{ij}) (R^x_j)^{-1}, \qquad x \in \{b, m\}, \tag{4.21}$$

and the inverse-orientation rule is preserved. Equations (2.47) and (4.20) are therefore the same passive law written with mutually inverse rechoice conventions. [ESTABLISHED]

If instead one structural link is constrained to be identical at every design point, an admissible frame rechoice must preserve that constraint: for every two design indices a, a' and every constrained edge,

$$h^x_{a,i}\Theta^x_{ij}(h^x_{a,j})^{-1} = h^x_{a',i}\Theta^x_{ij}(h^x_{a',j})^{-1}. \tag{4.22}$$

[DEFINITION]

At one fixed shared-link datum, define the admissible family to be the solution set of (4.22). Define the group that acts on the entire constrained submodel to be the setwise stabilizer of the shared-link constraint set. A larger transformation law requires an explicitly declared parameter compensation. [DEFINITION]

Proposition 4.7 (Shared-link admissibility need not be closed under composition). For $G = \mathrm{GL}(2,\mathbb{R})$, two design points, one constrained edge, and $\Theta_{ij} = I$, the solution set of (4.22) is not closed under pointwise multiplication. [ESTABLISHED]

Proof. Let $S = \mathrm{diag}(2,1)$ and $C = \left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$. The rechoice $h_{1,i} = h_{1,j} = I$, $h_{2,i} = h_{2,j} = S$ sends the shared link to I at both design points. The rechoice $\widetilde{h}_{a,i} = C$, $\widetilde{h}_{a,j} = I$ sends it to C at both points. For their pointwise product $h\widetilde{h}$, the two transformed links are C and $SCS^{-1} = \left(\begin{smallmatrix}1&2\\0&1\end{smallmatrix}\right)$, respectively, so the product violates (4.22). $\square$

Context-independent rechoices form a sufficient subgroup, but the first rechoice in the proof shows that they are not necessary. [ESTABLISHED]

At the unconstrained finite-design level, declaring the coordinate re-expressions independently gives the passive product $\prod_{a,i}(G \times G)$. If they must arise as values of smooth maps $g^x_i : \mathcal{C}_i \to G$, the realized group is the image of the corresponding evaluation map; it equals the displayed product whenever that evaluation map is surjective. For example, this holds for a smooth positive-dimensional base at finitely many distinct design points when G is connected, by smooth interpolation in disjoint coordinate neighborhoods. This product records two independently chosen trivializations of associated bundles induced from the same P; it is not a second principal structure group. The active principal gauge group is $\mathrm{Aut}_G(P)$ over $\mathrm{id}_{\mathcal{C}}$; one automorphism acts on both associated bundles, with local functions related by (2.12). Its image in the passive product is contained in the pointwise h_i-twisted diagonal and is contained in the ordinary diagonal in the common-frame specialization $h_i = e_G$. Equality with the corresponding pointwise diagonal holds only when the relevant active-gauge evaluation map is surjective; disconnected structure groups or global bundle constraints can obstruct that surjectivity. It is not the principal right G-action itself. A context-independent frame re-expression is the subgroup for which $g^x_{a,i}$ does not depend on a. [DEFINITION]

Hypothesis 4.8 (Kernel covariance). The transformed parameters (θ', X') are declared so that every nonroot factor obeys

$$P^m_{\theta',i}\left(\cdot \mid R^m_{\mathrm{pa}(i)} m_{\mathrm{pa}(i)}, X'\right) = (R^m_i)_\# P^m_{\theta,i}\left(\cdot \mid m_{\mathrm{pa}(i)}, X\right), \tag{4.23}$$

$$P^k_{\theta',i}\left(\cdot \mid R^b_{\mathrm{pa}(i)} k_{\mathrm{pa}(i)}, R^m_i m_i, X'\right) = (R^b_i)_\# P^k_{\theta,i}\left(\cdot \mid k_{\mathrm{pa}(i)}, m_i, X\right), \tag{4.24}$$

$$P^o_{\theta',i}\left(\cdot \mid R^b_i k_i, R^m_i m_i, X'\right) = P^o_{\theta,i}\left(\cdot \mid k_i, m_i, X\right), \tag{4.25}$$

with the analogous root identities. [HYPOTHESIS]

The second identity is the covariance requirement for the model-to-belief sample kernel. At a root it implies that the induced law map (4.2) obeys (2.15); at a nonroot it remains a parent-indexed kernel law unless an enlarged cross-bundle domain is declared. Neither route uses Φ.

Proposition 4.9 (Joint measure covariance and evidence invariance). Let $T(o, y) = (o, Ry)$, where $R = \bigoplus_{a,i}(R^b_{a,i} \oplus R^m_{a,i})$. Under Hypothesis 4.8,

$$P_{\theta'}(\cdot \mid X') = T_\# P_\theta(\cdot \mid X), \tag{4.26}$$

as probability measures, without a density or reference-measure hypothesis. Because T is the identity on observations, the observation marginals agree. Relative to the common observation reference measure this gives

$$p_{\theta'}(o \mid X') = p_\theta(o \mid X) \quad \text{for } \nu^O_D\text{-almost every } o. \tag{4.27}$$

If latent densities exist and $R_\# \nu^Y_D \ll \nu^Y_D$, define

$$j_R(y') := \frac{d(R_\# \nu^Y_D)}{d\nu^Y_D}(y'). \tag{4.28}$$

Then the density statement is

$$p_{\theta'}(o, y' \mid X') = p_\theta(o, R^{-1} y' \mid X) j_R(y') \quad \text{for } (\nu^O_D \otimes \nu^Y_D)\text{-almost every } (o, y'). \tag{4.29}$$

On the further differentiable equal-dimensional Euclidean tier, when every represented block is an invertible linear map and ν^Y_D is product Lebesgue measure, $j_R = |\det R|^{-1}$ and hence

$$p_{\theta'}(o, y' \mid X') = |\det R|^{-1} p_\theta(o, R^{-1} y' \mid X). \tag{4.30}$$

Equal dimensionality here is between the source and target coordinates within each channel; it does not impose $K = d_m$. The determinant formula is not asserted for mixed or atomic reference measures. [ESTABLISHED]

Proof. Apply (4.23)–(4.25) along the topological ordering. Each factor is the receiving-coordinate pushforward of the original factor, so the composed law is $T_\# P_\theta$. Projection to the observation coordinate commutes with this pushforward, so the observation measures agree and their densities satisfy (4.27). For a bounded measurable test function F on the joint observation–latent space,

$$\int F(o, y')p_{\theta'}(o, y' \mid X')\nu_D^O(do)\nu_D^Y(dy') = \int F(o, Ry)p_\theta(o, y \mid X)\nu_D^O(do)\nu_D^Y(dy),$$

by the measure identity. Rewriting the right side against $R_\# \nu_D^Y$ and applying (4.28) gives (4.29). On the stated Lebesgue tier, linear change of variables gives $d(R_\#\nu_D^Y)/d\nu_D^Y = |\det R|^{-1}$, proving (4.30). □

Evidence invariance is a redundancy constraint, not an empirical prediction: the action changes latent coordinates while leaving the observation law fixed. The linear-Gaussian chapter verifies Hypothesis 4.8 parameter by parameter and then displays the stronger common-frame reduction.

The passive coordinate action on the full parameter space can have nontrivial stabilizers, so its orbit space need not be a manifold. Subtracting the dimension of the coordinate group from a parameter count is unjustified without freeness and properness. Determining generic stabilizers and a regular quotient remains open; nothing later depends on that quotient. [OPEN]

Chapter 5

The Exact Single Evidence Bound

The previous chapter fixed one normalized joint. This chapter introduces one recognition law for the entire population and derives the exact measure-level identity that relates the two. Its primary form is an extended-real lower bound defined for every probability law; the familiar difference of expected log densities is a finite-domain representation of that functional. Conditional and local bounds used later are disintegrations or restrictions of this same construction, not competing definitions of evidence.

The chapter is organized around three things that are easy to get wrong. The recognition law is a joint law and is not recoverable from its coordinate marginals, so a calculation that substitutes marginals into an entropy is computing something else and the difference has a name and a sign. The identity holds under stated hypotheses, one of which, absolute continuity with respect to the posterior, is a genuine restriction and is exactly what later forbids degenerate recognition laws. And the exact coordinates of the alternating scheme are two specific objects; an optimizer direction is a proposal that must pass an acceptance test before it is entitled to the conclusions that attach to a coordinate.

There is also a measure-theoretic qualification that applies to every fixed-observation formula in this chapter. For fixed (θ, X), let $P^O_{\theta,X}$, $\Pi_{\theta,X}$, and the marginal-full set $\mathsf{O}^{\mathrm{reg}}_{\theta,X}$ be the observation marginal, regular conditional probability, and regular-observation set of Chapter 3. Unless an everywhere-defined density and conditional version has been declared as part of the model data, every posterior and ELBO statement below is a $P^O_{\theta,X}$-almost-everywhere statement, represented by selecting $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,X}$. Changing a density on a null slice can change exceptional point values but cannot change these almost-everywhere statements.

5.1 The correlated population recognition kernel

Definition 5.1 (Correlated population recognition kernel). Fix the finite design D, the observed value $o \in \mathsf{O}_D$, and the structural value $X \in \mathsf{X}$ of Chapter 3. Recognition is represented by one normalized Markov kernel

$$Q : (\mathsf{O}_D \times \mathsf{X}, \mathscr{O}_D \otimes \mathscr{X}) \longrightarrow \mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D), \qquad (o, X) \longmapsto Q_X(dY \mid o), \tag{5.1}$$

so that $Q_X(\mathsf{Y}_D \mid o) = 1$ and, for every $B \in \mathscr{Y}_D$, the map $(o, X) \mapsto Q_X(B \mid o)$ is $\mathscr{O}_D \otimes \mathscr{X}$-measurable. [DEFINITION]

The law is permitted to correlate agents with one another, the state channel with the model channel, and one design point with another. No factorization is imposed here; the restrictions that impose one are the subject of Chapter 15, and they are restrictions on this object rather than replacements for it. [DEFINITION]

Write $q_{i,a}^{o,X}$ and $s_{i,a}^{o,X}$ for the coordinate marginals defined in Definition 3.17. By Proposition 3.18, these marginals do not reconstruct $Q_X(\cdot \mid o)$ whenever the latent space contains at least two nondegenerate real coordinates.

The proposition is a statement about a single pair of coordinates, and the general situation is worse in a specific way. The set of joint laws with a prescribed family of marginals is the Fréchet class of those marginals; it is convex, it contains the product law, and in continuous Euclidean coordinates it is parameterized by the dependence structure, the copula, which the marginals leave entirely free [57]. When second moments exist, every cross-agent, cross-channel, and cross-design covariance block is unspecified by the marginals. The consequence for this chapter is not stylistic. Marginal recognition laws cannot be substituted for the population law in an exact entropy or evidence calculation, and the following makes the error exact rather than merely warning against it.

Theorem 5.2 (Extended total-correlation chain identity). Let $Y = (Y_b)_{b=1}^{B}$ take values in a finite product of standard Borel spaces, let Q be any joint probability law with marginals Q_b, let $(\rho_b)_{b=1}^{B}$ be arbitrary probability laws, and put $R = \bigotimes_b \rho_b$. With

$$\mathrm{TC}(Q) := D_{\mathrm{KL}}\left(Q \,\middle\|\, \bigotimes_b Q_b\right) \in [0, +\infty],$$

the identity

$$D_{\mathrm{KL}}\left(Q \,\middle\|\, \bigotimes_b \rho_b\right) = \mathrm{TC}(Q) + \sum_b D_{\mathrm{KL}}(Q_b \| \rho_b) \tag{5.2}$$

holds in $[0, +\infty]$. It includes both singular branches and never subtracts infinities. [ESTABLISHED]

Proof. For each block choose finite refining partitions $\mathcal{P}_{b,n}$ that generate its sigma-algebra, and let π_n be the resulting finite product quantization. On the finite product alphabet, direct summation gives

$$D_{\mathrm{KL}}(\pi_{n\#}Q \| \textstyle\bigotimes_b \pi_{b,n\#}\rho_b) = D_{\mathrm{KL}}(\pi_{n\#}Q \| \textstyle\bigotimes_b \pi_{b,n\#}Q_b) + \textstyle\sum_b D_{\mathrm{KL}}(\pi_{b,n\#}Q_b \| \pi_{b,n\#}\rho_b).$$

All displayed terms are nonnegative extended numbers, so this finite identity uses no signed-log cancellation. The refining-partition characterization of relative entropy, (3.17) applied to constant kernels, makes the left side increase to $D_{\mathrm{KL}}(Q \| \bigotimes_b \rho_b)$, the first term on the right increase to $\mathrm{TC}(Q)$, and each of the finitely many remaining terms increase to $D_{\mathrm{KL}}(Q_b \| \rho_b)$. Passing to the monotone limits proves (5.2) in $[0, +\infty]$.

For an explicit branch audit, if $Q \not\ll R$ then the left side is $+\infty$. If some $Q_b \not\ll \rho_b$, the corresponding marginal term is $+\infty$. Otherwise $\bigotimes_b Q_b \ll R$; transitivity shows that $Q \ll \bigotimes_b Q_b$ would contradict $Q \not\ll R$, so $\mathrm{TC}(Q) = +\infty$. Thus the right side is also $+\infty$ in every singular branch. $\square$

The infinite branch is realized by a continuous diagonal, not merely by a defective density calculation. If $U \sim \mathrm{Unif}[0,1]$ and $Q = \mathrm{law}(U,U)$, then both marginals are uniform but Q is singular with respect to $Q_1 \otimes Q_2$, so $\mathrm{TC}(Q) = +\infty$. Taking $\rho_b = Q_b$ makes (5.2) read $+\infty = +\infty + 0 + 0$ in the extended nonnegative reals.

Proposition 5.3 (Finite-density total correlation and the two signs). Partition the coordinates of Y into blocks indexed by b, let Q_X have density q_X with respect to ν_D^Y and block marginal densities q_b, and suppose the joint and all block differential entropies are finite. Let

$$\mathrm{TC}(Q_X) = D_{\mathrm{KL}}\left(Q_X \middle\| \bigotimes_b Q_{X,b}\right) \geq 0 \tag{5.3}$$

be the total correlation of the blocks. Then

$$\mathbb{E}_{Q_X}\left[\log q_X(Y \mid o)\right] = \sum_b \mathbb{E}_{Q_X}\left[\log q_b(Y_b \mid o)\right] + \mathrm{TC}(Q_X), \tag{5.4}$$

with equality of the two log-density expectations exactly when the blocks are independent under Q_X. For the following free-energy and pseudo-ELBO identities, assume in addition

$$\mathbb{E}_{Q_X}\left|\log p_\theta(o, Y \mid X)\right| < \infty,$$

so the common expected negative-log-joint term and both finite entropy terms define finite real quantities. Define the substituted free energy and its negative pseudo-ELBO by

$$\widetilde{\mathcal{F}} := \mathbb{E}_{Q_X}\left[-\log p_\theta(o, Y \mid X) + \sum_b \log q_b(Y_b \mid o)\right], \tag{5.5}$$

$$\widetilde{\mathcal{L}} := -\widetilde{\mathcal{F}}. \tag{5.6}$$

Then

$$\widetilde{\mathcal{F}} = \mathcal{F} - \mathrm{TC}(Q_X), \qquad \widetilde{\mathcal{L}} = \mathcal{L}(Q_X; X) + \mathrm{TC}(Q_X). \tag{5.7}$$

Thus it is the *substituted free energy* that is smaller by total correlation, while its negative *pseudo-ELBO* is larger by total correlation and need not be a lower bound on the log evidence. In the Gaussian case with joint covariance C and diagonal blocks C_{bb},

$$\mathrm{TC}(Q_X) = \tfrac{1}{2}\left(\sum_b \log\det C_{bb} - \log\det C\right). \tag{5.8}$$

[ESTABLISHED]

Proof. This is the finite-density corollary of Theorem 5.2 with $\rho_b = Q_b$, after the stated entropy hypotheses permit the logarithms to be split. Expanding gives $\mathrm{TC}(Q_X) = \mathbb{E}_{Q_X}[\log q_X] - \sum_b \mathbb{E}_{Q_X}[\log q_b]$, which is (5.4). Subtracting (5.5) from $\mathcal{F} = \mathbb{E}_{Q_X}[-\log p_\theta + \log q_X]$ gives $\mathrm{TC}(Q_X)$, and negation gives the second identity in (5.7). For (5.8), substitute the Gaussian differential entropy $\frac{1}{2}\log\det(2\pi e C)$ for the joint and for each block; the dimensional constants cancel because the block dimensions sum to d. □

The sign in Proposition 5.3 matters and is the reason the point is made here rather than in a remark. Substituting marginal entropies produces a smaller free-energy functional and, after negation, a larger pseudo-ELBO that is not a bound at all. A scheme that reports the latter as an evidence lower bound is reporting a quantity that can exceed the log evidence, and by an amount that grows with exactly the correlation the recognition law was introduced to represent.

5.2 Hypotheses

Hypothesis 5.4 (Evidence-domain hypotheses). The extended identity of Section 5.3 uses (H1)–(H2); its classical split into two expected log densities additionally uses (H3)–(H4). The selection is subject to the standing qualification $o \in \mathrm{O}^{\mathrm{reg}}_{\theta,X}$; equivalently, the measure-level theorem holds for $P^O_{\theta,X}$-almost every observation. Together (H1)–(H4) are the consolidated classical evidence domain and reproduce Hypotheses 3.15 and 3.16. [HYPOTHESIS]

(H1) Types and conditional version. The joint kernel $P_\theta(\cdot \mid X)$ of Chapter 4 and the recognition kernel $Q_X(\cdot \mid o)$ of Definition 5.1 are normalized and measurable as declared. The generative density $p_\theta(o, y \mid X)$ exists relative to the declared reference, and $P_\theta(\cdot \mid o, X)$ denotes the regular conditional probability $\Pi_{\theta,X}(o,\cdot)$ represented by (3.15) on $\mathrm{O}^{\mathrm{reg}}_{\theta,X}$. No density of Q_X is required at the extended tier. An everywhere pointwise version may replace the almost-everywhere convention only when that version is declared as model data.

(H2) Positive finite evidence. $0 < p_\theta(o \mid X) < \infty$. This excludes observed values that the model assigns zero density, at which the conditional posterior is undefined and there is nothing to approximate. It is used to divide by the evidence in (5.19).

(H3) Absolute continuity and recognition density for the classical split. $Q_X(\cdot \mid o) \ll \Pi_{\theta,X}(o,\cdot) = P_\theta(\cdot \mid o, X)$. Since the selected posterior slice is dominated by ν_D^Y, this implies $Q_X \ll \nu_D^Y$; write its RN density as $q_X(y \mid o)$. For parameterized recognition kernels, Theorem 3.8 and corollary 3.9 supplies a jointly measurable version under their stated hypotheses. This is a genuine restriction on the finite classical representation, but not on the extended functional, which assigns value $-\infty$ to the singular branch. Proposition 5.5 below makes the exclusion precise.

(H4) Log-integrability.

$$\mathbb{E}_{Q_X}\left[|\log p_\theta(o, Y \mid X)| + |\log q_X(Y \mid o)|\right] < \infty. \tag{5.9}$$

This is not implied by (H1) through (H3), and a witness settles the point rather than leaving it asserted: take $d = 1$ with a standard Gaussian conditional posterior and a standard Cauchy recognition law. Both have strictly positive Lebesgue densities, so (H1) through (H3) hold; but $\log p_\theta(o, y \mid X)$ differs from $-y^2/2$ by a constant, and the Cauchy law has no second moment, so (5.9) fails. Relative entropy and the extended ELBO remain well defined without (H4). What fails is only the split into two separately finite expectations: the formal difference can be an undefined $\infty - \infty$.

Hypothesis (H3) deserves its own statement, because it is the one with consequences later.

Proposition 5.5 (Subspace support violates absolute continuity). Suppose the conditional posterior $P_\theta(\cdot \mid o, X)$ is dominated by Lebesgue measure on $\mathbb{R}^d$, as it is in the linear-Gaussian realization of Chapter 4 with the positive definite covariances (13.9), and suppose $Q_X(\cdot \mid o)$ is supported on a proper affine subspace $\mathcal{S} \subsetneq \mathbb{R}^d$, meaning $Q_X(\mathcal{S} \mid o) = 1$ with $\dim \mathcal{S} < d$. Then (H3) fails and

$$D_{\mathrm{KL}}\left(Q_X(\cdot \mid o) \| P_\theta(\cdot \mid o, X)\right) = +\infty. \tag{5.10}$$

[ESTABLISHED]

Proof. A proper affine subspace of $\mathbb{R}^d$ is Lebesgue null, so $P_\theta(\mathcal{S} \mid o, X) = 0$ while $Q_X(\mathcal{S} \mid o) = 1$. Absolute continuity fails on the set $\mathcal{S}$. By the definition of relative entropy for measures that are not absolutely continuous, the divergence is $+\infty$ [22, 44]. ☐

This is the exact mechanism that forbids subspace-supported recognition laws, and its scope should be stated exactly. It does not say that constraining the recognition law is illegitimate; it says that a constraint which collapses the support onto a lower-dimensional set puts the law outside the family for which the bound is finite, so the resulting cost is not a large finite number to be traded against something else but an infinite one. Chapter 15 takes this up in the case that matters, where a coarse-graining is expressed by tying groups of coordinates to a common value: the tied law is supported on the range of an aggregation matrix, hence exactly of the form excluded here, and the admissible finite operation constrains the recognition mean instead, with a cost that the same chapter computes exactly. The forward reference is deliberate; the present chapter supplies the reason, not the application.

5.3 The exact evidence identity

Definition 5.6 (Slice measure and extended ELBO). Assume (H1)–(H2), fix $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,X}$, and abbreviate the finite latent slice measure, its mass, and its normalization by

$$M_o(B) := \int_B p_\theta(o, y \mid X)\nu_D^Y(dy) = z\Pi_{\theta,X}(o, B), \qquad z := M_o(\mathsf{Y}_D) = p_\theta(o \mid X) \in (0, \infty). \tag{5.11}$$

For *every* probability law Q on $(\mathsf{Y}_D, \mathscr{Y}_D)$, define

$$\mathcal{L}^{\mathrm{ext}}(Q; X, o) := \log z - D_{\mathrm{KL}}\big(Q \| \Pi_{\theta,X}(o, \cdot)\big) \in [-\infty, \log z] \tag{5.12}$$

and

$$\mathcal{F}^{\text{ext}}[Q;X,o] := -\mathcal{L}^{\text{ext}}(Q;X,o) = -\log z + D_{\text{KL}}\big(Q \| \Pi_{\theta,X}(o,\cdot)\big). \tag{5.13}$$

These are well-defined extended-real quantities because $\log z$ is finite and relative entropy lies in $[0,+\infty]$. [DEFINITION]

Theorem 5.7 (Extended ELBO gap identity). For every probability law Q on the latent space,

$$\log z - \mathcal{L}^{\text{ext}}(Q;X,o) = \mathcal{F}^{\text{ext}}[Q;X,o] + \log z = D_{\text{KL}}\big(Q \| \Pi_{\theta,X}(o,\cdot)\big). \tag{5.14}$$

Consequently $\mathcal{L}^{\text{ext}} \leq \log z$, with equality exactly at $Q = \Pi_{\theta,X}(o,\cdot)$. No unconditional formula of the form $\log z = \mathcal{L}^{\text{ext}} + D_{\text{KL}}$ is asserted, because its right side is undefined when $\mathcal{L}^{\text{ext}} = -\infty$ and $D_{\text{KL}} = +\infty$. [ESTABLISHED]

Proof. Both equalities are direct substitutions from (5.12)–(5.13); they only add or subtract the finite number $\log z$. Nonnegativity and the zero condition for relative entropy give the bound and equality case. □

Proposition 5.8 (Relative-log representation). If $Q \ll M_o$ and $r = dQ/dM_o$, then

$$\mathcal{L}^{\text{ext}}(Q;X,o) = -\int_{\mathsf{Y}_D} \log r(y) Q(dy), \qquad \int_{\{r<1\}} -\log r dQ = \int_{\{r<1\}} -r \log r dM_o \leq \frac{z}{e}. \tag{5.15}$$

Thus the integral is well defined in $(-\infty,+\infty]$ and its negative lies in $[-\infty,+\infty)$. If $Q \not\ll M_o$, then $\mathcal{L}^{\text{ext}}(Q;X,o) = -\infty$. [ESTABLISHED]

Proof. Because $\Pi_{\theta,X}(o,\cdot) = M_o/z$, the RN chain rule gives $dQ/d\Pi = zr$ when $Q \ll M_o$. The negative part of $\log r$ is Q-integrable by (5.15), since $-u\log u \leq 1/e$ on $0 \leq u \leq 1$ and $M_o(\mathsf{Y}_D) = z$. Hence

$$D_{\text{KL}}(Q\|\Pi) = \int \log(zr) dQ = \log z + \int \log r dQ$$

is a legitimate extended identity, and substitution into (5.12) proves the first branch. Since M_o and Π have the same null sets, $Q \not\ll M_o$ implies $D_{\text{KL}}(Q\|\Pi) = +\infty$, proving the second. □

Under the full classical domain Hypothesis 5.4, the relative-log functional has the separately integrable density representation

$$\mathcal{L}(Q_X;X) = \mathbb{E}_{Q_X}\left[\log p_\theta(o,Y \mid X) - \log q_X(Y \mid o)\right]. \tag{5.16}$$

The notation suppresses the fixed observation o and the generative parameter θ; the parameter subscript is restored as $\mathcal{L}_\vartheta$ whenever two parameter values are compared. The variational free energy is the opposite sign,

$$\mathcal{F}[Q_X;X,o] = -\mathcal{L}(Q_X;X). \tag{5.17}$$

In this domain $\mathcal{L}(Q_X;X) = \mathcal{L}^{\text{ext}}(Q_X;X,o)$ and $\mathcal{F} = \mathcal{F}^{\text{ext}}$; outside it only the extended definitions are available.

Theorem 5.9 (Classical finite exact evidence identity). Under Hypothesis 5.4, for $P^O_{\theta,X}$-almost every observation o (equivalently, for the selected $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,X}$),

$$\log p_\theta(o \mid X) = \mathcal{L}(Q_X; X) + D_{\mathrm{KL}}\left(Q_X(\cdot \mid o) \| P_\theta(\cdot \mid o, X)\right). \tag{5.18}$$

[ESTABLISHED]

Proof. For $B \in \mathscr{Y}_D$, the regular-observation construction (3.15), together with (H1) and (H2), gives the dominated conditional posterior

$$P_\theta(B \mid o, X) = \frac{1}{p_\theta(o \mid X)} \int_B p_\theta(o, y \mid X) \nu^Y_D(dy), \qquad \frac{dP_\theta(\cdot \mid o, X)}{d\nu^Y_D}(y) = \frac{p_\theta(o, y \mid X)}{p_\theta(o \mid X)}, \tag{5.19}$$

which is normalized because the denominator is the evidence. Writing $q_X(y \mid o) = dQ_X(\cdot \mid o)/d\nu^Y_D$ and using (H3) with the Radon–Nikodym chain rule gives, Q_X-almost surely,

$$\frac{dQ_X(\cdot \mid o)}{dP_\theta(\cdot \mid o, X)}(y) = \frac{q_X(y \mid o) p_\theta(o \mid X)}{p_\theta(o, y \mid X)}. \tag{5.20}$$

Hypothesis (H4) permits integration of its logarithm term by term, so

$$\begin{aligned} D_{\mathrm{KL}}\left(Q_X(\cdot \mid o) \| P_\theta(\cdot \mid o, X)\right) &= \int_{\mathsf{Y}_D} Q_X(dy \mid o) \log \frac{dQ_X(\cdot \mid o)}{dP_\theta(\cdot \mid o, X)}(y) \\ &= \mathbb{E}_{Q_X}\left[\log q_X(Y \mid o) - \log p_\theta(o, Y \mid X)\right] + \log p_\theta(o \mid X) \\ &= -\mathcal{L}(Q_X; X) + \log p_\theta(o \mid X). \end{aligned} \tag{5.21}$$

Rearranging gives (5.18). □

The almost-everywhere qualifier is part of the theorem, not a regularity detail suppressed by notation. If $\widetilde{p}_\theta$ is another joint-density version, Fubini's theorem gives equality of the two slices for ν^Y_D-almost every y outside a ν^O_D-null set of observations, and the two evidence and posterior versions therefore agree outside a $P^O_{\theta,X}$-null set. At an exceptional observation, the values of (5.19) and (5.16) can be changed without changing the joint measure. The theorem consequently makes no version-independent claim there.

Every corollary and coordinate statement below inherits this regular-observation qualification. For a comparison of finitely many parameter values, the observation must lie in the intersection of their marginal-full regular sets. An optimization over an uncountable parameter space requires either a common full-measure set proved for that family or jointly measurable pointwise density and conditional versions declared as part of the parameterized model; separate almost-everywhere versions for each parameter do not by themselves define a pointwise objective on the whole parameter space.

Corollary 5.10 (Lower bound and exact equality condition). Under Hypothesis 5.4, $\mathcal{L}(Q_X; X) \le \log p_\theta(o \mid X)$, and equality holds if and only if

$$Q_X(\cdot \mid o) = P_\theta(\cdot \mid o, X) \qquad \text{as probability measures on } (\mathsf{Y}_D, \mathscr{Y}_D). \tag{5.22}$$

Equality of all coordinate marginals is not sufficient. [ESTABLISHED]

Proof. The inequality is nonnegativity of relative entropy. For the equality case, strict convexity of $u \mapsto u \log u$ gives that the divergence vanishes exactly when the Radon–Nikodym derivative (5.20) equals one $P_\theta(\cdot \mid o, X)$-almost surely, which is equality of the two measures [22, 44]. Insufficiency of marginal agreement is Proposition 3.18, whose Gaussian witnesses have the same coordinate marginals but are distinct measures. □

5.4 The general restriction principle

Fix (θ, X), select $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,X}$ with $0 < p_\theta(o \mid X) < \infty$, and write $P^\star = P_\theta(\cdot \mid o, X)$. For every probability law Q on the latent space,

$$\mathcal{L}^{\mathrm{ext}}(Q; X, o) = \log p_\theta(o \mid X) - D_{\mathrm{KL}}(Q \| P^\star). \tag{5.23}$$

Proposition 5.11 (Restriction principle). Let $\mathcal{Q}' \subseteq \mathcal{Q}$ be arbitrary sets of probability laws. Then

$$\sup_{Q \in \mathcal{Q}'} \mathcal{L}^{\mathrm{ext}}(Q; X, o) = \log p_\theta(o \mid X) - \inf_{Q \in \mathcal{Q}'} D_{\mathrm{KL}}(Q \| P^\star), \tag{5.24}$$

and set inclusion gives

$$\sup_{Q \in \mathcal{Q}'} \mathcal{L}^{\mathrm{ext}}(Q; X, o) \le \sup_{Q \in \mathcal{Q}} \mathcal{L}^{\mathrm{ext}}(Q; X, o).$$

If both divergence infima below are finite, their optimized loss is the finite identity

$$\sup_{\mathcal{Q}} \mathcal{L}^{\mathrm{ext}} - \sup_{\mathcal{Q}'} \mathcal{L}^{\mathrm{ext}} = \inf_{\mathcal{Q}'} D_{\mathrm{KL}}(Q \| P^\star) - \inf_{\mathcal{Q}} D_{\mathrm{KL}}(Q \| P^\star) \ge 0. \tag{5.25}$$

The identities do not require attainment of the suprema or infima. [ESTABLISHED]

Proof. Equation (5.23) holds pointwise in Q. Taking suprema converts a finite constant minus a nonnegative extended divergence into that constant minus its infimum, including the value $-\infty$ when the infimum is $+\infty$. Set inclusion proves monotonicity. Only when both infima are finite is subtraction of the two supremum identities legitimate, and it gives (5.25). □

Constancy of the evidence is load bearing: the generative kernel is fixed once (θ, X) is fixed and does not read Q. If a proposed model changes with the recognition law, Proposition 4.3 and Proposition 4.4 show that no common evidence comparison follows. Thus every later closed-form "restriction cost" inherits its meaning from the fixed-target hypothesis.

Infinite-cost restrictions. Equation (5.25) is written only on its finite domain. If each $Q \in \mathcal{Q}'$ has a measurable set S_Q with

$$Q(S_Q) = 1, \qquad P^\star(S_Q) = 0, \tag{5.26}$$

then $Q \not\ll P^\star$ and $D_{\mathrm{KL}}(Q \| P^\star) = +\infty$. A proper closed support subset alone is insufficient: the uniform law on $[0, \frac{1}{2}]$ is absolutely continuous with respect to the uniform law on $[0, 1]$ and has divergence $\log 2$. The null-set condition, not proper closedness, is the obstruction. [ESTABLISHED]

Orientation, separation, and attainment. The infimum is a reverse information projection because Q is the first argument of the divergence. Excluding $P^\star$ does not ensure a positive or attained optimum: the family $\{\mathcal{N}(\mu, 1) : \mu > 0\}$ excludes $\mathcal{N}(0, 1)$, but its gaps $\mu^2/2$ have unattained infimum zero. A positive optimized gap requires $\inf_{Q\in\mathcal{Q}} D_{\mathrm{KL}}(Q\|P^\star) > 0$. Nor does uniqueness follow from convexity of a product family, because mixtures of product laws need not be products. The Gaussian restrictions of Chapter 15 instead prove convexity and coercivity in the moment parameters they optimize. [ESTABLISHED]

Corollary 5.12 (Independent channel-frame invariance). Let $T(o, y) = (o, Ry)$ be an independent belief–model frame re-expression as in Chapter 4, where $R = \bigoplus_{a,i}(R^b_{a,i} \oplus R^m_{a,i})$ is the assembled represented bimeasurable bijection of Y_D. Transport the entire slice by

$$M'_o = R_\# M_o, \qquad P'_{\theta'}(\cdot \mid o, X') = R_\# P_\theta(\cdot \mid o, X), \qquad Q'_{X'}(\cdot \mid o) = R_\# Q_X(\cdot \mid o).$$

Then $M'_o(\mathsf{Y}_D) = M_o(\mathsf{Y}_D) = z$ and

$$D_{\mathrm{KL}}\left(Q'_{X'} \middle\| P_{\theta'}(\cdot \mid o, X')\right) = D_{\mathrm{KL}}\left(Q_X \| P_\theta(\cdot \mid o, X)\right), \qquad \mathcal{L}'^{\mathrm{ext}}(Q'_{X'}; X', o) = \mathcal{L}^{\mathrm{ext}}(Q_X; X, o). \tag{5.27}$$

This includes the singular branch. Whenever both sides satisfy the classical split hypothesis, the same equality holds for $\mathcal{L}'$ and $\mathcal{L}$. [ESTABLISHED]

Proof. By the joint pushforward identity of Chapter 4 and the fact that T acts as the identity on the observation coordinate, the conditional posterior transforms as $P_{\theta'}(\cdot \mid o, X') = R_\# P_\theta(\cdot \mid o, X)$. Relative entropy is invariant under a common bimeasurable bijection. In the absolutely continuous branch, the RN derivative of the pushforwards is the original derivative composed with R^{-1}, and change of variables preserves its integral. In the singular branch, a posterior-null set carrying positive Q_X-mass maps to a pushed-posterior-null set carrying the same positive pushed-Q_X mass, so both divergences equal $+\infty$. The slice mass is invariant under pushforward, and substitution in (5.12) proves the second identity without subtracting infinities. The classical claim follows from its equality with the extended functional on the split domain. □

Corollary 5.12 is the statement that the bound is a bound on a frame-independent quantity, computed by a frame-independent procedure. It is used in Chapter 10 to rule out selection criteria that change under reframing, and it is the reason that any invariant offered later is stated in generalized-eigenvalue form rather than in terms of absolute eigenvalues.

5.5 The factorwise decomposition

Hypothesis 5.13 (Factorwise log-integrability). Restore the design index a suppressed in Chapter 4 and let $\mathcal{R} \subseteq V$ be the root set. To display the factor expectations separately, assume in this section that the absolute logarithm of every component density in the finite directed factorization is Q_X-integrable. [HYPOTHESIS]

This is strictly stronger than (H4), which controls only the complete log joint, and it excludes cancellation between individually undefined extended-real summands; a decomposition displayed without it can be a formal rearrangement of divergent quantities. [ESTABLISHED]

Proposition 5.14 (Exact factorwise decomposition). Under Hypotheses 5.4 and 5.13, substituting the normalized directed density (4.7) into (5.17) yields exactly one decomposition,

$$\begin{aligned}\mathcal{F}[Q_X;X,o] = &\sum_{a=1}^{M}\sum_{r\in\mathcal{R}}\mathbb{E}_{Q_X}\left[-\log p^m_{\theta,r,a}(m_{r,a}\mid X)\right]\\ &+\sum_{a=1}^{M}\sum_{r\in\mathcal{R}}\mathbb{E}_{Q_X}\left[-\log p^k_{\theta,r,a}(k_{r,a}\mid m_{r,a},X)\right]\\ &+\sum_{a=1}^{M}\sum_{i\in V\setminus\mathcal{R}}\mathbb{E}_{Q_X}\left[-\log p^m_{\theta,i,a}\left(m_{i,a}\mid m_{\mathrm{pa}(i),a},X\right)\right]\\ &+\sum_{a=1}^{M}\sum_{i\in V\setminus\mathcal{R}}\mathbb{E}_{Q_X}\left[-\log p^k_{\theta,i,a}\left(k_{i,a}\mid k_{\mathrm{pa}(i),a},m_{i,a},X\right)\right]\\ &+\sum_{a=1}^{M}\sum_{i\in V}\mathbb{E}_{Q_X}\left[-\log p^o_{\theta,i,a}\left(o_{i,a}\mid k_{i,a},m_{i,a},X\right)\right]\\ &+\mathbb{E}_{Q_X}\left[\log q_X(Y\mid o)\right].\end{aligned} \tag{5.28}$$

The generative sectors contain $3MN$ factors in total, and the final line is one joint recognition log-density, equivalently the negative joint differential entropy relative to ν_D^Y when that entropy is finite. It is not a sum of marginal entropies. [ESTABLISHED]

Proof. Write $\ell_{\theta,a}$ for the one-design-point log density obtained by taking the logarithm of (4.6), which is legitimate factor by factor under Hypothesis 5.13, and note that (4.7) gives $\log p_\theta(o,y\mid X) = \sum_{a=1}^{M}\ell_{\theta,a}(y_a;o_a,X)$. Taking $\mathbb{E}_{Q_X}$ of the negative of this and adding $\mathbb{E}_{Q_X}[\log q_X]$ reproduces the right side of (5.28) term by term. Denoting that right side by $\mathcal{F}^{\mathrm{fact}}$, the residual is

$$\begin{aligned}\mathcal{F}[Q_X;X,o]-\mathcal{F}^{\mathrm{fact}} = &\ \mathbb{E}_{Q_X}\left[\log q_X(Y\mid o)-\log p_\theta(o,Y\mid X)\right]\\ &-\left\{\mathbb{E}_{Q_X}\left[\log q_X(Y\mid o)\right]-\mathbb{E}_{Q_X}\left[\sum_{a=1}^{M}\ell_{\theta,a}(Y_a;o_a,X)\right]\right\}=0,\end{aligned} \tag{5.29}$$

where the same joint recognition term appears once on each side and cancels as a whole, and no step replaces it by marginal entropies. For the count, the root sector contributes $2M|\mathcal{R}|$ latent factors, the nonroot sector $2M(N-|\mathcal{R}|)$, and the observation sector MN, summing to $2MN+MN=3MN$. Root state bridges replace nonroot state transitions at roots; they are not additional factors. $\square$

Two readings of (5.28) must be resisted. The first is the substitution warned against in Proposition 5.3: the last line is a single number determined by the joint law, and replacing it by $\sum_b \mathbb{E}_{Q_X}[\log q_b]$ inflates the bound by the total correlation. The second is that the decomposition licenses independent optimization of its terms. Every expectation is taken under the same joint Q_X, so the terms are coupled through their common measure; the decomposition is an exact rewriting of one functional, not a decoupling of it into separately minimizable pieces.

5.6 Exact coordinates and accepted proposals

All coordinate statements below refer to the same normalized family P_θ of Chapter 4. Because Y_D is a finite product of Euclidean and discrete spaces it is a standard Borel space, so regular conditional distributions of the posterior exist and the relative entropy chain rule is available [42].

Proposition 5.15 (Exact E-coordinate in the extended domain). Let $\mathcal{B}$ be a block of latent coordinates with complement $-\mathcal{B}$, put

$$P := P_\theta(\cdot \mid o, X), \qquad P_- := P_{\theta,-\mathcal{B}}(\cdot \mid o, X), \qquad R := Q_{X,-\mathcal{B}}(\cdot \mid o),$$

with R an arbitrary probability law. Choose a Markov-kernel version $C(dY_\mathcal{B} \mid Y_{-\mathcal{B}})$ that agrees P_--almost everywhere with a regular conditional law of P, completing it arbitrarily off that full set. Then among *all* probability laws with complement marginal R, the extended-real divergence from P has minimum $D_{\mathrm{KL}}(R\|P_-) \in [0,+\infty]$, attained by

$$Q_X^{\mathrm{new}}(dY_\mathcal{B}, dY_{-\mathcal{B}} \mid o) = C\left(dY_\mathcal{B} \mid Y_{-\mathcal{B}}\right) R\left(dY_{-\mathcal{B}}\right). \tag{5.30}$$

It therefore maximizes $\mathcal{L}^{\mathrm{ext}}$ on that marginal class. The minimizer is unique as a joint measure when $D_{\mathrm{KL}}(R\|P_-) < \infty$; when this divergence is $+\infty$, every law in the class has divergence $+\infty$ and extended ELBO $-\infty$, so uniqueness is not asserted. If, in addition, the minimum is finite and the new law satisfies the log-integrability condition

$$\mathbb{E}_{Q_X^{\mathrm{new}}}\left[|\log p_\theta(o, Y \mid X)| + |\log q_X^{\mathrm{new}}(Y \mid o)|\right] < \infty, \tag{5.31}$$

then it satisfies (H3)–(H4) and maximizes the classical $\mathcal{L}$ among the laws with complement marginal R that satisfy Hypothesis 5.4. Finite complement relative entropy by itself does not imply (5.31). [ESTABLISHED]

Proof. Projection onto $Y_{-\mathcal{B}}$ and the extended data-processing inequality give

$$D_{\mathrm{KL}}(Q_X\|P) \geq D_{\mathrm{KL}}(R\|P_-)$$

for every law in the marginal class. If $D_{\mathrm{KL}}(R\|P_-) < \infty$, then $R \ll P_-$, the completion of C is irrelevant R-almost everywhere, and the relative-entropy chain rule on a standard Borel product gives

$$D_{\mathrm{KL}}\left(Q_X\|P\right) = D_{\mathrm{KL}}\left(R\|P_-\right) + \mathbb{E}_R\left[D_{\mathrm{KL}}\left(Q_{X,\mathcal{B}|-\mathcal{B}}\middle\|P_\theta(\cdot \mid Y_{-\mathcal{B}}, o, X)\right)\right]. \tag{5.32}$$

The second term is an integral of nonnegative extended-real quantities and is zero exactly when the two conditional laws agree for R-almost every $Y_{-\mathcal{B}}$. For (5.30) it vanishes, giving the minimum and uniqueness. Moreover, if $r = dR/dP_-$, direct disintegration gives

$$\frac{dQ_X^{\mathrm{new}}}{dP} = r(Y_{-\mathcal{B}}) \quad P\text{-almost everywhere},$$

so $Q_X^{\mathrm{new}} \ll P$ and its relative entropy is exactly $D_{\mathrm{KL}}(R\|P_-)$. If instead $D_{\mathrm{KL}}(R\|P_-) = +\infty$, data processing makes every candidate divergence $+\infty$, including the constructed law, so the extended optimum is attained but is nonunique. In both cases (5.12) converts divergence minimization into extended-ELBO maximization. Under the separate finite and log-integrability assumptions, Theorem 5.9 gives the classical statement. ☐

The classical extra obligation cannot be recovered from complement absolute continuity. First take P_- to be standard Gaussian and R standard Cauchy on $\mathbb{R}$. Then $R \ll P_-$, but $\log(dR/dP_-)(x)$ contains $x^2/2$ up to logarithmic and constant terms, and the Cauchy second moment is infinite; hence the extended coordinate is well defined but has value $-\infty$. Second, finite relative entropy still does not imply (H4). On the countable space $\{2, 3, \ldots\}$, let

$$P_-(\{n\}) = R(\{n\}) = \frac{c}{n(\log n)^2},$$

where c normalizes the series. Then $D_{\mathrm{KL}}(R\|P_-) = 0$, but for all sufficiently large n, $|\log P_-(\{n\})| \geq \frac{1}{2}\log n$, and therefore

$$\sum_{n\geq 2} P_-(\{n\})\,|\log P_-(\{n\})| \geq \frac{c}{2}\sum_{n \text{ large}} \frac{1}{n\log n} = +\infty.$$

With a trivial observation and conditional block this is precisely a $Q_X^{\mathrm{new}} = P$ of zero relative entropy for which the separate log expectations in (H4) are not integrable. Equation (5.31) is therefore an independent hypothesis, not a consequence of the KL chain rule.

The exact full E-step is the special case $\mathcal{B}$ equal to all coordinates, giving $Q_X^{\mathrm{new}} = P_\theta(\cdot \mid o, X)$ and a zero extended gap by Theorem 5.7; (H4) is needed only for its classical split. A product-family coordinate is a different object entirely: it optimizes within a restricted family, is not given by (5.30), and leaves a gap that Chapter 15 quantifies.

Proposition 5.16 (Exact M-coordinate). Hold Q_X fixed. Suppose the parameterized family supplies the common regular-observation set or the jointly measurable pointwise versions required above, and suppose Hypothesis 5.4 holds for Q_X at every candidate parameter being compared.

Specifically,

$$\mathbb{E}_{Q_X}\left[|\log q_X(Y \mid o)|\right] < \infty, \qquad \mathbb{E}_{Q_X}\left[|\log p_\vartheta(o, Y \mid X)|\right] < \infty$$

for every such ϑ. If a maximizer exists in the declared parameter space, then any

$$\theta^{\text{new}} \in \operatorname*{argmax}_{\vartheta} \mathbb{E}_{Q_X}\left[\log p_\vartheta(o, Y \mid X)\right] \tag{5.33}$$

also maximizes $\mathcal{L}_\vartheta(Q_X; X)$ over the same space, because $\mathbb{E}_{Q_X}[\log q_X(Y \mid o)]$ does not depend on ϑ. [ESTABLISHED]

The finiteness assumption is load-bearing. On $\{2, 3, \ldots\}$, let

$$q_n = \frac{c}{n(\log n)^2}, \qquad h_n = \mathbf{1}_{\{n=2\}} - q_2, \qquad r_n = 1 + \varepsilon h_n,$$

where c normalizes q and $0 < \varepsilon < 1/q_2$. Then r is bounded, strictly positive, nonconstant, and $\sum_n q_n r_n = 1$. With a trivial observation, set $p_0 = q$ and $p_1 = qr$. The entropy of q is infinite, so

$$\mathbb{E}_q[\log p_0] = \mathbb{E}_q[\log p_1] = -\infty.$$

An extended-real argmax of the expected complete log-likelihood therefore treats both candidates as maximizers. The separately integrated ELBO of (5.16) is undefined because (H4) fails, while its canonical relative-log extension distinguishes them:

$$\mathbb{E}_q\left[\log \frac{p_0}{q}\right] = 0, \qquad \mathbb{E}_q\left[\log \frac{p_1}{q}\right] = \mathbb{E}_q[\log r] = -D_{\mathrm{KL}}(q \| p_1) < 0.$$

Thus cancellation of a parameter-independent infinite entropy does not justify the M-coordinate claim.

The existence proviso in (5.33) is a hypothesis and not a formality. [HYPOTHESIS]

The covariance parameters of Chapter 4 range over an open cone, and the expected complete log-likelihood is unbounded above along sequences that drive a conditional covariance to the boundary whenever the corresponding expected residual second moment is singular. In that situation the supremum is not attained and (5.33) defines nothing. Under a nondegenerate Q_X the expected residual second moments are positive definite and the maximizer exists, so the hypothesis is mild in the Gaussian realization; it is stated because it is a hypothesis, not because it usually fails. [ESTABLISHED]

Definition 5.17 (Generalized M-coordinate acceptance). A generalized M-coordinate need not attain the maximum, but it must pass an acceptance condition:

$$\mathcal{L}^{\text{ext}}_{\theta^{\text{new}}}(Q_X; X, o) \geq \mathcal{L}^{\text{ext}}_{\theta^{\text{old}}}(Q_X; X, o). \tag{5.34}$$

The condition is a declaration about what counts as a step; it is checkable for arbitrary Q_X whenever each candidate has a selected positive-finite slice as in (H1)–(H2). No recognition density or (H4) is required. On the classical split domain it reduces to the same inequality for (5.16); in particular, the expected-complete-log-likelihood update of Proposition 5.16 is a finite-domain sufficient condition. [DEFINITION]

Proposition 5.18 (Evidence monotonicity). Suppose (H1)–(H2) hold for the fixed Q_X at both θ^{old} and θ^{new}; thus o lies in both regular-observation sets, or common pointwise density, slice, and conditional versions have been declared as model data. If the E-coordinate is exact at the old parameter, so that $Q_X = P_{\theta^{\text{old}}}(\cdot \mid o, X)$, and the M-coordinate satisfies (5.34), then

$$\log p_{\theta^{\text{new}}}(o \mid X) \geq \mathcal{L}^{\text{ext}}_{\theta^{\text{new}}}(Q_X; X, o) \geq \mathcal{L}^{\text{ext}}_{\theta^{\text{old}}}(Q_X; X, o) = \log p_{\theta^{\text{old}}}(o \mid X). \tag{5.35}$$

The first inequality is Theorem 5.7 at θ^{new}, the second is (5.34), and the final equality is the zero-gap case at θ^{old}. This proof never writes the undefined singular-branch sum $\mathcal{L}^{\text{ext}} + D_{\text{KL}}$. Under (H3)–(H4) at both candidates it reduces to the classical conclusion [24, 56]. [ESTABLISHED]

A natural-gradient direction and an ordinary optimizer direction are proposals, not accepted coordinates. The distinction is not pedantic, and the following states exactly what is and is not guaranteed.

Proposition 5.19 (Finite natural-gradient steps need not be monotone). Let $\mathsf{G} \succ 0$ be a preconditioner, write $\nabla\mathcal{L}$ for the gradient of $\vartheta \mapsto \mathcal{L}_\vartheta(Q_X; X)$, and consider the step $\vartheta^+ = \vartheta + \alpha\mathsf{G}^{-1}\nabla\mathcal{L}$. Define $\varphi(\alpha) = \mathcal{L}_{\vartheta^+}(Q_X; X)$. At a nonstationary point the direction is an ascent direction, since

$$\varphi'(0) = \nabla\mathcal{L}^\top \mathsf{G}^{-1}\nabla\mathcal{L} > 0. \tag{5.36}$$

This is an infinitesimal statement and does not extend to finite α. Suppose $\mathcal{L}_\xi = \mathcal{L}^\star - \frac{1}{2}(\xi - \theta^\star)^\top H(\xi - \theta^\star)$ with $H \succ 0$ throughout a neighborhood $\mathcal{U}$, let $d_{\max}$ be the largest generalized eigenvalue of the pair (H, G), and let v be a corresponding generalized eigenvector normalized so that $v^\top\mathsf{G}v = 1$. Then at any $\vartheta = \theta^\star + cv \in \mathcal{U}$ with $c \neq 0$, for every step whose endpoint ϑ^+ also lies in $\mathcal{U}$,

$$\mathcal{L}_{\vartheta^+} - \mathcal{L}_\vartheta = \tfrac{1}{2} d_{\max} c^2 \left[1 - (1 - \alpha d_{\max})^2\right] < 0 \qquad \text{whenever} \qquad \alpha > \frac{2}{d_{\max}}. \tag{5.37}$$

[ESTABLISHED]

Proof. Equation (5.36) is the chain rule with G^{-1} positive definite. For the second part write $e = \vartheta - \theta^\star$, so $\nabla\mathcal{L} = -He$ and $e^+ = (I - \alpha\mathsf{G}^{-1}H)e$. With $Hv = d_{\max}\mathsf{G}v$, $v^\top\mathsf{G}v = 1$, and $e = cv$ one gets $e^+ = (1 - \alpha d_{\max})cv$ and $\mathcal{L}_\vartheta = \mathcal{L}^\star - \frac{1}{2}d_{\max}c^2$. The endpoint hypothesis permits the same quadratic formula to be evaluated at ϑ^+. Substituting gives (5.37), and the bracket is negative exactly when $|1 - \alpha d_{\max}| > 1$, that is when $\alpha d_{\max} > 2$. □

The content of Proposition 5.19 is that monotonicity at finite step size is an additional requirement, supplied by a line search satisfying a sufficient-increase condition, by a trust region with an acceptance test, or by any other rule that verifies (5.34) on the same functional. Absent such a test the proposal is not entitled to (5.35), and the failure is not exotic: it occurs for a quadratic objective at any step length exceeding twice the reciprocal of the largest preconditioned curvature. Nor is the difficulty removed by improving the preconditioner, since (5.37) is stated in the preconditioned spectrum. Separately, optimizing a different objective, for instance a population energy that is not (5.16), supplies no guarantee for the fixed-joint evidence at all: two functionals with different stationary points can move in opposite directions under the same update, and only a bound with matched stationarity would transfer the conclusion.

5.7 What this chapter does not settle

Three limits of the development are recorded so that later chapters do not inherit more than was proved. Existence of a maximizer of (5.16) over a restricted recognition family is not established here and is not implied by Theorem 5.9; the theorem is an identity at a fixed pair of measures and says nothing about the attainability of any optimum. Tightness of the bound for any particular family is likewise not claimed, and the gap for the restrictions this document uses is computed in Chapter 15 rather than bounded here.

The third limit is a genuine open problem.

Open Problem 5.20 (Convergence of the alternating scheme). Whether the alternating scheme built from block E-coordinates (5.30) and accepted M-coordinates (5.34) converges to a stationary point of the common extended ELBO under the present hypotheses is not settled. [OPEN]

Block E-coordinates at fixed target and accepted parameter proposals make the compared objective values nondecreasing when their acceptance tests are evaluated on that same functional. They do not by themselves make the evidence nondecreasing. The evidence chain (5.35) additionally requires a full exact E phase, $Q_X = P_{\theta^{\mathrm{old}}}(\cdot \mid o, X)$, before each accepted M phase. Under that stronger schedule the evidence values are nondecreasing and converge when bounded above; convergence of the parameter sequence is a separate question requiring regularity conditions of the type identified for the classical algorithm [78], and global optimality does not follow even when they hold. [ESTABLISHED]

Closing the problem in the present setting requires either verifying those conditions for the declared parameter space, which is an open cone so that compactness of the relevant level sets must be checked rather than assumed, or exhibiting a counterexample within the declared family. [OPEN]

Nothing in Chapter 10 or Chapter 11 depends on the answer, because those chapters analyze the structure of the objective and its transformation under aggregation rather than the trajectory of any optimizer. [ESTABLISHED]

Chapter 6

Belief Fibers and Exponential Families

The objects carried by the belief and model associated bundles of Chapter 2 are normalized laws, while generative fibers carry kernels. Coordinates, smoothness, domination, and exponential-family structure are additional choices. This chapter states that hierarchy before selecting the finite-dimensional exponential subclass used by the general coarse theory.

6.1 Law fibers before coordinates

Definition 6.1 (Law and kernel fibers). For a standard Borel space $(\mathsf{Z}, \mathscr{Z})$, let

$$\mathcal{B}(\mathsf{Z}) \subseteq \mathcal{P}(\mathsf{Z}) \tag{6.1}$$

be a declared collection of normalized probability laws. For standard Borel spaces X, Y, let

$$\mathcal{K}(\mathsf{X}, \mathsf{Y}) \subseteq \mathrm{Markov}(\mathsf{X}, \mathsf{Y}) \tag{6.2}$$

be a declared collection of normalized Markov kernels. A law is not a kernel, and neither declaration supplies a common density, affine chart, manifold, Fisher metric, or convex cone. [DEFINITION]

The belief and model bundles may use different law fibers. Cross-bundle maps $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ and $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$ act between those associated bundles and are not iterations on either fiber. To prevent collision with that reserved notation, a generic endomorphism of a parameter set will be written Ψ.

Definition 6.2 (Partial realization). A parametrized model consists of a parameter space Θ, a domain $\Theta_{\mathrm{law}} \subseteq \Theta$, and a map

$$\mathfrak{r} : \Theta_{\mathrm{law}} \longrightarrow \mathcal{B}(\mathsf{Z}), \qquad \vartheta \longmapsto Q_\vartheta. \tag{6.3}$$

Outside Θ_{law}, a parameter may still denote an algebraic operator, but it does not denote a law through $\mathfrak{r}$. [DEFINITION]

A bimeasurable gauge action $g : \mathsf{Z} \to \mathsf{Z}$ acts on laws by

$$Q \longmapsto g_{\#} Q, \tag{6.4}$$

and on kernels by

$$(g_{\mathrm{out}} \cdot K \cdot g_{\mathrm{in}}^{-1})(x, B) = K(g_{\mathrm{in}}^{-1} x, g_{\mathrm{out}}^{-1} B). \tag{6.5}$$

These formulas type the action. [DEFINITION]

Closure of a declared fiber under the action is a separate hypothesis. [HYPOTHESIS]

6.2 Dominated and smooth tiers

Domination is not automatic. An uncountable family of Dirac laws on an uncountable standard Borel space admits no common sigma-finite dominating measure: any such measure would have to assign positive mass to uncountably many disjoint atoms, which a sigma-finite measure cannot do. [ESTABLISHED]

Hence density charts cannot be inferred from the general law fiber. [ESTABLISHED]

The differential tier used below is a finite-dimensional smooth parametrized-measure model. Differentiability in quadratic mean, square-integrable scores, and the interchange of differentiation and integration are hypotheses. The Fisher form is a Riemannian metric only after nondegeneracy is proved. Parametrized-measure models supply an intrinsic framework for this tier [8]. [HYPOTHESIS]

Stratified and infinite-dimensional tiers require their own tangent and integration theories; no universal global Fisher metric or natural chart is claimed. [NOT-CLAIMED]

6.3 Regular finite-dimensional exponential families

Definition 6.3 (Canonical family). Let ν be a nonzero sigma-finite measure on $(\mathsf{Z}, \mathscr{Z})$, let $T : \mathsf{Z} \to \mathsf{T}$ be measurable into a finite-dimensional inner-product space, and define

$$\mathsf{A}(\theta) = \log \int_{\mathsf{Z}} e^{\langle \theta, T(z) \rangle} \nu(dz), \qquad \mathsf{N} = \{\theta : \mathsf{A}(\theta) < \infty\}. \tag{6.6}$$

For $\theta \in \mathsf{N}$,

$$q_\theta(z) = \exp\{\langle \theta, T(z) \rangle - \mathsf{A}(\theta)\}, \qquad Q_\theta(dz) = q_\theta(z)\nu(dz). \tag{6.7}$$

Nothing is asserted for $\theta \notin \mathsf{N}$. [DEFINITION]

Proposition 6.4 (Finite-domain interface to ELBO coordinates). The map $(\theta, z) \mapsto q_\theta(z)$ in (6.7) is jointly measurable on $\mathsf{N} \times \mathsf{Z}$, and every law, density ratio, Fisher tensor, or ELBO coordinate built from this canonical representation is restricted to parameters in N. For a comparison of several canonical representations, either the comparison is made intrinsically at the measure level or the representations must have one declared common sigma-finite base measure; its density domain is the intersection of their natural domains. An extended-ELBO update additionally requires a positive-finite selected evidence slice at every compared parameter, while the classical expected-log-density representation separately requires Hypothesis 3.16. An algebraic operator that leaves N therefore ceases to define a probability or ELBO coordinate even if it remains an element of the operator cone. [ESTABLISHED]

Proof. Lower semicontinuity makes A Borel measurable, and measurability of T makes $(\theta, z) \mapsto \langle\theta, T(z)\rangle$ jointly measurable; exponentiation proves the first claim. Normalization is finite exactly on N by (6.6). The remaining statements are direct domain checks against Definitions 5.6 and 6.2 and hypothesis 3.16: a density comparison needs a fixed reference, the extended functional needs a normalized posterior slice with finite positive mass, and its classical split needs the additional support and log-integrability hypotheses. □

Proposition 6.5 (Convexity and lower semicontinuity). A is convex and lower semicontinuous as an extended-real function, and N is convex. [ESTABLISHED]

Proof. For $0 < t < 1$, Hölder's inequality gives

$$\int e^{\langle t\theta+(1-t)\vartheta, T\rangle} d\nu \le \left(\int e^{\langle\theta,T\rangle} d\nu\right)^t \left(\int e^{\langle\vartheta,T\rangle} d\nu\right)^{1-t}, \tag{6.8}$$

and taking logarithms proves convexity and convexity of the effective domain. Fatou's lemma applied to a convergent parameter sequence proves lower semicontinuity. □

Proposition 6.6 (Cumulants and Fisher information). At every interior parameter where local exponential domination permits differentiation under the integral,

$$\nabla \mathsf{A}(\theta) = \mathbb{E}_\theta[T], \qquad \nabla^2 \mathsf{A}(\theta) = \mathrm{Cov}_\theta(T). \tag{6.9}$$

The Hessian is the Fisher information in the natural chart. [ESTABLISHED]

Proof. Differentiate the normalizer once and twice. The score is $T - \mathbb{E}_\theta T$, so its second moment is the covariance in (6.9). Local exponential domination follows by enclosing the parameter in a compact subset of the interior and applying the elementary bound

$$|x|^r \le C_{r,\varepsilon}\left(e^{\varepsilon x} + e^{-\varepsilon x}\right) \tag{6.10}$$

to each coordinate of T. □

Hypothesis 6.7 (Regularity and minimality). The family is regular, so N is open, and minimal, so no nonzero $a \in \mathsf{T}$ makes $\langle a, T\rangle$ constant ν-almost everywhere. Regularity excludes natural-domain boundary points; minimality excludes affine redundancy. [HYPOTHESIS]

Proposition 6.8 (Minimality and nondegenerate information). Under the preceding hypothesis, $\nabla^2 \mathsf{A}(\theta)$ is positive definite, A is strictly convex, and $\theta \mapsto Q_\theta$ is injective. Failure of minimality makes the Hessian singular. [ESTABLISHED]

Proof. For every a,

$$a^\top \nabla^2 \mathsf{A}(\theta) a = \mathrm{Var}_{Q_\theta}(\langle a, T \rangle).$$

The variance vanishes exactly when the displayed statistic is almost surely constant. Since $q_\theta > 0$, Q_θ and ν have the same null sets, so minimality gives positive definiteness. Strict convexity follows. Equality of two densities makes $\langle \theta - \vartheta, T \rangle$ constant almost everywhere, and minimality then gives $\theta = \vartheta$. □

6.4 Operator and probability layers

Definition 6.9 (Operator layer). Let $\mathcal{C} \subseteq \mathsf{T}$ be a separately declared closed convex cone preserved by the algebraic maps under study. Its probability layer is $\mathcal{C} \cap \mathsf{N}$. The cone is chosen for algebraic closure, not normalization, and no such cone is asserted for an arbitrary law fiber. [DEFINITION]

Proposition 6.10 (A natural-domain boundary carries no law). Under regularity, if $\theta^\star \in \partial\mathsf{N}$, then

$$\int e^{\langle \theta^\star, T(z) \rangle} \nu(dz) = +\infty. \tag{6.11}$$

For $\theta_\tau = (1-\tau)\theta_0 + \tau\theta^\star$, with $\theta_0 \in \mathsf{N}$ and $w = \theta^\star - \theta_0$,

$$\mathsf{A}(\theta_\tau) \to +\infty, \qquad \langle w, \nabla \mathsf{A}(\theta_\tau) \rangle \to +\infty, \qquad \int_0^1 \mathrm{Var}_{Q_{\theta_\tau}}(\langle w, T \rangle) d\tau = +\infty. \tag{6.12}$$

[ESTABLISHED]

Proof. Regularity puts $\theta^\star$ outside N, proving (6.11). Convexity puts every θ_τ, $\tau < 1$, in N, while lower semicontinuity forces the first limit. For $g(\tau) = \mathsf{A}(\theta_\tau)$, convexity makes g' nondecreasing. If g' were bounded above, then g would remain bounded, a contradiction. The identities

$$g' = \langle w, \nabla \mathsf{A} \rangle, \qquad g'' = \mathrm{Var}_{Q_{\theta_\tau}}(\langle w, T \rangle)$$

and integration of g'' prove the remaining assertions. □

Thus the declared representation supplies no finite log partition, mean parameter, Fisher metric, relative entropy from the nonexistent endpoint law, or instance of the exact ELBO at $\theta^\star$. [ESTABLISHED]

This does not rule out a quotient, pin, new reference measure, or boundary stratum; each is a different declared object. [NOT-CLAIMED]

Proposition 6.11 (An operator fixed point need not be a law). Let $\Psi : \mathcal{C} \to \mathcal{C}$ be continuous. Every iterate may lie in $\mathcal{C} \cap \mathsf{N}$ while the unique fixed point lies on $\partial\mathsf{N}$. For $\mathsf{Z} = \mathbb{R}$, $T(z) = -z^2/2$, and Lebesgue ν,

$$\mathsf{A}(\theta) = -\tfrac{1}{2}\log\theta + \tfrac{1}{2}\log 2\pi, \qquad \mathsf{N} = (0,\infty), \qquad \mathcal{C} = [0,\infty), \tag{6.13}$$

the map $\Psi(\theta) = \theta/2$ has this property. [ESTABLISHED]

Proof. Starting at one gives $\theta_\ell = 2^{-\ell} > 0$, while $\theta_\ell \to 0$, the unique fixed point, and $\mathsf{A}(\theta_\ell) \to +\infty$. The constant kernel at zero has infinite Lebesgue integral. □

6.5 Declared repairs

Definition 6.12 (Pinning). Choose $\theta_{\mathrm{m}} \in \mathsf{N}$ and replace $\mathcal{C}$ by $\theta_{\mathrm{m}} + \mathcal{C}$; the pin does not flow. [DEFINITION]

Proposition 6.13 (Pinning under a recession condition). If $\theta_{\mathrm{m}} \in \operatorname{int}\mathsf{N}$ and every $\eta \in \mathcal{C}$ is a recession direction of N, then $\theta_{\mathrm{m}} + \mathcal{C} \subseteq \operatorname{int}\mathsf{N}$. [ESTABLISHED]

Proof. Translate an open ball around θ_{m} by η. The recession hypothesis keeps the translated ball inside N, so its center is interior. □

The pin changes the family, may change the fixed set, and introduces a parameter whose removal need not commute with the long-scale limit. Whether the mass sector is relevant, marginal, or irrelevant is an open spectral question for the declared scale map. [OPEN]

Definition 6.14 (Quotienting). Suppose $\mathsf{Z} = \mathbb{R}^n$, ν is Lebesgue measure, and the energy is invariant along a nonzero linear subspace $\mathcal{V}$. Declare the new sample space $\mathcal{W} = \mathcal{V}^\perp$, its Lebesgue measure, and the restricted statistic. [DEFINITION]

Proposition 6.15 (Quotient normalization is not full-space normalization). Writing $z = v + w$ in the orthogonal splitting,

$$\int_{\mathbb{R}^n} e^{\langle\theta,T(z)\rangle}\,dz = \lambda_{\mathcal{V}}(\mathcal{V})\int_{\mathcal{W}} e^{\langle\theta,T(w)\rangle}\,dw = +\infty. \tag{6.14}$$

The quotient law is normalized exactly when the second factor is finite. [ESTABLISHED]

Proof. The change of variables has unit Jacobian and the integrand is independent of v. Tonelli's theorem gives (6.14); the first factor is infinite. □

Definition 6.16 (Retaining a mass sector). Enlarge the state by a declared coercive parameter that remains in the natural domain under the scale map. This is neither pinning nor quotienting: the mass is a running coordinate and its scaling law must be stated. [DEFINITION]

A Moore–Penrose inverse and pseudodeterminant inserted into an interior full-space density formula do not repair normalization. They compute the finite quotient expression, whose dimension and reference measure differ, while the original integral remains infinite by (6.14). [ESTABLISHED]

6.6 Relative entropy and affine information projection

For a regular minimal family, let $\tau_\theta = \nabla \mathsf{A}(\theta)$ and let A^* be the convex conjugate. Direct expectation of the log-density ratio gives

$$D_{\mathrm{KL}}(Q_\vartheta \| Q_\theta) = \mathsf{A}(\theta) - \mathsf{A}(\vartheta) - \langle \nabla \mathsf{A}(\vartheta), \theta - \vartheta \rangle = D_{\mathsf{A}}(\theta, \vartheta), \tag{6.15}$$

and Legendre duality gives

$$D_{\mathrm{KL}}(Q_\vartheta \| Q_\theta) = D_{\mathsf{A}^*}(\tau_\vartheta, \tau_\theta). \tag{6.16}$$

This is the exact exponential-family divergence; in general it is not a finite-displacement Fisher quadratic [2, 19, 22]. [ESTABLISHED]

Proposition 6.17 (Affine mean constraints attain a unique projection). Assume the extended A is Legendre, as in a regular minimal canonical family, and let $\mathsf{M} = \operatorname{int} \operatorname{dom} \mathsf{A}^*$. If C is a closed affine subspace with $C \cap \mathsf{M} \neq \varnothing$, then for every $\tau_\theta \in \mathsf{M}$ there is a unique $\widehat{\tau} \in C \cap \mathsf{M}$ minimizing

$$\tau \longmapsto D_{\mathsf{A}^*}(\tau, \tau_\theta). \tag{6.17}$$

The corresponding natural parameter is uniquely $\widehat{\vartheta} = \nabla \mathsf{A}^*(\widehat{\tau})$. [ESTABLISHED]

Proof. The finite-dimensional Legendre Bregman projection theorem gives existence [9], uniqueness, and interiority on every nonempty closed convex feasible set meeting the interior domain. Strict convexity gives uniqueness, and essential smoothness excludes boundary attainment. For $C = \{\tau : B\tau = b\}$, the solution equivalently satisfies

$$B\widehat{\tau} = b, \qquad \nabla \mathsf{A}^*(\widehat{\tau}) - \theta + B^\top \lambda = 0, \tag{6.18}$$

which also proves the stated natural-parameter formula. □

Feasibility, closedness, and convexity are load bearing. For a Bernoulli mean domain $(0, 1)$, $C = \{0\}$ is infeasible; an open feasible interval may have a finite unattained infimum; and a closed nonconvex two-point set can have two minimizers. Gauge compatibility and commutation of this projection with a coarse parameter map remain open and require an intertwining theorem; existence alone does not provide them. [OPEN]

6.7 Boundary of this chapter

This chapter supplies the general law-fiber hierarchy, the exponential-family interior, its operator boundary, and the Bregman projection theorem. It does not contain graph aggregation, Gaussian determinants, Schur complements, or matrix pencils. Graph-exponential energy closure is developed in Chapter 10; the multivariate-Gaussian realization and its quadratic boundary rates appear later. An operator RG additionally requires a preserved parameter category, scale comparisons, rescaling, and topology. A law-level RG additionally requires normalized laws or kernels, compatible reference measures, and finite normalizers at every scale. A Fisher or ELBO reading additionally requires the hypotheses of the corresponding earlier chapters. [HYPOTHESIS]

Chapter 7

Local and Collective Free Energy

The bundle geometry of Chapter 2 is the inherited foundation of this chapter. The probabilistic construction below is additional structure. It is not inferred from the existence of the principal bundle, its group, or its two associated statistical fibers. The construction answers the local–global question by putting every interaction record in one fixed normalized joint law and then deriving each local objective by disintegration of that law. In this sense local and collective free energies are compatible, but they are not the same accounting object: the graph-local objectives below are coordinate potentials, and summing their incident-factor formulas overcounts shared records.

7.1 A normalized interaction-record model

Hypothesis 7.1 (Agent state and interaction kernels). Let $(\mathsf{X}, \mathscr{X})$ be a standard Borel context space, let V be an arbitrary finite set of agents, and let $\mathcal{A}$ be a finite set of interaction factors. Agent i has a standard Borel state space $(\mathsf{Y}_i, \mathscr{Y}_i)$. Its state may contain a belief-law point in $(\mathcal{E}_b)_{c_i}$, a model-law point in $(\mathcal{E}_m)_{c_i}$, and declared auxiliary latent variables. Write $(\mathsf{Y}, \mathscr{Y}) = \prod_{i \in V}(\mathsf{Y}_i, \mathscr{Y}_i)$ and let

$$P_0 : (\mathsf{X}, \mathscr{X}) \rightsquigarrow (\mathsf{Y}, \mathscr{Y}) \tag{7.1}$$

be a normalized probability kernel. Each factor a has a finite scope $\partial a \subseteq V$, a standard Borel record space $(\mathsf{O}_a, \mathscr{O}_a)$, and a normalized Markov kernel

$$K_a : (\mathsf{X} \times \mathsf{Y}_{\partial a}, \mathscr{X} \otimes \mathscr{Y}_{\partial a}) \rightsquigarrow (\mathsf{O}_a, \mathscr{O}_a). \tag{7.2}$$

For the density presentation used in this chapter, one fixed probability or sigma-finite measure ν_a dominates the whole family $\{K_a(X, y_{\partial a}, \cdot) : (X, y_{\partial a}) \in \mathsf{X} \times \mathsf{Y}_{\partial a}\}$, and a finite jointly measurable version

$$\ell_a : \mathsf{X} \times \mathsf{Y}_{\partial a} \times \mathsf{O}_a \longrightarrow [0, +\infty), \qquad K_a(X, y_{\partial a}, do_a) = \ell_a(o_a \mid X, y_{\partial a})\nu_a(do_a) \tag{7.3}$$

is fixed once and for all. Thus $\int \ell_a(o_a \mid X, y_{\partial a})\nu_a(do_a) = 1$ for every $(X, y_{\partial a})$. The common ν_a and the displayed joint measurability are family-level hypotheses, not recordwise choices. The kernels and P_0 are fixed by the generative parameters and structural data; none reads a recognition law, a variational parameter, or a live posterior. Their local coordinate presentations are required to transform by the common passive pushforward induced by a change of bundle frames, so the underlying measures and record probabilities are frame independent. [HYPOTHESIS]

This hypothesis permits correlated baseline states. A graph-local Markov blanket requires the additional condition that the regular conditionals of P_0 have the corresponding locality. Bundle covariance is imposed by requiring each K_a to be invariant under simultaneous changes of the incident agent frames and the factor frame. One concrete realization gives factor a its own principal frame and incidence-leg transports $U^x_{a\leftarrow i}$ in each channel $x \in \{b, m\}$; the likelihood is then a measurable function of the dressed variables $\rho_x(U^x_{a\leftarrow i})y^x_i$. The two representations remain distinct, and a belief–model comparison still requires the declared cross morphisms of Section 2.4. [DEFINITION]

Conditional independence of the records given the agent configuration defines the joint law

$$P_\theta(dy, do \mid X) = P_0(dy \mid X) \prod_{a\in\mathcal{A}} K_a(X, y_{\partial a}, do_a). \tag{7.4}$$

Proposition 7.2 (Normalization on arbitrary interaction cycles). The law in (7.4) is normalized for every finite hypergraph, including cyclic ones. [ESTABLISHED]

Proof. Integrate the record coordinates in any order. Each normalized kernel integrates to one, leaving the normalized baseline law $P_0(\cdot \mid X)$. The factor scopes are permitted to overlap because the record variables are conditionally independent children of one latent configuration. This argument would not validate an arbitrary cycle of separately prescribed agent conditionals, which need not be compatible with any joint law. □

For the remainder of this chapter fix one $X \in \mathsf{X}$. We suppress that fixed argument consistently: $P_0(dy) := P_0(dy \mid X)$, $K_a(y_{\partial a}, do_a) := K_a(X, y_{\partial a}, do_a)$, and $\ell_a(o_a \mid y_{\partial a}) := \ell_a(o_a \mid X, y_{\partial a})$. Every descendant symbol L_o, H_o, Z, Π_o and its block analogue is relative to this same fixed context unless an X argument is restored explicitly.

With $\nu = \bigotimes_{a\in\mathcal{A}} \nu_a$, the complete record density is the finite jointly measurable function

$$\ell(o \mid y) = \prod_{a\in\mathcal{A}} \ell_a(o_a \mid y_{\partial a}), \qquad P_{\theta,X}(dy, do) = P_0(dy)\ell(o \mid y)\nu(do). \tag{7.5}$$

For a fixed record o, define its likelihood before taking logarithms:

$$L_o(y) := \ell(o \mid y) = \prod_{a\in\mathcal{A}} \ell_a(o_a \mid y_{\partial a}) \in [0, +\infty). \tag{7.6}$$

Its extended interaction energy is

$$H_o(y) := -\log L_o(y) = \sum_{a\in\mathcal{A}} -\log \ell_a(o_a \mid y_{\partial a}), \qquad -\log 0 := +\infty. \tag{7.7}$$

Because each factor density is finite, H_o never equals $-\infty$, although individual negative log densities and their finite sum can be negative. A binary success record with $K_a(1 \mid y) = \exp[-E_a(y)]$ is a valid special case only when $E_a \in [0, +\infty]$. General observations are represented by the normalized kernels, not by treating an arbitrary exponential as a probability. [ESTABLISHED]

7.2 One collective VFE

Fix a regular record o for which

$$0 < Z(o) := \int_{\mathsf{Y}} L_o(y) P_0(dy) < +\infty, \tag{7.8}$$

and let

$$\Pi_o(dy) = Z(o)^{-1} L_o(y) P_0(dy) \tag{7.9}$$

be the conditional posterior. The regular-observation qualifications of Chapter 5 apply.

Theorem 7.3 (Collective interaction VFE). For every probability measure Q on Y, define the collective extended ELBO and VFE by

$$\boxed{\mathcal{F}_o^{\text{ext}}(Q) := -\mathcal{L}_o^{\text{ext}}(Q) := -\log Z(o) + D_{\text{KL}}(Q \| \Pi_o).} \tag{7.10}$$

This takes values in $\mathbb{R} \cup \{+\infty\}$ and is the specialization of Definition 5.6 and theorem 5.7. Equivalently, $\mathcal{L}_o^{\text{ext}}(Q) \leq \log Z(o)$ for every Q, with equality exactly when $Q = \Pi_o$ as measures. No domination or mean-field assumption on Q is needed. [ESTABLISHED]

Proof. Apply the extended gap theorem Theorem 5.7 with reference P_0 and likelihood L_o. Nonnegativity and the equality condition are the corresponding properties of relative entropy. □

If $Q \ll P_0$ and the separate complexity and energy terms are defined without an indeterminate $+\infty - \infty$, the same quantity has the familiar energy presentation

$$\mathcal{F}_o^{\text{ext}}(Q) = D_{\text{KL}}(Q \| P_0) + \mathbb{E}_Q H_o. \tag{7.11}$$

This is a derived representation, not the definition at singular Q. [ESTABLISHED]

There is one term in H_o for each actual interaction record. An undirected record shared by agents i and j occurs once. Writing both E_{ij} and E_{ji} in the global law asserts two directed records, not two descriptions of one record. [ESTABLISHED]

7.3 The conditional VFE of an agent or meta-agent

Let $B \subseteq V$ be nonempty and write $b = y_{B^c}$. The same construction therefore treats a singleton agent $B = \{i\}$ and a declared meta-agent block B without changing probability models. Define

$$\mathcal{A}_B = \{a \in \mathcal{A} : \partial a \cap B \neq \varnothing\}, \qquad g_{B,o}(y_B; b) = \prod_{a \in \mathcal{A}_B} \ell_a(o_a \mid y_{\partial a}), \tag{7.12}$$

and the likelihood of records whose scopes miss B,

$$L_{\bar{B},o}(b) = \prod_{a:\partial a \cap B = \varnothing} \ell_a(o_a \mid b_{\partial a}), \qquad L_o(y_B, b) = g_{B,o}(y_B; b) L_{\bar{B},o}(b). \tag{7.13}$$

The corresponding extended energies retain the useful additive notation

$$H_{B,o}(y_B; b) = -\log g_{B,o}(y_B; b) = \sum_{a \in \mathcal{A}_B} -\log \ell_a(o_a \mid y_{\partial a}) \tag{7.14}$$

and

$$H_{\bar{B},o}(b) = -\log L_{\bar{B},o}(b), \qquad H_o(y) = H_{B,o}(y_B; b) + H_{\bar{B},o}(b), \tag{7.15}$$

as an identity in $\mathbb{R} \cup \{+\infty\}$.

Fix once and for all a measurable regular-conditional version $b \mapsto P_{0,B}(\cdot \mid b)$ in the disintegration

$$P_0(dy_B, db) = P_{0,B^c}(db) P_{0,B}(dy_B \mid b). \tag{7.16}$$

Every pointwise formula in b below is relative to this declared version. Two baseline-conditional versions agree P_{0,B^c}-almost everywhere and hence, by (7.19), Π_{o,B^c}-almost everywhere. No pointwise version invariance is claimed beyond these almost-sure statements: their values and all descendant formulas at exceptional b are not determined by the joint law. Two different conditional likelihoods are needed. The incident-factor evidence is

$$Z_B(b) = \int g_{B,o}(y_B; b) P_{0,B}(dy_B \mid b), \tag{7.17}$$

whereas the full likelihood of the outside state is defined directly by

$$w_B(b) = \int L_o(y_B, b) P_{0,B}(dy_B \mid b). \tag{7.18}$$

The direct definition is valid even where the formal product $L_{\bar{B},o}(b) Z_B(b)$ would have the indeterminate form $0 \cdot \infty$. Tonelli's theorem gives

$$Z(o) = \int w_B(b) P_{0,B^c}(db), \qquad \Pi_{o,B^c}(db) = Z(o)^{-1} w_B(b) P_{0,B^c}(db). \tag{7.19}$$

Let $\mathsf{R}_{B,o}$ be the measurable set on which

$$0 < w_B(b) < +\infty, \quad 0 < L_{\bar{B},o}(b) < +\infty, \quad 0 < Z_B(b) < +\infty, \quad w_B(b) = L_{\bar{B},o}(b) Z_B(b). \tag{7.20}$$

Then $\Pi_{o,B^c}(\mathsf{R}_{B,o}) = 1$. Indeed, finite collective evidence makes $w_B < +\infty$ outside a P_{0,B^c}-null set, and the posterior outside is supported on $w_B > 0$. On that support the finite-factor hypothesis makes $L_{\bar{B},o}$ finite; positivity of w_B forces $L_{\bar{B},o} > 0$, after which ordinary factorization under the conditional integral gives $Z_B = w_B / L_{\bar{B},o} \in (0, +\infty)$. The regular set need not be P_{0,B^c}-full. [ESTABLISHED]

For $b \in \mathsf{R}_{B,o}$ relative to the fixed baseline-conditional version, the posterior conditional is

$$\Pi_{o,B}(dy_B \mid b) = Z_B(b)^{-1} g_{B,o}(y_B; b) P_{0,B}(dy_B \mid b), \tag{7.21}$$

and, after choosing any measurable probability-kernel extension off $\mathsf{R}_{B,o}$ (for example the fixed baseline conditional there), it is a version for which

$$\Pi_o(dy_B, db) = \Pi_{o,B^c}(db) \Pi_{o,B}(dy_B \mid b). \tag{7.22}$$

The posterior joint determines this kernel only Π_{o,B^c}-almost everywhere. In particular, neither the conditional nor the associated pointwise local VFE at an outside-null b is an invariant of the joint law. All almost-sure local/global statements below use $Q_{B^c} \ll \Pi_{o,B^c}$ and are therefore independent of these choices. [ESTABLISHED]

Theorem 7.4 (Local multi-agent ELBO). For $b \in \mathsf{R}_{B,o}$ and every probability measure $r_B(\cdot \mid b)$, define

$$\boxed{\mathcal{F}^{\mathrm{ext}}_{B,o}(r_B; b) := -\mathcal{L}^{\mathrm{ext}}_{B,o}(r_B; b) := -\log Z_B(b) + D_{\mathrm{KL}}\big(r_B(\cdot \mid b) \| \Pi_{o,B}(\cdot \mid b)\big).} \tag{7.23}$$

Thus $\mathcal{L}^{\mathrm{ext}}_{B,o}(r_B; b) \leq \log Z_B(b)$, with equality exactly at $r_B = \Pi_{o,B}$. The singleton $B = \{i\}$ is one agent's local VFE; an arbitrary nonempty B is the same construction for a meta-agent. No domination or mean-field condition on r_B is needed. [ESTABLISHED]

Proof. Apply Theorem 5.7 conditionally with reference $P_{0,B}(\cdot \mid b)$ and likelihood $g_{B,o}(\cdot; b)$. The equality condition follows from vanishing relative entropy. □

On the dominated energy tier, when its two terms are separately defined without an indeterminate infinity, the local functional is also

$$\mathcal{F}^{\mathrm{ext}}_{B,o}(r_B; b) = D_{\mathrm{KL}}\big(r_B(\cdot \mid b) \| P_{0,B}(\cdot \mid b)\big) + \mathbb{E}_{r_B(\cdot \mid b)} H_{B,o}(Y_B; b). \tag{7.24}$$

The exact relative-entropy chain rule is most safely written against the posterior. If $Q(dy_B, db) = Q_{B^c}(db) r_B(dy_B \mid b)$ and $Q_{B^c} \ll \Pi_{o,B^c}$, then, in $[0, +\infty]$,

$$D_{\mathrm{KL}}(Q \| \Pi_o) = D_{\mathrm{KL}}(Q_{B^c} \| \Pi_{o,B^c}) + \mathbb{E}_{Q_{B^c}} D_{\mathrm{KL}}\big(r_B(\cdot \mid Y_{B^c}) \| \Pi_{o,B}(\cdot \mid Y_{B^c})\big). \tag{7.25}$$

The baseline chain rule, whenever its regular conditional versions are used, likewise reads

$$D_{\mathrm{KL}}(Q\|P_0) = D_{\mathrm{KL}}(Q_{B^c}\|P_{0,B^c}) + \mathbb{E}_{Q_{B^c}} D_{\mathrm{KL}}\big(r_B(\cdot \mid Y_{B^c})\|P_{0,B}(\cdot \mid Y_{B^c})\big). \tag{7.26}$$

Theorem 7.5 (Exact local–global potential identity). Let $Q = Q_{B^c} r_B$ and $Q' = Q_{B^c} r'_B$ have the same outside marginal with $Q_{B^c} \ll \Pi_{o,B^c}$. Suppose $D_{\mathrm{KL}}(Q\|\Pi_o) < +\infty$ and $D_{\mathrm{KL}}(Q'\|\Pi_o) < +\infty$; the chain rule then makes both conditional-KL fields integrable, so the following subtraction is finite. Then

$$\mathcal{F}_o^{\mathrm{ext}}(Q') - \mathcal{F}_o^{\mathrm{ext}}(Q) = \mathbb{E}_{Q_{B^c}}\left[\mathcal{F}_{B,o}^{\mathrm{ext}}(r'_B; Y_{B^c}) - \mathcal{F}_{B,o}^{\mathrm{ext}}(r_B; Y_{B^c})\right]. \tag{7.27}$$

Thus every exact local conditional update is a coordinate update of the same collective VFE. [ESTABLISHED]

Proof. Apply (7.25) to both laws. Their outside KL terms agree. The common finite constant $-\log Z(o)$ cancels, as does the pointwise term $-\log Z_B(Y_{B^c})$ inside the two local functionals. The finite-difference hypotheses permit the remaining subtraction under the integral. □

When all baseline KL terms and positive and negative energy parts are integrable, the same result admits the familiar finite decomposition

$$\begin{aligned}\mathcal{F}_o^{\mathrm{ext}}(Q) = D_{\mathrm{KL}}(Q_{B^c}\|P_{0,B^c})\\ + \mathbb{E}_{Q_{B^c}}\left[H_{\bar{B},o}(Y_{B^c}) + \mathcal{F}_{B,o}^{\mathrm{ext}}(r_B; Y_{B^c})\right].\end{aligned} \tag{7.28}$$

This theorem licenses sequential exact coordinate minimization, damped updates accepted by the collective objective, and continuous block-gradient dynamics. Independently replacing all correlated full conditionals need not define any joint recognition law, so such a parallel prescription is not licensed. [ESTABLISHED]

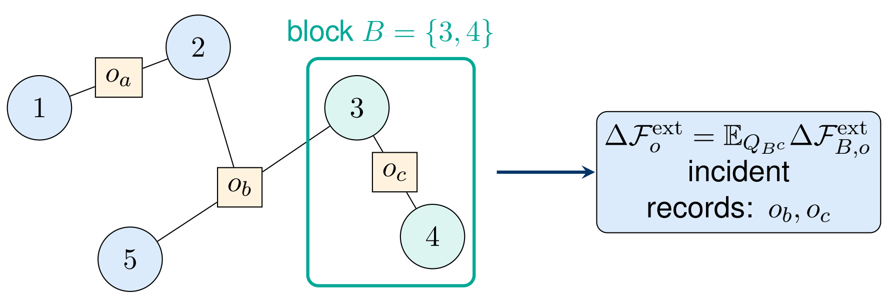

Figure 7.1: A block-local VFE contains every factor incident to the block. The factor o_b crosses the boundary and therefore enters the block conditional; o_a is outside and cancels. The equality is an exact-potential identity, not a sum of overlapping local objectives.

7.4 Additive accounting is a different construction

When $P_0 = \bigotimes_i \rho_i$, define the total correlation of an arbitrary joint recognition law Q by

$$\mathrm{TC}(Q) = D_{\mathrm{KL}}\left(Q \middle\| \bigotimes_i Q_i\right). \tag{7.29}$$

Without assuming mean-field factorization, the extended chain rule gives

$$D_{\mathrm{KL}}(Q\|P_0) = \mathrm{TC}(Q) + \sum_i D_{\mathrm{KL}}(Q_i\|\rho_i) \quad \text{in } [0, +\infty]. \tag{7.30}$$

Consequently, whenever the energy presentation (7.11) is defined without an indeterminate infinity,

$$\mathcal{F}_o^{\mathrm{ext}}(Q) = \mathrm{TC}(Q) + \sum_i D_{\mathrm{KL}}(Q_i\|\rho_i) + \mathbb{E}_Q\left[\sum_{a\in\mathcal{A}} E_{a,o}\right], \qquad E_{a,o} = -\log \ell_a(o_a \mid Y_{\partial a}). \tag{7.31}$$

[ESTABLISHED]

Equation (7.31) is an additive ledger: each interaction appears once inside the pointwise extended factor sum. It can be rewritten as $\sum_a \mathbb{E}_Q E_{a,o}$ when every $E_{a,o} \in L^1(Q)$, or under another condition excluding opposing infinities. Moreover, $\mathrm{TC}(Q) = 0$ exactly when $Q = \bigotimes_i Q_i$; mean field is thus a special case of the ledger, not a premise of it.

For the singleton conditional objectives actually defined above, assume the declared interaction factors have nonempty scopes. Their incident energies then obey the exact pointwise identity

$$H_{\{i\},o}(y_i; y_{-i}) = \sum_{a:i\in\partial a} E_{a,o}(y_{\partial a}), \qquad \sum_{i\in V} H_{\{i\},o}(y_i; y_{-i}) = \sum_{a\in\mathcal{A}} |\partial a| E_{a,o}(y_{\partial a}). \tag{7.32}$$

An empty-scope record, if separately permitted, is instead a label-independent global constant, and the right-hand sum must then be restricted to $|\partial a| > 0$. Because every $E_{a,o}$ lies in $\mathbb{R} \cup \{+\infty\}$ and both index sets are finite, the rearrangement is valid as an extended pointwise identity. Thus the specific local VFEs of Theorem 7.4 have overlapping incident-energy components and are not themselves the single-count ledger (7.31). This is a counting identity, not a universal no-go for arbitrary decompositions.

Ordered chain rules, a separate residual account, or nonlocal counting-number shares—including compensating negative shares assigned away from an incident coordinate—can restore an additive sum. Such a construction depends on extra bookkeeping and is not the conditional local VFE of Theorem 7.4. [ESTABLISHED]

7.5 Attention as a latent interaction label

Attention enters the fixed joint by a latent source label, not by allowing the generative law to inspect recognition beliefs. Let $\mathcal{I}_{\mathrm{att}} \subseteq V$ be the finite set of receiving agents. For each $i \in \mathcal{I}_{\mathrm{att}}$, let J_i augment the baseline independently as

$$P_0^{\mathrm{aug}}(dy, dj) = P_0^Y(dy) \prod_{i \in \mathcal{I}_{\mathrm{att}}} \pi_i(dj_i). \tag{7.33}$$

Suppose $\pi_{ij} > 0$ on the finite declared source set and that every $D_{ij} : \mathsf{Y} \to \mathbb{R}$ is finite and measurable. At the selected regular interaction record, suppose its density has source-dependent energy

$$\ell_i(o_i \mid y, J_i = j) = c_i(o_i, y) \exp[-D_{ij}(y)/\tau_i], \qquad \tau_i > 0, \tag{7.34}$$

where c_i is independent of j and the normalized observation kernel is part of the model data. The energy D_{ij} is a fixed gauge-invariant function of latent sample-level states and declared transports. It does not contain a live recognition distribution. Equation (7.34) is an identity on the selected observation slice; normalization of the kernel and its values away from that slice remain part of the declared generative model. More strongly, at the complete selected record the full augmented likelihood is required to factor as

$$L_o^{\mathrm{aug}}(y, j) = L_o^Y(y) \prod_{i \in \mathcal{I}_{\mathrm{att}}} c_i(o_i, y) \exp[-D_{ij_i}(y)/\tau_i], \tag{7.35}$$

where $L_o^Y : \mathsf{Y} \to [0, +\infty)$ is a finite measurable function independent of every source label and no other record or factor in the generative law reads any J_i. Equation (7.35) is a label-exclusivity hypothesis on the selected likelihood slice; normalization of the augmented observation kernel over all possible records remains declared model data. Without exclusivity, an additional J_i-dependent likelihood factor would enter Bayes' formula and generally change the displayed softmax below. [HYPOTHESIS]

The constant-row recognition calculation below additionally restricts the augmented recognition law to the full factorization

$$Q(dy, dj) = Q_Y(dy) \prod_{i \in \mathcal{I}_{\mathrm{att}}} \beta_i^Q(dj_i), \qquad \beta_i^Q(dj_i) \text{ independent of } y. \tag{7.36}$$

This assumption imposes both conditional independence among the source labels and independence of every label row from the latent state. It is not implied by the generative law. For the finite-valued formulas below, also assume $\mathbb{E}_{Q_Y}|D_{ij}| < +\infty$ on every active source and $\mathbb{E}_{Q_Y}|\log c_i(o_i, Y)| < +\infty$. [HYPOTHESIS]

Proposition 7.6 (Exact posterior and recognition attention). Under the generative label-exclusivity hypothesis (7.35), conditioning on y and the complete interaction record gives, on the positive-likelihood set $L_o^Y(y)\prod_{k\in\mathcal{I}_{\rm att}} c_k(o_k,y)>0$, the generative posterior row

$$\beta^P_{ij}(y)=\frac{\pi_{ij}\exp[-D_{ij}(y)/\tau_i]}{\sum_k \pi_{ik}\exp[-D_{ik}(y)/\tau_i]}. \tag{7.37}$$

and the same display may be chosen as its version off that set. Under the additional recognition restriction (7.36), the exact categorical contribution of row i to the collective VFE is

$$-\mathbb{E}_{Q_Y}\log c_i(o_i,Y)+\mathcal{F}_i^{\rm att}(\beta_i^Q), \tag{7.38}$$

where the first term is independent of β_i^Q and

$$\mathcal{F}_i^{\rm att}(\beta_i^Q)=D_{\rm KL}(\beta_i^Q\|\pi_i)+\frac{1}{\tau_i}\sum_j \beta^Q_{ij}\mathbb{E}_{Q_Y}D_{ij}. \tag{7.39}$$

Its unique interior minimizer on the positive-prior source simplex is

$$\beta^{Q\star}_{ij}=\frac{\pi_{ij}\exp[-\mathbb{E}_{Q_Y}D_{ij}/\tau_i]}{\sum_k \pi_{ik}\exp[-\mathbb{E}_{Q_Y}D_{ik}/\tau_i]}. \tag{7.40}$$

[ESTABLISHED]

Proof. The product label prior and the exclusive likelihood (7.35) make the conditional posterior over J factor by rows; every factor not containing j_i cancels. Bayes' formula then gives (7.37). With $Z_i^{\rm att}(y)=\sum_k \pi_{ik}\exp[-D_{ik}(y)/\tau_i]$, expanding the categorical relative entropy to that posterior gives

$$D_{\rm KL}(\beta_i^Q\|\beta_i^P)=D_{\rm KL}(\beta_i^Q\|\pi_i)+\tau_i^{-1}\sum_j \beta^Q_{ij}D_{ij}+\log Z_i^{\rm att}(y).$$

Subtracting $\log Z_i^{\rm att}(y)$ and the source-independent log likelihood $\log c_i(o_i,y)$, then taking the recognition expectation, proves (7.38) and (7.39). Lagrange multiplication under $\sum_j \beta^Q_{ij}=1$ proves (7.40). □

The two softmax rows answer different variational questions. $\beta_i^P(y)$ is the generative posterior at a fixed latent state, whereas $\beta_i^{Q\star}$ is the optimum after forcing one recognition row to be constant in y. In general $\beta_i^{Q\star}\neq\mathbb{E}_{Q_Y}[\beta_i^P(Y)]$; the two formulas agree pointwise when all differences $D_{ij}(Y)-D_{ik}(Y)$ are Q_Y-almost surely constant. [ESTABLISHED]

For completeness, still under (7.35), let an arbitrary recognition law disintegrate as $Q(dy,dj)=Q_Y(dy)Q_{J|Y}(dj\mid y)$. With $Q_{J_i|Y}$ denoting its conditional row marginals and the label-free term $-\mathbb{E}_{Q_Y}\log L_o^Y(Y)$ accounted for in the state sector, the exact attention-label contribution is

$$\begin{aligned}&-\sum_{i\in\mathcal{I}_{\rm att}}\mathbb{E}_{Q_Y}\log c_i(o_i,Y)+\mathbb{E}_{Q_Y}\,\mathrm{TC}(Q_{J|Y})\\&\quad+\sum_{i\in\mathcal{I}_{\rm att}}\mathbb{E}_{Q_Y}\left[D_{\rm KL}(Q_{J_i|Y}\|\pi_i)+\frac{1}{\tau_i}\mathbb{E}_{Q_{J_i|Y}}D_{iJ_i}(Y)\right],\end{aligned} \tag{7.41}$$

where $\mathrm{TC}(Q_{J|Y=y}) = D_{\mathrm{KL}}\left(Q_{J|Y=y} \middle\| \prod_{i \in \mathcal{I}_{\mathrm{att}}} Q_{J_i|Y=y}\right)$. Thus merely requiring every row marginal to be independent of y does not justify (7.39): a correlated conditional law retains the nonnegative conditional-total-correlation term. If conditional dependence is unrestricted, minimizing over all $Q_{J|Y=y}$ gives the product of the pointwise rows (7.37). Under a constrained coupled family, the coupling term or constraint enters the coordinate optimum, so a row softmax is not asserted. [ESTABLISHED]

The familiar row functional $\sum_j \beta_{ij} \mathbb{E}_{Q_Y} D_{ij} + \tau_i D_{\mathrm{KL}}(\beta_i \| \pi_i)$ is $\tau_i \mathcal{F}_i^{\mathrm{att}}$. It has the same row minimizer, but for $\tau_i \neq 1$ it is not an independently weighted sector of one standard global ELBO unless the entire objective is scaled or a different generalized free energy is declared. [ESTABLISHED]

Holding the latent recognition law Q_Y, and hence every $\mathbb{E}_{Q_Y} D_{ij}$, fixed during this row-coordinate flow, the Fisher natural gradient of (7.39) on the open simplex is the replicator equation

$$\dot{\beta}_{ij}^{Q} = -\gamma_i \beta_{ij}^{Q} \left(c_{ij} - \sum_k \beta_{ik}^{Q} c_{ik} \right), \qquad c_{ij} = \log \frac{\beta_{ij}^{Q}}{\pi_{ij}} + \frac{\mathbb{E}_{Q_Y} D_{ij}}{\tau_i}. \tag{7.42}$$

for $\gamma_i \geq 0$. Its partial contribution to the row-free-energy rate is

$$\left. \frac{d}{dt} \mathcal{F}_i^{\mathrm{att}} \right|_{\beta_i} = -\gamma_i \operatorname{Var}_{\beta_i^Q}(c_{ij}) \leq 0. \tag{7.43}$$

If Q_Y evolves simultaneously, its additional chain-rule term must be included. [ESTABLISHED]

7.6 Local evolution is collective descent

Let a regular product recognition family Q_η on an open parameter domain have coordinates $\eta = (\eta_i)_i$ and block-diagonal Fisher metric $G(\eta) = \bigoplus_i G_i(\eta_i)$, including categorical blocks when present. Assume $F(\eta) := \mathcal{F}_o^{\mathrm{ext}}(Q_\eta) \in \mathbb{R}$ is C^1, every $G_i(\eta_i)$ is positive definite, every $\gamma_i \geq 0$, and the trajectory remains in this domain. By Theorem 7.5, the gradient of the outside-averaged local VFE is the corresponding block gradient of F. Hence

$$\dot{\eta}_i = -\gamma_i G_i^{-1} \nabla_{\eta_i} F \tag{7.44}$$

obeys

$$\frac{d}{dt} F = -\sum_i \gamma_i (\nabla_{\eta_i} F)^\top G_i^{-1} (\nabla_{\eta_i} F) \leq 0. \tag{7.45}$$

Thus an agent following its exact local VFE follows a block of the collective VFE flow. [ESTABLISHED]

Block orthogonality is load bearing. With a nondiagonal Fisher metric, the global natural gradient mixes agents and independent local inversions are a different dynamics. Hard support boundaries and finite steps also require a separate projected flow, line search, damping, or acceptance test. [ESTABLISHED]

The variable t in (7.44) parameterizes the declared block flow. Passing from a regular solution curve to its oriented unparameterized orbit, and then measuring that orbit by Fisher length, are the separate constructions of Definition 9.4 and theorem 9.13. A common positive scalar mobility changes only the parameterization, whereas unequal block rates γ_i generally change the orbit. [ESTABLISHED]

No physical-time interpretation of t or of the resulting Fisher duration is asserted. [NOT-CLAIMED]

7.7 Are observations only agent–agent interactions?

Definition 7.7 (Operational environment node). An operational environment node is a standard Borel state space with a normalized state law and a measurable message kernel. This definition does not assert autonomy, self-modeling, or biological agency. [DEFINITION]

Theorem 7.8 (Observation–interaction kernel equivalence). Every normalized standard-Borel observation kernel $K(do \mid y)$ can be realized as a message from an operational environment node: there are $U \sim \mathrm{Unif}[0,1]$ and a measurable map F such that

$$O = F(Y,U), \qquad \Pr(F(y,U) \in A) = K(A \mid y). \tag{7.46}$$

Conversely, marginalizing any environment state and message policy produces a normalized observation kernel. [ESTABLISHED]

Proof. The forward statement is the randomization lemma for kernels on standard Borel spaces [39]. For the converse, if the environment has law $\rho(du)$ and policy $M(y,u,A)$, then $K(A \mid y) = \int M(y,u,A)\rho(du)$ is measurable in y, countably additive in A, and normalized. □

The theorem gives an exact operationally agent-only presentation when every datum is represented as a clamped state or message of a boundary node and all exogenous noise is included among those nodes. It does not eliminate the conditioning role. If O denotes the complete record map, the ELBO is conditioned on the retained sigma-algebra $\mathscr{O} = \sigma(O)$. Replacing the environment by boundary nodes preserves the same random variable O and hence the same $\mathscr{O}$; it does not delete it. In particular,

$$I(Y;O) = \int D_{\mathrm{KL}}\big(P(dY \mid o) \| P^Y(dY)\big) P^O(do), \tag{7.47}$$

with values in $[0,+\infty]$. This quantity measures how much the retained interaction record changes the latent law on average. [ESTABLISHED]

More generally, retaining only a statistic $S = T(O)$ gives the posterior $P(dY \mid S)$, not $P(dY \mid O)$. The deterministic-statistic chain rule

$$I(Y;O) = I(Y;S) + I(Y;O \mid S) \tag{7.48}$$

shows that this replacement is posterior-sufficient exactly when $I(Y;O \mid S) = 0$ (up to the usual almost-sure choice of conditional versions). Complete deletion makes S constant and leaves P^Y. A finite witness is $Y \sim \mathrm{Bernoulli}(1/2)$ with the deterministic message $O = Y$: then $I(Y;O) = \log 2$, $P(dY \mid O = o) = \delta_o$, while deletion returns the nondegenerate prior. Thus an agent-only redescription cannot remove the conditioning sigma-algebra unless it also accepts the corresponding loss of information. [ESTABLISHED]

Nor does the operational equivalence prove that a random seed is an autonomous agent. If physical agency requires persistent state, a Markov blanket, action, or its own local VFE, those are additional hypotheses. An Y-independent kernel is a boundary message with zero mutual information and therefore a null interaction in the informational sense. [ESTABLISHED]

The resulting answer is two-tiered. The probability theory can be written entirely as interactions among agent and environment nodes, while the ELBO still conditions on the sigma-algebra generated by those interaction records. Whether every environment node deserves the stronger ontological name "agent" is not determined by the principal bundle or by variational inference. [NOT-CLAIMED]

Chapter 8

Connection-Relative Informational Pullback Geometry

The preceding chapters separate the principal bundle, the two associated law bundles, and the normalized probabilistic model. This chapter adds a smooth information-geometric layer without identifying those objects. A chosen connection converts the derivative of an agent section into a vertical first jet. The Fisher metric and the Amari–Chentsov tensor can then be pulled back to the contextual base. The resulting tensors are invariant under passive changes of frame, but they depend on the selected connection and may be degenerate. These qualifications are part of the construction rather than defects to be removed by notation.

The same construction also supplies the differential interface to coarse-graining. A coarse morphism whose scale-connection defect form takes values in the isotropy subalgebra of the coarse section value intertwines the covariant first jets, while a parameter-independent Markov channel contributes a positive-semidefinite vertical Fisher defect. That isotropy condition is the only meaning this chapter attaches to compatibility of the two connections, and it is stated as (8.56) rather than left to the reader. The vertical defect composes by an exact unconditional cocycle identity, whereas its pullback to the base composes only up to an exact residual that this chapter displays. Deterministic energy precomposition and Galerkin aggregation do not satisfy the Markov hypothesis merely because they are called coarse maps.

8.1 Statistical tensors on the associated fibers

Hypothesis 8.1 (Regular channel models). For each channel $x \in \{b, m\}$, let $\mathcal{B}_x$ be a finite-dimensional smooth parametrized-measure model satisfying the selected smooth-tier hypothesis of Hypothesis 2.3. Assume differentiability in quadratic mean, a positive-definite Fisher form, third-power integrability of every score direction used below, smooth dependence of the resulting second- and third-order tensors, and domination sufficient to differentiate the divergence expressions below through third order. Assume also that the represented action $\widehat{\rho}_x(g)$ is induced by a parameter-independent bimeasurable change of sample coordinates and preserves $\mathcal{B}_x$. These assumptions exclude singular statistical strata, changing-support models without the stated differentiability, and arbitrary actions on parameter coordinates. [HYPOTHESIS]

At $p \in \mathcal{B}_x$, write ℓ_u for the score in direction $u \in T_p\mathcal{B}_x$. Define

$$g^F_{x,p}(u, v) = \mathbb{E}_p[\ell_u \ell_v], \tag{8.1}$$

$$\mathcal{T}_{x,p}(u, v, w) = \mathbb{E}_p[\ell_u \ell_v \ell_w]. \tag{8.2}$$

The first is the Fisher metric and the second is the symmetric Amari–Chentsov tensor. Their existence here is a statement about the selected parametrized-measure models, not about arbitrary law fibers. [DEFINITION]

Proposition 8.2 (Descent of the statistical tensors). Under Hypothesis 8.1, the represented G-action preserves g^F_x and $\mathcal{T}_x$. Consequently they descend to smooth vertical tensors, denoted by the same symbols, on $\mathcal{E}_x = P \times_{\widehat{\rho}_x} \mathcal{B}_x$. [ESTABLISHED]

Proof. Let r_g be the bimeasurable sample-coordinate change representing g. Pushforward defines the unitary map $L^2(p) \to L^2((r_g)_\# p)$ given by $f \mapsto f \circ r_g^{-1}$. Because r_g is independent of the statistical parameter, differentiability in quadratic mean identifies the score of the pushed-forward tangent direction with $\ell_u \circ r_g^{-1}$. The pushforward integration identity preserves the second and third score moments in (8.1) and (8.2). An invariant tensor on the fiber descends through the associated-bundle quotient. Explicitly, let $\iota_u : \mathcal{B}_x \to \mathcal{E}_x$ send β to the class $[u, \beta]$ of a principal frame u. The quotient convention (2.9) gives $\iota_{ug} = \iota_u \circ \widehat{\rho}_x(g)$, so a fiber tensor τ has frame-independent transport $\tau^u := (\iota_u^{-1})^*\tau$ exactly when $\widehat{\rho}_x(g)^*\tau = \tau$ for every g. Smoothness of the descended tensor follows by evaluating this construction on local smooth sections of P. □

The bimeasurable sample action in this proposition is load bearing. Closure of a set of parameter values under an arbitrary diffeomorphism of its parameter chart would not prove statistical isometry and would not define the same descended tensors. [ESTABLISHED]

8.2 Covariant first jets and pullback tensors

Fix one channel x temporarily and write $E = \mathcal{E}_x$, $\mathcal{B} = \mathcal{B}_x$, $\varpi = \varpi_x$, $g^F = g^F_x$, $\mathcal{T} = \mathcal{T}_x$, and $\omega = \omega_x$. The connection induces a horizontal subbundle $H^\omega E \subset TE$ and a vertical projection $\mathrm{ver}^\omega : TE \to VE$. For a smooth section $s : \mathcal{U} \to E$ over an open set $\mathcal{U} \subseteq \mathcal{C}$, define

$$D^\omega s := \mathrm{ver}^\omega \circ Ts : T\mathcal{U} \longrightarrow s^*VE. \tag{8.3}$$

The connection-split first jet is the pair $j^1_\omega s = (s, D^\omega s)$. For nonlinear statistical fibers, $D^\omega s$ is a vertical-valued one-form along s; it is not a linear covariant derivative on a vector space of sections. [DEFINITION]

Definition 8.3 (Covariant informational pullbacks). The connection-relative Fisher and Amari pullbacks of s are

$$h_s^\omega(X,Y) = g^F_{s(c)}(D^\omega s(X), D^\omega s(Y)), \tag{8.4}$$
$$c_s^\omega(X,Y,Z) = \mathcal{T}_{s(c)}(D^\omega s(X), D^\omega s(Y), D^\omega s(Z)), \tag{8.5}$$

for $X,Y,Z \in T_c\mathcal{C}$. Thus $h_s^\omega \in \Gamma(\mathrm{Sym}^2 T^*\mathcal{U})$ and $c_s^\omega \in \Gamma(\mathrm{Sym}^3 T^*\mathcal{U})$. The first is called a semimetric until its nondegeneracy is proved. [DEFINITION]

Theorem 8.4 (Passive gauge covariance of the covariant pullbacks). Under a passive local change of principal frame, with the connection and section coordinates transformed by their associated laws, both h_s^ω and c_s^ω are unchanged as tensors on $\mathcal{U}$. [ESTABLISHED]

Proof. In a local frame, the vertical jet transforms by the tangent representation,

$$D^{\omega'} s'(X) = T\widehat{\rho}_x(g^{-1}) D^\omega s(X), \tag{8.6}$$

where $g : \mathcal{U} \to G$ is the frame change and the inverse follows from the quotient convention (2.9). The connection law is (2.22) or (2.23), according to the channel. By Proposition 8.2, the tangent representation is an isometry of g^F and preserves $\mathcal{T}$. Substitution in (8.4) and (8.5) cancels the frame change in every argument. □

The word invariance in Theorem 8.4 means passive covariance: two coordinate descriptions of the same pair (ω, s) give the same tensor. It does not say that two different connections give the same pullback, and it does not say that the tensors are fixed by the active gauge group at a fixed connection. For the active statement, take the trivial principal $\mathbb{R}$-bundle over $\mathbb{R}$ with the normal location fiber, the zero connection, and the zero section, so that $h = 0$; the bundle automorphism $F(u(x)g) = u(x)(x+g)$ carries that section to $s'(x) = \mathcal{N}(x,1)$ and hence h to dx^2. [ESTABLISHED]

8.3 Exact dependence on the chosen connection

Let ω' be a second principal connection on P. Their difference is horizontal and Ad-equivariant; denote its descended form by

$$a \in \Omega^1(\mathcal{C}, \mathrm{Ad}(P)). \tag{8.7}$$

For the associated action, let $\vartheta_e : \mathrm{Ad}(P)_{\varpi(e)} \to V_eE$ denote the fundamental vertical map. Along s, set

$$R_a^s(X) = \vartheta_{s(c)}(a_c(X)). \tag{8.8}$$

The sign convention is fixed frame-free by (8.8): in a local trivialization the horizontal lift of X at (c,β) is $(X, -\zeta_{A(X)}\beta)$, where ζ_ξ is the fundamental vertical field of $\xi \in \mathfrak{g}$ for the represented action, so replacing A by $A + a$ adds $\vartheta(a(X))$ to the vertical jet. For a general law fiber $\mathcal{B} \subseteq \mathcal{P}(\mathsf{K})$ the fiber is not a vector space and the linear expression $\widehat{\rho}_{x*}(A)$ is undefined; it is available only in a declared fiberwise-linear associated-vector-bundle realization, where it agrees with $\zeta_{A(X)}$. The convention is compatible with (2.9). [DEFINITION]

Proposition 8.5 (Connection-change formulas). The two vertical jets and their Fisher pullbacks obey

$$D^{\omega'} s = D^{\omega} s + R^s_a, \tag{8.9}$$

$$h^{\omega'}_s(X,Y) = h^{\omega}_s(X,Y) + g^F(R^s_a X, D^{\omega} sY) + g^F(D^{\omega} sX, R^s_a Y) + g^F(R^s_a X, R^s_a Y). \tag{8.10}$$

The Amari pullback satisfies the exact polarized identity

$$c^{\omega'}_s(X,Y,Z) = \mathcal{T}((D^{\omega} s + R^s_a)X, (D^{\omega} s + R^s_a)Y, (D^{\omega} s + R^s_a)Z). \tag{8.11}$$

It therefore contains the original term, three terms with one occurrence of R^s_a, three terms with two occurrences, and one term with three occurrences. [ESTABLISHED]

Proof. In one local frame, replacing the horizontal lift $(X, -\zeta_{A(X)}\beta)$ by $(X, -\zeta_{A(X)+a(X)}\beta)$ shifts the vertical part of Ts by $\vartheta_{s(c)}(a(X)) = R^s_a(X)$, which proves (8.9). Bilinearity of g^F gives (8.10), and trilinearity of $\mathcal{T}$ gives (8.11). The formulas are independent of the selected local frame because a is an $\mathrm{Ad}(P)$-valued one-form and every displayed contraction is gauge invariant. □

The dependence can occur even in the simplest regular family. Let $\mathcal{C} = \mathbb{R}$, let P be the trivial principal translation bundle, and let $\mathcal{B} = \{\mathcal{N}(\mu,1) : \mu \in \mathbb{R}\}$. For the constant section $\mu = 0$, the zero connection gives $h = 0$, whereas the local connection $A' = a_0 dx$ gives

$$h^{A'} = a_0^2 dx^2. \tag{8.12}$$

This example refutes a connection-independent interpretation of h^{ω}_s; it does not refute the gauge invariance of a fixed connection–section pair. [ESTABLISHED]

8.4 Two channel geometries and their types

Restoring the channel labels gives the ordered pair

$$(h^{\omega_b}_{i,b}, h^{\omega_m}_{i,m}) \in \Gamma(\mathrm{Sym}^2\, T^*\mathcal{U}) \times \Gamma(\mathrm{Sym}^2\, T^*\mathcal{U}), \tag{8.13}$$

and an analogous pair of cubic tensors. This pair is determined by the two selected statistical models, sections, and connections. A single scalar geometry is not part of those data. [ESTABLISHED]

Hypothesis 8.6 (Weighted product geometry). If one joint geometry is required, choose constants $w_b, w_m > 0$ and define

$$h^{\mathrm{prod}} = w_b h^{\omega_b}_{i,b} + w_m h^{\omega_m}_{i,m}. \tag{8.14}$$

The weights fix the relative normalization of the two statistical channels. Unit weights are a specialization rather than a consequence of the common principal bundle. [HYPOTHESIS]

Whenever weighted products at two different scales are compared, two further declarations are consumed and neither follows from any fiberwise contraction theorem. Writing μ and $\bar{\mu}$ for the finite positive base measures used to integrate the two tensors and $w_x, \bar{w}_x$ for the two sets of channel weights, the comparison hypotheses are

$$\textbf{(X1)} \quad f_{\#}\mu = \bar{\mu}, \qquad\qquad \textbf{(X2)} \quad \bar{w}_x \circ f \leq w_x \quad \text{for } x \in \{b, m\}. \tag{8.15}$$

Under a channelwise comparison the two-channel difference is

$$h^{\mathrm{prod}} - f^*\bar{h}^{\mathrm{prod}} = \sum_{x \in \{b,m\}} \left[w_x \left(h_x - f^*\bar{h}_x \right) + \left(w_x - \bar{w}_x \circ f \right) f^*\bar{h}_x \right], \tag{8.16}$$

so independently declared coarse weights with $(\bar{w}_x \circ f) > w_x$ break positivity even when every channel has zero horizontal defect and a genuine Markov fiber map. Failure of (X1) reverses the integrated comparison just as directly: take $\mathcal{C} = \bar{\mathcal{C}}$ the disjoint union of two copies of $\mathbb{R}$ with $f = \mathrm{id}$, let h vanish on the first copy and equal dx^2 on the second, and let μ and $\bar{\mu}$ be unit point masses on the first and the second copy respectively; the fine integrated tensor is zero and the coarse one is one, with vanishing fiberwise defect throughout. [ESTABLISHED]

Proposition 8.7 (Radical of the weighted product). Under Hypotheses 8.1 and 8.6,

$$\operatorname{rad} h^{\mathrm{prod}} = \ker D^{\omega_b} q_i \cap \ker D^{\omega_m} s_i. \tag{8.17}$$

Hence the two degenerate channel tensors can jointly be nondegenerate exactly when their null distributions have zero intersection. [ESTABLISHED]

Proof. For every X,

$$h^{\mathrm{prod}}(X, X) = w_b g_b^F(D^{\omega_b} q_i X, D^{\omega_b} q_i X) + w_m g_m^F(D^{\omega_m} s_i X, D^{\omega_m} s_i X).$$

Both summands are nonnegative and both weights are positive. Their sum vanishes exactly when both vertical jets vanish. Polarization then identifies the radical. □

The common principal group acts separately through $\widehat{\rho}_b$ and $\widehat{\rho}_m$. It supplies no bilinear cross tensor on $V\mathcal{E}_b \times V\mathcal{E}_m$. Such a cross term requires a separately declared invariant bilinear form, and a comparison of channel points requires Φ or $\widetilde{\Phi}$ from Section 2.4. Neither the relative principal frame nor equal channel dimensions supplies either object. [ESTABLISHED]

8.5 Rank, radicals, and quotient geometry

Theorem 8.8 (Constant-rank quotient theorem). Let $s : \mathcal{U} \to E$ satisfy Hypothesis 8.1. Pointwise,

$$\operatorname{rad} h^\omega_s = \ker D^\omega s, \qquad \operatorname{rank} h^\omega_s = \operatorname{rank} D^\omega s. \tag{8.18}$$

If $D^\omega s$ has constant rank on $\mathcal{U}$, then $K^\omega_s = \ker D^\omega s$ is a smooth subbundle and

$$Q^\omega_s = T\mathcal{U}/K^\omega_s, \tag{8.19}$$

has a positive-definite metric and a symmetric cubic tensor defined by

$$\bar{h}^\omega_s([X],[Y]) = h^\omega_s(X,Y), \tag{8.20}$$
$$\bar{c}^\omega_s([X],[Y],[Z]) = c^\omega_s(X,Y,Z). \tag{8.21}$$

The induced map $Q^\omega_s \to \operatorname{im} D^\omega s$ is an isometric vector-bundle isomorphism. [ESTABLISHED]

Proof. If $D^\omega s X = 0$, then (8.4) gives $h^\omega_s(X,Y) = 0$ for all Y. Conversely, if X lies in the radical, then

$$0 = h^\omega_s(X,X) = g^F(D^\omega s X, D^\omega s X),$$

and positive definiteness of g^F forces $D^\omega s X = 0$. This proves (8.18). The applicable constant-rank statement is the one for morphisms of smooth vector bundles over a fixed base, not the constant-rank theorem for smooth maps of manifolds: $D^\omega s$ is a vector-bundle morphism $T\mathcal{U} \to s^*VE$ over id, and a vector-bundle morphism of locally constant rank has smooth kernel and image subbundles by local adaptation of frames. Replacing an argument of (8.20) or (8.21) by a representative differing by an element of K^ω_s leaves its vertical jet unchanged. The quotient tensors are therefore well defined. The factored map induced by $D^\omega s$ is injective onto its image and preserves the displayed tensors; the quotient metric is positive definite. □

The quotient in Theorem 8.8 is a vector bundle. It is not automatically the tangent bundle of a quotient manifold. For that interpretation, require a smooth Hausdorff quotient manifold $\overline{\mathcal{U}}$ and a surjective submersion $q_K : \mathcal{U} \to \overline{\mathcal{U}}$ whose fibers are exactly the null leaves and whose tangent kernel is $\ker Tq_K = K^\omega_s$. Consequently, the null distribution must be involutive,

$$D^\omega s([X,Y]) = 0 \quad \text{for all } X, Y \in \Gamma(K^\omega_s), \tag{8.22}$$

but involutivity alone supplies neither the quotient manifold nor the surjective-submersion quotient map. [ESTABLISHED]

Separately, the tensors must be basic for the null foliation. Their contraction with a null vector already vanishes by construction, but invariance along null leaves is an additional requirement:

$$\mathcal{L}_Z h^\omega_s = 0, \qquad \mathcal{L}_Z c^\omega_s = 0 \quad \text{for every } Z \in \Gamma(K^\omega_s). \tag{8.23}$$

Without this condition, even a smooth Hausdorff leaf-space quotient need not carry tensors whose pullbacks are h_s^ω and c_s^ω. [ESTABLISHED]

Basicness of the quadratic tensor is checkable directly on the jet, but the bundle-valued one-form $D^\omega s$ has no ordinary Lie derivative. Choose a g^F-compatible auxiliary connection ∇ on s^*VE and, for a s^*VE-valued one-form α, write

$$(\mathcal{L}_Z^\nabla \alpha)(X) = \nabla_Z(\alpha(X)) - \alpha([Z, X]). \tag{8.24}$$

Then for $Z \in \Gamma(K_s^\omega)$ and arbitrary X, Y, the Leibniz rule for the ordinary Lie derivative of h_s^ω gives

$$(\mathcal{L}_Z h_s^\omega)(X, Y) = g^F\left((\mathcal{L}_Z^\nabla D^\omega s)(X), D^\omega sY\right) + g^F\left(D^\omega sX, (\mathcal{L}_Z^\nabla D^\omega s)(Y)\right). \tag{8.25}$$

Metric compatibility supplies this identity directly, while $\iota_Z h_s^\omega = 0$ is already automatic from $Z \in \ker D^\omega s$. If $\nabla' = \nabla + B$ is another g^F-compatible connection, then B_Z is g^F-skew-adjoint, so the changes in the two displayed covariant summands cancel. Thus the displayed equality with the ordinary, canonical $\mathcal{L}_Z h_s^\omega$ is independent of the selected auxiliary metric connection, even though neither covariant summand is separately canonical. [ESTABLISHED]

Proposition 8.9 (A constant-rank nonintegrable radical). There is a regular one-dimensional statistical fiber, a smooth connection, and a section over $\mathbb{R}^3$ for which h_s^ω has constant rank one while $\ker D^\omega s$ is not involutive. [ESTABLISHED]

Proof. Use the normal location family and the trivial principal translation bundle over $\mathbb{R}^3$ with coordinates (x, y, z). In the constant section $\mu = 0$, choose the local connection form

$$\alpha = \mathrm{d}z - x\mathrm{d}y. \tag{8.26}$$

Then $D^\omega s = \alpha$ and

$$h_s^\omega = (\mathrm{d}z - x\mathrm{d}y)^2. \tag{8.27}$$

The rank is one everywhere, but

$$\alpha \wedge \mathrm{d}\alpha = -\mathrm{d}z \wedge \mathrm{d}x \wedge \mathrm{d}y \neq 0. \tag{8.28}$$

The Frobenius criterion therefore shows that $\ker \alpha$ is a contact distribution rather than a foliation. □

Constant rank is also necessary for the smooth quotient-bundle conclusion. For the same normal location family on $\mathcal{C} = \mathbb{R}$, with the zero connection and $s(x) = \mathcal{N}(x^2, 1)$,

$$h_s = 4x^2 dx^2, \tag{8.29}$$

has rank one away from the origin and rank zero at the origin. The pointwise quotients do not assemble into a vector bundle across that point. [ESTABLISHED]

8.6 Divergence jets and transported local comparison

Let $\mathscr{D} : \mathcal{B} \times \mathcal{B} \to \mathbb{R}$ be a C^3 contrast in a neighborhood of the diagonal: $\mathscr{D}(p, q) \geq 0$ and $\mathscr{D}(p, q) = 0$ exactly on the diagonal in that neighborhood. For tangent vectors below, assume additionally that $\mathscr{D}$ is invariant under the simultaneous represented action, $\mathscr{D}(\widehat{\rho}(g)p, \widehat{\rho}(g)q) = \mathscr{D}(p, q)$, so that it defines an unambiguous comparison of two points in one associated-bundle fiber. Without this invariance, the following jets are only local-frame objects. Assume also that the mixed second form defined below is positive definite on the selected regular tier. A contrast such as $\mathscr{D}(\mu, \nu) = (\mu - \nu)^4$ is positive off the diagonal but has a zero diagonal Hessian; it yields degenerate jets rather than a divergence- induced dualistic geometry. [HYPOTHESIS]

For tangent vectors $u, v, w \in T_p\mathcal{B}$, let $(u, 0)$ and $(0, u)$ denote their lifts to the first and second factors. In one common local chart, extend these vectors as constant coordinate vector fields and define the mixed diagonal jets

$$g_{\mathscr{D}}(u, v) = -[(u, 0)(0, v)\mathscr{D}]_{(p,p)}, \tag{8.30}$$

$$\Gamma_{\mathscr{D}}(u, v, w) = -[(u, 0)(v, 0)(0, w)\mathscr{D}]_{(p,p)}, \tag{8.31}$$

$$\Gamma^*_{\mathscr{D}}(u, v, w) = -[(0, u)(0, v)(w, 0)\mathscr{D}]_{(p,p)}, \tag{8.32}$$

$$\mathcal{T}_{\mathscr{D}} = \Gamma_{\mathscr{D}} - \Gamma^*_{\mathscr{D}}. \tag{8.33}$$

The individual third-order arrays are connection coefficients in a chart; their difference is a cubic tensor. The sign convention in (8.33) is chosen so that forward KL on an exponential family gives the score-third-moment tensor (8.2). [DEFINITION]

Proposition 8.10 (KL divergence jets). On a regular statistical model satisfying the integrability hypotheses of Hypothesis 8.1, the mixed second jet of $D_{\mathrm{KL}}(p\|q)$ is the Fisher metric, and the difference of the two mixed third jets is the Amari–Chentsov tensor. [ESTABLISHED]

Proof. Use a local parameter θ for the first argument and η for the second. Write $\ell_i = \partial_i \log p_\theta$ and let primed derivatives act on η. Normalization gives $\mathbb{E}_\theta[\ell_i] = 0$, while $\partial_i p_\theta = p_\theta \ell_i$ and $\partial_i \partial_j p_\theta = p_\theta(\ell_i \ell_j + \partial_i \ell_j)$. Direct differentiation at $\eta = \theta$ therefore gives

$$-\partial_i \partial'_j D_{\mathrm{KL}}(p_\theta \| p_\eta) = \mathbb{E}_\theta[\ell_i \ell_j], \tag{8.34}$$

$$-\partial_i \partial_j \partial'_k D_{\mathrm{KL}}(p_\theta \| p_\eta) = \mathbb{E}_\theta[(\ell_i \ell_j + \partial_i \ell_j)\ell_k], \tag{8.35}$$

$$-\partial'_i \partial'_j \partial_k D_{\mathrm{KL}}(p_\theta \| p_\eta) = \mathbb{E}_\theta[\ell_k \partial_i \ell_j]. \tag{8.36}$$

The score Hessian is symmetric in the chosen smooth chart, so subtracting the last line from the second cancels its score-derivative term and leaves $\mathbb{E}_\theta[\ell_i \ell_j \ell_k]$. Polarization gives the asserted formulas for arbitrary u, v, w. Differentiation under the integral is licensed by Hypothesis 8.1; these are the standard divergence-induced dualistic structures of information geometry [3, 7]. □

For a smooth base curve γ with $\gamma(0) = c$, let $\Omega^{\omega}_{0\to\epsilon}$ be the channel transport along $\gamma|_{[0,\epsilon]}$ and bring the section value back to the initial fiber by

$$\widehat{s}_{\gamma}(\epsilon) = (\Omega^{\omega}_{0\to\epsilon})^{-1} s(\gamma(\epsilon)). \tag{8.37}$$

Then

$$\frac{d}{d\epsilon}\widehat{s}_{\gamma}(\epsilon)\bigg|_{\epsilon=0} = D^{\omega} s(\dot{\gamma}(0)). \tag{8.38}$$

[ESTABLISHED]

Corollary 8.11 (Quadratic transported divergence). Let $g_{\mathscr{D}}$ be the metric induced by (8.30) and set $h^{\omega}_{s,\mathscr{D}} = (D^{\omega} s)^* g_{\mathscr{D}}$. Then

$$\mathscr{D}(s(c), \widehat{s}_{\gamma}(\epsilon)) = \frac{\epsilon^2}{2} h^{\omega}_{s,\mathscr{D}}(\dot{\gamma}(0), \dot{\gamma}(0)) + O(\epsilon^3). \tag{8.39}$$

If $g_{\mathscr{D}} = g^F$, in particular for the regular KL contrast of Proposition 8.10, then $h^{\omega}_{s,\mathscr{D}} = h^{\omega}_{s}$. [ESTABLISHED]

Proof. A contrast and its first derivative vanish on the diagonal. Its quadratic Taylor term in the second argument is the induced metric. Substituting (8.38) gives (8.39). □

The coefficient of ϵ^3 in a one-sided expansion such as (8.39) is not determined by $c^{\omega}_{s,\mathscr{D}} := (D^{\omega} s)^* \mathcal{T}_{\mathscr{D}}$ alone. It also contains the covariant acceleration of $\widehat{s}_{\gamma}$ and the connection coefficient selected by the one-sided divergence jet. The divergence-induced cubic is recovered from the difference of the two mixed third jets in (8.33), not from an arbitrary one-sided cubic Taylor coefficient. For regular KL, it is the Amari–Chentsov tensor and $c^{\omega}_{s,\mathscr{D}} = c^{\omega}_{s}$. [ESTABLISHED]

8.7 Vertical information length

For a smooth curve $\Gamma : J \to E$ over $\gamma = \varpi \circ \Gamma$, define its connection-relative vertical velocity by

$$v^{\omega}(\Gamma) = \mathrm{ver}^{\omega}(\dot{\Gamma}) = \dot{\Gamma} - H^{\omega}_{\Gamma}\dot{\gamma}, \tag{8.40}$$

and its vertical Fisher length by

$$L^{\omega}(\Gamma) = \int_J \sqrt{g^F(v^{\omega}(\Gamma), v^{\omega}(\Gamma))}\, d\lambda. \tag{8.41}$$

Here λ is an arbitrary curve parameter rather than a declared physical time. [DEFINITION]

If $\omega' = \omega + a$, define along the curve

$$R^{\Gamma}_{a}(\dot{\gamma}) = \vartheta_{\Gamma(\lambda)}\left(a_{\gamma(\lambda)}(\dot{\gamma}(\lambda))\right). \tag{8.42}$$

With the convention of (8.9), the exact curve identity is $v^{\omega'}(\Gamma) = v^{\omega}(\Gamma) + R^{\Gamma}_{a}(\dot{\gamma})$. [ESTABLISHED]

Proposition 8.12 (Vertical, horizontal, and section curves). The length L^ω is invariant under orientation-preserving C^1 reparameterization with positive derivative. If Γ is vertical, then $L^\omega(\Gamma)$ is independent of the connection. If Γ is ω-horizontal, then $L^\omega(\Gamma) = 0$. For a section curve $\Gamma = s \circ \gamma$,

$$L^\omega(s \circ \gamma) = \int_J \sqrt{h_s^\omega(\dot\gamma, \dot\gamma)} d\lambda. \tag{8.43}$$

For a mixed curve, changing ω to ω' changes the vertical velocity by $R_a^\Gamma(\dot\gamma)$ in the convention of (8.9); its length is therefore generally connection dependent. [ESTABLISHED]

Proof. Under a reparameterization $\lambda = \phi(r)$, vertical velocity is multiplied by $\phi'(r)$, and positive homogeneity of the square root converts that factor into the Jacobian in the one-dimensional change-of-variables formula. For a vertical curve, $\dot\gamma = 0$, so the horizontal correction vanishes for every connection. Horizontality is the equation $v^\omega(\Gamma) = 0$. If $\Gamma = s \circ \gamma$, then $v^\omega(\Gamma) = D^\omega s(\dot\gamma)$, and (8.43) follows from (8.4). The connection-change statement is (8.9) along the curve. □

The functional (8.41) measures only vertical statistical change. It assigns zero length to horizontal displacement and does not define a metric on TE without a separately chosen base metric. It also selects neither an inference orbit nor an orientation. Those require a separately declared configuration manifold together with an objective differential and a mobility or metric sharp map that defines its vector field. [NOT-CLAIMED]

A base curve γ orders contexts, and $s \circ \gamma$ evaluates one fixed section along that ordering; neither object is an agent inference history. Updating at one fixed c is instead a vertical curve in the relevant statistical fiber, while evolution of a complete contextual agent is a curve in a separately declared section configuration space. [DEFINITION]

The finite-dimensional length above applies pointwise or on a declared finite-design product. It does not construct a continuum section-space length or a probability law on a continuum space of sections, because no base measure, channel weighting, or gauge-quotient rule has been declared here. [NOT-CLAIMED]

8.8 Coarse-morphism naturality

Consider smooth associated bundles $E \to \mathcal{C}$ and $\bar E \to \bar{\mathcal{C}}$ with connections ω and $\bar\omega$. Let $f : \mathcal{C} \to \bar{\mathcal{C}}$ and let $\Psi : E \to \bar E$ be a smooth bundle morphism over f. Suppose sections s and $\bar s$ are related as follows. Before that relation is imposed, $\Psi \circ s$ is only a section of the pullback bundle $f^*\bar E$ along f; a fine section does not by itself define a section on $\bar{\mathcal{C}}$. [DEFINITION]

We adopt the descent hypothesis that there exists a smooth coarse section $\bar s$ satisfying

$$\Psi \circ s = \bar s \circ f. \tag{8.44}$$

For a surjective submersion f, this excludes fine sections whose Ψ-images vary along a fiber of f. [HYPOTHESIS]

Write $T^V\Psi : VE \to V\bar{E}$ for the vertical differential. Following the cross-channel defect of (2.26), define

$$(\mathcal{D}\Psi)_e(X) = A_\Psi(e;X) := \mathrm{ver}^{\bar{\omega}}\left(T_e\Psi(H^\omega_e X)\right) = T_e\Psi(H^\omega_e X) - H^{\bar{\omega}}_{\Psi(e)}(T_c f X). \tag{8.45}$$

The sign convention is fixed and load bearing: the transported fine horizontal lift comes first and the coarse horizontal lift is subtracted. The difference is vertical because $T\bar{\varpi} \circ T\Psi(H^\omega_e X) = T_c f X = T\bar{\varpi} \circ H^{\bar{\omega}}(T_c f X)$, and the second displayed form follows because the $\bar{\omega}$-horizontal part of $T\Psi(H^\omega_e X)$ is exactly $H^{\bar{\omega}}_{\Psi(e)}(T_c f X)$. Its type is $A_\Psi \in \Gamma(E; \varpi^* T^*\mathcal{C} \otimes \Psi^* V\bar{E})$, and along a section $A_\Psi(s;\cdot) \in \Gamma(\mathcal{U}; T^*\mathcal{U} \otimes f^*\bar{s}^* V\bar{E})$. The abbreviation A_Ψ is used below wherever the vertical values must be paired with $\bar{g}^F$. [DEFINITION]

Theorem 8.13 (Sharp section descent). Let f be a surjective smooth submersion, let Ψ be a smooth bundle morphism over f, and let $Q \in \Gamma(\mathcal{C}, E)$. Consider the three conditions: (P1) there exists $\bar{Q} \in \Gamma(\bar{\mathcal{C}}, \bar{E})$ with $\Psi \circ Q = \bar{Q} \circ f$; (P2) $\Psi \circ Q$ is constant on each fiber of f; and

$$\textbf{(P3)} \qquad T^V\Psi\left(D^\omega Q(X)\right) + A_\Psi\left(Q(c); X\right) = 0 \quad \text{for every } c \text{ and every } X \in \ker T_c f. \tag{8.46}$$

Then (P1) and (P2) are equivalent, both imply (P3), and (P3) implies (P2) when the fibers of f are connected. Under (P2) the descended $\bar{Q}$ is unique, is automatically smooth, and is automatically a section. [ESTABLISHED]

Proof. That (P1) implies (P2) is immediate. Conversely, a surjective smooth submersion is a smooth quotient map, so a smooth map constant on its fibers descends uniquely through it and the descended factor is smooth; this is where smoothness of $\bar{Q}$ becomes a theorem rather than an obligation. Surjectivity gives uniqueness, and $\bar{\varpi}\bar{Q}f = \bar{\varpi}\Psi Q = f\varpi Q = f$ with f surjective gives $\bar{\varpi}\bar{Q} = \mathrm{id}$. For (P2) implying (P3), take $X \in \ker T_c f$, split $T_c QX = H^\omega_{Q(c)}X + D^\omega Q(X)$, apply $T\Psi$, and use $T_c f X = 0$ to annihilate the coarse horizontal lift; what remains is $T(\Psi Q)(X) = A_\Psi(Q(c);X) + T^V\Psi(D^\omega Q(X))$, which fiber constancy forces to vanish. The same computation shows that (P3) makes $T(\Psi Q)$ annihilate $\ker Tf$, and a smooth map with vanishing derivative along a connected embedded submanifold is constant on it. □

The submersion hypothesis is load bearing rather than decorative. With $\mathcal{C} = \bar{\mathcal{C}} = \mathbb{R}$, $f(x) = x^3$ a smooth bijection that fails to be a submersion at the origin, the trivial bundle with fiber $\{\mathcal{N}(\mu, 1)\}$, Ψ the identity on fibers, and $Q(x) = \mathcal{N}(x, 1)$, the unique descent $\bar{Q}(y) = \mathcal{N}(y^{1/3}, 1)$ is continuous but not differentiable at $y = 0$. [ESTABLISHED]

Proposition 8.14 (A pointwise bundle morphism is not a map on configurations). Suppose f is a surjective submersion, $\ker T_{c_0} f \neq 0$ at some c_0, there are $e_0 \in E_{c_0}$ and $w \in V_{e_0}E$ with $T^V\Psi(w) \neq 0$, and a global smooth section Q_0 satisfies $Q_0(c_0) = e_0$. Then the projectable set $\Gamma_{\mathrm{proj}}(\Psi)$ of sections satisfying (P2) is a proper subset of $\Gamma(\mathcal{C}, E)$. [ESTABLISHED]

Proof. Choose a chart in which $\ker Tf$ is spanned by $\partial_1, \ldots, \partial_k$ with $k \geq 1$, and use the declared global section Q_0 through e_0. If Q_0 already fails (P2) there is nothing to prove. Otherwise let W

be a smooth vertical field in that local trivialization of E with $W(Q_0(c_0)) = w$ and local flow Φ^W, take $\chi \in C_c^\infty$ supported inside this trivializing chart with $\chi(c_0) = 0$ and $\partial_1\chi(c_0) \neq 0$, and define the global smooth section Q_ϵ by $Q_\epsilon(c) = \Phi^W_{\epsilon\chi(c)}(Q_0(c))$ on the support chart and $Q_\epsilon = Q_0$ off its support. Since $\chi(c_0) = 0$, both $A_\Psi(Q_\epsilon(c_0); \partial_1)$ and $T^V\Psi$ at that point are independent of ϵ, so differentiating the left-hand side of (8.46) at $\epsilon = 0$ gives $\partial_1\chi(c_0)T^V\Psi(w) \neq 0$ and (P3) fails for small nonzero ϵ. □

A concrete witness is the total collapse $\mathcal{C} = S^1$, $\bar{\mathcal{C}} = \{*\}$, which is a surjective submersion, with trivial bundles carrying the unit-variance normal location fiber, Ψ the identity on fibers, and $Q(x) = \mathcal{N}(\sin x, 1)$. Here $A_\Psi = 0$ and $\ker Tf = TS^1$, while $T^V\Psi(D^\omega Q\partial_x) = \cos x\partial_\mu$ is not identically zero, so no coarse section exists. In the linear realization with configuration space $L^2(S^1)$ the descendable set is exactly the constants: a closed subspace of infinite codimension and empty interior. This statement is scoped to a collapsing f; if f is a diffeomorphism every section descends, so no unscoped claim that a bundle morphism induces a configuration map nowhere is available. [ESTABLISHED]

Theorem 8.15 (Covariant first-jet chain rule). Under (8.44),

$$D^{\bar{\omega}}\bar{s}(TfX) = T^V\Psi(D^\omega sX) + (\mathcal{D}\Psi)_{s(c)}(X). \tag{8.47}$$

Hence the square of covariant first jets commutes exactly when $\mathcal{D}\Psi$ vanishes along s. In that case, every vertical covariant tensor $\bar{\tau}$ on $\bar{E}$ satisfies

$$f^*[(D^{\bar{\omega}}\bar{s})^*\bar{\tau}] = (D^\omega s)^*[(T^V\Psi)^*\bar{\tau}]. \tag{8.48}$$

[ESTABLISHED]

Proof. Differentiate (8.44) and decompose $TsX = H_s^\omega X + D^\omega sX$. Applying $T\Psi$ to the horizontal term gives the target horizontal lift plus $(\mathcal{D}\Psi)_{s(c)}(X)$ by (8.45); applying it to the vertical term gives $T^V\Psi(D^\omega sX)$. The vertical part of the other side is $D^{\bar{\omega}}\bar{s}(TfX)$, which proves (8.47). When the defect vanishes, applying $\bar{\tau}$ to the intertwined vertical jets proves (8.48). □

The base comparison of the two Fisher pullbacks can now be written exactly, with no compatibility hypothesis of any kind. Along s write $u_X = D^\omega sX$ for the fine vertical jet, $L = T^V\Psi$ for the vertical differential, $a_X = A_\Psi(s(c); X)$ for the horizontal defect, and $\bar{u}_X = D^{\bar{\omega}}\bar{s}(T_cfX) = Lu_X + a_X$ for the coarse vertical jet. With the vertical Fisher defect Δ_F^Ψ of (8.60) below, set

$$\delta_\Psi(X,Y) = \Delta_F^\Psi(u_X, u_Y), \tag{8.49}$$

$$\mathcal{X}_\Psi(X,Y) = \bar{g}^F(Lu_X, a_Y) + \bar{g}^F(a_X, Lu_Y), \tag{8.50}$$

$$\mathcal{Q}_\Psi(X,Y) = \bar{g}^F(a_X, a_Y). \tag{8.51}$$

Theorem 8.16 (Exact signed base Fisher comparison). Under (8.44) alone,

$$h_s^\omega - f^*\bar{h}_{\bar{s}}^{\bar{\omega}} = \delta_\Psi - \mathcal{X}_\Psi - \mathcal{Q}_\Psi \tag{8.52}$$

as symmetric two-tensors on the fine domain. Here $\mathcal{X}_\Psi$ is symmetric and sign indefinite while $\mathcal{Q}_\Psi \succeq 0$, so both retained tensors enter with a minus sign and neither may be summarized as one vertical mismatch term. [ESTABLISHED]

Proof. By (8.47), $f^*\bar{h}(X,Y) = \bar{g}^F(Lu_X + a_X, Lu_Y + a_Y) = \bar{g}^F(Lu_X, Lu_Y) + \mathcal{X}_\Psi(X,Y) + \mathcal{Q}_\Psi(X,Y)$, and by (8.60), $\bar{g}^F(Lu_X, Lu_Y) = g^F(u_X, u_Y) - \Delta_F^\Psi(u_X, u_Y)$. Subtracting from $h_s^\omega(X,Y) = g^F(u_X, u_Y)$ gives (8.52). □

Theorem 8.17 (Exact signed positivity criterion). Assume in addition the Markov hypotheses of Theorem 8.21, so that $\delta_\Psi \succeq 0$. Then at a fixed c the following are equivalent: the base comparison $h_s^\omega - f^*\bar{h}_{\bar{s}}^{\bar{\omega}}$ is positive semidefinite; the coarse jet is no longer than the fine jet, meaning $\|D^{\bar{\omega}}\bar{s}(T_c f X)\|_{\bar{g}^F} \le \|D^\omega s X\|_{g^F}$ for every X; and

$$2\bar{g}^F\left(T^V\Psi D^\omega sX, A_\Psi(s(c);X)\right) + \|A_\Psi(s(c);X)\|_{\bar{g}^F}^2 \le \delta_\Psi(X,X) \quad \text{for every } X. \tag{8.53}$$

Consequently vanishing of the horizontal defect is *sufficient* for base positivity and is *not necessary*: whenever the cross term $\bar{g}^F(Lu_X, a_X)$ is negative and large enough in modulus, (8.53) holds strictly with $a_X \neq 0$. The triangle-inequality margin $\|a_X\|_{\bar{g}^F} \le \sqrt{h(X,X)} - \sqrt{h(X,X) - \delta_\Psi(X,X)}$ is likewise sufficient and not necessary. [ESTABLISHED]

Proof. The first two conditions are (8.52) read on the diagonal, using $h(X,X) = \|u_X\|_{g^F}^2$ and $f^*\bar{h}(X,X) = \|Lu_X + a_X\|_{\bar{g}^F}^2$. Expanding that square and substituting $\|Lu_X\|_{\bar{g}^F}^2 = \|u_X\|_{g^F}^2 - \delta_\Psi(X,X)$ gives (8.53). □

Both failure directions are realized on the normal location fiber, and the arithmetic is exact. Take $\mathcal{C} = \bar{\mathcal{C}} = \mathbb{R}$, f the identity, $\mathcal{B} = \{\mathcal{N}(\mu,1)\}$ with $g^F = 1$, the sample kernel $N(x,\cdot) = \mathcal{N}(x,1)$ so that $\bar{\mathcal{B}} = \{\mathcal{N}(\mu,2)\}$ with $\bar{g}^F = \frac{1}{2}$ and $\Delta_F^\Psi = \frac{1}{2}$, the section $s(x) = \mathcal{N}(x,1)$, zero source connection, and a target connection contributing $a_X = b\partial_\mu$. Then $h_s^\omega - f^*\bar{h}_{\bar{s}}^{\bar{\omega}} = 1 - \frac{1}{2}(1+b)^2$, which equals $\frac{79}{200} > 0$ at $b = \frac{1}{10}$ and $\frac{23}{25} > 0$ at $b = -\frac{3}{5}$, both with nonzero horizontal defect, and equals $-\frac{1}{8} < 0$ at $b = \frac{1}{2}$. At $b = -\frac{3}{5}$ the margin above is violated, since $\|a_X\|_{\bar{g}^F} = \frac{3}{5\sqrt{2}}$ exceeds $1 - \frac{1}{\sqrt{2}}$, so the margin is genuinely only sufficient. Strict negativity survives even at zero information loss: with the identity kernel, so $\Delta_F^\Psi = 0$, constant related sections, source horizontal field ∂_x, and target horizontal field $\partial_x + a\partial_\mu$ with $a \neq 0$, one gets $u = 0$, $a_X = -a\partial_\mu$, and $h_s^\omega - f^*\bar{h}_{\bar{s}}^{\bar{\omega}} = -a^2 dx^2 \prec 0$. [ESTABLISHED]

The horizontal defect of an induced morphism is not an arbitrary vertical field. Suppose Ψ is induced by an equivariant principal scale map $\mathcal{P}$ over f with Lie-group homomorphism κ and law-fiber map q, in the sense of (11.15). Define the *scale-connection defect form*

$$\mathfrak{A}_\mathcal{P} = \mathcal{P}^*\bar{\omega} - \mathrm{d}\kappa \circ \omega \in \Omega^1(P, \bar{\mathfrak{g}}). \tag{8.54}$$

Theorem 8.18 (Isotropy criterion for a vanishing horizontal defect). The form $\mathfrak{A}_{\mathcal{P}}$ is horizontal and $\mathrm{Ad}\circ\kappa$-equivariant, hence descends to an element of $\Omega^1(\mathcal{C}; f^* \mathrm{Ad}\,\bar{P})$, and

$$A_\Psi(e;X) = \vartheta_{\Psi(e)}\left(\mathfrak{A}_{\mathcal{P}}(X)\right), \tag{8.55}$$

so the horizontal defect is a fundamental vertical field determined by a $\bar{\mathfrak{g}}$-valued one-form on the base rather than by the fiber point. Consequently, for related sections satisfying $\Psi\circ s = \bar{s}\circ f$, the horizontal defect vanishes along the fine section, meaning $A_\Psi(s(c);X) = 0$ for every $c\in\mathcal{C}$ and $X\in T_c\mathcal{C}$, if and only if

$$\mathfrak{A}_{\mathcal{P}}(X) \in \bar{\mathfrak{g}}_{\bar{s}(f(c))} \quad \text{for every } c \text{ and every } X, \tag{8.56}$$

the isotropy subalgebra of the coarse section value under $\widehat{\rho}$. The principal-level identity $\mathfrak{A}_{\mathcal{P}} = 0$ is sufficient, and is necessary only when the represented action is infinitesimally effective at $\bar{s}(f(c))$. Moreover $\mathcal{Q}_\Psi = \mathfrak{a}^*\mathfrak{k}_{\bar{s}(f(c))}$ with $\mathfrak{k}_{\bar{\beta}}(\xi,\eta) = \bar{g}^F(\bar{\zeta}_\xi\bar{\beta}, \bar{\zeta}_\eta\bar{\beta}) \succeq 0$, whose radical is exactly $\bar{\mathfrak{g}}_{\bar{\beta}}$. [ESTABLISHED]

Proof. Evaluating $\mathcal{P}^*\bar{\omega}$ on the fundamental field of ξ gives $\bar{\omega}(\bar{\zeta}_{\mathrm{d}\kappa\xi}) = \mathrm{d}\kappa\xi$, so the difference annihilates vertical vectors; equivariance follows from $R_g^*\mathcal{P}^*\bar{\omega} = \mathrm{Ad}_{\kappa(g)^{-1}}\mathcal{P}^*\bar{\omega}$ together with $\mathrm{d}\kappa\circ\mathrm{Ad}_{g^{-1}} = \mathrm{Ad}_{\kappa(g)^{-1}}\circ\mathrm{d}\kappa$. In local trivializations write $\Psi(c,\beta) = (f(c), \psi_c\beta)$ with $\psi_c = \widehat{\rho}(\varsigma(c))\circ q$. Using $H^\omega_{(c,\beta)}X = (X, -\zeta_{A(X)}\beta)$ and the differentiated intertwining $Tq\circ\zeta_\xi = \bar{\zeta}_{\mathrm{d}\kappa(\xi)}\circ q$, obtained from (11.14) at $g=\exp(t\xi)$, gives $A_\Psi(X) = \bar{\zeta}_{\mathfrak{a}(X)}(\psi_c\beta)$ with $\mathfrak{a} = \mathrm{Ad}_\varsigma(u^*\mathfrak{A}_{\mathcal{P}})$, which is (8.55). A fundamental field vanishes at $\bar{\beta}$ exactly when its generator lies in $\bar{\mathfrak{g}}_{\bar{\beta}}$, and conjugation by Ad_ς moves both the generator and the isotropy algebra together, which gives (8.56). The last statement is (8.55) substituted into (8.51). □

Equation (8.56) is the definition of connection compatibility used throughout this manuscript. No weaker or informal reading is intended, and no result below relies on a compatibility notion that has not been reduced to it. [DEFINITION]

Theorem 8.19 (Ordered composition of horizontal defects). For composable morphisms $E_0 \xrightarrow{\Psi_{01}} E_1 \xrightarrow{\Psi_{12}} E_2$ over f_{01}, f_{12}, with $\Psi_{02} = \Psi_{12}\circ\Psi_{01}$ and $f_{02} = f_{12}\circ f_{01}$,

$$A_{\Psi_{02}}(e;X) = T^V\Psi_{12}\big|_{\Psi_{01}(e)}\left(A_{\Psi_{01}}(e;X)\right) + A_{\Psi_{12}}\left(\Psi_{01}(e); T_c f_{01}X\right). \tag{8.57}$$

At the connection level this reads $\mathfrak{A}_{\mathcal{P}_{02}} = \mathcal{P}_{01}^*\mathfrak{A}_{\mathcal{P}_{12}} + \mathrm{d}\kappa_{12}\circ\mathfrak{A}_{\mathcal{P}_{01}}$. [ESTABLISHED]

Proof. Expand $T\Psi_{01}(H^{\omega_0}_e X) = H^{\omega_1}_{\Psi_{01}e}(Tf_{01}X) + A_{\Psi_{01}}(e;X)$ and apply $T\Psi_{12}$. The horizontal summand contributes $H^{\omega_2}_{\Psi_{02}e}(Tf_{02}X) + A_{\Psi_{12}}(\Psi_{01}e; Tf_{01}X)$ by (8.45) at the second stage, and the vertical summand contributes $T^V\Psi_{12}(A_{\Psi_{01}}(e;X))$ because a bundle morphism preserves verticality. Subtracting $H^{\omega_2}(Tf_{02}X)$ gives (8.57); the connection-level form follows by $T^V\Psi_{12}\circ\vartheta = \vartheta\circ\mathrm{d}\kappa_{12}$. □

The order in (8.57) is not cosmetic. The earlier defect takes values in $V\bar{E}_1$ and the later one in VE_2, so writing $A_{\Psi_{02}} = A_{\Psi_{01}} + A_{\Psi_{12}}$ is a type error before any question of correctness arises: the earlier defect must be pushed by the *later* vertical differential, and the later defect must be evaluated at the image point on the *pushed* base vector. Correspondingly, $A_{\Psi_{02}}(e;X)=0$ holds if and only if

$$T^V\Psi_{12}\big|_{\Psi_{01}(e)}A_{\Psi_{01}}(e;X)=-A_{\Psi_{12}}\left(\Psi_{01}(e);T_cf_{01}X\right). \tag{8.58}$$

Factorwise vanishing of $A_{\Psi_{01}}$ on $T\mathcal{C}_0$ and of $A_{\Psi_{12}}$ on $Tf_{01}(T\mathcal{C}_0)$ at the image points is sufficient, but it is not necessary because the two typed terms can cancel after the later vertical differential. On the abelian three-level instance $f_{01}(x)=2x$, $f_{12}(y)=3y$ with local connection forms 0, a_1dy, and a_2dz and identity fiber maps, the composition law reads $2a_1+2(3a_2-a_1)=6a_2$ identically in (a_1,a_2); thus $a_2=0$ and $a_1\neq 0$ give a vanishing composite defect while both factor defects are nonzero and cancel. Dropping the base pushforward gives a_1+3a_2, and dropping the vertical differential fails likewise. [ESTABLISHED]

$$\begin{tikzcd}
T_c\mathcal{C} \arrow[r, "D^\omega s"] \arrow[d, "T_cf"'] & V_{s(c)}E \arrow[d, "T^V\Psi"] \\
T_{f(c)}\bar{\mathcal{C}} \arrow[r, "D^{\bar\omega}\bar s"'] & V_{\bar s(f(c))}\bar E
\end{tikzcd}$$

$\mathcal{D}\Psi=0$: the first-jet square commutes

Figure 8.1: Naturality of the connection-split first jet. The square commutes exactly under the isotropy criterion (8.56). When the horizontal defect is nonzero, (8.47) records the missing vertical term and (8.52) records the two retained base tensors $-\mathcal{X}_\Psi$ and $-\mathcal{Q}_\Psi$ rather than suppressing them.

Corollary 8.20 (Null directions under a parallel coarse morphism). If $\mathcal{D}\Psi=0$ along s, then

$$Tf(\ker D^\omega s)\subseteq \ker D^{\bar\omega}\bar s. \tag{8.59}$$

Thus a coarse morphism can create additional null directions; first-jet naturality does not imply rank preservation. [ESTABLISHED]

Proof. Set $D^\omega sX=0$ in (8.47) and use $\mathcal{D}\Psi=0$. □

8.9 Fisher defects and their composition law

Suppose now that the fiber map underlying Ψ is the pushforward of a parameter-independent Markov kernel N between the two regular statistical experiments, normalized in the strong sense that $N(x,\bar{\mathsf{K}})=1$ for every x rather than for almost every x, and that it is equivariant under the represented gauge actions. Three further hypotheses are consumed here and none of them follows from affinity of the induced law map on the cone of measures. First, *family closure*: the induced law map $N_\star$ must satisfy $N_\star(\mathcal{B})\subseteq\bar{\mathcal{B}}$, so that $q:=N_\star|_{\mathcal{B}}$ is defined at all. Second, *smoothness of* q between the two declared parametrized-measure models, together with a jointly measurable and parameter-smooth version selection for $p \mapsto T_p^V \Psi$. Third, Hypothesis 8.1 is instantiated at the coarse scale as well as the fine one; its clause that the represented action is induced by a parameter-independent bimeasurable sample re-coordinatization preserving the model is exactly the $\widehat{\bar{\rho}}(\bar{G})$-invariance of $\bar{g}^F$, and that invariance is what cancels the context-dependent factor $\widehat{\bar{\rho}}(\varsigma(c))$ in the local representative $\psi_c = \widehat{\bar{\rho}}(\varsigma(c)) \circ q$ of Ψ. Only in a frame pair of the form $(u(c), \mathcal{P}(u(c)))$ is $T^V \Psi$ literally Tq; in an arbitrary frame pair the identification requires that cancellation. Finally, for the DQM transfer used below, assume a common σ-finite dominating measure μ for the selected fine family, with a fixed density version p_θ in the parameter neighborhood under discussion. This common-domination tier is sufficient for the argument below; no broader nondominated transfer theorem is asserted here. Jointly measurable, parameter-smooth conditional-score versions needed for pointwise bundle statements remain separate hypotheses. [HYPOTHESIS]

The coarse Fisher metric $\bar{g}^F$ can then be pulled back by $T^V \Psi$. Define the vertical Fisher defect by

$$\Delta_F^\Psi = g^F - (T^V \Psi)^* \bar{g}^F. \tag{8.60}$$

[DEFINITION]

Theorem 8.21 (Fisher defect pullback and contraction). Assume the preceding Markov hypotheses and $\mathcal{D}\Psi = 0$ along s. Then

$$\Delta_F^\Psi \succeq 0, \tag{8.61}$$
$$h_s^\omega - f^* h_{\bar{s}}^{\bar{\omega}} = (D^\omega s)^* \Delta_F^\Psi \succeq 0. \tag{8.62}$$

Equality on a base tangent vector X holds exactly when the fine score in the direction $D^\omega s X$ is measurable with respect to the coarse output. [ESTABLISHED]

Proof. Work in a frame pair $(u(c), \mathcal{P}(u(c)))$, where $T^V \Psi = Tq$. The step that carries an arbitrary quadratic-mean path through the channel is not a consequence of parameter independence by itself. Let $J_\theta(dx, dy) = P_\theta(dx) N(x, dy)$ and $\nu(dx, dy) = \mu(dx) N(x, dy)$. Then $\mathrm{d}J_\theta / \mathrm{d}\nu = p_\theta(x)$, so the fine DQM remainder has exactly the same squared $L^2(\nu)$ norm in the joint experiment. Apply Pollard's preservation theorem for measurable statistics [65] to the deterministic statistic $(x, y) \mapsto y$: the pushed family $P_\theta N$ is DQM with directional score $\bar{\ell}_w(y) = \mathbb{E}_{J_\theta}[\ell_w(X) \mid Y = y]$. The exact source and hypothesis map are recorded in the Task 10 DQM-transfer source map. Squared Hellinger contraction is compatible corroboration, but it alone does not establish this score identification. Given the joint-experiment reduction and Pollard's theorem, the law of total variance gives

$$\Delta_F^\Psi(u, u) = \mathbb{E}\,\mathrm{Var}(\ell_u \mid Y) \geq 0, \tag{8.63}$$

with equality exactly when ℓ_u is Y-measurable. Under the vanishing horizontal defect, (8.48) applied to $\bar{g}^F$ gives

$$f^* h_{\bar{s}}^{\bar{\omega}} = (D^\omega s)^* (T^V \Psi)^* \bar{g}^F.$$

Subtracting this identity from (8.4) proves (8.62). □

The guard $\mathcal{D}\Psi = 0$ in Theorem 8.21 may not be dropped silently, and what survives when it is dropped is stated exactly rather than gestured at. Without the guard the vertical conclusion (8.61) is unchanged, while the base conclusion (8.62) is replaced by the exact signed identity (8.52) and its criterion (8.53); positivity then holds on exactly the directions satisfying that criterion, and fails on the others. [ESTABLISHED]

Corollary 8.22 (Coarse or meta-agent perceived geometry). The coarse section $\bar{s}$ carries its own connection-relative semimetric

$$\bar{h}_{\bar{s}}^{\bar{\omega}} = (D^{\bar{\omega}}\bar{s})^*\bar{g}^F. \tag{8.64}$$

Under the exact section relation (8.44) and the compatibility condition $\mathcal{D}\Psi = 0$ along s,

$$f^*\bar{h}_{\bar{s}}^{\bar{\omega}} = (D^{\omega}s)^*(T^V\Psi)^*\bar{g}^F. \tag{8.65}$$

If the fiber map is also a genuine parameter-independent Markov channel, then the fine-minus-coarse perceived-geometry defect is

$$h_s^{\omega} - f^*\bar{h}_{\bar{s}}^{\bar{\omega}} = (D^{\omega}s)^*\Delta_F^{\Psi} \succeq 0. \tag{8.66}$$

[ESTABLISHED]

Proof. The first equation is the coarse-channel definition. The second is (8.48) with $\bar{\tau} = \bar{g}^F$, and the third is (8.62). □

The inequality in (8.62) is contravariant: the coarse tensor is pulled back to the fine tangent space. No canonical pushforward of a Riemannian metric is being used. If the coarse operation is a restriction along an embedding, an energy precomposition, a Galerkin map, or a parameter-dependent kernel, the Markov proof does not apply. [ESTABLISHED]

A perceived-geometry beta is not defined by subtracting tensors that live on different level bases. If the reference-space declarations of Chapter 11 include base diffeomorphisms $i_\ell : \mathcal{C}_\ell \to \mathcal{C}_\star$ and a block factor $b_\ell > 1$, then one may first define

$$\widetilde{h}_\ell = (i_\ell^{-1})^*h_\ell, \qquad \mathfrak{B}_{h,\ell} = \frac{\widetilde{h}_{\ell+1} - \widetilde{h}_\ell}{\log b_\ell} \quad \text{on } \mathcal{C}_\star. \tag{8.67}$$

This is the exact discrete perceived-geometry beta in that scheme. On a differentiable reference-space curve $s \mapsto \widetilde{h}_s$, its continuous counterpart is $\beta_h(s) = \partial_s\widetilde{h}_s$. Both definitions are conditional on the declared identifications and scale normalization. [DEFINITION]

Without those identifications, $h_{\ell+1} - h_\ell$ is ill typed rather than a beta function. [ESTABLISHED]

Theorem 8.23 (Vertical Fisher defect cocycle). Let $E_0 \xrightarrow{\Psi_{01}} E_1 \xrightarrow{\Psi_{12}} E_2$ be smooth bundle morphisms over base maps $f_{01} : \mathcal{C}_0 \to \mathcal{C}_1$ and $f_{12} : \mathcal{C}_1 \to \mathcal{C}_2$, carrying principal connections

$\omega_0, \omega_1, \omega_2$ and vertical Fisher metrics g_0^F, g_1^F, g_2^F. Whenever all three defects are defined,

$$\Delta_F^{\Psi_{12}\circ\Psi_{01}} = \Delta_F^{\Psi_{01}} + (T^V\Psi_{01})^*\Delta_F^{\Psi_{12}}. \tag{8.68}$$

This identity is unconditional: it uses neither the connections nor any section. If both arrows are parameter-independent Markov coarse maps, the two terms on the right are positive semidefinite. [ESTABLISHED]

Proof. Contravariance of tensor pullback gives

$$\begin{aligned}\Delta_F^{\Psi_{12}\circ\Psi_{01}} &= g_0^F - (T^V\Psi_{01})^*(T^V\Psi_{12})^*g_2^F \\ &= g_0^F - (T^V\Psi_{01})^*g_1^F + (T^V\Psi_{01})^*\left[g_1^F - (T^V\Psi_{12})^*g_2^F\right],\end{aligned}$$

which is (8.68). Positivity follows from (8.61) for each Markov arrow. □

Pulling (8.68) back to the base does *not* give an additive identity in general, and the exact residual is available in closed form. Fix sections s_0, s_1, s_2 related at each stage, write $\delta_{jk} = (D^{\omega_j}s_j)^*\Delta_F^{\Psi_{jk}}$ for the three base defects, and along s_0 write $v_X = T^V\Psi_{01}(D^{\omega_0}s_0X)$, $A_X = A_{\Psi_{01}}(s_0;X)$, and $\bar{u}_X = D^{\omega_1}s_1(Tf_{01}X) = v_X + A_X$.

Theorem 8.24 (Base Fisher defect cocycle: exact residual and sharp criterion). The residual $\mathcal{N} := \delta_{02} - \delta_{01} - f_{01}^*\delta_{12}$ has the three equivalent exact forms

$$\mathcal{N}(X,Y) = \Delta_F^{\Psi_{12}}(v_X, v_Y) - \Delta_F^{\Psi_{12}}(\bar{u}_X, \bar{u}_Y), \tag{8.69}$$

$$= -\left[\Delta_F^{\Psi_{12}}(\bar{u}_X, A_Y) + \Delta_F^{\Psi_{12}}(A_X, \bar{u}_Y)\right] + \Delta_F^{\Psi_{12}}(A_X, A_Y), \tag{8.70}$$

$$= -\left[\Delta_F^{\Psi_{12}}(v_X, A_Y) + \Delta_F^{\Psi_{12}}(A_X, v_Y)\right] - \Delta_F^{\Psi_{12}}(A_X, A_Y). \tag{8.71}$$

The sign of the quadratic term is coupled to the choice of argument in the cross terms: the coarse jet $\bar{u}$ carries $+\Delta_F^{\Psi_{12}}(A,A)$ and the pushed fine jet v carries $-\Delta_F^{\Psi_{12}}(A,A)$. Mixing the two conventions inflates the residual by exactly $2\Delta_F^{\Psi_{12}}(A_X, A_Y)$, which is two copies of the second-arrow vertical Fisher defect evaluated on the first-arrow horizontal defect. Consequently the sharp base cocycle $\delta_{02} = \delta_{01} + f_{01}^*\delta_{12}$ holds *if and only if*

$$\Delta_F^{\Psi_{12}}(v_X, v_X) = \Delta_F^{\Psi_{12}}(\bar{u}_X, \bar{u}_X) \quad \text{for every } X, \tag{8.72}$$

equivalently $\Delta_F^{\Psi_{12}}(A_X, 2v_X + A_X) = 0$ for every X. If the second arrow is a Markov coarse map, so that $\Delta_F^{\Psi_{12}} \succeq 0$, this is equivalently equality of the two $\Delta_F^{\Psi_{12}}$-seminorms. [ESTABLISHED]

Proof. Apply $(D^{\omega_0}s_0)^*$ to (8.68) to get $\delta_{02} = \delta_{01} + \Delta_F^{\Psi_{12}}(v_X, v_Y)$, and note that by definition $f_{01}^*\delta_{12}(X,Y) = \Delta_F^{\Psi_{12}}(\bar{u}_X, \bar{u}_Y)$. Subtracting gives (8.69). Substituting $\bar{u} = v + A$ in the second term gives (8.71), and substituting $v = \bar{u} - A$ in the first gives (8.70). Since $\mathcal{N}$ is symmetric, vanishing on the diagonal is vanishing, and (8.69) on the diagonal is exactly (8.72). □

Three conditions are sufficient for (8.72), in decreasing strength: the stage-one horizontal defect vanishes, which by Theorem 8.18 is the isotropy condition (8.56) at the first arrow; the second arrow is Fisher lossless, $\Delta_F^{\Psi_{12}} = 0$; or, weakest and new, $A_X \in \operatorname{rad} \Delta_F^{\Psi_{12}}$ for every X, so that the stage-one anomaly is invisible to the stage-two information loss. The third suffices because $\mathcal{N}$ is linear in the product $\Delta_F^{\Psi_{12}} A$. None of the three is necessary, since (8.72) is an equality of quadratic forms and admits accidental solutions: on the three-level normal location instance with fibers of variance $1, 2, 3$, sections $x \mapsto x$, $y \mapsto y/2$, $z \mapsto z/6$, base maps $f_{01}(x) = 2x$ and $f_{12}(y) = 3y$, and local connection forms 0, $a_1 dy$, $a_2 dz$, the residual is $\mathcal{N} = -\frac{2}{3} a_1(a_1 + 1)$, which vanishes at $a_1 = -1$ where the stage-one horizontal defect equals $-2 \neq 0$. On the same data the convention-mixed expression evaluates to $\frac{2}{3} a_1(a_1 - 1)$, exceeding $\mathcal{N}$ by $\frac{4}{3} a_1^2 = 2\Delta_F^{\Psi_{12}}(A, A)$, and at $a_1 = \frac{1}{10}$ the two numbers are $-\frac{11}{150}$ and $-\frac{3}{50}$. [ESTABLISHED]

Two distinct hypotheses are involved here and conflating them is the reading this chapter forbids. The sharp base cocycle needs only the stage-one condition (8.72). Reading the middle term δ_{12} as the stage-two information loss $h_1 - f_{12}^* h_2$ requires in addition that the stage-two horizontal defect vanish, by Theorem 8.16 applied at the second arrow. Only when both hold does (8.68) become an additive decomposition of information loss on the base. [ESTABLISHED]

The Amari defect

$$\Delta_{\mathcal{T}}^{\Psi} = \mathcal{T} - (T^V \Psi)^* \bar{\mathcal{T}}, \tag{8.73}$$

obeys the same algebraic cocycle identity. It has no analogous positive-semidefinite order because a nonzero cubic changes sign under $u \mapsto -u$. Exact Amari naturality therefore requires a $\mathcal{T}$-preserving statistical morphism; Markov coarse-graining alone does not supply a monotonicity statement for the cubic tensor. [ESTABLISHED]

8.10 Boundary of the construction

The chapter has constructed vertical tensors and their functorial defects. A scalar gauged sigma energy would additionally require a base cometric, a base density, channel weights, boundary conditions, and a decision about whether the connection is fixed or dynamical. None is selected by h_s^ω or c_s^ω. [NOT-CLAIMED]

Likewise, vertical Fisher length is invariant under reparameterization of a fixed regular curve, but it does not generate an orbit, identify a physical time coordinate, or compare independently evolved fine and coarse paths. Comparing such paths would require a semiconjugacy of their vector fields in addition to the pointwise contraction (8.62). That interface is developed in Chapter 9, where the semiconjugacy criterion, its sufficient natural-gradient conditions, and the exact duration relation are proved on a declared configuration tier; what remains open for the specific renormalization maps of this work is the obligation recorded at Proposition 9.28, namely exhibiting the functional compatibility and horizontal conformality of the declared coarse configuration map. [OPEN]

The scale index carried by (8.68) and by Theorem 8.24 is renormalization depth. It is not an inference-orbit coordinate and not a duration, and no result of this chapter converts it into either. [NOT-CLAIMED]

The general coarse-graining chapter uses the typed distinction exposed here. Its normalized Markov channels inherit the Fisher contraction theorem. Its energy precompositions and operator restrictions require their own probability and closure arguments and do not inherit that theorem by terminology. [ESTABLISHED]

Chapter 9

Relational Inference Histories and Fisher Information Clocks

The contextual base $\mathcal{C}$ is fixed data in the present theory. It organizes the domains on which agents are represented, but it carries no distinguished temporal parameter. Inference therefore has to be described as change in a statistical fiber or as change of a section, rather than as motion of the base. A curve in $\mathcal{C}$ orders contexts and can be used to probe a section; it is not, by that fact, an agent history. [HYPOTHESIS]

This chapter supplies two structures that the preceding pullback construction deliberately left open. A declared VFE gradient ray selects an oriented unparameterized inference orbit, and the Fisher line element assigns length to that already selected orbit. The resulting information clock is agent-relative and invariant under orientation-preserving reparameterization. It is neither a physical time coordinate nor a scalar that automatically exists on the whole configuration space. [DEFINITION]

9.1 Fixed-base kinematics

For agent i, restrict the belief and model bundles to its contextual domain $\mathcal{C}_i$ and form

$$\mathcal{E}_i = \mathcal{E}_b|_{\mathcal{C}_i} \times_{\mathcal{C}_i} \mathcal{E}_m|_{\mathcal{C}_i}, \qquad \varpi_i : \mathcal{E}_i \longrightarrow \mathcal{C}_i. \tag{9.1}$$

The pair $\boldsymbol{\omega}_i = (\omega_b, \omega_m)$ supplies a product horizontal distribution and a vertical projection $\mathrm{ver}^{\boldsymbol{\omega}_i}$. Its vertical Fisher form is not fixed until positive channel weights or a joint statistical metric are declared. The common bundle projection supplies neither choice. [DEFINITION]

Definition 9.1 (Vertical, horizontal, and mixed curves). Let J be a connected interval and let $\Gamma : J \to \mathcal{E}_i$ be C^1, with $\gamma = \varpi_i \circ \Gamma$. Relative to $\boldsymbol{\omega}_i$, its velocity has the unique splitting

$$\dot{\Gamma} = H_\Gamma^{\boldsymbol{\omega}_i} \dot{\gamma} + v_{\boldsymbol{\omega}_i}(\Gamma), \qquad v_{\boldsymbol{\omega}_i}(\Gamma) = \mathrm{ver}^{\boldsymbol{\omega}_i}(\dot{\Gamma}). \tag{9.2}$$

The primary labels are pointwise. At a parameter value λ the velocity is *stationary* when $\dot{\Gamma}(\lambda) = 0$; *vertical* when $\dot{\gamma}(\lambda) = 0$ and $\dot{\Gamma}(\lambda) \neq 0$; $\boldsymbol{\omega}_i$*-horizontal* when $v_{\boldsymbol{\omega}_i}(\Gamma)(\lambda) = 0$ and $\dot{\gamma}(\lambda) \neq 0$; and *mixed* when both $\dot{\gamma}(\lambda)$ and $v_{\boldsymbol{\omega}_i}(\Gamma)(\lambda)$ are nonzero. These four cases partition every parameter value. The interval-level labels are then defined by quantification: the curve is vertical, horizontal, or mixed on J when it has that pointwise type at every $\lambda \in J$. The interval-level labels are therefore *not* exhaustive, since a curve may change pointwise type along J. Every use of an interval label below is on a subinterval of one fixed pointwise type, so no downstream statement is affected. Verticality is intrinsic to the bundle projection; horizontality and the mixed decomposition depend on the selected connection. [DEFINITION]

Two further curve types occur in this chapter and neither is a curve in $\mathcal{E}_i$. A base curve $\gamma \subset \mathcal{C}_i$ carries no verticality or horizontality predicate at all, since those predicates are statements about a total-space velocity. A curve of configurations, defined in Section 9.2, lives in a section space; verticality statements about it are statements about its adjoint evaluation at a fixed context, by (9.4), and not about the configuration curve itself. Applying a verticality predicate directly to a base curve or to a configuration curve is a type error. [DEFINITION]

The definition agrees with the standard connection splitting of a bundle tangent space [43]. Since $T\varpi_i \circ H^{\boldsymbol{\omega}_i}$ is the identity and $T\varpi_i \circ \mathrm{ver}^{\boldsymbol{\omega}_i} = 0$, applying $T\varpi_i$ to (9.2) recovers $\dot{\gamma}$. Thus a vertical curve stays in one fiber, whereas a horizontal lift is a connection-selected comparison of points in different fibers. Neither operation moves the fixed base itself. [ESTABLISHED]

Proposition 9.2 (Horizontality is connection relative). The same nonvertical total-space curve can be mixed for one connection and horizontal for another. [ESTABLISHED]

Proof. Take the trivial line bundle $\mathbb{R}_x \times \mathbb{R}_y \to \mathbb{R}_x$ and the curve $\Gamma(r) = (r, r)$. For the horizontal distribution $H_0 = \mathrm{span}\{\partial_x\}$, its vertical velocity is ∂_y, so the curve is mixed. For $H_1 = \mathrm{span}\{\partial_x + \partial_y\}$, the same velocity is horizontal. The base trace $x = r$ is unchanged. □

9.2 Curves of sections and pointwise vertical change

Hypothesis 9.3 (Regular section configuration). Let $\mathfrak{S}_i$ be a declared regular space of pairs of smooth sections $Q = (q, s)$ of $\mathcal{E}_i \to \mathcal{C}_i$. Whenever a tangent-space calculation is used, assume either a finite-dimensional configuration submanifold or a specified smooth locally convex manifold structure for which evaluation is differentiable. No such structure follows merely from writing down all smooth sections. [HYPOTHESIS]

For a C^1 representative $Q_i : I \to \mathfrak{S}_i$, define its adjoint family by

$$\widehat{\Sigma}_i : I \times \mathcal{C}_i \longrightarrow \mathcal{E}_i, \qquad \widehat{\Sigma}_i(\lambda, c) = Q_i(\lambda)(c), \qquad \varpi_i \circ \widehat{\Sigma}_i(\lambda, c) = c. \tag{9.3}$$

Differentiation in the history parameter at fixed c gives

$$T\varpi_i \left(\partial_\lambda \widehat{\Sigma}_i(\lambda, c) \right) = 0, \qquad \partial_\lambda \widehat{\Sigma}_i(\lambda, c) \in V_{Q_i(\lambda)(c)}\mathcal{E}_i. \tag{9.4}$$

Thus a curve of full sections produces a vertical statistical update at every fixed context, even though its state is an entire section. [ESTABLISHED]

At each history position the instantaneous sections also induce a pair of connection-relative perceived geometries on the same fixed base:

$$\mathbf{h}_i(\lambda) = \left(h^{\omega_b}_{q_i(\lambda)}, h^{\omega_m}_{s_i(\lambda)}\right). \tag{9.5}$$

Consequently a section history determines an ordered family of base semimetrics without making the base itself evolve. Under $\widetilde{Q}_i = Q_i \circ \phi$, this family changes only by $\widetilde{\mathbf{h}}_i = \mathbf{h}_i \circ \phi$. The connections are held fixed in this statement; allowing them to change would add a separately declared connection history and the correction terms of Proposition 8.5. [ESTABLISHED]

Definition 9.4 (Oriented unparameterized history). Two regular representatives $Q_i : I \to \mathfrak{S}_i$ and $\widetilde{Q}_i : \widetilde{I} \to \mathfrak{S}_i$ represent the same oriented history when

$$\widetilde{Q}_i = Q_i \circ \phi, \qquad \phi : \widetilde{I} \longrightarrow I \text{ is an orientation-preserving } C^1 \text{ diffeomorphism.} \tag{9.6}$$

The resulting abstract oriented curve is denoted by $\mathscr{H}_i$. Its adjoint evaluation has the required type

$$\Sigma_i : \mathscr{H}_i \times \mathcal{C}_i \longrightarrow \mathcal{E}_b \times_{\mathcal{C}} \mathcal{E}_m, \qquad \varpi_i \circ \Sigma_i(r, c) = c. \tag{9.7}$$

If a representative is an embedding, $\mathscr{H}_i$ may be identified with its oriented image. With self-intersections, $\mathscr{H}_i$ retains traversal order and multiplicity and is not merely the set-theoretic image. [DEFINITION]

The distinction between a base probe and a section history is exposed by a diagonal evaluation. Let $\gamma : I \to \mathcal{C}_i$ and set $\Gamma(\lambda) = Q_i(\lambda)(\gamma(\lambda))$. Write

$$D^{\boldsymbol{\omega}_i} Q_i(\lambda) = \left(D^{\omega_b} q_i(\lambda), D^{\omega_m} s_i(\lambda)\right). \tag{9.8}$$

The chain rule and the connection splitting give

$$v_{\boldsymbol{\omega}_i}(\Gamma) = \partial_\lambda \widehat{\Sigma}_i(\lambda, \gamma(\lambda)) + D^{\boldsymbol{\omega}_i} Q_i(\lambda) \dot{\gamma}(\lambda). \tag{9.9}$$

The first term is history change at a fixed context. The second is the vertical response of one instantaneous section to a changing contextual probe. If γ is constant, only the pointwise history velocity remains; if Q_i is constant, only the section jet remains. When γ is constant, the diagonal curve is vertical exactly when the remaining vertical velocity is nonzero and is stationary when it vanishes. On an open subinterval where $\dot{\gamma} \neq 0$, the diagonal curve is mixed exactly when the sum in (9.9) is nonzero, and it is horizontal when that sum vanishes. [ESTABLISHED]

A smooth section is injective because $\varpi_i \circ Q = \mathrm{id}_{\mathcal{C}_i}$, and it is an embedding under the usual smooth-bundle hypotheses. A total-space curve can therefore be written as $Q \circ \gamma$ exactly when it lies in the section image, in which case $\gamma = \varpi_i \circ \Gamma$ is unique. For a general statistical parameterization that is not a section, such descent is only local under a local embedding hypothesis and is globally unique only under global injectivity. In particular, a nonconstant curve confined to one statistical fiber cannot be the pushforward of a nonconstant base curve through a section. [ESTABLISHED]

9.3 VFE selection of an oriented orbit

Hypothesis 9.5 (Regular metric inference domain). Let $\mathcal{Q}_i \subseteq \mathfrak{S}_i$ be a regular configuration manifold equipped with a positive-definite Fisher Riemannian metric G_i^F. In an infinite-dimensional realization, assume it is a strong metric so that the musical map is invertible on the covectors used below. Let $\mathcal{F}_i : \mathcal{Q}_i \to \mathbb{R}$ be C^2. A quotient-metric domain may replace $\mathcal{Q}_i$ only after the quotient has been constructed. [HYPOTHESIS]

A declared hypothesis with no exhibited model is vacuous, so the following construction exhibits one. It assumes no configuration manifold and no strong metric; it builds both from the bundle data already declared.

Definition 9.6 (An exhibited finite-dimensional configuration tier). Declare at scale ℓ a compact smooth base $\mathcal{C}_\ell$ with a finite positive Borel measure μ_ℓ; a trivial principal bundle $P = \mathcal{C}_\ell \times G$ with $G = (\mathbb{R}^K, +)$ acting on the fiber by translation of the mean and ω the flat connection; the fiber $\mathcal{B} = \{\mathcal{N}(m, \Sigma_0) : m \in \mathbb{R}^K\}$ with fixed $\Sigma_0 \succ 0$, whose fiber Fisher form is the constant $g^F = \Sigma_0^{-1}$; a measurable weight $w : \mathcal{C}_\ell \to (0, \infty)$ bounded above and below; smooth fields $\phi_1, \dots, \phi_N : \mathcal{C}_\ell \to \mathbb{R}^K$ that are linearly independent in $L^2(w\mu_\ell; \Sigma_0^{-1})$ and whose span contains every constant field $c \mapsto g$ for $g \in \mathbb{R}^K$; and

$$\mathcal{Q}_\ell = \left\{ s_\xi : \xi \in \mathbb{R}^N \right\}, \qquad s_\xi(c) = \mathcal{N}\left(\sum_a \xi_a \phi_a(c), \Sigma_0 \right), \tag{9.10}$$

Let $\mathsf{L}_\ell : \mathbb{R}^K \to \mathbb{R}^N$ be the unique coefficient map satisfying $\sum_a (\mathsf{L}_\ell g)_a \phi_a(c) = g$ for every c. Constant mean translations preserve the selected span and induce the affine coefficient action

$$\Theta_g^\ell(\xi) = \xi + \mathsf{L}_\ell g. \tag{9.11}$$

Thus the gauge action, its coefficient action, and its orbit directions are declared on $\mathcal{Q}_\ell$, rather than inferred from an arbitrary basis. Its configuration metric is

$$\mathsf{G}_\ell(V, V) = \int_{\mathcal{C}_\ell} w(c) g^F \left(\partial_V s_\xi(c), \partial_V s_\xi(c) \right) \mathrm{d}\mu_\ell(c), \tag{9.12}$$

declared as a *weighted product of marginal fiber metrics* and not as the pullback of a joint-law Fisher metric. [DEFINITION]

Theorem 9.7 (The exhibited tier satisfies the regular metric hypothesis). Under Definition 9.6: the configuration space $\mathcal{Q}_\ell$ is a nonempty finite-dimensional smooth manifold diffeomorphic to $\mathbb{R}^N$, with $\xi = 0$ the constant configuration $c \mapsto \mathcal{N}(0, \Sigma_0)$; the metric (9.12) is the constant Gram form

$$\mathsf{G}_\ell = \Phi, \qquad \Phi_{ab} = \int_{\mathcal{C}_\ell} \phi_a(c)^\top \Sigma_0^{-1} \phi_b(c) w(c) \mathrm{d}\mu_\ell(c), \tag{9.13}$$

independent of ξ, smooth, finite, and positive definite; any Riemannian metric on a finite-dimensional manifold is automatically strong; every C^1 functional has a unique gradient $\Phi^{-1}\nabla\mathcal{F}$, and for $\mathcal{F} \in C^2$

the field $X_\ell = -\Phi^{-1}\nabla\mathcal{F}$ is locally Lipschitz with locally unique integral curves; and the gauge orbit tangent generated by constant translations of the mean is $\mathsf{L}_\ell(\mathbb{R}^K)$, a linear subspace of $\mathbb{R}^N$, hence closed and complemented, so that the infimum in (9.38) is attained and equals the Φ-orthogonal projection. [ESTABLISHED]

Proof. The bundle is trivial and s_ξ is smooth jointly in (c,ξ). If $s_\xi = s_\eta$, then $\sum_a(\xi_a - \eta_a)\phi_a = 0$ in $L^2(w\mu_\ell;\Sigma_0^{-1})$, so the declared independence gives $\xi = \eta$; hence the selected section family is diffeomorphic to its coefficient space $\mathbb{R}^N$. Since $\partial_V s_\xi(c) = \sum_a v_a\phi_a(c)$ does not depend on ξ, the integrand of (9.12) is the quadratic form $v^\top\Phi v$ with Φ as displayed; finiteness is compactness of the base together with boundedness of w. The form Φ is positive definite: $v^\top\Phi v = 0$ forces $\sum_a v_a\phi_a = 0$ in $L^2(w\mu_\ell;\Sigma_0^{-1})$, and the fields are declared independent. Strongness in finite dimensions is the statement that every inner product on a finite-dimensional space induces the norm topology and that its musical map is a linear isomorphism. The gradient statement is invertibility of Φ. Finally, the span condition defines L_ℓ and gives $s_{\Theta_g^\ell(\xi)} = g\cdot s_\xi$; its finite-dimensional image is closed and complemented. □

Nondegeneracy is a rank condition that can be checked before any dynamics is written. With an M-atom design $\mu_\ell = \sum_{a=1}^M \rho_a\delta_{c_a}$ the Gram matrix has rank at most MK, so whenever $N > MK$ the metric is degenerate and, by Proposition 9.11, the natural-gradient equation has no solution or many. A worked instance is $\mathcal{C}_\ell = S^1$ with normalized arclength, $K = 1$, $\Sigma_0 = 1$, $w \equiv 1$, and $\{\phi_a\} = \{1, \cos\theta, \sin\theta\}$, for which $\Phi = \mathrm{diag}(1, \frac{1}{2}, \frac{1}{2})$ has determinant $\frac{1}{4}$ and eigenvalues $\{1, \frac{1}{2}\}$, so $\mathcal{Q}_\ell \cong \mathbb{R}^3$ carries a positive definite constant Gram metric. [ESTABLISHED]

The infinite-dimensional case has exactly two options and a boundary between them; the boundary is retained rather than removed. On $\mathcal{Q}^{L^2} = L^2(\mu;\mathbb{R}^K)$ with $\mathsf{G}^{L^2}(V,W) = \int V^\top\Sigma_0^{-1}W w\,\mathrm{d}\mu$, two-sided bounds $0 < w_- \le w \le w_+ < \infty$ and $\lambda_- I \preceq \Sigma_0^{-1} \preceq \lambda_+ I$ give $w_-\lambda_-\|V\|_{L^2}^2 \le \mathsf{G}^{L^2}(V,V) \le w_+\lambda_+\|V\|_{L^2}^2$, so the musical map is a topological isomorphism and the metric is strong; the price is that $\mathcal{F}$ must be C^1 on L^2, which excludes gradient-energy objectives. Either bound failing alone destroys the conclusion. On a Sobolev tier H^s with the same integrated metric the topology of G^{L^2} is strictly coarser, the musical map is injective and bounded but not surjective, and the metric is weak. The failure is realized: on $\mathcal{C} = S^1$ with $\mathcal{Q} = H^1(S^1)$ and $\mathcal{F}(Q) = \frac{1}{2}\int|Q'|^2$, the configuration $Q = \sum_{k\ge1} k^{-2}\sin k\theta$ has $\|Q\|_{H^1}^2$ finite while a gradient would be $-Q'' = \sum_k \sin k\theta$ with infinite L^2 norm, so no natural-gradient vector field exists at Q. The missing ingredient is Riesz representability of $\mathrm{d}\mathcal{F}_Q$, ensured for instance by $|\mathrm{d}\mathcal{F}_Q[V]| \le C_Q\|V\|_{L^2}$ on a dense domain. No third option is available: either the L^2 tier with a strong metric and an objective that is C^1 there, or a Sobolev tier with the Riesz hypothesis declared and this witness attached. [ESTABLISHED]

Hypothesis 9.8 (Standing configuration tier). Wherever this manuscript asserts a natural-gradient vector field or a Fisher duration on a configuration manifold, the manifold and its metric are those of Definition 9.6, or those of the L^2 tier above with its two-sided bounds and a C^1-on-L^2 objective. This is a restriction adopted by choice. It excludes the space of all smooth sections with no declared manifold topology, every weak-metric realization in which the musical map is not surjective, and every configuration metric labeled as an exact joint-law Fisher metric without the block-orthogonality or independence condition of (9.19). It is used at Sections 9.3, 9.4 and 9.8 to 9.10 and at every cross-scale duration comparison. Its model class is nonempty by Theorem 9.7, so nothing downstream is vacuous. Whether the recognition families a particular application declares can be brought to one of these two tiers is a separate modeling obligation and is not settled here. [HYPOTHESIS]

Hypothesis 9.9 (Exact multi-agent VFE lift). Calling the configuration objective an exact local or collective VFE requires more than its pair of marginal sections. Fix a block B, an outside marginal Q_{B^c}, and a declared manifold $\mathfrak{R}_B$ of admissible conditional recognition kernels. There must be a declared smooth marginalization or configuration-extraction map

$$\pi_i^{\mathrm{conf}} : \mathfrak{R}_B \longrightarrow \mathcal{Q}_i \tag{9.14}$$

on the admissible family. Supply a smooth right-inverse lift

$$\iota_i : \mathcal{Q}_i \longrightarrow \mathfrak{R}_B, \qquad Q_i \longmapsto r_B^{Q_i}(\cdot \mid b), \qquad \pi_i^{\mathrm{conf}} \circ \iota_i = \mathrm{id}_{\mathcal{Q}_i}, \tag{9.15}$$

so the lifted conditional law actually reproduces the displayed agent configuration. Equivalently, the full recognition law $Q_{B^c}(db) r_B^{Q_i}(dy_B \mid b)$ must have the declared marginals or section statistics encoded by Q_i. Without this commuting condition, ι_i is merely a parameterized family of laws, not a lift of the agent configuration.

For every Q_i in the inference domain, impose the support and finiteness hypotheses of Theorems 7.4 and 7.5: $Q_{B^c} \ll \Pi_{o,B^c}$, $Z_B \in (0, +\infty)$ Q_{B^c}-almost everywhere, $r_B^{Q_i}(\cdot \mid b) \ll P_{0,B}(\cdot \mid b)$, and integrability of the KL terms and positive and negative energy parts, so the displayed local, collective, and difference functionals are finite. Assume also that the lifted family is C^2 and satisfies the following chartwise domination condition. For every chart $\kappa : V \to \mathbb{R}^d$ of $\mathcal{Q}_i$ and every multi-index α with $|\alpha| \leq 2$, require a jointly measurable version in (ξ, b) of $\partial_\xi^\alpha \mathcal{F}_{B,o}(r_B^{\kappa^{-1}(\xi)}; b)$. For every compact $K \subset \kappa(V)$, there is a $G_K \in L^1(Q_{B^c})$ such that, for every multi-index α with $|\alpha| \leq 2$,

$$\sup_{\xi \in K} \left| \partial_\xi^\alpha \mathcal{F}_{B,o} \left(r_B^{\kappa^{-1}(\xi)}; b \right) \right| \leq G_K(b) \quad \text{for } Q_{B^c}\text{-almost every } b. \tag{9.16}$$

The case $|\alpha| = 0$ controls the objective integrand, and the cases $|\alpha| = 1, 2$ justify two differentiations under the Q_{B^c} expectation. Then define the outside-averaged objective

$$\overline{\mathcal{F}}_{B,o}(Q_i) = \mathbb{E}_{Q_{B^c}} \left[\mathcal{F}_{B,o}(r_B^{Q_i}; Y_{B^c}) \right]. \tag{9.17}$$

The lift is responsible for all correlation or copula data not determined by the displayed marginal sections. A fixed product or fixed-copula family is a special hypothesis that can supply it; the pair (q, s) alone cannot. Alternatively, a population configuration may be lifted directly to a full recognition law and the collective VFE pulled back along that lift. If G_i^F is called the exact recognition Fisher metric, also require

$$\mathsf{G}_i^F = \iota_i^* G_{\mathfrak{R}_B}^F \tag{9.18}$$

on a nondegenerate tier, or use its justified metric quotient. A weighted product of marginal-fiber metrics equals this joint pullback only under a separately proved block-orthogonality or fixed-dependence hypothesis. [HYPOTHESIS]

The criterion behind that last sentence is exact and worth stating, because it also shows that no order relation is available in either direction. Let L be the joint score in a configuration direction and let $L_b = \Pi_b L$ and $L_m = \Pi_m L$ be its orthogonal projections onto the centered $\sigma(Y_b)$- and $\sigma(Y_m)$-measurable subspaces, marginalization being a deterministic parameter-independent Markov kernel. Then

$$\|L\|^2 - \left(\|L_b\|^2 + \|L_m\|^2\right) = \|L - L_b - L_m\|^2 - 2\langle L_b, L_m\rangle, \tag{9.19}$$

whose first term is nonnegative and whose second is signed. Equality at unit weights therefore holds exactly when $\|L - L_b - L_m\|^2 = 2\langle L_b, L_m\rangle$; independence of the two channels is the clean sufficient structural condition, and under independence with independently excitable nondegenerate marginals the weights are forced to $w_b = w_m = 1$. A fixed non-independence copula does not suffice, since the derivative of its log density along the configuration parameter is generally nonzero. No Loewner ordering holds either way: with jointly Gaussian precision $\Lambda = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ the difference $\Lambda - (1-\rho^2)I_2 = \left(\begin{smallmatrix} \rho^2 & \rho \\ \rho & \rho^2 \end{smallmatrix}\right)$ has determinant $\rho^2(\rho^2-1) < 0$, taking the value $2\rho(1+\rho)$ along $(1,1)$ and $2\rho(\rho-1)$ along $(1,-1)$. Finally, distinct smooth right inverses of the same extraction map π_i^{conf} can, but need not, induce different pullback metrics. For the two-parameter normal-location family with Fisher metric $d\theta^2 + d\eta^2$ and extraction $\pi(\theta,\eta) = \theta$, the distinct lifts $\iota_0(x) = (x, 0)$ and $\iota_c(x) = (x, c)$, for $c \neq 0$, both induce dx^2, whereas $\iota_\Delta(x) = (x, x)$ induces $2dx^2$. Thus the configuration Fisher metric is not in general determined by the displayed configuration alone; a selected lift must be supplied unless a separately proved canonicity or lift-independence theorem removes that choice. [ESTABLISHED]

By Theorem 7.5, $\overline{\mathcal{F}}_{B,o}$ differs from the collective VFE restricted along $Q_i \mapsto Q_{B^c} r_B^{Q_i}$ only by a Q_i-independent outside term. Their differentials and natural-gradient rays therefore agree when the same block metric is used. A single realized conditional objective $\mathcal{F}_{B,o}(r_B; b)$ need not have that gradient, and independently moving outside marginals do not define the same fixed objective. [ESTABLISHED]

The Fisher natural gradient is characterized by

$$\mathsf{G}_i^F\left(\mathrm{grad}^F \mathcal{F}_i, Z\right) = d\mathcal{F}_i[Z] \quad \text{for every } Z \in T\mathcal{Q}_i. \tag{9.20}$$

At a noncritical configuration Q, define the positive descent ray

$$\mathcal{R}^-_{\mathcal{F}_i}(Q) = \left\{-a\,\mathrm{grad}^F \mathcal{F}_i(Q) : a > 0\right\}. \tag{9.21}$$

A natural-gradient history for the declared objective is an oriented unparameterized history whose regular representatives satisfy

$$\dot{Q}_i(\lambda) = -a_i(\lambda)\,\mathrm{grad}^F \mathcal{F}_i(Q_i(\lambda)), \qquad a_i(\lambda) > 0. \tag{9.22}$$

The objective alone does not select this ray: the metric, or an explicitly declared mobility replacing it, is load-bearing structure. When $\mathcal{F}_i$ is the pullback specified in Hypothesis 9.9, this is an exact VFE history; otherwise it is a history for a generic declared objective and no exact-VFE identification is asserted. [DEFINITION]

Proposition 9.10 (Scalar mobility preserves the oriented orbit). On a noncritical domain, multiplication of the natural-gradient vector field by any smooth positive scalar mobility changes only its parameterization. A positive-definite anisotropic mobility can change the unparameterized orbit. [ESTABLISHED]

Proof. If $X = -\operatorname{grad}^F \mathcal{F}_i$ and $X_a = aX$ with $a > 0$, then the integral curves have the same tangent line and orientation. Along an X-integral curve, the change of parameter defined by a positive integral of a^{-1} converts one differential equation into the other. For the second claim, use the Euclidean plane with

$$\mathcal{F}(x,y) = \frac{x^2+y^2}{2}, \qquad M = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}. \tag{9.23}$$

Ordinary gradient descent from (x_0, y_0) satisfies $y = (y_0/x_0)x$, whereas $(\dot{x}, \dot{y})^\top = -M\nabla\mathcal{F}$ satisfies

$$y = y_0 \left(\frac{x}{x_0}\right)^2. \tag{9.24}$$

For generic nonzero initial data these are different images. □

A VFE natural-gradient history is not generally a Fisher geodesic. On the same Euclidean Fisher plane, choose

$$\mathcal{F}(x,y) = \frac{x^2+2y^2}{2}. \tag{9.25}$$

Its natural-gradient flow obeys $\dot{x} = -x$ and $\dot{y} = -2y$, hence follows the curved trajectory in (9.24). Euclidean geodesics are straight lines, so gradient selection and geodesic selection are distinct even in this elementary Fisher geometry. [ESTABLISHED]

Proposition 9.11 (A semidefinite tensor cannot be silently inverted). If the proposed Fisher tensor on configuration space is semidefinite, the natural-gradient equation can fail to have a solution or can have nonunique solutions. [ESTABLISHED]

Proof. On $\mathbb{R}^2$ let $\mathsf{G} = dx^2$. For $\mathcal{F} = y$, the equation $\mathsf{G}(\operatorname{grad}\mathcal{F}, \cdot) = dy$ has no solution because the left side annihilates ∂_y. For $\mathcal{F} = x$, every vector $\partial_x + b\partial_y$ solves the equation. A justified metric quotient or a separately declared regularization is therefore required before an inverse or pseudoinverse is used. □

9.4 Fisher duration on a selected history

Definition 9.12 (Fisher length and information clock). For a C^1 representative $Q : I \to \mathcal{Q}_i$, define its Fisher speed, length, and cumulative clock from a chosen origin λ_0 by

$$\nu_F(\lambda) = \sqrt{\mathsf{G}_i^F(\dot{Q}(\lambda), \dot{Q}(\lambda))}, \tag{9.26}$$

$$L_F(Q; [a,b]) = \int_a^b \nu_F(\lambda) d\lambda, \tag{9.27}$$

$$\tau_{Q,\lambda_0}(\lambda) = \int_{\lambda_0}^{\lambda} \nu_F(u) du. \tag{9.28}$$

The origin is relational data attached to the history. The auxiliary parameter λ is not part of the resulting arc length. [DEFINITION]

Theorem 9.13 (Reparameterization invariance and unit-speed form). Let $\widetilde{Q} = Q \circ \phi$, where $\phi : \widetilde{I} \to I$ is an orientation-preserving C^1 diffeomorphism. Then corresponding subarcs have the same Fisher length and their cumulative clocks obey

$$\tau_{\widetilde{Q},\widetilde{\lambda}_0}(\widetilde{\lambda}) = \tau_{Q,\lambda_0}(\phi(\widetilde{\lambda})), \qquad \lambda_0 = \phi(\widetilde{\lambda}_0). \tag{9.29}$$

If $\nu_F > 0$, the cumulative clock is strictly increasing and gives the unit-speed representative of the oriented history. If Q satisfies (9.22), then on its noncritical part

$$\frac{d\tau}{d\lambda} = a_i \left\| \operatorname{grad}^F \mathcal{F}_i \right\|_F, \tag{9.30}$$

$$\frac{dQ}{d\tau} = -\frac{\operatorname{grad}^F \mathcal{F}_i}{\left\| \operatorname{grad}^F \mathcal{F}_i \right\|_F}, \tag{9.31}$$

$$\frac{d\mathcal{F}_i}{d\tau} = -\left\| \operatorname{grad}^F \mathcal{F}_i \right\|_F. \tag{9.32}$$

[ESTABLISHED]

Proof. The chain rule gives $\dot{\widetilde{Q}} = (\dot{Q} \circ \phi)\phi'$ and hence $\widetilde{\nu}_F = (\nu_F \circ \phi)\phi'$ because $\phi' > 0$. The substitution $u = \phi(\widetilde{\lambda})$ proves both length invariance and (9.29). Positive speed makes τ locally and globally invertible on the interval, and division by $d\tau/d\lambda$ gives unit speed. Substitution of (9.22) into (9.26) gives (9.30). The remaining identities follow from (9.20). □

This is the standard Riemannian arc-length construction applied to the selected statistical configuration manifold [3, 61]. Fisher geometry supplies duration; the VFE descent ray supplies orientation. A null segment in a semimetric or a critical stationary segment has $\nu_F = 0$, so the cumulative clock stalls and cannot be used as a coordinate there. [ESTABLISHED]

Proposition 9.14 (Endpoints do not determine an update duration). Let A and B lie in one path component of a regular Fisher manifold. Those endpoints alone determine neither an update curve nor its realized duration. For every actual update Q from A to B,

$$d_F(A,B) = \inf_{\substack{\Gamma(a)=A \\ \Gamma(b)=B}} L_F(\Gamma;[a,b]) \le L_F(Q;[a,b]), \tag{9.33}$$

and the inequality can be strict. [ESTABLISHED]

Proof. The first equality is the definition of Riemannian distance as the infimum of curve lengths. Since the realized update Q is one admissible curve, the inequality follows. In the Euclidean Fisher geometry of a two-parameter normal location family, a curved path between two points is longer than the straight geodesic joining them, which proves possible strictness. □

The information clock is also distinct from the amount of auxiliary algorithmic parameter used to approach a limit. In the unit-variance normal location family, the path $\mu(t) = e^{-t}$ has Fisher length

$$\int_0^\infty |\dot{\mu}(t)| dt = \int_0^\infty e^{-t} dt = 1, \tag{9.34}$$

although t ranges over an unbounded interval. No physical-time interpretation follows from the length construction. [NOT-CLAIMED]

9.5 Pointwise, design, continuum, and quotient clocks

At one fixed $c \in \mathcal{C}_i$, choose positive belief and model weights $w_b(c)$ and $w_m(c)$. The pointwise squared speed of a section history is

$$\nu_{i,c}^2 = w_b(c) g_b^F(\partial_\lambda q_i(c), \partial_\lambda q_i(c)) + w_m(c) g_m^F(\partial_\lambda s_i(c), \partial_\lambda s_i(c)). \tag{9.35}$$

Integration of $\nu_{i,c}$ defines a pointwise clock. This uses only vertical evaluation velocities from (9.4); it does not integrate over the base. [DEFINITION]

For a finite design $D = \{c_a\}_{a=1}^M$, declare positive design weights ρ_a and define

$$\nu_{i,D}^2 = \sum_{a=1}^M \rho_a \left[w_b(c_a) g_b^F(\partial_\lambda q_i(c_a), \partial_\lambda q_i(c_a)) + w_m(c_a) g_m^F(\partial_\lambda s_i(c_a), \partial_\lambda s_i(c_a)) \right]. \tag{9.36}$$

This is the product Fisher speed for the declared design and weights. It is not automatically a positive-definite metric on the full section space: any nonzero section variation vanishing at all c_a lies in its radical. On a smooth positive-dimensional base, a bump variation supported away from D is an explicit witness whenever the vertical fiber has a nonzero local direction there. [ESTABLISHED]

A continuum clock requires a finite positive base measure μ_i, measurable positive channel weights, square-integrable evaluation velocities, and a declared treatment of gauge redundancy. Under those hypotheses its proposed squared speed is

$$\nu_{i,\mu}^2 = \int_{\mathcal{C}_i} \left[w_b(c) g_b^F(\partial_\lambda q_i(c), \partial_\lambda q_i(c)) + w_m(c) g_m^F(\partial_\lambda s_i(c), \partial_\lambda s_i(c)) \right] d\mu_i(c). \tag{9.37}$$

The contextual base, the common principal bundle, and the VFE functional do not by themselves select μ_i, the relative channel weights, or a strong manifold topology. Without those declarations, (9.37) is not an available clock and only the pointwise or finite-design constructions are defined. [ESTABLISHED]

The single-integral form in (9.37) also presumes *contextual locality*: the declared recognition law carries no term coupling the section values at two distinct contexts $c \neq c'$. If it does, the pullback of its Fisher metric carries a double integral over $\mathcal{C}_i \times \mathcal{C}_i$ that no single integral represents, and (9.37) is then not the exact configuration metric but a declared local surrogate. [HYPOTHESIS]

On the finite-dimensional tier, let a finite-dimensional Lie group $\mathcal{G}_i$ act freely, properly, and isometrically on the regular Riemannian configuration manifold. The quotient then carries the Riemannian submersion metric. Its speed is

$$\|[\dot{Q}]\|_{\text{quot}}^2 = \inf_{\zeta \in T_Q(\mathcal{G}_i \cdot Q)} \mathsf{G}_i^F(\dot{Q} - \zeta, \dot{Q} - \zeta), \tag{9.38}$$

which equals the norm of the component orthogonal to the gauge orbit. A history-parameter-dependent coordinate representative can acquire spurious velocity tangent to that orbit, while (9.38) is representative independent. Passive re-trivialization is a change of coordinates, not an additional physical segment of the history. If the action is not free and proper, the orbit space can be stratified; a smooth quotient clock is then available only on a declared regular stratum or after a separate singular construction. [ESTABLISHED]

In an infinite-dimensional realization the decisive condition is *closedness of the orbit tangent*. In the strong Hilbert tier a closed orbit tangent gives attainment of (9.38) by orthogonal projection, and complementedness is then automatic because each tangent space is Hilbertable; requiring closed *and* complemented subbundles is the correct generalization only in the weak or Banach tier, where closed no longer implies complemented. Without closedness the infimum can vanish on a vector in the closure of a nonclosed orbit tangent without being attained, so no orthogonal gauge quotient clock has been constructed. The available witness is free and isometric rather than proper: on $\mathcal{Q} = \ell^2$ let $\mathcal{G} = h^1 = \{v : \sum n^2 v_n^2 < \infty\}$ act by translation, which is free and isometric with dense orbit tangent, so the quotient speed vanishes identically and defines no clock. Whether free, proper, and isometric action alone can fail is not settled by that witness, since properness would force the orbits to be closed; the statement carried here is therefore the closedness one, and the properness question is left explicitly open. [HYPOTHESIS]

9.6 The obstruction to a global unit-speed clock

On a connected noncritical open set $U \subseteq \mathcal{Q}_i$, set

$$N = \left\|\operatorname{grad}^F \mathcal{F}_i\right\|_F, \qquad U_F = -\frac{\operatorname{grad}^F \mathcal{F}_i}{N}, \qquad \alpha_F = U_F^\flat = -\frac{d\mathcal{F}_i}{N}. \tag{9.39}$$

The form α_F evaluates to one on the unit descent field U_F and annihilates the Fisher-orthogonal complement of that field. [DEFINITION]

Theorem 9.15 (Exact criterion for a global orthogonal clock). There is a smooth scalar $T : U \to \mathbb{R}$ satisfying

$$dT = \alpha_F, \tag{9.40}$$

if and only if α_F is exact. Equivalently, it must be closed and have vanishing period around every piecewise C^1 closed curve in U:

$$d\alpha_F = 0, \qquad \oint_\gamma \alpha_F = 0 \quad \text{for every closed } \gamma \subset U. \tag{9.41}$$

On a simply connected U, closedness suffices. Whenever T exists, every unit-speed VFE history in U obeys $dT/d\tau = 1$, and the level sets of T are Fisher-orthogonal to the descent field. [ESTABLISHED]

Proof. If (9.40) holds, exact forms are closed and have zero periods. Conversely, fix $Q_0 \in U$ and define $T(Q)$ as the integral of α_F from Q_0 to Q. Closedness and vanishing periods make this integral path independent, and the standard line-integral argument gives $dT = \alpha_F$. Since $\alpha_F(U_F) = \mathsf{G}_i^F(U_F, U_F) = 1$, the asserted unit rate follows. The kernel of $dT = U_F^\flat$ is the orthogonal complement of U_F. □

The local differential obstruction is explicit:

$$d\alpha_F = \frac{1}{N^2} dN \wedge d\mathcal{F}_i. \tag{9.42}$$

Thus the norm of the gradient must be locally constant along each connected regular level set of $\mathcal{F}_i$ for the normalized form to be closed. Even after local closedness is established, the period condition in (9.41) is the remaining global obligation. [ESTABLISHED]

For a concrete local obstruction, take the Euclidean plane with $\mathcal{F}(x, y) = xy$ on the first quadrant. Then $N = (x^2 + y^2)^{1/2}$ and

$$d\alpha_F = \frac{x^2 - y^2}{(x^2 + y^2)^{3/2}} dx \wedge dy, \tag{9.43}$$

which is not identically zero on any open neighborhood. Every individual regular orbit still has its cumulative arc-length clock, but those orbitwise origins do not assemble into the orthogonal scalar (9.40). At critical points $N = 0$ and α_F is undefined, so extension across critical strata is a separate problem. The objective $\mathcal{F}_i$ itself may remain a Lyapunov scalar, but it is not generally a unit Fisher clock. [ESTABLISHED]

9.7 Operational record clocks and exact Markov loss

Let $\{P_\theta\}$ be a regular fine statistical experiment and let $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ be a normalized, parameter-independent Markov kernel. Along a C^1 fine history $\theta = \theta(\lambda)$, write ℓ_λ^X for its directional score and define the record history by

$$P_\lambda^Y = P_\lambda^X K. \tag{9.44}$$

This equation identifies the record family as the image of the same fine path; it does not define a separately optimized record trajectory. [HYPOTHESIS]

Theorem 9.16 (Same-path record and meta-agent clock contraction). Under the preceding regularity and Markov hypotheses, the record score and the exact Fisher-speed defect are

$$\ell_\lambda^Y(Y) = \mathbb{E}_\lambda[\ell_\lambda^X(X) \mid Y], \tag{9.45}$$

$$(\nu_F^X)^2 - (\nu_F^Y)^2 = \mathbb{E}_\lambda \operatorname{Var}_\lambda(\ell_\lambda^X(X) \mid Y) \geq 0. \tag{9.46}$$

Consequently, on every common parameter interval,

$$L_F(P^Y) \leq L_F(P^X), \qquad \tau_Y(\lambda_1) - \tau_Y(\lambda_0) \leq \tau_X(\lambda_1) - \tau_X(\lambda_0). \tag{9.47}$$

Equality holds exactly when the fine directional score is measurable with respect to the record for almost every path position. The same result applies to a meta-agent only when its statistical law is the normalized parameter-independent Markov image of this same fine history. [ESTABLISHED]

Proof. Attach K to the fine experiment. Because K has no θ dependence, the joint score is $\ell_\lambda^X(X)$. Projection onto the record sigma algebra gives (9.45). The law of total variance gives (9.46), including its equality condition. Taking nonnegative square roots and integrating proves (9.47). This is the pathwise form of the Fisher score-projection theorem [8]. □

Parameter independence is load bearing. If a dominated channel has density $k_\lambda(y \mid x)$, its output score is instead

$$\ell_\lambda^Y(Y) = \mathbb{E}_\lambda\left[\ell_\lambda^X(X) + \partial_\lambda \log k_\lambda(Y \mid X) \,\middle|\, Y\right]. \tag{9.48}$$

The additional term can create Fisher information. For example, a parameter-independent input combined with a channel that emits a Bernoulli variable of success probability $(1 + e^{-\lambda})^{-1}$ has zero fine Fisher speed and positive output Fisher speed. A fitted coarse approximation, an energy precomposition, a Galerkin restriction, or a mismatch between generative and recognition parameters is likewise outside Theorem 9.16. [ESTABLISHED]

One apparent counterexample to Theorem 9.16 must be retired rather than answered. The independent b-fold replication $\{\mathcal{N}(x\mathbf{1}_b, I_{\mathbb{R}^b})\}$ has Fisher information b against the input family's value 1, which invites the reading that some coarse arrow increased information. There is no such arrow: if a parameter-independent normalized Markov kernel K satisfied $\mathcal{N}(x,1)K = \mathcal{N}(x\mathbf{1}_b, I_{\mathbb{R}^b})$ for every x, then (9.46) would force $b \le 1$, so no such K exists for $b \ge 2$. What the pair witnesses is a change of reference tangent space, in which the replication lift $\mathscr{I}_b$ of Proposition 12.9 has norm $\sqrt{b}$; the composite $\mathscr{L}_b = U_b \mathscr{I}_b$ therefore obeys $\|\mathscr{L}_b\| \le \sqrt{b}$ rather than $\|\mathscr{L}_b\| \le 1$, and no contradiction arises. The replication pair may not be cited as a Markov-contraction counterexample. [ESTABLISHED]

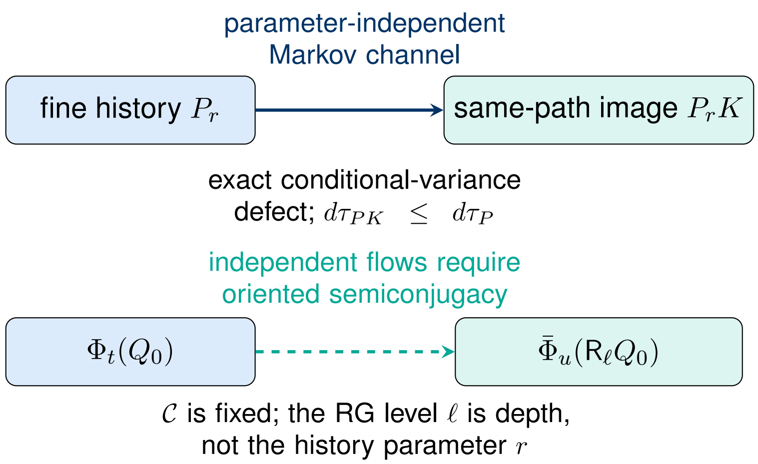

Figure 9.1: The upper arrow is the same-path comparison covered by Fisher contraction. The lower arrow compares independently generated flows and exists only after the additional semiconjugacy condition is proved.

9.8 The coarse configuration map

The coarse configuration map is separately declared data with its own symbol. Write

$$\mathsf{R}_\ell : \mathcal{Q}_\ell \longrightarrow \mathcal{Q}_{\ell+1}, \tag{9.49}$$

smooth, and never identified with the sample kernel K_ℓ, the induced law map, the associated-bundle morphism of Chapter 8, the nonlinear action map $\mathcal{R}^H$, the block measure-pair map $\mathcal{R}_b$, the derivative D_ℓ, the associated scale map $C_{\ell,s}$, the reference-space endomorphism $\widehat{\mathcal{R}}_\ell$, the conjugated retained projection $\widehat{R}_\ell$, the root-vertex set $\mathcal{R}$, or the descent ray $\mathcal{R}^-_{\mathcal{F}_i}$. The notation appendix carries the corresponding row. [DEFINITION]

A pointwise bundle morphism induces a map on configurations only on the projectable set of Theorem 8.13. Under the reachability hypotheses of Proposition 8.14, including a global smooth section through the witnessing fiber point, that set is a proper subset whenever the base map collapses a direction on which the vertical differential does not vanish; in the linear tier it can have empty interior. [ESTABLISHED]

No result here establishes that the projectable set is a smooth submanifold carrying a smooth induced arrow; that would require an additional transversality or elliptic-regularity hypothesis. The generic separately declared map R_ℓ is therefore neither constructed nor identified through section descent. [NOT-CLAIMED]

The averaging construction is the standard separately declared instance, and it carries an exact defect rather than an unconditional contraction. Let $f : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$ be measurable between standard Borel spaces, let μ be finite positive with $\bar{\mu} = f_{\#}\mu$, and let $\{\kappa_{\bar{c}}\}$ disintegrate μ over f. Suppose the coarse fiber $\bar{\mathcal{B}}$ is convex in a locally convex space, the structure group acts by restrictions of linear maps, and Ψ acts fiberwise as the restriction of a continuous affine map with linear part L, with the integrand Bochner integrable and the barycenter inside the fiber. Define

$$(\mathsf{R}_\ell s)\,(\bar{c}) = \int_{f^{-1}(\bar{c})} \Psi\,(s(c))\,\kappa_{\bar{c}}(\mathrm{d}c). \tag{9.50}$$

[DEFINITION]

For this averaging instance, if $\Psi \circ s$ is f-basic with unique descended section $\bar{s}$, then, for $\bar{\mu}$-almost every $\bar{c}$, the integrand equals $\bar{s}(\bar{c})$ for $\kappa_{\bar{c}}$-almost every point of the fiber. Normalization of the disintegration therefore gives $(\mathsf{R}_\ell s)(\bar{c}) = \bar{s}(\bar{c})$ for $\bar{\mu}$-almost every $\bar{c}$. [ESTABLISHED]

Hypothesis 9.17 (Chart joint convexity). In the declared affine chart of the coarse fiber, the map $(\bar{\beta}, v) \mapsto \bar{g}^F_{\bar{\beta}}(v,v)$ is jointly convex on $\bar{\mathcal{B}} \times \bar{W}$. The special case in which $\bar{g}^F$ is constant in the chart is written (JC-const). This hypothesis is a restriction on the chart, not a property of Fisher metrics: it holds in the location sector with fixed fiber covariance and in the law chart, and it fails in the covariance sector of the Gaussian moment chart. [HYPOTHESIS]

Theorem 9.18 (Exact averaging defect). Assume the preceding structure, Hypothesis 9.17, the fiberwise Fisher contraction $\Delta^\Psi_F = g^F - L^*\bar{g}^F L \succeq 0$, the weight condition $\bar{w} \circ f \le w$ of (8.15), and finiteness of every integral. Then $\mathsf{R}_\ell s$ is a section, R_ℓ is gauge equivariant because the transition maps act linearly and commute with barycenters, and for every $Z \in T_s\mathcal{Q}_\ell$ the defect

$$\Delta_{\mathrm{avg}}(Z) = \int_{\mathcal{C}_\ell} w g^F_{s(c)}(Z,Z)\mathrm{d}\mu - \int_{\mathcal{C}_{\ell+1}} \bar{w}\bar{g}^F_{(\mathsf{R}_\ell s)(\bar{c})}\,(T\mathsf{R}_\ell Z, T\mathsf{R}_\ell Z)\,\mathrm{d}\bar{\mu} \tag{9.51}$$

is nonnegative. Under (JC-const) it is exactly the sum of three separately nonnegative terms,

$$\Delta_{\mathrm{avg}}(Z) = \underbrace{\int_{\mathcal{C}_\ell} w\Delta^\Psi_F(Z,Z)\mathrm{d}\mu}_{\text{channel loss}} + \underbrace{\int_{\mathcal{C}_\ell} (w - \bar{w}\circ f)\left(L^*\bar{g}^F\right)(Z,Z)\mathrm{d}\mu}_{\text{weight gap}} + \underbrace{\int_{\mathcal{C}_{\ell+1}} \bar{w}\,\mathrm{Var}^{\bar{g}^F}_{\kappa_{\bar{c}}}(LZ)\mathrm{d}\bar{\mu}}_{\text{context gap}}, \tag{9.52}$$

where $\mathrm{Var}^{\bar{g}}_\kappa(V) = \int \bar{g}(V,V)\mathrm{d}\kappa - \bar{g}(\int V\mathrm{d}\kappa, \int V\mathrm{d}\kappa)$. [ESTABLISHED]

Proof. Affineness gives $T\mathsf{R}_\ell Z(\bar{c}) = \int LZ(c)\kappa_{\bar{c}}(\mathrm{d}c)$, so the pair $(\mathsf{R}_\ell s(\bar{c}), T\mathsf{R}_\ell Z(\bar{c}))$ is the $\kappa_{\bar{c}}$-barycenter of $(\Psi s(c), LZ(c))$. Joint convexity and Jensen's inequality give $\bar{g}^F_{\mathsf{R}_\ell s(\bar{c})}(T\mathsf{R}_\ell Z, T\mathsf{R}_\ell Z) \le \int \bar{g}^F_{\Psi s(c)}(LZ, LZ)\mathrm{d}\kappa_{\bar{c}}$. Multiplying by $\bar{w}$, integrating against $\bar{\mu}$, and using the disintegration converts the right-hand side into $\int(\bar{w}\circ f)(L^*\bar{g}^F)(Z,Z)\mathrm{d}\mu$; the weight condition and $\Delta^\Psi_F \succeq 0$ then give nonnegativity. When $\bar{g}^F$

is constant in the chart the Jensen step is an equality minus the variance term, and expanding $g^F = \Delta_F^\Psi + L^*\bar{g}^F$ separates the channel and weight contributions, which gives (9.52). □

Hypothesis 9.17 is not decorative, and the statement that averaging over the base loses information is false without it. Take two contexts with uniform κ, unit weights, Ψ the identity so that $\Delta_F^\Psi \equiv 0$, and the centered normal fiber $\{\mathcal{N}(0,\Sigma)\}$ in the moment chart, where $g^F(A,A) = A^2/(2\Sigma^2)$. At the configuration $(\Sigma_1,\Sigma_2) = (1,\delta)$ with tangent $Z = (1,0)$ the fine integrated metric is $\frac{1}{4}$ and the chart-barycenter image metric is $1/(2(1+\delta)^2)$, so their ratio is $2/(1+\delta)^2$, which equals $\frac{20000}{10201} = 1.9605921\ldots$ at $\delta = 10^{-2}$ and tends to the block size 2 as $\delta \to 0^+$. The defect is therefore strictly negative, tending to $-\frac{1}{4}$, and is negative exactly for $\delta < \sqrt{2}-1$. Every other hypothesis of Theorem 9.18 holds on this datum; only joint convexity fails, and the reason is exact: the Hessian of $(\Sigma, A) \mapsto A^2/(2\Sigma^2)$ has determinant $-A^2\Sigma^{-6} < 0$ for $A \neq 0$, whereas the law-chart integrand $\dot{p}^2/p$ has Hessian determinant zero and positive trace and is jointly convex. Accordingly no contraction statement may be attached to a generic averaging or variational coarse map. [ESTABLISHED]

Proposition 9.19 (Smoothness and gauge descent at the exhibited tier). Extend Definition 9.6 to scale $\ell+1$ with base $\mathcal{C}_{\ell+1}$, measure $\bar{\mu}$, weight $\bar{w}$, coarse fiber $\{\mathcal{N}(m,\bar{\Sigma}_0)\}$ with $\bar{\Sigma}_0 \succeq \Sigma_0$, and a coarse basis $\psi_1,\ldots,\psi_M$ linearly independent in $L^2(\bar{w}\bar{\mu};\bar{\Sigma}_0^{-1})$ whose span contains every constant field. Let $\mathsf{L}_{\ell+1} : \mathbb{R}^K \to \mathbb{R}^M$ represent those constants in the coarse basis as in (9.11). Suppose the fiberwise average of each fine basis field lies in the coarse span, $\int_{f^{-1}(\bar{c})} \phi_a \mathrm{d}\kappa_{\bar{c}} = \sum_b \mathsf{E}_{ba}\psi_b(\bar{c})$ for a constant matrix $\mathsf{E} \in \mathbb{R}^{M\times N}$ which intertwines the coefficient actions,

$$\mathsf{E}\mathsf{L}_\ell = \mathsf{L}_{\ell+1}. \tag{9.53}$$

Then the averaging map (9.50) reads $\mathsf{R}_\ell(\xi) = \mathsf{E}\xi$ in coordinates. It is linear, hence C^∞, with tangent map E everywhere; it is gauge equivariant for the affine coefficient actions because $\mathsf{E}\Theta_g^\ell = \Theta_g^{\ell+1}\mathsf{E}$; and its metric compatibility is exactly the matrix inequality

$$\mathsf{E}^\top\bar{\Phi}\mathsf{E} \preceq \Phi, \qquad \bar{\Phi}_{bc} = \int \psi_b^\top \bar{\Sigma}_0^{-1}\psi_c \bar{w}\mathrm{d}\bar{\mu}, \tag{9.54}$$

which is $\mathsf{R}_\ell^*\mathsf{G}_{\ell+1} \preceq \mathsf{G}_\ell$ and is implied by Theorem 9.18. [ESTABLISHED]

Proof. Averaging a section of the declared family produces the mean field $\bar{c} \mapsto \sum_a \xi_a \int \phi_a \mathrm{d}\kappa_{\bar{c}} = \sum_b (\mathsf{E}\xi)_b\psi_b(\bar{c})$, which is the coarse configuration $s_{\mathsf{E}\xi}$; smoothness and the tangent map are those of a linear map. For equivariance, the constant-field hypotheses give $\Theta_g^\ell(\xi) = \xi + \mathsf{L}_\ell g$ and $\Theta_g^{\ell+1}(\eta) = \eta + \mathsf{L}_{\ell+1} g$, while (9.53) gives $\mathsf{E}\Theta_g^\ell(\xi) = \Theta_g^{\ell+1}(\mathsf{E}\xi)$. Equivalently, $\int (m+g)\mathrm{d}\kappa_{\bar{c}} = \int m \mathrm{d}\kappa_{\bar{c}} + g$ because $\kappa_{\bar{c}}$ is a probability measure and both resulting fields remain in their declared spans. Compatibility is Theorem 9.18 evaluated on the finite-dimensional tangent, and it takes the displayed matrix form because both metrics are constant Gram forms. □

Metric compatibility is therefore carried as an explicit hypothesis discharged by Theorem 9.18, and it is not a corollary of any fiberwise contraction theorem. [HYPOTHESIS]

9.9 Independently recomputed flows

Let X_ℓ and $X_{\ell+1}$ be the declared fine and coarse natural-gradient vector fields on regular configuration manifolds $\mathcal{Q}_\ell$ and $\mathcal{Q}_{\ell+1}$, with local flows Φ_t and $\bar{\Phi}_u$. Pointwise contraction of $T\mathsf{R}_\ell$ does not state that R_ℓ maps an X_ℓ-orbit to an $X_{\ell+1}$-orbit. [ESTABLISHED]

Definition 9.20 (Oriented semiconjugacy). For an open $U \subseteq \mathcal{Q}_\ell$, the pair $(\mathsf{R}_\ell, a_\ell)$ is an oriented semiconjugacy of X_ℓ onto $X_{\ell+1}$ over U when $a_\ell : U \to (0, \infty)$ is continuous and

$$T_Q\mathsf{R}_\ell X_\ell(Q) = a_\ell(Q) X_{\ell+1}(\mathsf{R}_\ell Q) \quad \text{for every } Q \in U. \tag{9.55}$$

[DEFINITION]

Lemma 9.21 (The factor is determined and regular off the coarse critical set). If $X_{\ell+1}(\mathsf{R}_\ell Q) \neq 0$, then $a_\ell(Q)$ is unique and

$$a_\ell(Q) = \frac{\mathsf{G}_{\ell+1}\left(T_Q\mathsf{R}_\ell X_\ell(Q), X_{\ell+1}(\mathsf{R}_\ell Q)\right)}{\|X_{\ell+1}(\mathsf{R}_\ell Q)\|^2_{\mathsf{G}_{\ell+1}}}, \tag{9.56}$$

so, for any integer $k \geq 0$, a_ℓ is C^k wherever the maps $Q \mapsto T_Q\mathsf{R}_\ell X_\ell(Q)$, $Q \mapsto X_{\ell+1}(\mathsf{R}_\ell Q)$, and the pulled-back metric coefficients are C^k. Sufficient primitive hypotheses are $\mathsf{R}_\ell \in C^{k+1}$ and X_ℓ, $X_{\ell+1}$, and $\mathsf{G}_{\ell+1}$ of class C^k, with the displayed compositions defined. If instead $X_{\ell+1}(\mathsf{R}_\ell Q) = 0$, then (9.55) forces $T_Q\mathsf{R}_\ell X_\ell(Q) = 0$ and imposes no algebraic pointwise value on $a_\ell(Q)$. The definition still requires a positive continuous function on U: limits from the noncritical set may determine the value at Q, or may obstruct any continuous positive extension. [ESTABLISHED]

Proof. Pair (9.55) with $X_{\ell+1}(\mathsf{R}_\ell Q)$ and divide; strong nondegeneracy of $\mathsf{G}_{\ell+1}$ makes the denominator positive. $\square$

Continuity of a_ℓ is therefore a consequence on the noncritical part rather than an extra assumption, and the phrase noncritical must name the field it governs: the load-bearing reading is $X_{\ell+1} \neq 0$ on $\mathsf{R}_\ell(U)$. [DEFINITION]

Theorem 9.22 (Oriented semiconjugacy and maximal intervals). Assume (9.55) on an open U, with $X_{\ell+1}$ locally Lipschitz. Fix $Q \in U$, let J_Q be the connected component of 0 in the set of t for which $\Phi_s(Q) \in U$ for every s between 0 and t, and set $\sigma_Q(t) = \int_0^t a_\ell(\Phi_s Q)\mathrm{d}s$. Then σ_Q is a strictly increasing C^1 diffeomorphism of J_Q onto an open interval $\Sigma_Q \ni 0$, and

$$\Sigma_Q \subseteq \bar{J}^{\max}_{\mathsf{R}_\ell Q}, \qquad \mathsf{R}_\ell\left(\Phi_t(Q)\right) = \bar{\Phi}_{\sigma_Q(t)}\left(\mathsf{R}_\ell Q\right) \quad \text{for every } t \in J_Q, \tag{9.57}$$

where $\bar{J}^{\max}$ is the maximal interval of the coarse flow through $\mathsf{R}_\ell Q$. More explicitly, write $J_Q = (\alpha_Q, \beta_Q)$ and $\bar{J}^{\max}_{\mathsf{R}_\ell Q} = (\bar{\alpha}_Q, \bar{\beta}_Q)$, with endpoints in the extended real line. Then

$$\Sigma_Q = \left(\lim_{t \downarrow \alpha_Q} \sigma_Q(t), \lim_{t \uparrow \beta_Q} \sigma_Q(t) \right), \tag{9.58}$$

and the full maximal coarse orbit is traversed if and only if the two limits equal $\bar{\alpha}_Q$ and $\bar{\beta}_Q$, respectively. Conversely, differentiating an oriented flow semiconjugacy of this form at $t = 0$ yields (9.55). [ESTABLISHED]

Proof. Put $c(t) = \mathsf{R}_\ell(\Phi_t Q)$, so that $\dot{c} = a_\ell(\Phi_t Q) X_{\ell+1}(c)$ by (9.55). The integrand of σ_Q is continuous and strictly positive, so σ_Q is a C^1 diffeomorphism onto an open interval. Let $\theta = \sigma_Q^{-1}$ and $d = c \circ \theta$; then $d'(u) = X_{\ell+1}(d(u))$ on Σ_Q with $d(0) = \mathsf{R}_\ell Q$, so d is an integral curve of $X_{\ell+1}$, and by local Lipschitz continuity the maximal integral curve through $\mathsf{R}_\ell Q$ contains it. That gives both the domain inclusion and the flow identity. The order matters: the reparameterized curve is constructed first and the coarse flow is then shown to extend over Σ_Q, rather than the coarse flow being assumed defined at $\sigma_Q(t)$. The converse is the chain rule at $t = 0$. □

When the semiconjugacy holds along the entire complete fine orbit ($J_Q = \mathbb{R}$) and the coarse maximal interval is also $\mathbb{R}$, the endpoint criterion is equivalently

$$\int_0^{+\infty} a_\ell(\Phi_s Q)\mathrm{d}s = +\infty, \qquad \int_{-\infty}^{0} a_\ell(\Phi_s Q)\mathrm{d}s = +\infty. \tag{9.59}$$

A positive lower bound for a_ℓ is sufficient, but it is not necessary: for $X_\ell = \partial_x$, $X_{\ell+1} = \partial_y$, and $\mathsf{R}_\ell = \operatorname{arsinh}$ on $\mathbb{R}$, one has $a_\ell(x) = (1 + x^2)^{-1/2}$ and $\inf a_\ell = 0$, while $\sigma_0(t) = \operatorname{arsinh}(t)$ maps $\mathbb{R}$ onto $\mathbb{R}$. Nor does positivity alone guarantee full traversal: on $\mathbb{R}$ with $X_\ell = \partial_x$, $X_{\ell+1} = \partial_y$, and $\mathsf{R}_\ell = \arctan$, the factor $a_\ell = (1 + x^2)^{-1}$ is positive while $\Sigma_0 = (-\pi/2, \pi/2)$ is a proper subinterval of $\mathbb{R}$ even though both flows are complete. Positivity of the factor is also the entire orientation content: on $\mathbb{R}$ with $\mathcal{F}_\ell = x^2/2$, $\mathcal{F}_{\ell+1} = -y^2/2$, and $\mathsf{R}_\ell(x) = -x$, the orbit sets agree while $a_\ell \equiv -1$ and coarse descent becomes ascent, so requiring only $a_\ell \neq 0$ loses orientation. [ESTABLISHED]

Proposition 9.23 (Noncollapse is a separate hypothesis). Condition (9.55) alone permits total collapse: with $\mathsf{R}_\ell \equiv Q^*$ constant and $X_{\ell+1} \equiv 0$, both sides vanish identically, so it holds for every positive a_ℓ while every nonconstant fine orbit maps to a point and every coarse Fisher duration is zero. A nonconstant shared oriented history therefore additionally requires $T_Q \mathsf{R}_\ell X_\ell(Q) \neq 0$ on every nontrivial subarc, equivalently $X_{\ell+1}(\mathsf{R}_\ell Q) \neq 0$ there given $a_\ell > 0$, together with the maximal-interval condition of Theorem 9.22, namely the endpoint equalities in (9.58); for complete real-line flows these are the two divergence conditions in (9.59). Neither is implied by (9.55). [ESTABLISHED]

Theorem 9.24 (The semiconjugacy parameter rate cancels from geometric length). Under (9.55), write $\nu_\ell(r) = \|X_\ell(Q^{(\ell)}(r))\|_{\mathsf{G}_\ell}$ and $\nu^{\mathrm{img}}_{\ell+1}(r) = \|\frac{\mathrm{d}}{\mathrm{d}r} \mathsf{R}_\ell Q^{(\ell)}(r)\|_{\mathsf{G}_{\ell+1}}$. Then

$$\nu^{\mathrm{img}}_{\ell+1}(r) = a_\ell\left(Q^{(\ell)}(r)\right) \left\| X_{\ell+1}\left(\mathsf{R}_\ell Q^{(\ell)}(r)\right) \right\|_{\mathsf{G}_{\ell+1}}, \tag{9.60}$$

and for $r_0 < r_1$ in J_Q the accumulated coarse duration is

$$\tau^{(\ell+1)}(r_1) - \tau^{(\ell+1)}(r_0) = \int_{\sigma_Q(r_0)}^{\sigma_Q(r_1)} \left\| X_{\ell+1}\left(\bar{\Phi}_u(\mathsf{R}_\ell Q)\right)\right\|_{\mathsf{G}_{\ell+1}} \mathrm{d}u, \tag{9.61}$$

which is the intrinsic $\mathsf{G}_{\ell+1}$-arc length of the coarse orbit arc and does not depend on a_ℓ. [ESTABLISHED]

Proof. Equation (9.60) is (9.55) together with positive homogeneity of the norm and $a_\ell > 0$. For (9.61), substitute (9.60) and change variables $u = \sigma_Q(s)$, whose Jacobian is $\mathrm{d}u = a_\ell(\Phi_s Q)\mathrm{d}s$; the factor a_ℓ in the integrand cancels against that Jacobian exactly. □

The factor $a_\ell = \mathrm{d}\sigma_Q/\mathrm{d}r$ is the coarse-flow-parameter rate relative to the chosen fine orbit parameter. It is neither physical time nor a Fisher-duration rate, and it cancels from a final geometric arc length. For an unoriented relation with $a_\ell < 0$, ordinary arc length still uses $|a_\ell|$ and is unchanged by the orientation reversal; positivity is the extra condition selecting an oriented semiconjugacy. [ESTABLISHED]

Theorem 9.25 (Exact duration comparison criterion). Under (9.55), along a fixed orbit and from a common origin, the coarse and fine durations agree identically if and only if $\|T\mathsf{R}_\ell X_\ell\|_{\mathsf{G}_{\ell+1}} = \|X_\ell\|_{\mathsf{G}_\ell}$ almost everywhere along the orbit; they agree up to a constant factor $\kappa > 0$ if and only if the same holds with that factor; and the coarse duration has nonincreasing increments relative to the fine duration on every subinterval if and only if

$$\|T\mathsf{R}_\ell X_\ell\|_{\mathsf{G}_{\ell+1}} \leq \|X_\ell\|_{\mathsf{G}_\ell} \quad \text{pointwise along the orbit.} \tag{9.62}$$

[ESTABLISHED]

Proof. Both durations are integrals of continuous nonnegative densities from a common origin. Two such integrals agree for every upper limit exactly when their densities agree almost everywhere, by Lebesgue differentiation, and the scaled and ordered versions follow in the same way. □

The necessary and sufficient condition is thus the *scalar* inequality (9.62) evaluated on the single direction X_ℓ. The tensorial Loewner condition $\mathsf{R}_\ell^*\mathsf{G}_{\ell+1} \preceq \mathsf{G}_\ell$ along the curve is sufficient and strictly stronger; it is what one declares when the comparison must hold in every direction, and it is what Theorem 9.18 and proposition 9.19 deliver. Neither follows from any fiberwise Fisher contraction theorem, and two independent mechanisms show why. [ESTABLISHED]

The first mechanism is that the two configuration metrics are separately declared data. On $\mathcal{Q}_\ell = \mathcal{Q}_{\ell+1} = \mathbb{R}$ with $X_\ell = X_{\ell+1} = \partial_x$ and R_ℓ the identity, (9.55) holds with $a_\ell \equiv 1$, there is no collapse, and the map is a diffeomorphism; declaring $\mathsf{G}_\ell = dx^2$ and $\mathsf{G}_{\ell+1} = 4dx^2$ makes $\tau^{(\ell+1)} = 2\tau^{(\ell)}$. The first realization is one context with an identity fiber map, so that $\Delta_F^\Psi = 0$ exactly, and channel weights 1 and 4; this is the zero-information-loss metric-declaration witness. Separately, fine fiber $\{\mathcal{N}(\mu, 1)\}$ against coarse fiber $\{\mathcal{N}(\mu, \frac{1}{4})\}$ with the identity parameter map gives the non-Markov Fisher-increase witness $\Delta_F^\Psi = 1 - 4 = -3$, while the duration still doubles. The second mechanism is that a contraction theorem compares the image of the fine orbit, not an independently recomputed coarse orbit. On $\mathbb{R}^2$ with R_ℓ and Ψ both the identity, so that contraction is an equality and image durations agree, the objectives $\mathcal{F}_\ell = \frac{1}{2}(x^2 + y^2)$ and $\mathcal{F}_{\ell+1} = \frac{1}{2}(x^2 + 4y^2)$ give the fine history (e^{-t}, e^{-t}) of total duration $\sqrt{2}$ and the independently recomputed coarse history (e^{-t}, e^{-4t}), which is not a straight segment and has strictly greater length $\int_0^1 \sqrt{1 + 16u^6}\mathrm{d}u > \sqrt{2}$; and (9.55) itself fails there, since $(-x, -y) = a(-x, -4y)$ forces $a = 1$ and then $y = 4y$. The two mechanisms are logically independent and neither implies the other, which is the exact content of the statement that either condition without the other is insufficient. [ESTABLISHED]

Theorem 9.26 (When fiberwise contraction does lift). Assume the two scales share the base $\mathcal{C}_i$, the same finite base measure μ_i, and the same positive channel weights w_x; assume R_ℓ acts pointwise as $(\mathsf{R}_\ell Q)(c) = \Psi_c(Q(c))$ with each Ψ_c the pushforward of a normalized parameter-independent Markov kernel with vertical differential $T^V \Psi_c$; assume both configuration metrics are the corresponding weighted integrals of the fiber Fisher metrics; and assume every integral is finite. Then for every $Z \in T_Q \mathcal{Q}_\ell$,

$$\|Z\|^2_{\mathsf{G}_\ell} - \|T\mathsf{R}_\ell Z\|^2_{\mathsf{G}_{\ell+1}} = \int_{\mathcal{C}_i} \sum_x w_x \Delta_F^{\Psi_c}(Z_x(c), Z_x(c))\mathrm{d}\mu_i \geq 0, \tag{9.63}$$

so $\nu^{\mathrm{img}}_{\ell+1} \leq \nu_\ell$ pointwise and the coarse duration has nonincreasing increments, with equality on a subinterval exactly when the fine score in the direction $Z_x(c)$ is measurable with respect to the coarse output for μ_i-almost every c and every channel. [ESTABLISHED]

Proof. Differentiate the pointwise action, insert into the coarse metric, and subtract the fine one; the integrand is $\Delta_F^{\Psi_c}$, nonnegative by Theorem 8.21. Monotonicity of the integral and of the square root gives the speed inequality, and integrating in the orbit parameter gives the duration statement. □

Theorem 9.26 is the case of a shared base; Theorem 9.18 is an alternative sufficient extension that admits a collapsing base map, which is what a renormalization step performs. It additionally assumes affine-fiber, convexity, and barycenter hypotheses, so no general logical-strength comparison with the shared-base result is asserted. Its second hypothesis excludes this manuscript's own Galerkin aggregation, because by Proposition 16.5 that coarse operator is a restriction rather than a Markov pushforward, the two differing by a positive semidefinite Schur term with the restriction larger in the Loewner order. Its first hypothesis is exactly what the metric-declaration witness above violates. [ESTABLISHED]

9.10 Sufficient conditions for natural-gradient semiconjugacy

Throughout this section $X_\ell = -\operatorname{grad}^{\mathsf{G}_\ell} \mathcal{F}_\ell$ and $X_{\ell+1} = -\operatorname{grad}^{\mathsf{G}_{\ell+1}} \mathcal{F}_{\ell+1}$ with both objectives C^2 and both metrics strong. Nothing in this section is automatic; each conclusion consumes a stated hypothesis.

Equality of objectives does not intertwine natural gradients. The differential $\mathrm{d}\mathcal{F}$ is metric free while the gradient is not, so with R_ℓ the identity on $\mathbb{R}^2$, one objective $\frac{1}{2}(x^2+2y^2)$, and metrics $\mathrm{diag}(1,1)$ and $\mathrm{diag}(1,\kappa)$ with $\kappa \neq 1$, condition (9.55) would require $(x,2y) = a(x,2y/\kappa)$, forcing $a=1$ and then $\kappa = 1$; it therefore fails on the dense open set where $xy \neq 0$. [ESTABLISHED]

Hypothesis 9.27 (Functional compatibility). There is a C^1 function χ_ℓ with $\chi_\ell' > 0$ and an open $U \supseteq \mathscr{O}$ containing the fine orbit on which $\mathcal{F}_\ell = \chi_\ell \circ \mathcal{F}_{\ell+1} \circ \mathsf{R}_\ell$. This supplies orientation and noncollapse; it does not by itself supply (9.55). [HYPOTHESIS]

Proposition 9.28 (Metric sufficient conditions for semiconjugacy). Let R_ℓ be a surjective submersion with closed orbit-tangent splitting, let $\mathcal{H} = (\ker T\mathsf{R}_\ell)^{\perp_{\mathsf{G}_\ell}}$, and assume Hypothesis 9.27. If R_ℓ is a Riemannian submersion, meaning that $T\mathsf{R}_\ell|_{\mathcal{H}}$ is a linear isometry onto $T\mathcal{Q}_{\ell+1}$, then (9.55) holds with $a_\ell(Q) = \chi_\ell'(\mathcal{F}_{\ell+1}(\mathsf{R}_\ell Q)) > 0$. More generally, if R_ℓ is horizontally conformal with dilation $\varphi_\ell > 0$, then (9.55) holds with

$$a_\ell(Q) = \chi_\ell'\left(\mathcal{F}_{\ell+1}(\mathsf{R}_\ell Q)\right)\varphi_\ell(Q)^2 > 0, \qquad Q \in U. \tag{9.64}$$

[ESTABLISHED]

Proof. Under Hypothesis 9.27 the gradient $u = \mathrm{grad}^{\mathsf{G}_\ell}\mathcal{F}_\ell$ lies in $\mathcal{H}$. For $Z \in \mathcal{H}$, horizontal conformality gives $\mathsf{G}_{\ell+1}(T\mathsf{R}_\ell u, T\mathsf{R}_\ell Z) = \varphi_\ell^2 \mathsf{G}_\ell(u,Z) = \varphi_\ell^2 \mathrm{d}\mathcal{F}_\ell[Z] = \varphi_\ell^2\chi_\ell' \mathrm{d}\mathcal{F}_{\ell+1}[T\mathsf{R}_\ell Z] = \varphi_\ell^2\chi_\ell' \mathsf{G}_{\ell+1}(w, T\mathsf{R}_\ell Z)$ with $w = \mathrm{grad}^{\mathsf{G}_{\ell+1}}\mathcal{F}_{\ell+1} \circ \mathsf{R}_\ell$. Since $T\mathsf{R}_\ell(\mathcal{H}) = T\mathcal{Q}_{\ell+1}$, nondegeneracy gives $T\mathsf{R}_\ell u = \varphi_\ell^2\chi_\ell' w$; negating both sides gives the claim. The Riemannian-submersion case is $\varphi_\ell \equiv 1$. □

A sanity check and a nonempty model are both available. On Euclidean $\mathbb{R}$ with $\mathsf{R}_\ell(x) = \lambda x$ one has $\varphi_\ell = |\lambda|$ and $\mathcal{F}_\ell'(x) = \lambda\mathcal{F}_{\ell+1}'(\lambda x)$, so $T\mathsf{R}_\ell\,\mathrm{grad}\,\mathcal{F}_\ell = \lambda^2\mathcal{F}_{\ell+1}'(\lambda x)$, matching $a_\ell = \lambda^2$. On the exhibited tier of Definition 9.6 with $\mathcal{C}_\ell = S^1$, basis $\{1, \cos\theta, \sin\theta\}$, fine fiber $\{\mathcal{N}(m,1)\}$, coarse fiber $\{\mathcal{N}(m,2)\}$ realized by the sample kernel $N(x,\cdot) = \mathcal{N}(x,1)$, and $\mathcal{F}(\xi) = \frac{1}{2}|\xi|^2$, one has $\mathsf{G}_\ell = \Phi = \mathrm{diag}(1, \frac{1}{2}, \frac{1}{2})$, $\mathsf{G}_{\ell+1} = \frac{1}{2}\Phi$, R_ℓ the identity, hence $\varphi_\ell^2 = \frac{1}{2}$ and $\chi_\ell = \mathrm{id}$; the proposition predicts $a_\ell = \frac{1}{2}$, which is what the direct computation $X_\ell = -\Phi^{-1}\xi$ and $X_{\ell+1} = -2\Phi^{-1}\xi$ returns, with speed ratio $\nu_{\ell+1}^{\mathrm{img}}/\nu_\ell = 1/\sqrt{2}$ and both flows complete. [ESTABLISHED]

Where this manuscript's own coarse maps stand can now be stated exactly rather than gestured at. Under the hypotheses of Theorem 9.26, horizontal conformality with dilation φ_ℓ is equivalent to the integrated bilinear identity

$$\int_{\mathcal{C}_i}\sum_x w_x \Delta_F^{\Psi_c}\big(Z_x(c), W_x(c)\big)\,\mathrm{d}\mu_i(c) = (1-\varphi_\ell^2)\mathsf{G}_\ell(Z,W) \quad \text{for every } Z, W \in \mathcal{H}. \tag{9.65}$$

This is necessary and sufficient at the declared configuration tier. It does not imply the corresponding pointwise identity for a finite basis span: take a base which is the disjoint union of two equal-weight contexts, the shared constant-mean tangent $(1,1)$ in fine fibers $\mathcal{N}(\mu,1)$, and at the two contexts add independent Gaussian noises of variances 3 and $1/3$. The resulting Markov pushforwards have pulled-back location Fisher forms $1/4$ and $3/4$, respectively. Their integrated ratio is $\varphi_\ell^2 = 1/2$, but their pointwise ratios differ, so they are not pointwise conformal with that common dilation. The pointwise conclusion is available only after declaring a decomposable, localizable tangent tier (closed under compactly supported localization, with a full-support base measure), which permits the identity to be tested on every local pair. For a coarse objective defined by exact contraction, $\mathcal{F}_{\ell+1}[Q'] = \inf\{\mathcal{F}_\ell[q] : \mathsf{R}_\ell q = Q'\}$, one has $\mathcal{F}_{\ell+1}(\mathsf{R}_\ell q) \le \mathcal{F}_\ell(q)$ with equality exactly on the attaining set, so Hypothesis 9.27 holds with $\chi_\ell = \mathrm{id}$ there and nowhere else, and its differential consequence needs the orbit to lie in the interior of that set. A structured sufficient route to semiconjugacy for the manuscript's own renormalization histories is to exhibit χ_ℓ and an open U containing the orbit on which Hypothesis 9.27 holds, to prove for the exact-contraction objective that the orbit lies in the interior of the attaining set, and then to verify horizontal conformality of the declared R_ℓ. That route is sufficient only: it is not asserted necessary or minimal, and it does not establish (9.55) for the manuscript's own maps. It is stated in objects the manuscript already declares and is satisfiable in the exhibited tier. [OPEN]

9.11 RG depth is not inference time

A hierarchy of coarse objects may be indexed by

$$Q^{(\ell+1)} = \mathsf{R}_\ell(Q^{(\ell)}), \qquad \ell = 0, 1, 2, \dots. \tag{9.66}$$

The integer ℓ records RG depth. It is not a parameter on an inference history, and the sequence in (9.66) is not a temporal trajectory. A history can instead be present at every level, written $Q^{(\ell)}(r)$, with r labeling positions on its oriented orbit and $\tau^{(\ell)}(r)$ its accumulated Fisher duration at that level. Only when

$$Q^{(\ell+1)}(r) = \mathsf{R}_\ell(Q^{(\ell)}(r)), \tag{9.67}$$

and R_ℓ is realized by a parameter-independent Markov pushforward in the sense of Theorem 9.26 does the same-path clock contraction compare adjacent levels. Attaching that Markov predicate to a configuration arrow is a substantive hypothesis with content, not a typing remark, and it can fail: no parameter-independent Markov kernel realizes the joint mean-and-variance barycenter of two normal laws, although the centered variance barycenter and the fixed-common-variance location barycenter are both realizable. Independently recomputed levelwise histories additionally require Theorem 9.22 together with the noncollapse conditions of Proposition 9.23. [ESTABLISHED]

Proposition 9.29 (Typed non-identification of RG depth, orbit parameter, and duration). Scale depth ℓ, an orbit parameter r, and Fisher duration $\tau^{(\ell)}$ are distinct typed quantities: ℓ is a discrete scale index, r is a chosen oriented parameter on a selected orbit, and τ is metric-relative arc length from a chosen origin. After a metric, origin, orientation, and parameterized path have been chosen, τ can be a function of r. It is nevertheless not canonically determined by an unparameterized orbit or by scale depth: changing the metric rescales it, and changing the origin or orientation changes its parameterized representation. Thus no physical-time or canonical-clock identification follows from these objects. Strict monotonicity of $\tau^{(\ell)}$ does not make it a regular coordinate: with $h_s = 4x^2 dx^2$ one gets $\tau(r) = r^2$, whose inverse is not differentiable at the origin, while a zero-speed subinterval destroys strict monotonicity outright. [ESTABLISHED]

Proof. The non-identification follows from the definitions: only the extra metric, origin, orientation, and parameterized-path data produces a duration function. Metric relativity is homogeneity of the norm in (9.26); the regularity statement is the displayed integral $\int_0^R 2r\mathrm{d}r = R^2$ together with nondifferentiability of the square root at zero. □

Finally, $\tau^{(\ell)}$ is not a function on the contextual base and is not physical time. A regional clock potential requires the exactness and vanishing period conditions of Theorem 9.15, and those can fail locally, as (9.43) shows. No operational identification of $\tau^{(\ell)}$ with a clock reading is made here, and none follows from anything proved above. [NOT-CLAIMED]

Cross-scale covariant first jets, their horizontal-defect term, the unconditional vertical Fisher-defect cocycle, and the exact residual of its base pullback belong to Chapter 8. They compare typed geometry across a scale map; they do not turn the scale index into time. The construction in this chapter therefore yields oriented, parameter-free inference histories and agent-relative information duration while leaving any identification with physical time, any Lorentzian signature, any causal structure, and any canonical choice of connection outside the established theory. [NOT-CLAIMED]

Part II

General Coarse Maps and Renormalization

Chapter 10

General Coarse-Graining

Coarse-graining is not one operation. A normalized Markov channel pushes probability laws forward and obeys data processing. Identification of sample variables precomposes an unnormalized energy and requires a new reference measure and normalizer. A Galerkin map restricts an operator and becomes a propagator only after a reference form and signal maps are supplied. This chapter separates these types before any Gaussian matrix formula is used.

10.1 Markov coarse maps

Definition 10.1 (Coarse channel). For standard Borel spaces X, Y, a coarse channel is a parameter-independent Markov kernel

$$K : \mathsf{X} \rightsquigarrow \mathsf{Y}.$$

It sends $P \in \mathcal{P}(\mathsf{X})$ to

$$(PK)(B) = \int_{\mathsf{X}} K(x, B) P(dx). \tag{10.1}$$

A measurable statistic $c : \mathsf{X} \to \mathsf{Y}$ is the deterministic kernel $K_c(x, B) = \mathbf{1}_B(c(x))$. [DEFINITION]

Proposition 10.2 (Markov kernels form a category). If $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ and $L : \mathsf{Y} \rightsquigarrow \mathsf{Z}$, then

$$(KL)(x, C) = \int_{\mathsf{Y}} L(y, C) K(x, dy) \tag{10.2}$$

is a Markov kernel, composition is associative, the deterministic identity is an identity, and

$$P(KL) = (PK)L. \tag{10.3}$$

[ESTABLISHED]

Proof. Kernel integration gives measurability, and monotone convergence gives countable additivity. Tonelli's theorem gives, for measurable C,

$$[P(KL)](C) = \int_{\mathsf{X}} \int_{\mathsf{Y}} L(y, C) K(x, dy) P(dx) = \int_{\mathsf{Y}} L(y, C)(PK)(dy),$$

which proves (10.3); applying the identity to a third kernel proves associativity. □

A statistical experiment $\mathcal{E} = (\mathsf{X}, \{P_\theta\}_{\theta\in\Theta})$ is therefore sent functorially to $\mathcal{E}K = (\mathsf{Y}, \{P_\theta K\}_{\theta\in\Theta})$. Parameter independence is load bearing: a family K_θ is a parameter-dependent model transformation, not one channel between experiments. [DEFINITION]

10.2 KL contraction, equality, and recovery

Theorem 10.3 (Extended KL data processing). For arbitrary probability laws $P, Q \in \mathcal{P}(\mathsf{X})$ and every Markov kernel $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$,

$$D_{\mathrm{KL}}(PK\|QK) \le D_{\mathrm{KL}}(P\|Q) \qquad \text{in } [0, +\infty]. \tag{10.4}$$

No absolute-continuity or finiteness hypothesis is required. [ESTABLISHED]

Proof. If $P \not\ll Q$, the right side is $+\infty$ and the inequality is immediate. If $P \ll Q$, put $r = dP/dQ$ and form $\mathbb{Q}(dx, dy) = Q(dx)K(x, dy)$. The calculation against bounded test functions gives $d(PK)/d(QK) = \bar{r}$, where $\bar{r}(Y) = \mathbb{E}_{\mathbb{Q}}[r(X) \mid Y]$. For the nonnegative convex generator $\phi_0(t) = t \log t - t + 1$, conditional Jensen and monotone approximation give

$$D_{\mathrm{KL}}(PK\|QK) = \mathbb{E}_{\mathbb{Q}}\phi_0(\bar{r}(Y)) \le \mathbb{E}_{\mathbb{Q}}\phi_0(r(X)) = D_{\mathrm{KL}}(P\|Q),$$

including the case in which the last expectation is $+\infty$. No signed integral is rearranged. □

Theorem 10.4 (Data processing with its exact equality condition). Let $P, Q \in \mathcal{P}(\mathsf{X})$, $P \ll Q$, and $D_{\mathrm{KL}}(P\|Q) < \infty$. For a Markov kernel $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$, form

$$\mathbb{Q}(dx, dy) = Q(dx)K(x, dy), \qquad r = \frac{dP}{dQ}, \qquad \bar{r}(y) = \mathbb{E}_{\mathbb{Q}}[r(X) \mid Y = y].$$

Then

$$\frac{d(PK)}{d(QK)} = \bar{r}, \qquad D_{\mathrm{KL}}(PK\|QK) \le D_{\mathrm{KL}}(P\|Q), \tag{10.5}$$

and equality holds exactly when

$$r(X) = \bar{r}(Y) \quad \mathbb{Q}\text{-almost surely.} \tag{10.6}$$

[ESTABLISHED]

Proof. For bounded measurable f,

$$\int f d(PK) = \mathbb{E}_{\mathbb{Q}}[f(Y)r(X)] = \mathbb{E}_{\mathbb{Q}}[f(Y)\bar{r}(Y)],$$

which proves the Radon–Nikodym formula. Conditional Jensen for the strictly convex function $\phi(t) = t \log t$ gives

$$\mathbb{E}_{\mathbb{Q}}\phi(\bar{r}(Y)) \le \mathbb{E}_{\mathbb{Q}}\phi(r(X)).$$

Finite divergence and strict convexity make equality equivalent to (10.6). This is the information-loss theorem of Csiszár [22], Kullback and Leibler [44] in kernel form. □

Corollary 10.5 (Pairwise Bayes recovery). Let $R_Q(y, dx)$ be a regular conditional law of X given $Y = y$ under $\mathbb{Q}$. Then $QKR_Q = Q$, and under (10.6), $PKR_Q = P$. Conversely, any one kernel R recovering both P and Q forces equality in (10.5). [ESTABLISHED]

Proof. Disintegration gives the first identity. Under equality,

$$(PKR_Q)(A) = \int \bar{r}(y)R_Q(y, A)(QK)(dy) = \iint \mathbf{1}_A(x)r(x)\mathbb{Q}(dx, dy) = P(A).$$

The converse follows by applying data processing through K and then R. □

Corollary 10.6 (Infinite equality carries no recovery conclusion). The finiteness hypothesis of Theorem 10.4 cannot be removed from its equality characterization. Equality $D_{\mathrm{KL}}(PK\|QK) = D_{\mathrm{KL}}(P\|Q) = +\infty$ alone need not imply a common reverse kernel recovering P and Q. [ESTABLISHED]

Proof. Let X $= \{a, b, c\}$, Y $= \{u, v\}$, and let the deterministic channel send a to u and both b, c to v. Set

$$P = \tfrac{1}{2}\delta_a + \tfrac{1}{2}\delta_b, \qquad Q = \tfrac{1}{2}\delta_b + \tfrac{1}{2}\delta_c.$$

Both fine and coarse divergences $D_{\mathrm{KL}}(P\|Q)$ and $D_{\mathrm{KL}}(PK\|QK)$ are $+\infty$, because $P(a) > 0 = Q(a)$ and $(PK)(u) > 0 = (QK)(u)$. If one reverse kernel R recovered both, $QK = \delta_v$ would force $R(v, \cdot) = Q$. But then $(PK)R = \frac{1}{2}R(u, \cdot) + \frac{1}{2}Q$ assigns at least $1/4$ to c, contradicting $P(c) = 0$. □

This recovery is pairwise. It may depend on the chosen reference Q. For a dominated experiment, one common dominating probability P_0 and simultaneous finite equalities

$$D_{\mathrm{KL}}(P_\theta K\|P_0 K) = D_{\mathrm{KL}}(P_\theta\|P_0) \quad (\theta \in \Theta)$$

make the single Bayes kernel R_{P_0} recover every P_θ. Without those simultaneous hypotheses, experiment-level sufficiency means directly that one parameter-independent R satisfies $P_\theta KR = P_\theta$ for all θ. Pairwise equality does not prove that stronger statement. [ESTABLISHED]

10.3 Fisher contraction is local

Theorem 10.7 (Score projection and Fisher loss). Let $\{P_\theta\}_{\theta\in\Theta\subseteq\mathbb{R}^d}$ be differentiable in quadratic mean at θ_0, with score $\ell_{\theta_0} \in L^2_0(P_{\theta_0}; \mathbb{R}^d)$, and let K be a parameter-independent Markov kernel normalized for every input. Under the joint law $J_{\theta_0}(dx, dy) = P_{\theta_0}(dx)K(x, dy)$, the coarse family is differentiable in quadratic mean with score

$$\bar{\ell}_{\theta_0}(Y) = \mathbb{E}_{J_{\theta_0}}[\ell_{\theta_0}(X) \mid \sigma(Y)] \quad (P_{\theta_0}K)\text{-almost surely,} \tag{10.7}$$

Its Fisher information obeys

$$I_{\mathsf{Y}}(\theta_0) \preceq I_{\mathsf{X}}(\theta_0), \qquad I_{\mathsf{X}} - I_{\mathsf{Y}} = \mathbb{E}\,\mathrm{Cov}(\ell_{\theta_0}(X) \mid Y) \succeq 0. \tag{10.8}$$

Equality holds exactly when the score is Y-measurable. [ESTABLISHED]

Proof. Write Pollard's Le Cam DQM decomposition at θ_0 as $P_{\theta_0+h} = p_h P_{\theta_0} + P_h^{\perp}$, where $P_h^{\perp} \perp P_{\theta_0}$, $P_h^{\perp}(\mathsf{X}) = o(\|h\|^2)$, and

$$\int \left(\sqrt{p_h} - 1 - \tfrac{1}{2}h^{\top}\ell_{\theta_0}\right)^2 dP_{\theta_0} = o(\|h\|^2).$$

Choose a P_{θ_0}-null set A_h supporting $P_h^{\perp}$. The lifted joint family has the decomposition

$$J_{\theta_0+h} = p_h(x)J_{\theta_0} + J_h^{\perp}, \qquad J_h^{\perp} := P_h^{\perp}K.$$

The singular term is supported on $A_h \times \mathsf{Y}$, which is J_{θ_0}-null because the kernel is normalized, and its mass is $J_h^{\perp}(\mathsf{X} \times \mathsf{Y}) = P_h^{\perp}(\mathsf{X}) = o(\|h\|^2)$. The squared square-root remainder of the absolutely continuous term is unchanged by the lift, since integration against J_{θ_0} integrates the kernel to one. Thus the joint experiment is DQM with score $\ell_{\theta_0}(X)$, without a common dominating measure. Applying Pollard's preservation theorem for measurable statistics [65] to the deterministic statistic $(x, y) \mapsto y$ gives DQM of $P_\theta K$ and the displayed conditional score as an $L^2(P_{\theta_0}K; \mathbb{R}^d)$ equivalence class. Although the projected singular term can acquire an absolutely continuous part, its total mass remains $o(\|h\|^2)$. Taking second moments and using total covariance proves (10.8). Conditional expectation is an orthogonal projection, so equality holds precisely when the fine score is Y-measurable; in a direction v, equality holds precisely when $v^{\top}\ell_{\theta_0}(X)$ is Y-measurable. □

When the fine and coarse experiments form smooth statistical law bundles, the Markov channel is realized by a bundle morphism satisfying the exact isotropy criterion (8.56), and the fine and coarse sections satisfy the related-section condition (8.44), Theorem 8.21 pulls this same score-variance defect back to a positive-semidefinite defect between contextual-base tensors. Along one declared inference path, the corresponding record-level Fisher-clock contraction is Theorem 9.16. [ESTABLISHED]

Neither statement proves that a natural-gradient flow recomputed after coarse-graining traces the image of the fine orbit; that stronger comparison requires the oriented semiconjugacy conditions of Theorem 9.22. [NOT-CLAIMED]

Fisher equality at one parameter is local score sufficiency, not global recovery. Let $X = (A, B)$ with independent Bernoulli coordinates,

$$\Pr_{\theta}(A = 1) = \tfrac{1}{2} + \tfrac{\theta}{4}, \qquad \Pr_{\theta}(B = 1) = \tfrac{1}{2} + \tfrac{\theta^2}{4},$$

and let K discard B. At $\theta = 0$, the B-score vanishes, so the fine and coarse Fisher informations agree and are nonzero. Yet $\Pr_\theta(B = 1 \mid A)$ depends on θ, so no parameter-independent reverse kernel can recover the whole experiment. [ESTABLISHED]

10.4 The fixed-evidence ELBO consequence

Theorem 10.8 (Evidence-preserving coarse channel). Let O, X, and Y be standard Borel. Fix one normalized joint $P(do, dx)$ and fix a measurable regular-conditional kernel $o \mapsto P_o$ of X given O under P. Let $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ be a Markov channel satisfying $K(x, \mathsf{Y}) = 1$ for every $x \in \mathsf{X}$, and suppose that it does not read the arbitrary recognition law Q_o. Define

$$\bar{P}(do, dy) = \int K(x, dy)P(do, dx), \qquad \bar{P}_o := P_oK \text{ for every } o, \qquad \bar{Q}_o := Q_oK. \tag{10.9}$$

The declared kernel $o \mapsto \bar{P}_o$ is a regular conditional law of Y given O under $\bar{P}$, including its selected exceptional-point values. Fix a selected regular observation o and use the same declared evidence density representative $p(o)$ for the fine and coarse slices. Then $\bar{P}^O = P^O$ and the extended ELBO of Definition 5.6 satisfies

$$\bar{\mathcal{L}}^{\text{ext}}(Q_oK; o) = \log p(o) - D_{\text{KL}}(Q_oK \| P_oK) \geq \log p(o) - D_{\text{KL}}(Q_o \| P_o) = \mathcal{L}^{\text{ext}}(Q_o; o). \tag{10.10}$$

[ESTABLISHED]

Proof. Normalization of K preserves the observation marginal. Kernel integration makes $o \mapsto (P_oK)(B)$ measurable for every measurable $B \subseteq \mathsf{Y}$. Moreover, for measurable $A \subseteq \mathsf{O}$ and $B \subseteq \mathsf{Y}$,

$$\int_A (P_oK)(B)P^O(do) = \int_A \int_{\mathsf{X}} K(x, B)P_o(dx)P^O(do) = \bar{P}(A \times B).$$

A π–λ extension proves the regular-conditional identity. Thus the pointwise declaration $\bar{P}_o := P_oK$ selects a compatible version rather than applying an almost-sure identity at an arbitrarily fixed exceptional observation. Since the same evidence representative is used and $\bar{P}^O = P^O$, Equation (10.10) follows pointwise for the selected versions from Theorem 10.3; when $Q_o \not\ll P_o$ its rightmost value is $-\infty$, and the inequality remains valid without any undefined sum. □

Thus different latent inventories are not by themselves a prohibition. ELBOs are comparable under an evidence-preserving common Markov pushforward applied to the posterior and recognition law. Absent that relation, separately declared models may have different evidences, and an ELBO difference mixes evidence with variational tightness. [ESTABLISHED]

The increase in (10.10) reflects discarded information; it is not model improvement or EM evidence ascent. [NOT-CLAIMED]

10.5 Pushforward, energy precomposition, and identification

Let $\iota : \mathsf{Z} \to \mathsf{X}$ be a measurable injection. The formula

$$\bar{E}(z) = E(\iota z), \qquad \bar{P}(dz) = \frac{e^{-\bar{E}(z)}}{\bar{Z}} \bar{\nu}(dz) \tag{10.11}$$

defines a new model only after $\bar{\nu}$ and a finite $\bar{Z}$ are declared. It is not a channel from X to Z, and its pushforward can be supported on a fine-measure-null image. Therefore KL data processing, evidence preservation, and (10.10) do not follow. [DEFINITION]

Exact marginalization is a special Markov pushforward. Hard identification is energy precomposition. Restricting a recognition family is neither: it keeps the fixed joint and optimizes the same ELBO over a subset. A proof about any one of these operations cannot be transferred to another by calling all three coarse-graining. [ESTABLISHED]

10.6 Graph-exponential energy closure

Definition 10.9 (Graph-exponential energy). Let V be finite, Z_0 standard Borel with sigma-finite ν_0, and let $t : \mathsf{Z}_0 \to \mathsf{T}_0$ and $u : \mathsf{Z}_0^2 \to \mathsf{T}_1$ be measurable, with u symmetric. Set

$$\mathcal{E}_\theta(z) = \sum_{i \in V} \langle \alpha_i, t(z_i) \rangle + \sum_{\{i,j\}} \langle \beta_{ij}, u(z_i, z_j) \rangle. \tag{10.12}$$

This is an unnormalized energy until its integral relative to $\nu_0^{\otimes V}$ is finite. [DEFINITION]

For a partition $\mathcal{P}$, define

$$\iota_{\mathcal{P}} : \mathsf{Z}_0^{\mathcal{P}} \to \mathsf{Z}_0^V, \qquad (\iota_{\mathcal{P}} \bar{z})_i = \bar{z}_{I(i)}.$$

Assume the diagonal-affinity hypothesis

$$u(z, z) = \Upsilon t(z) + u_0 \tag{10.13}$$

for a linear $\Upsilon : \mathsf{T}_0 \to \mathsf{T}_1$. [HYPOTHESIS]

This excludes pair statistics whose diagonal restriction cannot be represented by the declared node statistic. [ESTABLISHED]

Theorem 10.10 (Graph-exponential trace closure). Under (10.13),

$$\mathcal{E}_\theta(\iota_{\mathcal{P}} \bar{z}) = \sum_I \langle \alpha_I, t(\bar{z}_I) \rangle + \sum_{I<J} \langle \beta_{IJ}, u(\bar{z}_I, \bar{z}_J) \rangle + c_{\mathcal{P}}(\theta), \tag{10.14}$$

where

$$\alpha_I = \sum_{i\in I} \alpha_i + \Upsilon^* \sum_{\{i,j\}\subseteq I} \beta_{ij}, \qquad \beta_{IJ} = \sum_{\substack{i\in I\\ j\in J}} \beta_{ij}, \qquad c_{\mathcal{P}}(\theta) = \left\langle \sum_I \sum_{\{i,j\}\subseteq I} \beta_{ij}, u_0 \right\rangle. \tag{10.15}$$

The induced parameter map is linear. A normalized coarse law exists if and only if

$$\bar{Z}_{\mathcal{P}}(\theta) = \int e^{\mathcal{E}_\theta(\iota_{\mathcal{P}}\bar{z})} \nu_0^{\otimes \mathcal{P}}(d\bar{z}) < \infty. \tag{10.16}$$

[ESTABLISHED]

Proof. Substitute the identified variables into (10.12). Cut edges sum into β_{IJ}. Internal edges use (10.13); the linear part joins α_I and the constant joins $c_{\mathcal{P}}$. This gives (10.14)–(10.15). The proof is algebraic and does not prove (10.16). □

If $u(z, z)$ is constant, one may choose $\Upsilon = 0$, so internal edges are annihilated and node parameters add. Translation invariance implies this constant-diagonal property, but the converse is false because diagonal data do not determine u off the diagonal. Uniqueness of the coordinate choice $\Upsilon = 0$ requires minimal effective node statistics or quotienting null parameter directions. [ESTABLISHED]

This theorem states conditional energy closure, not normalized-law closure. Its reference measure, normalizer, and any gauge action must be propagated separately. A compatible normalized family must satisfy

$$P_\theta K = \bar{P}_{\rho(\theta)}$$

for a declared channel K and parameter map ρ; energy precomposition alone does not prove that equation. [HYPOTHESIS]

10.7 Galerkin operators and message passing

Let $P : V_c \to V_f$ be injective, let $R \succ 0$ be a fine reference form, and let L be a fine bilinear form. Define

$$R_c = P^\top RP, \qquad C = R_c^{-1} P^\top R, \qquad H = R^{-1} L, \qquad H_c = CHP = R_c^{-1} P^\top LP. \tag{10.17}$$

Then $CP = I$. The form $P^\top LP$ is the Galerkin coarse form, but H_c is the typed coarse propagator. [DEFINITION]

Proposition 10.11 (Exact message-passing residual). The reduction and propagation obey

$$CH - H_c C = CH(I - PC). \tag{10.18}$$

Hence exact coarse message passing holds on all fine signals exactly when the right side vanishes. If H is self-adjoint in the R-inner product, $\operatorname{range} P$ being H-invariant is a sufficient and, for orthogonal projection PC, equivalent condition. [ESTABLISHED]

Proof. Substitute $H_c = CHP$ and use $CP = I$:

$$CH - H_c C = CH - CHPC = CH(I - PC).$$

For an R-self-adjoint operator, invariance of a closed subspace is equivalent to reduction by its R-orthogonal projector. □

Thus exact quadratic-form restriction, generalized spectral approximation, and message-passing preservation are different statements. The first follows from Galerkin algebra; the other two require additional invariant-subspace or error hypotheses. A neural pooling layer also needs a signal map and a partition rule. No GNN or oversmoothing theorem follows from energy closure alone. [NOT-CLAIMED]

10.8 Gauge and permutation equivariance

Proposition 10.12 (Equivariant Markov channels). Let a group G act bimeasurably on X, Y, and assume

$$K(gx, B) = K(x, g^{-1}B). \tag{10.19}$$

Then

$$(g_\# P)K = g_\#(PK), \tag{10.20}$$

and composites of equivariant kernels are equivariant. [ESTABLISHED]

Proof. For measurable B,

$$[(g_\# P)K](B) = \int K(gx, B)P(dx) = \int K(x, g^{-1}B)P(dx) = [g_\#(PK)](B).$$

Substitution in (10.2) proves closure under composition. □

For independently chosen belief and model frame coordinates, the passive coordinate group is $G \times G$, although the underlying principal bundle has structure group G. A joint coarse channel must intertwine the two represented coordinate actions. Under a genuine principal gauge transformation, those actions are the ρ_b- and ρ_m-images of the same element of G. KL equality does not automatically make its Bayes recovery equivariant; an equivariant conditional version is an additional hypothesis or theorem, especially for noncompact groups. [OPEN]

A data-dependent hard partition selector $S(X)$ must also be natural under node relabeling. If P permutes fine nodes and Q permutes coarse labels,

$$S(P \cdot X) = PS(X)Q^\top \tag{10.21}$$

is the required law. Gauge-carrying inputs must be reduced to invariant assignment scores, while represented messages transform equivariantly. These conditions do not select a unique partition; they only type an admissible selector. Constructing a selector satisfying them and the later holonomy conditions remains open. [OPEN]

10.9 Holonomy-conditioned marginal-law modes

Fix a channel $x \in \{b, m\}$, a finite connected cluster I, and one declared internal path groupoid. The paths may be edge-labeled graph paths for typed graph links or base curves for the selected channel connection; the two cases are not identified. Let each represented transport $T^x_\gamma : \mathsf{Z}^x_i \to \mathsf{Z}^x_j$, for $\gamma : i \to j$, be a bimeasurable bijection satisfying composition and inversion. Assume that the parent-law family $\mathscr{M}^x_i \subseteq \mathcal{P}(\mathsf{Z}^x_i)$ is equivariant: $(T^x_\gamma)_\# \mathscr{M}^x_i = \mathscr{M}^x_j$. Under a passive section rechoice $u^{x\prime}_i = u^x_i a^x_i$, write $R^x_i = \rho_x(a^x_i)^{-1}$ for the old-to-new sample-coordinate map; equivalently, the coordinate element is $g^x_i = (a^x_i)^{-1}$ and $R^x_i = \rho_x(g^x_i)$. Also assume that $\mathscr{M}^{x\prime}_i = (R^x_i)_\# \mathscr{M}^x_i$. [HYPOTHESIS]

Bimeasurability excludes lossy transports, and family equivariance excludes parent classes tied to one root or frame; both hypotheses are used in the invariance proof below. [ESTABLISHED]

Choose a root $r \in I$, paths $\gamma_i : i \to r$, marginal laws $q^x_i \in \mathcal{P}(\mathsf{Z}^x_i)$, and weights $w_i > 0$ with $\sum_{i \in I} w_i = 1$. Define

$$P^x_i = (T^x_{\gamma_i})_\# q^x_i, \qquad \mathfrak{H}^x_I(r) = \{T^x_\lambda : \lambda : r \to r\}, \tag{10.22}$$

$$\mathscr{Q}^x_{I,\mathrm{fix}}(r) = \{Q \in \mathscr{M}^x_r : (H)_\# Q = Q \text{ for every } H \in \mathfrak{H}^x_I(r)\}, \tag{10.23}$$

and the extended-real forward-KL score

$$\mathfrak{D}^x_I = \inf_{Q \in \mathscr{Q}^x_{I,\mathrm{fix}}(r)} \sum_{i \in I} w_i D_{\mathrm{KL}}(P^x_i \| Q), \qquad \inf \varnothing := +\infty. \tag{10.24}$$

Any minimizer, when one exists, is a holonomy-conditioned marginal-law mode. These are definitions, not existence assertions. [DEFINITION]

Theorem 10.13 (Invariant holonomy-conditioned marginal-law mode). The value in (10.24) is independent of the chosen paths, root, and passive gauge coordinates. If the infimum is attained, then

$$\mathfrak{D}^x_I = 0 \iff \text{there is } Q_\star \in \mathscr{Q}^x_{I,\mathrm{fix}}(r) \text{ such that } (T^x_{\gamma_i})_\# q^x_i = Q_\star \text{ for every } i \in I. \tag{10.25}$$

The right side is precisely a holonomy-stabilized parallel marginal-law section. If $\mathscr{Q}^x_{I,\mathrm{fix}}(r) = \varnothing$, then $\mathfrak{D}^x_I = +\infty$. If $\mathfrak{D}^x_I = 0$ but the infimum is not attained, the conclusion is only forward-KL approximability by admissible fixed parent laws; no member $Q_\star$ of that family has been produced. [ESTABLISHED]

Proof. For a bimeasurable bijection A, if $P \not\ll Q$, then $A_{\#}P \not\ll A_{\#}Q$, so both divergences below are infinite. If $P \ll Q$, then

$$\frac{d(A_{\#}P)}{d(A_{\#}Q)} = \frac{dP}{dQ} \circ A^{-1} \quad A_{\#}Q\text{-almost surely,} \tag{10.26}$$

and integration gives

$$D_{\mathrm{KL}}(A_{\#}P \| A_{\#}Q) = D_{\mathrm{KL}}(P \| Q), \tag{10.27}$$

as an identity in $[0, +\infty]$. If γ_i' is another path from i to r, then $H_i = T^x_{\gamma_i'} \circ (T^x_{\gamma_i})^{-1}$ belongs to $\mathfrak{H}^x_I(r)$. Hence, for every fixed parent Q,

$$D_{\mathrm{KL}}((H_i)_{\#}P^x_i \| Q) = D_{\mathrm{KL}}((H_i)_{\#}P^x_i \| (H_i)_{\#}Q) = D_{\mathrm{KL}}(P^x_i \| Q). \tag{10.28}$$

Thus every objective value, and therefore its infimum, is path independent.

For a new root r' and a path $\delta : r \to r'$, transport conjugates $\mathfrak{H}^x_I(r)$ onto $\mathfrak{H}^x_I(r')$. Equivariance of $\mathscr{M}^x$ therefore makes $Q \mapsto (T^x_\delta)_{\#}Q$ a bijection between the two fixed parent families, while the paths $\delta \circ \gamma_i$ send every P^x_i by that same bijection. Equation (10.27) preserves every term. Path independence then covers arbitrary paths to the new root.

Under the same passive section rechoices, use the old-to-new represented maps $R^x_i = \rho_x(a^x_i)^{-1}$ declared above. Then

$$T^{x\prime}_{\gamma_i} = R^x_r \circ T^x_{\gamma_i} \circ (R^x_i)^{-1}, \qquad q^{x\prime}_i = (R^x_i)_{\#}q^x_i. \tag{10.29}$$

Consequently all transported marginals are pushed forward by R^x_r, the holonomy group is conjugated by the same map, and parent-family equivariance carries the fixed family bijectively. A final application of (10.27) proves gauge invariance.

Relative entropy is nonnegative and vanishes exactly when its two probability measures agree. If a minimizer $Q_\star$ exists, positivity of every w_i makes its weighted sum zero exactly when $P^x_i = Q_\star$ for all i, which proves (10.25). Holonomy-fixity makes the reconstructed laws $((T^x_{\gamma_i})^{-1})_{\#}Q_\star$ independent of path, so they form the stated parallel section. The empty-family assertion is the convention in (10.24). Finally, if the zero infimum is not attained, its definition supplies, for every $n \geq 1$, some $Q_n \in \mathscr{Q}^x_{I,\mathrm{fix}}(r)$ with $\sum_i w_i D_{\mathrm{KL}}(P^x_i \| Q_n) < 1/n$, and supplies no minimizer. □

The score matches marginal laws after transport. The cluster and its weights are inputs, so it does not select a partition. It supplies neither a Markov kernel nor a normalized coarse channel, and it does not recover the dependence structure of a joint law. These stronger conclusions are not claimed. [NOT-CLAIMED]

For a statistical experiment $\{P_\theta\}_{\theta\in\Theta}$, exact common recovery instead requires parameter-independent Markov kernels C and R satisfying

$$P_\theta C R = P_\theta \qquad \text{for every } \theta \in \Theta. \tag{10.30}$$

Here C, R are local kernel notation, not the Galerkin reduction and reference form in (10.17). This is the common-recovery condition, not a consequence of (10.24). [DEFINITION]

Under the richness hypothesis of Proposition 3.18, distinct joint laws can have identical coordinate marginals. Identical transported marginals therefore do not imply (10.30). [ESTABLISHED]

Corollary 10.14 (Parallel cross morphisms preserve fixed marginal-law sections). Let $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ be a cross-associated-bundle morphism on law fibers that is parallel for the chosen path system. Writing $\widehat{T}^x_\gamma = (T^x_\gamma)_\#$, this means

$$\widehat{T}^m_\gamma \circ \Phi_i = \Phi_j \circ \widehat{T}^b_\gamma \qquad (\gamma : i \to j). \tag{10.31}$$

For base-connection paths this is ordinary parallelness. For independently declared graph paths, (10.31) is an additional link-intertwining hypothesis and does not follow from smooth parallelness. Assume also $\Phi_r(\mathscr{M}^b_r) \subseteq \mathscr{M}^m_r$. [HYPOTHESIS]

Then

$$\Phi_r\big(\mathscr{Q}^b_{I,\mathrm{fix}}(r)\big) \subseteq \mathscr{Q}^m_{I,\mathrm{fix}}(r). \tag{10.32}$$

Moreover, applying Φ_i fiberwise to a holonomy-stabilized parallel belief-law section gives such a model-law section. If $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$ is likewise parallel for the chosen path system and $\widetilde{\Phi}_r(\mathscr{M}^m_r) \subseteq \mathscr{M}^b_r$, then

$$\widetilde{\Phi}_r\big(\mathscr{Q}^m_{I,\mathrm{fix}}(r)\big) \subseteq \mathscr{Q}^b_{I,\mathrm{fix}}(r), \tag{10.33}$$

and the analogous model-to-belief statement holds. [ESTABLISHED]

Proof. If $Q \in \mathscr{Q}^b_{I,\mathrm{fix}}(r)$ and $\lambda : r \to r$, then

$$\widehat{T}^m_\lambda(\Phi_r(Q)) = \Phi_r(\widehat{T}^b_\lambda(Q)) = \Phi_r(Q). \tag{10.34}$$

Parent-family compatibility proves (10.32). If the belief marginals transport to Q, the same commutative square shows that their Φ_i-images transport to $\Phi_r(Q)$. Interchanging b, m and replacing Φ by $\widetilde{\Phi}$ proves (10.33). □

This corollary does not identify $\mathcal{E}_b$ with $\mathcal{E}_m$, does not require equal fiber dimensions or equivalent representations ρ_b, ρ_m. No KL-preservation hypothesis is imposed, so no equality or ordering between $\mathfrak{D}^b_I$ and $\mathfrak{D}^m_I$ is claimed. Existence, parallelness, and parent-family compatibility of each cross morphism are additional structure. [NOT-CLAIMED]

10.10 Projective systems of laws

Theorem 10.15 (Cylinder-level projective law). Let $\mathcal{I}$ be an index set and each coordinate space Z_i standard Borel. For every finite $D \subset \mathcal{I}$, let P_D be a probability law on $\mathsf{Z}_D = \prod_{i \in D} \mathsf{Z}_i$. If

$$(\pi_{E,D})_\# P_E = P_D \qquad (D \subset E), \tag{10.35}$$

then there is a unique law on the product cylinder sigma-algebra with finite-dimensional marginals P_D. Compatible finite-dimensional gauge equivariance passes to the limit. [ESTABLISHED]

Proof. This is the Kolmogorov extension theorem for standard Borel coordinate spaces [39]. The transformed and original laws have identical finite-dimensional marginals under compatible equivariance, so uniqueness gives the last assertion. □

Consistency alone does not put the limit on continuous or smooth sections, define a continuum reference density, justify an ELBO limit, or produce an RG flow. Those conclusions require support regularity, compatible kernels and rescalings, convergence or uniform integrability of the functionals, and a declared topology. [OPEN]

Physicist summary. A general coarse-graining is a noisy detector. Detectors compose, and they cannot increase KL distinguishability or Fisher information. Equality in KL means that a reverse detector recovers the particular laws; equality in Fisher means only that the infinitesimal score survived. Restricting an energy to block-constant fields is a different construction: it is a new model until its measure and normalizer are supplied. Galerkin restriction is exact for the quadratic form, but propagation needs a mass form and has the explicit residual (10.18). Holonomy determines the admissible parallel marginal-law modes. When the infimum is attained, transported forward KL tests whether the actual marginals occupy one; without attainment, a zero infimum establishes only approximability. Neither construction selects a partition or proves joint-law recovery. These distinctions are the general spine that the later multivariate-Gaussian chapters realize.

Chapter 11

General Renormalization

Coarse arrows alone form a scale diagram. They become a renormalization dynamics only after comparison and rescaling data identify what is being compared across levels. In the present theory the state includes one principal gauge bundle, two associated statistical bundles, their possibly distinct connections, and both cross-bundle morphisms. A scalar normalization of one operator cannot stand for that full state.

11.1 The scale category and comparison data

Definition 11.1 (Scale diagram). Let S be a category whose objects are resolutions ℓ. Choose a target category $\mathscr{K}$: sets and maps for a deterministic state, standard Borel spaces and Markov kernels for a stochastic state, an explicitly declared category of topological vector spaces and continuous linear maps for an operator state, or an explicitly declared product of such categories for a mixed state. In the analytic tier used below, the operator objects are Banach spaces and the morphisms are bounded linear maps. A state functor $F : \mathsf{S} \to \mathscr{K}$ assigns an object $\mathfrak{X}_\ell = F(\ell)$ and morphisms

$$C_{\ell k} \in \operatorname{Hom}_{\mathscr{K}}(\mathfrak{X}_\ell, \mathfrak{X}_k), \qquad C_{kr} \circ C_{\ell k} = C_{\ell r}. \tag{11.1}$$

Here $\circ$ denotes abstract composition in $\mathscr{K}$, with componentwise composition in a product category. Separately, for the stochastic component only, use right-acting kernel juxtaposition

$$K_{\ell r} := K_{\ell k} K_{kr}, \qquad (\mu K_{\ell k}) K_{kr} = \mu K_{\ell r}, \tag{11.2}$$

this is the kernel representing the abstract composite $K_{kr} \circ K_{\ell k}$. One symbol is never asked to be simultaneously a map, a kernel, and an operator. [DEFINITION]

Comparison requires more data. Either choose levelwise re-embeddings $\jmath_\ell : \mathfrak{X}_{\ell+1} \to \mathfrak{X}_\ell$, or identify each level with a reference object by declared isomorphisms $I_\ell : \mathfrak{X}_\ell \xrightarrow{\sim} \mathfrak{X}_\star$ in $\mathscr{K}$. Field, base, unit, and reference-measure rescalings are part of these maps. If an externally parameterized dynamical model is separately supplied, its parameter rescaling also belongs here; the scale diagram itself introduces no time variable, and RG depth is not the intrinsic inference duration of Chapter 9. Only the isomorphism option supplies the inverse used in the reference-space transformations

$$\widehat{\mathcal{R}}_\ell = I_{\ell+1} C_{\ell,\ell+1} I_\ell^{-1} : \mathfrak{X}_\star \longrightarrow \mathfrak{X}_\star. \tag{11.3}$$

[DEFINITION]

A positive scalar ζ_ℓ can be one component of this comparison; it cannot by itself identify changing fields, measures, or bundle objects. [ESTABLISHED]

If probability laws are present, every scale declares its sample space and reference measure. A law-level arrow is a normalized Markov kernel as in Chapter 10. An energy-level arrow additionally owes the coarse reference measure, its Radon–Nikodym or Jacobian rule, and a finite normalizer. Algebraic closure does not supply those data. [HYPOTHESIS]

11.2 Exact measure-pair arrows

Definition 11.2 (Scale-indexed measure pair). For each scale ℓ, let $(\mathsf{X}_\ell, \mathscr{X}_\ell)$ be a standard-Borel space, let ρ_ℓ be a probability measure, and let $m_\ell \ll \rho_\ell$ be a finite positive measure. Write

$$m_\ell(dx) = \exp[-H_\ell(x)]\rho_\ell(dx), \qquad 0 < M_\ell := m_\ell(\mathsf{X}_\ell) < \infty, \qquad \pi_\ell = \frac{m_\ell}{M_\ell}. \tag{11.4}$$

The extended-value convention is $\exp(-\infty) = 0$; thus the action is defined only ρ_ℓ-almost everywhere and no finite action is silently assumed. A scale arrow is a normalized, parameter-independent Markov kernel

$$K_\ell : \mathsf{X}_\ell \rightsquigarrow \mathsf{X}_{\ell+1}, \qquad \rho_{\ell+1} = \rho_\ell K_\ell, \qquad m_{\ell+1} = m_\ell K_\ell. \tag{11.5}$$

The downstream action is the Radon–Nikodym representative

$$H_{\ell+1} = -\log \frac{dm_{\ell+1}}{d\rho_{\ell+1}} \quad \rho_{\ell+1}\text{-almost everywhere.} \tag{11.6}$$

This is a definition of an exact law-level RG arrow. The requirement that the spaces be standard Borel is used later to choose regular conditional versions; it does not identify distinct scale spaces or produce an autonomous operator. [DEFINITION]

Proposition 11.3 (Exact composition and normalized law). For two composable normalized kernels K_ℓ and $K_{\ell+1}$, the measure-pair arrows compose exactly:

$$(\rho_\ell, m_\ell)(K_\ell K_{\ell+1}) = \big((\rho_\ell K_\ell)K_{\ell+1}, (m_\ell K_\ell)K_{\ell+1}\big). \tag{11.7}$$

Moreover $m_{\ell+1} \ll \rho_{\ell+1}$, $m_{\ell+1}(\mathsf{X}_{\ell+1}) = M_\ell$, and

$$\pi_{\ell+1} = \pi_\ell K_\ell. \tag{11.8}$$

Thus the normalized law and the unnormalized evidence mass are distinct components of the same exact construction. [ESTABLISHED]

Proof. Kernel composition and Tonelli's theorem give (11.7). If $(\rho_\ell K_\ell)(A) = 0$, then $K_\ell(x, A) = 0$ for ρ_ℓ-almost every x, hence also for m_ℓ-almost every x, which proves absolute continuity at the next scale. Normalization of K_ℓ preserves the mass of m_ℓ; division by that same mass gives (11.8). □

The subsequent finite-network realization takes each X_ℓ to be a finite product of agent coordinate spaces and supplies the action calculus, contraction theorems, and interaction content that this general arrow does not itself supply. The exact nonlinear interaction-coordinate map, its tilted derivative, and its retained-space residual are defined only after the additional product-equivalence premises of Section 12.3; they are not linear conditional-expectation maps away from the unperturbed action. In particular, a scale sequence is a cocycle until the comparison data of Section 11.1 identify its varying spaces. [ESTABLISHED]

11.3 The common-principal geometric state

Definition 11.4 (Geometric RG state). At level ℓ, declare a smooth base $\mathcal{C}_\ell$, a Lie group G_ℓ, and one principal right G_ℓ-bundle $\varpi_\ell : \mathscr{P}_\ell \to \mathcal{C}_\ell$. Let $\mathcal{B}_{\ell,b}$ and $\mathcal{B}_{\ell,m}$ be statistical fibers with possibly different left actions $\widehat{\rho}_{\ell,b}$ and $\widehat{\rho}_{\ell,m}$ of the same group. The belief and model bundles are

$$\mathcal{E}_{\ell,b} = \mathscr{P}_\ell \times_{\widehat{\rho}_{\ell,b}} \mathcal{B}_{\ell,b}, \qquad \mathcal{E}_{\ell,m} = \mathscr{P}_\ell \times_{\widehat{\rho}_{\ell,m}} \mathcal{B}_{\ell,m}. \tag{11.9}$$

Choose two principal connections $\omega_{\ell,b}, \omega_{\ell,m} \in \Omega^1(\mathscr{P}_\ell, \mathfrak{g}_\ell)$. They induce horizontal distributions H^b_ℓ, H^m_ℓ and parallel transports $\Omega_{\ell,\gamma}$ and $\widetilde{\Omega}_{\ell,\gamma}$ on the two associated bundles. Thus the general geometric state contains

$$\mathfrak{X}_\ell = \Big(\mathcal{C}_\ell;\ G_\ell, \mathscr{P}_\ell;\ \mathcal{E}_{\ell,b}, H^b_\ell;\ \mathcal{E}_{\ell,m}, H^m_\ell;\ \Phi_\ell, \widetilde{\Phi}_\ell;\ \ldots\Big). \tag{11.10}$$

Here

$$\Phi_\ell : \mathcal{E}_{\ell,b} \to \mathcal{E}_{\ell,m}, \qquad \widetilde{\Phi}_\ell : \mathcal{E}_{\ell,m} \to \mathcal{E}_{\ell,b}$$

are separately declared associated-bundle morphisms over the identity. They need not be invertible or mutual inverses. A fixed fiber map $f^{bm}_\ell : \mathcal{B}_{\ell,b} \to \mathcal{B}_{\ell,m}$ induces such a morphism only when it intertwines the actions,

$$f^{bm}_\ell \circ \widehat{\rho}_{\ell,b}(g) = \widehat{\rho}_{\ell,m}(g) \circ f^{bm}_\ell \quad \text{for every } g \in G_\ell, \tag{11.11}$$

and analogously in the reverse direction. Hence a common principal bundle does not manufacture either cross map. The ellipsis in (11.10) denotes law, kernel, or operator components declared by a realization. [DEFINITION]

When levelwise agent or meta-agent sections and regular vertical Fisher tensors are also declared, the geometric state may include their connection-relative perceived semimetrics. These components are not supplied by the associated bundles alone. If a scale arrow also supplies a bundle morphism, related fine and coarse sections, and a vanishing horizontal defect, their exact cross-scale naturality is Corollary 8.22. Its positive-semidefinite information-loss defect further requires that the fiber map be a normalized parameter-independent Markov channel. The discrete or continuous beta is defined only after the reference-base identifications in (8.67). Thus a meta-agent has its own pullback geometry, but no tensor on two different bases is subtracted before transport to the common reference space. [ESTABLISHED]

Local belief and model frames may nevertheless be chosen separately. On a common trivializing neighborhood, two sections $u^b_{\ell,i}, u^m_{\ell,i} : U_i \to \mathscr{P}_\ell$ determine a unique smooth relative principal-frame field

$$u^m_{\ell,i} = u^b_{\ell,i} \cdot h_{\ell,i}, \qquad h_{\ell,i} : U_i \to G_\ell. \tag{11.12}$$

Existence and uniqueness follow because each principal fiber is a free and transitive G_ℓ-torsor; the statement is independent of the representations. If the sections are re-chosen as $u^b_{\ell,i}a^b_{\ell,i}$ and $u^m_{\ell,i}a^m_{\ell,i}$, then $h_{\ell,i} \mapsto (a^b_{\ell,i})^{-1}h_{\ell,i}a^m_{\ell,i}$. When the representations differ, $\widehat{\rho}_{\ell,b}(h_{\ell,i})$ and $\widehat{\rho}_{\ell,m}(h_{\ell,i})$ act on different fibers; there is no matrix quotient of the associated frames and no cross-fiber identification without extra intertwining data such as (11.11). If only associated-frame data are given and either action is nonfaithful, the principal element $h_{\ell,i}$ need not be reconstructible without chosen lifts to $\mathscr{P}_\ell$. [ESTABLISHED]

Scale change must also be typed. Let $c_\ell : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$ be smooth, let $\kappa_\ell : G_\ell \to G_{\ell+1}$ be a Lie-group homomorphism, and let $\mathcal{P}_\ell : \mathscr{P}_\ell \to \mathscr{P}_{\ell+1}$ cover c_ℓ and obey

$$\mathcal{P}_\ell(p \cdot g) = \mathcal{P}_\ell(p) \cdot \kappa_\ell(g). \tag{11.13}$$

Existence of such a map is a genuine topological condition on the declared data, not a normalization: an equivariant $\mathcal{P}_\ell$ over c_ℓ exists if and only if the extended bundle $\mathscr{P}_\ell \times_{\kappa_\ell} G_{\ell+1}$ is isomorphic to the pullback $c^*_\ell \mathscr{P}_{\ell+1}$ as a principal $G_{\ell+1}$-bundle over $\mathcal{C}_\ell$. It can fail, for instance for the Hopf bundle over S^2 against the trivial bundle with c_ℓ the identity. This is a scope note on a declared datum rather than a defect of any theorem below, and every statement that uses $\mathcal{P}_\ell$ inherits it. [ESTABLISHED]

For $s \in \{b, m\}$, given a declared κ_ℓ-equivariant $\mathcal{P}_\ell$, a fiber map $q_{\ell,s} : \mathcal{B}_{\ell,s} \to \mathcal{B}_{\ell+1,s}$ descends to a bundle map if and only if

$$q_{\ell,s} \circ \widehat{\rho}_{\ell,s}(g) = \widehat{\rho}_{\ell+1,s}(\kappa_\ell(g)) \circ q_{\ell,s}. \tag{11.14}$$

It then induces

$$C_{\ell,s} : \mathcal{E}_{\ell,s} \to \mathcal{E}_{\ell+1,s}, \qquad C_{\ell,s}[p, z] = [\mathcal{P}_\ell(p), q_{\ell,s}(z)], \tag{11.15}$$

covering c_ℓ. The biconditional is a two-line computation on representatives. Two representatives of one point of $\mathcal{E}_{\ell,s}$ are (p, z) and $(pg, \widehat{\rho}_{\ell,s}(g)^{-1}z)$; their images under the displayed formula are $[\mathcal{P}_\ell(p), q_{\ell,s}(z)]$ and $[\mathcal{P}_\ell(p)\kappa_\ell(g), q_{\ell,s}(\widehat{\rho}_{\ell,s}(g)^{-1}z)]$, and the second equals $[\mathcal{P}_\ell(p), \widehat{\rho}_{\ell+1,s}(\kappa_\ell(g))q_{\ell,s}(\widehat{\rho}_{\ell,s}(g)^{-1}z)]$. These agree for every (p, z, g) exactly when (11.14) holds, which proves both directions. One degenerate case must be recorded rather than absorbed: if $\mathcal{B}_{\ell+1,s}$ reduces to a single $G_{\ell+1}$-fixed point, then (11.14) is vacuously true and is therefore not necessary in any informative sense. Two further hypotheses are consumed whenever the induced morphism is used at the information-geometric tier, and neither follows from affinity of the law pushforward on the cone of measures: family closure of the underlying kernel, meaning that the pushed laws remain in the declared coarse model, and smoothness of $q_{\ell,s}$ between the two declared parametrized-measure models. [ESTABLISHED]

More general associated-bundle maps may be declared directly, but they owe the same transition compatibility. For a genuine multi-step scale functor, the maps c, κ, $\mathcal{P}$, and $q_{\ell,s}$ also obey their respective composition laws. Explicitly, for composable steps the required identities are $c_{\ell+1} \circ c_\ell$, $\kappa_{\ell+1} \circ \kappa_\ell$, $\mathcal{P}_{\ell+1} \circ \mathcal{P}_\ell$, $q_{\ell+1,s} \circ q_{\ell,s}$, and consequently $C_{\ell+1,s} \circ C_{\ell,s}$, with the rightmost factor acting first at every level; at the law-fiber level the composite is the pushforward of the composed kernel by the Chapman–Kolmogorov identity. Identity arrows have identity components, so these laws make the family a functor on the thin category of the finite scale set. Adjacent maps without these laws form only a sequence of coarse steps. [ESTABLISHED]

If local comparison functions a_ℓ^s satisfy

$$\mathcal{P}_\ell(u_\ell^s(x)) = u_{\ell+1}^s(c_\ell x) \cdot a_\ell^s(x),$$

then equivariance of $\mathcal{P}_\ell$ forces

$$a_\ell^b \kappa_\ell(h_\ell) = (h_{\ell+1} \circ c_\ell) a_\ell^m. \tag{11.16}$$

Thus the two represented frame channels remain compatible even when their fiber dimensions or actions differ. [ESTABLISHED]

For general nonlinear associated fibers, the cross-scale morphism defects are the ordered pairs of maps

$$\mathbf{D}_\ell^\Phi = (C_{\ell,m} \circ \Phi_\ell, \Phi_{\ell+1} \circ C_{\ell,b}), \qquad \mathbf{D}_\ell^{\widetilde{\Phi}} = \left(C_{\ell,b} \circ \widetilde{\Phi}_\ell, \widetilde{\Phi}_{\ell+1} \circ C_{\ell,m}\right). \tag{11.17}$$

The defect vanishes when the two components of the corresponding pair are equal. For a base curve $\gamma : x \to y$, the cross-scale connection defects are likewise the pairs

$$\begin{aligned} \mathbf{D}_{\ell,\gamma}^b &= (C_{\ell,b,y} \circ \Omega_{\ell,\gamma}, \Omega_{\ell+1,c_\ell\gamma} \circ C_{\ell,b,x}), \\ \mathbf{D}_{\ell,\gamma}^m &= \left(C_{\ell,m,y} \circ \widetilde{\Omega}_{\ell,\gamma}, \widetilde{\Omega}_{\ell+1,c_\ell\gamma} \circ C_{\ell,m,x}\right). \end{aligned} \tag{11.18}$$

Independently, within-scale nonparallelness is measured by

$$\begin{aligned} \mathbf{A}_{\ell,\gamma}^\Phi &= \left(\widetilde{\Omega}_{\ell,\gamma} \circ \Phi_{\ell,x}, \Phi_{\ell,y} \circ \Omega_{\ell,\gamma}\right), \\ \mathbf{A}_{\ell,\gamma}^{\widetilde{\Phi}} &= \left(\Omega_{\ell,\gamma} \circ \widetilde{\Phi}_{\ell,x}, \widetilde{\Phi}_{\ell,y} \circ \widetilde{\Omega}_{\ell,\gamma}\right). \end{aligned} \tag{11.19}$$

[DEFINITION]

Subtraction of the two components is reserved for a declared vector or affine realization. [DEFINITION]

A sufficient principal-level condition for $\mathbf{D}_{\ell,\gamma}^s$ to vanish is

$$\mathcal{P}_\ell^* \omega_{\ell+1,s} = \mathrm{d}\kappa_\ell \circ \omega_{\ell,s}, \qquad s \in \{b, m\}. \tag{11.20}$$

If a representation is not faithful, vanishing of its represented transport defect need not imply this principal-level identity. [ESTABLISHED]

The infinitesimal object measuring failure of (11.20) is the scale-connection defect form $\mathfrak{A}_{\mathcal{P}_\ell} = \mathcal{P}_\ell^* \omega_{\ell+1,s} - \mathrm{d}\kappa_\ell \circ \omega_{\ell,s}$ of (8.54), and by Theorem 8.18 the represented horizontal defect vanishes along a coarse section exactly under the isotropy condition (8.56), which is strictly weaker than (11.20) whenever the represented action fails to be infinitesimally effective there. Across composable steps these defects obey an *ordered* law, not a sum:

$$\mathfrak{A}_{\mathcal{P}_{02}} = \mathcal{P}_{01}^* \mathfrak{A}_{\mathcal{P}_{12}} + \mathrm{d}\kappa_{12} \circ \mathfrak{A}_{\mathcal{P}_{01}}, \tag{11.21}$$

whose represented form is (8.57). Writing the composite defect as an unweighted sum of the two stage defects is a type error before it is an error of fact: the two take values in different vertical bundles, the earlier one must be pushed forward by the later vertical differential, and the later one must be evaluated at the image point on the pushed base tangent. [ESTABLISHED]

Proposition 11.5 (Covariance of the defect diagrams). Under admissible local-frame changes in the belief and model associated bundles, both components of every pair in (11.17)–(11.19) acquire the same endpoint coordinate changes. Consequently the equality locus of each pair is gauge invariant. In a fiberwise-linear vector-bundle realization, subtracting the two components gives the familiar Hom-bundle defect with the same left–right law. [ESTABLISHED]

Proof. Let $\mathscr{G}_{\ell,b,x}$ and $\mathscr{G}_{\ell,m,x}$ denote the fiber coordinate changes induced by the selected belief and model frame re-trivializations at x, represented through $\widehat{\rho}_{\ell,b}$ and $\widehat{\rho}_{\ell,m}$. For example,

$$\Phi_{\ell,x} \mapsto \mathscr{G}_{\ell,m,x} \circ \Phi_{\ell,x} \circ (\mathscr{G}_{\ell,b,x})^{-1}, \quad \Omega_{\ell,\gamma} \mapsto \mathscr{G}_{\ell,b,y} \circ \Omega_{\ell,\gamma} \circ (\mathscr{G}_{\ell,b,x})^{-1},$$

and a cross-scale map obeys

$$C_{\ell,b,x} \mapsto \mathscr{G}_{\ell+1,b,c_\ell(x)} \circ C_{\ell,b,x} \circ (\mathscr{G}_{\ell,b,x})^{-1},$$

with analogous model and endpoint laws. Substitution shows that both components of $\mathbf{A}^{\Phi}_{\ell,\gamma}$ acquire the same model coordinate change at y and inverse belief coordinate change at x. The equality of the two components is therefore independent of the chosen frames. The other pairs are identical calculations. In a linear realization, subtraction factors out the common endpoint actions. □

The defect data may be retained as running fields; the theory does not require them to vanish. A numerical size requires a declared metric or divergence on each target fiber. The compositions $\widetilde{\Phi}_\ell \Phi_\ell$ and $\Phi_\ell \widetilde{\Phi}_\ell$ are independent running endomorphisms and are not identities unless imposed. Smooth parallel transport remains distinct from separately declared graph links $\Theta^b_{ij}, \Theta^m_{ij}$. [ESTABLISHED]

The default common-principal construction does not impose a common local frame or a common connection. The stronger common-frame hypothesis is $u^b_{\ell,i} = u^m_{\ell,i}$, equivalently $h_{\ell,i} = e$, on the chosen patches. Independently, the equal-connection hypothesis is $\omega_{\ell,b} = \omega_{\ell,m}$. Even then, different representations generally produce different associated transports on different fibers: they represent the same principal holonomy but are not literally the same map. If a further associated-bundle isomorphism $\Xi_\ell : \mathcal{E}_{\ell,b} \xrightarrow{\sim} \mathcal{E}_{\ell,m}$ is declared, its existence is additional global data; equal fiber type is necessary but not sufficient in general. Its parallel and RG-preservation conditions are

$$\widetilde{\Omega}_{\ell,\gamma}\Xi_{\ell,x} = \Xi_{\ell,y}\Omega_{\ell,\gamma}, \qquad C_{\ell,m}\Xi_\ell = \Xi_{\ell+1}C_{\ell,b}.$$

The optional Ξ_ℓ is unavailable when the associated bundles are nonisomorphic, as can occur when ranks or fiber types differ. No such isomorphism is needed to define Φ_ℓ and $\widetilde{\Phi}_\ell$. [HYPOTHESIS]

Optional product-gauge extension. If an application truly requires independent gauge groups, independent principal-bundle topology, or independently acting gauge automorphisms, one may instead introduce

$$P_{\ell,b} \to \mathcal{C}_\ell, \qquad P_{\ell,m} \to \mathcal{C}_\ell, \qquad Q_\ell = P_{\ell,b} \times_{\mathcal{C}_\ell} P_{\ell,m},$$

where Q_ℓ is principal for $G_{\ell,b} \times G_{\ell,m}$. Each associated bundle then uses its own factor. This extension carries extra topological and gauge data and is not part of the minimal theory. In a category admitting the required mapping object, its belief-to-model maps are sections of

$$Q_\ell \times_{G_{\ell,b}\times G_{\ell,m}} \mathrm{Map}(\mathcal{B}_{\ell,b}, \mathcal{B}_{\ell,m}), \qquad (g_b, g_m)\cdot f = \widehat{\rho}_{\ell,m}(g_m) \circ f \circ \widehat{\rho}_{\ell,b}(g_b)^{-1}.$$

A constant fiber map descends only if it is fixed by this action, a generally restrictive condition for independently varying (g_b, g_m). Cross maps therefore still have to be supplied, and a diagonal reduction to one group requires additional group and bundle-identification data. [DEFINITION]

11.4 Autonomous maps and cocycles

Definition 11.6 (Dynamical type). The sequence $\widehat{\mathcal{R}}_\ell$ in (11.3) is: autonomous when it is one map $\widehat{\mathcal{R}}$; periodic when $\widehat{\mathcal{R}}_{\ell+p} = \widehat{\mathcal{R}}_\ell$; stationary random when it is a measurable stationary cocycle; and general otherwise. [DEFINITION]

The correct invariant object depends on this type. An autonomous fixed object satisfies $\widehat{\mathcal{R}}x = x$. A periodic scheme has a fixed object of its monodromy or an attracting p-cycle. A stationary random scheme may have a random invariant section. A general cocycle may have pullback or forward attracting sections. The term *fixed ray* additionally requires a positive cone and projectivization. [DEFINITION]

Proposition 11.7 (Uniform contraction need not give one fixed ray). Uniform Hilbert-projective contraction of a nonautonomous positive cocycle can erase initial conditions without selecting one scale-independent ray. [ESTABLISHED]

Proof. On $\mathbb{R}^2_{>0}$, let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Both are strictly positive and have a uniform projective contraction coefficient below one. Their Perron rays differ. Under the alternating cocycle, even iterates converge to the Perron ray $[1, \sqrt{2}]$ of

$$BA = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix},$$

while odd iterates converge to the distinct ray $A[1, \sqrt{2}]$. The attractor is period two, not one fixed ray. □

For maps satisfying $\tau(\widehat{\mathcal{R}}_\ell) \leq q < 1$, Birkhoff contraction gives

$$d_H(\widehat{\mathcal{R}}_{\ell-1} \cdots \widehat{\mathcal{R}}_0 x, \widehat{\mathcal{R}}_{\ell-1} \cdots \widehat{\mathcal{R}}_0 y) \leq q^\ell d_H(x, y), \tag{11.22}$$

where defined [15, 59]. This proves forgetting of initial rays, not time independence of the attracting section.

11.4.1 Ordered derivative cocycles and generalized cross-scale modes

A scale sequence by itself supplies no eigenvalue problem. It supplies an ordered family of maps between different spaces, and the invariant content of that family is a composition law together with compatible mode sections. This subsection fixes those types once, so that the interaction realization of Chapter 12 can instantiate them without inventing an ambient common space.

Definition 11.8 (Scale-indexed evolution and its derivative cocycle). Let $(X_\ell, \|\cdot\|_\ell)_{\ell \geq 0}$ be Banach spaces and let $T_\ell : \mathcal{U}_\ell \to X_{\ell+1}$ be continuously Fréchet differentiable on an open set $\mathcal{U}_\ell \subseteq X_\ell$. The ordered nonlinear evolution is

$$\Phi_{n \leftarrow \ell} = T_{n-1} \circ \cdots \circ T_\ell, \qquad \Phi_{\ell \leftarrow \ell} = \mathrm{id}_{X_\ell}, \tag{11.23}$$

defined at those points whose successive forward images remain in the successive domains. Along an exact orbit $x_{\ell+1} = T_\ell(x_\ell)$ contained in those domains, write

$$D_\ell = DT_\ell(x_\ell) \in \mathcal{L}(X_\ell, X_{\ell+1}), \qquad D_{n \leftarrow \ell} = D_{n-1} \cdots D_\ell, \qquad D_{\ell \leftarrow \ell} = I_{X_\ell}. \tag{11.24}$$

The rightmost factor acts first. Nothing is proved here; the display fixes domains, codomains, and order. [DEFINITION]

Proposition 11.9 (Ordered composition of the evolution and its derivative). For $\ell \leq n \leq r$ and an exact orbit as above,

$$\Phi_{r \leftarrow n} \circ \Phi_{n \leftarrow \ell} = \Phi_{r \leftarrow \ell}, \qquad D_{n \leftarrow \ell} = D\Phi_{n \leftarrow \ell}(x_\ell), \qquad D_{r \leftarrow n} D_{n \leftarrow \ell} = D_{r \leftarrow \ell}. \tag{11.25}$$

[ESTABLISHED]

Proof. The first identity is associativity of composition of maps. The second is proved by induction on $n-\ell$: it is the definition when $n=\ell$, and if it holds at n then the Fréchet chain rule applied to $\Phi_{n+1\leftarrow\ell} = T_n \circ \Phi_{n\leftarrow\ell}$ at x_ℓ, using $\Phi_{n\leftarrow\ell}(x_\ell) = x_n$, gives $D\Phi_{n+1\leftarrow\ell}(x_\ell) = DT_n(x_n)D_{n\leftarrow\ell} = D_n D_{n\leftarrow\ell}$. The third identity is the associativity of the ordered operator product, which is also the second identity applied to the first. □

The composition law in (11.25) is a two-parameter law, not a one-parameter semigroup. It reduces to a discrete semigroup only after declared isomorphisms $J_\ell : X_\star \to X_\ell$ make the identified maps $\widehat{T}_\ell = J_{\ell+1}^{-1} T_\ell J_\ell$ independent of ℓ; a p-periodic identified sequence produces a monodromy rather than a one-step autonomous map, in the sense of Definition 11.6. Writing a general segment as a power of one unnamed operator erases the base scale and, when the factors do not commute, also reverses the order: for $A = \left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$ one has $BA = \left(\begin{smallmatrix}1&1\\1&2\end{smallmatrix}\right)$ and $AB = \left(\begin{smallmatrix}2&1\\1&1\end{smallmatrix}\right)$, which differ. [ESTABLISHED]

For a mixed state in the sense of Definition 11.1, the composition law is owed in every declared component category. Composition of one linear derivative tier proves nothing about the base, principal, law-fiber, associated-bundle, action, or configuration components. [HYPOTHESIS]

Definition 11.10 (Compatible cross-scale mode line). A rank-one mode line of index a along the orbit is a family of nonzero vectors $v_{\ell,a} \in X_\ell$ together with scalars $\lambda_{\ell,a} \in \mathbb{C}$ satisfying

$$D_\ell v_{\ell,a} = \lambda_{\ell,a} v_{\ell+1,a}. \tag{11.26}$$

This is a compatibility equation between adjacent fibers. The expression $D_\ell v = \lambda v$ is not used: its two sides inhabit $X_{\ell+1}$ and X_ℓ, which are different spaces unless an identification has been declared. For a mode subbundle of finite rank d_a, choose injective frames $V_{\ell,a} : \mathbb{C}^{d_a} \to X_\ell$ and matrices $A_{\ell,a}$ with

$$D_\ell V_{\ell,a} = V_{\ell+1,a} A_{\ell,a}. \tag{11.27}$$

[DEFINITION]

Proposition 11.11 (Ordered product law for mode coefficients). Under (11.26),

$$D_{n\leftarrow\ell} v_{\ell,a} = \lambda_{n\leftarrow\ell,a} v_{n,a}, \qquad \lambda_{n\leftarrow\ell,a} = \prod_{k=\ell}^{n-1} \lambda_{k,a}, \tag{11.28}$$

with the empty product equal to one. Under (11.27) the corresponding finite-step matrix is the ordered product $A_{n-1,a} \cdots A_{\ell,a}$. If one factor vanishes, the propagated mode is annihilated from that step onward. The complex phase or real sign of each factor is retained separately from any growth exponent. [ESTABLISHED]

Proof. Induction on $n-\ell$ using (11.25) and (11.26): the case $n=\ell$ is the empty product, and $D_{n+1\leftarrow\ell} v_{\ell,a} = D_n D_{n\leftarrow\ell} v_{\ell,a} = \lambda_{n\leftarrow\ell,a} D_n v_{n,a} = \lambda_{n\leftarrow\ell,a} \lambda_{n,a} v_{n+1,a}$. The frame case is the same induction with matrix factors, whose order is fixed by (11.27). □

An ill-typed eigenvalue equation is not repaired by notation. With $X_0 = \mathbb{R}$, $X_1 = \mathbb{R}^2$, and $D_0 x = (x, 0)$, the expression $D_0 v = \lambda v$ equates an element of $\mathbb{R}^2$ with one of $\mathbb{R}$; the compatible line $v_0 = 1$, $v_1 = (1, 0)$, $\lambda_0 = 1$ is well typed and carries the same information. [ESTABLISHED]

Definition 11.12 (Cumulative log block scale and growth rates). Let $b_k > 1$ be the declared adjacent block ratios and set

$$\Delta s_k = \log b_k, \qquad s_{n\leftarrow\ell} = \sum_{k=\ell}^{n-1} \Delta s_k, \tag{11.29}$$

and assume $s_{n\leftarrow\ell} \to \infty$. For $u \in X_\ell \setminus \{0\}$ with $D_{n\leftarrow\ell} u \neq 0$, the finite-time growth rate relative to the declared norms is

$$\chi_{n\leftarrow\ell}(u) = \frac{\log\|D_{n\leftarrow\ell} u\|_n - \log\|u\|_\ell}{s_{n\leftarrow\ell}}, \tag{11.30}$$

with the convention that the rate is $-\infty$ once the image vanishes. Its upper and lower asymptotic rates are the corresponding limit superior and limit inferior; a Lyapunov exponent is asserted only when the two agree. For a mode line, the scalar finite-time exponent is

$$\upsilon_{n\leftarrow\ell,a} = \frac{\log|\lambda_{n\leftarrow\ell,a}|}{s_{n\leftarrow\ell}} = \frac{\sum_{k=\ell}^{n-1} \log|\lambda_{k,a}|}{s_{n\leftarrow\ell}}, \qquad \log 0 = -\infty. \tag{11.31}$$

A mode is called relevant, marginal, or irrelevant according as the declared asymptotic version of (11.31) is positive, zero, or negative. That trichotomy is a classification relative to the declared norms, the declared extensive normalization of the mode sections, the declared block ratios, and the declared comparison maps; it is not an intrinsic property of the scale sequence. [DEFINITION]

Proposition 11.13 (Scalar and norm growth agree for tempered mode sections). For a mode line and every $n > \ell$ with $\lambda_{n\leftarrow\ell,a} \neq 0$,

$$\chi_{n\leftarrow\ell}(v_{\ell,a}) = \upsilon_{n\leftarrow\ell,a} + \frac{\log\|v_{n,a}\|_n - \log\|v_{\ell,a}\|_\ell}{s_{n\leftarrow\ell}}. \tag{11.32}$$

The two rates therefore have the same asymptotic upper and lower values whenever the mode sections are normalized to unit norm, or more generally whenever $\log\|v_{n,a}\|_n$ is sublinear in $s_{n\leftarrow\ell}$. Under a scalar regauging $v'_{\ell,a} = c_{\ell,a} v_{\ell,a}$ with $c_{\ell,a} \neq 0$,

$$\lambda'_{\ell,a} = \lambda_{\ell,a} \frac{c_{\ell,a}}{c_{\ell+1,a}}, \qquad \lambda'_{n\leftarrow\ell,a} = \lambda_{n\leftarrow\ell,a} \frac{c_{\ell,a}}{c_{n,a}}, \tag{11.33}$$

so the scalar exponent is unchanged exactly when $|\log|c_{n,a}||/s_{n\leftarrow\ell} \to 0$. [ESTABLISHED]

Proof. Take norms in (11.28) and substitute into (11.30); the displayed remainder is exactly the endpoint norm difference divided by $s_{n\leftarrow\ell}$. Substituting $v'_{\ell,a} = c_{\ell,a} v_{\ell,a}$ into (11.26) and comparing coefficients gives the first equality in (11.33), and the telescoping product gives the second. Dividing $\log|\lambda'_{n\leftarrow\ell,a}|$ by $s_{n\leftarrow\ell}$ isolates the stated tempering condition. □

Theorem 11.14 (Exponent invariance under tempered comparison trivializations). Let $J_k : X_\star \to X_k$ be bounded linear isomorphisms with bounded inverses, and set

$$\widehat{D}_k = J_{k+1}^{-1} D_k J_k, \qquad \widehat{D}_{n\leftarrow\ell} = J_n^{-1} D_{n\leftarrow\ell} J_\ell. \tag{11.34}$$

If the complete trivializations and their inverses are tempered relative to the cumulative log block scale,

$$\frac{\log^+\|J_n\| + \log^+\|J_n^{-1}\|}{s_{n\leftarrow\ell}} \longrightarrow 0, \tag{11.35}$$

then for every $u \in X_\ell \setminus \{0\}$ the asymptotic upper and lower rates of u in the native norms coincide with those of $J_\ell^{-1}u$ computed for $\widehat{D}_{n\leftarrow\ell}$ in the reference norm. Together with the scalar tempering condition of Proposition 11.13, this preserves both the scalar mode exponent and the norm growth rate. [ESTABLISHED]

Proof. Write $x_n = D_{n\leftarrow\ell}u$ and $\widehat{x}_n = J_n^{-1}x_n = \widehat{D}_{n\leftarrow\ell}J_\ell^{-1}u$, which is the second identity in (11.34). Bounded invertibility gives

$$\frac{\|x_n\|_n}{\|J_n\|} \le \|\widehat{x}_n\|_\star \le \|J_n^{-1}\|\|x_n\|_n. \tag{11.36}$$

Taking logarithms changes the numerator of (11.30) by at most $\log^+\|J_n\| + \log^+\|J_n^{-1}\|$, and the fixed initial scale contributes the constant $|\log\|J_\ell\|| + |\log\|J_\ell^{-1}\||$. Dividing by $s_{n\leftarrow\ell}$ and using (11.35) and $s_{n\leftarrow\ell} \to \infty$ makes the difference of the two finite-time rates tend to zero, so their limits superior and inferior agree. The scalar statement is (11.33). $\square$

Proposition 11.15 (Superexponential distortion changes the apparent exponent). Tempering the scalar mode representatives alone is not sufficient. Let $X_k = \mathbb{R}$ with the native absolute value, let every D_k be the identity, let $b_k = e$, and let every scalar gauge factor be one, so that the native exponent is zero and the scalar normalization is tempered. Take $J_k u = e^{k^2}u$. Then $\widehat{D}_k = e^{-2k-1}I$ and $\widehat{D}_{n\leftarrow 0} = e^{-n^2}I$, so the apparent reference-space rate is $-n^2/n = -n \to -\infty$. The complete comparison trivialization and its inverse, not only the mode normalization, must satisfy (11.35). [ESTABLISHED]

Proof. $\widehat{D}_k = J_{k+1}^{-1}J_k = e^{k^2-(k+1)^2} = e^{-2k-1}$, and the telescoping product over $k = 0, \dots, n-1$ is e^{-n^2}. Here $s_{n\leftarrow 0} = n$, so (11.30) gives $-n$, while the native rate is zero. Condition (11.35) fails because $\log\|J_n\| = n^2$. $\square$

The definitions above are deliberately weaker than an existence statement. A multiplicative ergodic theorem would supply a measurable splitting into finitely or countably many exponent subspaces, but it requires data this scale diagram does not contain: a measure-preserving base flow, measurability of the cocycle in the base variable, integrability of $\log^+\|D\|$ and, for the two-sided splitting, of $\log^+\|D^{-1}\|$, and either finite dimension or a compactness or quasi-compactness hypothesis in the infinite-dimensional case [4]. A deterministic scale sequence supplies none of these by itself. This document therefore defines the growth rates above and does not assert an Oseledets splitting for the RG cocycle. [NOT-CLAIMED]

11.5 Scheme dependence and invariant candidates

Let $g(t)$ be running coordinates. Under a smooth reparameterization $g' = f(g)$ and scale change $t' = ct$,

$$\beta'(g') = \frac{1}{c} Df(g)\beta(g). \tag{11.37}$$

[ESTABLISHED]

Hence raw couplings, beta-function components, and a chosen blocking statistic are scheme dependent. They remain valid diagnostics when their scheme is stated, but are not universal observables. [ESTABLISHED]

For a differentiable homogeneous positive endomorphism at a fixed ray, let μ_0 be the radial eigenvalue and μ_a a transverse eigenvalue. The projective ratios and block-normalized dimensions are

$$\rho_a = \frac{\mu_a}{\mu_0}, \qquad \upsilon_a = \frac{\log|\rho_a|}{\log b}, \tag{11.38}$$

for declared block factor b. [DEFINITION]

They are invariant under smooth coordinate conjugacy at the ray once gauge, normalization, and other redundant directions have been quotiented and the same scale normalization is used. [ESTABLISHED]

Candidate cross-scheme quantities include conjugacy classes, invariant manifolds, projective eigenvalue ratios, dimensionless amplitude ratios, and matched spectral or heat exponents. Their existence, convergence, and completeness must be proved in the chosen realization. Universality is a statement about basins and invariant observables, not algebraic closure alone. [HYPOTHESIS]

11.6 Two-index thermodynamic and RG limits

A finite-volume RG family has two indices:

$$X_{n,\ell}, \qquad 0 \le \ell \le L(n), \tag{11.39}$$

where n controls system size and ℓ RG depth. Fixed-depth, maximal-depth, and diagonal limits are different:

$$\lim_{\ell\to\infty}\lim_{n\to\infty} X_{n,\ell}, \qquad \lim_{n\to\infty} X_{n,L(n)}, \qquad \lim_{n\to\infty} X_{n,\ell(n)}. \tag{11.40}$$

[DEFINITION]

Proposition 11.16 (Fixed-depth and maximal-depth limits need not agree). Fix an integer $b \ge 2$. For $n \ge 0$, put $m = b^n$ and let Path_m be the unweighted path on $\{1, \dots, m\}$, with combinatorial Laplacian L_m. At depth $0 \le \ell \le n$, form the consecutive blocks

$$B_j^{(\ell)} = \{(j-1)b^\ell + 1, \dots, jb^\ell\}, \qquad 1 \le j \le b^{n-\ell}. \tag{11.41}$$

Join two blocks exactly when a path edge crosses between them. The resulting simple quotient is exactly $\mathrm{Path}_{b^{n-\ell}}$. For any m divisible by b, let the unnormalized contiguous block-indicator prolongation be

$$S_{m,b} \in \mathbb{R}^{m\times(m/b)}, \qquad (S_{m,b})_{ij} = \mathbf{1}_{\{(j-1)b+1,\dots,jb\}}(i). \tag{11.42}$$

It realizes the exact operator quotient

$$S_{m,b}^{\mathsf{T}} L_m S_{m,b} = L_{m/b}. \tag{11.43}$$

Set $S_{m,1} = I_m$. At depth $1 \le \ell \le n$, define

$$S_{m,b^\ell} := S_{m,b} S_{m/b,b} \cdots S_{m/b^{\ell-1},b}, \tag{11.44}$$

this is the indicator of the blocks in (11.41) and obeys

$$S_{m,b^\ell}^{\mathsf{T}} L_m S_{m,b^\ell} = L_{m/b^\ell}. \tag{11.45}$$

The identities use unnormalized indicators. With orthonormalized one-step indicators $\widetilde{S}_{m,b} = b^{-1/2} S_{m,b}$, the left side of (11.43) is instead $b^{-1} L_{m/b}$, and the depth-ℓ factor is $b^{-\ell}$.

For every $m \ge 1$, the eigenvalues of L_m, including multiplicity, are

$$\lambda_{m,k} = 2 - 2\cos\left(\frac{\pi k}{m}\right), \qquad 0 \le k \le m-1. \tag{11.46}$$

Let

$$\mu_m = \frac{1}{m}\sum_{k=0}^{m-1} \delta_{\lambda_{m,k}}, \qquad N_m(\lambda) = \mu_m((-\infty,\lambda]), \qquad U_m(\lambda) = \frac{1}{m}\#\{1 \le k \le m-1 : \lambda_{m,k} \le \lambda\}. \tag{11.47}$$

These are right-continuous full and zero-mode-excluded counts, and $U_m = N_m - m^{-1}\mathbf{1}_{[0,\infty)}$. More explicitly, for $0 \le \lambda \le 4$, set $\theta(\lambda) = \arccos(1-\lambda/2)$; then

$$N_m(\lambda) = \frac{1+\min\{m-1, \lfloor m\theta(\lambda)/\pi \rfloor\}}{m}, \qquad U_m(\lambda) = \frac{\min\{m-1, \lfloor m\theta(\lambda)/\pi \rfloor\}}{m}. \tag{11.48}$$

Both counts vanish for $\lambda < 0$; for $\lambda > 4$, they equal 1 and $1 - m^{-1}$, respectively. Specifically, for $m \ge 2$, $N_m(\lambda) = m^{-1}$ on $[0, \lambda_{m,1})$, while U_m is zero there and jumps to m^{-1} at the included threshold $\lambda_{m,1}$; at the included upper endpoint $\lambda_{m,m-1}$, the counts are 1 and $1 - m^{-1}$.

For every fixed ℓ, $b^{n-\ell} \to \infty$ as $n \to \infty$, and both counts converge pointwise to the right-continuous function

$$N_\infty(\lambda) = U_\infty(\lambda) = \begin{cases} 0, & \lambda < 0, \\ \dfrac{1}{\pi}\arccos\left(1 - \dfrac{\lambda}{2}\right), & 0 \le \lambda \le 4, \\ 1, & \lambda > 4. \end{cases} \tag{11.49}$$

Thus the zero-mode-excluded lower-edge count obeys

$$U_\infty(\lambda) = \frac{1}{\pi}\arccos\left(1 - \frac{\lambda}{2}\right) \sim \frac{1}{\pi}\sqrt{\lambda} \qquad (\lambda \downarrow 0). \tag{11.50}$$

At maximal depth $\ell = n$, however, the exact quotient is Path_1 for every n, so $\mu_1 = \delta_0$, $N_1 = \mathbf{1}_{[0,\infty)}$, and $U_1 \equiv 0$. Hence the fixed-depth thermodynamic limit and the maximal-depth Path_1 limit are different. [ESTABLISHED]

Proof. Each block in (11.41) is connected, and exactly one original edge joins block j to block $j+1$, with no other interblock edge; this proves the exact quotient statement. More explicitly, for every $z \in \mathbb{R}^{m/b}$,

$$z^\mathsf{T} S_{m,b}^\mathsf{T} L_m S_{m,b} z = \sum_{i=1}^{m-1}\big((S_{m,b}z)_{i+1} - (S_{m,b}z)_i\big)^2 = \sum_{j=1}^{m/b-1}(z_{j+1} - z_j)^2 = z^\mathsf{T} L_{m/b} z.$$

Equality of the symmetric quadratic forms proves (11.43); iteration proves (11.45), and scalar normalization gives the stated factors. The discrete cosine eigenvectors of the path Laplacian give (11.46). Monotonicity of $k \mapsto \lambda_{m,k}$ then gives (11.48), including its threshold and upper endpoint. For every bounded continuous f,

$$\int f\,\mathrm{d}\mu_m = \frac{1}{m}\sum_{k=0}^{m-1} f\left(2 - 2\cos\frac{\pi k}{m}\right) \longrightarrow \frac{1}{\pi}\int_0^\pi f(2 - 2\cos\theta)\mathrm{d}\theta.$$

The limiting distribution has the continuous cumulative function in (11.49); removing the one zero atom changes a finite count by m^{-1} and hence has the same limit. Finally, $\arccos(1-\lambda/2) \sim \sqrt{\lambda}$, while $L_1 = 0$. □

Therefore a thermodynamic RG claim must specify a triangular family, projective consistency at fixed depth, uniform control of the coarse error, and a diagonal scaling $\ell(n)$. Operator, law, connection, and cross-morphism convergence are separate obligations. Proving an exchange of limits or a nontrivial diagonal limit remains open. [OPEN]

Put $U(\lambda) = N(\lambda) - N(0)$. Suppose U is nondecreasing and right-continuous, $U(0+) = 0$, $Z_+(t_0) < \infty$ for some $t_0 > 0$, and, for $c > 0$, $\alpha > 0$, and a function L slowly varying at infinity,

$$U(\lambda) \sim c\lambda^\alpha L(1/\lambda) \qquad (\lambda \downarrow 0).$$

Define the zero-mode-excluded heat moments

$$Z_+(t) = M_0(t) := \int_{(0,\infty)} e^{-t\lambda}\,dU(\lambda), \qquad M_k(t) := \int_{(0,\infty)} \lambda^k e^{-t\lambda}\,dU(\lambda).$$

The Abelian/Karamata direction of the regular-variation theorem [14, §1.7.1] gives

$$M_k(t) \sim c\alpha\Gamma(\alpha+k)t^{-(\alpha+k)}L(t) \qquad (t \to \infty;\ k = 0, 1, 2).$$

With $\pi_t(d\lambda) = e^{-t\lambda} dU(\lambda)/Z_+(t)$ and $S_+(t) = \log Z_+(t) + tM_1(t)/Z_+(t)$, the spectral susceptibility obeys

$$-\frac{dS_+}{d\log t} = t^2\left(\frac{M_2}{Z_+} - \left(\frac{M_1}{Z_+}\right)^2\right) \longrightarrow \alpha. \tag{11.51}$$

This is the Abelian forward direction; the converse Tauberian direction is not asserted here. [ESTABLISHED]

Dividing entropy by $\log|V_n|$ divides the susceptibility by the same factor and generally destroys a nonzero thermodynamic plateau. Zero-mode atoms and normalization must therefore be declared before interpreting a plateau as spectral dimension. [ESTABLISHED]

11.7 Projective laws and refinement

Compatible finite-dimensional laws obey the projective theorem of Section 10.10. An energy recursion alone does not provide those laws. A probabilistic RG needs normalized kernels $K_{\ell k}$ with

$$\mu_k = \mu_\ell K_{\ell k}, \qquad K_{\ell k}K_{kr} = K_{\ell r},$$

and corresponding rules for the two bundle sectors and their reference measures. Smooth-section support, continuum ELBO convergence, and connection convergence remain additional open obligations. [OPEN]

If the coarse map Q is noninjective, no deterministic two-sided inverse exists. A refinement kernel R can at most be a stochastic right section,

$$Q_\# R(c, \cdot) = \delta_c, \tag{11.52}$$

which redraws a compatible microstate. It cannot recover every discarded realized microstate. [ESTABLISHED]

Infinite divisibility may construct such refinements in distribution, but does not turn a many-to-one coarse-graining into a group inverse. [ESTABLISHED]

11.8 Typed comparison with Bayesian renormalization

Berman, Klinger, and Stapleton build on the dynamical Bayesian-inference flow of Berman et al. [11], using a centered-KL posterior flow on model-parameter space. In the regular late-data regime they approximate the posterior by a Gaussian with covariance $T^{-1}I^{-1}$. Set $\tau = 1/T$, and consider an inverse problem with a differentiable forward map G. The delta method gives the leading local predictive approximation

$$G(\theta) \mathrel{\dot\sim} \mathcal{N}\left(G(\mu_\tau), \quad \tau J_G(\mu_\tau) I(\mu_\tau)^{-1} J_G(\mu_\tau)^\top\right).$$

It is exact when G is affine, but a nonlinear Gaussian pushforward is not in general Gaussian. Accordingly, the pushed inverse Fisher cometric

$$D_\tau = J_G(\mu_\tau)I(\mu_\tau)^{-1}J_G(\mu_\tau)^\top \tag{11.53}$$

is the local diffusion tensor used in their ERG-form Fokker–Planck equation; the corresponding small-τ transition covariance is τD_τ [12]. Their continuation of this tensor beyond the late-data, small-τ regime is a renormalization construction, not an exact nonlinear pushforward theorem or a Bayesian asymptotic theorem. [ESTABLISHED]

This is an exact, carefully scoped literature statement. It does not identify an arbitrary recognition Fisher tensor with the BKS generative-model Fisher metric, transfer their stiff/sloppy eigenspaces to another operator sector, or prove that a graph aggregation runs in the opposite direction. [NOT-CLAIMED]

In the small-τ approximation, covariance broadens like τI^{-1} if I is held fixed, whereas a later Gaussian sufficient-statistic construction may add precision. That is a typed local contrast. A theorem of opposite flow directions would require a common state space, an explicit scale map, and a transported monotone including

$$\frac{d}{d\tau}\left[\tau I(\mu_\tau)^{-1}\right] = I^{-1} - \tau I^{-1}\frac{dI}{d\tau}I^{-1}.$$

No such bridge is asserted. [OPEN]

Theorem 10.13 gives a separate admissibility statement: represented holonomy restricts the marginal-law modes available inside an already proposed block. The BKS Fisher scale instead orders model-parameter directions by statistical distinguishability; it neither chooses node partitions nor tests holonomy. [ESTABLISHED]

Constructing a nondegenerate partition selector compatible with these separate structures remains open. [OPEN]

Mehta and Schwab give an exact variational-RG/deep-network correspondence under their RBM kernel and trace condition [53]. It is a historical exact precedent inside that construction, not a theorem that deterministic energy precomposition is a neural layer. Likewise, scalar Laplacian RG provides a heat-spectrum scale diagnostic and a heuristic real-space blocker; lifting its entrywise blocker covariantly to the two represented frame channels requires a new invariant statistic. These are constraints on applications, not definitions of the present general RG. [ESTABLISHED]

Physicist summary. Blocking removes detail; renormalization adds the maps that say how the blocked theory is compared with the original units. Here that comparison must carry one principal bundle, its two differently represented statistical bundles, two connection choices that may coincide, and two separately supplied cross morphisms. Separate belief and model frames are local views of the same principal geometry; their relative element lies in G_ℓ, even when its two representations act on unlike spaces. Failures to commute with blocking are covariant defect fields. Once all scales are identified, the dynamics may be autonomous, periodic, random, or a general cocycle, and each case has a different invariant object. System size and RG depth then form a two-index limit. Fixed points, universality, and continuum laws are meaningful only after these types and limits have been declared.

Chapter 12

Exact Renormalization of Agent Networks

This chapter constructs an effective variational theory for the interaction model of Chapter 7. Its primary objects are normalized measures and Markov kernels. Actions, couplings, and differential beta functions appear only after a reference and a scale identification have been declared. Exact closure is obtained by retaining induced hyperedges, marked attention events, complete holonomy data, and path-space memory. A pairwise, memoryless model with one scalar attention row and one averaged group link is therefore a truncation, not the general effective theory.

12.1 Law-level coarse-graining and the VFE identity

Let $P(do, dy)$ be the fixed normalized joint law on standard-Borel spaces, let o be regular, let $\Pi_o(dy)$ be its posterior, and let $Q_o \ll \Pi_o$ be a recognition law. A coarse channel between standard-Borel latent spaces is a fixed Markov kernel

$$C : \mathsf{Y} \rightsquigarrow \mathsf{Z} \tag{12.1}$$

that does not read Q_o and does not alter the observation coordinate. Define

$$P^c(do, dz) = \int_{\mathsf{Y}} C(y, dz) P(do, dy), \qquad \Pi_o^c = \Pi_o C, \qquad Q_o^c = Q_o C. \tag{12.2}$$

Theorem 12.1 (Exact coarse VFE and discarded conditional information). Let the fine and coarse VFE expressions be defined as extended-real sums, with the common finite evidence term separated from their KL gaps, and let

$$\widehat{Q}_o(dy, dz) = Q_o(dy) C(y, dz), \qquad \widehat{\Pi}_o(dy, dz) = \Pi_o(dy) C(y, dz). \tag{12.3}$$

Then P^c has the same observation evidence as P, and the extended-real additive identity is

$$\mathcal{F}_P(Q_o) = \mathcal{F}_{P^c}(Q_o^c) + \int_{\mathsf{Z}} D_{\mathrm{KL}}\big(\widehat{Q}_o(dy \mid z) \| \widehat{\Pi}_o(dy \mid z)\big) Q_o^c(dz). \tag{12.4}$$

In particular, the fine VFE is at least the coarse VFE. If the fine KL is finite, their ordinary real-valued difference equals the displayed conditional KL. The corresponding coarse ELBO rises only because a conditional inference gap has been discarded; the evidence has not increased. [ESTABLISHED]

Proof. The observation marginal is unchanged because $C(y, \mathsf{Z}) = 1$. Attaching the same channel to both measures preserves the fine relative entropy, $D_{\mathrm{KL}}(Q_o\|\Pi_o) = D_{\mathrm{KL}}(\widehat{Q}_o\|\widehat{\Pi}_o)$. Disintegration on z and the relative-entropy chain rule split the latter into $D_{\mathrm{KL}}(Q_o^c\|\Pi_o^c)$ plus the integral in (12.4). Substitute the exact evidence identities at the two scales. □

The theorem does not cover different generative and recognition channels, a fitted rather than pushed-forward P^c, or a simultaneous coarse-graining of the observation. Such operations generally change the comparison target or the evidence event. An invertible or evidence-sufficient observation channel is an exception, but its preserved event and posterior must be proved explicitly. [ESTABLISHED]

12.2 The exact effective likelihood and action

Fix one adjacent scale arrow and suppress its index temporarily. Its fine space is a finite product of standard-Borel agent spaces, so it is itself standard Borel. Let $\rho = P_0$ be a probability reference and let

$$m_o(dy) = L_o(y)\rho(dy), \qquad L_o(y) = \exp[-H_o(y)], \qquad 0 < M_o := m_o(\mathsf{Y}) < \infty, \qquad \pi(dy) = M_o^{-1}m_o(dy) \tag{12.5}$$

be the unnormalized evidence submeasure, its normalized law, and its strictly positive finite mass. The action may be extended valued; the displayed density convention is $e^{-\infty} = 0$ and all equalities involving actions are therefore almost-everywhere statements. Push both measures through the normalized, recognition-independent channel C:

$$\rho^c = \rho C, \qquad m_o^c = m_oC, \qquad \pi^c = \pi C = M_o^{-1}m_o^c. \tag{12.6}$$

Absolute continuity $m_o^c \ll \rho^c$ follows from $m_o \ll \rho$, and the channel preserves the evidence mass. Define

$$L_o^c(z) = \frac{dm_o^c}{d\rho^c}(z), \qquad H_o^c(z) = -\log L_o^c(z). \tag{12.7}$$

Theorem 12.2 (Conditional-partition formula and composition). Let $R_\rho(z, dy)$ be a reverse conditional of Y given $Z = z$ under $\rho(dy)C(y, dz)$. Then

$$\exp[-H_o^c(z)] = \int \exp[-H_o(y)]R_\rho(z, dy) \tag{12.8}$$

ρ^c-almost everywhere, and

$$Z(o) = \int \exp[-H_o^c(z)]\rho^c(dz). \tag{12.9}$$

For a second coarse channel D, the direct and staged effective likelihoods agree almost everywhere when their conditional versions are chosen from the same joint tower. [ESTABLISHED]

Proof. For bounded measurable f,

$$\int f(z)m_o^c(dz) = \iint f(z)L_o(y)C(y,dz)\rho(dy) = \int f(z)\mathbb{E}_\rho[L_o(Y) \mid Z = z]\rho^c(dz).$$

This identifies the Radon–Nikodym derivative and proves (12.8). Setting $f = 1$ proves (12.9). The tower property proves composition. □

For binary interaction successes, $0 \le L_o \le 1$ implies $0 \le L_o^c \le 1$, so one effective binary interaction record realizes the entire coarse action. For general record densities, L_o^c remains a valid density relative to the pushed probability reference, although it need not be bounded by one. An infinite Lebesgue reference cannot be pushed through an arbitrary projection without checking sigma-finiteness; this is why the probability pair (ρ, m_o) is primary. [ESTABLISHED]

12.2.1 Local bounded action chart and its derivatives

The unnormalized pair tracks the evidence mass, while the local action calculus is based at its normalized law. Because the fine and coarse spaces are standard Borel, fix a jointly measurable reverse kernel $\Pi(z, dy)$ for the normalized joint law:

$$\pi(dy)C(y,dz) = \pi^c(dz)\Pi(z,dy). \tag{12.10}$$

Values of Π on a π^c-null set are not used. Write $\mathfrak{B} = L^\infty(\pi;\mathbb{R})$ and choose $0 < \epsilon < \log 2$. The local chart is the open ball

$$\mathcal{U}_\epsilon = \{\varphi \in \mathfrak{B} : \|\varphi\|_\infty < \epsilon\}. \tag{12.11}$$

For every $\varphi \in \mathfrak{B}$, a perturbation acts on the complete measure pair by $m_o^\varphi = e^{-\varphi}m_o$. Its exact coarse action increment is

$$Q(\varphi)(z) = -\log \frac{d\big((e^{-\varphi}m_o)C\big)}{dm_oC}(z) = -\log \int e^{-\varphi(y)}\Pi(z,dy). \tag{12.12}$$

The last expression is finite and belongs to $L^\infty(\pi^c;\mathbb{R})$: it lies between $-\|\varphi\|_\infty$ and $\|\varphi\|_\infty$. Thus the Radon–Nikodym formula defines $Q : \mathfrak{B} \to L^\infty(\pi^c;\mathbb{R})$ on the entire bounded space. The real-analytic theorem below is asserted on $\mathcal{U}_\epsilon$, not on a general L^p neighborhood. Where both actions are finite, it equals the literal difference $\mathcal{R}^H[H_o + \varphi;\rho] - \mathcal{R}^H[H_o;\rho]$. For an extended-valued base action, the Radon–Nikodym ratio in (12.12) is the definition and no expression of the form $+\infty - (+\infty)$ is used. [DEFINITION]

Theorem 12.3 (Exact bounded action calculus). The map

$$Q : \mathcal{U}_\epsilon \longrightarrow L^\infty(\pi^c;\mathbb{R}) \tag{12.13}$$

is real analytic. If

$$(U\varphi)(z) = \int \varphi(y)\Pi(z, dy), \tag{12.14}$$

then its first two Fréchet derivatives at the origin are the bounded typed maps

$$DQ(0)[\varphi] = U\varphi, \tag{12.15}$$

$$D^2Q(0)[\varphi, \psi] = -\operatorname{Cov}_{\Pi(z,\cdot)}(\varphi, \psi). \tag{12.16}$$

More generally, with

$$\Pi^\varphi(z, dy) = \frac{e^{-\varphi(y)}\Pi(z, dy)}{\int e^{-\varphi(u)}\Pi(z, du)}, \tag{12.17}$$

the derivatives at $\varphi \in \mathcal{U}_\epsilon$ are

$$DQ(\varphi)[h](z) = \int h(y)\Pi^\varphi(z, dy), \tag{12.18}$$

$$D^2Q(\varphi)[h, k](z) = -\operatorname{Cov}_{\Pi^\varphi(z,\cdot)}(h, k). \tag{12.19}$$

All covariance expressions are in $L^\infty(\pi^c; \mathbb{R})$. [ESTABLISHED]

Proof. For $\|\varphi\|_\infty < \epsilon$, the exponential power series converges in $L^\infty(\pi)$, and positivity plus unitality of U give

$$\|U(e^{-\varphi}) - 1\|_\infty \leq e^\epsilon - 1 < 1. \tag{12.20}$$

The Banach-algebra logarithm series therefore converges in $L^\infty(\pi^c)$, proving analyticity of $Q = -\log U(e^{-\varphi})$. Differentiating the convergent series, or using the ordinary Fréchet chain rule for the exponential, bounded linear operator U, and logarithm, gives (12.18). Differentiating that normalized integral in direction k gives

$$\frac{d}{dt}\bigg|_{t=0} \int h d\Pi^{\varphi+tk} = -\int hk d\Pi^\varphi + \left(\int h d\Pi^\varphi\right)\left(\int k d\Pi^\varphi\right), \tag{12.21}$$

which is (12.19). Setting $\varphi = 0$ proves (12.15) and (12.16). The uniform lower bound $U(e^{-\varphi}) \geq e^{-\epsilon}$ and boundedness of h, k make every displayed derivative a bounded map into the stated target. □

Proposition 12.4 (Bounded recentering gives analyticity at every bounded action). For every $\varphi \in \mathfrak{B}$, define the positive unital operator

$$U^\varphi h := \frac{U(e^{-\varphi}h)}{U(e^{-\varphi})}. \tag{12.22}$$

The denominator lies between $e^{-\|\varphi\|_\infty}$ and $e^{\|\varphi\|_\infty}$, so this is a bounded $L^\infty(\pi) \to L^\infty(\pi^c)$ operator. For every bounded increment h,

$$Q(\varphi + h) - Q(\varphi) = -\log U^\varphi(e^{-h}). \tag{12.23}$$

Consequently, for each $0 < \epsilon < \log 2$, the left side is real analytic as a function of h on $\|h\|_\infty < \epsilon$, and

$$DQ(\varphi)[h] = U^\varphi h. \tag{12.24}$$

Thus Q is locally Fréchet analytic at every bounded action, even though (12.11) was the one origin-centered chart used in Theorem 12.3. [ESTABLISHED]

Proof. The bounds follow from positivity and unitality of U. Substitution in (12.12) gives

$$-\log U(e^{-\varphi-h}) + \log U(e^{-\varphi}) = -\log \frac{U(e^{-\varphi}e^{-h})}{U(e^{-\varphi})},$$

which is (12.23). Positivity and unitality of U^φ imply $\|U^\varphi(e^{-h}) - 1\|_\infty \le e^\epsilon - 1 < 1$. The same Banach-algebra exponential and logarithm series used in Theorem 12.3 therefore converge on this increment ball. Their first derivative at $h = 0$ is $U^\varphi h$. □

For $\varphi, \psi \in \mathfrak{B}$, the nonlinear map is order preserving and additively homogeneous. Indeed, $\varphi \le \psi + c$ implies $e^{-\varphi} \ge e^{-c}e^{-\psi}$, and positivity of U gives $Q(\varphi) \le Q(\psi) + c$. Applying this implication in both directions with $c = \|\varphi - \psi\|_\infty$ proves the exact bounded-action estimate

$$\|Q(\varphi) - Q(\psi)\|_\infty \le \|\varphi - \psi\|_\infty. \tag{12.25}$$

Under isometric cross-scale identifications, this prevents relevance from arising in the bounded measure-pair action sector. It does not constrain an extensive interaction lift or a separately proved weighted action space. [ESTABLISHED]

The full action space remembers the perturbed mass $m_o^\varphi(\mathsf{Y})$. If an application deliberately discards that scalar, it instead uses the projective action space

$$\overline{\mathfrak{B}} = \mathfrak{B}/\mathbb{R}1. \tag{12.26}$$

Since $Q(\varphi + c) = Q(\varphi) + c$ and $U1 = 1$, both Q and U descend to the quotient. The quotient is not used to erase an evidence constant in the normalized measure-pair theory. [ESTABLISHED]

For comparison, normalize the perturbed fine law itself:

$$\widehat{\pi}^\varphi = \frac{e^{-\varphi}\pi}{\pi(e^{-\varphi})}, \qquad \widehat{Q}(\varphi) := -\log \frac{d(\widehat{\pi}^\varphi C)}{d\pi^c} = Q(\varphi) + \log \pi(e^{-\varphi}). \tag{12.27}$$

The two action increments therefore agree only modulo a constant. Their derivatives make that distinction explicit:

$$D\widehat{Q}(0)[h] = Uh - \pi(h), \tag{12.28}$$
$$D^2\widehat{Q}(0)[h,k] = -\operatorname{Cov}_{\Pi(z,\cdot)}(h,k) + \operatorname{Cov}_\pi(h,k). \tag{12.29}$$

The score of the normalized path $t \mapsto \widehat{\pi}^{th}$ is $-h + \pi(h)$, and its coarse score is $-Uh + \pi(h)$. Consequently the defect in (12.31), applied to the centered score, is exactly the Fisher-information loss under the fixed Markov channel. This score statement remains an L^2 tangent result; it does not enlarge the nonlinear bounded action chart. [ESTABLISHED]

At a boundary where an action representative is infinite, or outside the bounded action space, neither a literal action difference nor a differential beta functional is supplied by this calculus. Within the bounded space, Proposition 12.4 supplies local analyticity at every center. At an extended-valued boundary the primary object remains the measure pair and its Radon–Nikodym action; an action beta requires the separately declared finite-valued vector space and subtraction hypotheses of Section 12.9. [ESTABLISHED]

12.2.2 Canonical contraction and the bounded Dobrushin certificate

Theorem 12.5 (Conditional-expectation contraction and exact defect). For every $1 \le p \le \infty$, the operator in (12.14) extends uniquely to a contraction

$$U : L^p(\pi;\mathbb{R}) \longrightarrow L^p(\pi^c;\mathbb{R}), \qquad \|U\varphi\|_{L^p(\pi^c)} \le \|\varphi\|_{L^p(\pi)}. \tag{12.30}$$

For every $1 \le p \le \infty$, it preserves the mean and hence maps centered tangent spaces to centered tangent spaces. At $p = 2$, the loss is exactly

$$\|\varphi\|^2_{L^2(\pi)} - \|U\varphi\|^2_{L^2(\pi^c)} = \int \mathrm{Var}_{\Pi(z,\cdot)}(\varphi)\pi^c(dz) \ge 0. \tag{12.31}$$

Equality holds exactly when there is a measurable g on the coarse space such that $\varphi(y) = g(z)$ for $\pi(dy)C(y,dz)$-almost every (y,z). For a deterministic coarse map $z = c(y)$, this is precisely $\varphi = g \circ c$ almost everywhere under π. [ESTABLISHED]

Proof. Conditional Jensen gives $|U\varphi|^p \le U|\varphi|^p$ for finite p; integrating the disintegration (12.10) proves (12.30). The $p = \infty$ case follows from the pointwise bound by the essential supremum. The same disintegration gives $\int U\varphi d\pi^c = \int \varphi d\pi$. Finally, the conditional variance identity

$$\int \varphi^2 d\pi = \int (U\varphi)^2 d\pi^c + \int \mathrm{Var}_{\Pi(z,\cdot)}(\varphi)\pi^c(dz) \tag{12.32}$$

proves (12.31). A nonnegative conditional variance integrates to zero exactly when it is zero π^c-almost everywhere, which is equivalent to the stated joint measurability condition. □

On $L^\infty(\pi)/\mathbb{R}1$, use

$$\|[\varphi]\|_{\mathrm{osc}} = \inf_{c\in\mathbb{R}} \|\varphi - c\|_\infty = \frac{1}{2}\left(\operatorname{esssup}\varphi - \operatorname{essinf}\varphi\right). \tag{12.33}$$

For the fixed reverse-kernel version, define its Dobrushin coefficient by

$$\delta = \operatorname*{esssup}_{(z,z')\sim(\pi^c)^{\otimes 2}} \sup_{A\in\mathscr{Y}} \left|\Pi(z,A) - \Pi(z',A)\right|. \tag{12.34}$$

Proposition 12.6 (Dobrushin contraction is a sufficient cocycle certificate). The quotient map satisfies

$$\|\overline{U}\|_{\mathrm{osc}\to\mathrm{osc}} \le \delta. \tag{12.35}$$

For a nonautonomous sequence of such arrows with block factors $b_k > 1$, set

$$B_{n\leftarrow\ell} = \prod_{k=\ell}^{n-1} b_k, \qquad \overline{U}_{n\leftarrow\ell} = \overline{U}_{n-1}\cdots\overline{U}_{\ell}. \tag{12.36}$$

If $B_{n\leftarrow\ell} \to \infty$ and

$$\limsup_{n\to\infty} \frac{\sum_{k=\ell}^{n-1} \log\delta_k}{\log B_{n\leftarrow\ell}} < 0, \qquad \log 0 = -\infty, \tag{12.37}$$

then every bounded projective perturbation has strictly negative upper growth rate in this oscillation norm. With a fixed isometric cross-scale identification, fixed $b > 1$, and $\delta < 1$, this gives the autonomous bound $\log\delta/\log b < 0$. The condition is sufficient only: failure of (12.37) proves neither marginality nor relevance. [ESTABLISHED]

Proof. For probability measures μ, ν, $\left|\int f d\mu - \int f d\nu\right|$ is at most $\mathrm{osc}(f)\sup_A |\mu(A) - \nu(A)|$. Apply this to the two reverse laws in (12.34), divide by two, and take the essential supremum to obtain (12.35). Iteration gives

$$\|\overline{U}_{n\leftarrow\ell}\|_{\mathrm{osc}\to\mathrm{osc}} \le \prod_{k=\ell}^{n-1} \delta_k. \tag{12.38}$$

Taking logarithms with the stated zero convention proves the claimed upper growth bound. Two controls show why neither one-step behavior nor failure of the displayed certificate can be promoted to a necessary condition. For the first, take reverse matrices

$$R_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1/2 & 1/2 \end{pmatrix}, \qquad R_1 = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{12.39}$$

with compatible laws $\pi_0 = (1/2, 1/2)$, $\pi_1 = (1/3, 1/3, 1/3)$, and $\pi_2 = (2/3, 1/3)$. Both one-step coefficients are one, but the two rows of R_1R_0 are both $(1/2, 1/2)$, so the composed coefficient is zero. Thus individual strict Dobrushin contraction is not necessary for complete two-step forgetting. For the opposite control, take $\pi_k = (1/2, 1/2)$ at every scale and let J have both rows $(1/2, 1/2)$. Each matrix below is then a reverse kernel compatible with these laws:

$$R_k = (1 - a_k)I + a_k J, \qquad a_k = 2^{-k-2}. \tag{12.40}$$

Then $\delta_k = 1 - a_k < 1$ at every step, but $\prod_k \delta_k > 0$. With fixed $b_k = b > 1$, the normalized accumulated logarithmic rate can therefore be zero. One-step strict contraction is not sufficient for the nonautonomous conclusion. The identity channel has $\delta = 1$ and leaves every bounded mode marginal, while a two-cycle has an eigenvalue -1; these controls also prevent a phase-blind conclusion. □

12.2.3 Score tangents, the extensive block lift, and an inhabited spectrum

Estimate (12.25) shows that the bounded measure-pair action sector is nonexpansive, so under isometric cross-scale identifications no bounded action mode in that sector can grow. A growing mode must therefore come from somewhere else, and this subsection identifies the first of the two places where it does: the score tangent carries an *extensive* replication factor that the one-arrow contraction theorem cannot see. The second place is the extensive interaction assembly of Section 12.3.

Definition 12.7 (Quadratic-mean differentiability and score). Let $(\mathsf{W}, \mathscr{W}, \mu)$ be a probability space and write

$$L_0^2(\mu) = \left\{ h \in L^2(\mu) : \int h\, d\mu = 0 \right\}. \tag{12.41}$$

A family $(\mu_t)_{|t|<t_0}$ of probability laws with $\mu_t \ll \mu$ and $\mu_0 = \mu$ is differentiable in quadratic mean at $t = 0$ with score $h \in L^2(\mu)$ when

$$\left\| \sqrt{\frac{d\mu_t}{d\mu}} - 1 - \frac{t}{2} h \right\|_{L^2(\mu)} = o(|t|), \qquad t \to 0, \tag{12.42}$$

where both one-sided limits are required. The Fisher information of the path at $t = 0$ is $\|h\|^2_{L^2(\mu)}$. Nothing is proved by this declaration. [DEFINITION]

Centering is forced rather than assumed. Writing $p_t = d\mu_t/d\mu$ and using $\int p_t\, d\mu = 1$ gives the exact identity $\int (\sqrt{p_t} - 1)\, d\mu = -\frac{1}{2} \int (\sqrt{p_t} - 1)^2\, d\mu$. Inserting (12.42) makes the left side $\frac{t}{2} \int h\, d\mu + o(|t|)$ and the right side $-\frac{t^2}{8} \|h\|_2^2 + o(t^2)$, so $\int h\, d\mu = 0$. The quadratic-mean tangent set is therefore contained in $L_0^2(\mu)$. [ESTABLISHED]

Lemma 12.8 (Every centered square-integrable score is realized two-sided). Let $h \in L_0^2(\mu)$ and put $a = \|h\|_2^2/4$. For every real t, define

$$p_t = \frac{(1 + th/2)^2}{1 + at^2}, \qquad \mu_t = p_t \mu. \tag{12.43}$$

Then every μ_t is a probability law, $\mu_0 = \mu$, and the path is differentiable in quadratic mean at $t = 0$ with score h, for $t \to 0^+$ and for $t \to 0^-$. No exponential moment, and no boundedness of h, is used. [ESTABLISHED]

Proof. Centering gives $\int (1 + th/2)^2\, d\mu = 1 + t\mu(h) + \frac{t^2}{4}\mu(h^2) = 1 + at^2$, so $p_t \geq 0$ integrates to one. Put $s_t = 1 + th/2$ and $N_t = 1 + at^2 \geq 1$, so that $\sqrt{p_t} = |s_t|/\sqrt{N_t}$ and

$$\sqrt{p_t} - s_t = \frac{|s_t| - s_t}{\sqrt{N_t}} + s_t \left(\frac{1}{\sqrt{N_t}} - 1 \right). \tag{12.44}$$

On $\{s_t < 0\}$ one has $th/2 < -1$, hence $|h| > 2/|t|$ and $|s_t| = |th/2| - 1 \le |th|/2$, so $|s_t| - s_t = 2|s_t| \le |th|$ there and the difference vanishes elsewhere. Since $N_t \ge 1$,

$$\frac{1}{t^2}\left\|\frac{|s_t| - s_t}{\sqrt{N_t}}\right\|_2^2 \le \int h^2 \mathbf{1}_{\{|h|>2/|t|\}}\, d\mu \longrightarrow 0 \tag{12.45}$$

by dominated convergence, because h^2 is integrable. For the second term, $\|s_t\|_2 = \sqrt{N_t}$, so its norm is exactly $|1 - \sqrt{N_t}| = O(t^2)$. Adding the two bounds and using $s_t = 1 + \frac{t}{2}h$ gives (12.42). □

Proposition 12.9 (Independent block replication of a differentiable path). Fix an integer $b \ge 1$. Let (μ_t) be any path satisfying (12.42) with score h, and form the product path $\mu_t^{\otimes b}$ at $\mu^{\otimes b}$. Then the product path is differentiable in quadratic mean at $t = 0$ with score

$$(\mathscr{I}_b h)(x_1, \dots, x_b) = \sum_{i=1}^{b} h(x_i), \tag{12.46}$$

and the replication map is a bounded linear injection

$$\mathscr{I}_b : L_0^2(\mu) \longrightarrow L_0^2(\mu^{\otimes b}), \qquad \|\mathscr{I}_b h\|_{L^2(\mu^{\otimes b})}^2 = b\|h\|_{L^2(\mu)}^2, \tag{12.47}$$

so $\|\mathscr{I}_b\| = \sqrt{b}$ and $b^{-1/2}\mathscr{I}_b$ is an isometry onto its range. The symbol $\mathscr{I}_b$ is the score replication lift; it is not the Hoeffding assembly E_ℓ of (12.74). [ESTABLISHED]

Proof. Write $r_t = \sqrt{d\mu_t/d\mu} = 1 + \frac{t}{2}h + t\varepsilon_t$ with $\|\varepsilon_t\|_{L^2(\mu)} \to 0$, which is (12.42). The product amplitude is $\prod_{i=1}^{b} r_t(x_i) = \prod_{i=1}^{b}(1 + u_i)$ with $u_i(x) = \frac{t}{2}h(x_i) + t\varepsilon_t(x_i)$. Expanding the finite product,

$$\prod_{i=1}^{b}(1 + u_i) = 1 + \sum_{i=1}^{b} u_i + \sum_{|S|\ge 2}\prod_{i\in S} u_i. \tag{12.48}$$

Because the coordinates are independent and each factor depends on one coordinate, $L^2(\mu^{\otimes b})$ norms of products factor, so $\|\prod_{i\in S} u_i\|_2 = \prod_{i\in S}\|u_i\|_2 = O(|t|^{|S|})$, and every term with $|S| \ge 2$ is $O(t^2)$. The linear part is $\frac{t}{2}\mathscr{I}_b h + t\sum_i \varepsilon_t(x_i)$, whose remainder has norm at most $b|t|\|\varepsilon_t\|_2 = o(|t|)$ by the triangle inequality. This proves quadratic-mean differentiability with score $\mathscr{I}_b h$. Independence and centering give $\mathbb{E}(\mathscr{I}_b h)^2 = \sum_i \mathbb{E}h(X_i)^2 + \sum_{i\ne j}\mathbb{E}h(X_i)\mathbb{E}h(X_j) = b\|h\|_2^2$, which is (12.47). □

The replication is extensive. Replacing (12.46) by the arithmetic mean without simultaneously rescaling the path parameter would change the Fisher information from $b\|h\|_2^2$ to $\|h\|_2^2/b$, which is a different theory. [ESTABLISHED]

Theorem 12.10 (Pushed score and the exact extensive Fisher defect). Let C_b : W$^b \rightsquigarrow$ Z be a normalized, parameter-independent Markov kernel between standard-Borel spaces, put $\mu^{(b)} = \mu^{\otimes b}$, $J_b(dx, dz) = \mu^{(b)}(dx)C_b(x, dz)$, and $\mu^{(b),c} = \mu^{(b)}C_b$. Let $U_b : L^2(\mu^{(b)}) \to L^2(\mu^{(b),c})$ be the reverse conditional expectation of Theorem 12.5 for this arrow. Then the pushed path $\mu_t^{\otimes b}C_b$ is differentiable in quadratic mean at $\mu^{(b),c}$ with score

$$\mathscr{L}_b h := U_b \mathscr{I}_b h = \mathbb{E}_{J_b}\left[\sum_{i=1}^{b} h(X_i) \,\middle|\, Z\right], \tag{12.49}$$

so that $\mathscr{L}_b : L_0^2(\mu) \to L_0^2(\mu^{(b),c})$ is linear with $\|\mathscr{L}_b\| \le \sqrt{b}$. Moreover the exact information budget is

$$b\|h\|_{L^2(\mu)}^2 - \|\mathscr{L}_b h\|_{L^2(\mu^{(b),c})}^2 = \int \mathrm{Var}_{J_b}\left(\mathscr{I}_b h \,|\, Z = z\right) \mu^{(b),c}(dz) \ge 0, \tag{12.50}$$

with equality exactly when $\mathscr{I}_b h$ agrees J_b-almost surely with a measurable function of Z. Equivalently $b^{-1/2}\mathscr{L}_b$ is a contraction of L_0^2 norms. [ESTABLISHED]

Proof. Suppose first that h is bounded and use the path (12.43). For $|t|$ small, $s_t \ge 1 - |t|\|h\|_\infty/2 > 0$, so $p_t = s_t^2/N_t = 1 + th + O(t^2)$ uniformly, and the product density is $\prod_i p_t(x_i) = 1 + t\mathscr{I}_b h + O(t^2)$ uniformly. Conditional expectation of a uniformly $O(t^2)$ remainder is uniformly $O(t^2)$, and

$$\frac{d(\mu_t^{\otimes b} C_b)}{d\mu^{(b),c}}(z) = \mathbb{E}_{J_b}\left[\frac{d\mu_t^{\otimes b}}{d\mu^{(b)}}(X) \,\middle|\, Z = z\right] = 1 + t\mathscr{L}_b h(z) + O(t^2) \tag{12.51}$$

uniformly, exactly as in (12.8). Since $\sqrt{1+w} = 1 + w/2 + O(w^2)$ uniformly for $|w| \le 1/2$, the square root of (12.51) is $1 + \frac{t}{2}\mathscr{L}_b h + O(t^2)$ uniformly, hence in $L^2(\mu^{(b),c})$. This is (12.42) for the pushed path.

For general $h \in L_0^2(\mu)$, choose bounded centered $h_n \to h$ in $L^2(\mu)$ and let $\mu_t^{(n)}$ be their paths (12.43). Squared Hellinger distance is an f-divergence with the convex $f(u) = (\sqrt{u} - 1)^2$, so conditional Jensen applied to the disintegration of the pushed joint law gives the data-processing bound

$$\left\|\sqrt{\frac{d(\mu_t^{\otimes b} C_b)}{d\mu^{(b),c}}} - \sqrt{\frac{d((\mu_t^{(n)})^{\otimes b} C_b)}{d\mu^{(b),c}}}\right\|_2 \le \left\|\sqrt{\frac{d\mu_t^{\otimes b}}{d\mu^{(b)}}} - \sqrt{\frac{d(\mu_t^{(n)})^{\otimes b}}{d\mu^{(b)}}}\right\|_2. \tag{12.52}$$

Both input paths are differentiable in quadratic mean with scores $\mathscr{I}_b h$ and $\mathscr{I}_b h_n$, so the right side divided by $|t|$ has limit superior $\frac{1}{2}\|\mathscr{I}_b(h - h_n)\|_2 = \frac{\sqrt{b}}{2}\|h - h_n\|_2$. Adding and subtracting the bounded-case expansion, and using $\|\mathscr{L}_b(h - h_n)\|_2 \le \sqrt{b}\|h - h_n\|_2$,

$$\limsup_{t\to 0} \frac{1}{|t|}\left\|\sqrt{\frac{d(\mu_t^{\otimes b} C_b)}{d\mu^{(b),c}}} - 1 - \frac{t}{2}\mathscr{L}_b h\right\|_2 \le \sqrt{b}\|h - h_n\|_2. \tag{12.53}$$

The left side does not depend on n, so letting $n \to \infty$ makes it zero, which is the general case.

Conditional Jensen gives $\|U_b\| \le 1$ and preservation of the mean, so $\mathscr{L}_b$ has the stated type and $\|\mathscr{L}_b\| \le \|\mathscr{I}_b\| = \sqrt{b}$. Finally (12.32) applied to $\mathscr{I}_b h$, together with (12.47), is exactly (12.50), and a nonnegative conditional variance integrates to zero precisely when it vanishes almost everywhere.
□

Two boundaries of (12.50) deserve a sentence. A channel that maps every block to one point sends every score to zero, so a fixed channel can annihilate the entire tangent. In the opposite direction, Fisher equality at one parameter is not recovery of the experiment: let A and B be independent with $\Pr_\theta(A=1)=\frac{1}{2}+\frac{\theta}{4}$ and $\Pr_\theta(B=1)=\frac{1}{2}+\frac{\theta^2}{4}$, and let the channel retain only A. At $\theta=0$ the B-score vanishes, so the full score is A-measurable and equality holds, yet the conditional law of B varies with θ and no parameter-independent reverse kernel reconstructs the experiment. [ESTABLISHED]

Proposition 12.11 (Bounded action classes embed isometrically in the Fisher tangent). For $\varphi\in\mathfrak{B}=L^\infty(\pi;\mathbb{R})$, the normalized exponential path $\widehat{\pi}^{t\varphi}$ of (12.27) is differentiable in quadratic mean with score

$$\mathscr{S}_\pi[\varphi]=-\varphi+\pi(\varphi), \tag{12.54}$$

which depends only on the class $[\varphi]\in\overline{\mathfrak{B}}$. Equip the action quotient with the Fisher norm

$$\|[\varphi]\|_F=\inf_{c\in\mathbb{R}}\|\varphi-c\|_{L^2(\pi)}=\|\varphi-\pi(\varphi)\|_{L^2(\pi)}. \tag{12.55}$$

Then $\mathscr{S}_\pi:(\overline{\mathfrak{B}},\|\cdot\|_F)\to L^2_0(\pi)$ is a linear isometry whose range is the bounded centered subspace; that range is dense but generally proper, so the Fisher-norm completion of the bounded action quotient is canonically $L^2_0(\pi)$. The two quotient norms are ordered by $\|[\varphi]\|_F\le\|[\varphi]\|_{\mathrm{osc}}$. Moreover the centering square commutes,

$$\mathscr{S}_{\pi^c}\big(\overline{U}[\varphi]\big)=U\mathscr{S}_\pi[\varphi], \tag{12.56}$$

and for the replicated block action $\Phi_\varphi(x_1,\dots,x_b)=\sum_i\varphi(x_i)$ one has $\mathscr{S}_{\mu^{(b)}}[\Phi_\varphi]=\mathscr{I}_b\mathscr{S}_\mu[\varphi]$ and, after the block arrow, $\mathscr{L}_b\mathscr{S}_\mu[\varphi]$. [ESTABLISHED]

Proof. Boundedness gives the uniform expansions $e^{-t\varphi}=1-t\varphi+O(t^2)$ and $\pi(e^{-t\varphi})=1-t\pi(\varphi)+O(t^2)$, so the density ratio of $\widehat{\pi}^{t\varphi}$ is $1-t[\varphi-\pi(\varphi)]+O(t^2)$ uniformly and its square root is $1+\frac{t}{2}\mathscr{S}_\pi[\varphi]+O(t^2)$, which is (12.42). Adding a constant to φ leaves (12.54) unchanged. The L^2 minimizer over constants is the mean, giving (12.55) and hence the isometry; $\inf_c\|\varphi-c\|_2\le\inf_c\|\varphi-c\|_\infty$ gives the norm ordering. Bounded functions are dense in $L^2(\pi)$ and centering is continuous, so the range is dense; it is proper whenever $L^2_0(\pi)$ contains an unbounded element, as for $\pi=N(0,1)$ and $h(x)=x$. Identity (12.56) is mean preservation of U together with linearity, and the replication statement is (12.46). □

The two norms on $\overline{\mathfrak{B}}$ do different work and must not be exchanged. The oscillation norm of (12.33) controls the nonlinear map Q; the Fisher norm controls the quadratic-mean tangent. Completing in the Fisher norm introduces unbounded scores on which the nonlinear bounded action map of Theorem 12.3 is not defined. A spectral statement proved on L^2_0 is therefore not a statement about the nonlinear bounded action chart, and conversely. [ESTABLISHED]

Two further separations belong here, and each is stated with the witness that makes it sharp. First, every class in the Fisher completion is realized as a two-sided quadratic-mean score by the *quadratic* path of Lemma 12.8, not by the exponential-action path $\widehat{\pi}^{t\varphi}$, which is defined only where its normalizer is finite. The witness that separates the two must therefore be a direction whose exponential normalizer diverges on *both* sides. The correct witness is an odd Hermite polynomial of degree at least three: for $\varphi = \mathrm{He}_3(x) = x^3 - 3x \in L^2_0(\gamma)$, the log integrand $-t\mathrm{He}_3(x) - x^2/2$ has an exact divergent tail on either side. For $t > 0$, substitute $x = -r$ to obtain

$$-t\mathrm{He}_3(-r) - \frac{r^2}{2} = tr^3 - \frac{r^2}{2} - 3tr \longrightarrow +\infty \qquad (r \to +\infty),$$

for $t < 0$, substitute $x = r$ to obtain

$$-t\mathrm{He}_3(r) - \frac{r^2}{2} = |t|r^3 - \frac{r^2}{2} - 3|t|r \longrightarrow +\infty.$$

The leading cubic is positive in both cases, so the exponential integrand is eventually at least one on an infinite tail and $N_3(t) = +\infty$ for *every* $t \neq 0$, while He_3 is a perfectly good Fisher tangent direction with $\mathbb{E}_\gamma[\mathrm{He}_3] = 0$ and $\mathbb{E}_\gamma[\mathrm{He}_3^2] = 6$. The degree-two direction $\varphi = -x^2$ is *not* such a witness and must not be cited as one: its normalizer is $\pi(e^{tx^2}) = (1-2t)^{-1/2}$, which is finite on the two-sided interval $(-\frac{1}{2}, \frac{1}{2})$, taking the values $\sqrt{6}/3$ at $t = -\frac{1}{4}$ and $\sqrt{2}$ at $t = \frac{1}{4}$. This is the classification already recorded in Proposition 16.7. [ESTABLISHED]

Second, the extensive replication lift is not a channel and no data-processing statement applies to it in the coarse-to-fine direction. The operator $\mathscr{I}_b$ of Proposition 12.9 maps $L^2_0(\pi)$ into $L^2_0(\pi^{\otimes b})$ with $\|\mathscr{I}_b\| = \sqrt{b} > 1$; it is a lift between two different reference tangent spaces, not a pushforward along any arrow. No parameter-independent normalized Markov kernel realizes b-fold replication for $b \geq 2$: if K satisfied $\mathcal{N}(x,1)K = \mathcal{N}(x\mathbf{1}_b, I_{\mathbb{R}^b})$ for every x, then Theorem 12.10 would force the output Fisher information b to satisfy $b \leq 1$. Consequently the replication pair may not be cited as a Markov-contraction counterexample or as evidence that Fisher information increased along a coarse-graining arrow, because there is no such arrow; the factor $\sqrt{b}$ belongs to the reference identifications I_ℓ of Definition 12.26 and to a declared configuration-metric normalization, and to nothing else. [ESTABLISHED]

Theorem 12.12 (Gaussian block spectrum in the normalized Hermite basis). Let $\gamma = N(0,1)$, let $b \geq 2$ be an integer, let $X_1, \ldots, X_b$ be independent with law γ, and let the block channel be the deterministic normalized sum

$$Z = b^{-1/2} \sum_{i=1}^{b} X_i, \tag{12.57}$$

whose law is again γ. It is that normalization, and not a general theorem, that makes source and target the same space and hence makes

$$\mathscr{L}_b = U_b \mathscr{I}_b : L^2_0(\gamma) \longrightarrow L^2_0(\gamma) \tag{12.58}$$

an endomorphism. Let He_k be the probabilists' Hermite polynomials and $e_k = \mathrm{He}_k/\sqrt{k!}$. Then $\{e_k\}_{k \geq 1}$ is a complete orthonormal basis of $L^2_0(\gamma)$ and

$$\mathscr{L}_b e_k = b^{1-k/2} e_k, \qquad k \geq 1. \tag{12.59}$$

Consequently $\mathscr{L}_b$ is self-adjoint, positive, and diagonal in that basis; it is Hilbert–Schmidt with $\sum_{k\geq 1} b^{2-k} = b^2/(b-1)$, hence compact; its norm and spectral radius are $\sqrt{b}$; and

$$\sigma(\mathscr{L}_b) = \left\{ b^{1-k/2} : k \geq 1 \right\} \cup \{0\}. \tag{12.60}$$

Every displayed nonzero value is a simple eigenvalue. The value 0 is *not* an eigenvalue: it is the unique accumulation point of the spectrum and belongs to the continuous spectrum. This is a linear tangent-space statement about the derivative at the Gaussian law; it is not a nonlinear convergence theorem. [ESTABLISHED]

Proof. Completeness of the normalized Hermite system in $L^2(\gamma)$ is standard, and deleting $e_0 = 1$ gives a complete orthonormal basis of the centered subspace. For each i, the pair (X_i, Z) is jointly centered Gaussian with unit variances and $\mathrm{Cov}(X_i, Z) = b^{-1/2}$, so $X_i \mid Z = z \sim N(z/\sqrt{b}, 1 - 1/b)$. Taking the conditional expectation of the generating function $\exp(tx - t^2/2)$ and completing the square,

$$\mathbb{E}\left[e^{tX_i - t^2/2} \,\middle|\, Z = z \right] = \exp\left(\frac{tz}{\sqrt{b}} + \frac{t^2}{2}\left(1 - \frac{1}{b}\right) - \frac{t^2}{2} \right) = \exp\left(sz - \frac{s^2}{2} \right), \quad s = \frac{t}{\sqrt{b}}. \tag{12.61}$$

The generating identity $\exp(sz - s^2/2) = \sum_{k\geq 0} \mathrm{He}_k(z) s^k/k!$ converges in $L^2(\gamma)$ locally uniformly in s, and conditional expectation is L^2 continuous, so the two power series in t may be compared coefficient by coefficient. This gives the Mehler regression identity $\mathbb{E}[\mathrm{He}_k(X_i) \mid Z] = b^{-k/2}\mathrm{He}_k(Z)$. Summing over the b replicated coordinates and dividing by $\sqrt{k!}$ proves (12.59); the argument is uniform in b and is not an induction from the binary case.

Since $\mathscr{L}_b$ is bounded and agrees with the displayed diagonal action on a complete orthonormal basis, it is that diagonal operator, hence self-adjoint and positive with eigenvalues $b^{1-k/2} \downarrow 0$. Square summability of the eigenvalues is the stated geometric series, so $\mathscr{L}_b$ is Hilbert–Schmidt and compact, and its norm is the largest eigenvalue $b^{1/2}$. The spectrum of a diagonal operator is the closure of its eigenvalue set, which is (12.60), and the eigenvalues are pairwise distinct so each is simple. Every eigenvalue is strictly positive, so $\ker \mathscr{L}_b = \{0\}$ and 0 is not an eigenvalue. The range contains every finite Hermite expansion and is therefore dense, but it is proper: the vector $y = \sum_{k\geq 1} b^{1-k/2} e_k$ lies in $L_0^2(\gamma)$ because the coefficients are square summable, while any preimage would have every Hermite coefficient equal to one, which is not square summable. An injective operator with dense, nonclosed range has 0 in its continuous spectrum. □

Definition 12.13 (Relevance conventions for the Gaussian block spectrum). Fix the replication sum (12.46), the block statistic (12.57), the same Fisher norm $\|\cdot\|_{L^2(\gamma)}$ at every scale, and the cumulative log block scale $\Delta s = \log b$ of (11.29). Then the exponent of the degree-k mode is $y_k = \log_b b^{1-k/2} = 1 - k/2$. Degree one is relevant with $y_1 = \frac{1}{2}$, degree two is marginal with $y_2 = 0$, and every degree $k \geq 3$ is irrelevant with $y_k = 1 - k/2 < 0$. The constant mode $k = 0$ would have eigenvalue b on the full space $L^2(\gamma)$ but is absent from the normalized probability tangent, where scores are centered. [DEFINITION]

This is the promised inhabited relevance spectrum, and it locates precisely where the growth comes from. The one-arrow operator U_b is a contraction by Theorem 12.5, and the bounded action sector is nonexpansive by (12.25). The eigenvalue $b^{1-k/2}$ exceeds one for $k = 1$ only because the replication factor $\|\mathscr{I}_b\| = \sqrt{b}$ of (12.47) is extensive while the norm is held fixed across scales. Relevance in this theory is therefore a joint property of the operator, the extensive normalization, the declared norms, and the block scale, in exactly the sense of Definition 11.12. Removing any one of the four removes the conclusion. [ESTABLISHED]

Proposition 12.14 (Convention dependence and forbidden extensions). The spectrum (12.60) is specific to the scalar independent Gaussian realization and its declared normalization. Four exact boundaries hold. First, if the coordinates are jointly centered Gaussian with unit variances and common pairwise correlation $\rho > -1/(b-1)$, and the statistic is normalized to unit variance as $Z = (\sum_i X_i)/\sqrt{b[1 + (b-1)\rho]}$, then $\operatorname{Corr}(X_i, Z) = \alpha = \sqrt{[1 + (b-1)\rho]/b}$ and

$$\mathscr{L}_{b,\rho} e_k = b\alpha^k e_k = b^{1-k/2} \left[1 + (b-1)\rho\right]^{k/2} e_k, \tag{12.62}$$

which differs from (12.59) for every $\rho \neq 0$; keeping the uncorrelated normalization instead destroys the property that the output law is γ, so the operator is then not an endomorphism at all. Second, for independent $N(0, I_d)$ coordinates and componentwise normalized sums the tensor Hermite functions satisfy $\mathscr{L}_b e_\alpha = b^{1-|\alpha|/2} e_\alpha$, so degree k has multiplicity $\binom{d+k-1}{k}$ rather than one. Third, a common orthogonal action commutes with the componentwise normalized sum and preserves each degree space, whereas a general $\mathrm{GL}(d)$ transformation changes the reference covariance and therefore requires an explicitly transported family of laws and tangent norms; no gauge link, holonomy, or curvature datum enters the scalar theorem. Fourth, (12.46) is the score of an independent product perturbation and is a restricted direction inside $L_0^2(\gamma^{\otimes b})$; the theorem says nothing about interaction scores or higher Hoeffding sectors, which are the subject of Section 12.3. [ESTABLISHED]

Proof. For the first statement, (X_i, Z) is again jointly centered Gaussian with unit variances and correlation α, so $X_i \mid Z = z \sim N(\alpha z, 1-\alpha^2)$; the computation (12.61) with $b^{-1/2}$ replaced by α gives $\mathbb{E}[\mathrm{He}_k(X_i) \mid Z] = \alpha^k \mathrm{He}_k(Z)$, and summing over i gives (12.62). Substituting $\alpha^2 = [1 + (b-1)\rho]/b$ gives the second form. For the second statement, tensor Hermite functions factor over coordinates and each factor obeys the scalar identity with the same $b^{-1/2}$, so the eigenvalue depends only on the total degree $|\alpha|$, and the number of multi-indices of total degree k in d variables is $\binom{d+k-1}{k}$. For the third, an orthogonal map preserves $N(0, I_d)$ and commutes with the componentwise sum, hence preserves each finite-dimensional degree space, while a general invertible linear map sends $N(0, I_d)$ to $N(0, QQ^\top)$, which is a different reference law. The fourth statement is the definition of $\mathscr{I}_b$. $\square$

The linearization also does not by itself establish nonlinear convergence. The spectrum controls the derivative at the Gaussian law, while a basin of attraction, a remainder estimate, and the behavior of the marginal direction are separate obligations recorded in Chapter B. [OPEN]

The same qualification is made in the primary source: Jona-Lasinio [38] obtains the binary case of (12.59) and states that the nonlinear terms required to complete a limit theorem are not pursued there. [ESTABLISHED]

Source scope. Section 2 of Jona-Lasinio [38] treats the binary normalized sum, linearizes the induced map on laws around a centered unit-variance Gaussian under normalization, centering, and variance constraints, and states the Hermite eigenvalues $2^{1-k/2}$; Section 5 identifies the linearized renormalization at a self-similar Gaussian fixed point with a conditional expectation and states the corresponding generalized Hermite eigenvalue equation; and Section 7 gives the two-tangent-space form of that eigenvalue equation together with its multiplicative composition law across scales. Those four items are what the citation supports. The arbitrary-integer statement (12.59), the quadratic-mean realization of Lemma 12.8, the extensive Fisher budget (12.50), the exact spectrum (12.60) including the status of 0, the correlated boundary (12.62), and the typed cocycle of Section 11.4.1 are proved here and are not attributed to that source. [ESTABLISHED]

12.2.4 Essential spectrum and topology dependence

Theorem 12.15 (Qualified positive-unital essential-spectrum theorem). Let X be a complex Banach lattice with quasi-interior unit $\mathbf{1} \in X_+$, let $U \in \mathcal{L}(X)$ be positive and unital, and write $q : \mathcal{L}(X) \to \mathcal{L}(X)/\mathcal{K}(X)$ for the Calkin quotient by the compact-operator ideal. Use the convention

$$r_{\text{ess}}(U) = r\big(q(U)\big). \tag{12.63}$$

Assume that whenever $r(U) > r_{\text{ess}}(U)$, the value $r(U)$ is a pole of the resolvent and U^* has a nonzero positive eigenfunctional λ satisfying $U^*\lambda = r(U)\lambda$. Then

$$r(U) > 1 \quad \Longrightarrow \quad r(U) = r_{\text{ess}}(U). \tag{12.64}$$

This does not assert quasi-compactness or a Perron theorem under weaker hypotheses. [ESTABLISHED]

Proof. The quotient map is contractive, so $r_{\text{ess}}(U) \le r(U)$. Suppose that $r(U) > 1$ and that the inequality is strict. The stated hypothesis gives $\lambda \ge 0$, $\lambda \ne 0$, and $U^*\lambda = r(U)\lambda$. Quasi-interiority implies $\lambda(\mathbf{1}) > 0$: otherwise positivity makes λ zero on the principal ideal generated by $\mathbf{1}$, hence on its dense closure. But unitality yields

$$r(U)\lambda(\mathbf{1}) = (U^*\lambda)(\mathbf{1}) = \lambda(U\mathbf{1}) = \lambda(\mathbf{1}), \tag{12.65}$$

a contradiction. Therefore strict inequality is impossible, and the two radii agree. □

Proposition 12.16 (Circle witness for norm-dependent radius). Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ carry its geodesic metric and Haar probability λ, let $D(z) = 2z \bmod 1$, and set

$$K(y, A) = \frac{1}{2} \sum_{z \in D^{-1}\{y\}} \mathbf{1}_A(z) \qquad (A \subseteq \mathbb{T} \text{ Borel}). \tag{12.66}$$

The kernel preserves Haar measure and its reverse is $\Pi(z, dy) = \delta_{D(z)}(dy)$, so the conditional-expectation operator is $Uf = f \circ D$. Its spectral radius is one on $L^\infty(\lambda)$. For every $0 < \alpha \le 1$, its spectral radius on complex periodic $C^\alpha(\mathbb{T})$ equipped with the sup-plus-Hölder norm is 2^α. At $\alpha = 1$, this means the Lipschitz space $C^{0,1}$, not C^1. The Hölder space with this standard norm is not the Banach lattice of Theorem 12.15; the witness is a separate topology comparison. Hence the same exact kernel has distinct spectral conclusions in two declared topologies. [ESTABLISHED]

Proof. For bounded f, Haar preservation gives $\|f \circ D^n\|_\infty = \|f\|_\infty$, so $\|U^n\| = 1$ and the first radius is one. With $[f]_\alpha$ the Hölder seminorm,

$$[f \circ D^n]_\alpha \le 2^{\alpha n}[f]_\alpha, \qquad \|U^n f\|_\infty \le \|f\|_\infty, \tag{12.67}$$

so the second radius is at most 2^α. For the fixed Fourier mode $f(t) = e^{2\pi i t}$, evaluate $f \circ D^n$ at 0 and $2^{-(n+1)}$. Their values differ by two while their circle distance is $2^{-(n+1)}$, whence

$$[f \circ D^n]_\alpha \ge 2^{1+\alpha(n+1)}. \tag{12.68}$$

After division by the fixed nonzero C^α norm of f, this gives the matching lower spectral-radius bound. □

12.3 Exact finite-network interaction coordinates

The following construction is an action-coordinate construction, not a claim that an arbitrary coarse channel preserves a product law. At every scale ℓ, let V_ℓ be an arbitrary finite set, let $\mathsf{X}_{\ell i}$ be standard Borel for $i \in V_\ell$, and put

$$\mathsf{X}_\ell = \prod_{i \in V_\ell} \mathsf{X}_{\ell i}, \qquad \nu_\ell = \bigotimes_{i \in V_\ell} \nu_{\ell i}, \qquad \pi_\ell \sim \nu_\ell. \tag{12.69}$$

Here each $\nu_{\ell i}$ is a probability law and $\sim$ denotes mutual absolute continuity. Thus $L^\infty(\pi_\ell)$ and $L^\infty(\nu_\ell)$ have the same equivalence classes and the same essential-supremum norm, without requiring bounded Radon–Nikodym ratios. The product structure of ν_ℓ, rather than equivalence alone, makes the coordinate integrals below representative-independent by Fubini. For a scale arrow, the target premise $\pi_{\ell+1} = \pi_\ell K_\ell \sim \nu_{\ell+1}$ is separately required; it does not follow from $\pi_\ell \sim \nu_\ell$. The evidence mass M_ℓ of Definition 11.2 is retained independently of these normalized references. [DEFINITION]

For $V_\ell = \varnothing$, the product is the one-point probability space, $\mathcal{G}_\ell = \{0\}$, and all assembly, extraction, and interaction maps below are the unique zero maps. The nontrivial estimates that divide by $3^{|V_\ell|} - 1$ are stated only when $V_\ell \neq \varnothing$.

Proposition 12.17 (Product equivalence is an admitted, not an automatic, scale premise). Let $\mathsf{X}_0 = \{0,1\}$ with its uniform law $\pi_0 = \nu_0$, and let K be the deterministic diagonal-cloning channel

$$K(x, \{(x,x)\}) = 1 \quad \text{from } \{0,1\} \text{ to } \{0,1\}^2. \tag{12.70}$$

Then $\pi_1 = \pi_0 K$ is supported on $\{(0,0),(1,1)\}$. There is no product probability $\nu_1 = \nu_{11} \otimes \nu_{12}$ on $\{0,1\}^2$ equivalent to π_1. In particular the uniform product reference gives positive mass to all four points and is not equivalent. No target Hoeffding decomposition relative to a product reference is admitted at this target. [ESTABLISHED]

Proof. Equivalence would make both diagonal atoms positive under ν_1. Thus each marginal would charge both 0 and 1, forcing both off-diagonal atoms to have positive product mass. But π_1 assigns them mass zero, contradicting equivalence. The uniform product is the stated special case. □

For $A \subseteq V_\ell$, let $C_{\ell,A}$ be conditional integration against the product complement, lifted back to X_ℓ:

$$(C_{\ell,A} f)(x_A) = \int f(x_A, y_{A^c}) \nu_{\ell,A^c}(dy_{A^c}), \qquad C_{\ell,\varnothing} f = \nu_\ell(f). \tag{12.71}$$

The operators commute and obey $C_{\ell,A} C_{\ell,B} = C_{\ell,A\cap B}$. Their Boolean-lattice Mobius projectors are

$$P_{\ell,A} = \sum_{B \subseteq A} (-1)^{|A|-|B|} C_{\ell,B} = \prod_{i \in A} (I - C_{\ell,V_\ell \setminus \{i\}}) C_{\ell,A}, \qquad A \subseteq V_\ell. \tag{12.72}$$

The second expression only records the commuting-product identity; the first is the definition. Let

$$\mathcal{H}_{\ell,A} = P_{\ell,A} L^\infty(\nu_\ell), \qquad \mathcal{G}_\ell = \bigoplus_{\varnothing \neq A \subseteq V_\ell}^{\ell^1} \mathcal{H}_{\ell,A}, \qquad \|g\|_{\mathcal{G}_\ell} = \sum_{\varnothing \neq A \subseteq V_\ell} \|g_A\|_\infty. \tag{12.73}$$

Thus g_A depends only on x_A and has zero product-reference mean in every active coordinate. This includes every nonempty hyperedge; it is finite at each ℓ, with no bound imposed on $|V_\ell|$. Define the action quotient and its assembly/extraction maps by

$$\overline{\mathfrak{B}}_\ell = L^\infty(\nu_\ell)/\mathbb{R}1, \qquad \phi_g := \sum_{\varnothing \neq A \subseteq V_\ell} g_A, \qquad E_\ell g = [\phi_g], \qquad \mathsf{H}_\ell[f] = (P_{\ell,A} f)_{\varnothing \neq A \subseteq V_\ell}. \tag{12.74}$$

Theorem 12.18 (Hoeffding assembly is an exact finite-network action isomorphism). For every finite V_ℓ, E_ℓ and H_ℓ in (12.74) are bounded inverse linear isomorphisms,

$$\mathsf{H}_\ell E_\ell = I_{\mathcal{G}_\ell}, \qquad E_\ell \mathsf{H}_\ell = I_{\overline{\mathfrak{B}}_\ell}, \qquad \|E_\ell\| \le 1, \qquad \|\mathsf{H}_\ell\| \le 3^{|V_\ell|} - 1. \tag{12.75}$$

The latter bound is valid for every finite network but is not dimension-free. Its exponential dependence is sharp in the worst case: for biased Rademacher product coordinates and $f(x) = \prod_i x_i$, the sum of component norms approaches the bound at the corresponding product scale as the bias tends to one. [ESTABLISHED]

Proof. The commuting conditional expectations give $P_{\ell,A}P_{\ell,B} = \mathbf{1}_{A=B}P_{\ell,A}$ and $\sum_{A\subseteq V_\ell} P_{\ell,A} = I$. Hence nonempty components recover g, while their sum is $f - C_{\ell,\varnothing}f$, which is the class $[f]$. The triangle inequality gives $\|E_\ell\| \le 1$. Since $P_{\ell,A}$ is an alternating sum of $2^{|A|}$ contractive coordinate expectations, its norm is at most $2^{|A|}$. Summing over nonempty A gives $\sum_{A\neq\varnothing} 2^{|A|} = 3^{|V_\ell|} - 1$. For independent $\{-1,1\}$-coordinates with $\Pr(X_i = 1) = p \ge 1/2$, set $n = |V_\ell|$ and $f = \prod_i X_i$. Direct expansion gives $\sum_{\varnothing\neq A\subseteq V_\ell}\|P_A f\|_\infty = (4p-1)^n - (2p-1)^n$, which tends to $3^n - 1$ as $p \uparrow 1$. □

Gauge covariance and its necessary realization hypothesis. Let a gauge transformation at scale ℓ be realized by a Borel coordinate permutation σ and coordinate Borel isomorphisms whose product map ϑ_ℓ sends ν_ℓ to $\widehat{\nu}_\ell$. Write $\Theta_\ell f = f \circ \vartheta_\ell^{-1}$ and let $\overline{\Theta}_\ell$ be its action on the quotient by constants. Fubini then gives

$$\Theta_\ell C^{\nu}_{\ell,A} = C^{\widehat{\nu}}_{\ell,\sigma A}\Theta_\ell, \qquad \Theta_\ell P^{\nu}_{\ell,A} = P^{\widehat{\nu}}_{\ell,\sigma A}\Theta_\ell, \qquad \overline{\Theta}_\ell E_\ell = \widehat{E}_\ell \Theta^{\mathcal{G}}_\ell, \qquad \Theta^{\mathcal{G}}_\ell \mathsf{H}_\ell = \widehat{\mathsf{H}}_\ell \overline{\Theta}_\ell. \tag{12.76}$$

Fubini proves the first identity and the rest follow by finite Mobius sums. Consequently coordinate expectations and interaction components are gauge-covariant under this componentwise/permutation realization. Product measure preservation alone is insufficient: a product-Haar-preserving shear of $\mathbb{T}^2$, for example $(x_1, x_2) \mapsto (x_1, x_1 + x_2)$, sends a singleton function of x_2 to one depending on both coordinates and mixes a singleton component into a pair component. A grading-intertwining hypothesis is therefore necessary for a claim that hyperedge degree is gauge covariant; componentwise/permutation realization is the declared sufficient hypothesis here, not the only logically possible one. If only almost-everywhere reverse-kernel equivariance is supplied, the valid statement is instead covariance of the L^∞ equivalence classes; it must not be promoted to pointwise covariance of a selected reverse-kernel version. [ESTABLISHED]

Proposition 12.19 (Radon–Nikodym covariance of the nonlinear interaction step). At adjacent scales, suppose $\vartheta_\ell : \mathsf{X}_\ell \to \widehat{\mathsf{X}}_\ell$ and $\vartheta_{\ell+1} : \mathsf{X}_{\ell+1} \to \widehat{\mathsf{X}}_{\ell+1}$ have the componentwise/permutation Borel realizations above. Transform the complete measure-pair data by

$$\widehat{\rho}_\ell = (\vartheta_\ell)_\# \rho_\ell, \qquad \widehat{m}_\ell = (\vartheta_\ell)_\# m_\ell, \qquad \widehat{\pi}_\ell = (\vartheta_\ell)_\# \pi_\ell, \tag{12.77}$$

and suppose the transformed normalized kernel obeys

$$\widehat{K}_\ell(\vartheta_\ell x, \vartheta_{\ell+1} B) = K_\ell(x, B) \tag{12.78}$$

for every Borel B. Then, as Radon–Nikodym equivalence classes,

$$\widehat{Q}_\ell(\Theta_\ell \varphi) = \Theta_{\ell+1} Q_\ell(\varphi), \tag{12.79}$$
$$\widehat{U}^{\Theta_\ell\varphi}_\ell \Theta_\ell h = \Theta_{\ell+1} U^{\varphi}_\ell h. \tag{12.80}$$

Consequently

$$\widehat{T}_\ell^{\mathcal{G}}\Theta_\ell^{\mathcal{G}} = \Theta_{\ell+1}^{\mathcal{G}} T_\ell^{\mathcal{G}}. \tag{12.81}$$

If the retained projections also intertwine, $\widehat{R}_k\Theta_k^{\mathcal{G}} = \Theta_k^{\mathcal{G}} R_k$ for $k = \ell, \ell+1$, then both residuals are covariant:

$$\widehat{r}_{\ell+1}^{\mathcal{G}}(\Theta_\ell^{\mathcal{G}} g) = \Theta_{\ell+1}^{\mathcal{G}} r_{\ell+1}^{\mathcal{G}}(g), \tag{12.82}$$

$$\widehat{\overline{r}}_{\ell+1}^{Q}(\Theta_\ell^{\mathcal{G}} g) = \overline{\Theta}_{\ell+1}\overline{r}_{\ell+1}^{Q}(g). \tag{12.83}$$

[ESTABLISHED]

Proof. Kernel covariance gives

$$\widehat{m}_\ell\widehat{K}_\ell = (\vartheta_{\ell+1})_\#(m_\ell K_\ell), \qquad (e^{-\Theta_\ell\varphi}\widehat{m}_\ell)\widehat{K}_\ell = (\vartheta_{\ell+1})_\#((e^{-\varphi}m_\ell)K_\ell).$$

The Radon–Nikodym derivative of two measures pushed through the same Borel isomorphism is the transported original derivative. Applying $-\log$ proves (12.79) without selecting a pointwise reverse conditional. Differentiate this identity at an arbitrary bounded φ, using Proposition 12.4, to obtain (12.80). Compose the first identity with the assembly and extraction intertwinings in (12.76) to prove (12.81). Finally insert the retained-projection intertwining into $I - R_{\ell+1}$ and into the exact residual identity (12.91); this proves the two residual formulas. □

12.3.1 Exact nonlinear interaction RG and retained projections

Let Q_ℓ be the RN-first bounded action map of Section 12.2.1; additive homogeneity makes its quotient map $\overline{Q}_\ell : \overline{\mathfrak{B}}_\ell \to \overline{\mathfrak{B}}_{\ell+1}$ well-defined on the full bounded domain. The exact nonlinear finite-network interaction step is

$$T_\ell^{\mathcal{G}} = \mathsf{H}_{\ell+1}\overline{Q}_\ell E_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}, \qquad g_{\ell+1}^{\mathrm{ex}} = T_\ell^{\mathcal{G}}(g_\ell). \tag{12.84}$$

It is exact as a change of the bounded action class, conditional on the product-equivalence premises at both scales; it does not assert that a pairwise or finite retained ansatz is closed. The displayed sum ϕ_g is the canonical zero-ν_ℓ-mean representative of $E_\ell g$, so the separately tracked perturbed evidence mass is

$$M_\ell(g) = M_\ell\pi_\ell\big(e^{-\phi_g}\big), \qquad M_{\ell+1}(g) = M_\ell(g), \tag{12.85}$$

where the second equality is normalization of K_ℓ, not an identification of the action quotient with a normalized probability path. [ESTABLISHED]

At g, define the tilted conditional-expectation operator on bounded classes by

$$U_\ell^{\phi_g}h = \frac{U_\ell(e^{-\phi_g}h)}{U_\ell(e^{-\phi_g})}. \tag{12.86}$$

It is unchanged when a constant is added to ϕ_g, and therefore descends to the action quotient. The chain rule and Proposition 12.4 give the correctly typed derivative

$$DT_\ell^{\mathcal{G}}(g) = \mathsf{H}_{\ell+1}\overline{U_\ell^{\phi_g}}E_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}. \tag{12.87}$$

Only at $g = 0$ may $U_\ell^{\phi_g}$ be replaced by the fixed reverse conditional expectation U_ℓ. Along an exact orbit these derivatives form a typed cocycle, rather than an ordinary eigenvalue equation across unequal network spaces. [ESTABLISHED]

Definition 12.20 (Interaction modes along an exact orbit). Instantiate Definition 11.8 with $X_\ell = \mathcal{G}_\ell$, $T_\ell = T_\ell^{\mathcal{G}}$, and an exact interaction orbit $g_{\ell+1} = T_\ell^{\mathcal{G}}(g_\ell)$, so that

$$D_\ell = DT_\ell^{\mathcal{G}}(g_\ell) = \mathsf{H}_{\ell+1}\overline{U_\ell^{\phi_{g_\ell}}}E_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}. \tag{12.88}$$

A generalized interaction mode of index a is a compatible line $D_\ell v_{\ell,a} = \lambda_{\ell,a} v_{\ell+1,a}$ in the sense of Definition 11.10, and its finite-step coefficient is the ordered product (11.28) taken with respect to the cumulative log block scale (11.29) of the same blocking sequence. No eigenvalue equation is written inside one unnamed interaction space. [DEFINITION]

Corollary 12.21 (Temperedness of interaction comparison trivializations). Suppose the declared comparison isomorphisms $J_\ell : \mathcal{G}_\star \to \mathcal{G}_\ell$ obey the growth bound

$$\log^+\|J_n\| + \log^+\|J_n^{-1}\| \le c\,(1 + |V_n|) \tag{12.89}$$

for a constant c independent of n. This bound holds for every trivialization assembled from the maps of (12.75) together with a bounded gauge realization, because $\|E_\ell\| \le 1$ and $\|\mathsf{H}_\ell\| \le 3^{|V_\ell|} - 1$. Then the tempering hypothesis (11.35) of Theorem 11.14 holds whenever $|V_n|/s_{n\leftarrow\ell} \to 0$. In particular it holds for every blocking sequence whose vertex counts do not increase and whose block ratios satisfy $b_k \ge b > 1$. [ESTABLISHED]

Proof. Insert (12.89) into the left side of (11.35) and divide by $s_{n\leftarrow\ell}$; the constant term vanishes because $s_{n\leftarrow\ell} \to \infty$. If the vertex counts do not increase then $|V_n| \le |V_\ell|$ while $s_{n\leftarrow\ell} \ge (n-\ell)\log b \to \infty$. The stated operator bounds are (12.75). □

This corollary is the point at which the sharp finite-size extraction bound of Theorem 12.18 becomes a hypothesis of the mode theory rather than a curiosity: the bound is exponential in $|V_\ell|$, so a scheme in which the network grows fast enough with depth can violate (11.35) and change the reported exponents, exactly as in Proposition 11.15. [ESTABLISHED]

Let $R_\ell : \mathcal{G}_\ell \to \mathcal{G}_\ell$ be a declared bounded idempotent retained projection which intertwines the componentwise/permutation gauge action in (12.76). Define the exact and retained coordinate outputs and the coordinate/action-quotient residuals:

$$g^{\mathrm{ret}}_{\ell+1} = R_{\ell+1}T^{\mathcal{G}}_{\ell}(g_\ell), \qquad r^{\mathcal{G}}_{\ell+1}(g_\ell) = (I - R_{\ell+1})T^{\mathcal{G}}_{\ell}(g_\ell), \tag{12.90}$$

$$\overline{r}^{Q}_{\ell+1}(g_\ell) = \overline{Q}_\ell E_\ell g_\ell - E_{\ell+1}R_{\ell+1}\mathsf{H}_{\ell+1}\overline{Q}_\ell E_\ell g_\ell \qquad = E_{\ell+1}r^{\mathcal{G}}_{\ell+1}(g_\ell). \tag{12.91}$$

The second line is the exact projected-residual identity in the action quotient. Since $E_{\ell+1}$ is injective, $\overline{r}^{Q}_{\ell+1}(g) = 0$ if and only if $r^{\mathcal{G}}_{\ell+1}(g) = 0$. Consequently

$$T^{\mathcal{G}}_{\ell}(\operatorname{Ran} R_\ell) \subseteq \operatorname{Ran} R_{\ell+1} \iff r^{\mathcal{G}}_{\ell+1}(g) = 0 \quad \text{for every } g \in \operatorname{Ran} R_\ell. \tag{12.92}$$

Thus a retained update is exact precisely under this exact-image-invariance condition; otherwise it is a projection scheme with the displayed residual. Moreover extraction and assembly give the exact coordinate identity and the two-sided finite-network norm control, for $V_{\ell+1} \neq \varnothing$,

$$\mathsf{H}_{\ell+1}\overline{r}^{Q}_{\ell+1} = r^{\mathcal{G}}_{\ell+1}, \qquad \frac{\|r^{\mathcal{G}}_{\ell+1}\|_{\mathcal{G}_{\ell+1}}}{3^{|V_{\ell+1}|} - 1} \leq \|\overline{r}^{Q}_{\ell+1}\|_{\mathfrak{B}_{\ell+1}} \leq \|r^{\mathcal{G}}_{\ell+1}\|_{\mathcal{G}_{\ell+1}}. \tag{12.93}$$

Under the full transformed-measure, kernel, and retained-projection intertwining hypotheses of Proposition 12.19, both residuals are gauge covariant. [ESTABLISHED]

12.4 Exact closure generates hyperedges

Suppose the complete nonnegative density of the law, including the baseline, is factorized against a product or block-product reference. Marginalizing an internal variable multiplies every incident factor, including baseline factors, and integrates their product. The result is one factor on the union of the remaining incident scopes. Iterating this operation represents the normalized law by generally unnormalized factors on the elimination cliques; the inherited total normalizer, not each factor, normalizes their product. Tonelli's theorem makes the resulting joint law independent of elimination order even though the displayed factorization is scheme dependent. For an arbitrary correlated baseline without such a factorization, retain the baseline as one global factor and use the law-level pushforward instead. [ESTABLISHED]

For a finite meta-agent set $\mathcal{P}$ and a finite-valued effective action, choose an anchor z° and define

$$\Phi^c_A(z_A) = \sum_{B\subseteq A} (-1)^{|A|-|B|} H^c_o(z_B, z^\circ_{B^c}), \qquad A \subseteq \mathcal{P}. \tag{12.94}$$

Mobius inversion gives

$$H^c_o(z) = \sum_{A\subseteq\mathcal{P}} \Phi^c_A(z_A). \tag{12.95}$$

Thus the full hypergraph family is exactly closed. If H^c_o is extended valued, retaining it as one top-order factor gives exact closure without forming undefined alternating sums. [ESTABLISHED]

The anchor is a covariant background, not a coordinatewise constant under frame changes. Consequently each displayed factor obeys $\Phi^c_A(gz_A; gz^\circ) = \Phi^c_A(z_A; z^\circ)$ when the action is gauge invariant. If an anchor is held fixed only in one coordinate chart, gauge covariance is asserted only for the reconstructed sum (12.95). [ESTABLISHED]

Pairwise closure is false in general. Eliminating the center s_0 of an Ising star produces

$$\sum_{s_0=\pm 1} \exp\left[s_0\left(h_0 + \sum_{r=1}^{3} J_r s_r\right)\right] = 2\cosh\left(h_0 + \sum_{r=1}^{3} J_r s_r\right), \tag{12.96}$$

whose negative log has cubic coefficient $2\,\mathrm{sech}^2(h_0)\tanh(h_0)J_1J_2J_3 + O(J^5)$ when $h_0J_1J_2J_3 \neq 0$; it is therefore nonzero for all sufficiently small nonzero J_r with $h_0 \neq 0$. [ESTABLISHED]

12.5 Fine–meta bridge kernels

The posterior bridge

$$B_o(dy, dz) = \Pi_o(dy)C(y, dz) \tag{12.97}$$

defines the full reverse kernel $R_o(z, dy) = B_o(dy \mid z)$. For fine agent j and meta-agent I, take the (Y_j, Z_I) marginal and disintegrate it in both directions:

$$\Pi_{o,j}(dy_j)K_{I\leftarrow j}(y_j, dz_I) = \Pi^c_{o,I}(dz_I)K_{j\leftarrow I}(z_I, dy_j). \tag{12.98}$$

These normalized kernels are the exact agent-to-meta and meta-to-agent pair-marginal association, or bridge, kernels. They are not thereby causal generative interactions. For bounded observables f and g, they obey

$$\int f(y_j)(K_{I\leftarrow j}g)(y_j)\Pi_{o,j}(dy_j) = \int (K_{j\leftarrow I}f)(z_I)g(z_I)\Pi^c_{o,I}(dz_I). \tag{12.99}$$

[ESTABLISHED]

Equation (12.98) does not say that Z_I alone reconstructs the full fine posterior. The exact full reverse generally reads all meta variables. A product baseline and blockwise coarse channel factorize the baseline reverse, but a likelihood coupling blocks can destroy that factorization in the posterior. Posterior-local refinement additionally requires the conditional-independence property $Y_I \perp Z_{-I} \mid (Z_I, o)$, or another separately proved sufficient condition. [ESTABLISHED]

For a deterministic statistic $z = c(y)$, the observable pullback and posterior conditional expectation are

$$(\mathsf{P}f)(y) = f(c(y)), \qquad (\mathsf{C}g)(z) = \mathbb{E}_{\Pi_o}[g(Y) \mid c(Y) = z]. \tag{12.100}$$

They satisfy $\mathsf{CP} = I$ in $L^p(\Pi^c_o)$, equivalently Π^c_o-almost everywhere, and compose by the tower property along a fixed nested filtration. Values of a conditional kernel at unused coarse points are immaterial. For a randomized channel, the corresponding operators remain adjoint but $\mathsf{CP} = I$ is not automatic. [ESTABLISHED]

12.6 Gauge-covariant cross-scale operators

For an arbitrary block partition $\mathcal{P}$, define the component-indexed meta-agent set

$$\widehat{\mathcal{P}} = \{\widehat{I} = (I, c) : I \in \mathcal{P}, c \in \pi_0(\Gamma_I)\}. \tag{12.101}$$

Choose a rooted spanning tree inside each connected component $\widehat{I}$ and require successive trees to be nested. From this point through the rooted operator formulas, I, J denote elements of $\widehat{\mathcal{P}}$ and r_I denotes their single component root. The state of an original disconnected block is the product over its component meta-agents; collapsing that product to one root, row, or feature is an additional declared coarse channel. For channel $x \in \{b, m\}$, let $\tau^x_{I\leftarrow i}$ be the ordered product of the channel-specific graph links from i to the root r_I. Under the passive section rechoices $a^b_i = a_i$ and $a^m_i = b_i$ of Equation (2.10), it transforms as

$$\tau^{x\prime}_{I\leftarrow i} = (a^x_{r_I})^{-1}\tau^x_{I\leftarrow i}a^x_i. \tag{12.102}$$

These are two passive coordinate rechoices in one principal G-bundle, not an active $G \times G$ gauge symmetry. For one active automorphism, the two local functions instead obey $k^m_i = h^{-1}_i k^b_i h_i$.

Every non-tree edge is retained through the channel-typed based holonomy map

$$H^x_I : \pi_1(\Gamma_I, r_I) \longrightarrow G, \qquad H^{x\prime}_I(\gamma) = (a^x_{r_I})^{-1}H^x_I(\gamma)a^x_{r_I}. \tag{12.103}$$

For each marked boundary edge $e : j \to i$ from block J to block I, retain

$$V^x_e = \tau^x_{I\leftarrow i}\Theta^x_e(\tau^x_{J\leftarrow j})^{-1}, \qquad V^{x\prime}_e = (a^x_{r_I})^{-1}V^x_e a^x_{r_J}. \tag{12.104}$$

The exact datum is the simultaneous root-framed equivariant state $(\bar{z}^x_I, H^x_I, \{V^x_e\})$, or its one simultaneous root-gauge orbit; separately quotienting each holonomy loses its orientation relative to the root features and boundary legs. For noncompact G, a naive conjugacy quotient need not be standard Borel, so the measure tier retains the raw root-framed G-space unless a proper-action or standard-Borel quotient theorem is supplied. Different trees change the presentation by Nielsen transformations; nested trees with the compatibility below give strict composition, while arbitrary choices give a root-gauge-equivariant isomorphism of presentations. [ESTABLISHED]

On a separately declared linear feature fiber V_x, let $R_x : G \to \mathrm{GL}(V_x)$ be a linear representation. Choose $w_{Ii} \geq 0$ with $\sum_{i\in I} w_{Ii} = 1$ and set

$$(\mathsf{C}_x z)_I = \sum_{i\in I} w_{Ii}R_x(\tau^x_{I\leftarrow i})z_i, \qquad (\mathsf{P}_x\bar{z})_i = R_x(\tau^x_{I\leftarrow i})^{-1}\bar{z}_I. \tag{12.105}$$

Then $\mathsf{C}_x\mathsf{P}_x = I$. With $R_{x,f} = \bigoplus_i R_x(a^x_i)$ and $R_{x,c} = \bigoplus_I R_x(a^x_{r_I})$, passive covariance is

$$\mathsf{C}'_x = R^{-1}_{x,c}\mathsf{C}_x R_{x,f}, \qquad \mathsf{P}'_x = R^{-1}_{x,f}\mathsf{P}_x R_{x,c}. \tag{12.106}$$

At nested levels $0 \to 1 \to 2$, let $\sigma^{x,12}_{A\leftarrow I}$ be the ordered interroot link from the root of I to the root of A. Strict composition additionally requires

$$\tau^{x,02}_{A\leftarrow i} = \sigma^{x,12}_{A\leftarrow I}\tau^{x,01}_{I\leftarrow i}, \qquad w^{02}_{Ai} = w^{12}_{AI}w^{01}_{Ii}. \tag{12.107}$$

It then follows that $\mathsf{C}^{02}_x = \mathsf{C}^{12}_x\mathsf{C}^{01}_x$ and $\mathsf{P}^{02}_x = \mathsf{P}^{01}_x\mathsf{P}^{12}_x$; nested forests alone would not imply either equality. If A_x is a fine message operator with $A'_x = R^{-1}_{x,f}A_xR_{x,f}$, the fine-to-meta, meta-to-fine, and meta-to-meta message operators are

$$A^x_{I\leftarrow j} = (\mathsf{C}_xA_x)_{Ij}, \qquad A^x_{j\leftarrow I} = (A_x\mathsf{P}_x)_{jI}, \qquad A^x_{I\leftarrow J} = (\mathsf{C}_xA_x\mathsf{P}_x)_{IJ}. \tag{12.108}$$

They transform between the corresponding endpoint frames; in particular,

$$(\mathsf{C}_xA_x\mathsf{P}_x)' = R^{-1}_{x,c}(\mathsf{C}_xA_x\mathsf{P}_x)R_{x,c}.$$

[ESTABLISHED]

These message formulas are not an energy marginalization theorem. A hard-tied quadratic energy uses $\mathsf{P}^*_xH_x\mathsf{P}_x$; an exact Gaussian marginal uses a Schur complement; a general law uses Theorem 12.2. The identity $\mathsf{C}_x\mathsf{P}_x = I$ alone does not make $\mathsf{C}_xH_x\mathsf{P}_x$ positive. [ESTABLISHED]

If $\Phi_f : V_{b,f} \to V_{m,f}$ is a separately declared linear cross-channel morphism on this tier, usually $\bigoplus_i \Phi_i$, with $\Phi'_f = R^{-1}_{m,f}\Phi_fR_{b,f}$, its compressed map is

$$\Phi^c = \mathsf{C}_m\Phi_f\mathsf{P}_b, \qquad \Phi^{c\prime} = R^{-1}_{m,c}\Phi^cR_{b,c}. \tag{12.109}$$

The common principal bundle does not create Φ, and the two representations must never be identified merely because both use G. [ESTABLISHED]

12.7 Exact attention between meta-agents

A row-stochastic matrix β_{ij} is a conditional law of a source given a receiver. It cannot be coarse-grained without the receiver law. Let $\alpha_i(y)$ be a normalized receiver occupancy and define the joint marked edge-event law

$$\eta_{ij}(y) = \alpha_i(y)\beta_{ij}(y), \qquad \sum_{i,j}\eta_{ij}(y) = 1. \tag{12.110}$$

The event probabilities are gauge-invariant scalars: $\eta'_{ij} = \eta_{ij}$. For a posterior bridge and node partition, set

$$\eta^c_{IJ}(z) = \mathbb{E}_{B_o}\left[\sum_{i\in I}\sum_{j\in J}\eta_{ij}(Y)\,\middle|\,Z = z\right], \quad \alpha^c_I(z) = \sum_J \eta^c_{IJ}(z), \quad \beta^c_{IJ}(z) = \frac{\eta^c_{IJ}(z)}{\alpha^c_I(z)}. \tag{12.111}$$

The last ratio is used on $\{\alpha^c_I > 0\}$; a row on a zero-occupancy receiver is immaterial. Pushing η and then disintegrating is normalized and associative under nested partitions by the tower property.

The next scale must push the joint law η^c, not β^c alone. Consequently η^c, α^c, and β^c are gauge invariant. [ESTABLISHED]

In an augmented receiver–source-label version of Section 7.5, the fixed recognition-independent label baseline is

$$P_0^{\mathrm{att}}(dy, di, dj) = P_0^Y(dy) a_i \pi_{ij}, \qquad \sum_i a_i = 1, \quad \sum_j \pi_{ij} = 1. \tag{12.112}$$

Fix the latent state y and define the row partition function $Z_i(y) = \sum_j \pi_{ij} \exp[-D_{ij}(y)/\tau]$. At the selected record, the conditional prior inside a block with $a_I = \sum_{k \in I} a_k > 0$ is $a_{i|I} = a_i / a_I$. Define the conditional block evidence weight $W_I(y) = \sum_{k \in I} a_{k|I} c_k(o_k, y) Z_k(y)$. On $\{W_I(y) > 0\}$, its posterior is evidence weighted:

$$\alpha_i'(y) = \frac{a_{i|I} c_i(o_i, y) Z_i(y)}{W_I(y)}, \qquad \beta_{IJ}^c(y) = \sum_{i \in I} \alpha_i'(y) \sum_{j \in J} \beta_{ij}. \tag{12.113}$$

The formula holds posterior-almost everywhere. On a zero-evidence block the receiver row is an arbitrary conditional version and is removed from the effective support. The equality $\alpha_i' = a_{i|I}$ holds exactly when $c_i(o_i, y) Z_i(y)$ is constant on the conditional support in I. This guarantees equality with the pre-observation coarse mixture, although equality can also occur accidentally after source coarsening—for example, when all receivers induce the same block-source row. If all receiver labels are retained, the exact meta-label is their vector or multiset; replacing it by one categorical row is a further coarse channel whose conditional KL appears in (12.4). [ESTABLISHED]

For one categorical row and a source partition $j \mapsto J$, the exact coarse prior and energy are

$$\pi_J^c = \sum_{j \in J} \pi_j, \qquad E_J^c = -\tau \log \left[\frac{1}{\pi_J^c} \sum_{j \in J} \pi_j \exp[-E_j/\tau] \right]. \tag{12.114}$$

for every block with $\pi_J^c > 0$; a zero-prior block has zero posterior mass and is removed from the effective support. This log-sum-exp rule associates for nested partitions of the same row. It does not merge independently normalized receiver rows without (12.113). [ESTABLISHED]

The transported meta-message in channel x is represented exactly by marked parallel edges. Let the fine mark $U_{i \leftarrow j}^x$ obey the passive endpoint law and define its dressed version $V_{ij}^x = \tau_{I \leftarrow i}^x U_{i \leftarrow j}^x (\tau_{J \leftarrow j}^x)^{-1}$. On the linear feature tier, define the following moment only under the conditional Bochner-integrability condition

$$\mathbb{E}_{B_o} \left[\sum_{i \in I, j \in J} \eta_{ij}(Y) \| R_x(V_{ij}^x) \| \middle| Z \right] < \infty \quad \text{almost everywhere.} \tag{12.115}$$

Then

$$\begin{aligned}
\mathsf{T}^x_{IJ}(z) &= \mathbb{E}_{B_o}\left[\sum_{i\in I, j\in J} \eta_{ij}(Y) R_x(V^x_{ij}) \middle| Z = z\right], \\
\overline{\mathsf{U}}^x_{IJ}(z) &= \frac{\mathsf{T}^x_{IJ}(z)}{\eta^c_{IJ}(z)} \quad \text{on } \{\eta^c_{IJ} > 0\}, \\
\mathsf{A}^{x,\mathrm{att}}_{IJ}(z) &= \frac{\mathsf{T}^x_{IJ}(z)}{\alpha^c_I(z)} \quad \text{on } \{\alpha^c_I > 0\}.
\end{aligned} \tag{12.116}$$

On $\{\eta^c_{IJ} > 0\}$, $\mathsf{A}^{x,\mathrm{att}}_{IJ} = \beta^c_{IJ}\overline{\mathsf{U}}^x_{IJ}$. On $\{\alpha^c_I > 0, \eta^c_{IJ} = 0\}$, the attention-weighted block is zero and $\overline{\mathsf{U}}^x_{IJ}$ is left undefined as a null-event conditional. For passive root-frame rechoices with no node relabeling $(\sigma = \mathrm{id})$, corresponding transported conditional versions, and the transformed coarse Borel state $z' = \vartheta_{\ell+1}(z)$, covariance holds Π^c_o-almost everywhere in z (equivalently, under the transformed coarse law, almost everywhere in z'): $\mathsf{T}^{x\prime}_{IJ}(z') = R_x(a^x_{r_I})^{-1}\mathsf{T}^x_{IJ}(z)R_x(a^x_{r_J})$, so these operators transform from the J root frame to the I root frame. The symbol $R_{x,c}$ remains the direct-sum coordinate operator on the separately declared linear feature fiber in (12.106); it is not an action on the Borel state z. The conditional mean need not belong to $R_x(G)$: the average of rotations by $\pi/2$ and $-\pi/2$ is the zero matrix. Moreover, this Hom operator is exact only for its stated first operator moment on a common source input. If marks and source features are correlated, their means do not determine $\mathbb{E}[Uz]$; for example, equal-probability marks/features (I, v) and $(-I, -v)$ have zero separate means but mean product v. Full closure therefore retains the marked operator–feature kernel or all marked edges, with (12.116) as a derived moment, rather than a fictitious averaged group element. [ESTABLISHED]

At a fixed outside context, define $\beta^Q_I(J \mid z_{-I}) = Q(J \mid z_{-I})$ and $K^Q_{I\leftarrow J}(dz_I \mid z_{-I}) = Q(dz_I \mid J, z_{-I})$, and define the corresponding generative conditionals under $P(\cdot \mid z_{-I}, o)$. When the recognition categorical and conditional kernels are absolutely continuous with respect to their generative counterparts, with the extended-value convention, the conditional KL chain rule is

$$\begin{aligned}
&D_{\mathrm{KL}}\big(Q(J, Z_I \mid z_{-I}) \| P(J, Z_I \mid z_{-I}, o)\big) \\
&\quad = D_{\mathrm{KL}}\big(\beta^Q_I(\cdot \mid z_{-I}) \| \beta^P_I(\cdot \mid z_{-I}, o)\big) \\
&\qquad + \sum_J \beta^Q_I(J \mid z_{-I}) D_{\mathrm{KL}}\big(K^Q_{I\leftarrow J}(\cdot \mid z_{-I}) \| K^P_{I\leftarrow J}(\cdot \mid z_{-I}, o)\big).
\end{aligned} \tag{12.117}$$

Thus meta-attention and meta-agent interaction kernels are two typed parts of one conditional ELBO, not interchangeable names for the same object. [ESTABLISHED]

12.8 Ordered-path closure and memory

For a finite ordered path interval, apply the pointwise coarse channel to the full joint path law. Its order parameter is auxiliary: it is neither a coordinate on the fixed base $\mathcal{C}$ nor the intrinsic information clock of Chapter 9. Theorem 12.1 then applies on path space without change. The

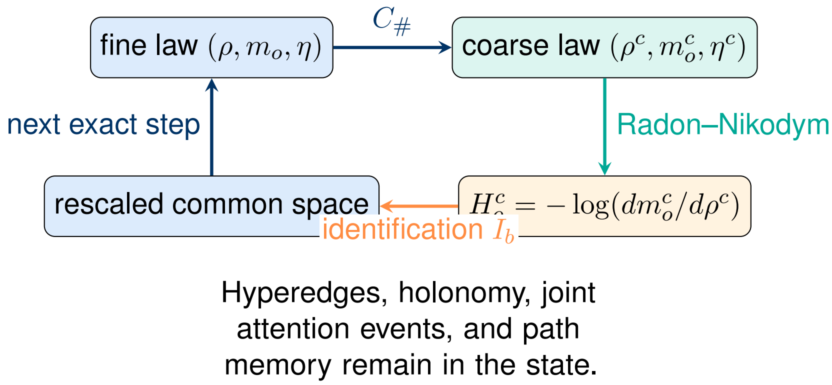

Figure 12.1: The exact RG acts first on normalized measures. An effective action and a beta function are introduced only after a pushed reference and a declared identification return the changing scale spaces to one comparison space.

resulting coarse path law is normalized and gauge covariant when the required spatial transports and, for a separately declared dynamical model, parameter-direction transports are included. It need not be first-order Markov. [ESTABLISHED]

Theorem 12.22 (Standard-Borel strong lumpability). Let $(\mathsf{Y}, \mathscr{Y})$ and $(\mathsf{Z}, \mathscr{Z})$ be standard Borel, let $c : \mathsf{Y} \to \mathsf{Z}$ be Borel and surjective, and let T be a Markov kernel on Y. Consider

$$T(y, c^{-1}B) = T(y', c^{-1}B) \quad \text{whenever } c(y) = c(y'), \qquad \text{for every } B \in \mathscr{Z}. \tag{12.118}$$

If a Markov kernel T^c on Z satisfies

$$c_\#(\mu T) = (c_\#\mu)T^c \qquad \text{for every probability } \mu \text{ on } \mathsf{Y}, \tag{12.119}$$

then (12.118) holds and T^c is unique. Conversely, if (12.118) holds and c admits a Borel right inverse $\varsigma : \mathsf{Z} \to \mathsf{Y}$ with $c \circ \varsigma = \mathrm{id}$, then

$$T^c(z, B) = T\big(\varsigma(z), c^{-1}B\big) \tag{12.120}$$

is a Markov kernel satisfying (12.119). A Borel right inverse exists in the two cases used in this chapter: a countable coarse space, and a product projection $\mathsf{Y} = \mathsf{Z} \times \mathsf{Y}'$ with Y' nonempty, for which one fixes any point of Y'. Without a measurable selection, (12.120) still defines the coarse transition on the range of c and the selection must be declared separately. [ESTABLISHED]

Proof. Assume (12.119) and take $\mu = \delta_y$. Then $c_\#(\delta_y T)(B) = T(y, c^{-1}B)$ while $((c_\#\delta_y)T^c)(B) = T^c(c(y), B)$, so

$$T(y, c^{-1}B) = T^c(c(y), B) \qquad \text{for every } y \in \mathsf{Y},\ B \in \mathscr{Z}, \tag{12.121}$$

whose right side depends on y only through $c(y)$; this is (12.118). Surjectivity of c and (12.121) determine T^c at every z, so it is unique.

Conversely assume (12.118) and a Borel ς. For fixed z, the set map $B \mapsto T(\varsigma(z), c^{-1}B)$ is a probability measure, because c^{-1} commutes with countable disjoint unions and $c^{-1}\mathsf{Z} = \mathsf{Y}$. For fixed B, the map $z \mapsto T(\varsigma(z), c^{-1}B)$ is the composition of the Borel map ς with the measurable map $y \mapsto T(y, c^{-1}B)$, hence measurable. So (12.120) is a Markov kernel. Since $c(\varsigma(c(y))) = c(y)$, hypothesis (12.118) gives $T^c(c(y), B) = T(y, c^{-1}B)$ for every y. Therefore, for any probability μ,

$$c_\#(\mu T)(B) = \int T(y, c^{-1}B)\mu(dy) = \int T^c(c(y), B)\mu(dy) = \int T^c(z, B)(c_\#\mu)(dz),$$

which is (12.119). □

Strong lumpability is exactly the every-initial-law condition, and it is not necessary for a single initial law. Take $\mathsf{Y} = \{1, 2, 3\}$ with $c(1) = c(2) = a$ and $c(3) = \beta$, and let T send 1 to 3, send 2 to 1 or 2 with equal probability, and send 3 to 1 or 3 with equal probability. Then $T(1, \{1, 2\}) = 0$ while $T(2, \{1, 2\}) = 1$, so (12.118) fails. Started at δ_3, however, the chain never leaves $\{1, 3\}$, on which c is injective, so the coarse process is a Markov chain with transitions $a \mapsto \beta$ surely and $\beta \mapsto a$ or β with equal probability. Weak lumpability at a selected initial law is therefore a strictly weaker statement, and it is the only one available when (12.118) fails. [ESTABLISHED]

The finite-state form of the biconditional is classical and is the lumpability criterion of Kemeny and Snell [41, Ch. 6]; that source is cited only at its finite-state scope, and the standard-Borel statement above is proved here. [ESTABLISHED]

Theorem 12.23 (Exact discrete projection-memory recurrence). Let X and W be vector spaces, let $T : \mathsf{X} \to \mathsf{X}$, $\mathsf{C} : \mathsf{X} \to \mathsf{W}$, and $\mathsf{P} : \mathsf{W} \to \mathsf{X}$ be linear with $\mathsf{CP} = I_{\mathsf{W}}$, and set

$$\Pi^{\mathrm{res}} = \mathsf{PC}, \qquad \mathsf{Q} = I - \Pi^{\mathrm{res}}, \tag{12.122}$$

so that Π^{res} is idempotent. For the fine recursion $x_{n+1} = Tx_n$ write $w_n = \mathsf{C}x_n$, so that $\Pi^{\mathrm{res}}x_n = \mathsf{P}w_n$. Then for every $n \geq 0$

$$\mathsf{Q}x_n = (\mathsf{Q}T\mathsf{Q})^n\mathsf{Q}x_0 + \sum_{k=0}^{n-1}(\mathsf{Q}T\mathsf{Q})^{n-1-k}\mathsf{Q}T\mathsf{P}w_k, \tag{12.123}$$

and the resolved coordinate obeys the exact closed recurrence

$$w_{n+1} = \mathsf{C}T\mathsf{P}w_n + \sum_{k=0}^{n-1}\mathsf{C}T\mathsf{Q}(\mathsf{Q}T\mathsf{Q})^{n-1-k}\mathsf{Q}T\mathsf{P}w_k + \mathsf{C}T\mathsf{Q}(\mathsf{Q}T\mathsf{Q})^n\mathsf{Q}x_0. \tag{12.124}$$

The operators

$$\mathsf{C}T\mathsf{Q}(\mathsf{Q}T\mathsf{Q})^m\mathsf{Q}T\mathsf{P}, \qquad m \geq 0, \tag{12.125}$$

are the memory kernels and the last term of (12.124) is the unresolved initial-state term. Equation (12.124) is exact; no closure or Markov assumption is used. [ESTABLISHED]

Proof. Apply Π^{res} and Q to $x_{n+1} = Tx_n$ and use $x_n = \Pi^{\text{res}}x_n + \mathsf{Q}x_n = \mathsf{P}w_n + \mathsf{Q}x_n$ together with $\mathsf{Q}\mathsf{Q}x_n = \mathsf{Q}x_n$. This gives the coupled system

$$w_{n+1} = \mathsf{C}T\mathsf{P}w_n + \mathsf{C}T\mathsf{Q}\,\mathsf{Q}x_n, \qquad \mathsf{Q}x_{n+1} = \mathsf{Q}T\mathsf{Q}\,\mathsf{Q}x_n + \mathsf{Q}T\mathsf{P}w_n,$$

where both lines used idempotence of Q to write $\mathsf{Q}x_n = \mathsf{Q}(\mathsf{Q}x_n)$. Solving the second line by induction on n gives (12.123): it is an identity at $n = 0$, and applying $\mathsf{Q}T\mathsf{Q}$ to the case n and adding $\mathsf{Q}T\mathsf{P}w_n$ reproduces the case $n + 1$ with the index shifted. Substituting (12.123) into the first line gives (12.124). □

Corollary 12.24 (Autonomy of the resolved recursion). Let $\mathcal{X}_0$ be an admitted initial class. The resolved recursion is autonomous on $\mathcal{X}_0$, meaning $w_{n+1} = \mathsf{C}T\mathsf{P}w_n$ for every $n \geq 0$ and every $x_0 \in \mathcal{X}_0$, if and only if the total correction in (12.124) vanishes for every such n and x_0. Two sufficient conditions are that all memory kernels (12.125) vanish and that

$$\mathsf{C}T\mathsf{Q}(\mathsf{Q}T\mathsf{Q})^n\mathsf{Q}x_0 = 0, \qquad n \geq 0, \tag{12.126}$$

throughout $\mathcal{X}_0$. On the resolved class $\mathsf{Q}x_0 = 0$ the family (12.126) is vacuous, and there $\mathsf{Q}T\mathsf{P} = 0$ is sufficient but not necessary: on $\mathsf{X} = \mathbb{R}^2$ with $\mathsf{W} = \mathbb{R}$, $\mathsf{C}(x, y) = x$, $\mathsf{P}(w) = (w, 0)$, and $T(x, y) = (x, x)$, one has $\mathsf{Q}T\mathsf{P}w = (0, w) \neq 0$ for $w \neq 0$ while $\mathsf{C}T\mathsf{Q} = 0$, so every memory kernel vanishes and the resolved recursion $w_{n+1} = w_n$ is exactly autonomous. [ESTABLISHED]

Proof. The biconditional is (12.124) read as an identity. The two displayed families are exactly the two correction sums, so their vanishing is sufficient. For the witness, $\mathsf{Q}(x, y) = (0, y)$ and $T(0, y) = (0, 0)$ give $\mathsf{C}T\mathsf{Q} = 0$, while $T(w, 0) = (w, w)$ gives $\mathsf{Q}T\mathsf{P}w = (0, w)$. □

Recurrence (12.124) is proved here for the discrete linear observable tier. The projection-operator method of eliminating unresolved coordinates in favor of a memory kernel and an initial-state term originates with Nakajima [55] and Zwanzig [81], who work in continuous time on ensemble densities; those sources are cited as the historical antecedent of the method and not as proofs of (12.124). [ESTABLISHED]

12.9 The RG transformation and beta functions

Let $b > 1$ label one blocking ratio. A genuine RG step consists of a coarse channel C_b and a declared rescaling/identification kernel I_b that returns the target to a common measurable state space. Define the typed rescaled kernel $K_b = C_b I_b$. Compatibility means, after the declared canonical identifications,

$$K_{b_1 b_2} = K_{b_1} K_{b_2}. \tag{12.127}$$

On the primary measure pair, define

$$\mathcal{R}_b(\rho, m_o) = \big((\rho C_b)I_b, (m_o C_b)I_b\big) = (\rho K_b, m_o K_b). \tag{12.128}$$

Write $\mathcal{R}_b^{\rho}\rho = (\rho C_b)I_b = \rho K_b$ for its reference component. Equation (12.127) makes $\mathcal{R}_{b_1 b_2} = \mathcal{R}_{b_2}\mathcal{R}_{b_1}$; otherwise the sequence is a typed cocycle rather than an autonomous semigroup. [ESTABLISHED]

On a dominated tier, the action map retains its reference argument:

$$\mathcal{R}_b^H[H;\rho] := -\log \frac{d\big((e^{-H}\rho)K_b\big)}{d(\rho K_b)}. \tag{12.129}$$

On a declared vector space of finite-valued actions that is closed under this map for the declared reference trajectory and on which subtraction exists, the exact discrete step beta functional is

$$\mathfrak{B}_b^H[H;\rho] = \frac{\mathcal{R}_b^H[H;\rho] - H}{\log b}. \tag{12.130}$$

If the evidence mass is deliberately forgotten and finite unnormalized actions are quotiented by additive constants, the same formula is interpreted in that quotient. For extended-valued actions, including points at which the numerator would be $+\infty - (+\infty)$, only the measure-pair RG is primary. A beta function on a truncated coupling vector exists only after invariance of that finite family is proved; otherwise omitted hyperedges contribute an explicit residual. [ESTABLISHED]

Proposition 12.25 (Change of reference is an inhomogeneous law for the action beta). Fix one arrow with rescaled kernel K_b and reverse conditional $R_\rho(z, dy)$ as in Theorem 12.2. Let H be an action with $m = e^{-H}\rho$ finite and positive, let Δ be bounded and measurable with $\rho(e^{-\Delta}) = 1$, and set

$$\rho' = e^{-\Delta}\rho, \qquad H' = H - \Delta, \qquad \text{so that} \qquad e^{-H'}\rho' = e^{-H}\rho = m. \tag{12.131}$$

Then ρ' is again a probability reference for the same measure pair, and $(\rho' K_b)$-almost everywhere

$$\mathcal{R}_b^H[H';\rho'] = \mathcal{R}_b^H[H;\rho] - \mathcal{R}_b^H[\Delta;\rho]. \tag{12.132}$$

On a declared finite-valued action vector space containing H, Δ, and their images, the discrete beta functional of (12.130) therefore obeys the inhomogeneous law

$$\mathfrak{B}_b^H[H';\rho'] = \mathfrak{B}_b^H[H;\rho] - \mathfrak{B}_b^H[\Delta;\rho]. \tag{12.133}$$

The added term is not a linear reparameterization of the beta functional, and it vanishes exactly when the reference-change increment is itself a fixed action, $\mathcal{R}_b^H[\Delta;\rho] = \Delta$. A beta functional is therefore a joint statement about the action and the declared reference trajectory. [ESTABLISHED]

Proof. Since $\rho(e^{-\Delta}) = 1$ and Δ is bounded, ρ' is a probability measure equivalent to ρ, and (12.131) holds by construction. Both ρK_b and $\rho' K_b$ dominate mK_b, and the chain rule for Radon–Nikodym derivatives gives

$$\frac{d(mK_b)}{d(\rho' K_b)} = \frac{d(mK_b)}{d(\rho K_b)} \cdot \left[\frac{d(\rho' K_b)}{d(\rho K_b)}\right]^{-1}$$

almost everywhere. By Theorem 12.2 applied to the pair $(\rho, e^{-\Delta}\rho)$, the bracketed factor equals $\int e^{-\Delta(y)} R_\rho(z, dy) = \exp(-\mathcal{R}_b^H[\Delta;\rho])$, which is strictly positive and finite because Δ is bounded. Taking $-\log$ of the display and using (12.129) twice gives (12.132). Subtracting $H' = H - \Delta$, dividing by $\log b$, and regrouping gives (12.133). The vanishing criterion is $\mathcal{R}_b^H[\Delta;\rho] - \Delta = 0$. □

Definition 12.26 (Typed discrete interaction beta). Let $\mathcal{G}_\star$ be a declared reference interaction space and let

$$J_\ell : \mathcal{G}_\star \xrightarrow{\sim} \mathcal{G}_\ell \tag{12.134}$$

be bounded isomorphisms with bounded inverses. This is the inverse orientation of the reference identifications I_ℓ of (11.3), so $J_\ell = I_\ell^{-1}$ whenever both are declared. The pulled exact interaction map and the exact discrete interaction beta are

$$\widehat{T}_\ell^{\mathcal{G}} = J_{\ell+1}^{-1} T_\ell^{\mathcal{G}} J_\ell : \mathcal{G}_\star \longrightarrow \mathcal{G}_\star, \qquad \beta_\ell^{\mathrm{ex}}(g) = \frac{\widehat{T}_\ell^{\mathcal{G}}(g) - g}{\Delta s_\ell}, \qquad \Delta s_\ell = \log b_\ell. \tag{12.135}$$

Both terms of the numerator lie in $\mathcal{G}_\star$; the subtraction $T_\ell^{\mathcal{G}}(g) - g$ is not formed, because its two terms inhabit $\mathcal{G}_{\ell+1}$ and $\mathcal{G}_\ell$. The symbol $\widehat{T}_\ell^{\mathcal{G}}$ is not the retained projection R_ℓ of (12.90), and no beta functional is written before (12.134) is declared. If the coupling object is a manifold rather than a Banach space, the difference must be replaced by a declared chart difference, logarithm, or retraction into one tangent space. [DEFINITION]

Proposition 12.27 (Exact, retained, and residual interaction beta). With $\widehat{R}_\ell = J_\ell^{-1} R_\ell J_\ell$ for the bounded idempotent retained projections of (12.90), define for $g \in \operatorname{Ran} \widehat{R}_\ell$

$$\beta_\ell^{\mathrm{ret}}(g) = \frac{\widehat{R}_{\ell+1} \widehat{T}_\ell^{\mathcal{G}}(g) - g}{\Delta s_\ell}, \qquad \delta\beta_\ell(g) = \beta_\ell^{\mathrm{ex}}(g) - \beta_\ell^{\mathrm{ret}}(g). \tag{12.136}$$

Then the omitted part of the flow is exactly the transported coordinate residual divided by the log block step,

$$\delta\beta_\ell(g) = \frac{(I - \widehat{R}_{\ell+1}) \widehat{T}_\ell^{\mathcal{G}}(g)}{\Delta s_\ell} = \frac{J_{\ell+1}^{-1} r_{\ell+1}^{\mathcal{G}}(J_\ell g)}{\Delta s_\ell}, \tag{12.137}$$

and its action-quotient partner is

$$\delta\overline{\beta}_{\ell+1}^{Q}(g) = \frac{E_{\ell+1} r_{\ell+1}^{\mathcal{G}}(J_\ell g)}{\Delta s_\ell}, \qquad \mathsf{H}_{\ell+1} \delta\overline{\beta}_{\ell+1}^{Q}(g) = \frac{r_{\ell+1}^{\mathcal{G}}(J_\ell g)}{\Delta s_\ell}. \tag{12.138}$$

Consequently $\delta\beta_\ell$ vanishes on the whole retained sector if and only if the exact image of the retained sector is retained,

$$\delta\beta_\ell\big|_{\operatorname{Ran} \widehat{R}_\ell} = 0 \quad \Longleftrightarrow \quad T_\ell^{\mathcal{G}}(\operatorname{Ran} R_\ell) \subseteq \operatorname{Ran} R_{\ell+1}, \tag{12.139}$$

which is (12.92). Boundedness and idempotence of R_ℓ do not imply this invariance, so a retained beta is called exact only after (12.139) is proved, and otherwise carries $\delta\beta_\ell$ with the two-sided norm control of (12.93). [ESTABLISHED]

Proof. Subtracting the two displays in (12.136) cancels g and leaves $(I - \widehat{R}_{\ell+1})\widehat{T}^{\mathcal{G}}_{\ell}(g)/\Delta s_{\ell}$. Substituting $\widehat{R}_{\ell+1} = J_{\ell+1}^{-1} R_{\ell+1} J_{\ell+1}$ and $\widehat{T}^{\mathcal{G}}_{\ell} = J_{\ell+1}^{-1} T^{\mathcal{G}}_{\ell} J_{\ell}$ gives $J_{\ell+1}^{-1}(I - R_{\ell+1})T^{\mathcal{G}}_{\ell}(J_{\ell} g)/\Delta s_{\ell}$, which is (12.137) by the definition of $r^{\mathcal{G}}_{\ell+1}$. Applying $E_{\ell+1}$ and $\mathsf{H}_{\ell+1}$ and using the inverse identities of Theorem 12.18 together with (12.91) gives (12.138). Since $J_{\ell+1}^{-1}$ is injective and $\Delta s_{\ell} > 0$, $\delta\beta_{\ell}(g) = 0$ is equivalent to $r^{\mathcal{G}}_{\ell+1}(J_{\ell} g) = 0$, and J_{ℓ} maps $\operatorname{Ran}\widehat{R}_{\ell}$ onto $\operatorname{Ran} R_{\ell}$; quantifying over that sector and using (12.92) gives (12.139). □

A projected fixed point is not an exact fixed point. On $\mathcal{G}_{\star} = \mathbb{R}^2$ with $b = e$, take $R(x, y) = (x, 0)$ and $T(x, y) = (x, x)$. For every retained input $g = (x, 0)$ one has $\beta^{\mathrm{ret}}(g) = (0, 0)$ while $\beta^{\mathrm{ex}}(g) = (0, x)$, so the retained flow reports a line of fixed points although the exact step generates an omitted component whenever $x \neq 0$. [ESTABLISHED]

The interaction beta is also scheme dependent, and its scheme dependence is larger than the coupling reparameterization of (11.37). If $J'_{\ell} = J_{\ell} S_{\ell}$ for bounded isomorphisms S_{ℓ} of $\mathcal{G}_{\star}$, then $\widehat{T}'_{\ell} = S_{\ell+1}^{-1}\widehat{T}_{\ell} S_{\ell}$ and

$$\beta'_{\ell}(g) = \frac{S_{\ell+1}^{-1}\widehat{T}^{\mathcal{G}}_{\ell}(S_{\ell} g) - g}{\Delta s_{\ell}}, \tag{12.140}$$

which reduces to the usual pushforward relation only for a scale-independent $S_{\ell} \equiv S$. Even the identity native step acquires a nonzero beta in a moving comparison frame: on $\mathcal{G}_{\ell} = \mathbb{R}$ with $T_{\ell} = \mathrm{id}$, $b_{\ell} = e$, and $J_{\ell} u = a_{\ell} u$, formula (12.135) gives $\beta_{\ell}(g) = (a_{\ell}/a_{\ell+1} - 1)g$, which is zero for $a_{\ell} \equiv 1$ and nonzero otherwise. The comparison scheme is therefore part of the statement of any reported beta function. [ESTABLISHED]

Definition 12.28 (Smooth scale bundle, scale connection, and continuous beta). A continuous beta function is a construction on a separately declared tier and is not obtained from a discrete sequence. Declare an oriented one-dimensional C^1 scale manifold S with coordinate s, a C^1 Banach vector bundle $\pi_S : \mathscr{G} \to S$ whose fibers are the coupling spaces, and a scale connection, that is a covariant derivative ∇^{scale} on sections of $\mathscr{G}$ satisfying the usual linearity and Leibniz rules. For a C^1 running coupling section $g(s) \in \mathscr{G}_s$, the continuous beta is

$$\beta^{\mathrm{scale}}(s) = \nabla^{\mathrm{scale}}_{\partial_s} g(s) \in \mathscr{G}_s. \tag{12.141}$$

In a local trivialization $I_s : \mathscr{G}_s \to \mathcal{G}_{\star}$ with connection form $A_s \in \mathcal{L}(\mathcal{G}_{\star})$ and $\widetilde{g}(s) = I_s g(s)$,

$$I_s \beta^{\mathrm{scale}}(s) = \partial_s \widetilde{g}(s) + A_s \widetilde{g}(s), \qquad A'_s = S_s A_s S_s^{-1} - (\partial_s S_s) S_s^{-1} \ \text{ under } I'_s = S_s I_s. \tag{12.142}$$

A raw coordinate derivative does not obey that transformation law. On the trivial line bundle with $g(s) = 1$ parallel in a frame with $A_s = 0$, the frame change $S_s = e^{s^2}$ gives raw derivative $2se^{s^2} \neq 0$ while $A'_s = -2s$ and the covariant derivative $\partial_s(e^{s^2}) - 2se^{s^2}$ is zero. The scale connection is a different object from the two contextual principal connections ω_b, ω_m of Definition 11.4: the former has base S and compares coupling fibers at different resolutions, the latter have base $\mathcal{C}_{\ell}$ at one fixed resolution and compare associated statistical fibers along directions in $T\mathcal{C}_{\ell}$. It is also distinct from the inference-orbit parameter and the Fisher duration of Chapter 9. Nothing is proved by this declaration. [DEFINITION]

One realization of that tier is the following density calculation. Suppose that scale is a genuine continuous semigroup and that, relative to one common dominating measure, the positive densities $r_t, p_t \in \mathrm{Dom}(\mathcal{A}^*)$ are differentiable in a topology that permits pointwise ratios and the following dual pairings, with $p_t = r_t \exp[-H_t]$, $\dot{r}_t = \mathcal{A}^* r_t$, and $\dot{p}_t = \mathcal{A}^* p_t$. Then

$$\boxed{\dot{H}_t = -\frac{\mathcal{A}^*(r_t \exp[-H_t])}{r_t \exp[-H_t]} + \frac{\mathcal{A}^* r_t}{r_t}} \tag{12.143}$$

almost everywhere wherever the displayed ratios and their dual pairings are defined. If the reference is stationary, the second term vanishes. [ESTABLISHED]

Proof. Differentiate $H_t = -\log(p_t/r_t)$ and substitute the two forward equations. □

On a declared Banach action space with a Schauder basis $\{\psi_A\}$ and continuous coordinate functionals $\{\psi_A^*\}$, or a Hilbert space with a Riesz basis, suppose the expansion $H_t = \sum_A g_A(t)\psi_A$ converges and differentiation and coefficient extraction can be interchanged. The component beta functions are then the coefficients of the right side of (12.143):

$$\dot{g}_A = \left\langle \psi_A^*, -\frac{\mathcal{A}^*(r_t e^{-H_t})}{r_t e^{-H_t}} + \frac{\mathcal{A}^* r_t}{r_t} \right\rangle. \tag{12.144}$$

This is exact for the stated complete convergent basis. A truncated basis defines a projection and leaves an explicit residual. [ESTABLISHED]

Proposition 12.29 (Discrete endpoints do not determine a continuous beta). Let a scale evolution family on the tier of Definition 12.28 consist of maps

$$\mathsf{V}(s,t) : \mathscr{G}_t \longrightarrow \mathscr{G}_s, \qquad \mathsf{V}(r,s) \circ \mathsf{V}(s,t) = \mathsf{V}(r,t), \qquad \mathsf{V}(t,t) = I_{\mathscr{G}_t}, \tag{12.145}$$

which is the continuous analogue of (11.25) and is again two-parameter. Restriction to integer endpoints does not determine the family, hence does not determine (12.141). On $S = \mathbb{R}$ with the trivial line bundle and zero connection form, the two families

$$\mathsf{V}^{(0)}(s,t) = 1, \qquad \mathsf{V}^{(\epsilon)}(s,t) = \exp\{\epsilon[\sin(2\pi s) - \sin(2\pi t)]\} \tag{12.146}$$

both satisfy (12.145) and agree at every pair of integer endpoints, while their generators are 0 and $2\pi\epsilon\cos(2\pi s)$. The one-parameter semigroup form, in which $\mathsf{V}(s,t)$ depends only on $s-t$, additionally requires a stationary trivialization, a stationary connection, an autonomous vertical field, and a global existence domain; it is not a consequence of (12.145). [ESTABLISHED]

Proof. Both displayed families satisfy the composition and identity laws by inspection, because each is of the form $f(s)/f(t)$ with f positive. At integer s and t the sine terms vanish, so $\mathsf{V}^{(\epsilon)}(s,t) =$

$1 = \mathsf{V}^{(0)}(s,t)$. Differentiating $\mathsf{V}^{(\epsilon)}(s,t)$ in s at $s=t$ gives $2\pi\epsilon\cos(2\pi t)$, which is not identically zero for $\epsilon \neq 0$. □

The smooth scale base, the bundle, the interpolation, and the connection are therefore additional declared input. A discrete blocking sequence supplies (12.135) and nothing more. [ESTABLISHED]

Attention flows primarily through the joint event law. Along a differentiable path on $\{\alpha_I > 0\}$, since $\beta_{IJ} = \eta_{IJ}/\alpha_I$,

$$\dot{\beta}_{IJ} = \frac{\dot{\eta}_{IJ} - \beta_{IJ}\dot{\alpha}_I}{\alpha_I}. \tag{12.147}$$

For a differentiable single softmax row on a fixed strictly positive prior support, with $\tau > 0$ and $\beta_J = \pi_J \exp[-E_J/\tau]/Z$, this becomes the replicator beta function

$$\dot{\beta}_J = \beta_J\left(u_J - \sum_K \beta_K u_K\right), \qquad u_J = \partial_t \log \pi_J - \frac{\dot{E}_J}{\tau} + \frac{E_J\dot{\tau}}{\tau^2}. \tag{12.148}$$

At a support change or on $\{\alpha_I = 0\}$, the joint law η is the well-defined flow variable; neither the quotient row nor its logarithmic parametrization is used. [ESTABLISHED]

12.10 Fixed points, flows, and scaling operators

Theorem 12.30 (Exhaustive fixed-point equations). A normalized fixed theory is exactly a rescaled invariant pair

$$\mathcal{R}_b(\rho_*, m_*) = (\rho_*, m_*) \quad \text{for every declared } b. \tag{12.149}$$

On the dominated normalized tier this is equivalent to the conjunction

$$\mathcal{R}_b^{\rho}\rho_* = \rho_*, \qquad \mathcal{R}_b^{H}[H_*;\rho_*] = H_*, \tag{12.150}$$

almost everywhere with respect to the fixed reference. If the evidence mass is discarded and actions are considered only projectively, the weaker fixed-ray equation is

$$\mathcal{R}_b^{H}[H_*;\rho_*] = H_* + c_{\mathrm{ray}}(b). \tag{12.151}$$

It represents the normalized pair in (12.149) only when the reference is also invariant and normalization fixes $c_{\mathrm{ray}}(b) = 0$. A fixed attention law is a fixed rescaled event measure η_*. A fixed conditional row alone does not make the attention law fixed: η is fixed exactly when its receiver marginal α is fixed and its disintegrated row β is fixed α-almost everywhere. [ESTABLISHED]

Proof. The first equation is the definition of invariance on the common rescaled measure space. Invariance of its first component and taking the Radon–Nikodym derivative of its second component give the two conditions in (12.150). Quotienting by the action's additive gauge gives (12.151), but restoring the tracked evidence mass removes that freedom. Disintegration of a fixed η gives the final statement. □

At a fixed pair, additionally require H_* to lie in the declared finite-valued action Banach space. Set

$$m_*(dy) = e^{-H_*(y)}\rho_*(dy), \qquad M_* = m_*(\mathsf{Y}), \qquad \pi_* = M_*^{-1}m_*, \qquad \pi_*^c = \pi_* K_b, \tag{12.152}$$

and fix the standard-Borel reverse-kernel version Π_* defined by

$$\pi_*(dy)K_b(y, dz) = \pi_*^c(dz)\Pi_*(z, dy). \tag{12.153}$$

Also require

$$0 < L_*^c(z) = \mathbb{E}_{\rho_*}[e^{-H_*} \mid Z = z] < \infty \quad \rho_*^c\text{-almost everywhere.} \tag{12.154}$$

For bounded perturbations $\varphi \in L^\infty(\pi_*)$, the first two variations of one exact conditional-partition step are, π_*^c-almost everywhere,

$$D_H \mathcal{R}_b^H\big|_{(H_*,\rho_*)}[\varphi](z) = \mathbb{E}_{\Pi_*(dy|z)}\varphi(Y), \qquad D_H^2 \mathcal{R}_b^H\big|_{(H_*,\rho_*)}[\varphi,\varphi](z) = -\operatorname{Var}_{\Pi_*(dy|z)}\varphi(Y), \tag{12.155}$$

for the composite kernel $K_b = C_b I_b$, which already includes the declared rescaling. The bounded-chart derivation of these two variations is Theorem 12.3; the displayed conditional-expectation and covariance formulas are exact on that stated domain. [ESTABLISHED]

Definition 12.31 (Spectral and cocycle relevance conventions). An ordinary spectrum is used only after a fixed point, an autonomous blocking ratio $b > 1$, and declared bounded isometric identifications turn the linearization into one bounded endomorphism T of one complex Banach action space. An isolated eigenoperator $T\psi_a = \lambda_a \psi_a$ then has the defined exponent $\upsilon_a = \log|\lambda_a|/\log b$; it is called relevant, marginal, or irrelevant according as this number is positive, zero, or negative, while its sign or complex phase is retained. Generalized eigenspaces are used only in finite dimensions or for isolated point spectrum with the required spectral projections.

Without those identifications, the derivative is a typed cocycle between different action spaces. A generalized mode and its Lyapunov growth rate must then be stated with the comparison maps and the chosen norms, rather than written as an eigenvalue equation in one unnamed space. These are classification definitions; no autonomous spectrum, mode, or exponent is asserted for a general nonautonomous sequence. [DEFINITION]

Definition 12.32 (Typed invariant and fixed objects). Write $Y_\ell^{(\tau)}$ for the object of one declared tier τ and $F_\ell^{(\tau)} : Y_\ell^{(\tau)} \to Y_{\ell+1}^{(\tau)}$ for its declared step. The general invariant object of a nonautonomous scale diagram is an *invariant section*, namely a sequence with

$$y_{\ell+1} = F_\ell^{(\tau)}(y_\ell), \tag{12.156}$$

which needs no identification. After declared isomorphisms $J_\ell : Y_\star^{(\tau)} \to Y_\ell^{(\tau)}$, define the typed reference-space step

$$\widehat{F}_\ell^{(\tau)} = J_{\ell+1}^{-1} \circ F_\ell^{(\tau)} \circ J_\ell : Y_\star^{(\tau)} \longrightarrow Y_\star^{(\tau)}. \tag{12.157}$$

A *reference fixed object* is a $y_\star \in Y_\star^{(\tau)}$ with

$$\widehat{F}_\ell^{(\tau)}(y_\star) = y_\star \qquad \text{for every } \ell, \tag{12.158}$$

which in an autonomous scheme is the ordinary equation $\widehat{F}^{(\tau)}(y_\star) = y_\star$. If the identified maps are p-periodic, the base-ℓ reference monodromy is the endomorphism

$$\widehat{\mathcal{M}}_\ell^{(\tau)} = \widehat{F}_{\ell+p-1}^{(\tau)} \circ \cdots \circ \widehat{F}_\ell^{(\tau)} : Y_\star^{(\tau)} \longrightarrow Y_\star^{(\tau)}, \tag{12.159}$$

and a *reference monodromy fixed object* satisfies $\widehat{\mathcal{M}}_\ell^{(\tau)}(y_{\star,0}) = y_{\star,0}$. Its reference p-cycle is

$$y_{\star,j+1} = \widehat{F}_{\ell+j}^{(\tau)}(y_{\star,j}) \quad (0 \le j < p), \qquad y_{\star,p} = y_{\star,0},$$

entirely in $Y_\star^{(\tau)}$; its points need not be fixed by any one-step reference map. Without a common object the expression $F_\ell^{(\tau)}(y) = y$ is ill typed, and no subtraction across tiers or scales is formed. A fixed ray additionally requires a declared cone and projectivization. [DEFINITION]

The tiers of this chapter are the following, and each carries its own step and its own invariance equation. The law tier carries the normalized law π_ℓ, the tracked measure pair (ρ_ℓ, m_ℓ) of Definition 11.2, and the marked attention event law η_ℓ of (12.110); its fixed objects are characterized by Theorem 12.30. The action tier carries H_ℓ relative to ρ_ℓ and separately the additive class $[H_\ell]$; the two are related by (12.150) and (12.151), and by Proposition 12.25 the action tier is meaningful only together with its reference trajectory. The full interaction tier carries $g_\ell \in \mathcal{G}_\ell$ with step $T_\ell^{\mathcal{G}}$, and its reference fixed objects use $\widehat{T}_\ell^{\mathcal{G}}$ of (12.135). The retained interaction tier carries $\operatorname{Ran} R_\ell$, and a retained fixed object is exact only under (12.139). The bundle tier carries the geometric state (11.10), and the configuration tier carries the section manifolds of Chapter 9 with their own smooth comparison maps; neither is the law, action, or interaction map. [DEFINITION]

Proposition 12.33 (Fixedness does not transfer between tiers). None of the following implications holds, and each is refuted by a finite witness. A fixed normalized law does not give a fixed reference-dependent action: on a two-point set hold $\mu = (1/2, 1/2)$ fixed while the reference alternates between $\rho_0 = (1/4, 3/4)$ and $\rho_1 = (3/4, 1/4)$, so that the actions $H_i = -\log(d\mu/d\rho_i)$ are interchanged. A fixed action class does not give a fixed tracked pair: H and $H + c$ are the same class, while $e^{-H}\rho$ and $e^{-c}e^{-H}\rho$ have evidence masses differing by the factor e^{-c}, which is why (12.151) is strictly weaker than (12.149) until mass preservation forces $c_{\text{ray}}(b) = 0$. A fixed retained interaction does not give a fixed exact interaction: the witness $R(x, y) = (x, 0)$ and $T(x, y) = (x, x)$ of Proposition 12.27 has zero retained beta at every $(x, 0)$ and nonzero exact residual for $x \neq 0$. A fixed law does not give a fixed configuration: take the configuration manifold S^1 with antipodal step $q \mapsto -q$ and a constant configuration-to-law extraction, so that the law is fixed while the configuration step has no fixed point. A fixed conditional attention row does not give a fixed event law: holding β fixed while the receiver occupancy α_ℓ alternates makes $\eta_\ell = \alpha_\ell \beta$ alternate, which is the disintegration clause of Theorem 12.30. Finally a monodromy object is not a one-step fixed object: the period-two maps $F_0(x) = x + 1$ and $F_1(x) = x - 1$ on $\mathbb{R}$ have monodromy $F_1 \circ F_0 = \mathrm{id}$, which fixes every point, while neither one-step map has a fixed point. [ESTABLISHED]

Proof. Each clause is the displayed computation. For the first, $d\mu/d\rho_0 = (2, 2/3)$ and $d\mu/d\rho_1 = (2/3, 2)$ are distinct. For the second, $(e^{-c}e^{-H}\rho)(\mathsf{Y}) = e^{-c}m(\mathsf{Y})$. The third is (12.137) evaluated at $g = (x, 0)$, which gives $(0, x)$. For the fourth, a constant extraction sends both q and $-q$ to the same law, while $q \mapsto -q$ has no fixed point on S^1. The fifth is $\eta_\ell = \alpha_\ell \beta$ with α_ℓ nonconstant in ℓ. For the sixth, $F_1(F_0(x)) = x$ while $F_0(x) = x$ and $F_1(x) = x$ have no solutions. □

Two prohibitions follow. First, relevance, marginality, and irrelevance are properties of a mode relative to declared norms, a declared extensive normalization, the declared block ratios, and the declared comparison maps, in the sense of Definition 11.12 and Definition 12.13; Proposition 11.15 and Corollary 12.21 show that changing the comparison data alone can change the reported exponent. Second, none of the objects above is a physical law. A fixed measure pair, a fixed action class, a fixed interaction coordinate, a monodromy object, and a fixed configuration are mathematical invariants of a declared scale diagram; identifying any of them with an empirical regularity requires the separate apparatus listed at Open Problem B.1, which this document does not supply. [NOT-CLAIMED]

The equations characterize all fixed points without claiming that every model class admits a closed-form enumeration. The following exact sectors illustrate them. The identity channel fixes every law. A terminal channel followed by its one-point identification has the constant infrared fixed theory. For independent additive vector states and the block statistic $b^{-1/\alpha} \sum_{i=1}^{b} Y_i$, every strictly α-stable baseline with constant likelihood $m_o = Z(o)\rho$ gives a fixed measure pair; the centered Gaussian is the $\alpha = 2$ case. At one fixed integer b this list is not exhaustive because b-semistable laws can also be fixed. Exhaustiveness by strict stability requires invariance over all declared block sizes (or a continuum of scales) with the usual centering and regularity hypotheses. [ESTABLISHED]

Flat graph connection data form an invariant sector under rooted contraction, but become a fixed connection only after a self-similar graph identification has been declared. Uniform attention is fixed-form only when receiver occupancies, block sizes, source labels, and self-edge conventions are also self-similar. One-hot rows are invariant boundary sectors of (12.148). These qualifications prevent an invariant form from being mislabeled as a complete fixed theory. [ESTABLISHED]

Along every exact coarse path, (12.4) makes the recognition gap nonincreasing and the ELBO nondecreasing at fixed evidence. This monotone flow is information loss under resolution, not a proof of approach to a nontrivial critical fixed point. Attraction requires spectral control of (12.155) on the declared common space. [ESTABLISHED]

12.11 Closure theorem and infinite extensions

Theorem 12.34 (Complete finite law-level gauge-VFE theory). For every finite standard-Borel agent hypergraph, suppose the model supplies the common principal bundle and its two associated statistical bundles; normalized gauge-covariant baseline and interaction-record kernels with positive finite evidence; channel-specific incidence and boundary links U^x, Θ^x obeying $U^{x\prime}_{a\leftarrow i} = (a^x_a)^{-1} U^x_{a\leftarrow i} a^x_i$ and the separate belief and model representations; normalized recognition laws satisfying the stated absolute-continuity and log-integrability conditions; structural coarse channels that are recognition independent and gauge equivariant; globally gauge-equivariant jointly measurable versions of every disintegration used for the displayed bridge kernels; pushed probability references; component-rooted forests together with their simultaneous raw root-framed holonomy maps, root features, and dressed boundary generators; joint marked attention-event laws; every induced effective hyperedge; full path laws or their exact memory kernels; and rescaling kernels satisfying (12.127). Then the global and local ELBOs, agent–meta bridge kernels, meta-agent interaction kernels, meta-attention, effective likelihoods, measure-pair scale composition, and measure-pair fixed equations defined in this and the preceding chapter are simultaneously normalized, gauge covariant, and exact. [ESTABLISHED]

Proof. Normalization and the collective/local identities are Proposition 7.2 and theorems 7.3 and 7.5. The common-pushforward chain rule is Theorem 12.1; the conditional-partition and tower identities are Theorem 12.2. Mobius inversion or the retained top-order factor closes the hypergraph. Regular disintegration gives an almost-everywhere version family of bridge kernels, and the supplied equivariant-version hypothesis selects the displayed globally covariant versions. Rooted dressing gives the separate gauge-covariant channel maps while the full holonomy representation preserves the information lost by a tree presentation. Pushforward and disintegration of η give exact attention. Path pushforward supplies dynamic closure, and the explicit common-space kernel identity types and composes the measure-pair RG. Each operation is closed in the stated collection, so their finite composition is closed. □

Corollary 12.35 (Exact analytic action, beta, and scaling tier). In addition to the hypotheses of Theorem 12.34, suppose the pushed measures admit the declared common dominators. On the linear cross-scale tier, additionally supply the vector features and linear representations R_x, multiplicative weights and inter-root transports in (12.107), the extra fine cross morphism Φ_f, and the full marked operator–feature kernels. These hypotheses make the fine–meta, meta–fine, meta–meta, cross-channel, and attention operators exact and compositional on their stated domains; the Hom moments also require (12.115). For the discrete beta require a declared reference trajectory and a finite-valued action vector space closed under the reference-dependent action map; a change of that reference trajectory then obeys the inhomogeneous law (12.133). For the discrete interaction beta require in addition the comparison isomorphisms (12.134), and report the residual (12.137) unless (12.139) is proved. For the continuous beta require the separately declared smooth scale bundle and scale connection of Definition 12.28, together with the generator-domain, differentiability, positivity, and dual-pairing hypotheses stated before (12.143). For component beta functions require the stated Schauder or Riesz basis and interchange conditions. For action fixed points retain invariance of the reference component. For linearization require (12.154) and use the bounded perturbations in $L^\infty(\pi_*)$ stated before (12.155). Then the effective actions, discrete and continuous beta functionals, component flows, action fixed-point equations, and linearized scaling operators are exact on their declared analytic domains. [ESTABLISHED]

Proof. The Radon–Nikodym formula gives the action. Ordinary subtraction on the finite action space gives the discrete beta; differentiation in the declared generator domain gives the continuous beta. Valid coefficient extraction gives the component flow. Invariance of both the reference and likelihood ratio is equivalent to pair invariance, and dominated differentiation under the conditional partition gives the first two variations. □

The theorem is uniform in the number and arrangement of agents; it is not a verification on one fixed graph size. For a countable agent or time index, suppose instead that the normalized finite-cylinder joints form a projectively consistent family on standard Borel coordinate spaces. Kolmogorov extension then supplies the full normalized joint [39], and the stated identities hold for every regular finite-cylinder observation and finite block. Projective consistency alone does not make one prescribed infinite record positive probability: an infinite sequence of independent positive-cost binary successes can still be null. Conditioning on an infinite record therefore requires a chosen regular conditional at an almost-everywhere record or a proved DLR, Doob-transform, or finite-volume posterior limit. An infinite-law KL pushforward additionally requires compatible $Q \ll \Pi$ and an infinite coarse channel. Existence and uniqueness of a thermodynamic DLR state, convergence of free-energy densities, and interchange of volume and RG limits remain the separate open obligations recorded in Chapter B. [ESTABLISHED]

Part III

The Multivariate-Gaussian Realization

Chapter 13

A Linear-Gaussian Generative Realization

This chapter realizes the fixed directed law of Chapter 4 in two associated bundles induced from the same principal G-bundle, using possibly different representations and separately chosen local frames. The Gaussian is a tractable subclass, not the ambient theory. The specialization assumes that at a root the model-to-belief kernel induces a map from model laws to belief laws that may realize $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$. The sample matrix inside that kernel is not itself the associated-bundle morphism. At a nonroot the kernel is parent-indexed unless its domain is enlarged. The existence of $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ creates no reverse conditional.

13.1 Directed forest and conditional laws

Take

$$k_i \in \mathbb{R}^K, \qquad m_i \in \mathbb{R}^{d_m}, \qquad o_i \in \mathbb{R}^{d_{o,i}}, \tag{13.1}$$

and specialize the directed acyclic graph to a forest, so each nonroot i has one parent $j = \mathrm{par}(i)$. This is a displayed subclass; the general DAG replaces each parent term by a finite sum. [DEFINITION]

Fix one design point c_a. Let $\Theta^b_{ij} := \Theta^b_{a,ij} \in G$ and $\Theta^m_{ij} := \Theta^m_{a,ij} \in G$ abbreviate the design-indexed same-channel graph-link copies of Proposition 4.7. A submodel that constrains one common link across all design points retains the stabilizer of that constraint, namely the frame rechoices for which $h^x_{a,i}\Theta^x_{ij}(h^x_{a,j})^{-1}$ is independent of a. Context-independent rechoices form a sufficient, but not generally maximal, subgroup; any larger law must be declared explicitly. For a nonroot,

$$m_i \mid m_j, X \sim \mathcal{N}\left(F^m_i \rho_m(\Theta^m_{ij}) m_j + \eta^m_i, \Sigma^m_i\right), \tag{13.2}$$

$$k_i \mid k_j, m_i, X \sim \mathcal{N}\left(F^b_i \rho_b(\Theta^b_{ij}) k_j + B_i m_i + \eta^b_i, \Sigma^b_i\right), \tag{13.3}$$

$$o_i \mid k_i, m_i, X \sim \mathcal{N}\left(H^b_i k_i + H^m_i m_i + \eta^o_i, \Sigma^o_i\right). \tag{13.4}$$

The parameter shapes are

$$\rho_b(\Theta^b_{ij}), F^b_i \in \mathbb{R}^{K\times K}, \qquad \rho_m(\Theta^m_{ij}), F^m_i \in \mathbb{R}^{d_m\times d_m}, \qquad B_i \in \mathbb{R}^{K\times d_m}, \tag{13.5}$$
$$H^b_i \in \mathbb{R}^{d_{o,i}\times K}, \qquad H^m_i \in \mathbb{R}^{d_{o,i}\times d_m}, \qquad \eta^b_i \in \mathbb{R}^K, \quad \eta^m_i \in \mathbb{R}^{d_m}, \quad \eta^o_i \in \mathbb{R}^{d_{o,i}}. \tag{13.6}$$

The represented links are invertible. The conditional endomorphisms F^b_i, F^m_i need not be: positive definiteness of the receiving covariance, not invertibility of the mean map, normalizes a Gaussian kernel.

The bridge matrix $B_i : \mathbb{R}^{d_m} \to \mathbb{R}^K$ acts on sample coordinates. It is one parameter of a Gaussian kernel, not a map between probability-law fibers. At a nonroot, the kernel also depends on the parent belief coordinate. Only after fixing that parent, or enlarging the domain to include it, does integration against a model law define a belief law. The left-right covariance law below is a parameter transformation that makes this induced law map covariant; it does not identify B_i with $\widetilde{\Phi}$. No reading supplies a belief-to-model factor.

Each root r has the proper laws

$$m_r \mid X \sim \mathcal{N}(\bar{m}_r, \Sigma^m_{r,0}), \tag{13.7}$$
$$k_r \mid m_r, X \sim \mathcal{N}(B_r m_r + \eta^b_{r,0}, \Sigma^b_{r,0}), \tag{13.8}$$

followed by (13.4). Every receiving covariance is positive definite:

$$\Sigma^m_i, \Sigma^m_{r,0} \in \mathbb{S}^{d_m}_{++}, \qquad \Sigma^b_i, \Sigma^b_{r,0} \in \mathbb{S}^K_{++}, \qquad \Sigma^o_i \in \mathbb{S}^{d_{o,i}}_{++}. \tag{13.9}$$

[HYPOTHESIS]

Proposition 13.1 (Normalization and root moments). Under (13.9), every kernel above is a nondegenerate normalized density and the assembled directed law is a probability measure. The root belief coordinate has

$$\bar{k}_r = B_r \bar{m}_r + \eta^b_{r,0}, \qquad \Sigma^{b,\mathrm{marg}}_{r,0} = \Sigma^b_{r,0} + B_r \Sigma^m_{r,0} B_r^\top. \tag{13.10}$$

[ESTABLISHED]

More generally, if a root model law is $s = \mathcal{N}(\mu_m, C_m)$, then the law map induced by the root kernel is

$$\widetilde{\phi}_r(s) = \mathcal{N}\left(B_r\mu_m + \eta^b_{r,0}, \quad \Sigma^b_{r,0} + B_r C_m B_r^\top\right). \tag{13.11}$$

This map, rather than B_r alone, is the candidate local realization of $\widetilde{\Phi}$. [ESTABLISHED]

Proof. A Gaussian density with positive definite covariance integrates to one for every mean, so the normalization proposition of Chapter 4 applies. The tower property gives the first moment and the law of total covariance gives the second identity. Its added term is positive semidefinite, so the marginal covariance dominates the conditional covariance. □

The Gaussian observation kernel may be replaced by any normalized Markov kernel from $\mathbb{R}^K \times \mathbb{R}^{d_m}$ to the observation space. That changes the likelihood but not the exact ELBO identity, which uses only the fixed normalized joint. [ESTABLISHED]

13.2 Independent channel-frame covariance

For the global coordinate formulas in this section, invoke Hypothesis 2.11 on a declared region $\mathcal{C}_0$ containing the contexts under study. Choose $\sigma_0 : \mathcal{C}_0 \to P|_{\mathcal{C}_0}$ and restrict each local frame to $\mathcal{C}_i \cap \mathcal{C}_0$. Then write the independent frame coordinates as

$$U_i^b : \mathcal{C}_i \cap \mathcal{C}_0 \to G, \qquad U_i^m : \mathcal{C}_i \cap \mathcal{C}_0 \to G, \qquad u_i^x = \sigma_0 \cdot (U_i^x)^{-1}. \tag{13.12}$$

Thus σ_0 is common only on the explicitly trivialized region; no global triviality of P is inferred. The corresponding relative principal-frame field is

$$h_i = U_i^b (U_i^m)^{-1}, \qquad u_i^m = u_i^b h_i. \tag{13.13}$$

This G-valued field exists even when ρ_b and ρ_m are inequivalent or have different dimensions. Its represented actions $\rho_b(h_i)$ and $\rho_m(h_i)$ live in different endomorphism spaces and do not define the bridge B_i or either associated-bundle morphism. An independent pair of channel-frame re-expressions is $(g_i^b, g_i^m) \in G \times G$, with represented matrices $R_i^b = \rho_b(g_i^b)$ and $R_i^m = \rho_m(g_i^m)$ as in (4.18). The pair records two choices of local trivialization of bundles associated to the same P; it does not replace the principal structure group G by $G \times G$. In the convention of (2.10), the corresponding principal-section rechoices are $u_i^{b\prime} = u_i^b \cdot (g_i^b)^{-1}$ and $u_i^{m\prime} = u_i^m \cdot (g_i^m)^{-1}$. This accounts for the opposite-sided inverse in the geometric local-map law and makes the formulas below identical to (2.15).

The same-channel links obey (4.20). The conditional maps and offsets transform as

$$F_i^{b\prime} = R_i^b F_i^b (R_i^b)^{-1}, \qquad F_i^{m\prime} = R_i^m F_i^m (R_i^m)^{-1}, \tag{13.14}$$

$$B_i' = R_i^b B_i (R_i^m)^{-1}, \qquad \eta_i^{b\prime} = R_i^b \eta_i^b, \qquad \eta_i^{m\prime} = R_i^m \eta_i^m. \tag{13.15}$$

The left-right law for B_i, together with the covariance and offset laws, makes the induced root law map (13.11) obey the local law of $\widetilde{\Phi}$. The parameter B_i alone is not that morphism, and the two frames remain independent. Receiving covariances transform by their own congruences:

$$\Sigma_i^{b\prime} = R_i^b \Sigma_i^b (R_i^b)^\top, \qquad \Sigma_i^{m\prime} = R_i^m \Sigma_i^m (R_i^m)^\top. \tag{13.16}$$

For either channel,

$$\Lambda_i = \Sigma_i^{-1}, \qquad \Sigma_i' = R_i \Sigma_i R_i^\top, \qquad \Lambda_i' = R_i^{-\top} \Lambda_i R_i^{-1}. \tag{13.17}$$

At a root,

$$\bar{m}'_r = R^m_r \bar{m}_r, \qquad \Sigma^{m\prime}_{r,0} = R^m_r \Sigma^m_{r,0} (R^m_r)^\top, \tag{13.18}$$

$$B'_r = R^b_r B_r (R^m_r)^{-1}, \qquad \eta^{b\prime}_{r,0} = R^b_r \eta^b_{r,0}, \qquad \Sigma^{b\prime}_{r,0} = R^b_r \Sigma^b_{r,0} (R^b_r)^\top. \tag{13.19}$$

Gauge-inert observations require

$$H^{b\prime}_i = H^b_i (R^b_i)^{-1}, \qquad H^{m\prime}_i = H^m_i (R^m_i)^{-1}, \qquad \eta^{o\prime}_i = \eta^o_i, \qquad \Sigma^{o\prime}_i = \Sigma^o_i. \tag{13.20}$$

[DEFINITION]

Proposition 13.2 (Gaussian kernel covariance). The transformation laws above satisfy the three kernel pushforward identities (4.23)–(4.25), and hence the normalized joint and its evidence obey (4.26)–(4.27) under all independent channel-frame re-expressions. [ESTABLISHED]

Proof. For the belief kernel, the transformed conditional mean evaluated at transformed inputs is

$$F^{b\prime}_i \rho_b(\Theta^{b\prime}_{ij})(R^b_j k_j) + B'_i(R^m_i m_i) + \eta^{b\prime}_i = R^b_i \left(F^b_i \rho_b(\Theta^b_{ij}) k_j + B_i m_i + \eta^b_i \right). \tag{13.21}$$

The model-channel calculation is the same without a bridge. Congruence (13.16) therefore makes each transformed Gaussian the receiving-frame pushforward of the original. At the observation kernel,

$$H^{b\prime}_i (R^b_i k_i) + H^{m\prime}_i (R^m_i m_i) + \eta^{o\prime}_i = H^b_i k_i + H^m_i m_i + \eta^o_i, \tag{13.22}$$

and its covariance is fixed. The root calculation is identical. Composition now invokes the frame-covariance result of Chapter 4. □

Equation (13.10) transforms by belief-channel congruence, because both of its summands do. Positive definiteness is preserved independently in each channel.

Group-valued frame coordinates remain primary. An expression $U = \exp \phi$ is valid only inside an exponential chart; for example, $\mathrm{diag}(-1, -2) \in \mathrm{GL}^+(2)$ has no real logarithm because each negative eigenvalue occurs in one Jordan block [23]. [ESTABLISHED]

13.3 Common-frame specialization

Under the common-frame specialization of Section 2.11, impose $u^b_i = u^m_i$ and use one frame re-expression $g_i \in G$ with

$$R^b_i = \rho_b(g_i), \qquad R^m_i = \rho_m(g_i). \tag{13.23}$$

If the graph links are also identified, Θ^b_{ij} and Θ^m_{ij} are the same group link seen through the two representations. Equations (13.14)–(13.20) then reduce to the familiar common-frame, two-representation formulas. [ESTABLISHED]

This reduction ties the gauge choices but does not identify the latent dimensions, make B_i invertible, turn Φ and $\widetilde{\Phi}$ into inverses, or add a reverse generative factor. Compatibility of a cross map with the two connections remains the parallelness condition of Chapter 2; an equal connection is a further hypothesis.

Finally, the assembled Gaussian precision need not have the Laplacian-plus-self-terms form used later. A general directed transition can contribute an off-diagonal block that is neither symmetric nor sign-correct. The interaction family is therefore a separately declared Gaussian operator subfamily, not a consequence of linear-Gaussian directedness. [NOT-CLAIMED]

Chapter 14

The Gaussian Information Form and the Interaction Family

The preceding chapters first allow general belief-law and model-law fibers, with smooth manifolds selected only at the differential tier, and then identify regular exponential families as a tractable dominated subclass. Nothing in that construction selects the multivariate Gaussian. This chapter makes that additional choice. It develops one finite-dimensional Gaussian realization in information form, where the primitive coordinates are a vector and a matrix of natural parameters rather than a mean and a covariance, and it then isolates the still narrower subfamily of Gaussian precisions on which the renormalization half of this document acts. None of the matrix identities below is asserted for an arbitrary belief fiber.

Two things are worth separating at the outset, because conflating them is the standing hazard of the whole subject. Writing a Gaussian in information form is a change of coordinates and costs nothing. Restricting the precision to the Laplacian-plus-self-terms shape of Section 14.3 is a substantive restriction that costs a great deal, and it does not follow from the generative constructions of the preceding part. Section 14.4 proves that it does not, exhibits the exact obstruction, and then states precisely the parameter condition under which the shape is recovered. The restriction is carried forward as a declared hypothesis, and every later result that uses it says so.

Throughout, the development is written for the state sector, with agents $i \in V = \{1, \dots, N\}$, state latents $k_i \in \mathbb{R}^K$, and stacked latent Y of dimension $n := NK$. The model sector, with latents $m_i \in \mathbb{R}^{d_m}$, is identical with K replaced by d_m, and a partition mixing the two sectors is admissible wherever a partition is required. The design $D = \{c_a\}_{a=1}^M$ plays no role in the algebra below; either fix one design point, or read the index i as ranging over agent-design pairs, and every statement stands unchanged.

14.1 The recognition family in information form

Let ν be Lebesgue measure on $\mathbb{R}^n$. The Gaussian family in *information form* is parameterized by a vector $h \in \mathbb{R}^n$ and a symmetric positive definite matrix $J \in \mathbb{S}^n_{++}$ through the density

$$q(y \mid h, J) = \exp\left(h^\top y - \tfrac{1}{2} y^\top J y - \mathsf{A}(h, J)\right), \qquad y \in \mathbb{R}^n. \tag{14.1}$$

The pair (h, J) are the natural parameters, J is the precision, and A is the log normalizer. The symbol A is reserved for the log normalizer and never for a self term; the self terms of Section 14.3 are written A_i with a subscript and are $K \times K$ matrices.

Proposition 14.1 (Log normalizer, moments, and convexity). For every $(h, J) \in \mathbb{R}^n \times \mathbb{S}^n_{++}$ the density in Equation (14.1) is normalizable, with

$$\mathsf{A}(h, J) = \tfrac{1}{2} h^\top J^{-1} h - \tfrac{1}{2} \log\det J + \tfrac{n}{2} \log 2\pi. \tag{14.2}$$

Its differential is

$$d\mathsf{A} = \mu^\top dh - \tfrac{1}{2} \operatorname{Tr}\left[\left(C + \mu\mu^\top\right) dJ\right], \qquad \mu = J^{-1}h, \qquad C = J^{-1}, \tag{14.3}$$

so the mean and covariance of $q(\cdot \mid h, J)$ are μ and C, and the Hessian of A in h at fixed J is C. Regarded as a function of the natural parameter pair (h, Θ) with $\Theta = -\frac{1}{2}J$, the log normalizer is strictly convex on the open convex domain $\mathbb{R}^n \times \{\Theta \prec 0\}$. [ESTABLISHED]

Proof. Complete the square in the exponent,

$$h^\top y - \tfrac{1}{2} y^\top J y = -\tfrac{1}{2}\left(y - J^{-1}h\right)^\top J \left(y - J^{-1}h\right) + \tfrac{1}{2} h^\top J^{-1} h, \tag{14.4}$$

and integrate the Gaussian kernel, whose value is $(2\pi)^{n/2}(\det J)^{-1/2}$ because $J \succ 0$. Exponentiating and taking logarithms gives Equation (14.2). Differentiating Equation (14.2) termwise, $d(\frac{1}{2}h^\top J^{-1}h) = h^\top J^{-1} dh - \frac{1}{2}\operatorname{Tr}(J^{-1}hh^\top J^{-1} dJ)$ and $d(-\frac{1}{2}\log\det J) = -\frac{1}{2}\operatorname{Tr}(J^{-1}dJ)$, which is Equation (14.3). The identification of $\nabla_h \mathsf{A}$ with the mean and $\nabla_h^2 \mathsf{A}$ with the covariance is the standard exponential-family cumulant property, valid here because the family is regular: the domain is open and differentiation under the integral is licensed on the interior of the natural domain [19]. Convexity is the convexity of a log-Laplace transform, and strictness follows because the family is minimal, the statistics $(y, yy^\top)$ being affinely independent [19, 75]. Since $J \mapsto -\frac{1}{2}J$ is affine, convexity in (h, Θ) transfers to convexity in (h, J) on $\{J \succ 0\}$. □

Two consequences of Equation (14.3) organize everything that follows. The first is that (h, J) and (μ, C) are two coordinate systems on one family, related by the Legendre correspondence of a minimal exponential family, so no information is lost by working in either. The second is that the Fisher information of the family in the natural parameter h is exactly $C = J^{-1}$, while the Fisher information in the mean parameter μ at fixed covariance is exactly J. The precision is therefore simultaneously a coupling matrix and a metric, and the renormalization chapters use both readings.

14.2 What an off-diagonal precision block is

Let $\mathfrak{B}$ be a finite partition of the n coordinates into blocks, with block dimensions n_b summing to n, and write J_{bc} for the corresponding submatrices. The meaning of J_{bc} for $b \neq c$ is fixed by the following statement, which is standard but is stated here in full because the rest of the document depends on reading it correctly.

Proposition 14.2 (Conditional reading of the precision). Let $Y \sim \mathcal{N}(\mu, J^{-1})$ with $J \succ 0$. For any block b,

$$\mathbb{E}\left[Y_b \mid Y_{-b} = y_{-b}\right] = \mu_b - J_{bb}^{-1} \sum_{c \neq b} J_{bc}\left(y_c - \mu_c\right), \qquad \mathrm{Cov}\left(Y_b \mid Y_{-b}\right) = J_{bb}^{-1}. \tag{14.5}$$

For any two distinct blocks b and c, the variables Y_b and Y_c are conditionally independent given all remaining coordinates if and only if $J_{bc} = 0$. [ESTABLISHED]

Proof. Fix y_{-b} and read the exponent of Equation (14.1) as a function of y_b alone. It is $-\frac{1}{2} y_b^\top J_{bb} y_b$ plus a linear term, so the conditional law is Gaussian with precision J_{bb}, and completing the square in y_b around μ gives the displayed conditional mean. For the second claim, condition on the coordinates outside $b \cup c$ and repeat the argument: the conditional law of (Y_b, Y_c) is Gaussian with precision $\left(\begin{smallmatrix} J_{bb} & J_{bc} \\ J_{cb} & J_{cc} \end{smallmatrix}\right)$, and a Gaussian density factorizes into a function of y_b times a function of y_c if and only if the quadratic form has no cross term, that is if and only if $J_{bc} = 0$. This is the Gaussian case of the pairwise Markov property [47, Ch. 5]; the same statement organizes the Gaussian Markov random field literature [66, Ch. 2]. □

Equation (14.5) also fixes the sign convention that Section 14.3 will exploit. A negative semidefinite off-diagonal block $J_{bc} \preceq 0$ makes the conditional mean of Y_b move *toward* y_c as y_c departs from its own mean, so negative off-diagonal precision is attractive coupling. The interaction family is the family in which every off-diagonal block is attractive in this sense.

Precision blocks are not covariance values. A zero off-diagonal precision block is a statement about conditional dependence and carries no implication about the marginal covariance. The tridiagonal example

$$J = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \qquad J^{-1} = \frac{1}{4}\begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \tag{14.6}$$

has $J_{13} = 0$ and $(J^{-1})_{13} = \frac{1}{4} \neq 0$. In general the marginal covariance of a block is the inverse of a Schur complement,

$$\left(J^{-1}\right)_{bb} = \left(J_{bb} - J_{b,-b} J_{-b,-b}^{-1} J_{-b,b}\right)^{-1}, \tag{14.7}$$

so it records the effect of every path through the remaining coordinates, whereas J_{bb}^{-1} records only the conditional dispersion of Equation (14.5) [80]. The two coincide exactly when $J_{b,-b} = 0$. [ESTABLISHED]

The distinction exhibited by (14.6) is the reason this document works in information form. A restriction imposed on the recognition family is a restriction on conditional dependence, and it is stated cleanly only in coordinates in which conditional dependence is a coordinate. The exact price is computed in Chapter 15.

14.3 The declared interaction subfamily

Definition 14.3 (Interaction family). Let $\Lambda \in \mathbb{S}^{NK}$ be partitioned into $K \times K$ blocks indexed by agents. Say that Λ belongs to the *interaction family* $\mathcal{I}(V)$ when there exist matrices $A_i \in \mathbb{S}^K_+$ and $W_{ij} = W_{ji} \in \mathbb{S}^K_+$ for $i \neq j$ with

$$\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}, \qquad \Lambda_{ij} = -W_{ij} \quad (i \neq j). \tag{14.8}$$

The A_i are the *self terms*, the W_{ij} the *edge weights*, and $L := \Lambda - \operatorname{blockdiag}(A_1, \ldots, A_N)$ the *Laplacian part*. Nothing is proved by this definition; it declares a type. [DEFINITION]

The edge weights are matrices, so L is a matrix-weighted graph Laplacian rather than a scalar-weighted one. That object, and the ways in which it fails to behave like its scalar counterpart, are the subject of the matrix-weighted consensus literature [69], and one of those failures appears immediately below.

Proposition 14.4 (Energy and kernel). For $\Lambda \in \mathcal{I}(V)$ and any $y = (y_1, \ldots, y_N)$,

$$y^\top \Lambda y = \sum_{i \in V} y_i^\top A_i y_i + \sum_{i<j} (y_i - y_j)^\top W_{ij} (y_i - y_j). \tag{14.9}$$

Consequently $\Lambda \succeq 0$ for every admissible choice of parameters; the consensus subspace $\mathbf{1} \otimes \mathbb{R}^K$ lies in $\ker L$; and $\Lambda \succ 0$ if and only if the only y satisfying $A_i y_i = 0$ for all i together with $W_{ij}(y_i - y_j) = 0$ for all $i < j$ is $y = 0$. A necessary condition for $\Lambda \succ 0$ is $\sum_i A_i \succ 0$. [ESTABLISHED]

Proof. Expand the right-hand side of Equation (14.9). The self sum contributes $\sum_i y_i^\top A_i y_i$ to the diagonal blocks. Each unordered pair contributes $y_i^\top W_{ij} y_i + y_j^\top W_{ij} y_j - 2 y_i^\top W_{ij} y_j$, which places W_{ij} on the ii and jj diagonal blocks and $-W_{ij}$ on the ij and ji off-diagonal blocks. Summing reproduces Equation (14.8). Every term of Equation (14.9) is nonnegative because $A_i \succeq 0$ and $W_{ij} \succeq 0$, so $\Lambda \succeq 0$; and a sum of nonnegative terms vanishes only when each vanishes, which for a positive semidefinite matrix A and vector v means $v^\top A v = 0$ if and only if $Av = 0$. Taking $y = \mathbf{1} \otimes v$ makes every difference vanish, so the consensus subspace lies in $\ker L$, and evaluating Equation (14.9) there gives $v^\top (\sum_i A_i) v$, whence the necessary condition. □

The kernel criterion in Proposition 14.4 is where the matrix-weighted setting departs from the scalar one. In the scalar case a connected interaction graph forces $\ker L$ to be exactly the consensus subspace. With matrix weights, a singular W_{ij} leaves the difference $y_i - y_j$ free along $\ker W_{ij}$, so $\ker L$ can be strictly larger than the consensus subspace even on a connected graph, and connectivity alone is not a definiteness criterion [69]. The criterion stated in the proposition is exact and replaces it.

Proposition 14.5 (The interaction family is thin). For $K \geq 2$ and $N \geq 2$, $\mathcal{I}(V)$ is contained in a proper linear subspace of $\mathbb{S}^{NK}$ of codimension at least $\binom{N}{2}\frac{K(K-1)}{2}$, hence has Lebesgue measure zero in $\mathbb{S}^{NK}$. [ESTABLISHED]

Proof. Symmetry of a matrix Λ already forces $\Lambda_{ji} = \Lambda_{ij}^\top$ for every ordered pair, so this identity carries no information about the family. Membership additionally requires each off-diagonal block Λ_{ij} to be symmetric *as a $K \times K$ matrix*, since $W_{ij} = W_{ji}$ and $W_{ji} = W_{ij}^\top$ together give $W_{ij} = W_{ij}^\top$. That is $\frac{K(K-1)}{2}$ independent linear equations per unordered pair, and there are $\binom{N}{2}$ pairs. The remaining requirements are conic inequalities, which can only shrink the set further. □

Proposition 14.5 is the quantitative form of the claim that the interaction shape is a restriction in a fixed product chart rather than a normalization. It is not invariant under arbitrary independent block-diagonal coordinate changes, so membership may be created or destroyed by such a change and no coordinate-orbit exclusion follows from thinness. An exact witness starts from the interaction precision with $N = 2$, $K = 2$, $A_1 = A_2 = I_2$, and $W_{12} = I_2$. Congruence by $G = \operatorname{diag}(I_2, S)$, $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, changes the off-diagonal block from $-I_2$ to $-S$, which is not symmetric and hence is outside the family; applying G^{-1} to that transformed precision creates membership again. The invariant statement is only that a declared trivialization must be fixed before Equation (14.8) is imposed. Section 14.4 asks when the generative construction supplies such a distinguished chart.

14.4 The interaction form is a hypothesis and not a consequence

14.4.1 Trivialized coordinates and flat comparison

In this Gaussian belief-channel subsection, write $\Theta_{ij} := \rho_b(\Theta_{ij}^b) \in \mathrm{GL}^+(K)$ for the represented graph link. This abbreviation does not identify it with a base-manifold parallel transport.

The state latents carry frames. A residual between two agents is not $k_i - k_j$, which compares vectors in different fibers, but $k_i - \Theta_{ij}k_j$, where Θ_{ij} is the represented belief-channel graph link. The interaction shape of Definition 14.3 involves plain differences, so it can only be a statement in coordinates in which the graph links have been removed. When that removal is possible is settled exactly by the following.

Proposition 14.6 (Simultaneous trivialization is exactly coboundary). Let $G = (V, E)$ be a connected graph and let $\{\Theta_{ij}\}_{(i,j)\in E} \subset \mathrm{GL}^+(K)$ satisfy $\Theta_{ji} = \Theta_{ij}^{-1}$. There exist frames $\{U_i\} \subset \mathrm{GL}^+(K)$ with $U_i^{-1}\Theta_{ij}U_j = I$ for every edge if and only if $\Theta_{ij} = U_iU_j^{-1}$ for those frames, and this holds if and only if the ordered product of graph links around every cycle of G is the identity. When it holds, the frames are unique up to a common right factor $U_i \mapsto U_i g$ with $g \in \mathrm{GL}^+(K)$ independent of i. [ESTABLISHED]

Proof. The condition $U_i^{-1}\Theta_{ij}U_j = I$ is literally $\Theta_{ij} = U_iU_j^{-1}$, which gives the first equivalence. For the second, suppose such frames exist and let $i_0, i_1, \ldots, i_r = i_0$ be a cycle. Then $\prod_t \Theta_{i_t i_{t+1}} = \prod_t U_{i_t}U_{i_{t+1}}^{-1} = U_{i_0}U_{i_0}^{-1} = I$ by telescoping. Conversely, assume every cycle product is the identity, fix a spanning tree and a root r, set $U_r = I$, and define U_i as the ordered product of graph links along the tree path from r to i. The relation $\Theta_{ij} = U_i U_j^{-1}$ then holds on every tree edge by construction, and on a non-tree edge it holds precisely because the cycle formed by that edge and the tree path between its endpoints has trivial product. For uniqueness, if $U_i U_j^{-1} = U_i' U_j'^{-1}$ on every edge then $U_i'^{-1} U_i = U_j'^{-1} U_j$ for adjacent i, j, so this quantity is constant on a connected graph. □

Hypothesis 14.7 (Flat comparison). The declared graph links are a coboundary, $\Theta_{ij} = U_i U_j^{-1}$, so that the trivialization $z_i = U_i^{-1} \mu_i$ of the means, and the same map $z_i = U_i^{-1} k_i$ applied to the latents themselves, turn every residual into a plain difference,

$$\mu_i - \Theta_{ij} \mu_j = U_i \left(z_i - z_j \right). \tag{14.10}$$

This is a choice, not a derivation. By Proposition 14.6 it excludes exactly the graph-link assignments with nontrivial holonomy around some cycle, which are permitted in Regime II. It is used in two places: it makes the interaction shape of Definition 14.3 a statement in one fixed coordinate system, and it supplies a clean sufficient domain for closure under aggregation. In the flat chart the represented link $\widehat{\Theta}_{ij} := U_i^{-1} \Theta_{ij} U_j$ equals I, so merging two agents by $z_i = z_j = z$ annihilates their internal residual. More generally, in an arbitrary common chart the internal edge energy becomes

$$z^\top (I - \widehat{\Theta}_{ij})^\top W_{ij} (I - \widehat{\Theta}_{ij}) z,$$

which vanishes for every z exactly when $W_{ij}^{1/2}(I - \widehat{\Theta}_{ij}) = 0$. Positive-definite W_{ij}, or faithfulness of the weight on every represented direction, reduces this condition to $\widehat{\Theta}_{ij} = I$; with merely positive-semidefinite weights, a nontrivial represented link can survive in invisible null directions. Thus Proposition 14.6 remains the exact coboundary theorem, while the converse from energy annihilation to a trivial represented link requires the stated nondegeneracy. The coarse-graining chapter carries out the weighted computation and separates the sufficient trivializing domain from degenerate exceptions. [HYPOTHESIS]

Two remarks fix the status of Equation (14.10). First, the change of variables $y = Uz$ with $U = \mathrm{blockdiag}(U_1, \ldots, U_N)$ acts on precisions by congruence, $\Lambda^{(y)} = U^{-\top} \Lambda^{(z)} U^{-1}$, which preserves inertia by Sylvester's law [34, Ch. 4] but does not preserve membership in $\mathcal{I}(V)$, since the transformed off-diagonal block $U_i^{-\top} \Lambda_{ij}^{(z)} U_j^{-1}$ need not be symmetric when $U_i \neq U_j$. Membership in the interaction family is therefore a frame-dependent assertion. Second, that frame dependence is exactly why the invariants of the renormalization chapters are stated as generalized eigenvalues of the pair (L, Λ) rather than as eigenvalues of either matrix alone: the pair transforms by the same congruence and the generalized spectrum is unchanged. [ESTABLISHED]

14.4.2 What a directed linear-Gaussian model actually contributes

Consider the state sector of a directed linear-Gaussian model of the kind used in the preceding part: each non-root agent i has a single parent $j = \mathrm{pa}(i)$ and receiving kernel

$$k_i \mid k_j \sim \mathcal{N}\left(G_i \Theta_{ij} k_j, \Lambda_i^{-1} \right), \qquad \Lambda_i \in \mathbb{S}_{++}^K, \qquad G_i \in \mathbb{R}^{K \times K}, \tag{14.11}$$

and each root r carries a proper prior $\mathcal{N}(0, \Lambda_r^{-1})$, a nonzero prior mean shifting only the information vector h and not the precision. The gain is written G_i rather than A_i because the latter symbol is reserved for the self terms of Definition 14.3. Unlike the represented graph link Θ_{ij}, the conditional mean map G_i need not be invertible: positive definiteness of the innovation covariance normalizes the receiving kernel for every matrix G_i, exactly as Chapter 4 declares. A single subscript on Λ denotes the innovation precision of one receiving kernel; a double subscript denotes a block of the assembled precision. When an invertible-gain submodel is wanted it will be named explicitly.

Proposition 14.8 (Assembled precision blocks). Write $T_i := G_i \Theta_{i\,\mathrm{pa}(i)}$ and $\mathrm{ch}(i) = \{\ell : \mathrm{pa}(\ell) = i\}$. The joint precision of the model above has blocks

$$\Lambda_{i\,\mathrm{pa}(i)} = -\Lambda_i T_i, \qquad \Lambda_{ii} = \Lambda_i + \sum_{\ell \in \mathrm{ch}(i)} T_\ell^\top \Lambda_\ell T_\ell, \tag{14.12}$$

and $\Lambda_{ij} = 0$ whenever i and j are not adjacent in the parent graph. [ESTABLISHED]

Proof. The negative log joint density is $\frac{1}{2}\sum_i (k_i - T_i k_{\mathrm{pa}(i)})^\top \Lambda_i (k_i - T_i k_{\mathrm{pa}(i)})$ plus terms not depending on k, with the parent term absent at a root. Expanding one summand gives $\frac{1}{2} k_i^\top \Lambda_i k_i - k_i^\top \Lambda_i T_i k_{\mathrm{pa}(i)} + \frac{1}{2} k_{\mathrm{pa}(i)}^\top T_i^\top \Lambda_i T_i k_{\mathrm{pa}(i)}$. The precision is the Hessian of this function, and collecting the three contributions of each factor gives Equation (14.12). □

By Equation (14.12), membership in the interaction family requires the matrix

$$W_{i\,\mathrm{pa}(i)} := \Lambda_i T_i = \Lambda_i G_i \Theta_{i\,\mathrm{pa}(i)} \tag{14.13}$$

to be symmetric and positive semidefinite. Neither property is automatic, and both fail on explicit parameter values.

Two sources of failure must be kept apart, and a convention does it. Under Hypothesis 14.7 the trivialized residual is $z_i - \widetilde{T}_i z_j$ with

$$\widetilde{T}_i := U_i^{-1} T_i U_j = U_i^{-1} G_i U_i, \qquad \widetilde{\Lambda}_i := U_i^\top \Lambda_i U_i, \tag{14.14}$$

the second equality in the first display using $\Theta_{ij} = U_i U_j^{-1}$, so that (14.12) holds verbatim in trivialized coordinates with (Λ_i, T_i) replaced by $(\widetilde{\Lambda}_i, \widetilde{T}_i)$. The graph links have then already been removed and any remaining obstruction is attributable to the gain alone. For the rest of this section the tildes are dropped and (Λ_i, T_i) denote the trivialized quantities.

Proposition 14.9 (Two global-condition witnesses). There exist admissible parameters for which no interaction-family representation of the assembled precision exists.

$$\Lambda_i = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right) \succ 0, \qquad T_i = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in \mathrm{GL}^+(2) \qquad \Longrightarrow \qquad -\Lambda_i T_i = \left(\begin{smallmatrix} 1 & -2 \\ 3 & -1 \end{smallmatrix}\right), \tag{14.15}$$

which differs from its own transpose by a matrix of Frobenius norm $5\sqrt{2}$, so the off-diagonal block is not symmetric and no W exists at all. Separately, taking $K = 1$ with $G_i = -1$ and $\Theta_{ij} = \Lambda_i = 1$

gives a symmetric block with $W = -1 < 0$, so the sign requirement fails; the same construction with $G_i = -I$, $\Theta_{ij} = I$ and $\Lambda_i = I$ gives $W = -I \prec 0$ at every K. [ESTABLISHED]

Proof. Direct computation. For the first, $\Lambda_i T_i = \left(\begin{smallmatrix} -1 & 2 \\ -3 & 1 \end{smallmatrix}\right)$, so $-\Lambda_i T_i$ is as displayed and $-\Lambda_i T_i - (-\Lambda_i T_i)^\top = \left(\begin{smallmatrix} 0 & -5 \\ 5 & 0 \end{smallmatrix}\right)$ with Frobenius norm $\sqrt{50}$. Positive definiteness of Λ_i follows from $\det = 5 > 0$ and trace $5 > 0$; invertibility of T_i from $\det T_i = 1 > 0$. For the second, (14.13) evaluates to $\Lambda_i G_i \Theta_{ij} = -I$, and $-I$ is not positive semidefinite for any $K \geq 1$. □

Proposition 14.9 settles the status of the interaction shape. The gain matrix G_i is a free generative parameter in the construction of the preceding part, not required to be symmetric, positive, or even close to the identity, and for the displayed values the assembled precision has no representation of the form of Equation (14.8). Whatever else may be said for the interaction family, it is not a consequence of the general linear-Gaussian directed model.

14.4.3 Exactly when the form does hold

Having shown that the form fails, the obligation is to say when it holds. Two statements are available, one global and one local, and they answer different questions.

Proposition 14.10 (Exact global interaction condition). The assembled precision of Equation (14.12) lies in $\mathcal{I}(V)$ with the edge assignment of Equation (14.13) if and only if, for every non-root i,

$$W_i := \Lambda_i T_i \in \mathbb{S}_+^K, \tag{14.16}$$

and, for every agent i,

$$A_i := \Lambda_i - W_i + \sum_{\ell \in \mathrm{ch}(i)} \left(W_\ell \Lambda_\ell^{-1} W_\ell - W_\ell \right) \succeq 0, \tag{14.17}$$

with the term $-W_i$ omitted at a root. [ESTABLISHED]

Proof. Condition (14.16) is exactly the requirement that the off-diagonal blocks have the form $-W$ with W symmetric positive semidefinite. Given it, $T_i = \Lambda_i^{-1} W_i$ and $T_\ell^\top \Lambda_\ell T_\ell = W_\ell \Lambda_\ell^{-1} \Lambda_\ell \Lambda_\ell^{-1} W_\ell = W_\ell \Lambda_\ell^{-1} W_\ell$, using symmetry of W_ℓ. The self term is then the residual $A_i = \Lambda_{ii} - \sum_{j \sim i} W_{ij}$, where the edges at i are the one to its parent, of weight W_i, and one to each child ℓ, of weight W_ℓ. Substituting Equation (14.12) gives Equation (14.17), and family membership requires exactly that this residual be positive semidefinite. □

Condition (14.17) is a balance at each node and permits cancellation: a parent deficit $\Lambda_i - W_i \prec 0$ may be offset by child surpluses. That freedom disappears if one asks the stronger and more natural question, namely whether each generative factor separately lies in the family, so that the interaction structure can be read off the model factor by factor rather than after a global cancellation. The answer is sharp.

The cancellation is not merely formal. Take one scalar receiving factor $j \to i$ with $\Lambda_i = 1$, $T_i = \frac{1}{2}$, and root precision $\Lambda_j = 1$. In the block order (i, j), Equation (14.12) gives

$$\Lambda_{\text{joint}} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{5}{4} \end{pmatrix} = \begin{pmatrix} A_i + W & -W \\ -W & A_j + W \end{pmatrix}, \qquad W = \tfrac{1}{2}, \quad A_i = \tfrac{1}{2}, \quad A_j = \tfrac{3}{4}. \tag{14.18}$$

Thus the assembled precision belongs to $\mathcal{I}(\{i,j\})$, although $T_i^2 = \frac{1}{4} \neq \frac{1}{2} = T_i$, so the receiving factor does not belong locally: its parent-side residual is $T_i^2\Lambda_i - W = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$, and the root prior offsets it to $A_j = 1 - \frac{1}{4} = \frac{3}{4}$. The matrix is positive definite because its leading principal minors are 1 and 1. This exact witness proves that factorwise membership is sufficient but not necessary for global membership. [ESTABLISHED]

Proposition 14.11 (Edge-local characterization with singular gains allowed). Consider a single receiving factor $j \to i$ in isolation, contributing Λ_i to the ii block, $T_i^\top \Lambda_i T_i$ to the jj block, and $-\Lambda_i T_i$ to the ij block. This factor lies in $\mathcal{I}(\{i,j\})$ if and only if

$$X := \Lambda_i^{-1/2} (\Lambda_i T_i) \Lambda_i^{-1/2} \text{ is an orthogonal projection,} \tag{14.19}$$

equivalently if and only if $\Lambda_i T_i$ is symmetric and T_i is idempotent, equivalently if and only if T_i is a Λ_i-orthogonal projection. In that case the self terms are $A_i = \Lambda_i^{1/2}(I - X)\Lambda_i^{1/2}$ and $A_j = 0$. The full-rank case $X = I$ is $T_i = I$, in which the factor is the pure agreement factor whose mean is the parent's own value in the trivialized coordinates and whose edge weight is $W = \Lambda_i$. [ESTABLISHED]

Proof. Membership requires three things: $W := \Lambda_i T_i \succeq 0$ and symmetric; the i-side residual $\Lambda_i - W \succeq 0$; and the j-side residual $T_i^\top \Lambda_i T_i - W \succeq 0$. Assume W symmetric and set $X = \Lambda_i^{-1/2} W \Lambda_i^{-1/2}$, which is symmetric. Then $W \succeq 0$ reads $X \succeq 0$; $\Lambda_i - W \succeq 0$ reads $X \preceq I$; and, using $T_i = \Lambda_i^{-1} W$ so that $T_i^\top \Lambda_i T_i = W \Lambda_i^{-1} W = \Lambda_i^{1/2} X^2 \Lambda_i^{1/2}$, the third reads $X^2 \succeq X$. Now $0 \preceq X \preceq I$ implies $X^2 \preceq X$, since X and X^2 are simultaneously diagonalizable and $\lambda^2 \leq \lambda$ for $\lambda \in [0,1]$. Combining, $X^2 = X$, so X is a symmetric idempotent, that is an orthogonal projection. Conversely any orthogonal projection satisfies all three conditions, and then $T_i = \Lambda_i^{-1/2} X \Lambda_i^{1/2}$ satisfies $T_i^2 = T_i$ and $\Lambda_i T_i = \Lambda_i^{1/2} X \Lambda_i^{1/2}$ is symmetric positive semidefinite. The self terms follow by substitution, the j-side residual being $\Lambda_i^{1/2}(X^2 - X)\Lambda_i^{1/2} = 0$. □

Corollary 14.12 (Invertible-gain factor). If the receiving gain T_i is required to be invertible, the isolated factor lies in $\mathcal{I}(\{i,j\})$ if and only if $T_i = I$. Indeed, Proposition 14.11 makes T_i idempotent, and multiplying $T_i^2 = T_i$ by T_i^{-1} gives $T_i = I$; the converse is the full-rank case of that proposition. [ESTABLISHED]

Proposition 14.11 describes the stronger factor-local requirement, not global family membership. With singular gains allowed, a proper projection constrains only the components of the residual inside $\mathrm{range}(X)$ and leaves the complement to the receiving-side self term. In the invertible-gain submodel, Corollary 14.12 removes every proper projection and leaves only pure agreement.

The deterministic replacement check CHK-GAUSS-PROJECTION tests 4000 generic gains and exact controls of ranks 0 through 3 under a fully specified protocol; its current output corroborates the characterization but does not prove generic measure zero. [NUMERICAL]

Hypothesis 14.13 (Global interaction form). The renormalization half of this document acts on recognition precisions that belong globally to Equation (14.8), in trivialized coordinates, and this is adopted as a declared restriction. For a precision assembled from the directed model of Equation (14.11), the restriction is exactly Proposition 14.10; it does *not* require every receiving factor to satisfy Proposition 14.11, because the balance terms in Equation (14.17) may cancel as Equation (14.18) proves. It is used in the aggregation closure statement, in the definition of the flow, and in the fixed-point analysis; each of those results names it. The document does not claim that it is forced by anything, and Proposition 14.5 records that it is a measure-zero restriction on the space of Gaussian recognition laws. [HYPOTHESIS]

Hypothesis 14.14 (Optional factorwise-agreement realization). When the interaction parameters are required to be readable from the generative model one factor at a time, every receiving factor is separately assumed to satisfy Proposition 14.11. This is a strictly stronger hypothesis than global membership: it permits Λ_i-orthogonal projections in the relaxed singular-gain model and only $T_i = I$ in the invertible-gain model, while it excludes the globally admissible witness (14.18). The operator-level renormalization results require Hypothesis 14.13, not this stronger factorwise realization, unless they explicitly invoke a factor interpretation. [HYPOTHESIS]

14.5 Nonemptiness, and a note on conditioning

A declared subfamily is worthless if it is empty or numerically degenerate, so both should be checked. Nonemptiness is a construction, not an experiment.

Proposition 14.15 (Constructive nonemptiness and exact kernel). Under Hypothesis 14.7, let each edge of a graph $G = (V, E)$ carry an agreement factor with residual $\mu_i - \Theta_{ij}\mu_j$ and link covariance $R_{ij} \in \mathbb{S}^K_{++}$, and let each vertex carry a quadratic self term $A_i \succeq 0$ in the trivialized coordinates. A self term is the precision of a proper full-dimensional Gaussian prior at vertex i only when $A_i \succ 0$. The assembled trivialized energy is

$$E(z) = \tfrac{1}{2}\sum_i z_i^\top A_i z_i + \tfrac{1}{2}\sum_{i<j} (z_i - z_j)^\top W_{ij} (z_i - z_j), \qquad W_{ij} = U_i^\top R_{ij}^{-1} U_i, \tag{14.20}$$

where the edge sum is over E. The associated precision Λ lies in $\mathcal{I}(V) \cap \mathbb{S}^{NK}_+$, every represented edge weight is positive definite, and

$$\ker \Lambda = \{z : A_i z_i = 0 \text{ for every } i, \quad z_i = z_j \text{ for every } (i,j) \in E\}. \tag{14.21}$$

If V_α are the connected components of G, then equivalently

$$\ker \Lambda = \bigoplus_{\alpha} \left\{ \mathbf{1}_{V_\alpha} \otimes v : v \in \bigcap_{i \in V_\alpha} \ker A_i \right\}. \tag{14.22}$$

Consequently $\Lambda \succ 0$ exactly when $\bigcap_{i \in V_\alpha} \ker A_i = \{0\}$ for every connected component. In particular, one positive-definite self term in each component suffices; taking every $A_i \succ 0$ makes every vertex self term a proper prior and constructs an element of $\mathcal{I}(V) \cap \mathbb{S}^{NK}_{++}$. [ESTABLISHED]

Proof. By (14.10) the residual is $U_i(z_i - z_j)$, so the quadratic form $(\mu_i - \Theta_{ij}\mu_j)^\top R_{ij}^{-1}(\mu_i - \Theta_{ij}\mu_j)$ equals $(z_i - z_j)^\top U_i^\top R_{ij}^{-1} U_i (z_i - z_j)$. The matrix $U_i^\top R_{ij}^{-1} U_i$ is symmetric and positive definite because R_{ij}^{-1} is and U_i is invertible. Comparing with (14.9) identifies the parameters and proves positive semidefiniteness. Equality in the energy holds exactly when $A_i z_i = 0$ at every vertex and, because every $W_{ij} \succ 0$, when $z_i = z_j$ on every edge. The latter conditions make z constant on each connected component, which proves (14.21) and (14.22). A positive-semidefinite matrix is positive definite exactly when its kernel is trivial, yielding the final criterion and the stated proper witnesses. □

Proposition 14.15 also makes the role of Hypothesis 14.7 concrete by contrast. Without trivialization the same agreement factor contributes the off-diagonal block $-R_{ij}^{-1}\Theta_{ij}$, which is symmetric only in special position. The deterministic replacement check `CHK-GAUSS-TRIVIALIZATION` emits its matrices and compares that asymmetric raw block with the symmetric positive-definite trivialized weight. It corroborates the proposition and is not part of its proof. [NUMERICAL]

Conditioning is a different question and admits only protocol-specific computational evidence. The deterministic replacement check `CHK-GAUSS-CONDITIONING` specifies and runs a 200-draw interaction ensemble together with a zero-self consensus control and 200 unstructured positive-definite controls. Its output concerns only that sampler. Positive semidefiniteness and the general null-space criterion are proved by Proposition 14.4; Proposition 14.15 specializes that criterion to the constructed positive-definite edge weights and supplies explicit proper witnesses. Thinness is proved by Proposition 14.5; no universal conditioning theorem is claimed. [NUMERICAL]

14.6 Marginalization belongs to the Gaussian coarse chapter

Aggregation identifies variables, whereas exact Gaussian marginalization eliminates them by a Schur complement. The interaction cone is closed under the former and not under unrestricted matrix-weighted Kron reduction. The exact positive-definite $N = 3, K = 2$ rational counterexample, including the asymmetric manufactured coupling, is stated and proved once in Chapter 17. [ESTABLISHED]

The same chapter gives the positive result in its sharp current form: every fixed congruence-diagonal cone $\mathcal{C}_H = \{HDH^\top : D \succeq 0 \text{ diagonal}\}$ is closed under every defined node-block Schur elimination. This strictly contains the common-orthogonal-eigenbasis case because congruence-diagonal matrices need not commute. Maximality is proved only for rich, independently parameterized closed convex product cones with an SPD order unit and closure under every defined one-node elimination. Correlated, nonconvex, variable-chart, rank-changing, and nonflat families remain outside that theorem. [ESTABLISHED]

Chapter 15

Gaussian Recognition Restrictions

Fix the general restriction principle of Proposition 5.11. This chapter specializes it to nondegenerate multivariate-Gaussian laws, computes the exact costs of block factorization and mean restriction, and states when comparisons between recognition structures are meaningful.

15.1 The block restriction and its exact optimum

For the rest of this chapter the exact conditional is Gaussian and nondegenerate,

$$P^\star = \mathcal{N}\left(\mu, J^{-1}\right), \qquad J = J^\top \succ 0, \qquad \mu = J^{-1}h, \tag{15.1}$$

on $\mathbb{R}^n$ with $n = NK$ in the state sector of the previous chapter. A partition $\mathfrak{B}$ of the n coordinates into blocks of dimensions n_b is declared, and the restricted family is

$$\mathcal{Q}_{\mathfrak{B}} = \left\{\mathcal{N}\left(\mu_Q, C_Q\right) : \mu_Q \in \mathbb{R}^n, \quad C_Q = \operatorname{blockdiag}_{b\in\mathfrak{B}}\left(C_b\right), \quad C_b \in \mathbb{S}^{n_b}_{++}\right\}. \tag{15.2}$$

The blocks are whatever the declared partition says they are. A block may collect one agent's coordinates, may split one agent's state and model channels, or may collect several agents; nothing below identifies a matrix block with an agent, and every statement is relative to $\mathfrak{B}$.

Proposition 15.1 (Gaussian reverse relative entropy). For $Q = \mathcal{N}(\mu_Q, C_Q)$ with $C_Q \succ 0$ and $P^\star$ as in Equation (15.1),

$$D_{\mathrm{KL}}\left(Q \| P^\star\right) = \frac{1}{2}\left[\operatorname{Tr}\left(JC_Q\right) + \left(\mu_Q - \mu\right)^\top J\left(\mu_Q - \mu\right) - n - \log\det J - \log\det C_Q\right]. \tag{15.3}$$

[ESTABLISHED]

Proof. Write $D_{\mathrm{KL}}(Q\|P^\star) = \mathbb{E}_Q[\log q] - \mathbb{E}_Q[\log p^\star]$. The first term is the negative differential entropy of a Gaussian, $-\frac{1}{2}\log\det(2\pi e C_Q)$. For the second, $-\log p^\star(y) = \frac{1}{2}(y-\mu)^\top J(y-\mu) + \frac{n}{2}\log 2\pi - \frac{1}{2}\log\det J$, and taking the expectation under Q uses $\mathbb{E}_Q[(Y-\mu)^\top J(Y-\mu)] = \operatorname{Tr}(JC_Q) + (\mu_Q - \mu)^\top J(\mu_Q - \mu)$, which is the standard expansion of a quadratic form about a shifted center. Adding the two and canceling the $\frac{n}{2}\log 2\pi$ terms gives Equation (15.3). □

Equation (15.3) separates: the mean enters only through the quadratic term and the covariance only through the trace and the log determinant, with no cross term. That separation is used repeatedly below and is the reason the two kinds of restriction studied in this chapter have exactly the costs they have.

Theorem 15.2 (Exact block optimum). Over the family of Equation (15.2) the objective of Equation (15.3) has a unique minimizer, given by

$$\mu_Q^\star = \mu, \qquad C_b^\star = J_{bb}^{-1} \quad \text{for every } b \in \mathfrak{B}. \tag{15.4}$$

The optimal recognition mean is exact even though the recognition covariance is restricted. [ESTABLISHED]

Proof. On the block-diagonal set, $\mathrm{Tr}(JC_Q) = \sum_b \mathrm{Tr}(J_{bb}C_b)$, because only the diagonal blocks of J meet a block-diagonal C_Q under the trace, and $\log\det C_Q = \sum_b \log\det C_b$. The objective is therefore

$$D_{\mathrm{KL}} = \frac{1}{2}\left(\mu_Q - \mu\right)^\top J \left(\mu_Q - \mu\right) + \frac{1}{2}\sum_{b\in\mathfrak{B}} \left[\mathrm{Tr}\left(J_{bb}C_b\right) - \log\det C_b\right] + \text{const}, \tag{15.5}$$

a sum of a strictly convex quadratic in μ_Q, since $J \succ 0$, and one term per block. Each block term is strictly convex on $\mathbb{S}_{++}^{n_b}$, since $C_b \mapsto \mathrm{Tr}(J_{bb}C_b)$ is linear and $C_b \mapsto -\log\det C_b$ is strictly convex on the positive definite cone [18, §3.1.5]. The parameter set is convex, being $\mathbb{R}^n$ times a product of positive definite cones, so the minimizer is unique if it exists. Differentiating on the open cone gives

$$\nabla_{\mu_Q} D_{\mathrm{KL}} = J\left(\mu_Q - \mu\right), \qquad \nabla_{C_b} D_{\mathrm{KL}} = \tfrac{1}{2}\left(J_{bb} - C_b^{-1}\right), \tag{15.6}$$

whose simultaneous zero is Equation (15.4), admissible because $J_{bb} \succ 0$ as a principal submatrix of a positive definite matrix. □

The same optimum arrives from coordinate ascent, which is worth recording because the two derivations fail in different ways and agreeing is informative. Holding all factors but b fixed, the exact coordinate maximizer of the bound is Gaussian with covariance J_{bb}^{-1} and mean

$$\mu_b^{\mathrm{new}} = J_{bb}^{-1}\left(h_b - \sum_{c\neq b} J_{bc}\mu_c\right), \tag{15.7}$$

which is Equation (14.5) read as an update rule. Any fixed point of the full sweep satisfies $J\mu_Q = h$, hence $\mu_Q = \mu$ [16, 17]. The covariance is pinned in one step and the mean only at a fixed point, which is the standard asymmetry of Gaussian coordinate ascent.

Theorem 15.3 (Determinant gap). Let $\mathcal{L}_{\mathfrak{B}}^\star$ denote the bound at the optimum of Theorem 15.2. Then

$$\log p_\theta(o \mid X) - \mathcal{L}_{\mathfrak{B}}^\star = \frac{1}{2}\left[\sum_{b\in\mathfrak{B}} \log\det J_{bb} - \log\det J\right] \geq 0, \tag{15.8}$$

with equality if and only if $J_{bc} = 0$ for every pair of distinct blocks of $\mathfrak{B}$. [ESTABLISHED]

Proof. At the optimum, $\mathrm{Tr}(JC_Q^\star) = \sum_b \mathrm{Tr}(J_{bb}J_{bb}^{-1}) = \sum_b n_b = n$ and $\log\det C_Q^\star = -\sum_b \log\det J_{bb}$, while the quadratic mean term vanishes. Substituting into Equation (15.3) leaves

$$\inf_{Q\in\mathcal{Q}_{\mathfrak{B}}} D_{\mathrm{KL}}\left(Q\|P^\star\right) = \frac{1}{2}\left[\sum_{b\in\mathfrak{B}} \log\det J_{bb} - \log\det J\right], \tag{15.9}$$

and Equation (5.23) converts this into Equation (15.8).

For nonnegativity and the equality condition, two lemmas suffice. The first is that $0 \prec S \preceq T$ implies $\det S \le \det T$, with equality only when $S = T$. To see it, set $R := T^{-1/2}ST^{-1/2}$, which is symmetric with eigenvalues in $(0,1]$ and with $\det R = \det S/\det T$. That ratio is a product of numbers in $(0,1]$, hence at most one, and it equals one only when every eigenvalue equals one, which for a symmetric matrix forces $R = I$ and so $S = T$. The second lemma is that, for $J \succ 0$ partitioned into a first block and its complement,

$$\det J = \det J_{11}\cdot\det\left(J_{22} - J_{21}J_{11}^{-1}J_{12}\right), \tag{15.10}$$

with the Schur complement positive definite [80]. Since $J_{21}J_{11}^{-1}J_{12} \succeq 0$, that Schur complement is $\preceq J_{22}$, so the first lemma gives $\det J \le \det J_{11}\det J_{22}$, with equality if and only if $J_{21}J_{11}^{-1}J_{12} = 0$, which since $J_{11}^{-1} \succ 0$ holds if and only if $J_{11}^{-1/2}J_{12} = 0$, that is if and only if $J_{12} = 0$. Applying this with the first block of $\mathfrak{B}$ against the union of the rest, and then inducting on the number of blocks, gives Equation (15.8) and the stated equality condition. This is Fischer's determinant inequality for positive definite matrices [34, §7.8]. □

Canonical-correlation form and telescoping identity. For a two-block precision, let $A \in \mathbb{S}^p_{++}$, $D \in \mathbb{S}^q_{++}$, and $C \in \mathbb{R}^{p\times q}$, and write

$$J = \begin{pmatrix} A & C \\ C^\top & D \end{pmatrix}, \qquad A = L_AL_A^\top, \qquad D = L_DL_D^\top, \qquad R := L_A^{-1}CL_D^{-\top}. \tag{15.11}$$

Both diagonal blocks are positive definite. The Schur complement factorizes as

$$D - C^\top A^{-1}C = L_D\left(I_q - R^\top R\right)L_D^\top, \qquad \det J = (\det A)(\det D)\det\left(I_q - R^\top R\right). \tag{15.12}$$

If ρ_a are the singular values of R, positive definiteness of the Schur complement gives $0 \le \rho_a < 1$, and the two-block increment is exactly

$$\Delta(A,D;C) = -\frac{1}{2}\log\det\left(I_q - R^\top R\right) = -\frac{1}{2}\sum_{a=1}^{\min(p,q)}\log\left(1-\rho_a^2\right). \tag{15.13}$$

For ordered blocks $B_1,\dots,B_m$, define the cumulative coordinate sets $U_k := \bigcup_{j=1}^k B_j$. The left-fold increment at step $k \ge 2$ is

$$\delta_k = \frac{1}{2}\left[\log\det J_{U_{k-1}U_{k-1}} + \log\det J_{B_kB_k} - \log\det J_{U_kU_k}\right]. \tag{15.14}$$

Consequently the intermediate cumulative determinants cancel exactly:

$$\sum_{k=2}^{m} \delta_k = \frac{1}{2}\left[\sum_{k=1}^{m} \log\det J_{B_kB_k} - \log\det J\right]. \tag{15.15}$$

The total is order independent in exact arithmetic; the increments and the finite-precision trajectory can depend on the declared order. [ESTABLISHED]

Operational stable realization. The ordinary numerical path computes L_A and L_D by Cholesky factorization, obtains R by triangular solves, computes its singular values, and evaluates each summand as `-0.5*log1p(-rho**2)`. It forms neither A^{-1} nor D^{-1} and does not subtract large floating-point log determinants. Validation and ordinary linear algebra use exact power-of-two normalization and range-safe residuals. If that scaling loses a nonzero entry, or a binary64 Cholesky, solve, SVD, or condition estimate is unusable, an adaptive-precision exact-dyadic Cholesky calculation certifies positive definiteness and supplies the log-determinant gap. A separate boundary policy handles a squared singular value within an operational $64n\epsilon_{\mathrm{mach}}$ neighborhood of one. There the exact stored binary64 entries are reevaluated at a minimum of 200 decimal digits, including the determinant-gap value; a second binary64 solve backend makes the diagnostic the maximum excursion observed across the two declared binary64 backends. The code rejects an exact-input value outside the Schur domain and rejects a raw binary64 excursion above one when its discrepancy exceeds that operational guard. It also uses the exact-input value path when $\kappa_2(J)\epsilon_{\mathrm{mach}} \geq 10^{-4}$, an explicitly operational conditioning trigger selected for the frozen protocol rather than asserted as a universal error threshold. A nonfinite binary64 condition estimate is also a fallback trigger, not a rejection criterion. An ill-conditioned but in-domain value is reevaluated rather than rejected merely for triggering that path. An early high-precision fallback selected before residual evaluation records the Cholesky residual, solve residual, residual tolerance, and backward-error diagnostics as `null` because they were not evaluated. If the residual-health gate is reached, every finite measurement is retained. A field is `null` only when it has no finite usable numeric value, either because an attempted measurement is nonfinite or because a derived diagnostic requires an unavailable constituent; finite sibling fields remain numeric. The evaluation method distinguishes a finite threshold excursion from nonfinite Cholesky, solve, or tolerance diagnostics and lists multiple nonfinite causes in that fixed order. Thus `null` denotes neither failure nor zero, and the zero-step one-block case retains its exact zero diagnostics. The `GapStep` API retains this fieldwise quartet; the `FactorizationCaseRecord` protocol surface stores only the aggregate backward diagnostic, maximum Cholesky residual, and ordered evaluation-method tuple, not per-step solve or tolerance values. Despite the compatibility name `backward_error_bound`, the numeric quantity is a one-ULP-up summary of already rounded local residual measurements, not a certified perturbation or forward-error bound. These guarded fallbacks are declared finite-precision procedures, not theorems that Cholesky or solve residuals universally control canonical-correlation or gap forward error. [DEFINITION]

Proposition 15.4 (The optimum is dominated by the exact marginal). For every block b,

$$C_b^\star = J_{bb}^{-1} \preceq \left(J^{-1}\right)_{bb}, \tag{15.16}$$

with equality if and only if $J_{b,-b} = 0$. [ESTABLISHED]

Proof. By Equation (14.7), $(J^{-1})_{bb} = S_b^{-1}$ with $S_b = J_{bb} - J_{b,-b}J_{-b,-b}^{-1}J_{-b,b}$, and $S_b \preceq J_{bb}$ because the subtracted matrix is positive semidefinite. Inversion reverses the Loewner order on positive definite matrices: if $0 \prec S \preceq T$ then $T^{-1/2}ST^{-1/2} \preceq I$, so the eigenvalues of $T^{1/2}S^{-1}T^{1/2}$ are at least one, giving $S^{-1} \succeq T^{-1}$ [34, §7.7], [13, Ch. V]. Applying this to $S_b \preceq J_{bb}$ gives Equation (15.16). Equality forces $S_b = J_{bb}$, hence $J_{b,-b}J_{-b,-b}^{-1}J_{-b,b} = 0$, hence $J_{b,-b} = 0$ as in the proof of Theorem 15.3. □

Proposition 15.4 is the exact form of a statement usually made loosely, that a factorized reverse-relative-entropy approximation understates posterior uncertainty [16, 17, 70]. Here it is a Loewner inequality with a proof and an equality condition, and it holds blockwise rather than merely in some scalar summary. It also fixes what the restriction is and is not.

Conditional dependence, not marginal covariance. The family of Equation (15.2) constrains the recognition covariance, hence the recognition precision, to be block diagonal. By Proposition 14.2 that is a statement that the recognition law makes distinct blocks conditionally independent given the rest. It is not, and does not imply, the assertion that the target's marginal covariance blocks $(J^{-1})_{bc}$ vanish; (14.6) shows that those can be nonzero while $J_{bc} = 0$. The reverse-relative-entropy projection changes only the candidate Q: it sets $C_{bc}^\star = 0$ for $b \neq c$ and $C_{bb}^\star = J_{bb}^{-1} \preceq (J^{-1})_{bb}$. It neither leaves nor replaces any covariance block of the fixed target $P = \mathcal{N}(\mu, J^{-1})$. Reading the block restriction as a claim about the target's marginal correlation is a category error, and it is the error that makes the determinant gap look like a correlation-strength measurement rather than what Theorem 15.3 says it is, a coupling-structure measurement. [ESTABLISHED]

15.2 Observation independence, and its exact scope

Theorem 15.3 expresses the cost as a pure log-determinant functional of J. Whether that makes the cost independent of the realized observation depends on where J comes from, and the answer differs for the two kinds of restriction this document uses. Getting the difference right matters because a later chapter compares recognition structures across data and would draw the wrong conclusion from the wrong version.

Proposition 15.5 (Conditional precision is observation free). Suppose the joint law of (o, Y) given X is Gaussian with precision $\mathcal{J}$ and information vector η, both functions of (θ, X) alone. Then the exact conditional law of Y given o has precision and mean

$$J = \mathcal{J}_{YY}, \qquad \mu = J^{-1}\left(\eta_Y - \mathcal{J}_{Yo}o\right), \tag{15.17}$$

so J does not depend on the realized value of o while μ is an affine function of it. [ESTABLISHED]

Proof. Read the exponent of the joint density as a function of y at fixed o. The quadratic part is $-\frac{1}{2}y^\top \mathcal{J}_{YY}y$ and the linear part is $(\eta_Y - \mathcal{J}_{Yo}o)^\top y$, so by Equation (14.1) the conditional law is Gaussian in information form with the displayed natural parameters, and $\mu = J^{-1}h$. □

Corollary 15.6 (Observation independence of the determinant gap). Under the hypothesis of Proposition 15.5, the gap of Theorem 15.3 is a function of (θ, X) and the declared partition alone, and does not depend on the realized observation. [ESTABLISHED]

Proof. The gap depends on Q and o only through J, by Theorem 15.3, and through nothing else, because the mean sector contributes exactly zero at the optimum by Theorem 15.2. Proposition 15.5 makes J observation free. □

Two hypotheses are doing the work in Corollary 15.6 and both are load bearing. The first is joint Gaussianity with an observation-independent joint precision, which excludes, for instance, a model whose observation noise covariance is itself a function of the realized observation; without it Equation (15.17) fails and J carries data. The second, and the one more easily overlooked, is that the restriction is a *factorization*, so that the optimal recognition mean coincides with the target mean and the quadratic term in Equation (15.3) vanishes identically at the optimum. A restriction that constrains the mean does not have that property, and the next proposition computes exactly what survives.

Proposition 15.7 (Cost of a mean restriction). Let $[B\ B_\perp]$ be an orthogonal matrix with $B \in \mathbb{R}^{n\times r}$, and restrict the recognition family to

$$\mathcal{Q}_B = \left\{\mathcal{N}\left(\mu_Q, C_Q\right) : \mu_Q \in \mathrm{range}(B), \quad C_Q \in \mathbb{S}^n_{++}\right\}, \tag{15.18}$$

with the covariance unrestricted. Writing $\mu_\perp := B_\perp^\top \mu$ for the component of the target mean orthogonal to the retained subspace, the exact minimum of the relative entropy over $\mathcal{Q}_B$ is

$$\inf_{Q\in\mathcal{Q}_B} D_{\mathrm{KL}}\left(Q \| P^\star\right) = \frac{1}{2}\mu_\perp^\top \left(B_\perp^\top J^{-1} B_\perp\right)^{-1} \mu_\perp, \tag{15.19}$$

attained at $C_Q^\star = J^{-1}$ and $\mu_Q^\star = \mu - J^{-1}B_\perp(B_\perp^\top J^{-1}B_\perp)^{-1}\mu_\perp$. The matrix $(B_\perp^\top J^{-1}B_\perp)^{-1}$ is the *marginal* precision of the restricted directions, that is the inverse of their exact marginal covariance. [ESTABLISHED]

Proof. By the separation noted after Equation (15.3), the covariance sector is unconstrained and its minimum is attained at $C_Q^\star = J^{-1}$, where it contributes zero. What remains is to minimize $\frac{1}{2}(\mu_Q - \mu)^\top J(\mu_Q - \mu)$ subject to $B_\perp^\top \mu_Q = 0$. Put $v = \mu_Q - \mu$, so the constraint reads $B_\perp^\top v = -\mu_\perp$. Stationarity of the Lagrangian gives $Jv = B_\perp \lambda$, hence $v = J^{-1}B_\perp\lambda$; imposing the constraint gives $\lambda = -(B_\perp^\top J^{-1}B_\perp)^{-1}\mu_\perp$, which is well defined because $B_\perp$ has full column rank and $J^{-1} \succ 0$. The objective at that point is $\frac{1}{2}v^\top Jv = \frac{1}{2}\lambda^\top(B_\perp^\top J^{-1}B_\perp)\lambda$, which is Equation (15.19), and the minimizer is as displayed. The final identification holds because $\mathrm{Cov}(B_\perp^\top Y) = B_\perp^\top J^{-1}B_\perp$ under $P^\star$. □

Corollary 15.8 (A mean restriction carries the data). Under the hypothesis of Proposition 15.5, the cost in Equation (15.19) is a quadratic function of the realized observation, since $\mu_\perp = B_\perp^\top J^{-1}(\eta_Y - \mathcal{J}_{Yo}o)$ is affine in o. It is observation free if and only if $B_\perp^\top J^{-1}\mathcal{J}_{Yo} = 0$. Thus the observation independence of Corollary 15.6 does not extend to restrictions of the mean, and a construction that reports one cost as though it were the other is reporting a data-dependent quantity as a structural one. [ESTABLISHED]

Proposition 15.9 (Marginal precision against restricted precision). With $[B\ B_\perp]$ as above,

$$B_\perp^\top JB_\perp - \left(B_\perp^\top J^{-1}B_\perp\right)^{-1} = B_\perp^\top JB\left(B^\top JB\right)^{-1}B^\top JB_\perp \succeq 0, \tag{15.20}$$

so the marginal precision of the restricted directions is dominated in the Loewner order by the restriction $B_\perp^\top JB_\perp$ of the precision to those directions, the difference has rank at most r, and the two coincide if and only if $B^\top JB_\perp = 0$, that is if and only if the retained and restricted subspaces are J-orthogonal. [ESTABLISHED]

Proof. Since $[B\ B_\perp]$ is orthogonal, conjugating by it gives $\widetilde{J} := [B\ B_\perp]^\top J[B\ B_\perp]$ and also $[B\ B_\perp]^\top J^{-1}[B\ B_\perp] = \widetilde{J}^{-1}$. Hence $B_\perp^\top J^{-1}B_\perp$ is the lower right block of $\widetilde{J}^{-1}$, and by the block inversion formula its inverse is the Schur complement

$$\widetilde{J}_{\perp\perp} - \widetilde{J}_{\perp\|}\widetilde{J}_{\|\|}^{-1}\widetilde{J}_{\|\perp}, \tag{15.21}$$

where $\|$ and $\perp$ index the two groups of columns [80]. Subtracting Equation (15.21) from $\widetilde{J}_{\perp\perp} = B_\perp^\top JB_\perp$ leaves exactly the displayed product, which is positive semidefinite and of rank at most $\mathrm{rank}(B^\top JB_\perp) \le r$. It vanishes if and only if $(B^\top JB)^{-1/2}B^\top JB_\perp = 0$, that is if and only if $B^\top JB_\perp = 0$. □

Proposition 15.9 is the reason Equation (15.19) must be written with the marginal precision and not with the restriction of the precision. The two are different matrices, ordered but not equal, and substituting one for the other overstates the cost of a mean restriction by the rank-r correction of Equation (15.20) except in the J-orthogonal case. The deterministic replacement check `CHK-RESTRICTION-SCHUR` independently reconstructs the block-inversion and constrained-optimization endpoints and includes the orthogonal equality control. It corroborates the proved identities and is not evidence for them. [NUMERICAL]

Corollary 15.10 (Costs of combined restrictions add). Restricting the mean to $\mathrm{range}(B)$ and the covariance to the block-diagonal set of Equation (15.2) simultaneously has exact cost

$$\frac{1}{2}\left[\sum_{b\in\mathfrak{B}}\log\det J_{bb} - \log\det J\right] + \frac{1}{2}\mu_\perp^\top\left(B_\perp^\top J^{-1}B_\perp\right)^{-1}\mu_\perp, \tag{15.22}$$

the first term structural and the second data dependent. [ESTABLISHED]

Proof. Equation (15.3) has no term coupling μ_Q to C_Q, and the two constraint sets act on disjoint parameter blocks, so the minimizations are independent and their values add. Each value is supplied by Theorem 15.3 and Proposition 15.7 respectively. □

15.3 Comparing recognition structures at fixed model

Equation (15.8) assigns a number to a partition, which invites the question of whether one recognition structure is better than another. The question is well posed only under conditions that are easy to violate, and this section states both the affirmative result and its boundary.

Proposition 15.11 (Refinement monotonicity for every comparable pair). Let $\mathfrak{B}$ and $\mathfrak{B}'$ be any two partitions such that $\mathfrak{B}'$ refines $\mathfrak{B}$, meaning every block of $\mathfrak{B}'$ is contained in a block of $\mathfrak{B}$. Then $\mathcal{Q}_{\mathfrak{B}'} \subseteq \mathcal{Q}_{\mathfrak{B}}$ and

$$\mathcal{L}^{\star}_{\mathfrak{B}} - \mathcal{L}^{\star}_{\mathfrak{B}'} = \sum_{b \in \mathfrak{B}} \frac{1}{2} \left[\sum_{b' \in \mathfrak{B}' :\, b' \subseteq b} \log\det J_{b'b'} - \log\det J_{bb} \right] \geq 0, \tag{15.23}$$

each summand being the determinant gap of the positive definite matrix J_{bb} under the partition induced on b, and vanishing exactly when J_{bb} is block diagonal under that induced partition. Thus the coarser structure dominates the finer one for every comparable pair in the refinement partial order. The mechanism is the Loewner ordering of Theorem 15.3: within each coarse block, a Schur complement is dominated by the corresponding diagonal block. [ESTABLISHED]

The selected hierarchy invokes this proposition only on its declared edges or, by composition, on comparable nodes of its declared chain or rooted tree. It imposes no ordering on nonnested partitions. [DEFINITION]

Proof. A block-diagonal matrix under $\mathfrak{B}'$ is block diagonal under $\mathfrak{B}$ whenever $\mathfrak{B}'$ refines $\mathfrak{B}$, giving the inclusion of families. Subtracting two instances of Equation (15.8) cancels $\log\det J$ and regroups the remaining terms by coarse block, giving Equation (15.23). Each bracket is Theorem 15.3 applied to $J_{bb} \succ 0$ with the induced partition, hence nonnegative with the stated equality condition. □

Proposition 15.12 (Non-nested structures are not ordered). Refinement is a partial order, and outside it the bounds are not ordered by any structural feature such as the number or the sizes of the blocks. For $n = 3$ take

$$J(a,b) = \begin{pmatrix} 1 & a & 0 \\ a & 1 & b \\ 0 & b & 1 \end{pmatrix}, \qquad a^2 + b^2 < 1, \tag{15.24}$$

which is positive definite, and compare the two partitions $\mathfrak{B}_1 = \{\{1,2\},\{3\}\}$ and $\mathfrak{B}_2 = \{\{1\},\{2,3\}\}$, which have identical block-size multisets and neither of which refines the other. Their gaps are $\frac{1}{2}\log\frac{1-a^2}{1-a^2-b^2}$ and $\frac{1}{2}\log\frac{1-b^2}{1-a^2-b^2}$, so the ordering is decided by whether $|a|$ exceeds $|b|$ and reverses when they are exchanged. At $(a,b) = (0.8, 0.2)$ the values are 0.0589 and 0.5493; at $(0.2, 0.8)$ they are exchanged. [ESTABLISHED]

Proof. Positive definiteness follows from the leading minors 1, $1-a^2$ and $1-a^2-b^2$. The gaps follow from Equation (15.8) with $\det J = 1-a^2-b^2$, $\det J_{\{1,2\}\{1,2\}} = 1-a^2$, $\det J_{\{2,3\}\{2,3\}} = 1-b^2$ and the remaining diagonal entries equal to one. The numerical values are $\frac{1}{2}\log(0.36/0.32)$ and $\frac{1}{2}\log(0.96/0.32)$. $\square$

Proposition 15.12 says that a recognition structure is well chosen relative to a given coupling matrix, not well chosen in the abstract, and it is what makes any partition-selection rule an optimization over structures rather than a reading off of a scalar. It also disposes of the temptation to treat block count as a proxy for quality.

Proposition 15.13 (The gap does not select a structure). Along any declared nested refinement chain containing the one-block partition, the gap of Equation (15.8) is nonincreasing under coarsening by Proposition 15.11, and it attains its minimum value zero at that one-block partition for every $J \succ 0$. This monotonicity supplies no ordering of nonnested partitions. Minimizing the gap therefore selects the unrestricted family and returns no information about structure. [ESTABLISHED]

Proof. For the one-block partition, $\sum_b \log\det J_{bb} = \log\det J$ and the gap is zero; nonnegativity is Theorem 15.3, while monotonicity holds along the declared chain by Proposition 15.11. Proposition 15.12 supplies the nonnested boundary. $\square$

Proposition 15.14 (ELBO values do not order unrelated joints). Let P_θ and $P_{\theta'}$ be normalized joints over the same observation but possibly different latent inventories, with recognition families $\mathcal{Q}$ and $\mathcal{Q}'$ on their respective latent spaces. Assume the selected o belongs to the regular-observation set of each joint, or that pointwise versions have been declared for both. If no evidence-preserving channel relating the two joints is given, then

$$\mathcal{L}^\star - \mathcal{L}'^\star = [\log p_\theta(o \mid X) - \log p_{\theta'}(o \mid X)] - \left[\inf_{\mathcal{Q}} D_{\mathrm{KL}} - \inf_{\mathcal{Q}'} D_{\mathrm{KL}}\right], \tag{15.25}$$

which combines an evidence difference, a model-comparison quantity, with a tightness difference, a recognition-quality quantity. Neither bracket is identifiable from the value of the left side, so the sign of that value orders neither the models nor the recognition families. [ESTABLISHED]

Proof. Apply Equation (5.24) to each model and subtract. To show that both failures occur within normalized models, let $\ell \le 0$, $g \ge 0$, take $O \in \{o, \bar{o}\}$, $Y \in \{0,1\}$, and define a normalized joint by

$$P_{\ell,g}(O=o, Y=1) = e^{\ell-g}, \qquad P_{\ell,g}(O=o, Y=0) = e^{\ell}(1-e^{-g}), \qquad P_{\ell,g}(O=\bar{o}, Y=0) = 1-e^{\ell},$$

with the remaining atom assigned mass zero. Then

$$\log P_{\ell,g}(O=o)=\ell, \qquad P_{\ell,g}(Y=1\mid O=o)=e^{-g}.$$

For the singleton recognition family $\mathcal{Q}=\{\delta_1\}$, the minimized gap is g, so the optimized bound is $\ell-g$. To make the two latent inventories literally different, append an unused deterministic binary coordinate $Z=0$ to either second model and replace its recognition law by $\delta_1\otimes\delta_0$; taking a product with this point mass changes neither normalization, evidence, relative entropy, nor the optimized ELBO. The pairs

$$(\ell_1,g_1)=(-1,0), \quad (\ell_2,g_2)=(0,2)$$

give $\mathcal{L}_1^\star=-1>\mathcal{L}_2^\star=-2$ although the evidence ordering is $\ell_1<\ell_2$. The pairs

$$(\ell_1,g_1)=(0,1), \quad (\ell_2,g_2)=(-2,0)$$

again give $\mathcal{L}_1^\star=-1>\mathcal{L}_2^\star=-2$, although the first recognition family has the larger gap and hence the less tight bound. Thus a bound ordering determines neither bracket in Equation (15.25). □

Proposition 15.14 is a statement about unrelated joints, not about latent inventory by itself. If a normalized Markov kernel $K:x\rightsquigarrow z$ defines the coarse joint $\bar{P}=(\mathrm{id}_O\otimes K)_\# P$ and the coarse recognition law is $\bar{Q}=QK$, then the observation marginal and evidence are unchanged and data processing gives

$$\bar{\mathcal{L}}(\bar{Q};o)=\log p(o)-D_{\mathrm{KL}}(QK\|P_oK)\geq \log p(o)-D_{\mathrm{KL}}(Q\|P_o)=\mathcal{L}(Q;o).$$

Thus different latent spaces can carry ordered bounds when both laws are related by the same evidence-preserving channel. Absent such a relation, an ELBO difference mixes the evidence difference with the tightness difference as in (15.25). The general hypotheses and equality conditions are stated in Chapter 10. [ESTABLISHED]

Open Problem 15.15 (An intrinsic criterion). Propositions 15.13 and 15.14 leave a gap that this document does not close. There is no criterion here, internal to a single evidence bound at a fixed model, that selects a nondegenerate recognition structure: the determinant gap is minimized at the trivial partition, the mean-restriction cost of Equation (15.19) is a cost at fixed structure, and an arbitrary replacement joint supplies no common evidence. Evidence-preserving Markov pushforwards are the important exception, but their data-processing order still does not select a nondegenerate partition. What would settle the problem positively is a functional on admissible channels or partitions, expressible relative to one fixed joint and not monotone to an endpoint, whose optimizer is nondegenerate. A negative result would have to prove endpoint degeneracy for a precisely stated class of functionals. Absent either, the choice of structure remains a model-selection or RG-scheme declaration. [OPEN]

Chapter 16

Gaussian Information Geometry

Chapter 6 distinguishes general law fibers from smooth statistical manifolds, regular exponential families, and the multivariate-Gaussian realization. This chapter works only in the last tier. It identifies the dual Gaussian charts, computes their Fisher metrics, and separates frame-invariant generalized spectral data from ordinary chart-dependent eigenvalues. It does not assert that an RG flow contracts any information-geometric quantity.

16.1 Canonical and expectation coordinates

Use the nondegenerate Gaussian family and log normalizer of Proposition 14.1. For the stacked latent $Y \in \mathbb{R}^n$, $n = NK$, write

$$q(y) = \exp\left(h^\top y - \tfrac{1}{2} y^\top J y - \mathsf{A}(h, J)\right), \qquad h \in \mathbb{R}^n, \quad J \in \mathbb{S}^n_{++}, \tag{16.1}$$

where

$$\mathsf{A}(h, J) = \tfrac{1}{2} h^\top J^{-1} h - \tfrac{1}{2} \log\det J + \tfrac{n}{2} \log 2\pi. \tag{16.2}$$

For sufficient statistic $T(y) = (y, yy^\top)$, the canonical natural and expectation coordinates are

$$\eta = \left(h, -\tfrac{1}{2} J\right), \tag{16.3}$$

$$\tau = \nabla_\eta \mathsf{A} = (\mu, M_2), \qquad M_2 = C + \mu\mu^\top = \mathbb{E}_q[YY^\top], \tag{16.4}$$

The information-form chart (h, J) ranges over $\mathbb{R}^n \times \mathbb{S}^n_{++}$, whereas the canonical matrix coordinate $-J/2$ is negative definite. The family is regular and minimal, so the Legendre map is a bijection on its natural domain [19]. [ESTABLISHED]

The Fisher matrix in natural coordinates is the Hessian of the potential and equals the covariance of the sufficient statistic,

$$g_\eta = \nabla^2_\eta \mathsf{A}(\eta) = \mathrm{Cov}_q\left(T(Y)\right), \tag{16.5}$$

which is the standard exponential-family fact [2, 3, 19]. Because $\tau = \nabla_\eta \mathsf{A}$, the Jacobian of the coordinate change is $\partial\tau/\partial\eta = g_\eta$, and the Fisher metric expressed in the expectation coordinates is

$$g_\tau = \left(\frac{\partial\eta}{\partial\tau}\right)^\top g_\eta \left(\frac{\partial\eta}{\partial\tau}\right) = g_\eta^{-1}, \tag{16.6}$$

the Hessian of the convex conjugate $\mathsf{A}^*(\tau) = \sup_\eta(\langle\eta,\tau\rangle - \mathsf{A}(\eta))$, which for this family is the negative differential entropy of q [3, 58]. [ESTABLISHED]

Four coordinate presentations must be kept apart. The canonical natural chart is $\eta = (h, -J/2)$; the affinely equivalent information-form chart is (h, J); the expectation chart is $\tau = (\mu, M_2)$; and the moment chart is (μ, C). The last is useful for sampling and covariance transport but is not Legendre dual to the natural chart, because $C = M_2 - \mu\mu^\top$. [DEFINITION]

Symmetric-matrix coordinates carry one further convention. Derivatives in the matrix argument are taken with respect to the Frobenius pairing $\langle X, Z\rangle = \mathrm{Tr}(X^\top Z)$ on the space of symmetric matrices. Concretely this is the chart whose coordinates are the diagonal entries together with the off-diagonal entries scaled by $\sqrt{2}$, for which the Euclidean inner product of coordinate vectors equals the Frobenius pairing of the matrices they represent. With the unscaled half-vectorization instead, every off-diagonal coordinate derivative acquires a factor of two and the matrix representing (16.6) changes accordingly. All matrix blocks displayed below are in the Frobenius-paired chart. [DEFINITION]

When the recognition precision has the interaction form of Chapter 14, $J = \Lambda$, with $\Lambda_{ii} = A_i + \sum_{j\neq i} W_{ij}$, $\Lambda_{ij} = -W_{ij}$, and $C = \Lambda^{-1}$. Below, Λ emphasizes the interaction operator and J the natural parameter; they are the same matrix.

16.2 The chart correction

It is often asserted that the mean sector of the Fisher metric of a Gaussian family *is* the precision. The assertion is true for the mean block in the moment chart. In the expectation chart it is false for the restriction to the coordinate mean directions, while the metric induced on the quotient by the second-moment directions is again exactly the precision. Thus "the mean sector" is ambiguous in the expectation chart until restriction or quotient is named. Both readings are computed below.

Proposition 16.1 (Fisher metric in the moment chart). *In the chart (μ, C) the Fisher metric of the Gaussian family is block diagonal, and for tangent vectors (u, A) and (v, B) with $u, v \in \mathbb{R}^n$ and A, B symmetric,*

$$g_{(\mu,C)}\big((u,A),(v,B)\big) = u^\top \Lambda v + \tfrac{1}{2}\,\mathrm{Tr}\,(\Lambda A \Lambda B)\,, \qquad \Lambda = C^{-1}. \tag{16.7}$$

[ESTABLISHED]

Proof. Write $e = y - \mu$ and $\log q(y) = -\frac{1}{2}e^\top\Lambda e + \frac{1}{2}\log\det\Lambda - \frac{n}{2}\log 2\pi$. A displacement (u, A) in the moment chart moves μ by u and C by A, hence Λ by $-\Lambda A\Lambda$, and the directional score is

$$\partial_{(u,A)}\log q = u^\top\Lambda e + \tfrac{1}{2}e^\top\Lambda A\Lambda e - \tfrac{1}{2}\,\mathrm{Tr}(\Lambda A). \tag{16.8}$$

Under $e \sim \mathcal{N}(0, C)$ the first term has second moment $u^\top\Lambda C\Lambda u = u^\top\Lambda u$. The cross term between the linear and quadratic pieces is a third central moment of a Gaussian and vanishes, which is exactly why the metric is block diagonal in this chart. Setting $S = \Lambda A \Lambda$, the remaining piece is $\frac{1}{4}\operatorname{Var}(e^\top S e) = \frac{1}{2}\operatorname{Tr}(SCSC) = \frac{1}{2}\operatorname{Tr}(\Lambda A \Lambda A)$. Polarization gives (16.7). $\square$

Expression (16.7) is the classical Fisher–Rao metric of the multivariate normal model; it is the metric whose geodesic equations and Rao distance are studied by Skovgaard [67] and by Atkinson and Mitchell [5], and it appears in the survey of Pinele et al. [64] as their equation (7), in the general reparameterized form $g_{ij} = \partial_i \mu^\top \Sigma^{-1} \partial_j \mu + \frac{1}{2}\operatorname{Tr}\left(\Sigma^{-1}\partial_i \Sigma \Sigma^{-1} \partial_j \Sigma\right)$. The printed equation in that reference carries a typographical slip, repeating the index i in place of j in the second factor of the trace; the intended and correct expression is the one displayed here, and it agrees with (16.7) on setting $\theta = (\mu, C)$. That reference also records, as its equation (8), that the affine action $(\mu, \Sigma) \mapsto (Q\mu + c, Q\Sigma Q^\top)$ is an isometry of this metric for every $Q \in \mathrm{GL}_n(\mathbb{R})$ and $c \in \mathbb{R}^n$, a fact used again in Section 16.4.

Proposition 16.2 (Fisher metric in the expectation chart). *In the chart (μ, M_2) with $M_2 = C + \mu\mu^\top$, the Fisher metric has mean block*

$$g_{[\mu\mu]} = \left(1 + \mu^\top \Lambda \mu\right) \Lambda + \Lambda \mu \mu^\top \Lambda, \tag{16.9}$$

cross block

$$g_{[\mu M_2]}(u, B) = -u^\top \Lambda B \Lambda \mu, \tag{16.10}$$

and second block $g_{[M_2 M_2]}(A, B) = \frac{1}{2}\operatorname{Tr}(\Lambda A \Lambda B)$, unchanged from (16.7). Hence the metric is not block diagonal in this chart unless $\mu = 0$. [ESTABLISHED]

Proof. The chart change is $C = M_2 - \mu\mu^\top$, so a displacement (u, B) in the expectation chart induces

$$\delta\mu = u, \qquad \delta C = B - \left(u\mu^\top + \mu u^\top\right), \tag{16.11}$$

and the metric in the new chart is the pullback of (16.7) along (16.11). For the mean block take $B = 0$, so $\delta C = -(u\mu^\top + \mu u^\top)$. Then

$$\begin{aligned} \operatorname{Tr}\left(\Lambda \delta C \Lambda \delta C\right) &= \operatorname{Tr}\left[\left(\Lambda u \mu^\top + \Lambda \mu u^\top\right)\left(\Lambda u \mu^\top + \Lambda \mu u^\top\right)\right] \\ &= 2\left(u^\top \Lambda \mu\right)^2 + 2\left(\mu^\top \Lambda \mu\right)\left(u^\top \Lambda u\right), \end{aligned} \tag{16.12}$$

each of the four traces being evaluated by cycling. Adding the mean contribution $u^\top \Lambda u$ from (16.7) and halving (16.12) gives the quadratic form

$$u^\top \Lambda u + \left(\mu^\top \Lambda \mu\right)\left(u^\top \Lambda u\right) + \left(u^\top \Lambda \mu\right)^2 = u^\top \left[\left(1 + \mu^\top \Lambda \mu\right)\Lambda + \Lambda \mu \mu^\top \Lambda\right] u, \tag{16.13}$$

which is (16.9). For the cross block pair $(u, 0)$ with $(0, B)$: the mean parts do not meet, and the covariance parts contribute $\frac{1}{2}\operatorname{Tr}\left(\Lambda\left[-(u\mu^\top + \mu u^\top)\right]\Lambda B\right) = -u^\top \Lambda B \Lambda \mu$, which is (16.10). For the second block take $u = 0$, so $\delta C = B$ and nothing changes. $\square$

Corollary 16.3 (Restriction and quotient mean sectors). *Decompose the expectation-chart tangent space as $\mathbb{R}^n \oplus \mathbb{S}^n$, with the second summand tangent to M_2. The metric induced on the quotient by the second-moment directions is the Schur complement*

$$g_\mu^{\text{quot}} = g_{[\mu\mu]} - g_{[\mu M_2]} g_{[M_2 M_2]}^{-1} g_{[M_2\mu]} = \Lambda. \tag{16.14}$$

Equivalently, it is the restriction of g_τ to the g_τ-orthogonal complement of the second-moment directions, expressed through the quotient coordinate μ. [ESTABLISHED]

Proof. The Fisher metric is positive definite on this minimal Gaussian family, so the second block and its Schur complement are invertible. The block-inverse identity gives

$$\left(g_\mu^{\text{quot}}\right)^{-1} = \left(g_\tau^{-1}\right)_{[hh]}.$$

By (16.6), $g_\tau^{-1} = g_\eta$, and its hh block is $\mathrm{Cov}_q(Y) = C = \Lambda^{-1}$ by (16.5). Inverting proves (16.14). The Schur complement is the standard formula for the metric on the orthogonal quotient. □

Corollary 16.4 (Size of the discrepancy and the exact equality case). *Write $v = \Lambda^{1/2}\mu$. Then*

$$g_{[\mu\mu]} = \Lambda^{1/2}\left[\left(1 + \|v\|^2\right) I_n + vv^\top\right]\Lambda^{1/2}, \tag{16.15}$$

so the eigenvalues of $\Lambda^{-1}g_{[\mu\mu]}$ are $1 + \mu^\top\Lambda\mu$ with multiplicity $n-1$ and $1 + 2\mu^\top\Lambda\mu$ with multiplicity one. Consequently $g_{[\mu\mu]} \succeq \Lambda$ always, and $g_{[\mu\mu]} = \Lambda$ if and only if $\mu = 0$. [ESTABLISHED]

Proof. Equation (16.15) is (16.9) rewritten. The bracket is $\left(1 + \|v\|^2\right) I + vv^\top$, whose eigenvalues are $1 + \|v\|^2 + \|v\|^2$ along v and $1 + \|v\|^2$ on $v^\perp$; since $\Lambda^{-1}g_{[\mu\mu]} = \Lambda^{-1/2}\left[\cdot\right]\Lambda^{1/2}$ is similar to the bracket, its spectrum is that list. For the equality case, $g_{[\mu\mu]} - \Lambda = \left(\mu^\top\Lambda\mu\right)\Lambda + \Lambda\mu\mu^\top\Lambda$ is a sum of two positive semidefinite matrices, so it vanishes only if both vanish; the first forces $\mu^\top\Lambda\mu = 0$ and hence $\mu = 0$ because $\Lambda \succ 0$. The converse is immediate. □

The discrepancy between the two *restriction* mean blocks is therefore governed by the squared Mahalanobis norm of the mean, $\mu^\top\Lambda\mu$, and it is unbounded. It is not a normalization constant and it does not disappear in any limit that keeps the mean away from the origin. This statement does not concern the quotient metric, which equals Λ exactly by Corollary 16.3.

The mechanism behind the discrepancy is worth stating plainly, because it is what makes the error easy to commit. Both charts call their first coordinate μ, but they do not have the same coordinate vector fields. The vector $\partial/\partial\mu$ at fixed C moves the mean while holding the covariance fixed, so M_2 changes; the vector $\partial/\partial\mu$ at fixed M_2 moves the mean and compensates in C. These are different tangent vectors at the same point of the manifold, and a metric assigns them different lengths. Writing $g_{[\mu\mu]}$ without saying which of the two is meant does not denote anything.

The same observation resolves the phrase "the mean sector" in the expectation chart. A metric tensor has a well defined restriction to a subspace of the tangent space only once that subspace is declared; what is not chart independent is the *complement*. In the moment chart the covariance directions are Fisher-orthogonal to the mean directions, by the vanishing third-moment argument in the proof of Proposition 16.1, so “the mean sector” can be taken to mean either the restriction to the mean directions or the orthogonal quotient, and the two agree. In the expectation chart (16.10) is nonzero, the mean and second-moment directions are not Fisher-orthogonal. The restriction is (16.9), whereas the orthogonal quotient is (16.14); the former differs from Λ away from $\mu = 0$, and the latter equals Λ at every point. Absent a declared reading, “the mean sector” in that chart names no unique object.

The chart discipline. A coordinate identity that mixes the mean and covariance sectors of a Gaussian family must name the chart and whether it uses restriction or quotient. Identity (16.7) holds in (μ, C); identities (16.9) and (16.10) give the expectation-chart restriction and cross blocks; and (16.14) gives its quotient mean metric. By Corollary 16.4 the two restriction mean blocks agree only at $\mu = 0$, while the quotient mean metric equals Λ everywhere. [ESTABLISHED]

Verification

The statements are proved above in general dimension; the machine checks reported here test those proofs and are not a substitute for them. Two symbolic routes and two numerical routes were run, all in exact or double precision on the CPU.

The first symbolic route is independent of Proposition 16.1: the Hessian of (16.2) was formed by computer algebra in the natural chart $(h, \mathrm{svec}(-J/2))$ with the Frobenius-paired symmetric coordinates, evaluated at exact rational (μ, Λ), inverted exactly as in (16.6), and its blocks compared with (16.9) and (16.10). At dimensions $n = 1, 2, 3$ the difference was the exact zero matrix in both blocks, and at $n = 2$ the cross block became exactly zero when $\mu = 0$ was substituted and was nonzero otherwise, as Proposition 16.2 requires. The second symbolic route carries general symbols: at dimensions $n = 2$ and $n = 3$, with μ a vector of free symbols and Λ a general symmetric matrix of free symbols, the pullback of (16.7) along (16.11) was simplified to zero difference against all three blocks of Proposition 16.2. Those two runs are proofs at those dimensions, not samples.

The deterministic numerical route `CHK-IG-EXPECTATION-METRIC` uses the closed-form Gaussian relative entropy and its Fisher quadratic limit in six directions at a fully specified base point. It compares the expectation-chart block with Equation (16.9), uses the moment-chart precision as a negative control against chart conflation, and checks the predicted generalized spectrum. The computation is reproducible corroboration, not a proof. [NUMERICAL]

16.3 Sample space and parameter space

The renormalization chapters will import vocabulary from a literature that computes Fisher metrics of likelihoods with respect to *model* parameters and reads their spectra as a scale [12]. The object developed above is not that object, and keeping the two typed apart is a precondition for importing anything at all.

Four spaces are in play. The latent sample space is $\mathbb{R}^n$, where Y takes its values; the interaction energy $y \mapsto \frac{1}{2}y^\top \Lambda y$ is a quadratic form on it. The recognition parameter manifold is the set of laws (16.1), coordinatized by any of the four charts of Section 16.1; the Fisher metric (16.7) is a Riemannian metric on it. The model parameter space is where θ lives and indexes the generative kernel. For a regular conditional model and a declared fixed input X, its conditional Fisher tensor is

$$I_X(\theta)(u,v) = \mathbb{E}_\theta\left[\partial_u \log p_\theta(O \mid X)\partial_v \log p_\theta(O \mid X) \;\middle|\; X\right]. \tag{16.16}$$

If inputs are randomized or a design ensemble is used, a declared probability law $r(dX)$ supplies the averaged tensor

$$I_r(\theta) = \int I_X(\theta) r(dX). \tag{16.17}$$

Different choices of X or r can change both the tensor and its rank. Only where the selected tensor is nondegenerate is it a Riemannian metric on the model parameter space; that space has the dimension of θ, which has no relation to $n = NK$ [1, 3]. Finally, the contextual base $\mathcal{C}$ indexes law-bundle sections. Once a connection and a statistical section are chosen, Definition 8.3 gives a generally semidefinite tensor on $T\mathcal{C}$; it is not any of the preceding three objects. [ESTABLISHED]

What makes the confusion easy is an exact but scoped component identity in this family, and stating it precisely is more useful than warning against it. The recognition family is a location family in its mean, so the mean parameter takes values in the same vector space $\mathbb{R}^n$ as the sample variable, and the mean block (16.7) of the parameter-space metric has the same components as the sample-space form Λ. The equality is of components in matched charts, not of objects: one is a metric on the tangent spaces of a manifold of probability laws, the other is a bilinear form on the latent vector space. For a family that is not a location family the two have different dimensions and no comparison arises. The identity is therefore a feature of the Gaussian mean sector and cannot be used as evidence that a construction defined on one space transfers to the other. [ESTABLISHED]

There is a corresponding exact but scoped relation to the interaction Laplacian L. If the transported endpoint laws on an edge (i, j) have means in $\mathbb{R}^K$ and share one fixed covariance $C_{ij} = J_{ij}^{-1} \in \mathbb{S}^K_{++}$ within the mean submodel, then

$$D_{\mathrm{KL}}\big(\mathcal{N}(\mu_i, C_{ij}) \| \mathcal{N}(\mu_j, C_{ij})\big) = \frac{1}{2}(\mu_i - \mu_j)^\top J_{ij}(\mu_i - \mu_j) = \frac{1}{2} g^{\mathrm{mean}}_{(\mu, C_{ij})}(\mu_i - \mu_j, \mu_i - \mu_j). \tag{16.18}$$

Thus pairwise KL is exactly one half of the associated mean-sector Fisher quadratic. Summing these typed edge quadratics defines the corresponding $NK \times NK$ connection-Laplacian energy; the assembled matrix L is not inserted into a single K-dimensional edge term. For a regular non-Gaussian family, freezing the Fisher tensor at one parallel base law gives the analogous form only as the local second-order term of transported KL [3]. Outside these Gaussian or frozen-local hypotheses, a sample-space L is not a recognition Fisher tensor or a natural-gradient operator. [ESTABLISHED]

For a smooth section of the full-Gaussian law bundle, the Fisher metric proved in Proposition 16.1 is the vertical tensor pulled back by the connection-relative first jet in Definition 8.3. The resulting base tensor depends on the section and chosen connection, and its rank is the rank seen by that first jet. The Gaussian component formulas do not identify the contextual base with the sample or parameter spaces. [ESTABLISHED]

The word "pullback" consequently has three typed uses in this manuscript. First, the connection-relative construction of Chapter 8 pulls a vertical statistical tensor back to the contextual base by $D^\omega s$. Second, the restriction below pulls a tensor back along an injective map of parameter manifolds. Third, the pullback attractor terminology in Section 11.4 describes backward asymptotic dependence for a nonautonomous cocycle; it is not a tensor operation. No theorem transfers between these uses merely because the same word appears. [DEFINITION]

The distinction has a sharp consequence for coarse-graining, which the next chapters will need. Restricting the recognition mean to a subspace and marginalizing onto that subspace are different operations and give different metrics.

Proposition 16.5 (Pullback versus pushforward). *Let $\Lambda \succ 0$ on $\mathbb{R}^n$, let $B \in \mathbb{R}^{n\times k}$ have orthonormal columns, and let $B_\perp$ span the orthogonal complement. The Fisher metric of the submodel obtained by constraining the mean to $\operatorname{range}(B)$ is the restriction $B^\top \Lambda B$; the Fisher metric of the mean of the pushforward law under $y \mapsto B^\top y$ is the marginal precision $\left(B^\top \Lambda^{-1} B\right)^{-1}$. They satisfy*

$$B^\top \Lambda B - \left(B^\top \Lambda^{-1} B\right)^{-1} = \left(B^\top \Lambda B_\perp\right)\left(B_\perp^\top \Lambda B_\perp\right)^{-1}\left(B_\perp^\top \Lambda B\right) \succeq 0, \tag{16.19}$$

with equality if and only if $B^\top \Lambda B_\perp = 0$, that is if and only if the retained and discarded subspaces are Λ-orthogonal. [ESTABLISHED]

Proof. The matrix $[B \quad B_\perp]$ is orthogonal, so in that basis Λ has blocks $\Lambda_{11} = B^\top \Lambda B$, $\Lambda_{12} = B^\top \Lambda B_\perp$, $\Lambda_{22} = B_\perp^\top \Lambda B_\perp$. The block inversion formula gives $B^\top \Lambda^{-1} B = \left(\Lambda_{11} - \Lambda_{12}\Lambda_{22}^{-1}\Lambda_{21}\right)^{-1}$ [34], and inverting again yields (16.19). The right side is a congruence of $\Lambda_{22}^{-1} \succ 0$ and is therefore positive semidefinite, and it vanishes exactly when $\Lambda_{12} = 0$. The identification of the two sides as pullback and pushforward metrics is the Gaussian conditional and marginal precision respectively [47]. □

Corollary 16.6 (Full-rank inclusion and averaging). *Let $S \in \mathbb{R}^{n\times k}$ have full column rank, set $M_S = S^\top S$, and let $S^\dagger = M_S^{-1} S^\top$. The mean submodel $\mu = S\bar{\mu}$ has Fisher matrix $S^\top \Lambda S$, while the mean of the pushforward under $y \mapsto S^\dagger y$ has Fisher matrix*

$$\Lambda_{\mathrm{av}} = M_S \left(S^\top \Lambda^{-1} S\right)^{-1} M_S. \tag{16.20}$$

Writing $B = S M_S^{-1/2}$, so that $B^\top B = I_k$, and completing B to an orthogonal matrix $[B\ B_\perp]$ gives

$$S^\top \Lambda S - \Lambda_{\mathrm{av}} = M_S^{1/2}\left[B^\top \Lambda B - \left(B^\top \Lambda^{-1} B\right)^{-1}\right] M_S^{1/2} \succeq 0. \tag{16.21}$$

Equality holds exactly when $B^\top \Lambda B_\perp = 0$, equivalently when $\operatorname{range}(S)$ and its Euclidean orthogonal complement are Λ-orthogonal. [ESTABLISHED]

Proof. Pulling the mean metric back along $\bar{\mu} \mapsto S\bar{\mu}$ gives $S^\top \Lambda S$. The pushforward covariance is

$$S^\dagger \Lambda^{-1} (S^\dagger)^\top = M_S^{-1} S^\top \Lambda^{-1} S M_S^{-1},$$

whose inverse is (16.20). Substituting $S = BM_S^{1/2}$ into both matrices yields (16.21), and Proposition 16.5 supplies its positivity and equality condition. □

Restriction is a pullback along an injective map of parameters, and it makes the metric larger in the Loewner order; marginalization is a pushforward along a map of the sample space, and it makes the metric smaller. The monotonicity theorems of information geometry govern the second operation, not the first, a point taken up in Section 16.5. The deterministic replacement check `CHK-IG-PULLBACK-PUSHFORWARD` tests the Schur identity and Loewner order over a specified ensemble and includes the Λ-orthogonal equality control. Computation is not proof and the proof is above. [NUMERICAL]

Whether the scale-setting construction of the model-space literature transfers to Λ is therefore not settled by observing that both are called Fisher metrics. The transfer requires a declaration that makes the coarse recognition mean a parameter of the generative law fitted against the bound, after which the relevant Fisher metric is the one with respect to that parameter and the import becomes a statement rather than an analogy. This section records the requirement and does not discharge it. What would close it is that declaration together with a check that the resulting parameter-space metric is nondegenerate, since the construction in that literature is written throughout with an inverted metric and does not treat the singular case. [OPEN]

16.3.1 Ambient score tangents and exponential action charts

A third distinction is needed before the renormalization chapters use the word "tangent". At a fixed Gaussian law three objects are in play, and they are nested rather than equal. The first is the finite-dimensional tangent space of the declared regular exponential family, whose metric is (16.5); its dimension is that of the chart (h, J). The second is the ambient centered score Hilbert space $L_0^2(\gamma)$ of (12.41), which contains every quadratic-mean score of every path through the law, whether or not that path stays in the Gaussian family. The third is the bounded action quotient $L^\infty/\mathbb{R}\mathbf{1}$, which is the domain of the nonlinear action calculus of Theorem 12.3. By Lemma 12.8 every element of the second space is an actual two-sided score, and by Proposition 12.11 the third embeds isometrically into the second with dense but generally proper range. The first is a finite-dimensional subspace of the second, of dimension $n + n(n+1)/2$ by (16.8), and is a proper subspace whenever $L_0^2(\gamma)$ is infinite dimensional, which holds for every nondegenerate Gaussian on $\mathbb{R}^n$. [DEFINITION]

The gap between the second and third objects is not a technicality, and the Hermite directions show exactly where it opens.

Proposition 16.7 (Exact two-sided domain of a Hermite exponential action). *Let $\gamma = N(0,1)$, let He_k be the probabilists' Hermite polynomial of degree $k \geq 1$, and set*

$$N_k(t) = \int e^{-t\mathrm{He}_k(x)}\gamma(dx) \in (0, +\infty]. \tag{16.22}$$

Then $N_1(t) < \infty$ for every real t; $N_2(t) < \infty$ exactly for $t > -\frac{1}{2}$, where $N_2(t) = e^t(1+2t)^{-1/2}$; for odd $k \geq 3$, $N_k(t) = +\infty$ for every $t \neq 0$; and for even $k \geq 4$, $N_k(t) < \infty$ exactly for $t \geq 0$. Consequently every He_k with $k \geq 1$ lies in $L^2_0(\gamma)$ and is realized as a two-sided quadratic-mean score, while for $k \geq 3$ no two-sided exponential action neighborhood of the origin contains it. [ESTABLISHED]

Proof. For $k = 1$ the integral is the Gaussian moment-generating function $e^{t^2/2}$, finite for every t. For $k = 2$, $\mathrm{He}_2(x) = x^2 - 1$, so $N_2(t) = e^t \int e^{-tx^2}\gamma(dx)$, and the Gaussian integral $\int e^{-tx^2}\gamma(dx)$ converges exactly when $t + \frac{1}{2} > 0$, with value $(1+2t)^{-1/2}$. For $k \geq 3$ write $\mathrm{He}_k(x) = x^k + O(x^{k-2})$, so the exponent of the integrand of (16.22) against the Gaussian density is $-t\mathrm{He}_k(x) - x^2/2$, whose leading behavior is $-tx^k$ because $k \geq 3$ dominates degree two. For odd k and any $t \neq 0$, exactly one of the two tails has $-tx^k \to +\infty$, so the integrand grows superpolynomially there and the integral diverges. For even k and $t < 0$, both tails have $-tx^k \to +\infty$ and the integral diverges; for $t > 0$ both tails have $-tx^k \to -\infty$, so the integrand is eventually bounded by e^{-cx^k} for some $c > 0$ and the integral converges; and $N_k(0) = 1$. Membership in $L^2_0(\gamma)$ holds because Hermite polynomials are square integrable against γ and orthogonal to the constants, and the realization is Lemma 12.8. ☐

The two Gaussian score directions of (16.8) sit inside the benign part of this classification. At $n = 1$, $\mu = 0$, and $\Lambda = 1$, that score reads $ux + \frac{A}{2}(x^2 - 1) = u\mathrm{He}_1 + \frac{A}{2}\mathrm{He}_2$, so the mean and variance directions occupy Hermite degrees one and two, and by Proposition 16.7 both admit two-sided exponential neighborhoods; the constraint $t > -\frac{1}{2}$ in the variance direction is the natural-domain boundary $J \succ 0$ seen through the exponential chart. Every ambient direction of degree three or higher behaves differently. A complete Hermite basis of $L^2_0(\gamma)$ is therefore not a chart of the Gaussian model and not a set of directions of the bounded action space. In particular the spectral statement of Theorem 12.12 is a theorem about the L^2_0 tangent, and it must not be read as a statement about the nonlinear bounded action map, whose domain is the third object above. [ESTABLISHED]

16.4 Gauge invariants of the interaction precision

A local reframing is a change of local trivialization of the fibers, and it changes the components of every object while changing no law. Let

$$T = \mathrm{diag}\,(T_1, \ldots, T_N)\,, \qquad T_i \in \mathrm{GL}^+(K), \tag{16.23}$$

act on the assembled latent coordinate by $y \mapsto Ty$. Every quadratic form in y then transforms by inverse congruence,

$$\Lambda \longmapsto T^{-\top}\Lambda T^{-1}, \qquad L \longmapsto T^{-\top}LT^{-1}, \qquad A \longmapsto T^{-\top}AT^{-1}, \tag{16.24}$$

where L is the Laplacian part of the interaction precision and $A = \operatorname{diag}(A_1, \ldots, A_N)$ collects the self terms, so that $\Lambda = L + A$. Two hypotheses are being used and should be visible. First, the splitting $\Lambda = L + A$ is a declared decomposition of the assembled precision, made once in the frame-dependent chart; (16.24) then says that congruence acts on the two summands separately, which is immediate from linearity. Second, the reframing is block diagonal with respect to the agent partition, which is what a change of local trivialization is; a coordinate change mixing agents would not preserve the block structure that makes A block diagonal. [HYPOTHESIS]

Under a nonconstant reframing the image $T^{-\top}LT^{-1}$ need not itself be a matrix-weighted Laplacian, since the row sums that define that normal form are not preserved when neighboring blocks are reframed differently. This is a statement about normal forms, not about the object: L is a bilinear form on the latent space, and (16.24) is how a bilinear form reads in a new chart. The invariants below compare forms and are therefore unaffected.

Proposition 16.8 (Frame invariance of the generalized spectrum). *Let $\Lambda \succ 0$ and $L = L^{\top}$, and let T be invertible. The generalized eigenvalue problem*

$$Lx = d\Lambda x \tag{16.25}$$

and the reframed problem $\left(T^{-\top}LT^{-1}\right)x' = d'\left(T^{-\top}\Lambda T^{-1}\right)x'$ have the same eigenvalues, with eigenvectors related by $x' = Tx$. [ESTABLISHED]

Proof. Suppose $Lx = d\Lambda x$ and set $x' = Tx$. Then $T^{-\top}LT^{-1}x' = T^{-\top}Lx = dT^{-\top}\Lambda x = d\left(T^{-\top}\Lambda T^{-1}\right)x'$. The map $x \mapsto Tx$ is a bijection, so the correspondence is onto, and the argument run with T^{-1} gives the converse. $\square$

Because $\Lambda \succ 0$, the pair (L, Λ) is a symmetric-definite, hence regular, pencil: it is simultaneously diagonalizable by congruence, all n generalized eigenvalues are real, and they obey a Rayleigh characterization [30, 34]. The positive-definiteness hypothesis is load bearing. Congruence covariance alone also holds for a singular pencil, but it does not imply that the determinant polynomial has degree n or even that it is nonzero. For example, at a common singular ray $(L, \Lambda) = (L, L)$ with $\ker L \neq \{0\}$,

$$\det(L - d\Lambda) = \det\big((1-d)L\big) \equiv 0,$$

so the full-space determinant does not define an n-element generalized spectrum. A quotient, pin, or mass repair must first be declared and the resulting pencil proved regular before the conclusions below can be invoked. Thus Propositions 16.8 to 16.10 apply to the Gaussian probability interior

$\Lambda \succ 0$ (or to a separately proved regular repair), not automatically to an operator-layer boundary point. That qualification is the pencil-level instance of the hierarchy in Chapter 6.

Proposition 16.9 (Localization and endpoints). *If in addition $L \succeq 0$ and $A = \Lambda - L \succeq 0$, then every generalized eigenvalue of (16.25) lies in $[0, 1]$. The value 0 is attained exactly on $\ker L$, and the value 1 exactly on $\ker A$. If, further, L is the matrix-weighted Laplacian part of (14.8), with $W_{ij} \succeq 0$, then every configuration constant across agents lies in $\ker L$, and hence $\dim \ker L \geq K$.* [ESTABLISHED]

Proof. By the Rayleigh characterization, each generalized eigenvalue is a value of $d(x) = x^\top Lx/x^\top \Lambda x$ at a stationary point, and $x^\top \Lambda x = x^\top Lx + x^\top Ax$ with both terms nonnegative and the sum positive. Hence $0 \leq d(x) \leq 1$ for all $x \neq 0$, with $d(x) = 0$ iff $x^\top Lx = 0$, which for $L \succeq 0$ is $x \in \ker L$, and $d(x) = 1$ iff $x \in \ker A$. Under the additional interaction-Laplacian hypothesis, the difference form

$$x^\top Lx = \sum_{i<j} (x_i - x_j)^\top W_{ij} (x_i - x_j)$$

vanishes on every $x = \mathbf{1} \otimes w$, $w \in \mathbb{R}^K$. These K independent configurations lie in $\ker L$, proving the dimension clause. □

The generalized eigenvalue is therefore readable, and its reading is frame independent: d_k is the fraction of the total precision along the k-th pencil direction that is supplied by interaction rather than by self terms. The consensus directions sit at $d = 0$ because interaction supplies no precision there; a direction that no self term constrains sits at $d = 1$.

Proposition 16.10 (Frame dependence of absolute spectra and subspace determinants). *(i) For $T = cI_n$ with $c > 0$, which is a legitimate reframing since $\det(cI_K) = c^K > 0$, the ordinary spectrum obeys $\operatorname{spec}\left(T^{-\top}\Lambda T^{-1}\right) = c^{-2}\operatorname{spec}(\Lambda)$. (ii) Let $U \subset \mathbb{R}^n$ have orthonormal basis B, let $U' = TU$ have orthonormal basis B', and let $\Lambda' = T^{-\top}\Lambda T^{-1}$. Then*

$$\log\det\left(B'^\top \Lambda' B'\right) = \log\det\left(B^\top \Lambda B\right) - \log\det\left(B^\top T^\top T B\right), \tag{16.26}$$

so a subspace log-determinant is frame independent if and only if T restricted to U has unit determinant in absolute value; every orthogonal T satisfies this condition. [ESTABLISHED]

Proof. Part (i) is immediate. For (ii), write $TB = B'S$ with S invertible, so $B' = TBS^{-1}$; orthonormality of B' gives $S^{-\top}\left(B^\top T^\top TB\right)S^{-1} = I$ and hence $\det(S)^2 = \det\left(B^\top T^\top TB\right)$. Then $B'^\top \Lambda' B' = S^{-\top}B^\top T^\top T^{-\top}\Lambda T^{-1}TBS^{-1} = S^{-\top}\left(B^\top \Lambda B\right)S^{-1}$, and taking log determinants gives (16.26). □

The consequence is a prohibition, and it is the reason this section exists. A reframing changes no law; it changes only the components in which the law is written. A quantity computed from those components that is not invariant under (16.24) is therefore a property of the chart. By Proposition 16.10(i), any threshold criterion applied to the absolute eigenvalues of Λ can be made to accept or to reject any given direction by choosing c, so no such criterion states anything about the model. Any admissible criterion must be phrased in generalized-eigenvalue form, that is as a statement about a pencil in which both members transform by the same congruence. By Proposition 16.10(ii) the same warning applies to determinant criteria evaluated on a subspace: they are invariant under orthogonal reframings and not otherwise, so a construction that uses one must either declare its frame or carry the Jacobian term in (16.26) explicitly. [ESTABLISHED]

Verification

Proposition 16.8 is proved above and needs no support. The deterministic replacement check `CHK-GENERALIZED-SPECTRUM` measures the generalized spectrum over twelve fully specified block congruences and uses the moving ordinary spectrum as its negative control. It establishes numerical usability only for that protocol; congruence invariance is proved above. [NUMERICAL]

16.5 What is not claimed

The Fisher metric is used here as a declared choice of geometry on the recognition manifold, and the classical uniqueness results are not strong enough to make that choice forced in this setting. On a finite normalized simplex the Fisher metric is, up to a positive scale, the unique Riemannian metric invariant under congruent Markov embeddings; this is Chentsov's theorem [73]. Campbell's extension to the positive cone of nonnormalized measures is a characterization, but not uniqueness up to scale: writing $s = \sum_i x_i$, every metric in the characterized family has coordinate components

$$g_x(\partial_i, \partial_j) = A(s) + \delta_{ij}\frac{sB(s)}{x_i}, \qquad B(s) > 0, \qquad A(s) + B(s) > 0, \tag{16.27}$$

for two otherwise arbitrary smooth functions A and B, and conversely every such pair defines an invariant family [20]. On the normalized simplex, tangent vectors satisfy $\sum_i u_i = \sum_i v_i = 0$, so the A term vanishes and (16.27) restricts to $B(1)\sum_i u_i v_i / x_i$, a constant multiple of Fisher. Thus simplex uniqueness and the larger nonnormalized cone family are separate statements. [ESTABLISHED]

The latent space here is $\mathbb{R}^n$, so neither finite-sample statement applies directly. Extensions to general sample spaces exist and are established, but they carry their own hypotheses: Ay et al. [6] and Lê [48] obtain uniqueness under a strong-continuity assumption on the information metric, and the monograph treatment collects the required regularity [7]. [ESTABLISHED]

The development therefore adopts the Fisher metric because it is the Hessian of the exponential-family potential (16.5) and because that is the structure the calculations use, and it does not claim that no other invariant metric is available in the continuous setting. [HYPOTHESIS]

No Markov/data-processing monotonicity theorem is asserted for the deterministic Galerkin aggregation map. [NOT-CLAIMED]

The established general statement is that the Fisher metric contracts under a Markov kernel: pushing a family forward through a stochastic map cannot increase Fisher information [3, 7]. [ESTABLISHED]

That score-projection theorem yields the contextual-base tensor defect in Theorem 8.21 only when the channel is realized by a bundle morphism satisfying the isotropy criterion (8.56) and the fine and coarse sections obey (8.44); without that criterion the exact signed comparison (8.52) replaces it and positivity holds only in the directions meeting (8.53). [ESTABLISHED]

The theorem does not cover the aggregation operation of the following chapters, because by Proposition 16.5 the coarse operator is a restriction, a pullback along an inclusion of mean submodels, and not a pushforward along a Markov kernel; the two differ by the positive semidefinite Schur term in (16.19), and the restriction is the larger of the two in the Loewner order. [ESTABLISHED]

That same distinction is why the pointwise-Markov hypothesis of Theorem 9.26 is not satisfied by the Galerkin aggregation, so no configuration-level Fisher contraction or duration comparison may be attached to it by that route. [ESTABLISHED]

Whether a monotone functional decreases under the full declared aggregation/RG map is open. Closing it requires either exhibiting a Markov kernel whose pushforward realizes the map, after which the classical theorem applies verbatim, or exhibiting a Lyapunov functional and proving its decrease directly. [OPEN]

The later fixed-covariance Gaussian relaxation in Section 17.5.1 is a different continuous-time dynamics: its quadratic energy has an exact dissipation identity, and that scoped flow is a Fisher natural-gradient flow. [ESTABLISHED]

It does not turn the coarse map itself into a Markov contraction. [NOT-CLAIMED]

Nothing here should be read as importing a uniqueness statement from the quantum setting. There, the metrics monotone under completely positive trace-preserving maps form a family parameterized by operator monotone functions rather than a single metric [63], so the phrase “the monotone metric” does not denote in that context, and an argument that relies on uniqueness cannot be transported from it. [ESTABLISHED]

No Markov-morphism contraction theorem for the deterministic aggregation maps is asserted. [NOT-CLAIMED]

Finally, this section supplies a metric and not a dynamics. No natural-gradient identification should be inferred merely from the appearance of Λ as a preconditioner: an update is a natural gradient only when its metric is the Fisher metric of the family being updated, in a named chart, and with the objective differential typed in the matching chart [1, 51]. The fixed-covariance Gaussian mean flow in Section 17.5.1 satisfies those requirements and is identified there. [ESTABLISHED]

No broader identification for the remaining updates is claimed. [NOT-CLAIMED]

Even for a declared natural-gradient flow, the metric does not turn its curve parameter into a clock. The oriented-history and Fisher-duration constructions of Chapter 9 quotient that parameterization explicitly. [ESTABLISHED]

No physical-time identification is claimed. [NOT-CLAIMED]

Chapter 17

Multivariate-Gaussian Coarse-Graining

This chapter realizes the three operations of Chapter 10 in the multivariate-Gaussian interaction family. It does not repeat the Markov or graph-exponential theory. Its claims concern hard identification, Gaussian marginalization and recognition restrictions, all at finite dimension. Unless a channel index is displayed, the symbols $K, \Theta, \Lambda, L, A, W$ in this chapter denote the belief/recognition channel. A model-channel realization is separate and uses its own $K_m, \Theta^m, \Lambda_m, L_m, A_m, W_m$; no equality of the two channels is implied. [DEFINITION]

Within the connection-Laplacian realization, Proposition 17.7 applies only to invertible represented transports with inverse reverse transports and positive-definite represented internal weights. Under those hypotheses, trivial represented internal holonomy is the criterion for a connected cluster to retain a full K-dimensional, state-independent structural fixed section; nontrivial holonomy may still leave a lower-dimensional fixed sector. [ESTABLISHED]

Marginal similarity is not identified with this structural criterion. [NOT-CLAIMED]

17.1 Hard aggregation of the interaction precision

Assume the declared interaction family of Chapter 14,

$$\Lambda_{ii} = A_i + \sum_{j \neq i} W_{ij}, \qquad \Lambda_{ij} = -W_{ij} \quad (i \neq j), \qquad A_i \succeq 0, \quad W_{ij} = W_{ji} \succeq 0. \tag{17.1}$$

In flat trivialized coordinates its energy is

$$\mathcal{E}(z) = \sum_i z_i^\top A_i z_i + \sum_{i<j} (z_i - z_j)^\top W_{ij} (z_i - z_j). \tag{17.2}$$

[HYPOTHESIS]

This form is a Gaussian subfamily, not a consequence of an arbitrary linear-Gaussian directed model. [NOT-CLAIMED]

For a hard partition $\mathcal{P}$, let

$$S = \widehat{S} \otimes I_K, \qquad \widehat{S}_{iI} = \mathbf{1}_{\{i \in I\}}, \qquad \Lambda_c = S^\top \Lambda S.$$

The columns of S are orthogonal but not normalized:

$$S^\top S = \operatorname{diag}(n_I) \otimes I_K. \tag{17.3}$$

[DEFINITION]

Proposition 17.1 (Exact Gaussian aggregation). For every hard partition,

$$(\Lambda_c)_{IJ} = -\sum_{\substack{i\in I\\ j\in J}} W_{ij} \quad (I \neq J), \qquad (\Lambda_c)_{II} = \sum_{i\in I} A_i + \sum_{\substack{i\in I\\ j\notin I}} W_{ij}. \tag{17.4}$$

Thus $A_I = \sum_{i\in I} A_i$, $W_{IJ} = \sum_{i\in I, j\in J} W_{ij}$, and every internal edge is annihilated. [ESTABLISHED]

Proof. Substitute one coordinate z_I for every z_i in a cluster. An internal edge contributes zero. A cut edge contributes its difference energy between z_I and z_J, and self terms add. Finite sums preserve positive semidefiniteness; independently, $\Lambda_c \succeq 0$ follows by congruence. □

This is the Galerkin coarse operator with a piecewise-constant prolongator, not a new numerical construction [49, 72]. The parameter map is exact for hard identification. A separately fitted coupling is outside that exact trace construction, while learned or soft pooling is a different approximate model. [NOT-CLAIMED]

17.2 Kron reduction: a counterexample and the closed cone

Gaussian marginalization eliminates nodes by a Schur complement. It is not the aggregation above and is not closed on the unrestricted matrix-weighted interaction family.

Proposition 17.2 (Unrestricted Kron reduction leaves the family). Take $N = 3$, $K = 2$, $A_1 = A_2 = A_3 = I_2$, and

$$W_{12} = 0, \qquad W_{13} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad W_{23} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{17.5}$$

Then the assembled Λ is positive definite. Eliminating node 3 manufactures

$$-\left(\Lambda^{\mathrm{Kron}}\right)_{12} = W_{13}\Lambda_{33}^{-1}W_{23} = \frac{1}{19}\begin{pmatrix} 9 & 3 \\ 4 & 14 \end{pmatrix}, \tag{17.6}$$

which is not symmetric and hence is not a positive-semidefinite interaction weight. [ESTABLISHED]

Proof. $\Lambda_{33} = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 5 \end{smallmatrix}\right)$ has determinant 19, and direct multiplication gives (17.6). The leading principal minors of the full precision are $2, 6, 18, 48, 102, 233$, so Sylvester's criterion gives $\Lambda \succ 0$. The full Schur complement is symmetric; its two mirrored off-diagonal blocks are transposes. Membership in (17.1) additionally requires each individual block to be symmetric, which fails. □

The scalar loopy-Laplacian family is closed under every defined elimination [25]. The matrix analog is closed on a larger subfamily than common orthogonal diagonalization.

Definition 17.3 (Congruence-diagonal cone). For $H \in \mathrm{GL}(K)$, define

$$\mathcal{C}_H = \{HDH^\top : D = \mathrm{diag}(d_1, \ldots, d_K),\ d_a \geq 0\}. \tag{17.7}$$

[DEFINITION]

Theorem 17.4 (Congruence-diagonal Kron closure). If every A_i and W_{ij} lies in one fixed $\mathcal{C}_H$, then every defined node-block Schur complement remains in the interaction family with all reduced parameters in $\mathcal{C}_H$. [ESTABLISHED]

Proof. Factor

$$\Lambda = (I_N \otimes H)\Lambda_D(I_N \otimes H^\top), \tag{17.8}$$

where, after grouping coordinates by channel, Λ_D is a direct sum of K scalar loopy Laplacians. For retained nodes R and eliminated nodes E,

$$\mathrm{Sc}_E(\Lambda) = (I_R \otimes H)\Big[(\Lambda_D)_{RR} - (\Lambda_D)_{RE}(\Lambda_D)_{EE}^{-1}(\Lambda_D)_{ER}\Big](I_R \otimes H^\top). \tag{17.9}$$

The middle operator eliminates channel by channel. Each scalar Schur complement is another loopy Laplacian: its off-diagonals remain nonpositive because the eliminated block is a nonsingular M-matrix, and its row sums remain nonnegative. Reassembling the channels gives reduced self terms and weights in $\mathcal{C}_H$. □

This family strictly contains the common-orthogonal-eigenbasis family. For

$$H = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad X = H\,\mathrm{diag}(1,2)H^\top, \quad Y = H\,\mathrm{diag}(3,1)H^\top,$$

one has $XY - YX = \left(\begin{smallmatrix} 0 & -5 \\ 5 & 0 \end{smallmatrix}\right) \neq 0$, so X, Y cannot be simultaneously orthogonally diagonalized, while they belong to the same $\mathcal{C}_H$. [ESTABLISHED]

Theorem 17.5 (Qualified maximality). Let $\mathcal{C} \subseteq \mathbb{S}_+^K$ be a closed convex cone with an SPD order unit $M \in \mathrm{relint}\,\mathcal{C}$. Assume that anchors and incident edge coefficients range independently over $\mathcal{C}$, arbitrary positive edge scaling is allowed, every defined one-node elimination must remain in the family, and every manufactured off-diagonal block must be minus a symmetric member of $\mathcal{C}$. Then $\mathcal{C} \subseteq \mathcal{C}_H$ for some H, and a maximal cone under these assumptions equals $\mathcal{C}_H$. [ESTABLISHED]

Proof. For $X, Y \in \mathcal{C}$, relative interior gives an $\varepsilon > 0$ such that

$$A_e = M - \varepsilon(X + Y) \in \mathcal{C}.$$

Use $\varepsilon X, \varepsilon Y$ as the two incident weights and M as retained anchors. The eliminated diagonal is M, and the manufactured coupling is $\varepsilon^2 X M^{-1} Y$. Its required symmetry forces

$$XM^{-1}Y = YM^{-1}X.$$

After whitening by $M^{-1/2}$, every pair in the whitened cone commutes. Commuting real symmetric matrices are simultaneously orthogonally diagonalizable, so

$$\mathcal{C} \subseteq \{M^{1/2} O D O^\top M^{1/2} : D \succeq 0 \text{ diagonal}\} = \mathcal{C}_H.$$

The preceding theorem proves that $\mathcal{C}_H$ itself is closed, yielding maximality. □

The assumptions exclude correlated or nonconvex admissible sets, cones without an SPD order unit, parameter-dependent constraints, variable congruence charts, and nonflat link data. Classification in those domains remains open. [OPEN]

17.3 Nonflat links and effective support

For a graph edge $e = (i, j)$, write the twisted energy as

$$(z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j). \tag{17.10}$$

An internal edge under hard identification contributes

$$(I - \Theta_e)^\top W_e (I - \Theta_e), \tag{17.11}$$

which vanishes exactly when

$$W_e^{1/2}(I - \Theta_e) = 0. \tag{17.12}$$

[ESTABLISHED]

Positive-definite W_e forces $\Theta_e = I$; a singular weight may hide a nonidentity twist. [ESTABLISHED]

For $N_e = \ker W_e$ and quotient map $q_e : \mathbb{R}^K \to \mathbb{R}^K / N_e$,

$$W_e^{1/2}(I - \Theta_e) = 0 \quad \Longleftrightarrow \quad q_e \Theta_e = q_e. \tag{17.13}$$

[ESTABLISHED]

This is edge-local. A single common quotient $\mathbb{R}^K/N$ on which every induced weight is positive definite exists exactly when

$$\ker W_e = N, \qquad \Theta_e N = N \quad \text{for every relevant edge.} \tag{17.14}$$

On that quotient ordinary group-valued holonomy is defined. When the kernels differ, there is no canonical product of the edge-local quotients and no single “represented support.” [ESTABLISHED]

A triangle makes the boundary exact:

$$\begin{aligned} W_{12} &= \operatorname{diag}(1,0), & \Theta_{12} &= \operatorname{diag}(1,2), \\ W_{23} &= \operatorname{diag}(0,1), & \Theta_{23} &= \operatorname{diag}(2,1), \\ W_{31} &= I, & \Theta_{31} &= I. \end{aligned}$$

Every edge satisfies (17.12), yet the loop product is $2I$, fixing no nonzero vector. The local kernels differ, so no common effective quotient exists. [ESTABLISHED]

17.4 Cut links and rectangular sheaf closure

For positive-definite cut weights between two already trivialized clusters, define

$$C = \sum W_e, \qquad B = \sum W_e\Theta_e, \qquad A = \sum \Theta_e^\top W_e \Theta_e.$$

One group-valued coarse pair reproduces the cut exactly if and only if all cut twists coincide. Indeed, with $\bar{\Theta} = C^{-1}B$,

$$\Delta = A - B^\top C^{-1} B = \sum_e (\Theta_e - \bar{\Theta})^\top W_e (\Theta_e - \bar{\Theta}) \succeq 0, \tag{17.15}$$

and $\Delta = 0$ exactly under common twists. [ESTABLISHED]

The weighted mean has the correct gauge law but can be singular; equal weights with twists I_2 and $-I_2$ give $\bar{\Theta} = 0$. [ESTABLISHED]

The exact closed category is larger than one invertible link.

Theorem 17.6 (Rectangular endpoint-map closure). Choose a minimal factor

$$W_e = C_e^\top C_e, \qquad C_e : \mathbb{R}^K \to E_e.$$

Then

$$(z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j) = \|R_{i,e} z_i - R_{j,e} z_j\|_{E_e}^2, \qquad R_{i,e} = C_e, \quad R_{j,e} = C_e\Theta_e. \tag{17.16}$$

For partial injections $S_{iI} : V_I \to \mathbb{R}^K$, the coarse endpoint maps are

$$R^c_{I,e} = R_{i,e} S_{iI}, \qquad R^c_{J,e} = R_{j,e} S_{jJ}. \tag{17.17}$$

They reproduce $S^\top \Lambda S$ exactly, parallel cut edges compress losslessly by the direct sum of their E_e, and nested blocking is associative. [ESTABLISHED]

Proof. Expanding (17.16) gives the original pair form. Precomposition gives (17.17); its cross block is

$$-(R^c_{I,e})^\top R^c_{J,e} = -S_{iI}^\top W_e \Theta_e S_{jJ},$$

and the diagonal blocks agree by the same expansion. Vertical stacking in $\bigoplus_e E_e$ sums the energies without loss. A second injection T gives $(RS)T = R(ST)$, proving repeated closure. □

This is the degree-zero cellular-sheaf Laplacian category [31]. Vertex spaces may have unequal dimensions and automorphism groups $\mathrm{GL}(V_I)$; edge basis changes act on both endpoint maps and the edge metric. It is gauge covariant and strictly larger than the fixed-K group-link family. Open questions are canonical minimal compression, characterization of compression to one invertible transporter and one SPD weight, and compatibility with normalized laws and RG rescaling. [OPEN]

17.5 Partial fixed-section aggregation

For each undirected internal edge, choose one orientation $e = (i, j)$, so Θ_e transports from j to i. Assume $\Theta_e \in \mathrm{GL}(K)$, $\Theta_{\bar{e}} = \Theta_e^{-1}$, and $W_e \succ 0$. Positive definiteness is the nondegeneracy assumption used below. [HYPOTHESIS]

The cluster connection Laplacian L_I is defined by

$$z^\top L_I z = \sum_{e=(i,j)\in E_I} (z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j). \tag{17.18}$$

[DEFINITION]

With semidefinite weights, the exact condition remains the edgewise constraint in (17.12), and the resulting effective-support problem is not replaced by the following proposition. [ESTABLISHED]

Let $(I_\alpha)_\alpha$ be the connected components of the internal graph. Choose a root r_α and a spanning tree in each component. For a tree path from r_α to v, let $P_{v\leftarrow r_\alpha}$ be the ordered product of its represented transports. Let Hol_{r_α} be the group of represented transport products around all closed walks based at r_α, and define

$$\mathrm{Fix}(\mathrm{Hol}_{r_\alpha}) = \{u \in \mathbb{R}^K : Hu = u \text{ for every } H \in \mathrm{Hol}_{r_\alpha}\}. \tag{17.19}$$

[DEFINITION]

Proposition 17.7 (Kernel–holonomy fixed-space isomorphism). On every connected component I_α, root evaluation is a linear isomorphism

$$\mathrm{ev}_{r_\alpha} : \ker L_{I_\alpha} \xrightarrow{\sim} \mathrm{Fix}(\mathrm{Hol}_{r_\alpha}), \qquad z \longmapsto z_{r_\alpha}. \tag{17.20}$$

Consequently,

$$\ker L_I \cong \bigoplus_\alpha \mathrm{Fix}(\mathrm{Hol}_{r_\alpha}), \qquad f_I := \dim\ker L_I = \sum_\alpha \dim \mathrm{Fix}(\mathrm{Hol}_{r_\alpha}). \tag{17.21}$$

[ESTABLISHED]

Proof. By (17.18) and $W_e \succ 0$, a vector is in $\ker L_{I_\alpha}$ exactly when $z_i = \Theta_e z_j$ on every edge. Such a parallel section is determined by its root value, so root evaluation is injective. Transporting its value around any rooted closed walk returns the same value, placing its image in $\mathrm{Fix}(\mathrm{Hol}_{r_\alpha})$.

Conversely, take $u \in \mathrm{Fix}(\mathrm{Hol}_{r_\alpha})$ and define $z_v = P_{v\leftarrow r_\alpha} u$ along the spanning tree. Tree-edge constraints hold by construction. For a non-tree edge $e = (i, j)$, the tree path from r_α to j, followed by e, followed by the reverse tree path from i to r_α, has holonomy

$$H_e = P_{r_\alpha\leftarrow i}\Theta_e P_{j\leftarrow r_\alpha}. \tag{17.22}$$

Since $H_e u = u$ and $P_{r_\alpha\leftarrow i} = P^{-1}_{i\leftarrow r_\alpha}$, multiplication by $P_{i\leftarrow r_\alpha}$ gives $\Theta_e z_j = z_i$. Thus every edge constraint holds, proving surjectivity. The full Laplacian is the direct sum over connected components, which yields (17.21). □

For a connected cluster, $f_I = K$ exactly when every represented holonomy element is the identity. This is the full fixed sector. The case $0 < f_I < K$ is a partial fixed sector, while $f_I = 0$ leaves no retained cluster degree. A nonfaithful representation can make represented holonomy trivial even when principal holonomy is not. [ESTABLISHED]

This rank is a state-independent structural admissibility criterion. It assigns no partition cost, singletons remain admissible, and it does not require the constituent belief means to coincide. [NOT-CLAIMED]

Define the abstract fixed fiber

$$\mathsf{F}_I = \bigoplus_\alpha \mathrm{Fix}(\mathrm{Hol}_{r_\alpha}). \tag{17.23}$$

After choosing bases, let $\iota_I : \mathsf{F}_I \hookrightarrow (\mathbb{R}^K)^I$ be the full-column matrix that extends each root value by the spanning-tree construction in the proof. Its range is independent of those choices and equals $\ker L_I$. For a partition $\mathcal{P}$, put $\iota_\mathcal{P} = \mathrm{blkdiag}_{I\in\mathcal{P}}\, \iota_I$. Then

$$L_I \iota_I = 0, \qquad \Lambda_\mathrm{c} = \iota_\mathcal{P}^\top \Lambda \iota_\mathcal{P} \succeq 0, \qquad \Lambda_\mathrm{c} \succ 0 \iff \mathrm{range}(\iota_\mathcal{P}) \cap \ker \Lambda = \{0\}. \tag{17.24}$$

[ESTABLISHED]

Internal edges vanish, while cut edges become the rectangular endpoint data of (17.17). An ordinary smooth vector bundle over varying parameters additionally requires locally constant f_I; rank jumps produce strata rather than a smooth bundle. [ESTABLISHED]

Proposition 17.8 (Nested fixed sections compose exactly). Let a larger cluster A be a union of first-level clusters, and let

$$\iota_{A\leftarrow\mathcal{P}} = \mathrm{blkdiag}_{I\subset A}\, \iota_I, \qquad L_A^{(1)} = \iota_{A\leftarrow\mathcal{P}}^\top L_A \iota_{A\leftarrow\mathcal{P}}. \tag{17.25}$$

Then the fine and first-level kernels obey

$$\ker L_A = \iota_{A\leftarrow\mathcal{P}} \ker L_A^{(1)}. \tag{17.26}$$

If $\iota_A^{(2)} : \mathsf{F}_A \hookrightarrow \bigoplus_{I \subset A} \mathsf{F}_I$ has range $\ker L_A^{(1)}$, then the fine-level fixed-section injection is

$$\iota_A^{\mathrm{fine}} = \iota_{A \leftarrow \mathcal{P}} \iota_A^{(2)}. \tag{17.27}$$

[ESTABLISHED]

Proof. Every zero mode of L_A has zero energy in each positive-semidefinite edge summand. Its restriction to every first-level cluster therefore belongs to $\ker L_I$, so it lies in $\mathrm{range}(\iota_{A \leftarrow \mathcal{P}})$. Write it as $\iota_{A \leftarrow \mathcal{P}} y$. Since $L_A \succeq 0$,

$$y \in \ker L_A^{(1)} \iff y^\top L_A^{(1)} y = 0 \iff \iota_{A \leftarrow \mathcal{P}} y \in \ker L_A. \tag{17.28}$$

This proves (17.26); substituting the range of $\iota_A^{(2)}$ gives (17.27). Matrix composition is associative, so the same statement iterates through further levels. □

Thus the vertex fixed spaces and sheaf endpoint maps are exactly functorial under nested partitions. [ESTABLISHED]

Probability normalization, rescaling, and minimal group-link compression remain separate obligations. [OPEN]

17.5.1 Constant-metric Gaussian relaxation

Proposition 17.9 (Relaxation onto the fixed sector). Let $R = R^\top \succ 0$ and $L = L^\top \succeq 0$ be constant matrices on $\mathbb{R}^d$. If the columns of N_0 form a basis of $\ker L$, define the R-orthogonal projector

$$\Pi_0^R = N_0 (N_0^\top R N_0)^{-1} N_0^\top R, \tag{17.29}$$

and set $\Pi_0^R = 0$ when $\ker L = \{0\}$. The flow

$$R\dot{z} = -Lz, \qquad z(0) = z_0, \tag{17.30}$$

converges to $\Pi_0^R z_0$. If the positive spectrum is nonempty and

$$\lambda_+ = \lambda_{\min}^+ (R^{-1/2} L R^{-1/2}), \tag{17.31}$$

then

$$\|z(t) - \Pi_0^R z_0\|_R \le e^{-\lambda_+ t} \|z_0 - \Pi_0^R z_0\|_R. \tag{17.32}$$

For $\mathcal{E}(z) = \frac{1}{2} z^\top L z$, the exact dissipation identity is

$$\frac{d}{dt} \mathcal{E}(z(t)) = -\dot{z}^\top R \dot{z} = -z^\top L R^{-1} L z \le 0. \tag{17.33}$$

[ESTABLISHED]

Proof. Set $y = R^{1/2}z$ and $A = R^{-1/2}LR^{-1/2} \succeq 0$. Then $\dot{y} = -Ay$. The spectral theorem gives convergence to the Euclidean projection of y_0 onto $\ker A = R^{1/2}\ker L$, with transverse norm decaying at the smallest positive eigenvalue of A. Transforming back gives (17.29) and (17.32). Direct differentiation gives (17.33). □

For a constant passive reframing $z' = Tz$, $T \in \mathrm{GL}(d)$, covariance of the equation requires the congruence laws

$$R' = T^{-\top}RT^{-1}, \qquad L' = T^{-\top}LT^{-1}, \qquad \Pi_0^{R'} = T\Pi_0^R T^{-1}. \tag{17.34}$$

[ESTABLISHED]

The choice $R = I_d$ is therefore equivariant only under orthogonal reframings. If $T = T(t)$, the ordinary derivative acquires the term $\dot{T}T^{-1}z'$. Covariance requires a time covariant derivative; when the original temporal connection is zero, it is

$$D_t'z' = \dot{z}' - \dot{T}T^{-1}z' = T\dot{z}, \qquad R'D_t'z' = -L'z'. \tag{17.35}$$

[ESTABLISHED]

The information-geometric identification is exact on the fixed-covariance Gaussian mean family $q_z = \mathcal{N}(z, R^{-1})$. Its mean Fisher matrix is R, and

$$D_{\mathrm{KL}}(q_z \| q_w) = \frac{1}{2}(z-w)^\top R(z-w). \tag{17.36}$$

Thus (17.30) is exactly the Fisher natural-gradient flow of $\mathcal{E}$ on that mean family. For a general regular statistical family, only the local expansion

$$D_{\mathrm{KL}}(q_\theta \| q_{\theta+d\theta}) = \frac{1}{2}d\theta^\top F(\theta)d\theta + o(\|d\theta\|^2), \tag{17.37}$$

is available under the standard regularity hypotheses [3]. [ESTABLISHED]

Freezing $F(\theta_0) = R$ yields a local linearization. The development does not identify a generic, state-dependent Fisher flow with the constant global projector and rate above. [NOT-CLAIMED]

The solution of (17.30) is a parameterized dynamical trajectory. Away from stationary points, its oriented unparameterized orbit is a history in the sense of Definition 9.4, and its Fisher arclength is invariant under orientation-preserving reparameterization by Theorem 9.13. The exponential rate and the dissipation per unit t are properties of this chosen parameterization, not of the unparameterized history. [ESTABLISHED]

Here t is a declared flow parameter; it is neither a coordinate on the timeless contextual base, an RG-depth index, nor physical time. No physical-time identification is claimed. [NOT-CLAIMED]

17.5.2 Gaussian marginal-law specialization

Specialize Theorem 10.13 after transporting every constituent marginal to one root. For this specialization, declare the admissible parent family to be the full nondegenerate Gaussian family. Trivial represented holonomy then makes the holonomy-fixed admissible subfamily equal to that declared parent; it does not itself select the parent family. Write the transported marginals as $q_i^{(r)} = \mathcal{N}(m_i, C_i)$, with $C_i \succ 0$, and choose weights $a_i > 0$ satisfying $\sum_i a_i = 1$. [HYPOTHESIS]

Proposition 17.10 (Full-Gaussian forward-KL barycenter). The unique minimizer of

$$\mathcal{D}_I(m, C) = \sum_i a_i D_{\mathrm{KL}}\big(\mathcal{N}(m_i, C_i)\|\mathcal{N}(m, C)\big), \tag{17.38}$$

matches the first two moments:

$$\bar{m} = \sum_i a_i m_i, \qquad \bar{C} = \sum_i a_i \left[C_i + (m_i - \bar{m})(m_i - \bar{m})^\top\right]. \tag{17.39}$$

Its minimized score is

$$\mathcal{D}_I^\star = \frac{1}{2}\left[\log\det\bar{C} - \sum_i a_i \log\det C_i\right]. \tag{17.40}$$

[ESTABLISHED]

Proof. The Gaussian KL formula gives

$$\begin{aligned} 2\mathcal{D}_I(m, C) &= \mathrm{Tr}(C^{-1}M(m)) - K + \log\det C - \sum_i a_i \log\det C_i, \\ M(m) &= \sum_i a_i \left[C_i + (m_i - m)(m_i - m)^\top\right]. \end{aligned} \tag{17.41}$$

Completing the weighted square gives $M(m) = \bar{C} + (m - \bar{m})(m - \bar{m})^\top$, so $m = \bar{m}$ is the unique minimizer. At that value, $\mathrm{Tr}(C^{-1}\bar{C}) - \log\det(C^{-1}\bar{C}) \geq K$, with equality exactly when $C = \bar{C}$. Substitution proves (17.40). □

Under nontrivial represented holonomy, the moment-matched Gaussian need not belong to the holonomy-fixed parent family. [ESTABLISHED]

The applicable optimization is then the constrained one in Theorem 10.13. The development does not claim that (17.39) removes that constraint. [NOT-CLAIMED]

For an explicit witness, let $K = 2$, $\mathcal{H} = \{I_2, -I_2\} \subset \mathrm{SO}(2)$, and take the one-law ensemble $\mathcal{N}(u, I_2)$ with $u \neq 0$. Its unconstrained moment-matched Gaussian is itself and is not $\mathcal{H}$-invariant. The Haar-constrained optimizer below is instead $\mathcal{N}(0, I_2 + uu^\top)$, with score $\frac{1}{2}\log(1 + \|u\|^2) > 0$. Thus the holonomy constraint can be active even when there is no pairwise disagreement among the transported constituents.

Corollary 17.11 (Compact-closure holonomy Gaussian barycenter). Let $\mathcal{H} \subset \mathrm{GL}(K)$ be a represented holonomy group whose closure $\overline{\mathcal{H}}$ is compact in $\mathrm{GL}(K)$, and let $\mu_{\overline{\mathcal{H}}}$ be normalized Haar probability measure on $\overline{\mathcal{H}}$. A nondegenerate Gaussian is $\mathcal{H}$-invariant exactly when it is $\overline{\mathcal{H}}$-invariant. Define

$$a = \sum_i a_i m_i, \qquad M_2 = \sum_i a_i (C_i + m_i m_i^\top), \tag{17.42}$$

and their Haar projections

$$a_{\overline{\mathcal{H}}} = \int_{\overline{\mathcal{H}}} h a d\mu_{\overline{\mathcal{H}}}(h), \qquad M_{2,\overline{\mathcal{H}}} = \int_{\overline{\mathcal{H}}} h M_2 h^\top d\mu_{\overline{\mathcal{H}}}(h), \qquad C_{\overline{\mathcal{H}}} = M_{2,\overline{\mathcal{H}}} - a_{\overline{\mathcal{H}}} a_{\overline{\mathcal{H}}}^\top. \tag{17.43}$$

Then $C_{\overline{\mathcal{H}}} \succ 0$, and $\mathcal{N}(a_{\overline{\mathcal{H}}}, C_{\overline{\mathcal{H}}})$ is the unique minimizer of (17.38) over all $\mathcal{H}$-invariant nondegenerate Gaussians. Its score retains the log-determinant form

$$\mathcal{D}^\star_{I,\overline{\mathcal{H}}} = \frac{1}{2}\left[\log\det C_{\overline{\mathcal{H}}} - \sum_i a_i \log\det C_i\right]. \tag{17.44}$$

[ESTABLISHED]

Proof. The stabilizer of a Gaussian under the continuous $\mathrm{GL}(K)$ pushforward action is closed. Hence invariance under $\mathcal{H}$ is equivalent to invariance under $\overline{\mathcal{H}}$. For every such invariant candidate q, invariance of KL under a common invertible pushforward gives

$$\sum_i a_i D_{\mathrm{KL}}(q_i^{(r)} \| q) = \int_{\overline{\mathcal{H}}} \sum_i a_i D_{\mathrm{KL}}(h_\# q_i^{(r)} \| q) d\mu_{\overline{\mathcal{H}}}(h). \tag{17.45}$$

The orbit-averaged source ensemble on the right has first moment $a_{\overline{\mathcal{H}}}$ and second moment $M_{2,\overline{\mathcal{H}}}$. Haar invariance makes both moments $\overline{\mathcal{H}}$-invariant, so its moment-matched Gaussian is an admissible candidate. Its covariance is positive definite because it contains the Haar average of the positive-definite matrices $hC_i h^\top$. The calculation in Proposition 17.10 therefore proves uniqueness. Every element of the compact matrix group $\overline{\mathcal{H}}$ has $|\det h| = 1$, so $\log\det(hC_i h^\top) = \log\det C_i$, which yields (17.44). □

Compactness cannot be dropped from this averaging formula. If the represented holonomy contains $2I_K$, an invariant Gaussian would require $m = 2m$ and $C = 4C$; hence $m = 0$ and $C = 0$, contradicting nondegeneracy. The generated expanding holonomy group therefore admits no invariant SPD Gaussian. [ESTABLISHED]

There is a separate exact identity when every transported constituent and the parent are restricted to one fixed covariance $C_0 \succ 0$. Define the symmetrized convention

$$D_{\mathrm{SKL}}(p, q) := \frac{1}{2}\left[D_{\mathrm{KL}}(p\|q) + D_{\mathrm{KL}}(q\|p)\right]. \tag{17.46}$$

For $q_i^{(r)} = \mathcal{N}(m_i, C_0)$ and $\bar{q}_{C_0} = \mathcal{N}(\bar{m}, C_0)$, the fixed-covariance barycenter score satisfies

$$\sum_i a_i D_{\mathrm{KL}}(q_i^{(r)} \| \bar{q}_{C_0}) = \sum_{i<j} a_i a_j D_{\mathrm{SKL}}(q_i^{(r)}, q_j^{(r)}) = \frac{1}{2} \sum_{i,j} a_i a_j D_{\mathrm{SKL}}(q_i^{(r)}, q_j^{(r)}). \tag{17.47}$$

[ESTABLISHED]

This follows from the weighted variance identity in the C_0^{-1} inner product. It is not the free-covariance optimization of Proposition 17.10, whose covariance includes the between-mean term in (17.39). [ESTABLISHED]

Ordinary pairwise KL between untransported local marginals is not a cross-fiber quantity. After choosing a path, a second path can change the transported law by a loop-holonomy action. Stabilization by that action guarantees path independence; without it the two KL values need not agree. For example, using the holonomy H in (17.49) and $q_i = q_j = \mathcal{N}(e_2, I_3)$, the identity path gives KL zero, whereas a path differing by H gives KL equal to 2. Moreover, an unpenalized sum of nonnegative within-cluster pairwise KL values is minimized by the singleton partition, where it vanishes. It therefore supplies no nondegenerate partition selector by itself. It also contains only marginal information and does not test the common recovery-kernel condition for joint recovery in Section 10.2. [ESTABLISHED]

Two elementary examples separate the structural and marginal statements. On a two-vertex tree with identity transport and $W = I_K$, represented holonomy is trivial, yet the marginals $\mathcal{N}(-ae_1, I_K)$ and $\mathcal{N}(ae_1, I_K)$ have transported divergence

$$D_{\mathrm{KL}}\big(\mathcal{N}(-ae_1, I_K) \| \mathcal{N}(ae_1, I_K)\big) = 2a^2, \tag{17.48}$$

which is arbitrarily large. Conversely, take a triangle with $K = 3$, unit edge weights, identity transports on two edges, and closing transport

$$H = \mathrm{diag}(1, -1, -1) \in \mathrm{SO}(3). \tag{17.49}$$

Its represented holonomy is nontrivial and $\mathrm{Fix}(H) = \mathrm{span}(e_1)$, so the structural fixed sector has $f_I = 1 < K$. Nevertheless, H preserves $\mathcal{N}(0, \sigma^2 I_3)$; assigning that marginal at every vertex makes all transported marginal KL values zero. [ESTABLISHED]

Transported KL therefore measures marginal occupation of the available law modes, not the dimension of the state-independent structural fixed section. [ESTABLISHED]

17.6 Quotient volume for hard membership

Let $P = \widehat{S} \in \{0,1\}^{n \times m}$, with nonempty cluster sizes $s_1, \ldots, s_m$, $D = \mathrm{diag}(s_i)$, $s = (s_i)$, and $n = \sum_i s_i$. For a positive-semidefinite matrix, pdet denotes the product of its positive eigenvalues.

On the translation quotient, the metric induced from the fine Euclidean quotient is

$$H = D - \frac{ss^\top}{n}, \qquad \ker H = \operatorname{span}(\mathbf{1}_m), \qquad \operatorname{pdet} H = \frac{m\prod_i s_i}{n}. \tag{17.50}$$

[ESTABLISHED]

Proof. For a coarse class $[x]$,

$$\min_a \|Px + a\mathbf{1}_n\|^2 = x^\top \left(D - \frac{ss^\top}{n}\right) x.$$

Deleting row and column r, the matrix determinant lemma gives the principal cofactor $\prod_i s_i/n$. Since a rank-$(m-1)$ Laplacian with kernel $\mathbf{1}_m$ has pseudodeterminant m times any principal cofactor, (17.50) follows. □

For K channels, standard quotient Lebesgue measure and fine-induced Hausdorff measure are related by

$$\mu_{\text{ind}} = J_{\mathcal{P}}\mu_{\text{std}}, \qquad J_{\mathcal{P}} = \left(\frac{m\prod_i s_i}{n}\right)^{K/2}. \tag{17.51}$$

If

$$\operatorname{range}(S) \cap \ker \Lambda = \mathbf{1}_n \otimes \mathbb{R}^K,$$

the quotient Gaussian is proper and its generalized determinant satisfies

$$\det_{H\otimes I_K}(\Lambda_{\mathrm{c}}) := \frac{\operatorname{pdet}\Lambda_{\mathrm{c}}}{(\operatorname{pdet} H)^K} = \frac{\operatorname{pdet}\Lambda_{\mathrm{c}}}{J_{\mathcal{P}}^2}. \tag{17.52}$$

This is the determinant of the precision relative to the quotient metric $H \otimes I_K$. For the common unnormalized factor $\exp(-x^\top\Lambda_{\mathrm{c}}x/2)$, let $d_q = (m-1)K$. Its normalizers are

$$\begin{aligned} Z_{\text{std}} &= (2\pi)^{d_q/2}(\operatorname{pdet}\Lambda_{\mathrm{c}})^{-1/2}, \\ Z_{\text{ind}} &= (2\pi)^{d_q/2}\det_{H\otimes I_K}(\Lambda_{\mathrm{c}})^{-1/2} = J_{\mathcal{P}}Z_{\text{std}}. \end{aligned} \tag{17.53}$$

The factor cancels from the normalized law because the measure changes by the same amount. [ESTABLISHED]

Raw partition functions differ by $\log J_{\mathcal{P}}$, so a partition score needs a fixed volume convention, an exact correction, or a compensating prior. [ESTABLISHED]

17.7 Recognition-side costs

Let $B \in \mathbb{R}^{NK\times mK}$ have orthonormal columns spanning $\operatorname{range} S$, let $B_\perp \in \mathbb{R}^{NK\times r}$, $r = NK - mK$, complete it so that $[B\ B_\perp]$ is orthogonal, and write $p = \mathcal{N}(\mu, \Lambda^{-1})$. These definitions apply throughout this section.

The admissible mean-only family leaves covariance unrestricted. For $q = \mathcal{N}(\nu, \Sigma)$, $\nu \in \operatorname{range} S$, its exact optimized cost is

$$\mathcal{G}_{\text{tie}} = \inf_{\nu \in \operatorname{range} S,\ \Sigma \succ 0} D_{\text{KL}}\big(\mathcal{N}(\nu, \Sigma) \| p\big) = \frac{1}{2} m_\perp^\top \left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1} m_\perp, \qquad m_\perp = B_\perp^\top \mu. \tag{17.54}$$

[ESTABLISHED]

Proof. Optimizing the Gaussian KL over covariance gives Λ^{-1}, leaving $\frac{1}{2}(\nu - \mu)^\top \Lambda (\nu - \mu)$. Minimizing over $\nu = Ba$ gives the Λ-projection of μ. In the orthogonal basis $[B\ B_\perp]$, its remaining block is the Schur complement

$$\Lambda_{22} - \Lambda_{21} \Lambda_{11}^{-1} \Lambda_{12} = \left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1}.$$

This proves (17.54). □

A recognition law supported exactly on $\operatorname{range} S$ is singular against p, so its relative entropy to p is $+\infty$. The following declared full-rank regularization makes the divergence and every finite term explicit. Let $\nu_\star$ be the mean optimizer above and set

$$\Sigma_\varepsilon = B(B^\top \Lambda B)^{-1} B^\top + \varepsilon B_\perp B_\perp^\top, \qquad q_\varepsilon = \mathcal{N}(\nu_\star, \Sigma_\varepsilon), \qquad \varepsilon > 0. \tag{17.55}$$

Then, exactly,

$$\begin{aligned} \mathcal{G}_\varepsilon := D_{\text{KL}}(q_\varepsilon \| p) = \mathcal{G}_{\text{tie}} + \frac{r}{2} \log \frac{1}{\varepsilon} - \frac{r}{2} + \frac{1}{2} \log \det(B_\perp^\top \Lambda^{-1} B_\perp) \\ + \frac{\varepsilon}{2} \operatorname{Tr}(B_\perp^\top \Lambda B_\perp). \end{aligned} \tag{17.56}$$

[ESTABLISHED]

Proof. In the orthogonal basis $[B\ B_\perp]$, the covariance in (17.55) has determinant $\det(B^\top \Lambda B)^{-1} \varepsilon^r$, while

$$\operatorname{Tr}(\Lambda \Sigma_\varepsilon) = mK + \varepsilon \operatorname{Tr}(B_\perp^\top \Lambda B_\perp).$$

The Gaussian KL formula and the optimized mean term give all terms in (17.56) after using

$$\det \Lambda = \det(B^\top \Lambda B) \det \left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1},$$

which is the block determinant identity. □

Thus $\mathcal{G}_\varepsilon \to +\infty$ as $\varepsilon \downarrow 0$: identification is a generative construction, not a finite recognition restriction.

The marginal precision in (17.54) obeys

$$\left(B_\perp^\top \Lambda^{-1} B_\perp\right)^{-1} \preceq B_\perp^\top \Lambda B_\perp, \tag{17.57}$$

with equality exactly when $B^\top \Lambda B_\perp = 0$. Using the restriction precision instead therefore overstates the cost except under Λ-orthogonality. [ESTABLISHED]

If only covariance factorization across a partition is imposed, the exact Gaussian optimum has gap

$$\mathcal{G}_{\text{fact}} = \frac{1}{2}\left[\sum_{I}\log\det\Lambda_{II} - \log\det\Lambda\right]. \tag{17.58}$$

[ESTABLISHED]

The mean-tie cost is nonnegative and vanishes at the finest partition; the factorization gap decreases under merging by Fischer's determinant inequality and vanishes at the coarsest partition. The quotient volume is convention dependent across partitions. Therefore these terms do not supply a nondegenerate intrinsic scale without an additional coefficient or a different nonmonotone functional. Whether such a selector exists remains open. [OPEN]

The corrected cross-inventory boundary is the general theorem of Section 10.4: a common evidence-preserving Markov pushforward makes ELBOs comparable and ordered. The hard energy trace analyzed here is not such a pushforward unless a separate theorem proves it, so its cross-partition difference can mix evidence and tightness. [ESTABLISHED]

17.8 The frame sector

If a coarse Gaussian (U_I, μ_I, Σ_I) is sent to constituent i through $U_i U_I^{-1}$, then

$$\mathcal{N}(U_i U_I^{-1}\mu_I, U_i U_I^{-1}\Sigma_I U_I^{-\top} U_i^{\top}) = \mathcal{N}(U_i\bar{m}, U_i\bar{S}U_i^{\top}), \tag{17.59}$$

with $\bar{m} = U_I^{-1}\mu_I$ and $\bar{S} = U_I^{-1}\Sigma_I U_I^{-\top}$. Thus the coarse frame cancels. With faithful representation the level set is exactly one gauge orbit; without faithfulness only unidentifiability follows. [ESTABLISHED]

For $K \geq 2$, $n \geq 2$, no map $F : \mathrm{GL}^+(K)^n \to \mathrm{GL}^+(K)$ is both left equivariant and permutation symmetric. Choose an order-n rotation $v \in \mathrm{SO}(K)$ and inputs $(I, v, \ldots, v^{n-1})$. Left multiplication by v cyclically permutes the tuple. Symmetry keeps F fixed, while equivariance sends it to vF; hence $(I - v)F = 0$, making F singular. [ESTABLISHED]

Physicist summary. Hard Gaussian aggregation is exact Galerkin restriction: internal springs disappear and cut springs add. Gaussian marginalization is a different operation and mixes matrix channels; it fails in general but closes exactly when all parameters decouple in one congruence metric. Nonflat or rank-changing links close in the larger sheaf category of endpoint measurement ports. With positive-definite represented internal weights, holonomy determines the available parallel modes: trivial holonomy retains the full K-dimensional sector, while nontrivial holonomy can retain only a partial sector or none. When the infimum is attained, transported KL measures whether the marginals occupy an available mode; without attainment, a zero infimum establishes only approximability. It is not a structural-holonomy test or a partition selector, and joint recovery remains separate. Quotient volumes, mean tying, and determinant gaps are explicit, but none selects an intrinsic blocking scale by itself.

Chapter 18

Multivariate-Gaussian Operator Renormalization

Chapter 11 separates a genuine scale diagram from a sequence of differently sized objects. This chapter instantiates that diagram on the multivariate-Gaussian (MVG) interaction sector of Chapter 17. The flowed object is an operator parameter, not automatically a normalized Gaussian law: a law-level flow also needs a coarse reference measure, an information-vector rule, and a finite normalizer at every scale. Likewise, the hard aggregation below is only one admissible coarse arrow. The effective-support, sheaf, and Kron reductions of Chapter 17 give other arrows on narrower domains. Unless a channel index is displayed, this chapter flows the belief/recognition operator sector; a model-channel flow is a separate typed construction with its own dimension, links, and parameters. [DEFINITION]

18.1 The MVG operator arrow

At level ℓ, let $\Xi_\ell^{\mathrm{MVG}} = ((A_i^{(\ell)}), (W_{ij}^{(\ell)}))$ with $A_i^{(\ell)}, W_{ij}^{(\ell)} \in \mathbb{S}_+^K$, and assemble

$$(\Lambda_\ell)_{ii} = A_i^{(\ell)} + \sum_{j \neq i} W_{ij}^{(\ell)}, \qquad (\Lambda_\ell)_{ij} = -W_{ij}^{(\ell)}. \tag{18.1}$$

For a hard partition, write $S_\ell = \widehat{S}_\ell \otimes I_K$. Its unrescaled aggregation map is

$$\mathcal{A}_{S_\ell}(\Lambda_\ell) = S_\ell^\top \Lambda_\ell S_\ell, \qquad \mathcal{A}_{S_\ell}(\Xi_\ell^{\mathrm{MVG}}) = \left(\left(\sum_{i \in I} A_i^{(\ell)} \right)_I, \left(\sum_{\substack{i \in I \\ j \in J}} W_{ij}^{(\ell)} \right)_{I \neq J} \right). \tag{18.2}$$

[DEFINITION]

The second identity is exact on the hard-aggregation domain proved in Section 17.1; nonflat data must first satisfy the effective-support or rectangular endpoint conditions of Sections 17.3 and 17.4. [ESTABLISHED]

Proposition 18.1 (Aggregation is a positive semigroup action). If S_1 maps fine nodes to intermediate clusters and S_2 maps those clusters to coarse clusters, then

$$\mathcal{A}_{S_2}\mathcal{A}_{S_1} = \mathcal{A}_{S_1 S_2}. \tag{18.3}$$

Each $\mathcal{A}_S$ is linear and positive on the product cone of node and edge weights. It is generally noninjective because every edge internal to a cluster is annihilated. [ESTABLISHED]

Proof. Congruence gives $S_2^\top(S_1^\top \Lambda S_1)S_2 = (S_1S_2)^\top \Lambda (S_1S_2)$. The parameter formula follows by associativity of finite sums. Sums preserve positive semidefiniteness, so the map is positive. Changing an internal weight while holding every cut weight and self term fixed leaves $\mathcal{A}_S(\pi)$ unchanged, proving noninjectivity.

An RG arrow additionally declares a positive rescaling:

$$\Lambda_{\ell+1} = \zeta_\ell^{-1}\mathcal{A}_{S_\ell}(\Lambda_\ell), \qquad \zeta_\ell > 0. \tag{18.4}$$

This is the MVG component of the comparison data required by Section 11.1. The source and target usually have different dimensions. A fixed ray is therefore typed only after comparison maps identify a repeated hierarchy with one fixed parameter space. Without that identification, (18.4) is a cocycle of positive maps, not an autonomous dynamical system.

Equal-block spectral bounds. If every cluster has b nodes, then $S^\top S = bI$. For every symmetric Λ,

$$b\lambda_{\min}(\Lambda) \le \lambda_{\min}(S^\top \Lambda S), \qquad \lambda_{\max}(S^\top \Lambda S) \le b\lambda_{\max}(\Lambda). \tag{18.5}$$

Indeed, congruence preserves the Loewner inequalities $\lambda_{\min}(\Lambda)I \preceq \Lambda \preceq \lambda_{\max}(\Lambda)I$. These are finite-level bounds. Divergence or convergence as $\ell \to \infty$ additionally requires an infinite hierarchy and a declared normalization. [ESTABLISHED]

18.2 The biadditive closed family

Let each node carry $x_i \ge 0$, and fix $A, M \in \mathbb{S}_+^K$. The family

$$A_i = x_i A, \qquad W_{ij} = x_i x_j M \tag{18.6}$$

is exactly closed under every hard partition: with $x_I = \sum_{i\in I} x_i$,

$$A_I = x_I A, \qquad W_{IJ} = x_I x_J M. \tag{18.7}$$

This follows from distributivity and proves closure, not attraction. [ESTABLISHED]

Proposition 18.2 (Positive additive rules are forced to be linear). Suppose $\alpha : \mathbb{R}_+ \to \mathbb{S}_+^K$ is additive and $w : \mathbb{R}_+^2 \to \mathbb{S}_+^K$ is symmetric and separately additive. Then there are $A, M \in \mathbb{S}_+^K$ such that

$$\alpha(x) = xA, \qquad w(x, y) = xyM. \tag{18.8}$$

Thus (18.6) is the maximal positive family among rules depending only on additive scalar node masses and obeying those additivity hypotheses. [ESTABLISHED]

Proof. If $y \geq x$, additivity gives $\alpha(y) - \alpha(x) = \alpha(y - x) \succeq 0$; hence α is Loewner monotone. For every v, the scalar function $x \mapsto v^\top \alpha(x) v$ is additive, nonnegative, and therefore monotone. A monotone additive function on $\mathbb{R}_+$ is linear, so $v^\top \alpha(x) v = x v^\top \alpha(1) v$. Polarization yields $\alpha(x) = x\alpha(1)$. For fixed y, the same argument gives $w(x, y) = xw(1, y)$; applying it to the second argument gives $w(1, y) = yw(1, 1)$. Take $A = \alpha(1)$ and $M = w(1, 1)$.

On a homogeneous complete graph with equal blocks of size b, the unrescaled map acts on the common pair (A, W) by

$$A \longmapsto bA, \qquad W \longmapsto b^2 W. \tag{18.9}$$

The dense normalization $\zeta = b^2$ fixes the coupling sector and contracts the self sector by $1/b$. Consequently the candidate limiting ray has $A = 0$. It is a boundary operator: its consensus directions lie in the kernel, so it does not define a full-space normalized Gaussian density. Pinning, quotienting the kernel, and retaining an external positive mass are different repairs and define different probability problems. A pseudoinverse inserted into the original Lebesgue density is not a fourth repair.

The exponent 2 is not topology independent. For a typical block $B(b)$ and coarse cut $\partial B(b)$, define the weighted sums

$$W_\partial(b) = \sum_{e \in \partial B(b)} W_e, \qquad A_B(b) = \sum_{i \in B(b)} A_i.$$

If, in a declared matrix norm, $\|W_\partial(b)\| \asymp b^{s_W}$ and $\|A_B(b)\| \asymp b^{s_A}$, then a scalar normalization that keeps the coupling sector nondegenerate scales as b^{s_W}, while the normalized self sector scales as $b^{s_A - s_W}$. Edge count alone does not determine either exponent when the fine weights vary with b.

The count-based exponents follow only under explicit nonzero-average assumptions. If $n_{\mathrm{cut}}(b) \asymp b^s$ and

$$\frac{W_\partial(b)}{n_{\mathrm{cut}}(b)} \longrightarrow \overline{W} \neq 0, \qquad \frac{A_B(b)}{b} \longrightarrow \overline{A} \neq 0,$$

then $s_W = s$, $s_A = 1$, and the coupling and normalized-self exponents are respectively s and $1 - s$. For the homogeneous complete-graph scheme above, a cut between two $\Theta(b)$-sized blocks has $s = 2$; consecutive blocks in a nearest-neighbor chain have $s = 0$. Relevance of the self term is therefore a property of the declared weighted graph family and normalization scheme, not of (18.4) alone. [ESTABLISHED]

18.3 Cone dynamics: what Birkhoff does and does not prove

Let

$$\mathcal{K}_N = \left(\prod_i \mathbb{S}_+^K\right) \times \left(\prod_{i<j} \mathbb{S}_+^K\right) \tag{18.10}$$

be the full MVG parameter cone. On the interior of a proper cone $\mathcal{K}$, Hilbert's projective metric is

$$d_{\mathrm{H}}(x, y) = \log\left(\frac{\inf\{\beta > 0 : x \preceq_{\mathcal{K}} \beta y\}}{\sup\{\alpha > 0 : \alpha y \preceq_{\mathcal{K}} x\}}\right). \tag{18.11}$$

A positive endomorphism of finite projective diameter contracts this metric; a primitive finite-dimensional positive matrix is the familiar special case [15, 59]. The hypotheses fail for sum-only MVG aggregation on the full cone.

Proposition 18.3 (The full-cone route is blocked). After a hierarchical identification makes $\mathcal{A}_S$ an endomorphism, sum-only aggregation is not primitive on $\mathcal{K}_N$. Moreover, the dense candidate ray with $A = 0$ lies on $\partial\mathcal{K}_N$, whereas a full-cone Birkhoff contraction has an interior Perron ray. Hence Birkhoff's full-cone theorem cannot prove attraction to that candidate. [ESTABLISHED]

Proof. An absent coarse cut remains a zero edge coordinate, so the map can meet the boundary even from positive data on the represented fine edges. More sharply, define for a subspace $U \subseteq \mathbb{R}^K$

$$\mathcal{F}_U = \{(W_e)_e : \operatorname{ran} W_e \subseteq U \text{ for every } e\}. \tag{18.12}$$

Each $\mathcal{F}_U$ is an invariant face because the range of a sum of positive semidefinite matrices contained in U is still contained in U. Proper invariant faces rule out primitivity. Finally, $A = 0$ is a boundary coordinate of $\prod_i \mathbb{S}_+^K$, while the Perron ray of a strict interior contraction is interior.

The coupling cone alone removes the vanishing self coordinate but not the matrix obstruction. Suppose an identified spatial map $B \geq 0$ has a positive eigenvector $c = (c_e)_e$. Because aggregation only adds weights, for every $M \in \mathbb{S}_+^K$

$$W_e = c_e M \tag{18.13}$$

is an eigenray with the same spatial eigenvalue. Different matrix rays are not mixed; on the raw product cone their projective separation can be infinite. Thus sum-only aggregation cannot select a fiber direction. It can at most contract a spatial cone after the fiber ray has been declared.

Conjecture 18.4 (Fixed-B scalarized attraction). Fix $M_0 \succ 0$ and restrict to

$$\mathcal{K}_{M_0}^W = \{(c_e M_0)_e : c_e \geq 0\}. \tag{18.14}$$

If a hierarchical identification yields one fixed spatial endomorphism B, some power B^q is strictly positive, and its Perron vector factorizes as $c_{ij} = x_i x_j$, then Birkhoff's theorem gives a unique attracting spatial ray $W_{ij} = x_i x_j M_0$ on this scalarized cone. [CONJECTURE]

This statement freezes every matrix direction. It is not a theorem about the full coupling cone, a nonautonomous sequence B_ℓ, or the common-principal RG state of Section 11.3. Alternating positive maps may contract projective distances while approaching a periodic orbit, as Section 11.4 shows. The missing obligations are a fixed hierarchical identification, primitivity of the resulting B, and the factorization of its Perron vector.

Proposition 18.5 (Residual fiber label). For $M \in \mathbb{S}_+^K$, the orbit under residual global reframing $M \mapsto h^\top M h$, $h \in \mathrm{GL}(K)$, is determined exactly by $\operatorname{rank} M$. Hence the biadditive ansatz has $K+1$ internal congruence classes and no continuous internal modulus. [ESTABLISHED]

Proof. Sylvester's law of inertia classifies real symmetric matrices under congruence [34]. A positive semidefinite matrix has no negative eigenvalues, so its inertia is determined by its rank. In particular, every $M \succ 0$ is congruent to I_K, witnessed by $h = M^{-1/2}$.

Rank does not classify the spatial graph, the limiting spectral measure, corrections to scaling, basin, or scheme. Nor does the proposition extend unchanged to a nonflat pair (W, Θ), whose residual action combines congruence on W with conjugation on Θ.

18.4 An exact finite-sector gate

The homogeneous calculation (18.9) is exact. On $\mathbb{S}^K \oplus \mathbb{S}^K$, its eigenvalues are b^2 on the coupling sector and b on the self sector, each with multiplicity $K(K+1)/2$. The spectral ratio outside the dominant eigenspace is therefore

$$\frac{b}{b^2} = \frac{1}{b}, \tag{18.15}$$

equal to 0.5 at $b = 2$. Comparing the two largest entries of the sorted spectrum would incorrectly return one because the dominant eigenvalue is degenerate. [ESTABLISHED]

Equation (18.15) is the complete finite-sector result. Attraction, uniqueness, and scheme independence remain open because they require a declared invariant cone, hierarchy, and limiting protocol. [OPEN]

18.5 Gauge-invariant spectral data

Let L be the interaction Laplacian obtained from (18.1) by setting all $A_i = 0$. Both L and $\Lambda = L + A$ are bilinear forms. Represent a local reframing by the invertible new-to-old coordinate map

$$G : \mathbb{R}^{NK}_{\text{new}} \xrightarrow{\sim} \mathbb{R}^{NK}_{\text{old}}, \qquad G = \operatorname{diag}(g_1, \ldots, g_N).$$

Pullback to the new coordinates sends both forms by congruence, $(L, \Lambda) \mapsto (G^\top L G, G^\top \Lambda G)$. Ordinary eigenvalues of either matrix are therefore frame dependent. The generalized spectrum of the pair is invariant when the pencil is regular.

Proposition 18.6 (Regular-pencil invariance). Suppose $\det(L - d\Lambda)$ is not the zero polynomial. Then (L, Λ) and $(G^\top LG, G^\top \Lambda G)$ have the same generalized eigenvalues with the same multiplicities. If Λ is invertible, there are exactly NK finite roots. [ESTABLISHED]

Proof.

$$\det(G^\top LG - dG^\top \Lambda G) = \det(G)^2 \det(L - d\Lambda). \tag{18.16}$$

The nonzero factor does not change roots or regularity. When Λ is invertible, the coefficient of d^{NK} is $(-1)^{NK} \det \Lambda \neq 0$.

Proposition 18.7 (Regularity criterion). For $L, A \succeq 0$ and $\Lambda = L + A$, the pencil (L, Λ) is regular if and only if

$$\ker L \cap \ker A = \{0\}. \tag{18.17}$$

In that case $\Lambda \succ 0$. [ESTABLISHED]

Proof. Positive semidefiniteness gives $\ker(L + A) = \ker L \cap \ker A$. A nonzero common-kernel vector annihilates $L - d\Lambda$ for every d, so the determinant is identically zero. Conversely, if the intersection is trivial then $L + \Lambda = 2L + A \succ 0$; evaluating at $d = -1$ proves that the determinant polynomial is not identically zero. The same intersection identity gives $\ker \Lambda = \{0\}$.

At the dense candidate ray $A = 0$, one has $\Lambda = L$ and $\det(L - dL) \equiv 0$. Passing to a complement of $\ker L$ produces the constant generalized spectrum $\{1\}$, not a graph-sensitive label. Pinning and quotienting therefore do not preserve the same spectral object.

Proposition 18.8 (Mass-pencil transfer). Let $R \succ 0$, $a > 0$, and $\Lambda = L + aR$. Generalized eigenvalues of (L, R) and (L, Λ) are related by

$$d = \frac{\lambda}{\lambda + a}, \qquad \lambda = \frac{ad}{1 - d}. \tag{18.18}$$

The correspondence preserves eigenvectors, multiplicities, and ordering. Consequently,

$$N_d(t) = N_\lambda \left(\frac{at}{1 - t} \right), \qquad 0 \le t < 1. \tag{18.19}$$

If $N_\lambda(s) - N_\lambda(0) \sim Cs^\alpha$ near zero, then at fixed $a > 0$, $N_d(t) - N_d(0) \sim Ca^\alpha t^\alpha$. [ESTABLISHED]

Proof. The equation $Lv = d(L + aR)v$ is equivalent, for $d \neq 1$, to $Lv = (ad/(1 - d))Rv$. The case $d = 1$ would require $Rv = 0$, which is impossible. Monotonicity of the fractional map gives the counting identity and its asymptotic consequence.

Mass removal and the infrared limit do not commute:

$$\lim_{a \downarrow 0} \lim_{\lambda \downarrow 0} \frac{\lambda}{\lambda + a} = 0, \qquad \lim_{\lambda \downarrow 0} \lim_{a \downarrow 0} \frac{\lambda}{\lambda + a} = 1. \tag{18.20}$$

Thus a positive mass regularizes each finite problem but does not supply a mass-independent fixed-ray spectrum.

Thermodynamic spectral labels. For a declared sequence (V_n, L_n, R_n), define

$$N_n(t) = \frac{1}{|V_n|K} \#\{k : \lambda_k^{(n)} \leq t\}. \tag{18.21}$$

A cumulative exponent $N(t) - N(0) \sim Ct^\alpha$ is meaningful only after proving convergence $N_n \to N$ in a stated mode. A fixed finite graph has an atomic measure, so a density exponent is not defined there. RG depth ℓ and thermodynamic size n form the two-index family of Section 11.6; neither the existence nor the commutation of those limits follows from the finite congruence identity. [OPEN]

Equality of one exponent is not equality of universality classes. The full limiting measure, amplitudes, corrections to scaling, invariant faces, basin, and blocking scheme may still differ. Projective eigenvalue ratios, block-normalized scaling dimensions, and matched heat/IDS exponents are candidate invariant observables; the raw running matrices are scheme-dependent coordinates.

18.6 Finite stochastic refinement on matrix cones

Section 11.7 proves the general typing statement: a stochastic right section samples a fine preimage and is not an inverse of the realized coarse map. The matrix cone sharpens both the obstruction and the available finite construction.

The central Wishart family has Laplace transform $\det(I + \sigma\Theta)^{-p}$. Apart from the degenerate law δ_0, it is infinitely divisible exactly when $\operatorname{rank}\sigma = 1$ [52]. Therefore a full-rank Wishart prior cannot provide convolution roots of every order for $K \geq 2$; the surviving rank-one case is an embedded scalar Gamma law on a proper face. This does not mean full-cone convolution semigroups are absent. Define the trace-one cone

$$\mathbb{S}^K_{+,1} = \{U \succeq 0 : \operatorname{tr} U = 1\}.$$

Let α be a finite positive Borel measure on $\mathbb{S}^K_{+,1}$, and let $\beta : \mathbb{S}^K_{+,1} \to (0, \infty)$ be measurable with

$$\int_{\mathbb{S}^K_{+,1}} \log\left(1 + \beta(U)^{-1}\right) \alpha(dU) < \infty.$$

For $\Theta \succeq 0$, the corresponding matrix-Gamma law on $\mathbb{S}^K_+$ has Laplace transform

$$\mathcal{L}_{\alpha,\beta}(\Theta) = \exp\left[-\int_{\mathbb{S}^K_{+,1}} \log\left(1 + \frac{\operatorname{tr}(U\Theta)}{\beta(U)}\right) \alpha(dU)\right], \tag{18.22}$$

and, for fixed β, replacing α by $t\alpha$, $t \geq 0$, produces a convolution semigroup; $t = 0$ is δ_0 [62]. Generic full-cone draws need not lie in one Kron-closed chart, however.

Theorem 18.9 (Congruence-diagonal Gamma–Dirichlet bridge). Fix the Kron-closed cone $\mathcal{C}_H$ from Section 17.2, channel parameters $a_r, \beta_r > 0$, and intensities $t_e > 0$. Independently draw

$$X_{e,r} \sim \mathrm{Gamma}(t_e a_r,\ \text{rate } \beta_r), \qquad W_e = H\,\mathrm{diag}(X_{e,1}, \dots, X_{e,K})H^\top. \tag{18.23}$$

For a coarse coordinate f formed from constituents E_f, $\sum_{e \in E_f} W_e$ has the same form with intensity $\sum_{e \in E_f} t_e$. Conditioned on

$$W_f = H\,\mathrm{diag}(c_1, \dots, c_K)H^\top,$$

sample independently over channels

$$(P_{e,r})_{e \in E_f} \sim \mathrm{Dirichlet}((t_e a_r)_{e \in E_f}), \qquad X_{e,r} = c_r P_{e,r}. \tag{18.24}$$

Then every $W_e \in \mathcal{C}_H$ and $\sum_{e \in E_f} W_e = W_f$ exactly. Applied independently to coarse self terms and cut edges, this is a stochastic right section of any fixed finite aggregation map on the represented coordinates. [ESTABLISHED]

Proof. Scalar Gamma variables with a common rate add by addition of their shape parameters. Conditional on their sum, their proportions have the Dirichlet law in (18.24). The identities hold channelwise, and multiplication by the fixed congruence $H(\cdot)H^\top$ preserves both $\mathcal{C}_H$ and addition.

Internal fine edges are absent from the coarse coordinates and require a separate prior. The theorem neither recovers the discarded realization nor constructs a two-sided inverse. An infinite graph-level construction still owes compatible internal-edge priors, nested bridge kernels, gauge-covariant chart data, cylinder consistency, and tightness or another projective-limit theorem. [OPEN]

18.7 Relations to Bayesian, Laplacian, network, and learned RG

Bayesian renormalization. For a Gaussian mean family with fixed precision Λ, the Fisher metric in the mean coordinates is Λ, and the inverse Fisher cometric is Λ^{-1}. This local identity makes an MVG comparison with Bayesian renormalization natural. Berman, Klinger, and Stapleton, however, begin with a posterior over model parameters. In a late-data local Gaussian regime, the delta method pushes an inverse Fisher cometric through a forward map to obtain a sample-space diffusion tensor, exactly for an affine map and to leading local order for a nonlinear map [10, 12]. Howard, Klinger, Maiti, and Stapleton subsequently apply Bayesian RG to neural network field theories [35]. [ESTABLISHED]

Hard energy precomposition $\Lambda \mapsto S^\top \Lambda S$ is not that posterior flow. [ESTABLISHED]

No theorem identifies their scale orientation, connection data, or global continuation with (18.4); the comparison therefore supplies no “opposite-flow” theorem between this operator coarsening and Bayesian learning. [NOT-CLAIMED]

Laplacian RG. With a positive reference form R, the generalized heat generator $R^{-1}L$ transforms by similarity, so its spectrum and heat trace are gauge invariant. The scalar Laplacian-RG construction motivates the entropy scale diagnostic and the positive susceptibility

$$C(\tau) = -\frac{dS}{d\log\tau} \tag{18.25}$$

[74]. Raw off-diagonal heat-kernel blocks transform as $K_{ij} \mapsto g_i^{-1} K_{ij} g_j$, however, and are not invariant scalar node affinities. [ESTABLISHED]

A gauge-invariant real-space block selector, zero-mode treatment, and entropy normalization remain open. The heat scale diagnostic does not by itself prove a unique or scheme-independent partition. [OPEN]

Network RG. Exact closure occurs in the independent-edge multiscale family of Garuccio, Lalli, and Garlaschelli, whose hidden variables aggregate additively [28]. It is a characterized family, not a theorem for arbitrary hidden-variable networks. Interaction closure after a separately specified elimination is a different result. Interacting-network RG can generate higher-order interactions outside restricted coordination classes [21]; it does not imply exact closure of arbitrary matrix-weighted node aggregation. Conversely, (18.2) does not integrate out edge variables and should not be presented as that network transformation. The broader network-RG taxonomy is reviewed by Gabrielli et al. [27]. [ESTABLISHED]

Graph learning and variational RG. The Galerkin identity $L_c = S^\top L S$ preserves the restricted quadratic energy. It does not imply spectral approximation, message-passing commutation, or a graph neural network realization; those require the approximation and residual controls separated in Section 10.7 [37, 50]. DiffPool instead learns soft cluster assignments end to end [79]; that modeling choice does not by itself provide an exact RG semigroup or a closure theorem. Oono and Suzuki prove conditional exponential information loss for deep graph convolutional networks under spectral and weight hypotheses [60]; their result is an oversmoothing boundary, not a Galerkin-commutation theorem. Likewise, Mehta and Schwab's exact variational-RG/deep-network correspondence assumes a hidden-variable kernel satisfying its trace condition [53]. A deterministic identification $y = S\widetilde{y}$ does not supply that kernel or trace condition, so their exact mapping is a precedent within its own construction, not a proof that (18.4) is a deep network. [ESTABLISHED]

18.8 Remaining MVG obligations

The operator recursion is exact on its declared domain, but the following extensions remain open. Partial fixed-section fibers and rectangular cellular-sheaf endpoint maps are already structurally closed under nested blocking by Sections 17.4 and 17.5. [ESTABLISHED]

This structural result does not supply a canonical compression to one invertible link and one SPD weight or an RG rescaling across variable-rank fibers. A full MVG law flow must specify the information vector, reference measure, and normalizer and prove that every iterate stays in the Gaussian natural domain. Attraction needs a typed autonomous endomorphism or a cocycle theorem, a basin, and a matrix mixing mechanism or an explicit scalarized restriction. Universality needs scheme-invariant observables and the two-index convergence required by Section 11.6. A gauge-invariant, nondegenerate block selector remains open. Stochastic inverse RG needs graph-level projective priors, including priors for annihilated internal edges. [OPEN]

Physicist summary. The MVG interaction precision has a clean positive coarse map: cut weights and self terms add, and a declared scalar restores the chosen units. On dense homogeneous graphs the coupling sector scales like b^2 while the self sector scales like b, giving an exact relative factor $1/b$. That algebra produces a boundary operator ray, not a normalized Gaussian fixed law. Birkhoff contraction becomes plausible only after freezing the matrix direction and constructing one primitive spatial endomorphism; sum-only aggregation preserves matrix faces and cannot select that direction. Generalized spectra and Gamma–Dirichlet finite refinements are exact gauge-compatible tools, but neither solves scheme independence, nonflat hierarchy, or the infinite projective-limit problem.

Part IV

Boundaries, Evidence, and Interpretation

Chapter 19

Obstructions

The construction developed in this document is deliberately narrow. Its general belief-law and model-law layers fix one normalized joint and one exact bound, with manifolds introduced only on selected finite-dimensional tiers; the concrete obstructions in this chapter concern particular Gaussian realizations and must not be promoted to no-go theorems for arbitrary belief or model fibers. Each statement below names its architecture, and the weaker constructions that survive it are kept explicit.

Every discrete map in this chapter is a graph-link datum. We write Θ_e^b and Θ_e^m for belief- and model-sector links on an oriented edge copy e; the Gaussian examples suppress the sector superscript. Reverse orientation of the same edge copy obeys $\Theta_{\bar{e}} = \Theta_e^{-1}$. The reciprocal-pair example instead uses two *distinct parallel edge copies* $e \neq f$, traversed in opposite directions, so their link values are not required to be inverses. These symbols are distinct from the base-connection transports $\Omega_\gamma, \widetilde{\Omega}_\gamma$ and from the cross-bundle morphisms $\Phi, \widetilde{\Phi}$. No graph link is identified with base parallel transport without a curve assignment and an equality hypothesis.

The material organizes around one temptation. A tower of scales in which every level's prior is supplied by the level above must terminate, and its top level has no level above. It is natural to try to close the tower by a reciprocal Gaussian agreement pair. The exact result is narrower than "cycles fail": the *flat, unanchored reciprocal pair* is singular, and an objective that appends its energy while omitting a required global normalizer is not an evidence bound. Cyclic globally normalized Markov random fields with proper anchors are legitimate probability models; directed auxiliary-latent models and cycles of inference updates are legitimate as well. Sections 19.1 through 19.2 isolate the failed architecture, Section 19.3 separates it from a distinct Bayesian-renormalization construction, and Sections 19.4 and 19.5 prove that an anchored Gaussian star supplies a unique, geometrically attracting coordinate-ascent fixed point. Section 19.6 finally shows that nontrivial graph-link holonomy can remove the kernel of this reciprocal pair, while making no claim about cyclic models in general.

19.1 The flat unanchored reciprocal Gaussian fold is singular

Consider two latents $u, v \in \mathbb{R}^K$ joined by distinct parallel edge copies $e : v \to u$ and $f : u \to v$. This is the minimal reciprocal fold: v supplies a penalty for u, and u supplies one for v. The two residuals contribute the assembled precision

$$J = E_e^\top R_e^{-1} E_e + E_f^\top R_f^{-1} E_f, \qquad E_e = \begin{pmatrix} I & -\Theta_e \end{pmatrix}, \quad E_f = \begin{pmatrix} -\Theta_f & I \end{pmatrix}, \tag{19.1}$$

with $R_e, R_f \succ 0$ and $\Theta_e, \Theta_f \in \mathrm{GL}^+(K)$. Write $H = \Theta_f \Theta_e$ for the graph holonomy of the parallel-edge two-cycle based at v. The typed inverses $\Theta_{\bar{e}} = \Theta_e^{-1}$ and $\Theta_{\bar{f}} = \Theta_f^{-1}$ exist but do not appear in this loop.

Proposition 19.1 (Kernel of a reciprocal pair). For every pair of positive definite link covariances,

$$\ker J = \big\{(\Theta_e v, v) : v \in \ker(H - I)\big\}, \qquad \dim\ker J = \dim\ker(H - I). \tag{19.2}$$

The kernel does not depend on R_e or R_f: its dimension is the multiplicity of the eigenvalue one in the loop holonomy. [ESTABLISHED]

Proof. J is a sum of two positive semidefinite matrices, so $w \in \ker J$ if and only if w lies in the kernel of each summand. For positive definite R, the form $w \mapsto w^\top E^\top R^{-1} E w$ vanishes exactly when $Ew = 0$, so $\ker J = \ker E_e \cap \ker E_f$. Writing $w = (u, v)$, the first condition is $u = \Theta_e v$ and the second is $v = \Theta_f u$; substituting gives $v = \Theta_f \Theta_e v = Hv$, and conversely any v fixed by H produces a solution with $u = \Theta_e v$. The map $v \mapsto (\Theta_e v, v)$ is injective, so the dimensions agree.

Corollary 19.2 (A flat parallel-edge fold is singular). If the two distinct edge copies give the same flat endpoint comparison, equivalently $\Theta_f = \Theta_e^{-1}$, then $H = I$ and

$$\ker J = \big\{(\Theta_e v, v) : v \in \mathbb{R}^K\big\}, \qquad \dim\ker J = K. \tag{19.3}$$

The assembled precision is singular for every positive definite pair of link covariances, and its kernel is exactly the set of transport-consistent configurations.[ESTABLISHED]

Boundary of the no-go. Corollary 19.2 is not a theorem against cycles. If $P_u, P_v \succ 0$, then

$$J_{\text{anch}} = J + \operatorname{diag}(P_u, P_v) \succ 0, \tag{19.4}$$

because a positive definite matrix added to a positive semidefinite matrix is positive definite. Hence

$$p(u, v) = Z^{-1} \exp\Big[-\tfrac{1}{2}(u, v)^\top J_{\text{anch}}(u, v) + h^\top (u, v)\Big]$$

is a legitimate globally normalized Gaussian model built from the reciprocal parallel-edge energy, with finite $Z = (2\pi)^K (\det J_{\text{anch}})^{-1/2} \exp(\frac{1}{2} h^\top J_{\text{anch}}^{-1} h)$. The same proof covers a finite cycle with any number of vertices: its Gaussian edge energies sum to a positive semidefinite precision, and an SPD block anchor at every vertex makes the total precision positive definite. Such a Markov random field loses the convenience of a product of locally normalized directed kernels, not existence as a probability law. Likewise, Proposition 19.1 shows that the unanchored parallel-edge pair is definite when $1 \notin \mathrm{spec}(H)$. The obstruction is therefore exactly the conjunction of the reciprocal-pair energy, no SPD anchor, and flat parallel-edge holonomy $H = I$. [ESTABLISHED]

Within that scope the direction of the degeneracy is important. A Gaussian energy whose precision has a K-dimensional kernel does not define a normalized Lebesgue density, so the exact identity of Chapter 5 has no joint to which it can apply. The flat direction consists precisely of those configurations in which u and v agree under transport. The unanchored flat pair charges exactly nothing there, so it supplies no coercivity along its intended agreement direction.

There are two independent ways to restore definiteness. Equation (19.4) adds declared coercivity and remains valid under flat parallel-edge holonomy. Alternatively, (19.2) makes the unanchored pair positive definite exactly when 1 is not an eigenvalue of H, which requires abandoning the Regime-I coboundary and declaring at least one of the two parallel edge copies independently. That route leaves the fixed-K flat interaction family used by the standard aggregation flow: an internal twisted cycle leaves a self residue, and the cut sector needs the enlarged nonflat rule left open in Chapter 17. This is a limitation of that flow, not a no-go for globally normalized cyclic models. [ESTABLISHED]

An objective-only reading fails for a different reason. If the reciprocal energy is appended outside the declared joint, the evidence $\log p_\theta(o \mid X)$ is determined by the factors that are in the joint and is unaffected, so the appended term has nothing to bound. If instead the energy is included in a globally normalized Markov random field, it is part of a legitimate joint whenever its global partition function is finite, but that partition function must be retained. Thus there are three distinct cases: the flat unanchored energy is improper; the properly anchored or non-flat energy with its global normalizer is a valid undirected model; and the energy appended while its normalizer is omitted is not an evidence bound. Chapter 5 applies to the second case and not to the other two. [ESTABLISHED]

19.2 A proper anchor restores definiteness and makes the global normalizer essential

Adding a proper prior to each folded latent replaces J by $J + \mathrm{diag}(P_0, P_0)$ with $P_0 \succ 0$. This restores definiteness, since the added term is positive definite and the sum of a positive semidefinite and a positive definite matrix is positive definite. What it does not restore is the ability to treat the normalization as a constant.

For a Gaussian in information form the log normalizer is

$$\mathsf{A}(h, J) = \tfrac{1}{2} h^\top J^{-1} h - \tfrac{1}{2} \log\det J + \tfrac{n}{2} \log 2\pi, \tag{19.5}$$

so any dependence of J on the transports passes directly into A. In (19.1) the transports appear inside E_e and E_f, and the question is whether their contribution to $\det J$ cancels. It does not.

Proposition 19.3 (The global normalizer depends on the model graph link). Take $K = 1$, unit link variances, flat parallel-edge values $\Theta_e = a$ and $\Theta_f = a^{-1}$ with $a > 0$, and a coordinate-isotropic anchor precision $p_0 > 0$. Then

$$\det\bigl(J + p_0 I\bigr) = p_0^2 + p_0\bigl(a + a^{-1}\bigr)^2, \tag{19.6}$$

which is a nonconstant function of a, uniquely minimized on $\mathrm{GL}^+(1) = (0, \infty)$ at $a = 1$, with value $p_0^2 + 4p_0$.[ESTABLISHED]

Proof. With $E_e = (1, -a)$ and $E_f = (-a^{-1}, 1)$, the two rank-one contributions sum to

$$J = \begin{pmatrix} 1 + a^{-2} & -(a + a^{-1}) \\ -(a + a^{-1}) & 1 + a^2 \end{pmatrix},$$

whose determinant is $(1 + a^{-2})(1 + a^2) - (a + a^{-1})^2 = (2 + a^2 + a^{-2}) - (a^2 + 2 + a^{-2}) = 0$, confirming Corollary 19.2 in this case. Adding $p_0 I$ gives $\det(J + p_0 I) = p_0^2 + p_0\bigl[(1 + a^{-2}) + (1 + a^2)\bigr] = p_0^2 + p_0(2 + a^2 + a^{-2})$, and $2 + a^2 + a^{-2} = (a + a^{-1})^2$. Since $a + a^{-1} \geq 2$ for $a > 0$, with equality only at $a = 1$, the stated minimizer is unique.

The witness settles nonconstancy, because variation in one admissible instance refutes a constant-normalizer claim. For zero information vector, write

$$D(a) = p_0^2 + p_0(a + a^{-1})^2, \qquad \mathsf{A}(a) = -\tfrac{1}{2}\log D(a) + \text{const}.$$

Then

$$\mathsf{A}'(1) = 0, \qquad \mathsf{A}''(1) = -\frac{4}{p_0 + 4} < 0. \tag{19.7}$$

Thus the identity is a strict *maximum* of the isolated log-normalizer contribution, not a minimum. Minimizing the corresponding isolated contribution to the negative log density pushes away from $a = 1$; the quadratic and information-vector terms also vary with a, so no universal direction follows for the complete objective. What is universal in this witness is only that the global term $-\frac{1}{2}\log\det(J + p_0 I)$ cannot be absorbed into per-agent constants or dropped from a normalized joint. [ESTABLISHED]

This variation is a change of model, not a pure coordinate change. It holds the coordinate matrices $p_0 I$, $R_e = R_f = I$, and the reference Lebesgue coordinates fixed while varying a. Under a genuine reframing, all covariances, precisions, information vectors, and the reference-density Jacobian transform together, and Proposition 4.9 leaves the probability measure and evidence invariant. Proposition 19.3 therefore proves transport-parameter dependence of a declared anchored model; it does not prove a force along a gauge orbit. [ESTABLISHED]

19.3 Bayesian posterior flow is a different construction

Bayesian renormalization, as formulated by Berman et al. [12], declares a prior and a likelihood and studies posterior and model-space descriptions indexed by an information-theoretic scale. A posterior is the output of that declared Bayesian model; the construction does not, merely by allowing the posterior to flow, make a generative kernel read its own posterior. The typing prohibition of Chapter 4 therefore does not refute Bayesian renormalization. [ESTABLISHED]

The two frameworks nevertheless cannot be identified without an additional construction. If one takes a posterior from one scale and inserts it as the generative top factor at another, then one has declared either a new model at each scale or an explicit transition rule between models. That can be legitimate, but the evidence and ELBO at different scales then refer to different declared joints unless a common target and an exact comparison map are supplied. Proposition 4.3 gives the precise warning: a family of exact pointwise identities does not itself designate one member or one evidence, and comparing two bounds certifies an improvement in one fixed quantity only on an evidence level set. This is a condition on a proposed composition with the present model, not a criticism of a posterior flow considered on its own. [ESTABLISHED]

The resulting boundary is narrower than a prohibition on comparing different latent inventories. If two normalized joints are declared independently, with no evidence-preserving channel between them, their optimized ELBO difference mixes an evidence difference with a tightness difference and supplies no universal order. By contrast, a parameter-independent probability kernel $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ satisfying $K(x, \mathsf{Y}) = 1$ for every x, applied to both the posterior and recognition law through the joint channel $(o, x) \mapsto \delta_o \otimes K(x, \cdot)$, preserves the observation evidence and orders the bounds by data processing. The normalization equation is the hypothesis that makes the pushforward preserve total mass; a sub-Markov kernel is outside this statement. The arbitrary-model caveat applies only in the absence of that relation. [ESTABLISHED]

The asymptotic covariance $C_T = T^{-1}I^{-1} + o(T^{-1})$ is a $T \to \infty$ statement and supplies no bound, limit, or monotonicity as $T \downarrow 0$. Properness at zero information depends on the declared prior and normalizer; with a proper prior, the no-data posterior is that prior. Hence this expansion supports no improper-endpoint claim about Bayesian renormalization. [NOT-CLAIMED]

Nor does a verbal contrast between “posterior widening” and “precision addition” establish that the two scale directions are opposites. The first depends on the declared Bayesian path, and the second is an unrescaled Gaussian aggregation identity. Relating them would require an explicit map between their scale parameters and an operator comparison in a common theory space; none is supplied here. [NOT-CLAIMED]

19.4 What survives

Two constructions remain, and the second is the interesting one.

The first is a declared top prior on a truncated tower. The tower is finite, nothing is folded, and the top level carries a prior that is part of the declared parameters. This is fully typed, it satisfies the hypotheses of Chapter 5 without amendment, and it costs the closure ambition entirely: the top is given, not constituted.[DEFINITION]

The second replaces the fold by an apex latent inside the joint. Let $b \in \mathbb{R}^K$ carry a declared proper prior $\mathcal{N}(0, P_0^{-1})$ with $P_0 \succ 0$, and let the constituents be conditionally independent given it,

$$b \sim \mathcal{N}(0, P_0^{-1}), \qquad y_i \mid b \sim \mathcal{N}(\Theta_i b, R_i), \qquad i = 1, \ldots, n. \tag{19.8}$$

The scale graph is a star, hence a tree, so no cycle can form and Proposition 19.1 has nothing to act on. Three properties make this the right replacement rather than merely an admissible one.

Proposition 19.4 (Definiteness of the anchored star). The assembled precision $J_\star$ of (19.8) is positive definite for every choice of $\Theta_i \in \mathrm{GL}^+(K)$ and $R_i \succ 0$.[ESTABLISHED]

Proof. The quadratic form is $b^\top P_0 b + \sum_i (y_i - \Theta_i b)^\top R_i^{-1} (y_i - \Theta_i b)$, a sum of positive semidefinite terms. It vanishes only if $b^\top P_0 b = 0$ and each residual vanishes; the first forces $b = 0$ because $P_0 \succ 0$, and then $y_i = \Theta_i b = 0$. The kernel is trivial.

Proposition 19.5 (The exact mean-field coordinate is precision addition). Under a mean-field recognition family factorizing over b and the constituents, the optimal factor for b at fixed remaining factors is Gaussian with precision

$$P_b = P_0 + \sum_{i=1}^{n} \Theta_i^\top R_i^{-1} \Theta_i, \tag{19.9}$$

independent of the remaining factors, and with mean $P_b^{-1} \sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$.[ESTABLISHED]

Proof. The coordinate-ascent optimum satisfies $\log q^*(b) = \mathbb{E}_{-b}[\log p(b, y)] +$ const. Only terms quadratic in b contribute to the precision, and taking expectations over the other factors leaves them unchanged, so the coefficient of $-\frac{1}{2} b^\top (\cdot) b$ is the bb block of $J_\star$, which is $P_0 + \sum_i \Theta_i^\top R_i^{-1} \Theta_i$. The linear coefficient is $\sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$.

Equation (19.9) is the transported precision-pooling rule, obtained here as an exact consequence of the joint rather than posited as an aggregation convention. It is also the sufficiency statement of Chapter 10 read in the recognition direction: the statistic that the constituents supply to the apex is $\sum_i \Theta_i^\top R_i^{-1} y_i$, and (19.9) is the precision that statistic carries.

Proposition 19.6 (The declared prior supplies exactly the missing coercivity). Let $w = (v, \Theta_1 v, \ldots, \Theta_n v)$ be a transport-consistent configuration of (19.8). Then

$$w^\top J_\star w = v^\top P_0 v \tag{19.10}$$

exactly, for every $v \in \mathbb{R}^K$ and every choice of transports and link covariances.[ESTABLISHED]

Proof. Every constituent residual $y_i - \Theta_i b$ vanishes on such a configuration, so the only surviving term of the quadratic form is $v^\top P_0 v$.

The three propositions together locate precisely what the flat unanchored pair lacks and what the star supplies. By Corollary 19.2 the pair charges nothing on the transport-consistent ray; by (19.10) the star charges exactly $v^\top P_0 v$ there, no more and no less. The deterministic check `CHK-OBS-STAR-FOLD-NEW-PROTOCOL` constructs star and flat-pair instances and verifies the precision-addition identity and coercivity contrast under its declared protocol. The propositions, not the computation, establish the claim. [NUMERICAL]

That a prior must be declared somewhere is not a shortcoming of this particular construction, and it is worth recording as a general fact rather than as a concession.

Proposition 19.7 (A declared root is unavoidable). In any finite directed acyclic generative model there is at least one latent with no parents, whose marginal law is therefore part of the declared parameters. Hence no finite acyclic construction can have every prior constituted by other latents.[ESTABLISHED]

Proof. A finite directed acyclic graph admits a topological order, and its first element has no incoming edges. Its factor in the joint is unconditional, so it is declared.

The ambition behind the fold was to eliminate the declared root. Proposition 19.7 says that within finite acyclic models the ambition is unachievable for structural reasons rather than for want of ingenuity, and Corollary 19.2 says that the particular flat unanchored reciprocal-pair escape is improper. Properly anchored globally normalized cyclic models remain available, but their anchors and partition functions are declarations rather than derivations from the population.

19.5 The participatory content survives in the inference

What the fold was reaching for is nonetheless available, and it is available exactly where Propositions 19.5 and 19.7 jointly place it: in the inference rather than in the model.

At a variational fixed point of (19.8), the apex factor has the mean $P_b^{-1} \sum_i \Theta_i^\top R_i^{-1} \mathbb{E}[y_i]$ of Proposition 19.5, which is a transported pooling of every constituent's current belief; and each constituent's effective prior, in its own coordinate update, is that apex belief transported back down. Each agent's effective coordinate input is therefore constituted by the apex update, and the apex input is constituted by the agents' updates. This is a loop in an inference algorithm, not a cycle in the factorization of the model. Every step is a coordinate update of a recognition law against a joint that is fixed, acyclic, and normalized, so nothing reads a posterior into a generative kernel and the prohibition of Chapter 4 is respected throughout.[ESTABLISHED]

The distinction being drawn is not verbal. A cycle in the model concerns the factorization or global potential of a joint; Corollary 19.2 rejects only one flat unanchored reciprocal Gaussian potential. A cycle in the inference is a property of an iterative scheme applied to a fixed joint. What the star gives up is the ambition that the apex prior be derived; what it keeps is a bidirectional dependence of exact mean-field coordinates during inference.

Theorem 19.8 (Exact Gaussian-star fixed point and contraction). Allow arbitrary fixed information vectors $r_b, r_1, \dots, r_n$ by adding $r_b^\top b + \sum_i r_i^\top y_i$ to the log density of (19.8); the displayed centered star is the case $r_b = r_i = 0$. Define

$$B := \sum_{i=1}^{n} \Theta_i^\top R_i^{-1} \Theta_i, \qquad P_b := P_0 + B, \qquad M := P_b^{-1} B. \tag{19.11}$$

Consider the exact block schedule that updates every constituent factor from the current apex factor, in parallel or in any order, and then updates the apex factor. The factor covariances become R_i and P_b^{-1}, respectively, on their first updates, and the means obey

$$m_i^{(t+1)} = \Theta_i m_b^{(t)} + R_i r_i, \tag{19.12}$$

$$m_b^{(t+1)} = P_b^{-1} \left(r_b + \sum_i \Theta_i^\top R_i^{-1} m_i^{(t+1)} \right) = c + M m_b^{(t)}, \qquad c = P_b^{-1} \left(r_b + \sum_i \Theta_i^\top r_i \right). \tag{19.13}$$

There is a unique fixed point,

$$m_b^\star = P_0^{-1} \left(r_b + \sum_i \Theta_i^\top r_i \right), \qquad m_i^\star = \Theta_i m_b^\star + R_i r_i, \tag{19.14}$$

and, in the norm $\|x\|_{P_b} = (x^\top P_b x)^{1/2}$,

$$\|m_b^{(t+1)} - m_b^\star\|_{P_b} \le \rho \|m_b^{(t)} - m_b^\star\|_{P_b}, \qquad \rho = \lambda_{\max} \left(P_b^{-1/2} B P_b^{-1/2} \right) < 1. \tag{19.15}$$

Thus the exact schedule converges geometrically from every initial collection of means. [ESTABLISHED]

Proof. Taking the expectation of the joint log density over all factors except q_i leaves a quadratic in y_i with precision R_i^{-1} and information vector $R_i^{-1}\Theta_i m_b^{(t)} + r_i$, which gives (19.12). Proposition 19.5 with the added information vector gives the first expression in (19.13); substitution gives the affine recursion. Since

$$I - M = P_b^{-1}(P_b - B) = P_b^{-1} P_0,$$

the fixed-point equation $(I - M)m_b^\star = c$ has the unique solution (19.14), because $P_0 \succ 0$.

The matrix M is similar to the symmetric matrix

$$S := P_b^{-1/2} B P_b^{-1/2} = I - P_b^{-1/2} P_0 P_b^{-1/2}.$$

Both $B \succeq 0$ and $P_0 \succ 0$, so every eigenvalue of S lies in $[0, 1)$. If $e_t = m_b^{(t)} - m_b^\star$, then

$$\|e_{t+1}\|_{P_b} = \|S P_b^{1/2} e_t\|_2 \le \lambda_{\max}(S) \|e_t\|_{P_b},$$

which proves (19.15). After the first updates, each factor covariance already equals its fixed-point covariance. Consequently $D_{\mathrm{KL}}(q_b^{(t)} \| q_b^\star) = \frac{1}{2}\|e_t\|_{P_b}^2$ contracts by at least ρ^2 per full sweep, and the sum of constituent divergences after their next update is $\frac{1}{2}e_t^\top B e_t \le \frac{1}{2}e_t^\top P_b e_t$. This also proves convergence of the entire product factor in relative entropy. □

The strict anchor $P_0 \succ 0$ is essential. If it is weakened to $P_0 \succeq 0$, the symmetric matrix S can have eigenvalue one, and neither uniqueness nor geometric contraction follows. When $B = 0$, $\rho = 0$ and the apex reaches its fixed point in one update. Delayed, noisy, asynchronous, or inexact updates are not covered by Theorem 19.8 and remain a separate convergence problem. [OPEN]

If a one-parameter family is genuinely wanted, for a construction that interpolates between an uninformative and a data-driven description, a legitimate one is available and does not repeat the failures of Section 19.3. Declare a tempering path $p_\beta \propto p(Y)p(o \mid Y)^\beta$ for $\beta \in [0, 1]$, whose endpoints are the prior and the posterior of one fixed model, both proper, and whose evidence satisfies the exact path-sampling identity

$$\log p(o) = \int_0^1 \mathbb{E}_{p_\beta}\big[\log p(o \mid Y)\big]\,\mathrm{d}\beta \tag{19.16}$$

[29]. The family is indexed by a declared temperature and built from one declared prior–likelihood pair; its normalizing constant is retained at every β, both endpoints are proper when the prior and posterior are proper, and (19.16) is an identity rather than a bound. The identity itself is established and cited; the construction is not developed here, and it is not identified with the information-scale construction of Section 19.3.[ESTABLISHED]

19.6 In the reciprocal-pair architecture, the kernel is a flatness artifact

Proposition 19.1 measures the obstruction of Section 19.1 by one number, $\dim\ker(H - I)$, and Corollary 19.2 evaluates that number under flat parallel-edge holonomy, where it is K. Reading the corollary as a statement about cyclicity is a misreading, and the correct reading inverts the relation between the two halves of this document.

Corollary 19.9 (Nontrivial holonomy shrinks the reciprocal-pair kernel). Under the hypotheses of Proposition 19.1, with $H = \Theta_f\Theta_e$ the holonomy of the parallel-edge two-cycle,

$$\begin{aligned} \dim\ker J = K &\iff H = I, \\ H \neq I &\implies \dim\ker J \le K - 1, \\ \ker J = \{0\} &\iff 1 \notin \operatorname{spec}(H). \end{aligned} \tag{19.17}$$

The set of $H \in \mathrm{GL}^+(K)$ at which the last condition fails is the intersection of $\mathrm{GL}^+(K)$ with the zero set of $\det(H - I)$, a polynomial that is not identically zero since $H = 2I$ gives $\det(H - I) = 1$. It is therefore a proper algebraic subset, relatively closed in $\mathrm{GL}^+(K)$, with dense complement there, and has Lebesgue measure zero in the ambient matrix space. The flat case $H = I$ is the maximally degenerate point of this reciprocal-pair family, while a generic H gives no kernel.[ESTABLISHED]

Proof. All three statements are read off (19.2), which gives $\dim \ker J = \dim \ker(H - I)$ with no dependence on the link covariances. The kernel of $H - I$ is all of $\mathbb{R}^K$ exactly when $H - I = 0$; if $H \neq I$ then $H - I$ has rank at least one, so its kernel has dimension at most $K - 1$; and the kernel is trivial exactly when 1 is not an eigenvalue of H, that is when H fixes no nonzero vector. For the genericity statement, $\det(H - I)$ is a polynomial in the entries of H, and a nonzero polynomial has a closed, nowhere-dense, Lebesgue-null zero set in the ambient Euclidean matrix space; intersecting with $\mathrm{GL}^+(K)$ gives the stated relative properties.

The same computation controls the normalization, and in a form that makes the transport dependence of Section 19.2 exact rather than exhibited at one witness.

Proposition 19.10 (Graph-link dependence of the determinant factors through holonomy). For the reciprocal pair of (19.1),

$$\det J = \frac{\big(\det(I - H)\big)^2}{\det R_e \det R_f}. \tag{19.18}$$

The graph links enter the determinant only through the holonomy of the loop, and J is singular exactly when $\det(I - H) = 0$, in agreement with Proposition 19.1.[ESTABLISHED]

Proof. Stack the two residual maps as $\widetilde{E} = \begin{pmatrix} I & -\Theta_e \\ -\Theta_f & I \end{pmatrix}$ and write $\widetilde{R} = \mathrm{diag}(R_e, R_f)$, so that $J = \widetilde{E}^\top \widetilde{R}^{-1} \widetilde{E}$ by (19.1). Hence $\det J = (\det \widetilde{E})^2 / (\det R_e \det R_f)$, and the Schur formula gives $\det \widetilde{E} = \det(I - \Theta_f \Theta_e) = \det(I - H)$.

What holonomy removes. Exactly one defect of this exact architecture: the unanchored pair's kernel. When $1 \notin \mathrm{spec}(H)$ the assembled precision is positive definite, so the reciprocal energy with its global normalizer defines a Gaussian law and the failure recorded in Corollary 19.2 does not occur. By (19.2), the transport-consistent zero modes are $\{(\Theta_e v, v) : v \in \ker(H - I)\}$, which collapse to the origin. Cyclicity did not produce that kernel; the flat loop product did. This statement concerns the two-residual precision (19.1) and is not generalized to other cyclic potentials. [ESTABLISHED]

What holonomy does not remove. First, it does not turn the reciprocal energy into a pair of locally normalized directed kernels. It can instead define an undirected Gaussian potential, and then its *global* partition function is part of the model. Section 19.2 exhibits its parameter dependence in an anchored scalar model; Proposition 19.10 gives the unanchored non-flat determinant exactly. By (19.5) the latter contains

$$-\tfrac{1}{2} \log \det J = -\log\big|\det(I - H)\big| + \tfrac{1}{2} \log \det R_e + \tfrac{1}{2} \log \det R_f, \tag{19.19}$$

a nonconstant model term that cannot be absorbed into per-agent constants. It diverges as H approaches the singular flat locus, exactly where the unanchored law ceases to exist. Retaining this term gives a legitimate normalized model; omitting it changes the objective. Proposition 19.3 and Equation (19.19) both vary declared model transports while holding other coordinate parameters fixed, so neither is a statement about motion along a pure gauge orbit. [ESTABLISHED]

Second, nontrivial H does not eliminate declarations. A loop with $H \neq I$ requires at least one link declared independently of the pointwise frames, because a Regime-I loop product telescopes to the identity. Replacing a declared prior by a declared edge link relocates the declaration. [ESTABLISHED]

A conditional incompatibility with the standard flat flow. If one insists simultaneously on an *unanchored* reciprocal pair and on the fixed-K flat aggregation flow, the two requirements conflict. The standard flow requires trivial graph-link holonomy on every cluster and, for exact cut closure, on every union of two clusters; Corollary 19.9 says the unanchored pair is definite only when $1 \notin \operatorname{spec}(H)$. In that scoped architecture,

$$\begin{aligned} \text{standard coarsening needs} \quad & \operatorname{Hol}(\gamma) = e_G \quad \text{on the cycles in its declared domain,} \\ \text{the unanchored reciprocal pair needs} \quad & 1 \notin \operatorname{spec}(H). \end{aligned} \tag{19.20}$$

An SPD anchor removes the right-hand demand and therefore avoids this incompatibility, at the cost of explicitly declaring the coercive top term. The star avoids it for the same reason and also has no cycles. Equation (19.20) is not a constraint on all cyclic models or on the general belief-law and model-law theory. [ESTABLISHED]

Placing the reciprocal cycle permanently between clusters does not evade the scoped conflict along a finite flow to the coarsest resolution. Admissibility is inherited under refinement, while each nontrivial blocking strictly decreases the number of clusters. At a level with two nonempty clusters, exact fixed-K cut closure is the condition $\Delta = 0$ in (17.15); after the two clusters have been trivialized, that equality forces their cut twists to coincide and hence makes their union, the whole population, trivializing. With one cluster, whole-population triviality is already the cluster condition. Thus a non-flat reciprocal cycle can remain outside the standard flow only by stopping at a finite maximal resolution or by changing the coarse family. [ESTABLISHED]

Remaining non-flat compression problem. The rectangular endpoint-map or cellular-sheaf category constructed in Chapter 17 is already gauge covariant and closed under repeated linear-quadratic blocking. [ESTABLISHED]

What remains open is compression of that richer cut datum to one invertible fixed-rank graph link, together with a normalized probability/RG lift that retains the global normalizer. A solution must preserve closure under another merge; it is not an open question about anchored cyclic Gaussian models, which (19.4) settles positively. [OPEN]

Both new statements are proved above. The deterministic replacements `CHK-OBS-HOLONOMY-DET-KERNEL` and `CHK-OBS-NORMALIZER-WITNESS` construct their transports and covariances explicitly, check the determinant identity and prescribed fixed-space dimensions, and include a normalized anchored control. Rank tolerance remains part of the numerical protocol; the proofs settle the statements. [NUMERICAL]

Chapter 2 distinguishes three holonomies, and everything in this section uses only graph-link holonomy: the ordered product of a declared comparison assignment on a population graph. Neither side of (19.20) refers to the base holonomy of a smooth connection along curves of $\mathcal{C}$. No conclusion about bundle topology or base curvature follows from the reciprocal-pair calculation. [ESTABLISHED]

Chapter 20

Empirical Commitments and Interpretation

This chapter proves no mathematical result. It distinguishes declared readings, hypotheses consumed by the mathematics, and extensions that would require new evidence. A reading is declared. [DEFINITION]

The sole interpretive restriction used later is observational closure. [HYPOTHESIS]

Unresolved empirical or interpretive consequences remain open. [OPEN]

Availability is not support, and no preference below changes a theorem proved in the preceding chapters. [NOT-CLAIMED]

20.1 What the base is, and what it is not

The general construction has distinct belief and model associated bundles, $\mathcal{E}_b$ and $\mathcal{E}_m$, over the contextual base $\mathcal{C}$. An agent carries sections such as

$$q_i^b \in \Gamma\left(\mathcal{C}_i, \mathcal{E}_b|_{\mathcal{C}_i}\right), \qquad q_i^m \in \Gamma\left(\mathcal{C}_i, \mathcal{E}_m|_{\mathcal{C}_i}\right), \tag{20.1}$$

whose values are declared probability laws in their respective fibers. The cross maps $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ and $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$ do not identify those bundles, and neither is assumed inverse or connection-parallel. The base indexes the sections; the finite design is a declared subset, not a random sample from a law on $\mathcal{C}$. No expectation over contexts is used. The projective-limit chapter separately identifies what is established for cylinder laws and what remains open for measures supported on smooth section spaces. [DEFINITION]

The informational pullbacks of Chapter 8 can give this fixed base an agent-, section-, and connection-relative positive-semidefinite tensor. Passive gauge invariance of that tensor does not make it independent of the chosen connection, and degeneracy can leave contextual directions unmeasured. [ESTABLISHED]

This construction supplies no Lorentzian signature, causal cones, spacetime measure, gravitational field equation, or identification of $\mathcal{C}$ with physical spacetime. [NOT-CLAIMED]

The word “timeless” has an equally narrow meaning in Chapter 9: an oriented inference history retains order while quotienting away its arbitrary orientation-preserving parameterization, and Fisher length supplies a nonnegative information-geometric duration. [DEFINITION]

It does not assert that physical time is unreal or emergent, nor identify Fisher duration with relativistic proper time, thermal time, or a quantum clock. [NOT-CLAIMED]

Two readings remain compatible with this typing. On the first, the base is Kantian scaffolding: a form under which appearances are ordered, while the fiber variables are what the agents infer [40]. On the second, the base and bundles are noumenal structure, approached only through agent-side representations. [DEFINITION]

The absence of a probability law on the base discriminates neither reading. Noumenal objects are not ordinary uncertain latents, but scaffolding is likewise not an inferential target. [DEFINITION]

No equation selects between the two. [NOT-CLAIMED]

20.2 When a noumenal reading earns its keep

This chapter adopts an explicit idle-wheel criterion: a posit with no trace in any declared observable is removed by parsimony. [DEFINITION]

No result in this chapter derives the declared rule. [NOT-CLAIMED]

For attributional scope in this chapter, “constructive empiricism” denotes the position combining a literal reading of scientific theories with the claim that accepting a theory involves belief only in its empirical adequacy [71, 11–12]. This is a declared use of the cited position, not a theorem of the manuscript. [DEFINITION]

The manuscript does not claim that constructive empiricism entails its removal rule. [NOT-CLAIMED]

The criterion keeps three kinds of typed geometric or comparison data separate. Discrete population links are Θ^b_{ij} and Θ^m_{ij}, connection parallel transports along base curves are Ω_γ and $\widetilde{\Omega}_\gamma$, and the cross-bundle maps are Φ and $\widetilde{\Phi}$. An edge-labeled graph loop $\gamma = (e_0, \ldots, e_{r-1})$ determines, for example,

$$H^b_\gamma = \Theta^b_{e_0}\Theta^b_{e_1}\cdots\Theta^b_{e_{r-1}}, \qquad [H^b_\gamma]_{\mathrm{conj}}, \tag{20.2}$$

and analogously in the model sector. [DEFINITION]

Under vertex reframing, the loop product is conjugated by the rechoice at its base vertex, so its conjugacy class is invariant. [ESTABLISHED]

The graph-link observable is not claimed to be evidence of base curvature or bundle topology. [NOT-CLAIMED]

In the flat graph-link specialization every loop product telescopes. [ESTABLISHED]

Under the declared idle-wheel criterion, the resulting identity observable gives the noumenal reading no additional work. A declared nontrivial graph-link holonomy supplies a model-internal referent, but only within the interaction complex. [DEFINITION]

Identifying Θ_e with Ω_{γ_e} requires an embedded interaction complex, assigned piecewise-smooth base curves, and equality with the path-ordered exponential of a specified connection. Even that would establish a connection-holonomy statement, not topological nontriviality. [OPEN]

A separate open route is to ground the reading in a specified base connection whose holonomy has a population-level operational trace. [OPEN]

A positive result requires a named principal bundle and connection, an assigned base loop, a gauge-invariant population-record statistic, a typed map from the connection data to its record law, and two admissible connection data sets for which every declared non-holonomy input is fixed, the represented loop holonomies have distinct conjugacy classes, and the statistic's induced distributions differ. A stronger sufficient formulation requires the controlled record-law map to factor nontrivially through the conjugacy class of the represented loop holonomy. No such tuple or empirical evidence is presently supplied. A theorem that every statistic in a declared admissible controlled class is insensitive to that holonomy class would close the route negatively. [OPEN]

Proposition 8.5 proves exact change formulas for the covariant Fisher and Amari–Chentsov pullbacks when a chosen connection changes. [ESTABLISHED]

Theorem 8.4 proves their invariance under passive reframing of a fixed connection–section pair. [ESTABLISHED]

The manuscript does not derive a canonical connection from either result. [NOT-CLAIMED]

It also does not construct a population observable sensitive to base holonomy. [NOT-CLAIMED]

20.3 Which realism this is

The desktop analogy belongs to the interface theory of perception, whose evolutionary argument permits an arbitrary measurable relation between representation and world [32, 33]. The present formalism is narrower: agents are sections of associated law bundles, and a global reference frame exists only under the corresponding principal-bundle trivialization hypothesis. Local frames can exist when a global one does not, and a pointwise graph coboundary does not provide a smooth Čech trivialization. [DEFINITION]

What survives a change of frame is structural data, schematically

$$\mathcal{I}_{\mathrm{struct}} = \Big\{ \text{ELBO and KL values},\ \operatorname{rank} M,\ \operatorname{spec}(L,\Lambda),\ [h^{\omega_b}_{q_i^b}],\ [h^{\omega_m}_{q_i^m}],\ [H^b_\gamma]_{\mathrm{conj}},\ [H^m_\gamma]_{\mathrm{conj}} \Big\}, \tag{20.3}$$

on the exact domains established earlier. This supports an epistemic structural-realist reading in Worrall's sense [77]. Here the brackets on the pullback tensors denote their passive-coordinate equivalence classes for the stated sections and fixed connections, not connection-independent canonical geometries. [DEFINITION]

Ontic structural realism in the eliminativist sense, in which structure is all that exists, is available but unsupported [45, 46]. [OPEN]

Supporting it would require an argument from formal unidentifiability to ontological elimination and discriminating evidence for that bridge. [OPEN]

In this chapter, "moderate structural realism" denotes Esfeld and Lam's position on which objects and relations occupy the same ontological footing and objects are characterized only by their relations [26]. This is a declared use of the cited position, not a theorem of the manuscript. [DEFINITION]

The manuscript proposes, but does not establish, that its invariance and unidentifiability results fit that moderate reading. [OPEN]

Establishing the fit would require a typed interpretation relating the formalism's objects and relations to the source's spacetime ontology and discriminating evidence for that interpretation. [OPEN]

The manuscript does not claim that either structural reading follows from its invariance results. [NOT-CLAIMED]

20.4 Two agreements, and the cost of conflating them

For a graph loop and a configuration of Gaussian means, the two relevant conditions are

$$H^b_\gamma = I \qquad \text{and} \qquad \mu_i = \Theta^b_e \mu_j \quad \text{on each edge copy } e : j \to i. \tag{20.4}$$

The first is frame agreement: the graph links admit a common trivialization on the cluster. The second is belief agreement: a particular configuration is link-consistent. [DEFINITION]

They are independent. A graph coboundary constrains links but leaves the means free. Conversely, under nontrivial holonomy, belief agreement can survive on the fixed subspace

$$\mu_{i_0} \in \ker(H^b_\gamma - I). \tag{20.5}$$

[ESTABLISHED]

The manuscript therefore avoids using "consensus" for both. Common graph trivialization is an admissibility condition for the declared fixed-rank coarse recursion, exact under positive-definite represented weights and sufficient with semidefinite weights. Belief agreement is a configuration and a null direction of the interaction energy. Aggregation never consults the realized means, so belief agreement does not select a partition. [DEFINITION]

Semidefinite null-direction exceptions belong to the typed effective-support problem rather than to an interpretive identification of the two agreements. [DEFINITION]

20.5 The participatory reading and its exact cost

Operational observational closure declares that every datum directly available to an interior agent is a clamped state or message of another interior or boundary environment node. [HYPOTHESIS]

The kernel-level equivalence is proved in Theorem 7.8. [ESTABLISHED]

The interpretive hypothesis is the decision to call every such boundary node an agent. [HYPOTHESIS]

It does not eliminate the conditioning record, require the boundary node to be autonomous, exclude an unobserved common cause, prove convergence of an RG orbit, or identify a fixed object with a physical law. [NOT-CLAIMED]

The obstruction chapter locates the participatory loop in inference rather than in a self-referential generative kernel. In the proper Gaussian-star model, the apex update pools the constituents' beliefs and each constituent update receives the apex belief; the joint remains fixed, acyclic, and normalized. Reading that reciprocal inferential dependence as the content of the construction resembles Wheeler's participatory proposal [76], but is neither derived from nor evidence for it. [DEFINITION]

Likewise, a relational inference history tracks a recognition-state orbit against the fixed normalized joint and conditioning record. [DEFINITION]

Quotienting its parameterization does not feed the orbit back into the generative kernel or make the joint self-referential. [NOT-CLAIMED]

Three commitments often hidden in that slogan must be separated. Operational observational closure is the declared model restriction above. [HYPOTHESIS]

Strong agent-only ontology additionally requires a criterion of agency, such as persistent state, action, a Markov blanket, or a local VFE, for every boundary node. The formalism supplies no unique such criterion. [NOT-CLAIMED]

Identifying a repaired limiting object with a physical law is the independent semantic claim B.1; it lacks a target system, validation criterion, probability-level closure, and discriminating evidence. [OPEN]

Ontological closure, the claim that nothing exists behind the appearances, is not claimed. [NOT-CLAIMED]

The formalism does not exclude an unobserved environment affecting all agents without becoming the argument of any agent's belief. [NOT-CLAIMED]

The present graph-holonomy reading and Gaussian RG also occupy different domains. The former becomes non-idle under nontrivial graph-link holonomy, whereas the standard fixed-rank iterated flow uses its declared trivializing domain. Rectangular endpoint-map closure resolves the linear-quadratic blocking problem, but compression to one invertible link, a normalized probability lift, effective-support typing across rank changes, and external scale selection remain unsettled. [OPEN]

A base-holonomy reading is a different extension and needs the operational bridge stated above. The formalism does not adjudicate between these readings. [NOT-CLAIMED]

20.6 What would make the interpretation testable

The current mathematics permits an internal admissibility check: compute fundamental graph-loop products and test whether a proposed cluster lies in the trivializing domain; with semidefinite weights, also test weighted invisibility on the represented support. [DEFINITION]

This checks declared model data. It is not empirical detection of principal-bundle topology or base curvature. [NOT-CLAIMED]

A cross-scale empirical test would require a preregistered target system, sampling units, noise model, and estimator for gauge-typed parameters with uncertainty. It would also require at least two admissible partitions, declared blocking and rescaling maps, gauge alignment, a held-out statistic, an acceptance tolerance, and Gaussian-graphical and spectral baselines. One would fit only at fine resolution, push the result through each blocking without coarse refitting, and compare the predicted held-out coarse statistic both with data and across admissible blockings. Failure would falsify that added cross-scale hypothesis, not the finite-dimensional theorems. [OPEN]

No such target, estimator, tolerance, baseline margin, attraction theorem, or scheme-robust invariant has been supplied. The present empirical status is therefore inconclusive. A viable interpretation must continue to distinguish graph-link holonomy, base-connection holonomy, bundle topology, and gauge redundancy, and must state which of them its evidence actually bears on.

Appendix A

Typed Notation Contract

This appendix is a type checker, not a second development of the theory. Superscripts b and m always denote the belief and model channels. They may have different representations, fiber dimensions, connections, and coarse maps, but the ambient theory induces both channels from one principal G-bundle.

Symbol	Type	Meaning and prohibited identification
$\mathcal{C}$	smooth base manifold	Context space. It is neither the population graph nor a probability sample space.
$P \xrightarrow{\pi} \mathcal{C}$	principal G-bundle	The common source of belief and model associated bundles. It is not either statistical bundle.
ρ_b, ρ_m	actions on sample fibers	Both are representations of the same G; they need not be equivalent, faithful, or of equal dimension. $\rho_x(g)$ acts on a sample coordinate in channel x.
$\widehat{\rho}_b, \widehat{\rho}_m$	actions on laws	$\widehat{\rho}_x(g)q = (\rho_x(g))_{\#}q$. A hatted action is not an action on a sample vector.
$\mathcal{E}_b = P \times_{\widehat{\rho}_b} \mathcal{B}_b, \quad \mathcal{E}_m = P \times_{\widehat{\rho}_m} \mathcal{B}_m$	associated law bundles over $\mathcal{C}$	Their fibers contain belief and model laws. They are not principal bundles.
$u_i^b, u_i^m;\ h_i$ with $u_i^m = u_i^b h_i$	two local sections of P; unique map $h_i : \mathcal{C}_i \to G$	The relative principal frame exists independently of ρ_b, ρ_m. It does not define a map between their representation spaces.
$u_i^{x\prime} = u_i^x a_i^x; \quad R_i^x = \rho_x(a_i^x)^{-1} = \rho_x(g_i^x)$, with $g_i^x = (a_i^x)^{-1}$	passive section rechoice and old-to-new sample-coordinate map in channel x	If $v_x' = R_i^x v_x$, then $q_i^{x\prime} = (R_i^x)_{\#} q_i^x$ and a same-channel transport obeys $T_{r\leftarrow i}^{x\prime} = R_r^x \circ T_{r\leftarrow i}^x \circ (R_i^x)^{-1}$. A linear cross-channel map transforms by $F_i' = R_i^m F_i (R_i^b)^{-1}$, covariance by $\Sigma_i' = R_i^x \Sigma_i (R_i^x)^\top$, and precision by inverse congruence. This R_i^x is neither a belief–model map nor an RG restriction.
$\Phi : \mathcal{E}_b \to \mathcal{E}_m$	belief-to-model associated-bundle morphism	Covers $\mathrm{id}_{\mathcal{C}}$. It is not a gauge automorphism or a same-channel transport.
$\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$	model-to-belief associated-bundle morphism	Covers $\mathrm{id}_{\mathcal{C}}$. It is independent of Φ and is not presumed to be its inverse.
ω_b, ω_m, or H^b, H^m	two principal connections on P, or induced horizontal distributions	They are separate geometric data on the same principal bundle. Vector-bundle covariant derivatives are a linear specialization.
$\Omega_\gamma, \widetilde{\Omega}_\gamma$	$(\mathcal{E}_b)_{\gamma(0)} \to (\mathcal{E}_b)_{\gamma(1)}$ and $(\mathcal{E}_m)_{\gamma(0)} \to (\mathcal{E}_m)_{\gamma(1)}$	Same-channel parallel transports induced by the two connections along a piecewise-smooth base curve $\gamma \subset \mathcal{C}$.

$\varpi_x : \mathcal{E}_x \to \mathcal{C}, \quad V\mathcal{E}_x$	associated law-bundle projection and its vertical tangent bundle	The vertical space is tangent to a statistical fiber. It is neither a sample space nor the tangent space of the contextual base.
$g^F_x, \mathcal{T}_x$	vertical symmetric tensors of orders two and three on $\mathcal{E}_x$	The descended Fisher and Amari–Chentsov tensors exist under the declared regular-model hypotheses. They are not tensors on $\mathcal{C}$ until a section and connection are supplied.
$D^{\omega_x} s = \mathrm{ver}^{\omega_x} \circ Ts$	map $T\mathcal{U} \to s^*V\mathcal{E}_x$ for a smooth section $s : \mathcal{U} \to \mathcal{E}_x$	Connection-relative vertical first jet. For nonlinear statistical fibers it is not a linear covariant derivative on a vector space of sections.
$h^{\omega_x}_s, c^{\omega_x}_s$	sections of $\mathrm{Sym}^2\, T^*\mathcal{U}$ and $\mathrm{Sym}^3\, T^*\mathcal{U}$	Pullbacks of g^F_x and $\mathcal{T}_x$ by $D^{\omega_x} s$. They are passive-gauge covariant but connection-relative, and they are not fixed by the active gauge group at fixed connection; $h^{\omega_x}_s$ is only a semimetric until nondegeneracy is proved.
$N;\ N_\star;\ q = N_\star\|_{\mathcal{B}}$	sample kernel $\mathsf{K} \times \bar{\mathscr{K}} \to [0,1]$; affine law map $\mathcal{P}(\mathsf{K}) \to \mathcal{P}(\bar{\mathsf{K}})$; law-fiber map $\mathcal{B} \to \bar{\mathcal{B}}$	Three distinct objects. N determines $N_\star$, and $N_\star$ restricts to q only under the declared family closure $N_\star(\mathcal{B}) \subseteq \bar{\mathcal{B}}$; neither arrow reverses. Sample-level equivariance of N implies the law-level intertwining, and not conversely. None of the three is the associated-bundle morphism.
$\Psi;\ T^V\Psi$	associated-bundle morphism $E \to \bar{E}$ over f; its vertical differential $VE \to V\bar{E}$	Ψ is induced by $(\mathcal{P}, \kappa, q)$ exactly under (11.14). For a fixed principal map $\mathcal{P}$ and associated-bundle morphism Ψ, the representative $q = \iota^{-1}_{\mathcal{P}(u)} \Psi \iota_u$ is unique. Replacing the target frame changes that representative by the full represented target gauge action, not merely by image isotropy; Ψ never determines N. $T^V\Psi$ equals Tq only in a frame pair $(u, \mathcal{P}(u))$; in any other frame pair the identification uses coarse-side invariance of $\bar{g}^F$.
$A_\Psi = \mathcal{D}\Psi;\ \mathfrak{A}_{\mathcal{P}}$	section of $\varpi^*T^*\mathcal{C} \otimes \Psi^*V\bar{E}$; element of $\Omega^1(\mathcal{C}; f^*\,\mathrm{Ad}\,\bar{P})$	Horizontal defect and scale-connection defect form, related by (8.55). The sign convention is transported fine horizontal minus coarse horizontal lift. They compose by the ordered law (8.57); an unweighted sum of two stage defects is a type error.
$\Delta^\Psi_F;\ \delta_\Psi;\ \mathcal{X}_\Psi;\ \mathcal{Q}_\Psi$	vertical form on $\mathrm{Sym}^2\, V^*E$; its section pullback and the two anomaly tensors on $\mathrm{Sym}^2\, T^*\mathcal{U}$	Δ^Ψ_F is the fiber Fisher defect and δ_Ψ its base pullback; they are different objects on different bundles. The base comparison retains *two* tensors, $-\mathcal{X}_\Psi$ with $\mathcal{X}_\Psi$ signed and $-\mathcal{Q}_\Psi$ with $\mathcal{Q}_\Psi \succeq 0$; neither may be summarized as one mismatch term.
$\mathcal{E}_i, \varpi_i;\ \boldsymbol{\omega}_i$	$\mathcal{E}_b\|_{\mathcal{C}_i} \times_{\mathcal{C}_i} \mathcal{E}_m\|_{\mathcal{C}_i} \to \mathcal{C}_i$; product connection	Agent i's paired law bundle and the pair $\boldsymbol{\omega}_i = (\omega_b, \omega_m)$. The common projection does not select relative channel weights or a joint Fisher metric.
$v^\omega(\Gamma), L^\omega(\Gamma)$	vertical velocity and vertical Fisher length of a curve $\Gamma : J \to E$	The parameter on J is arbitrary. This length selects neither an inference orbit nor a physical time.
$\mathfrak{S}_i$	declared regular configuration space of belief–model sections for agent i	Its tangent vectors are section variations evaluated vertically at each fixed context. It is not the contextual base or an undeclared infinite-dimensional probability space.

$\mathscr{H}_i$	oriented unparameterized curve in $\mathfrak{S}_i$	An equivalence class under orientation-preserving reparameterization. Its orientation is selected by the VFE descent ray only after a regular metric or mobility is declared.
$\Sigma_i : \mathscr{H}_i \times \mathcal{C}_i \to \mathcal{E}_i$	adjoint evaluation map of an inference history	$\varpi_i \circ \Sigma_i(r, c) = c$: history changes the section value while the base context remains fixed.
$\pi_i^{\mathrm{conf}}, \iota_i$	configuration extraction from an admissible conditional/full-law family and its smooth right-inverse	$\pi_i^{\mathrm{conf}} \circ \iota_i = \mathrm{id}$ is the commuting condition that makes ι_i a genuine joint-law lift of the displayed agent configuration, rather than an unrelated parameterized family.
$\overline{\mathcal{F}}_{B,o}$	outside-averaged conditional block VFE at fixed Q_{B^c}	Under the local–global support, finiteness, integrability, and differentiation hypotheses, it differs from the restricted collective VFE only by a configuration-independent term.
$\mathsf{G}_i^F = \iota_i^* G^F_{\mathfrak{R}_B}$	exact Fisher metric induced from the lifted conditional/full recognition law	A weighted marginal Fisher sum has this status only under separately proved block orthogonality or fixed dependence; degeneracy requires a justified quotient.
$\mathcal{Q}_\ell;\ \mathsf{G}_\ell$	declared configuration manifold at scale ℓ; its *strong* configuration metric	Determined by the declared base measure, channel weights, basis, and topology, not by the bundle. G_ℓ is a section of $\mathrm{Sym}^2 T^*\mathcal{Q}_\ell$ and is neither the fiber metric g_x^F nor the base semimetric $h_s^{\omega_x}$. It carries a scale or agent subscript always, and is not the unsubscripted M-step preconditioner G of Chapter 5. A labeled weighted product equals a joint-law pullback only under the criterion (9.19).
$\mathsf{R}_\ell : \mathcal{Q}_\ell \to \mathcal{Q}_{\ell+1}$	smooth coarse configuration map	Separately declared data. It is not the sample kernel K_ℓ, the law map $N_\star$, the bundle morphism Ψ or $C_{\ell,s}$, the nonlinear action map $\mathcal{R}^H$, the block measure-pair map $\mathcal{R}_b$, the derivative D_ℓ, the reference-space endomorphism $\widehat{\mathcal{R}}_\ell$, the conjugated retained projection $\widehat{R}_\ell$, the root-vertex set $\mathcal{R}$, or the descent ray $\mathcal{R}^-_{\mathcal{F}_i}$. A pointwise bundle morphism induces it only on the projectable set, which can have empty interior.
$a_\ell;\ Q^{(\ell)}(r);\ \tau^{(\ell)}(r)$	positive semiconjugacy factor; configuration at depth ℓ and orbit position r; accumulated Fisher duration at that depth	Distinct typed quantities plus a parameter-conversion datum. Depth ℓ is a discrete scale index; r is a chosen oriented orbit parameter; and $\tau^{(\ell)}$ is metric-relative arc length from a chosen origin. After choosing metric, origin, orientation, and path, τ can be a function of r, but it is not canonically determined by an unparameterized orbit or by scale depth. The factor $a_\ell = \mathrm{d}\sigma/\mathrm{d}r$ is the coarse-flow-parameter rate relative to the fine parameter, not a physical-time or Fisher-duration rate; ordinary arc length uses $\lvert a_\ell \rvert$, and positivity selects the oriented semiconjugacy.
$\nu_F, L_F, \tau_{Q,\lambda_0}$	Fisher speed, realized path length, and cumulative clock from a chosen origin	They are defined on a selected regular history. Length and cumulative duration are invariant under orientation-preserving reparameterization; neither is physical time.

$T^b_{ij}(c), T^m_{ij}(c)$	elements of G at one $c \in \mathcal{C}_i \cap \mathcal{C}_j$	Transition functions computed from two frame atlases of the same P. Their cocycles are cohomologous descriptions of one principal-bundle class. They use no nonconstant curve and are not graph links.
Θ^b_e, Θ^m_e	oriented-edge-copy elements of G on $\mathfrak{I} = (V, E, ^-)$	Channel-specific graph-link data, transforming under discrete vertex frame changes h^x_i, with $\Theta^x_{\bar{e}} = (\Theta^x_e)^{-1}$. Parallel copies retain distinct labels. Equality with represented base transport requires vertex contexts, assigned curves, $h^x_i = g^x_i(c_i)$, and the explicit link–transport hypothesis. The superscripted contextual re-expression h^x_i is not the relative principal-frame field h_i.
$\mathcal{H}, \overline{\mathcal{H}}$	represented graph holonomy group and its closure in the represented matrix group	A finitely generated $\mathcal{H}$ need not be closed. Haar averaging in the compact-closure theorem is over $\overline{\mathcal{H}}$, after explicitly assuming that this closure is compact.
$\nu_{\mathrm{mix}} = \lambda + \sum_n 2^{-n}\delta_{a_n}$	fixed sigma-finite reference for a mixed family	Exists in this form exactly when the union of all positive atomic supports is countable, assuming no singular-continuous components. It is fixed for the family, not selected separately at each parameter.
$M_o;\ z = M_o(\mathsf{Y}_D);\ \Pi_o = M_o/z$	finite evidence-slice measure; positive-finite mass; posterior probability	The slice is primary at a selected regular observation. M_o is not itself a probability unless $z = 1$, and normalization requires $0 < z < \infty$.
$\mathcal{L}^{\mathrm{ext}}, \mathcal{F}^{\mathrm{ext}}$	maps on every latent probability law with values in $[-\infty, \log z]$ and $[-\log z, +\infty]$	$\mathcal{L}^{\mathrm{ext}} = \log z - D_{\mathrm{KL}}(Q\|\Pi_o)$ and $\mathcal{F}^{\mathrm{ext}} = -\mathcal{L}^{\mathrm{ext}}$. The safe gap identity is $\log z - \mathcal{L}^{\mathrm{ext}} = D_{\mathrm{KL}}$; the sum $\mathcal{L}^{\mathrm{ext}} + D_{\mathrm{KL}}$ is not formed in the singular branch.
$\mathcal{L}, \mathcal{F}$	finite classical split of the extended functionals	The expected-log-joint minus expected-log-recognition formula is used only under absolute continuity and separate log-integrability. It is not the definition on arbitrary Q.
$\mathrm{TC}(Q) = D_{\mathrm{KL}}(Q\| \bigotimes_b Q_b)$	extended nonnegative total correlation	May equal $+\infty$. Its product-reference chain identity holds in $[0, +\infty]$ without subtracting entropies; marginal-entropy formulas are finite-domain corollaries.
$K : \mathsf{X} \rightsquigarrow \mathsf{Y}$	normalized, parameter-independent Markov kernel	It satisfies $K(x, \mathsf{Y}) = 1$ for every x, pushes laws forward, and obeys data processing. For right-acting scale kernels, $K_{\ell r} := K_{\ell k}K_{kr}$ means $(\mu K_{\ell k})K_{kr} = \mu K_{\ell r}$, representing the abstract composite $K_{kr} \circ K_{\ell k}$. A kernel is not an energy restriction or a Galerkin matrix.
$\mathscr{K}_{\mathrm{op}}$	declared operator category	Its objects are topological vector spaces and its morphisms are continuous linear maps; the analytic tier uses Banach spaces and bounded linear maps. Abstract arrows compose with $\circ$, not with right-acting kernel juxtaposition.
$I_{\mathbb{R}^b};\ I_b$	the b-dimensional identity covariance; declared rescaling/identification kernel	The covariance $I_{\mathbb{R}^b}$ occurs in replicated Gaussian laws. The plain I_b is a normalized kernel composed with C_b in the RG measure-pair map; it is neither that covariance nor the score lift $\mathscr{I}_b$.

$\iota_{\mathcal{P}}$	coarse-to-fine deterministic embedding	Imposes a hard block constraint on an energy family. Normalization and an induced probability channel require separate proofs.
C, P	compression and prolongation on a declared linear feature tier	Here CP $= I$. A message propagator may be compressed as CAP, but a hard-tied quadratic energy uses P$^*H$P, an exact Gaussian marginal uses a Schur complement, and a general law uses Markov pushforward.
$K_a, E_{a,o}$	normalized interaction-record kernel and its fixed-record negative log density	One actual record is one global factor. It may occur in several incident coordinate-local VFEs without being duplicated in the joint.
$\alpha_i, \beta_{ij}, \eta_{ij} = \alpha_i\beta_{ij}$	receiver occupancy, conditional source attention, and joint edge-event law	Exact meta-attention pushes η and then disintegrates. A row β alone does not determine receiver weights.
$\mathscr{P}_\ell \xrightarrow{\varpi_\ell} \mathcal{C}_\ell$	scale-ℓ principal bundle and its principal projection	$\mathscr{P}_\ell$ is distinct from the root-level notation P, from the normalized law π_ℓ, and from every interaction-coordinate operator. The projection is ϖ_ℓ, not π_ℓ.
$\boldsymbol{\Xi}_\ell^{\mathrm{MVG}} = ((A_i^{(\ell)}), (W_{ij}^{(\ell)}))$	scale-ℓ MVG operator-parameter tuple	It determines the assembled interaction precision in Chapter 18. It is neither the normalized law π_ℓ nor a bundle projection.
$(\rho_\ell, m_\ell);\ M_\ell, \pi_\ell$	scale-ℓ probability reference, finite evidence measure, its mass, and normalized law	Here $M_\ell = m_\ell(\mathsf{X}_\ell) \in (0,\infty)$ and $\pi_\ell = m_\ell/M_\ell$. The measure pair is the primary law-level RG state; normalization does not discard the separately tracked evidence mass.
$c_{\mathrm{ray}}(b);\ n_{\mathrm{cut}}(b)$	additive fixed-action-ray shift; cut-edge count for the declared block ratio b	It is present only on the projective action tier of the b-step map $\mathcal{R}_b^H$. A fixed tracked measure pair restores mass preservation and forces $c_{\mathrm{ray}}(b) = 0$. The count $n_{\mathrm{cut}}(b)$ occurs only in the operator-scale counting law.
$B(b);\ \partial B(b)$	typical block at ratio b; its coarse cut	Local notation for the weighted scaling comparison. Neither object is fixed by the block ratio alone; both belong to the declared weighted graph family.
$W_\partial(b), A_B(b);\ s_W, s_A$	cut-coupling and block-self sums; their declared norm-growth exponents	$W_\partial(b) = \sum_{e\in\partial B(b)} W_e$ and $A_B(b) = \sum_{i\in B(b)} A_i$. The exponents satisfy $\\|W_\partial(b)\\| \asymp b^{s_W}$ and $\\|A_B(b)\\| \asymp b^{s_A}$ in the declared norm; edge count alone does not determine them.
$b > 1;\ C_b,\ I_b,\ K_b = C_bI_b$	block-RG structural coarse kernel, rescaling/identification kernel, and composite kernel	C_b is the structural coarse Markov kernel and I_b the declared rescaling/identification kernel returning its target to the common measurable state space; their right-acting composite is $K_b = C_bI_b$.
$\mathcal{R}_b, \mathcal{R}_b^\rho;\ \mathcal{R}_b^H, \mathfrak{B}_b^H$	block-RG measure-pair and reference maps; finite-action map and action beta	$\mathcal{R}_b(\rho, m) = (\rho K_b, mK_b)$ is the primary measure-pair map and $\mathcal{R}_b^\rho\rho = \rho K_b$ its reference component. On the dominated, finite-action tier, $\mathcal{R}_b^H[H;\rho]$ is the reference-dependent action map and $\mathfrak{B}_b^H[H;\rho] = (\mathcal{R}_b^H[H;\rho] - H)/\log b$. Neither the action nor its beta is defined outside its stated finite-action domain.
$K_\ell;\ \Pi_\ell(z, dx)$	normalized forward scale kernel; reverse conditional for $\pi_\ell(dx)K_\ell(x, dz)$	K_ℓ pushes ρ_ℓ and m_ℓ forward together. Π_ℓ is a fixed version defined only $\pi_{\ell+1}$-almost everywhere; it is not an inverse of K_ℓ.

$Q_\ell,\ U_\ell = DQ_\ell(0)$	bounded conditional log-Laplace action increment and its linearization	$Q_\ell(\varphi) = -\log U_\ell(e^{-\varphi})$ is finite on bounded actions and locally analytic at every bounded center by the recentering identity, whereas $U_\ell\varphi = \mathbb{E}[\varphi(X_\ell) \mid X_{\ell+1}]$ extends as an L^p contraction. The larger linear domain is not a nonlinear action domain.
$\mathfrak{B}_\ell, \overline{\mathfrak{B}}_\ell$	$L^\infty(\pi_\ell;\mathbb{R})$ and its quotient by $\mathbb{R}\mathbf{1}$	The full space tracks additive evidence-mass changes. The quotient is used only on the explicitly projective tier and carries the oscillation norm.
$\nu_\ell = \bigotimes_i \nu_{\ell i};\ C_{\ell,A}, P_{\ell,A}$	product interaction reference; coordinate expectation and Boolean-lattice Hoeffding projector	The interaction tier separately assumes $\nu_\ell \sim \pi_\ell$ at every admitted scale. A normalized Markov kernel need not preserve this premise; the diagonal-cloning channel is the exact obstruction.
$\mathcal{G}_\ell, \phi_g;\ E_\ell, \mathsf{H}_\ell$	full nonempty-subset interaction Banach space and canonical representative; quotient assembly and Hoeffding extraction	Here $\phi_g = \sum_{A\neq\varnothing} g_A$, $E_\ell g = [\phi_g]$, and $\mathsf{H}_\ell E_\ell = E_\ell \mathsf{H}_\ell = I$ on their typed spaces. The extraction bound $3^{\lvert V_\ell\rvert} - 1$ is finite for every finite network but is not dimension-free.
$T_\ell^{\mathcal{G}};\ U_\ell^{\phi_g}$	exact nonlinear interaction RG and its tilted conditional-expectation factor	$T_\ell^{\mathcal{G}} = \mathsf{H}_{\ell+1}\overline{Q}_\ell E_\ell$, while $DT_\ell^{\mathcal{G}}(g) = \mathsf{H}_{\ell+1}\overline{U_\ell^{\phi_g}}E_\ell$. The untilted U_ℓ occurs only at $g = 0$.
$R_\ell; r^{\mathcal{G}}_{\ell+1}, \overline{r}^{Q}_{\ell+1}$	bounded idempotent retained projection; coordinate and action-quotient residuals	The retained map is projected unless the exact-image invariance condition holds. This indexed projection is unrelated to the stochastic refinement kernel R below; the residual norm is always reported when nonzero.
$\delta_\ell,\ B_{n\leftarrow\ell}, \overline{U}_{n\leftarrow\ell}$	essential Dobrushin coefficient, accumulated block factor, and ordered quotient cocycle	The accumulated log-δ inequality is a sufficient irrelevance certificate, with $\log 0 = -\infty$; its failure is not a relevance or marginality theorem.
$r_{\mathrm{ess}}(U) = r(U + \mathcal{K}(X))$	essential spectral radius in the Calkin algebra $\mathcal{L}(X)/\mathcal{K}(X)$	This convention quotients by all compact operators. It is not a finite-rank quotient, and the qualified positive-unital theorem additionally requires its stated Banach-lattice, quasi-interior-unit, pole, and positive dual-eigenfunctional hypotheses.
$L_0^2(\mu);\ \mathscr{S}_\pi[\varphi] = -\varphi + \pi(\varphi)$	centered square-integrable score tangent; score of the normalized exponential action path	$\mathscr{S}_\pi$ is an isometry from the action quotient with the Fisher norm $\|[\varphi]\|_F$ onto the bounded centered subspace, whose Fisher completion is $L_0^2(\pi)$. The Fisher norm is not the oscillation norm, and the completion is not a domain of the nonlinear action map Q_ℓ.
$\mathscr{I}_b;\ U_b;\ \mathscr{L}_b = U_b\mathscr{I}_b$	iid score replication lift; block reverse conditional expectation; score renormalization operator	$\mathscr{I}_b h = \sum_{i=1}^{b} h(x_i)$ with $\|\mathscr{I}_b\| = \sqrt{b}$. The replication $\mathscr{I}_b$ is not the Hoeffding assembly E_ℓ; $\mathscr{L}_b$ is not the ELBO $\mathcal{L}$. $\mathscr{L}_b$ is an endomorphism only when the block statistic is normalized so that its law equals the input law.
$\mathrm{He}_k;\ e_k = \mathrm{He}_k/\sqrt{k!}$	probabilists' Hermite polynomials and their normalized orthonormal basis	Complete orthonormal in $L^2(\gamma)$; deleting e_0 gives a basis of $L_0^2(\gamma)$. Every e_k is unbounded, so a Hermite basis is not a basis of $L^\infty/\mathbb{R}\mathbf{1}$ and, for $k \geq 3$, admits no two-sided exponential action neighborhood.

$\Delta s_\ell = \log b_\ell;\ s_{n\leftarrow\ell} = \sum_{k=\ell}^{n-1} \Delta s_k$	adjacent and cumulative log block scale	$s_{n\leftarrow\ell} = \log B_{n\leftarrow\ell}$. It is a scale index, not a duration; every reported growth rate is divided by it.
$D_\ell = DT_\ell(x_\ell);\ D_{n\leftarrow\ell} = D_{n-1}\cdots D_\ell;\ \widehat{D}_{n\leftarrow\ell}$	derivative along an exact orbit and its ordered composition	The rightmost factor acts first. The hatted cocycle is the one transported by the declared comparison trivializations. These are maps between different spaces; an ordinary spectrum requires a declared identification.
$v_{\ell,a}, \lambda_{\ell,a};\ \lambda_{n\leftarrow\ell,a}$	compatible cross-scale mode line and its coefficients	Defined by $D_\ell v_{\ell,a} = \lambda_{\ell,a} v_{\ell+1,a}$ and the ordered product law. The equation $D_\ell v = \lambda v$ is ill typed and is not used.
$\chi_{n\leftarrow\ell}(u);\ \upsilon_{n\leftarrow\ell,a}$	finite-time norm growth rate and scalar mode exponent	Both are relative to the declared norms, the declared extensive normalization, the block ratios, and the comparison maps. A Lyapunov exponent is asserted only when the upper and lower asymptotic rates agree; no Oseledets splitting is claimed. The exponent υ_a is distinct from the generative latent variable y_a.
$J_\ell : \mathcal{G}_\star \to \mathcal{G}_\ell;\ \widehat{T}_\ell^{\mathcal{G}} = J_{\ell+1}^{-1} T_\ell^{\mathcal{G}} J_\ell$	comparison trivialization and the pulled exact interaction map	This is the inverse orientation of I_ℓ, so $J_\ell = I_\ell^{-1}$ when both are declared. Exponent invariance requires tempered $\lVert J_n \rVert$ and $\lVert J_n^{-1} \rVert$, not only tempered mode normalization.
$\beta_\ell^{\mathrm{ex}}, \beta_\ell^{\mathrm{ret}}, \delta\beta_\ell$	exact, retained, and omitted discrete interaction beta	All three live in $\mathcal{G}_\star$ and divide by Δs_ℓ. $\delta\beta_\ell$ is the transported coordinate residual; it vanishes on the retained sector exactly under exact-image invariance. Neither is the action beta $\mathfrak{B}_b^H$, whose reference change is inhomogeneous.
$\nabla^{\mathrm{scale}}, \beta^{\mathrm{scale}};\ \mathsf{V}(s,t)$	scale connection on a declared C^1 Banach bundle over a scale manifold, its covariant scale derivative, and a scale evolution family	The scale connection has base the scale manifold and is not either contextual principal connection ω_b, ω_m, nor the inference-orbit parameter, nor Fisher duration. Discrete endpoint data do not determine it.
$\widehat{\mathcal{M}}_\ell^{(\tau)}$	base-ℓ reference monodromy of a p-periodic identified scale sequence	It is an endomorphism of the declared reference space $Y_\star^{(\tau)}$. A reference monodromy fixed object and all points of its p-cycle lie in that space and need not be fixed by any one-step reference map.
$\Pi^{\mathrm{res}} = \mathsf{PC};\ \mathsf{Q} = I - \Pi^{\mathrm{res}}$	resolved projection and its complement on the linear observable tier	Idempotent by $\mathsf{CP} = I$. This projection is not the reverse kernel Π_ℓ, not the posterior Π_o, and not the retained interaction projection R_ℓ.
$R_\rho(z, dy)$	reverse conditional of the probability reference ρ under the coarse channel	Used by the conditional-partition formula and by the reference-change law for the action beta. It is neither the retained projection R_ℓ nor the stochastic refinement kernel R below.
$C_{\ell k} : \mathfrak{X}_\ell \to \mathfrak{X}_k$	arrow in the scale diagram	Composes abstractly as $C_{kr} \circ C_{\ell k} = C_{\ell r}$. Right-acting kernel juxtaposition is the separate convention stated above. A renormalization dynamics also needs comparison and rescaling maps.
$c_\ell, C_\ell^b, C_\ell^m$	base and channelwise adjacent-scale maps	C_ℓ^x covers c_ℓ. Their failures to commute with $\Phi_\ell, \widetilde{\Phi}_\ell$ or parallel transport are covariant defect diagrams.

Q, R	coarse map and stochastic refinement kernel	For noninjective Q, $Q_{\#}R(c,\cdot) = \delta_c$ is only a stochastic right section; it does not recover the discarded realized microstate.
$P_b, P_m;\ G_b \times G_m$	optional product-gauge extension	Used only when independent principal topology or independent physical gauge symmetries are intended. It is not required for separate frames, representations, or connections in the ambient theory.

Three transports that must remain distinct. Ω_γ and $\widetilde{\Omega}_\gamma$ move laws between fibers over different base points in one channel. $T^x_{ij}(c)$ compares two coordinates in the same principal fiber. Θ^x_e is a declared edge-copy variable on the interaction complex. Only an explicit hypothesis may identify the represented Θ^x_e with parallel transport along an assigned base curve.

Three uses of "pullback." The informational tensor pullback of Chapter 8 uses the connection-relative vertical jet $D^\omega s$ to obtain tensors on the contextual base. The parameter restriction of Proposition 16.5 is an ordinary tensor pullback along an injective map of statistical parameter manifolds. A pullback attractor in Section 11.4 is instead a backward-asymptotic object for a nonautonomous cocycle. These are three different typed constructions; their shared word transfers no theorem.

Four notions of order and path. A base curve $\gamma \subset \mathcal{C}$ orders contexts and carries no verticality or horizontality predicate at all. A bundle curve Γ has stationary, vertical, horizontal, or mixed velocity at each parameter value relative to a connection, and the interval labels obtained by quantifying those pointwise types are not exhaustive. An inference history $\mathscr{H}_i$ is an oriented unparameterized orbit in a section configuration space; its evaluation at each fixed context is vertical, and verticality predicates apply to that evaluation rather than to the configuration curve. An ordered interaction-record path retained by exact coarse-graining is a sample-space coordinate, while RG depth is a scale index. None is identified with physical time.

Finite-design link copies. The full context-dependent product of frame-coordinate changes uses $\Theta^x_{a,e}$ with discrete vertex action $h^x_{a,i} = g^x_i(c_a)$, abbreviated as $\Theta^x_{a,ij}$ when the edge copy is unique. A submodel that constrains one shared link across all a retains the stabilizer for which $h^x_{a,i}\Theta^x_{ij}(h^x_{a,j})^{-1}$ is independent of a. Context-independent rechoices are sufficient but need not exhaust that stabilizer; any larger law requires a declared compensation.

Two uses of "coarse." A Markov channel discards sample information while preserving normalization. A hard restriction substitutes a lower-dimensional configuration into an energy. A Galerkin map compresses an operator. These operations can agree in a special realization, but no shared word changes their types or transfers a theorem from one to another.

Appendix B

Central Open-Obligation Ledger

This appendix collects the manuscript's unresolved obligations in one place. It does not upgrade any of them. A *conjecture* is a favored, falsifiable statement that has not been proved; an *open problem* has no presumed answer. The local theorem statements and hypotheses remain authoritative.

B.1 Mathematical and probabilistic closure

Continuum law theory (open). Construct a refining projective family of normalized finite laws, prove tightness in a declared topology on sections, control gauges and reference measures, and pass the ELBO, connections, and observables to the limit. Finite-dimensional consistency alone does not do this. [OPEN]

Pointwise-dominated and nondominated likelihood families (open). The finite interaction theory uses one family-level dominating measure for each record kernel and a finite jointly measurable density version. If domination exists only separately at each latent state, pointwise Radon–Nikodym versions need not supply a common measurable likelihood $L_o(y)$. Extend the construction directly at the joint-measure and regular conditional-law level, or prove hypotheses giving jointly measurable versions, and specify how record-null version choices propagate through local disintegrations and coarse maps. [OPEN]

Regular frame-coordinate quotient (open). Determine generic stabilizers of the passive $G \times G$ action generated by independent belief and model trivializations, and identify strata on which the orbit space is a regular quotient. This coordinate action is not the principal structure group. A dimension subtraction is invalid without freeness and properness. [OPEN]

Optimization, scheduling, and projection (open). The finite conditional-gap and local–global potential identities establish objective compatibility, not convergence. For each proposed algorithm, declare the sequential, asynchronous, or parallel schedule and give sufficient compact-sublevel, lower-semicontinuity, differentiability, and exactness or line-search/damping hypotheses for convergence to a stationary point, or give a counterexample in the declared family. Parallel replacement of correlated full conditionals additionally requires proof that the replacements define one joint law. Separately, prove when Bregman projection is gauge compatible and commutes with a coarse parameter map, and determine whether a positive mass pin is relevant, marginal, or irrelevant under that map. [OPEN]

Joint-law lift (open). An exact VFE history requires the commuting conditional/full-law lift of Hypothesis 9.9, including the dependence coordinates omitted by marginal belief–model sections. Existence of such a smooth right inverse, nondegeneracy of its Fisher pullback, and preservation under coarse graining must be proved for each declared recognition family; they do not follow from the two associated bundles alone. [OPEN]

Global relational information clock (open). The finite-dimensional Fisher length of one regular oriented history is defined and reparameterization invariant. A scalar clock shared across agents or patches additionally requires the exactness and period conditions isolated in Theorem 9.15, compatibility across critical and null strata, and, for a continuum of sections, the missing base measure and function-space theory. None of those follow from local dissipation alone. [OPEN]

B.2 General coarse maps and renormalization

Partition selection and experiment-level recovery (open). Theorem 10.13 proves that the marginal-law distortion is invariant under gauge, path, and root choices. It characterizes its attained zero as a holonomy-stabilized parallel marginal section. The normalized evidence-preserving joint-law push-forward is now supplied by Theorem 12.1; what remains open is one common recovery kernel for an entire statistical experiment and a relabeling-natural, gauge-compatible nondegenerate partition selector. [OPEN]

Infinite-volume RG limit (open). The finite exact scale theory supplies normalized kernels, pushed measure pairs, a local analytic bounded-action calculus, its L^p conditional expectation extension, the exact conditional-variance loss, full power-set Hoeffding interaction coordinates, and an exact nonlinear interaction map with a typed projection residual. The interaction theorem separately assumes an equivalent product reference at every admitted scale; an arbitrary normalized Markov arrow need not preserve that tier. None of these finite-step theorems supplies a thermodynamic limit. An infinite-volume limit still requires projectively consistent cylinder laws, boundary conditions or a DLR specification, tightness in the selected state topology, convergence of free-energy densities, and compatible limits of the connection and cross-morphism components. [OPEN]

Two-index limits and universality (open). Prove a nontrivial diagonal limit or an exchange theorem for system size n and RG depth ℓ. Then establish convergence of invariant observables, basin independence, and blocking-scheme robustness. A finite congruence identity or one matched exponent is insufficient. [OPEN]

Extensive relevance beyond the scalar Gaussian realization (open). Theorem 12.12 inhabits all three relevance classes for independent scalar Gaussian coordinates under the declared replication sum, block statistic, and fixed Fisher norm. Proposition 12.14 shows that correlated coordinates, unrestricted multivariate covariance, and general $\mathrm{GL}(d)$ actions each leave that theorem. Obtaining the corresponding spectrum for a correlated, anisotropic, or gauge-mixed block requires a declared transported family of laws and tangent norms together with a separate diagonalization. [OPEN]

Nonlinear attraction at the Gaussian tangent (open). The Hermite spectrum controls the derivative of the score map at the Gaussian law. A basin of attraction, a remainder estimate for the nonlinear map, and the fate of the marginal degree-two direction do not follow from a linear spectral statement and are not supplied here. [OPEN]

Multiplicative ergodic structure of the interaction cocycle (open). Definition 11.10 and Theorem 11.14 define compatible cross-scale modes and give exponent invariance under tempered comparison data. An Oseledets splitting would additionally require a measure-preserving base flow, measurability of the cocycle in the base variable, integrability of $\log^+\|D\|$ and of $\log^+\|D^{-1}\|$, and either finite dimension or a compactness hypothesis. A deterministic blocking sequence supplies none of these, so the growth rates remain definitions rather than an existence theorem. [OPEN]

Smooth scale tier and continuous beta (open). Definition 12.28 declares the scale manifold, Banach bundle, and scale connection that a continuous beta requires, and Proposition 12.29 shows that a discrete blocking sequence does not determine them. Selecting a canonical interpolation, or proving that a reported continuous beta is independent of the choice inside a declared class, remains open. [OPEN]

Nontrivial invariant retained subspace (open). (12.139) settles exactly when a retained beta function is exact. Whether any retained subspace other than the zero and full ones is invariant under $T_\ell^{\mathcal{G}}$ for an admitted network and blocking channel is not settled here, so every reported projected flow carries its residual $\delta\beta_\ell$. [OPEN]

Bayesian-RG bridge (open). To compare this flow with Bayesian renormalization, place both on one state space, transport the relevant Fisher tensors, declare a common scale map, and prove a monotone. Similar Gaussian formulas do not prove opposite flow directions. [OPEN]

Fine–coarse history semiconjugacy for the declared RG maps (open). Theorem 9.16 compares Fisher duration along a single path pushed through a parameter-independent record channel, and Theorems 9.22, 9.24 and 9.25 and proposition 9.23 settle what oriented semiconjugacy does and does not give once it holds. What is not proved is that this manuscript's own independently recomputed coarse natural-gradient flow satisfies it. One structured sufficient route is the one isolated at Proposition 9.28: exhibit χ_ℓ with $\chi_\ell' > 0$ and an open set containing the fine orbit on which Hypothesis 9.27 holds, prove for the exact-contraction coarse objective that the orbit lies in the interior of the attaining set, and then verify horizontal conformality of the declared R_ℓ. This route is not asserted necessary or minimal. Behavior at critical and singular strata remains part of the open work. The route is satisfiable, since the exhibited tier of Definition 9.6 meets it with $a_\ell = \frac{1}{2}$. [OPEN]

Configuration tier for the declared recognition family (open). Definition 9.6 and Theorem 9.7 exhibit a nonempty finite-dimensional configuration manifold with a strong constant Gram metric, a rank test for nondegeneracy, and a closed gauge-orbit tangent, and the infinite-dimensional L^2 and Sobolev tiers are separated with the failure witness attached. What is not settled is whether the manuscript's own declared recognition family can be brought to one of those tiers at every scale at which a natural-gradient field or a Fisher duration is asserted. Until that is done, the history theorems hold on the exhibited tier and are conditional elsewhere. [OPEN]

Smooth structure of the projectable set (open). Theorem 8.13 characterizes which fine sections descend, and Proposition 8.14 shows the descendable set is proper and can have empty interior. Whether that set is a smooth submanifold of a declared configuration manifold carrying a smooth induced arrow needs a transversality or elliptic-regularity hypothesis that is not supplied. This manuscript avoids the question by declaring R_ℓ separately; the pointwise-induced route remains open for anyone who prefers it. [OPEN]

Proper, free, and isometric gauge action in infinite dimensions (open). The available witness for failure of the orthogonal quotient speed is free and isometric with a dense, hence nonclosed, orbit tangent. Properness would force closed orbits, so that witness does not settle whether free, proper, and isometric action alone suffices. Either a proper witness or a proof under properness is required. [OPEN]

B.3 Multivariate-Gaussian realization

Scalarized attraction (conjecture). Conjecture 18.4 predicts a unique attracting spatial ray only after the matrix direction M_0 and one primitive spatial map B are fixed and the Perron vector factorizes. Primitivity, factorization, and the fixed hierarchy still require proof in any application. The claim does not cover the full matrix cone or a nonautonomous cocycle. [CONJECTURE]

Admissible cone classification (open). Extend the exact classification beyond the fixed congruence-diagonal cone to correlated or nonconvex constraints, variable charts, parameter-dependent domains, and nonflat link data. [OPEN]

Nonflat-link compression (open). Characterize when the rectangular endpoint data of a blocked cut compress to one invertible transporter and one SPD weight, remains closed under another merge, and admits a normalized probability and rescaling lift. [OPEN]

Intrinsic scale selection (open). Find a nondegenerate criterion internal to one fixed evidence problem, or prove endpoint degeneracy for a stated class. A successful alternative may be a normalized Markov pushforward or a proved Lyapunov functional; the determinant gap and quotient-volume convention do not supply one. [OPEN]

Information-geometric transfer (open). Identify the coarse recognition parameter with a parameter of the fitted generative law, prove nondegeneracy of the relevant Fisher metric, and then test the proposed scale construction. Recognition-space and model-space Fisher tensors are not interchangeable by name. [OPEN]

Stochastic inverse RG (open). Choose compatible priors for discarded internal edges, nested refinement kernels, chart data, and a projective-limit theorem. Infinite divisibility can redraw compatible microstates in distribution; it cannot invert a many-to-one realized blocking. [OPEN]

Update robustness (open). Extend the anchored Gaussian convergence theorem, or find its boundary, for delayed, noisy, asynchronous, or inexact updates. The present contraction proof covers only its stated synchronous exact regime. [OPEN]

B.4 Geometric and physical interpretation

Operational base holonomy (open). Choose a named principal bundle and connection, an assigned base loop, a gauge-invariant population-record statistic, and a typed map from connection data to its record law; then construct two admissible connection data sets with every declared non-holonomy input fixed, distinct represented loop-holonomy conjugacy classes, and differing statistic laws, or prove insensitivity for that same declared controlled class. A stronger sufficient formulation requires the record-law map to factor nontrivially through the represented holonomy conjugacy class. Graph-link holonomy alone does not settle this question, and no such operational tuple or empirical evidence is presently supplied. [OPEN]

Graph-to-base identification (open). Embed the interaction complex in $\mathcal{C}$, assign oriented piecewise-smooth curves, and prove $\widehat{\rho}_b(\Theta_e^b) = \Omega_{\gamma_e}$ and $\widehat{\rho}_m(\Theta_e^m) = \widetilde{\Omega}_{\gamma_e}$. Even then the result concerns connection holonomy, not bundle topology. [OPEN]

Canonical nondegenerate pullback geometry (open). Theorem 8.4 proves passive gauge covariance for a fixed section and chosen connection, while Theorem 8.8 identifies the conditional quotient geometry. A canonical or empirically specified connection, relative channel weights, constant rank, integrability of the radical, a regular Hausdorff leaf space, basicness of both induced tensors, and a physical interpretation remain separate obligations. No canonical connection is selected anywhere in this manuscript. Every pullback tensor, and every mixed or section-induced length that uses a horizontal–vertical split, stays connection relative; every Fisher duration stays relative to its declared metric. By contrast, the vertical Fisher length of a curve confined to one fixed statistical fiber uses only the vertical Fisher metric and is connection independent, although it remains metric relative. [OPEN]

Necessity in the base Fisher cocycle and the positivity criterion (open). Theorem 8.24 gives the exact residual of the base Fisher-defect cocycle and shows that the sharp cocycle holds exactly under the quadratic-form equality (8.72) (an equal-seminorm criterion only under the Markov/positive-semidefinite hypothesis), for which three sufficient conditions are recorded and none is necessary. Similarly, Theorem 8.17 shows that a vanishing horizontal defect is sufficient and not necessary for base positivity. Classifying the accidental solutions of either criterion, in terms of intrinsic data rather than by inspection of a chart, is open. [OPEN]

Physical-time identification (open). Fisher duration on an oriented inference history is unchanged by an orientation-preserving change of parameter. It is a statistical length, not a physical clock. A physical-time claim would require a named target clock, a calibration map and uncertainty model, causal and compositional consistency, and discriminating evidence against alternative parameterizations. None is supplied here. [OPEN]

Open Problem B.1 (Physical-law identification). Identifying a repaired limiting object with a physical law requires a named target system, an observable and estimator, a frozen validation and uncertainty protocol, a baseline margin, probability-level closure, and held-out evidence robust across admissible blocking schemes. None is supplied here. Accordingly, the present mathematical theory neither proves a physical law nor supports ontological closure. [OPEN]

Appendix C

Numerical Provenance

The reproducibility package is part of the manuscript record:

`verification/claims.json`	claim inventory, seeds, sample counts, tolerances, mappings, and open obligations
`verification/run_checks.py`	single deterministic CPU runner
`verification/current-results.json`	current environment, source and protocol hashes, individual outcomes, residuals, and overall result
`verification/VERIFICATION.md`	human-readable protocol and interpretation

From the repository root, the declared run is

```
& "C:\Python314\python.exe" manuscripts/gauge_vfe_rg/verification/run_checks.py
```

The generated JSON record is the authority for exact counts and numerical values; this appendix intentionally does not duplicate seed lists or result tables that can drift.

Frozen factorization-gap stress protocol. The check `CHK-CG-FACTOR-GAP-STRESS-3138` is a new deterministic protocol, not a reconstruction of an undocumented historical run. Seed 20260803 fixes exactly $3{,}138$ case identities: $2{,}400$ general, 200 exactly block diagonal, 200 nearly decoupled, 120 scale, 120 permutation, 80 nested-refinement, and 18 100-decimal-digit control cases. Every stratum covers dimensions 2 through 16 and condition-number tiers $1, 10^2, \ldots, 10^{14}$ as nominal pre-transform generator labels. The legacy field `condition_number` stores that nominal label, while every digest-bound record also stores `achieved_condition_number`, defined as `numpy.linalg.cond` of the regenerated matrix. The observed global range is $[1, 1.015265553314175 \times 10^{14}]$, but this is only a global coverage claim. The achieved ranges are:

Stratum	Minimum	Maximum
Global	1	$1.015265553314175 \times 10^{14}$
General	1	$1.015265553314175 \times 10^{14}$
Exactly block diagonal	1	$1.0075938532153164 \times 10^{14}$
Nearly decoupled	1.0000000000000002	$9.273791153286738 \times 10^{8}$
Scale	1	$1.0036120669923527 \times 10^{14}$
Permutation	1	$1.0071935024197442 \times 10^{14}$
Nested refinement	1	$1.0077056571347422 \times 10^{14}$
100-digit controls	1.0000000000000004	$1.0048132000124055 \times 10^{14}$

The two scalar blocks in the dimension-two exactly-block-diagonal cases cannot realize their nominal targets, and the nearly-decoupled transformation does not attain its high nominal tiers. Each record binds the regenerated binary64 matrix and partition by SHA-256, and repeated executions must agree exactly. The named runtime check separately records all $3{,}138$ attempted schedule cases, successfully evaluated 100-decimal-digit exact-binary64 references, and completed production-reference comparisons. PASS requires $3{,}138$ in each category and zero failures; the 18 designated controls remain a named stratum rather than the sole accuracy evidence. [NUMERICAL]

The ordinary value path uses Cholesky triangular solves, canonical-correlation singular values, and `log1p(-rho**2)`. Near the floating-point boundary, including a raw excursion above one, the runner compares two binary64 solve backends, applies a $64n\epsilon_{\mathrm{mach}}$ operational rejection guard, and uses a 200-decimal-digit evaluation of the exact stored binary64 input for the reported gap. The value path also falls back when $\kappa_2(J)\epsilon_{\mathrm{mach}} \geq 10^{-4}$; this is a frozen-protocol conditioning trigger, not a proved perturbation radius. A nonfinite binary64 condition estimate also triggers exact-input evaluation, rather than rejection, after high-precision Cholesky certification of positive definiteness. The public diagnostic name `residual_derived_clip_bound` is retained for protocol compatibility, but on this boundary path its value is the outward-rounded discrepancy between the maximum raw binary64 ρ^2 and the high-precision exact-input ρ^2. It is not presented as a general residual perturbation bound. The positive excursion witness is a separate focused regression fixture, not an outcome of the ordinary $3{,}138$-case schedule. Its matrix digest is `45310a74550d3759fed0f83f71a6cf3b0f45942499361d723a2248bbe243e2e3`, its 200-digit exact-input gap rounds to 22.777105858844084, its raw excursion is $4.440892098500626 \times 10^{-16}$, and its outward allowance is $9.432369402326953 \times 10^{-16}$. The check mechanically recomputes all four quantities and requires PASS. Its value comes from the exact-input fallback rather than from clipping ρ^2 below one. The compatibility fields `clipping_applied` and `clipping_amount` record that a positive cross-backend excursion was admitted and its maximum excess above one; they do not say that this amount was substituted into the returned value. Exact-input domain violations and raw excursions whose discrepancies exceed the guard are rejected. Cholesky and solve residuals remain local backward diagnostics; the protocol does not promote them to canonical-correlation or gap, relative-entropy, monotonicity, scale, or oracle-comparison forward-error bounds, and no forward tolerance includes them. Forward comparisons instead use independent exact-binary64 high-precision references plus separately declared endpoint-roundoff or protocol-acceptance policies. Their finite-health gate is $64\gamma_n$, with $\gamma_n = nu/(1-nu)$ and binary64 unit roundoff $u = \epsilon_{\mathrm{mach}}/2$. Early high-precision fallbacks selected before residual evaluation serialize the residual, tolerance, and backward-error diagnostics as `null` because they were not evaluated. A residual-health fallback instead retains each finite binary64 measurement, including finite values that exceed the health gate. A field is `null` only when it has no finite usable numeric value, either because an attempted measurement is nonfinite or because a derived diagnostic requires an unavailable constituent; finite sibling fields remain numeric. The evaluation method distinguishes the finite threshold case from nonfinite Cholesky, solve, or tolerance causes and lists multiple causes in that fixed order. These retained measurements remain local backward diagnostics and do not enter a forward-error tolerance. In every branch, `null` denotes neither failure nor zero. The exact zero diagnostics for a one-block, zero-step result remain numeric zero. The `GapStep` API retains the fieldwise residual quartet. A `FactorizationCaseRecord` serializes only the aggregate backward diagnostic, maximum

Cholesky residual, and ordered evaluation-method tuple, not per-step solve or tolerance values. The compatibility field `backward_error_bound` is a sum of one-ULP-up summaries of already rounded local residual measurements; it is not a certified perturbation or forward-error bound. [NUMERICAL]

What a passing run means. It means that the current deterministic, double-precision CPU implementations satisfy their declared finite checks and controls within their recorded tolerances. Those checks corroborate algebraic identities, exercise finite witnesses, and detect regressions in the encoded endpoints. They do not prove a theorem, genericity, an asymptotic limit, universality, or a physical interpretation. Analytic claims rest on derivations or eligible cited results, not on numerical agreement.

Freshness and scope. The result artifact records interpreter and dependency versions and binds the discovered manuscript sources and verification protocol by SHA-256. Any source, protocol, input, dependency, or environment change requires a new run before the numerical evidence is current. The suite uses no GPU and makes no GPU claim. A missing protocol for an empirical statement remains inconclusive even when every implemented check passes.

The checked-in `verification/current-results.json` is the older 29-check artifact. It is not evidence for this 30-check package. A governed 30-check regeneration remains required before that evidence is current. This appendix does not claim that deferred artifact is current.

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